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Freier et al.

(54) **METHODS FOR TREATING HYPERCHOLESTEROLEMIA**

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(21) Appl. No.: **17/987,064**

(22) Filed: **Nov. 15, 2022**

Related U.S. Application Data

(63) Continuation of application No. 17/032,550, filed on Sep. 25, 2020, now Pat. No. 11,530,410, which is a continuation of application No. 16/057,471, filed on Aug. 7, 2018, now Pat. No. 10,858,657, which is a continuation of application No. 15/483,992, filed on Apr. 10, 2017, now abandoned, which is a continuation of application No. 14/539,831, filed on Nov. 12, 2014, now Pat. No. 9,650,636, which is a continuation of application No. 14/152,825, filed on Jan. 10, 2014, now Pat. No. 8,912,160, which is a continuation of application No. 13/302,070, filed on Nov. 22, 2011, now Pat. No. 8,664,190, which is a continuation of application No. 12/478,488, filed on Jun. 4, 2009, now Pat. No. 8,093,222, which is a continuation-in-part of application No. 12/516,457, filed on Jan. 20, 2010, now Pat. No. 8,084,437, filed as application No. PCT/US07/24369 on Nov. 27, 2007.

(60) Provisional application No. 60/867,395, filed on Nov. 27, 2006, provisional application No. 60/912,892, filed on Apr. 19, 2007, provisional application No. 60/986,286, filed on Nov. 7, 2007, provisional application No. 60/988,074, filed on Nov. 14, 2007, provisional application No. 61/058,707, filed on Jun. 4, 2008, provisional application No. 61/059,169, filed on Jun. 5, 2008.

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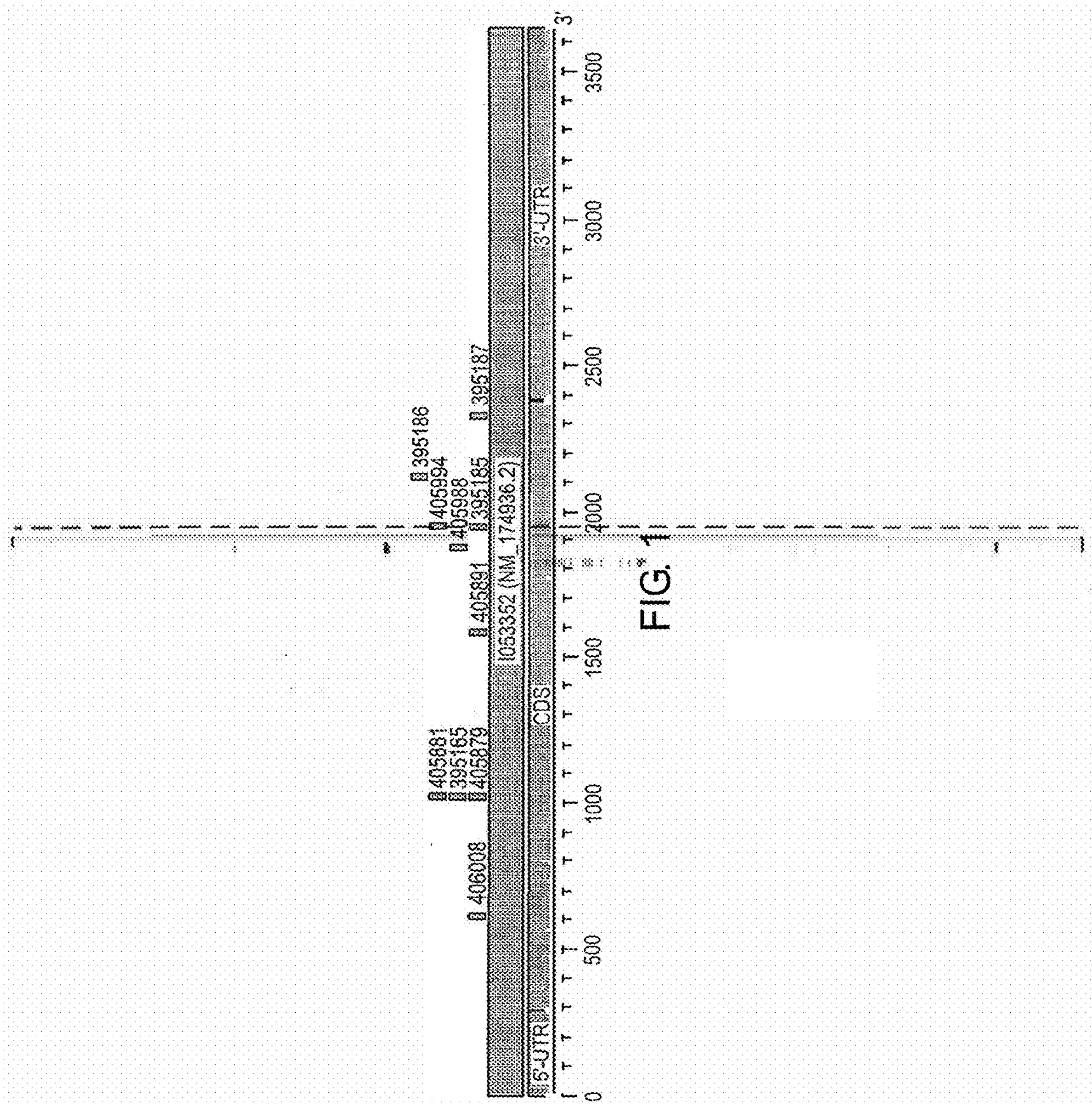
A61K 45/06 (2006.01)

(52) **U.S. Cl.**
CPC ***C12N 15/1137*** (2013.01); ***A61K 45/06*** (2013.01); *A61K 48/00* (2013.01)

(57) **ABSTRACT**

Disclosed herein are antisense compounds and methods for decreasing LDL-C in an individual having elevated LDL-C. Additionally disclosed are antisense compounds and methods for treating, preventing, or ameliorating hypercholesterolemia and/or atherosclerosis. Further disclosed are antisense compounds and methods for decreasing coronary heart disease risk. Such methods include administering to an individual in need of treatment an antisense compound targeted to a PCSK9 nucleic acid. The antisense compounds administered include gapmer antisense oligonucleotides.

Specification includes a Sequence Listing.



395186
405994
405988
395185
395187
405891
053352 (NM_174936.2)
405881
395165
405879
406008
5'-UTR
CDS
3'-UTR
0
500
1000
1500
2000
2500
3000
3500
3'

FIG. 1

FIG. 2

5-point Dose Response with Human PCSK9 Leads in HEK-293 Cells Treated for 24 Hr

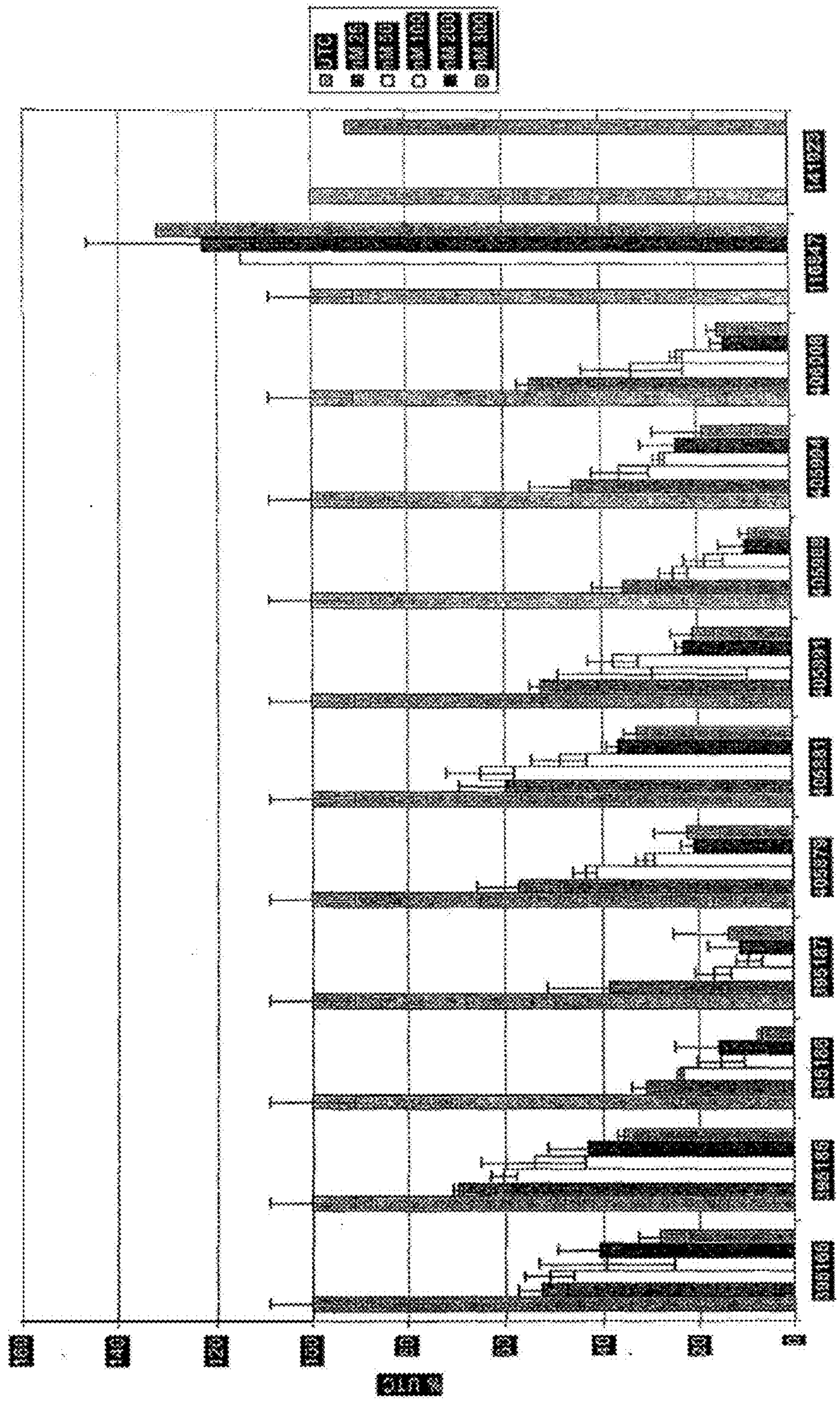


FIG. 3

PCSK9 mRNA levels in Cynomolgus Monkey Primary Hepatocytes 24 Hr After ASO Transfection

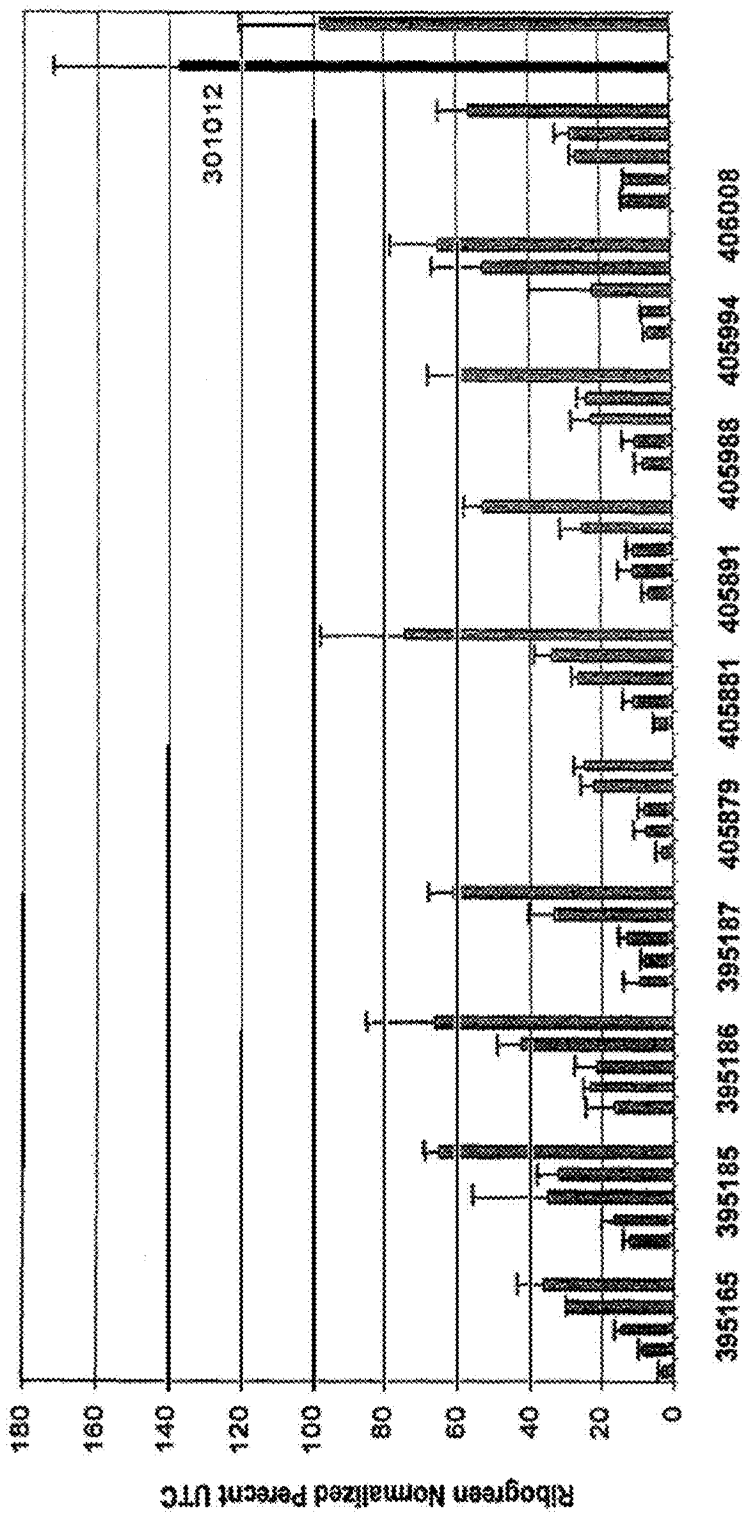


FIG. 4

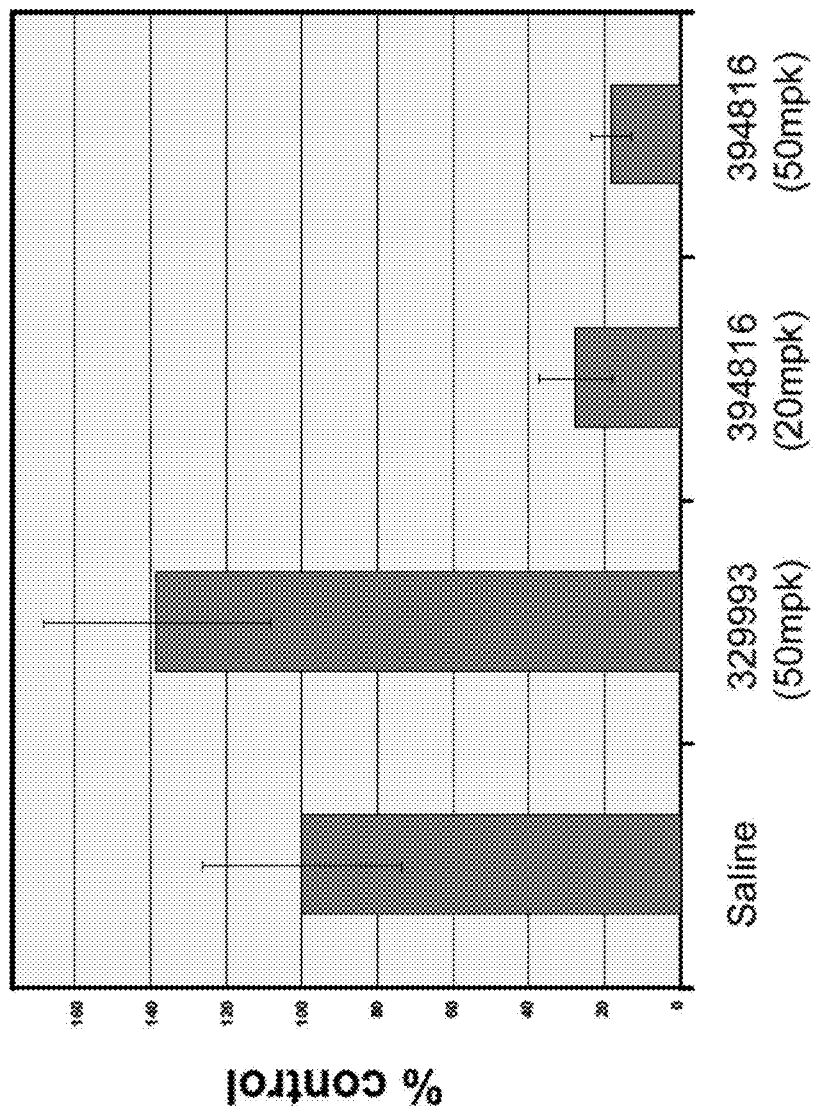


FIG. 5

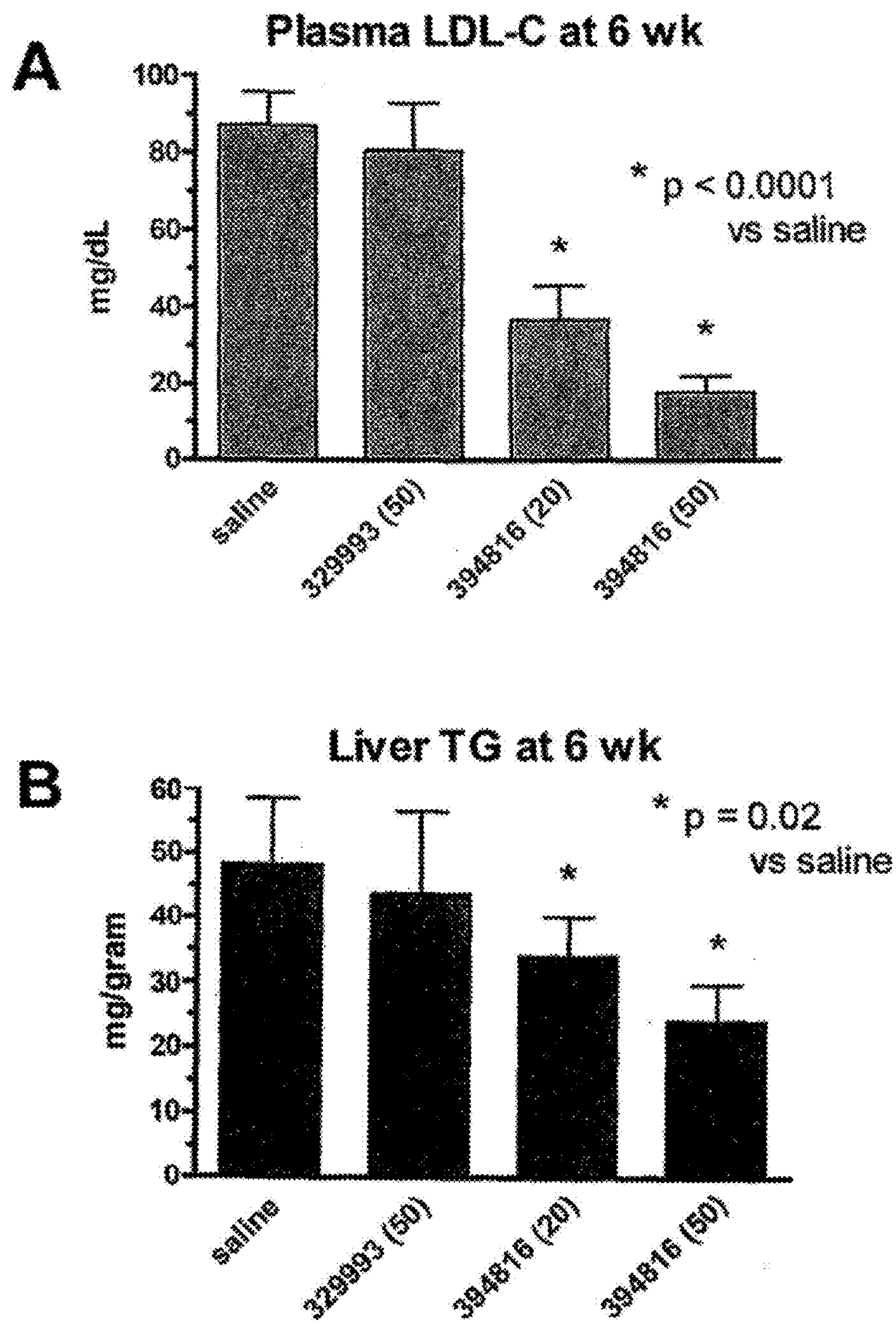

A
Plasma LDL-C at 6 wk
mg/dL
100
80
60
40
20
0
* p < 0.0001
vs saline
saline
329993 (50)
394816 (20)
394816 (50)
B
Liver TG at 6 wk
mg/gram
60
50
40
30
20
10
0
* p = 0.02
vs saline
saline
329993 (50)
394816 (20)
394816 (50)

FIG. 6

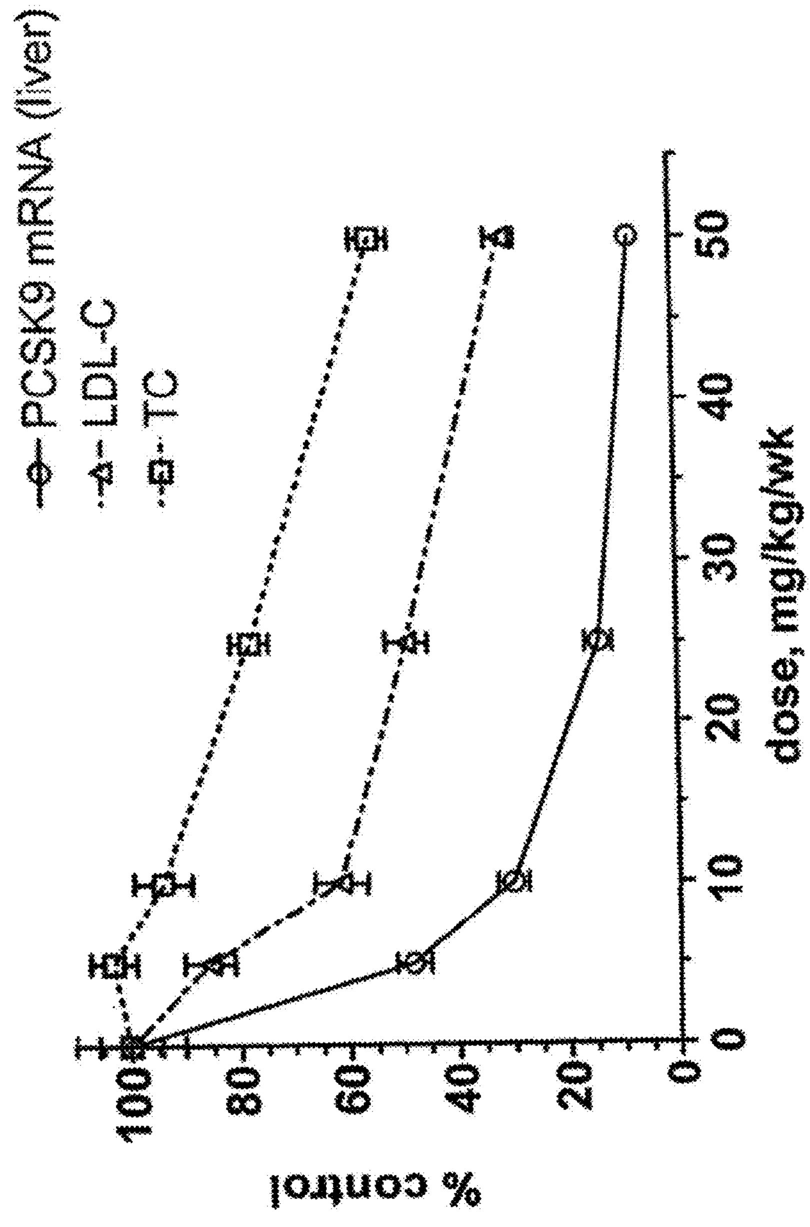

PCSK9 mRNA (liver)
LDL-C
TC
% control
100
80
60
40
20
0
0
10
20
30
40
50
dose, mg/kg/wk

FIG. 7

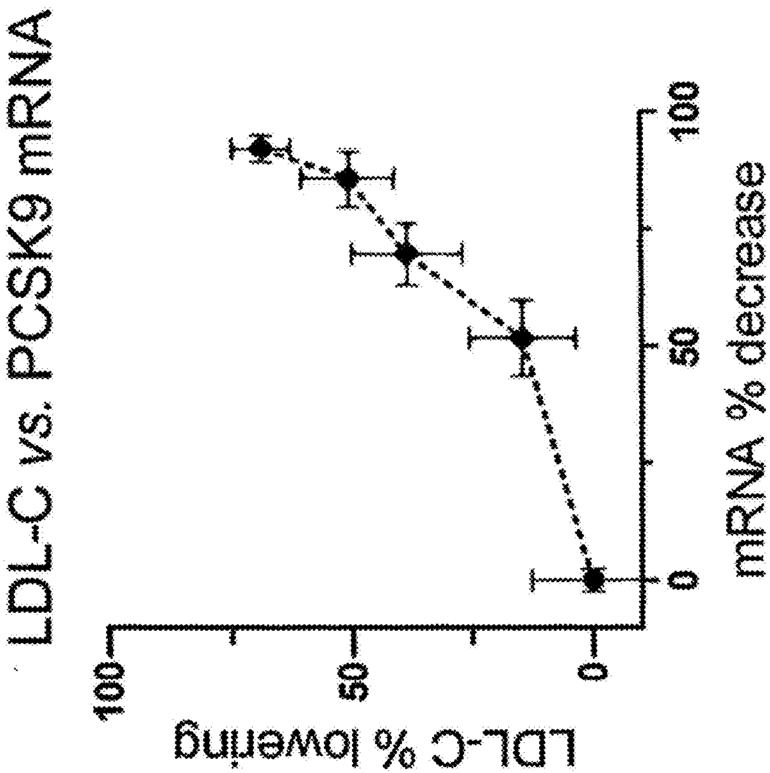

LDL-C vs. PCSK9 mRNA
LDL-C % lowering
100
50
0
mRNA % decrease
0
50
100

FIG. 8

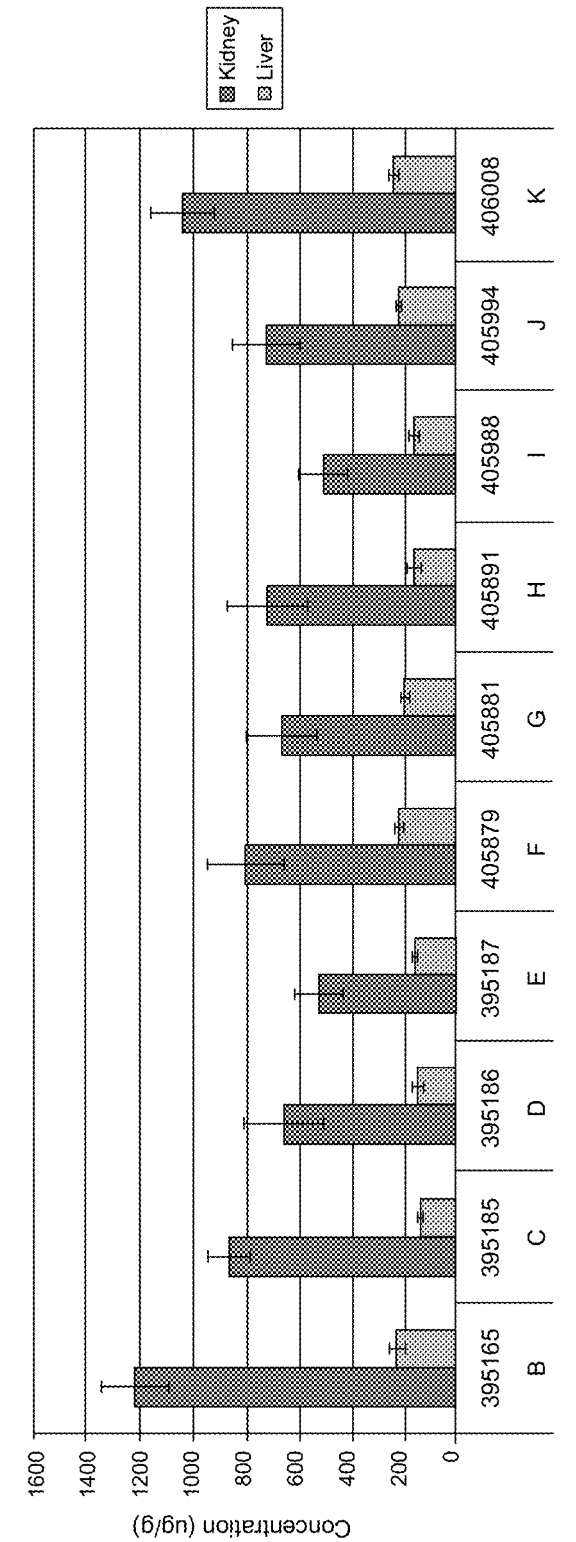

Kidney
Liver
Concentration (ug/g)
1600
1400
1200
1000
800
600
400
200
0
395165
B
395185
C
395186
D
395187
E
405879
F
405881
G
405891
H
405988
I
405994
J
406008
K

FIG. 9

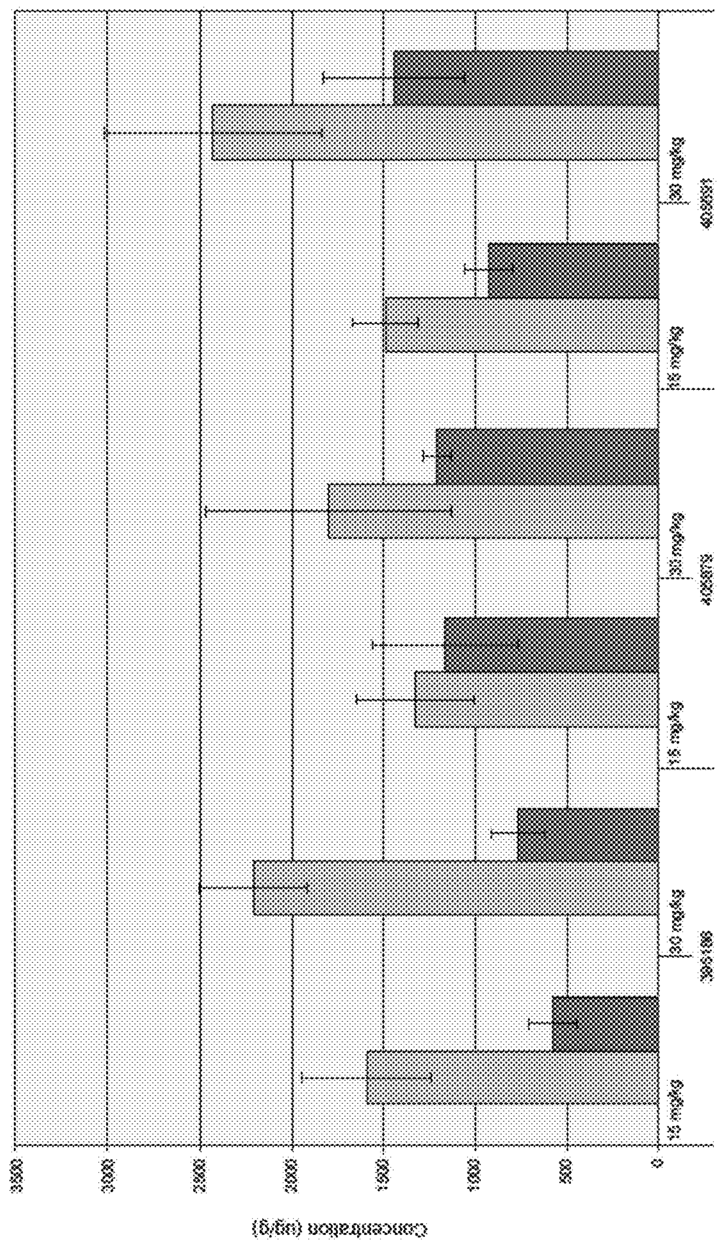

Concentration (ug/g)
0
500
1000
1500
2000
2500
3000
3500
15 mg/kg
30 mg/kg
15 mg/kg
30 mg/kg
15 mg/kg
30 mg/kg

FIG. 10

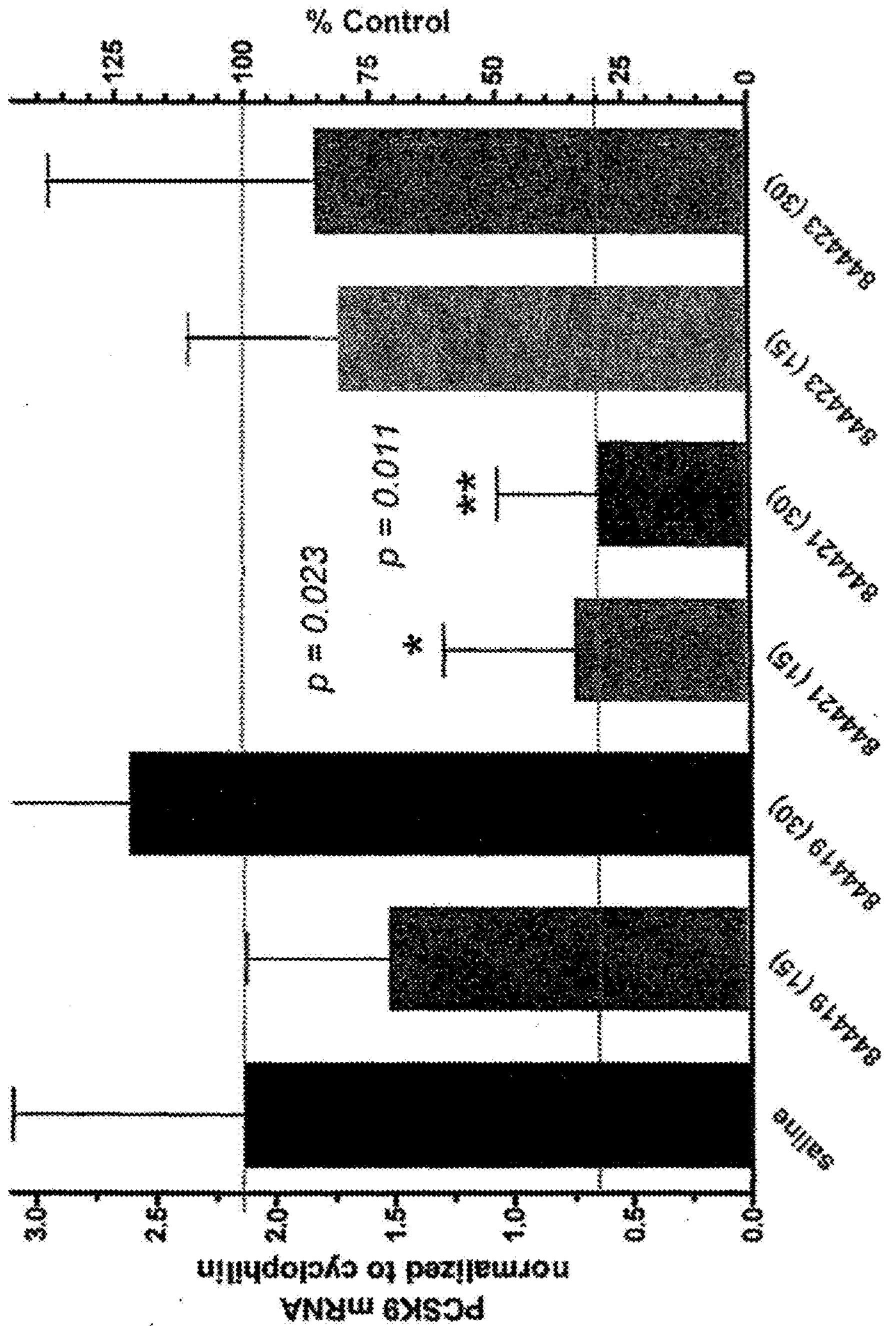

% Control
125
100
75
50
25
0
PCSK9 mRNA
normalized to cyclophilin
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Saline
844419 (15)
844419 (30)
844421 (15)
844421 (30)
844423 (15)
844423 (30)
p = 0.023
p = 0.011
*
**

FIG. 11

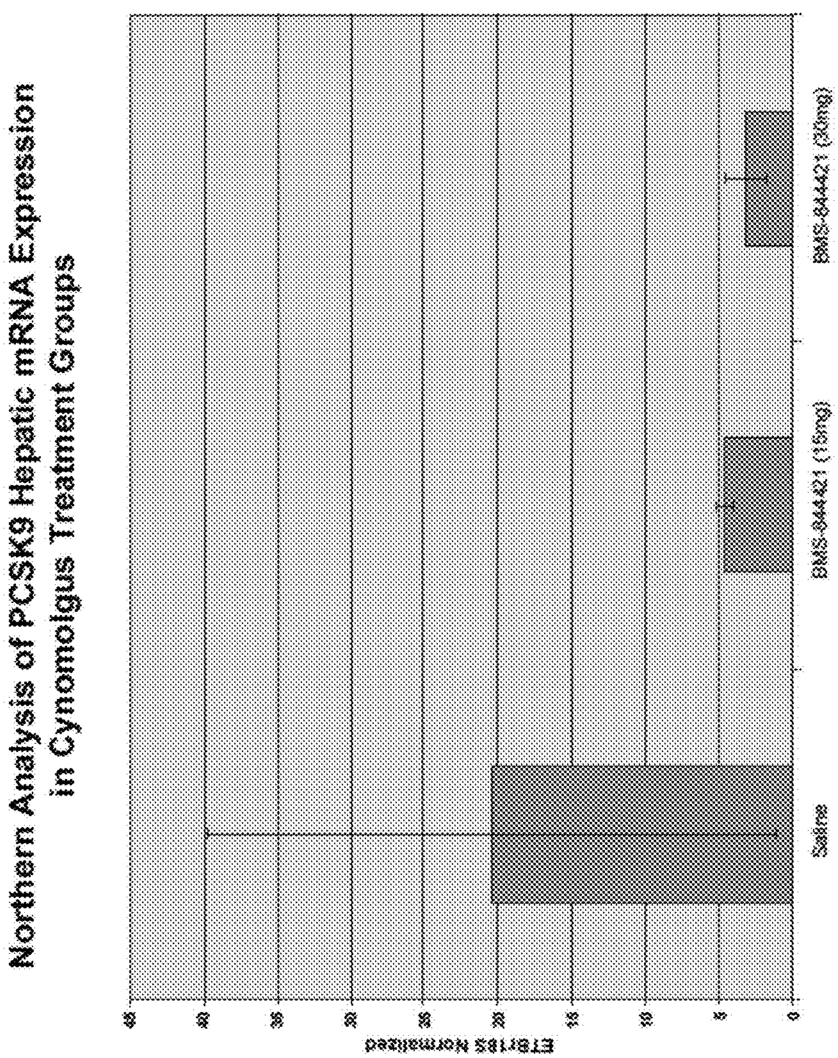

Northern Analysis of PCSK9 Hepatic mRNA Expression
in Cynomolgus Treatment Groups
ETBr18S Normalized
0
5
10
15
20
25
30
35
40
45
Saline
BMS-844421 (15mg)
BMS-844421 (30mg)

FIG. 12

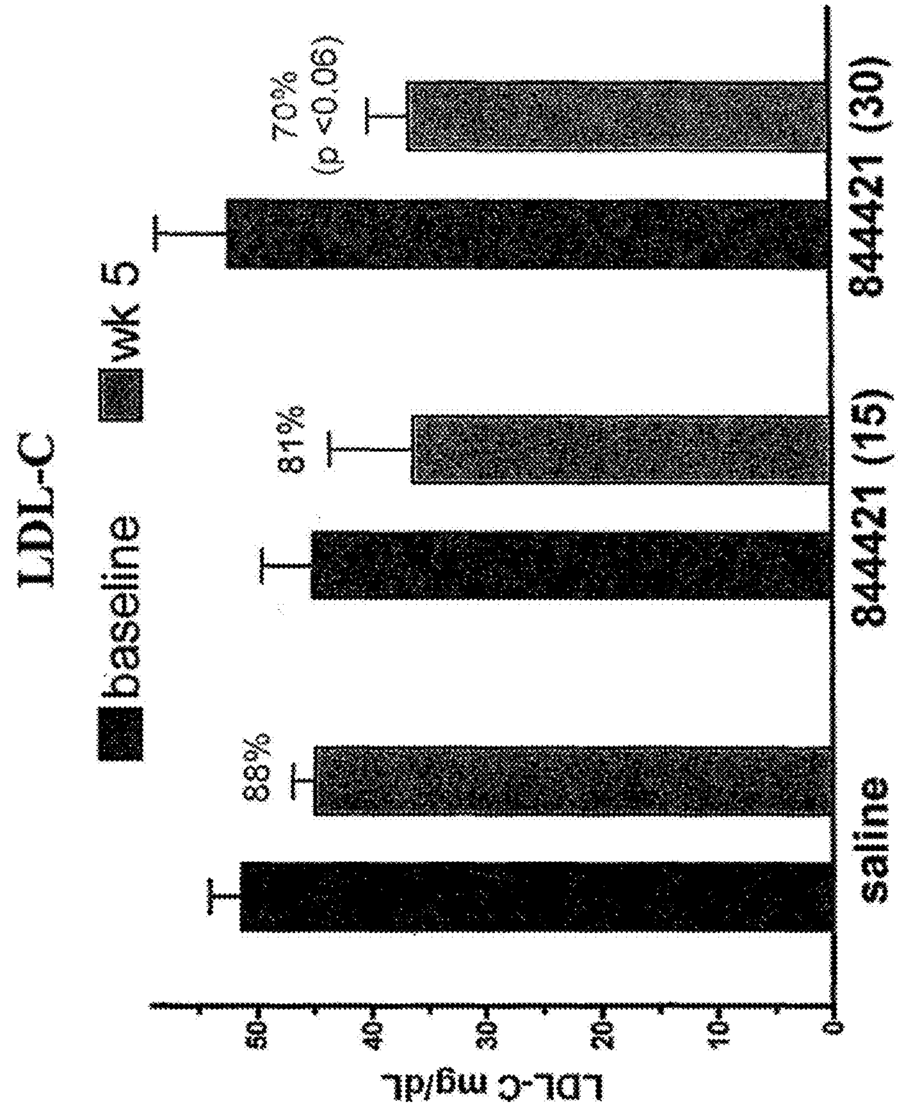
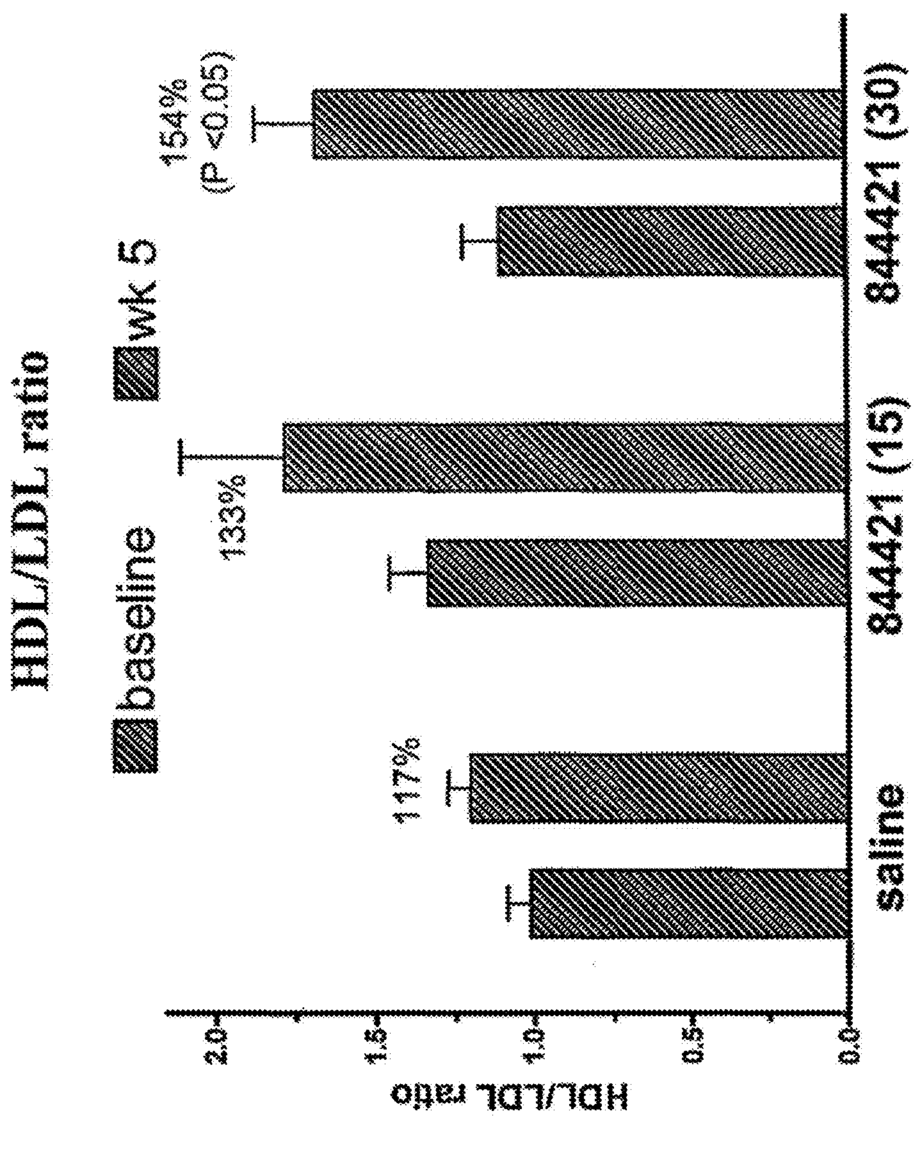

LDL-C
baseline
wk 5
LDL-C mg/dL
0
10
20
30
40
50
88%
81%
70%
(p <0.06)
saline
844421 (15)
844421 (30)

FIG. 13


HDL/LDL ratio
baseline
wk 5
HDL/LDL ratio
2.0
1.5
1.0
0.5
0.0
117%
133%
154%
(P <0.05)
saline
844421 (15)
844421 (30)

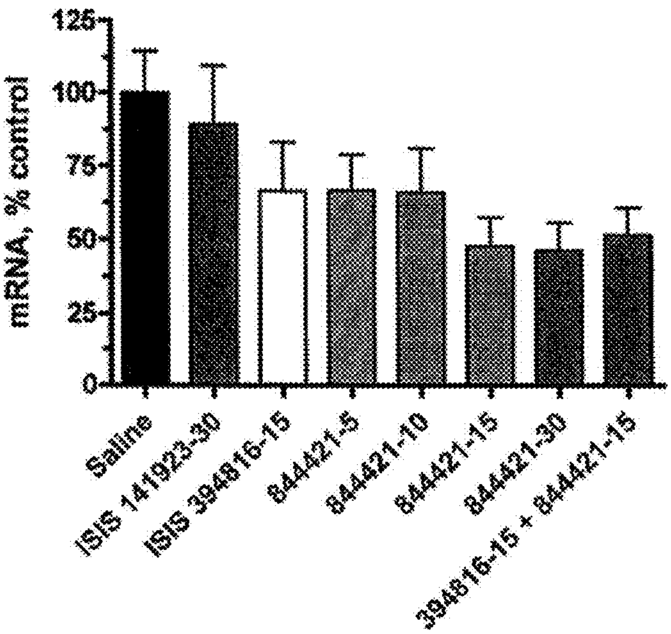

mRNA, % control
125
100
75
50
25
0
Saline
ISIS 141923-30
ISIS 394816-15
844421-5
844421-10
844421-15
844421-30
394816-15 + 844421-15

FIG. 14

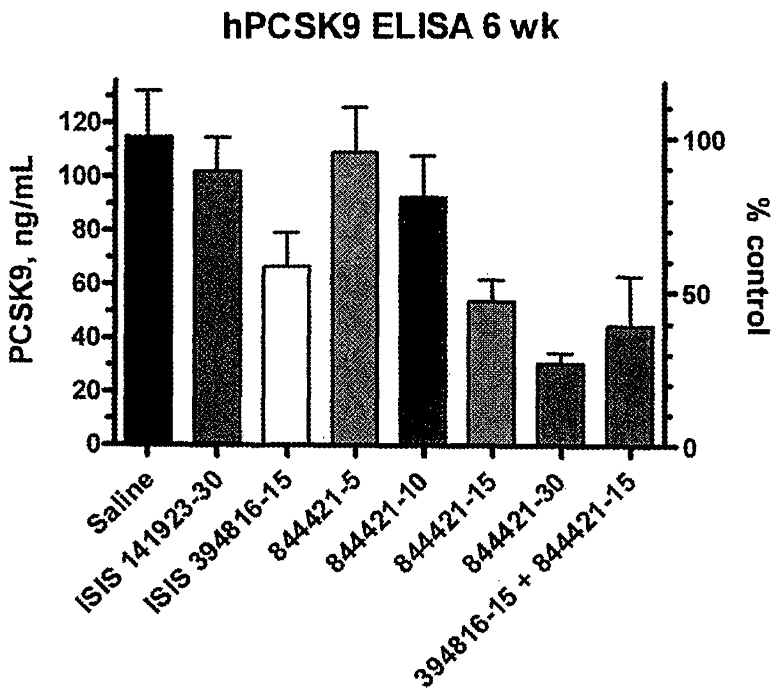

hPCSK9 ELISA 6 wk
PCSK9, ng/mL
% control
0
20
40
60
80
100
120
0
50
100
Saline
ISIS 141923-30
ISIS 394816-15
844421-5
844421-10
844421-15
844421-30
394816-15 + 844421-15

FIG. 15

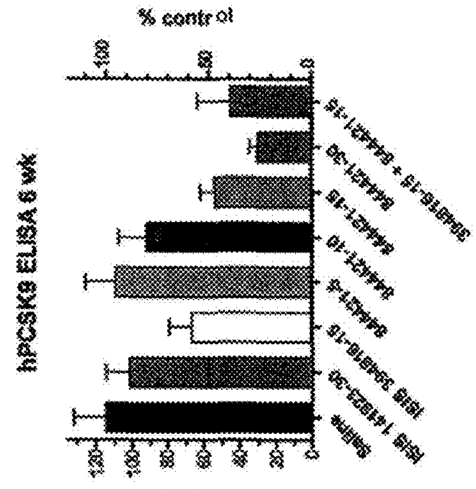


FIG. 16C

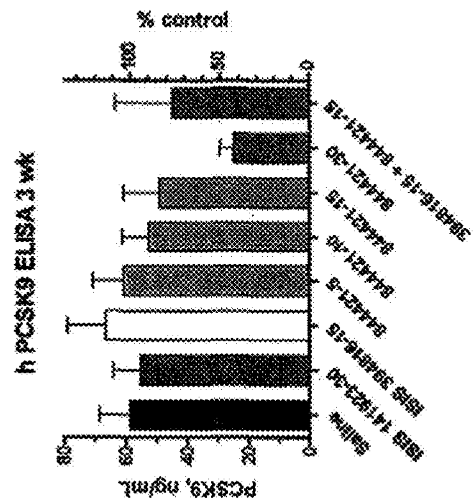


FIG. 16B

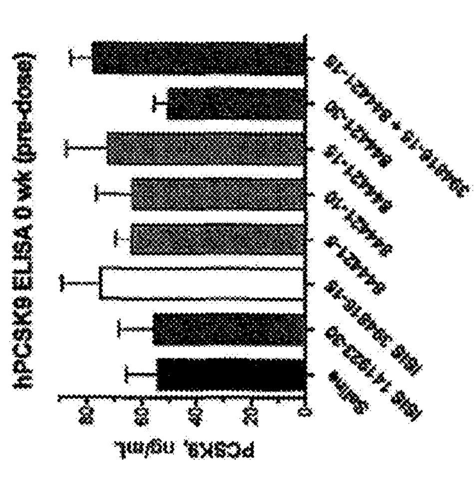


FIG. 16A

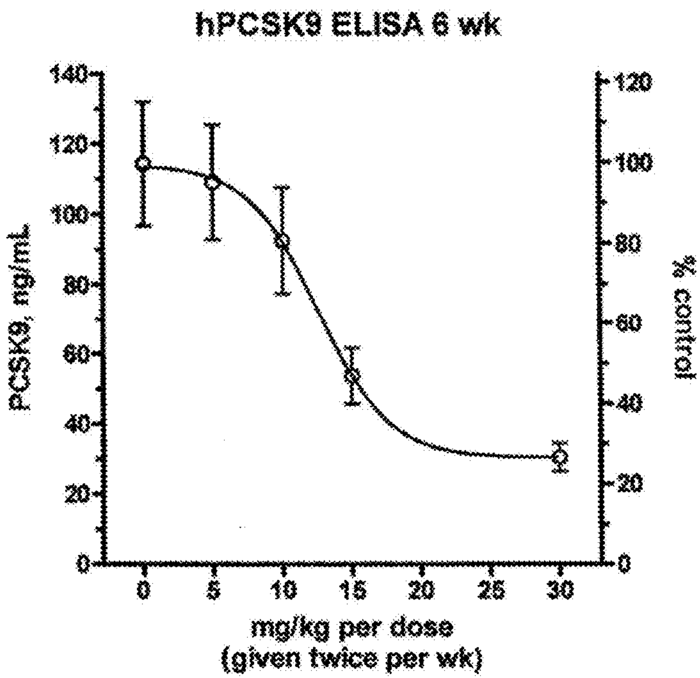

hPCSK9 ELISA 6 wk
PCSK9, ng/mL
140
120
100
80
60
40
20
0
% control
120
100
80
60
40
20
0
0
5
10
15
20
25
30
mg/kg per dose
(given twice per wk)

FIG. 17

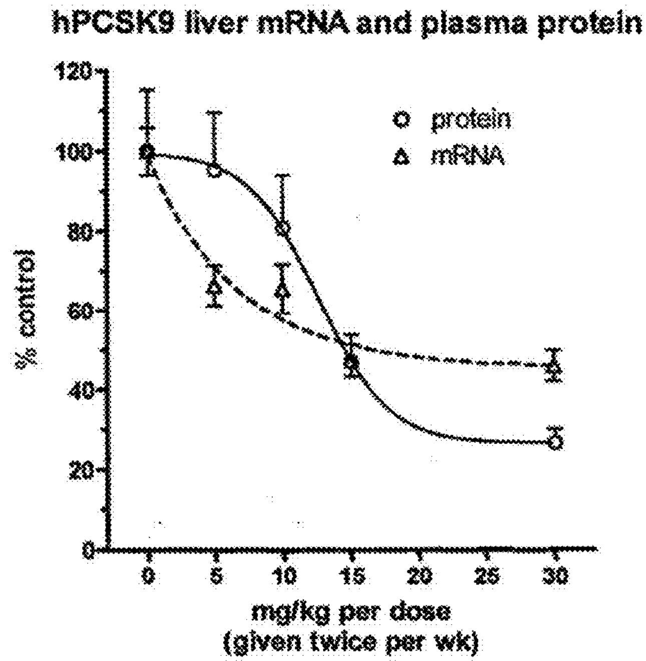

hPCSK9 liver mRNA and plasma protein
120
100
80
60
40
20
0
% control
protein
mRNA
0
5
10
15
20
25
30
mg/kg per dose
(given twice per wk)

FIG. 18

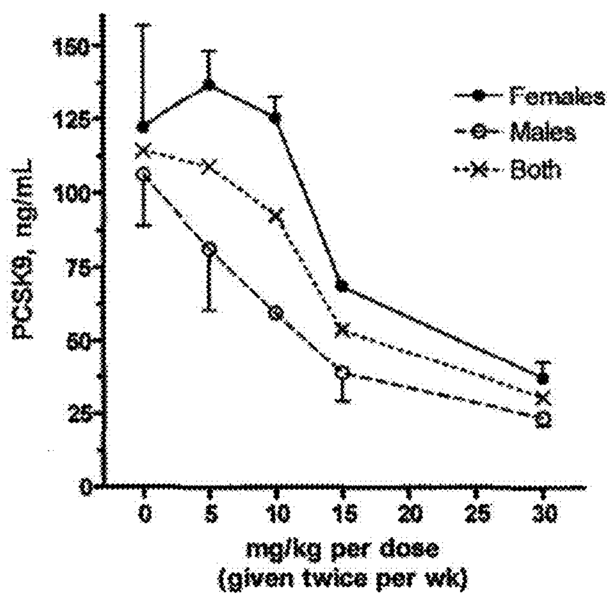

150
125
100
75
50
25
0
PCSK9, ng/mL
0
5
10
15
20
25
30
mg/kg per dose
(given twice per wk)
Females
Males
Both

FIG. 19

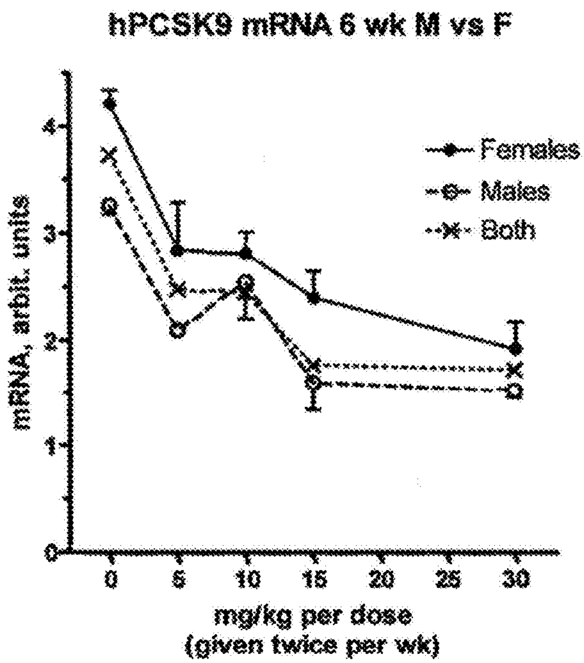

hPCSK9 mRNA 6 wk M vs F
mRNA, arbit. units
0
1
2
3
4
0
5
10
15
20
25
30
mg/kg per dose
(given twice per wk)
Females
Males
Both

FIG. 20

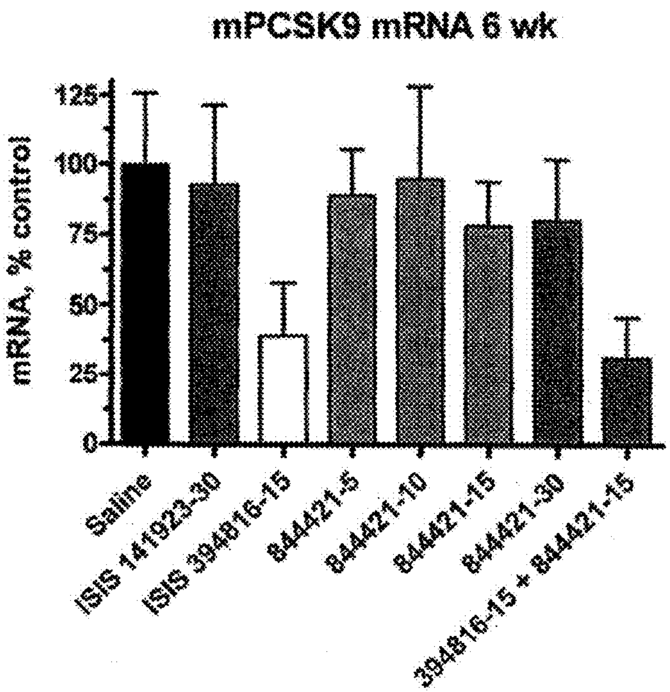

mPCSK9 mRNA 6 wk
mRNA, % control
125
100
75
50
25
0
Saline
ISIS 141923-30
ISIS 394816-15
844421-5
844421-10
844421-15
844421-30
394816-15 + 844421-15

FIG. 21

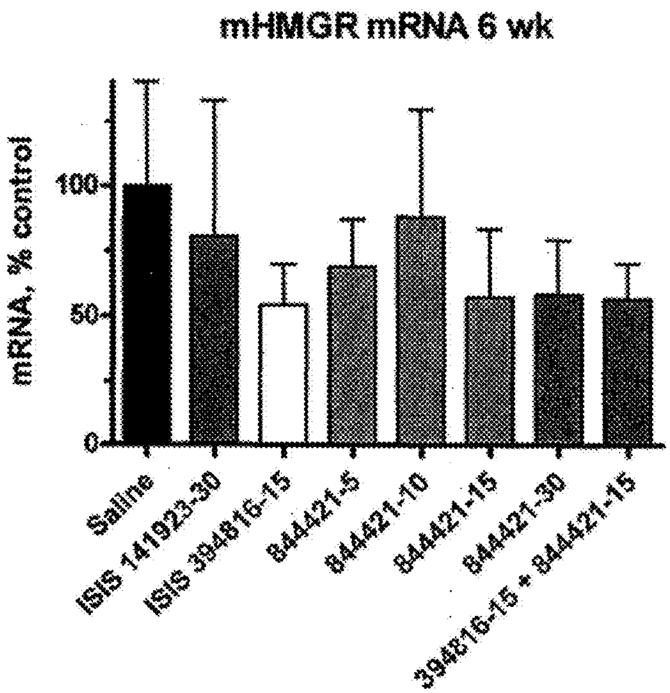

mHMGR mRNA 6 wk
mRNA, % control
100
50
0
Saline
ISIS 141923-30
ISIS 394816-15
844421-5
844421-10
844421-15
844421-30
394816-15 + 844421-15

FIG. 22

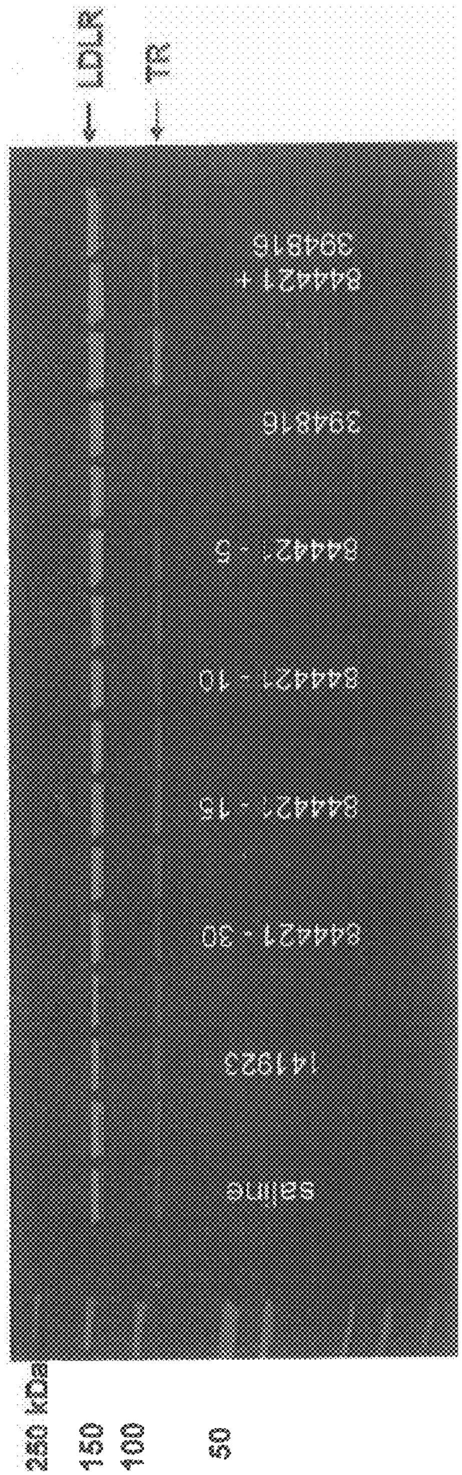


FIG. 23

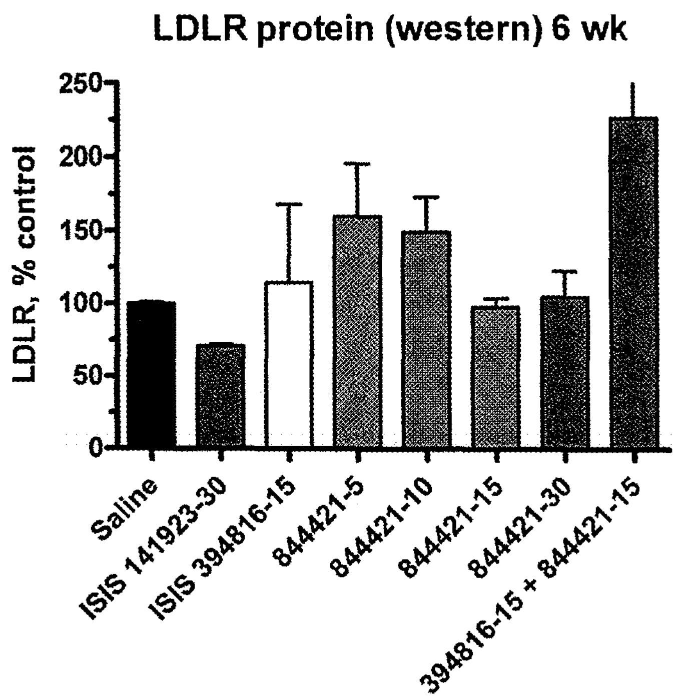
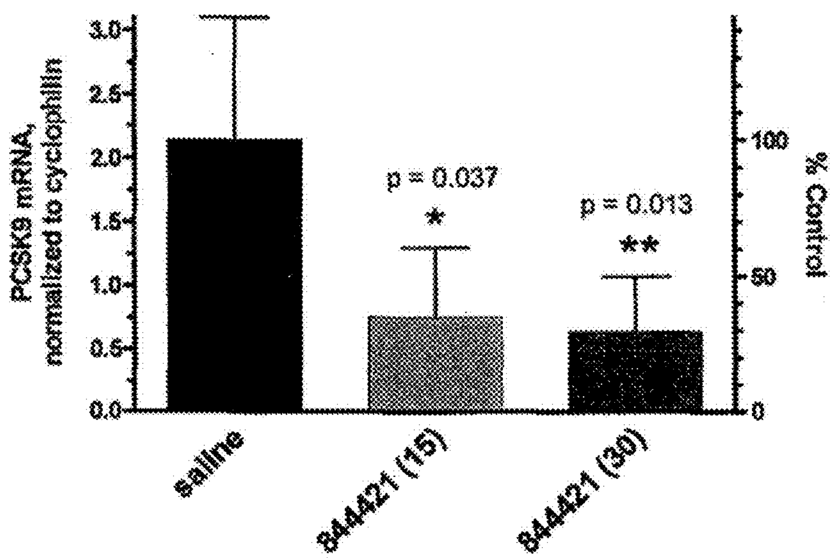

LDLR protein (western) 6 wk
LDLR, % control
250
200
150
100
50
0
Saline
ISIS 141923-30
ISIS 394816-15
844421-5
844421-10
844421-15
844421-30
394816-15 + 844421-15

FIG. 24



Data are mean +/- SD for male and females combined.

FIG. 25

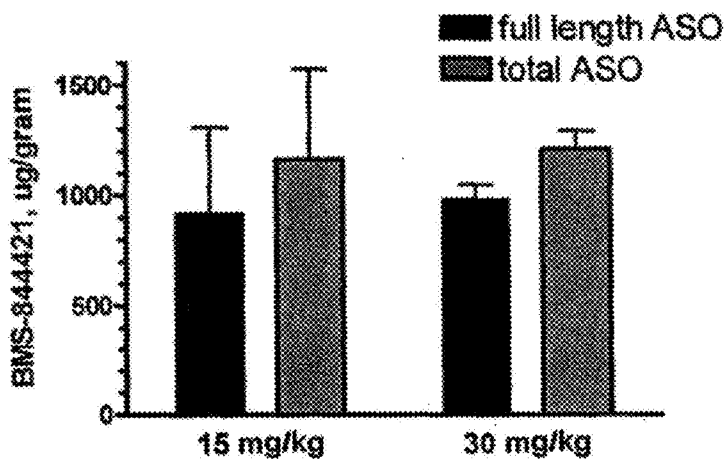

full length ASO
total ASO
BMS-844421, ug/gram
1500
1000
500
0
15 mg/kg
30 mg/kg

FIG. 26

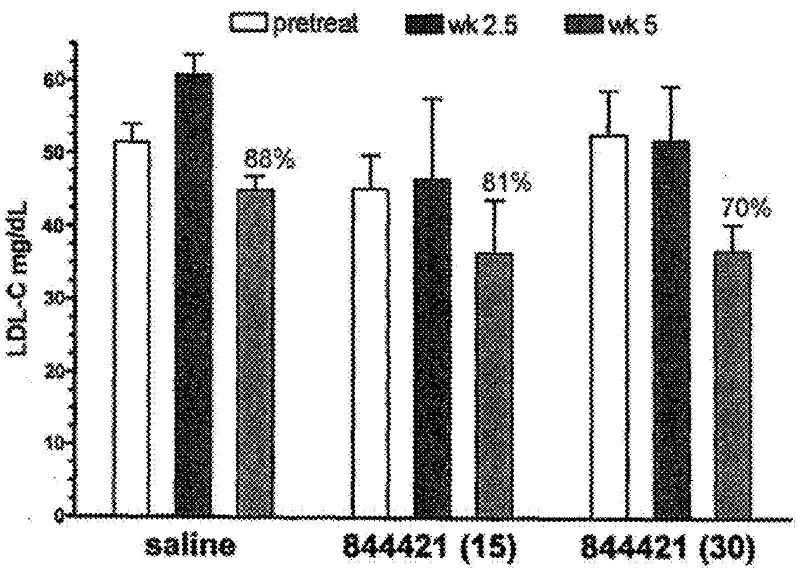

pretreat
wk 2.5
wk 5
LDL-C mg/dL
0
10
20
30
40
50
60
88%
81%
70%
saline
844421 (15)
844421 (30)

FIG. 27

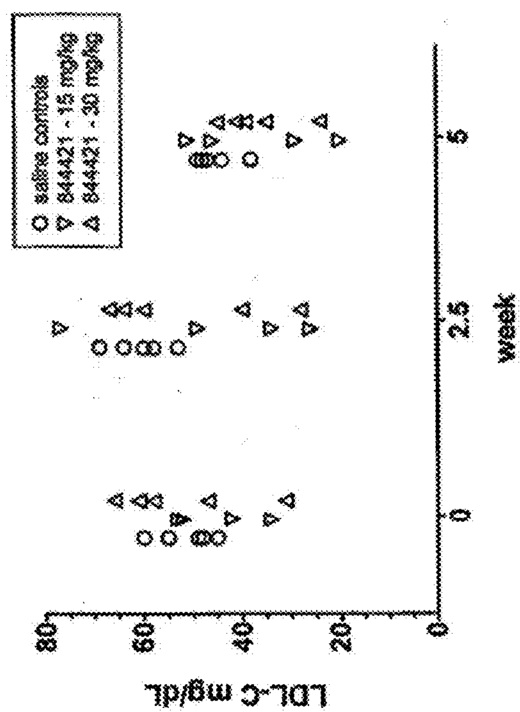


FIG. 28A

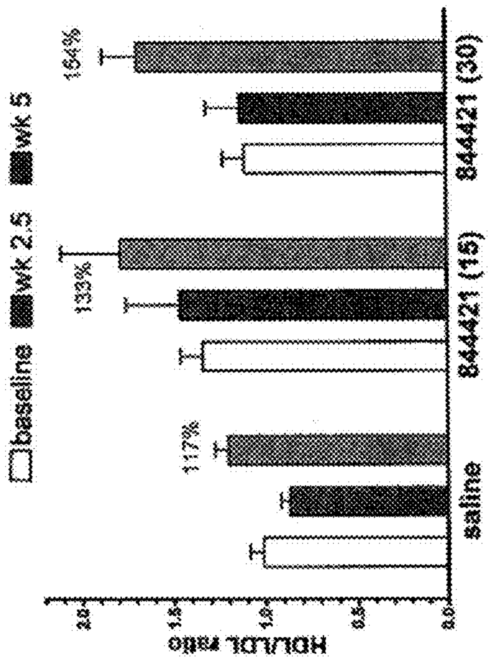


FIG. 28B

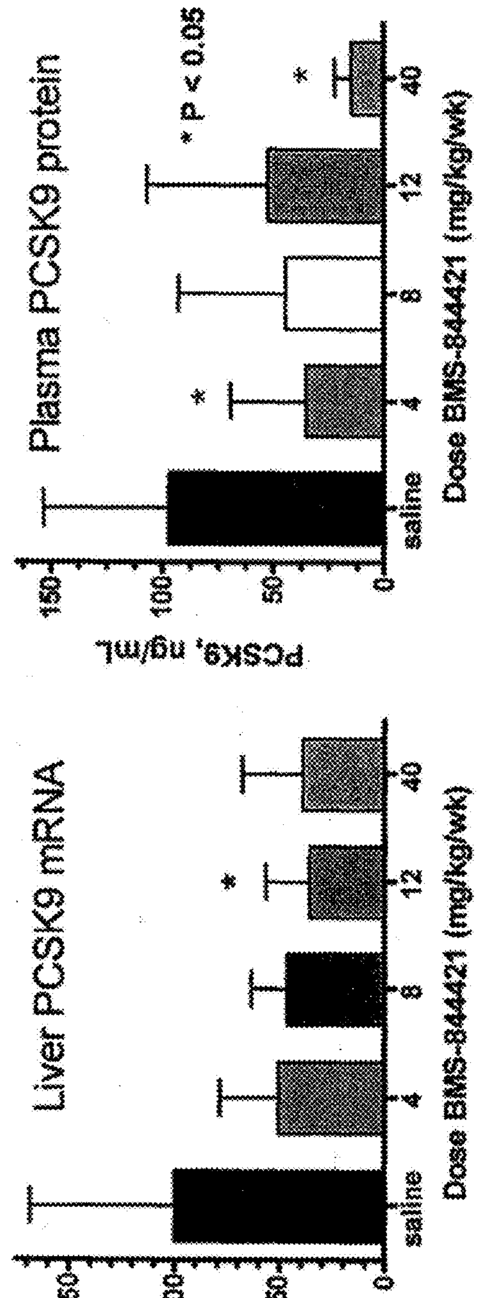

Liver PCSK9 mRNA
% control
150
100
50
0
saline
4
8
12
40
Dose BMS-844421 (mg/kg/wk)
*
Plasma PCSK9 protein
PCSK9, ng/mL
150
100
50
0
saline
4
8
12
40
Dose BMS-844421 (mg/kg/wk)
* P < 0.05
*
*

FIG. 29

METHODS FOR TREATING HYPERCHOLESTEROLEMIA

SEQUENCE LISTING

[0001] The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled 201155 Sequence Listing.xml, created on Nov. 4, 2022 which is 616.1 Kb in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] Methods are provided for lowering LDL-C levels in an individual having elevated LDL-C levels. Such methods are further useful to treat hypercholesterolemia, and to reduce the risk of coronary heart disease.

BACKGROUND

[0003] Atherosclerosis is a complex, polygenic disease of arterial degeneration that can lead to coronary heart disease (CHD). In Western societies, complications arising from atherosclerosis are the most common causes of death. Several risk factors for CHD have been well-established, and include elevated low density lipoprotein-cholesterol (LDL-C), low levels of high density lipoprotein cholesterol (HDL-C), cigarette smoking, hypertension, age, and family history of early CHD. Given the abundant clinical data and epidemiological studies indicating that lowering LDL-C is beneficial for the prevention of adverse coronary events, the primary target of pharmacological intervention in the treatment and prevention of CHD is lowered LDL-C.

[0004] In individuals with autosomal dominant hypercholesterolemia (ADH), elevated LDL-C levels have been linked to mutations in the genes encoding LDL-receptor (LDL-R), apolipoprotein B (apoB), or proprotein convertase subtilisin/kexin type 9 (PCSK9) (Abifadel et al., Nat. Genet., 2003, 34:154-156). The PCSK9 gene (also known as FH3; NARC1; NARC-1; and HCHOLA3) was mapped to Chromosome 1 at location 1p32.3. PCSK9 was identified as a third locus associated with ADH when gain-of-function mutations in PCSK9 were found to be linked to elevated LDL-C levels. ApoB-100 participates in the intracellular assembly and secretion of triglyceride-rich lipoproteins and is a ligand for the LDL-R. PCSK9 is proposed to reduce LDL-R expression levels in the liver. Reduced LDL-R expression results in reduced hepatic uptake of circulating apoB containing lipoproteins, which in turn leads to elevated cholesterol.

[0005] Familial hypercholesterolemia (FH) is caused by hundreds of different mutations in the LDL-R, and is phenotypically characterized by elevated plasma LDL-C levels and deposits of LDL-C in tendons, skin and arteries, leading to premature cardiovascular disease. Homozygous and heterozygous mutations in the LDL-R are associated with FH. Likewise, heterozygous and homozygous mutations in the ligand-binding domain of PCSK9 are associated with the FH phenotype. Mutations in this gene have been associated with a third form of autosomal dominant FH.

[0006] In mice genetically deficient in PCSK9, a marked increase in LDL-R expression and increased plasma LDL clearance rate were observed. Conversely, overexpression in mice of wild-type PCSK9, or certain PCSK9 mutants, promotes decreased LDL-R expression and elevated LDL-C levels.

SUMMARY OF THE INVENTION

[0007] Provided herein are methods for the treatment of hypercholesterolemia and/or atherosclerosis, as well as methods for the reduction of elevated LDL-C levels and CHD risk. Methods for treating hypercholesterolemia comprise administering to an individual an antisense compound targeted to a PCSK9 nucleic acid. In such methods, an antisense compound targeted to a PCSK9 nucleic acid may be targeted to sequences as set forth in GENBANK® Accession No. NM_174936.2, nucleotides 25475000 to 25504000 of GENBANK® Accession No. NT_032977.8, or GENBANK® Accession No. AK124635.1.

[0008] Methods for treating and/or preventing atherosclerosis comprise administering to an individual an antisense compound targeted to a PCSK9 nucleic acid. Methods for reducing LDL-C levels comprise administering to an individual an antisense compound targeted to a PCSK9 nucleic acid. Methods for reducing CHD risk comprise administering to an individual an antisense compound targeted to a PCSK9 nucleic acid. Such methods may comprise the administration of a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid.

[0009] Methods are provided for reducing LDL-C levels in an individual having elevated LDL-C levels, comprising administering to the individual a therapeutically effective amount of an antisense oligonucleotide targeted to a PCSK9 nucleic acid, thereby reducing LDL-C levels. Also provided are methods for reducing LDL-C levels in an individual comprising selecting an individual having elevated LDL-C levels, and administering to the individual a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid, and additionally monitoring LDL-cholesterol levels. Further provided are methods for treating atherosclerosis in an individual, comprising selecting an individual diagnosed with atherosclerosis, administering to the individual a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid, and monitoring atherosclerosis. Also provided are methods for reducing coronary heart disease risk, comprising selecting an individual having elevated LDL-C levels and one or more additional indicators of coronary heart disease, administering to the individual a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid, and monitoring LDL-C levels. Additionally provided are methods for treating hypercholesterolemia, comprising administering to an individual diagnosed with hypercholesterolemia a therapeutically effective amount of an antisense oligonucleotide targeted to a PCSK9 nucleic acid, thereby reducing cholesterol levels.

[0010] Further provided is a use of an effective amount of one or more of the disclosed antisense compounds for treating atherosclerosis or hypercholesterolemia in an individual in need of such treatment. Also provided is a use of an effective amount of one or more of the disclosed antisense compounds for reducing LDL-C levels or reducing CHD risk in an individual in need of such treatment. Also provided is the use of one or more of the disclosed antisense compounds in the manufacture of a medicament for the treatment of atherosclerosis or hypercholesterolemia. Further provided is the use of one or more of the disclosed antisense compounds in the manufacture of a medicament for reducing LDL-C levels or for reducing CHD risk.

[0011] In any of the methods provided, a PCSK9 nucleic acid may be the sequence set forth in SEQ ID NO: 1. Thus, the antisense compound may be targeted to a PCSK9 nucleic acid as set forth in SEQ ID NO: 1.

[0012] Any of the methods provided herein may further comprise monitoring LDL-C levels.

[0013] An individual may be selected for administration of an antisense compound targeted to a PCSK9 nucleic acid when the individual exhibits an LDL-C level above 100 mg/dL, above 130 mg/dL, above 160 mg/dL, or above 190 mg/dL. Administration of an antisense compound targeted to a PCSK9 nucleic acid may result in an LDL-C level below 190 mg/dL, below 160 mg/dL, below 130 mg/dL, below 100 mg/dL, below 70 mg/dL, or below 50 mg/dL.

[0014] In any of the aforementioned methods, administration of the antisense compound may comprise parenteral administration. The parenteral administration may further comprise subcutaneous or intravenous administration.

[0015] In any of the methods provided herein, the antisense compound may have least 80%, at least 90%, or at least 95% complementarity to SEQ ID NO: 1. Alternatively, the antisense compound may have 100% complementarity to SEQ ID NO: 1.

[0016] The antisense compounds provided herein and employed in any of the described methods may be 8 to 80 subunits in length, 12 to 50 subunits in length, 12 to 30 subunits in length, 15 to 30 subunits in length, 18 to 24 subunits in length, 19 to 22 subunits in length, or 20 subunits in length. Further, the antisense compounds employed in any of the described methods may be antisense oligonucleotides 8 to 80 nucleotides in length, 12 to 50 nucleotides in length, 12 to 30 nucleotides in length 15 to 30 nucleotides in length, 18 to 24 nucleotides in length, 19 to 22 nucleotides in length, or 20 nucleotides in length.

[0017] In any of the methods provided, the antisense compound may be an antisense oligonucleotide. Moreover, the antisense oligonucleotide may be a gapmer antisense oligonucleotide. The gapmer antisense oligonucleotide may comprise a gap segment of ten 2'-deoxynucleotides positioned between wing segments of five 2'-MOE nucleotides. The gapmer antisense oligonucleotide may be a gap-widened antisense oligonucleotide, comprising a gap segment of fourteen 2'-deoxynucleotides positioned between wing segments of three 2'-MOE nucleotides.

[0018] In any of the methods provided, the antisense compounds may have at least one modified internucleoside linkage. Additionally, each internucleoside linkage may be a phosphorothioate internucleoside linkage. Each cytosine may be a 5-methyl cytosine.

[0019] Also provided herein are antisense compounds targeted to a PCSK9 nucleic acid. Further provided are antisense oligonucleotides targeted to a PCSK9 nucleic acid. The antisense compounds, including antisense oligonucleotides, may be targeted to PCSK9 nucleic acids, which include the sequences as set forth in GENBANK® Accession No. NM_174936.2, nucleotides 25475000 to 25504000 of GENBANK® Accession No. NT_032977.8, or GENBANK® Accession No. AK124635.1. The antisense compounds, including antisense oligonucleotides, may have at least 70%, at least 80%, at least 90%, or at least 95% complementarity to a PCSK9 nucleic acid. The antisense compounds, including antisense oligonucleotides, may have 99% complementarity to a PCSK9 nucleic acid. The antisense compounds, including antisense oligonucleotides, may have 100% complementarity to a PCSK9 nucleic acid. For any of the antisense compounds provided, including antisense oligonucleotides, the PCSK9 nucleic acid may be the sequence set forth in SEQ ID NO: 1.

[0020] Antisense compounds targeted to a PCSK9 nucleic acid may be 8 to 80 subunits in length, 12 to 50 subunits in length, 12 to 30 subunits in length, 15 to 30 subunits in length, 18 to 24 subunits in length, 19 to 22 subunits in length, or 20 subunits in length. Antisense oligonucleotides targeted to a PCSK9 nucleic acid may be 8 to 80 nucleotides in length, 12 to 50 nucleotides in length, 12 to 30 nucleotides in length, 15 to 30 nucleotides in length, 18 to 24 nucleotides in length, 19 to 22 nucleotides in length, or 20 nucleotides in length.

[0021] Moreover, the antisense compound may be a gapmer antisense oligonucleotide. The gapmer antisense oligonucleotide may comprise a gap segment of ten 2'-deoxynucleotides positioned between wing segments of five 2'-MOE nucleotides. The gapmer antisense oligonucleotide may be a gap-widened antisense oligonucleotide, comprising a gap segment of fourteen 2'-deoxynucleotides positioned between wing segments of three 2'-MOE nucleotides.

[0022] Antisense compounds, including antisense oligonucleotides, targeted to a PCSK9 nucleic acid may have at least one modified internucleoside linkage. Additionally, each internucleoside linkage may be a phosphorothioate internucleoside linkage. Each cytosine may be a 5-methyl cytosine.

[0023] Antisense compounds, including antisense oligonucleotides, targeted to a PCSK9 nucleic acid may have at least one modified nucleoside. In certain embodiments, a modified nucleoside is a sugar-modified nucleoside. In certain such embodiments, the sugar-modified nucleosides can further comprise a natural or modified heterocyclic base moiety and/or a natural or modified internucleoside linkage and may include further modifications independent from the sugar modification. In certain embodiments, a sugar modified nucleoside is a 2'-modified nucleoside, wherein the sugar ring is modified at the 2' carbon from natural ribose or 2'-deoxy-ribose.

[0024] In certain embodiments, a 2' modified nucleoside has a 2'-F, 2'-OCH_2 (2'-OMe) or a 2'-$O(CH_2)_2$—OCH_3 (2'-O-methoxyethyl or 2'-MOE) substituent group

[0025] In certain embodiments, a 2'-modified nucleoside has a bicyclic sugar moiety. In certain such embodiments, the bicyclic sugar moiety is a D sugar in the alpha configuration. In certain such embodiments, the bicyclic sugar moiety is a D sugar in the beta configuration. In certain such embodiments, the bicyclic sugar moiety is an L sugar in the alpha configuration. In certain such embodiments, the bicyclic sugar moiety is an L sugar in the beta configuration.

[0026] In certain embodiments, the bicyclic sugar moiety comprises a bridge group between the 2' and the 4'-carbon atoms. In certain such embodiments, the bridge group comprises from 1 to 8 linked biradical groups. In certain embodiments, the bicyclic sugar moiety comprises from 1 to 4 linked biradical groups. In certain embodiments, the bicyclic sugar moiety comprises 2 or 3 linked biradical groups. In certain embodiments, the bicyclic sugar moiety comprises 2 linked biradical groups. In certain embodiments, a linked biradical group is selected from —O—, —S—, —N(R1)-, —C(R1)(R2)-, —C(R1)=C(R1)-, —C(R1)=N—,

—C(═NR1)-, —Si(R1)(R2)-, —S(═O)2-, —S(═O)—, —C(═O)— and —C(═S)—; where each R1 and R2 is, independently, H, hydroxyl, C1-C12 alkyl, substituted C1-C12 alkyl, C2-C12 alkenyl, substituted C2-C12 alkenyl, C2-C12 alkynyl, substituted C2-C12 alkynyl, C5-C20 aryl, substituted C5-C20 aryl, a heterocycle radical, a substituted hetero-cycle radical, heteroaryl, substituted heteroaryl, C5-C7 alicyclic radical, substituted C5-C7 alicyclic radical, halogen, substituted oxy (—O—), amino, substituted amino, azido, carboxyl, substituted carboxyl, acyl, substituted acyl, CN, thiol, substituted thiol, sulfonyl (S(═O)2-H), substituted sulfonyl, sulfoxyl (S(═O)—H) or substituted sulfoxyl; and each substituent group is, independently, halogen, C1-C12 alkyl, substituted C1-C12 alkyl, C2-C12 alkenyl, substituted C2-C12 alkenyl, C2-C12 alkynyl, substituted C2-C12 alkynyl, amino, substituted amino, acyl, substituted acyl, C1-C12 aminoalkyl, C1-C12 aminoalkoxy, substituted C1-C12 aminoalkyl, substituted C1-C12 aminoalkoxy or a protecting group.

[0027] In some embodiments, the bicyclic sugar moiety is bridged between the 2' and 4' carbon atoms with a biradical group selected from —O—(CH2)p-, —O—CH2-, —O—CH2CH2-, —O—CH(alkyl)-, —NH—(CH2)p-, —N(alkyl)-(CH2)p-, —O—CH(alkyl)-, —(CH(alkyl))-(CH2)p-, —NH—O—(CH2)p-, —N(alkyl)-O—(CH2)p-, or —O—N(alkyl)-(CH_2)p-, wherein p is 1, 2, 3, 4 or 5 and each alkyl group can be further substituted. In certain embodiments, p is 1, 2 or 3.

[0028] In one aspect, each of said bridges is, independently, —[C(R1)(R2)]n-, —[C(R1)(R2)]n-O—, —C(R1R2)-N(R1)-O— or —C(R1R2)-O—N(R1)-. In another aspect, each of said bridges is, independently, 4'-(CH2)3-2', 4'-(CH2)2-2', 4'-CH2-O-2', 4'-(CH2)2-O-2', 4'-CH2-O—N(R1)-2' and 4'-CH2-N(R1)-O-2'- wherein each R1 is, independently, H, a protecting group or C1-C12 alkyl.

[0029] The present disclosure further discloses subsets of the antisense compounds above that show superior properties relating to pharmacodynamics and/or pharmacokinetics, among other properties. Exemplary antisense compounds reduce PCSK9 expression in a cultured cell model system, e.g., Hep3B cells, cynomolgus monkey hepatocytes, or human primary hepatocytes. In particular embodiments, antisense compounds reduce PCSK9 expression in a dose-dependent manner in a cell culture system, where the antisense compounds have an IC_{50} in the nanomolar range.

[0030] Exemplary antisense compounds also reduce the expression of a PCSK9 nucleic acid in an animal model, such as a transgenic mouse that expresses a human PCSK9 nucleic acid (HuPCSK9Tg mice). In particular embodiments, antisense compounds include those that reduce PCSK9 express in an animal model with a physiology that correlates closely with humans, such as monkey, e.g., the cynomolgus monkey.

[0031] Exemplary antisense compounds display minimal side effects. Side effects include responses to the administration of the antisense compound unrelated to the targeting of a PCSK9 nucleic acid, such as an inflammatory response in the host individual. Exemplary antisense compounds are well tolerated by the host individual. Tolerability may be determined though histological analysis of various cell types, and includes examination of such markers as cytoplasmic swelling from multifocal apoptosis in liver cells, follicular hyperplasia in spleen cells, macrophage infiltration, and the like. In particular embodiments, antisense compounds produce minimal signs of host intolerance.

[0032] Exemplary antisense compounds further display favorable pharmacokinetics. In particular embodiments, antisense compounds accumulate at a relatively high ratio in the liver versus other sensitive organs, such as kidneys. In particular embodiments, antisense compounds exhibit relatively high half-lives in relevant biological fluids or tissues, e.g., liver tissue, reflecting higher stability and resistance to nucleases, for example.

[0033] In particular embodiments, antisense compounds target identical sequences in human and animal PCSK9 sequences. In other embodiments, antisense compounds target PCSK9 sequences without a known single nucleotide polymorphism (SNP), have a high relative G-content, have minimal secondary structure, have greatly reduced effets on the host's serum chemistry, body weight, or histopathology, compared to other antisense compounds, and/or induce a minimal adverse immunohistocompatibility reaction.

[0034] In a particular embodiment, the antisense compound is selected from the group of Isis 405881, Isis 399819, Isis 395165, Isis 405879, Isis 406008, Isis 405891, Isis 395186, Isis 405988, Isis 405994, Isis 406023, Isis 395187, Isis 395185, Isis 406033, Isis 405923, Isis 399900, Isis 405995, Isis 405991, Isis 406005, Isis 399793, and Isis 395152. The aforementioned antisense compounds effectively repress PCSK9 expression in a Hep3B cell culture system. In another embodiment, antisense compounds effectively inhibit PCSK9 expression with a low IC_{50} in a battery of cell culture systems. Exemplary antisense compounds include those selected from the group of Isis 395165, Isis 395185, Isis 395186, Isis 395187, Isis 405879, Isis 405881, Isis 405891, Isis 405988, Isis 405994, and Isis 406008. In yet another embodiment, antisense compounds exhibit minimal or no adverse histopathological effects when administered at particularly high doses to an individual. Exemplary antisense compounds include those selected from the group consisting of Isis 395185, Isis 395186, Isis 395187, Isis 405879, Isis 405891, and Isis 405988. In yet another embodiment, antisense compounds affect therapeutic endpoints, e.g., reduction of plasma LDL-C, liver TG, or liver PCSK9 mRNA, in an animal model. Exemplary antisense compounds include Isis 405879. In yet another embodiment, antisense compounds display combinations of the characteristics above and reduce liver PCSK9 mRNA expression in an animal model with high efficiency, e.g., Isis 405879.

Definitions

[0035] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by on of skill in the art to which the invention(s) belong.

[0036] Unless specific definitions are provided, the nomenclature utilized in connection with, and the procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well known and commonly used in the art. Standard techniques may be used for chemical synthesis, chemical analysis, pharmaceutical preparation, formulation and delivery, and treatment of subjects. Certain such techniques and procedures may be found for example in "Antisense Drug Technology: Principles, Strategies, and Applications." by Stanley Crooke, Boca Raton: Taylor & Francis Group, 2008; "Carbohydrate

Modifications in Antisense Research" Edited by Sangvi and Cook, American Chemical Society, Washington D.C., 1994; and "Remington's Pharmaceutical Sciences," Mack Publishing Co., Easton, Pa., 18th edition, 1990; and which is hereby incorporated by reference for any purpose. Where permitted, all patents, patent applications, published applications and publications, GENBANK sequences, websites and other published materials referred to throughout the entire disclosure herein, unless noted otherwise, are incorporated by reference in their entirety. All of the GENBANK® Accession Nos. along with their associated sequence and structural data pertaining to such sequences including gene organization and structural elements and SNP information that may be found in sequence databases such as the National Center for Biotechnology Information (NCBI) are incorporated herein by reference in their entirety. In the event that there is a plurality of definitions for terms herein, those in this section prevail. Where reference is made to a URL or other such identifier or address, it is understood that such identifiers can change and particular information on the internet can come and go, but equivalent information can be found by searching the internet. Reference thereto evidences the availability and public dissemination of such information.

[0037] A "pharmaceutical composition" means a mixture of substances suitable for administering to an individual. For example, a pharmaceutical composition may comprise an antisense oligonucleotide and a sterile aqueous solution.

[0038] "Administering" means providing a pharmaceutical agent to an individual, and includes, but is not limited to administering by a medical professional and self-administering.

[0039] "Individual" means a human or non-human animal selected for treatment or therapy.

[0040] "Duration" means the period of time during which an activity or event continues. In certain embodiments, the duration of treatment is the period of time during which doses of a pharmaceutical agent are administered.

[0041] "Parenteral administration," means administration through injection or infusion. Parenteral administration includes, but is not limited to, subcutaneous administration, intravenous administration, or intramuscular administration.

[0042] "Subcutaneous administration" means administration just below the skin. "Intravenous administration" means administration into a vein.

[0043] "Dose" means a specified quantity of a pharmaceutical agent provided in a single administration. In certain embodiments, a dose may be administered in two or more boluses, tablets, or injections. For example, in certain embodiments, where subcutaneous administration is desired, the desired dose requires a volume not easily accommodated by a single injection. In such embodiments, two or more injections may be used to achieve the desired dose. In certain embodiments, a dose may be administered in two or more injections to minimize injection site reaction in a individual.

[0044] "Dosage unit" means a form in which a pharmaceutical agent is provided. In certain embodiments, a dosage unit is a vial containing lyophilized antisense oligonucleotide. In certain embodiments, a dosage unit is a vial containing reconstituted antisense oligonucleotide.

[0045] "Pharmaceutical agent" means a substance provides a therapeutic benefit when administered to a individual. For example, in certain embodiments, an antisense oligonucleotide targeted to PCSK9 is pharmaceutical agent.

[0046] "Diluent" means an ingredient in a composition that lacks pharmacological activity, but is pharmaceutically necessary or desirable. For example, in drugs that are injected the diluent may be a liquid, e.g., saline solution.

[0047] "Active pharmaceutical ingredient" means the substance in a pharmaceutical composition that provides a desired effect.

[0048] "Pharmaceutically acceptable carrier" means a medium or diluent that does not interfere with the structure of the oligonucleotide. Certain of such carriers enable pharmaceutical compositions to be formulated as, for example, tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspension and lozenges for the oral ingestion by a subject.

[0049] "Therapeutically effective amount" means an amount of a pharmaceutical agent that provides a therapeutic benefit to an individual. In certain embodiments, a therapeutically effective amount of antisense compound targeted to a PCSK9 nucleic acid is an amount that decreases LDL-C in the individual.

[0050] "Hypercholesterolemia" means a condition characterized by elevated serum cholesterol.

[0051] "Hyperlipidemia" means a condition characterized by elevated serum lipids.

[0052] "Hypertriglyceridemia" means a condition characterized by elevated serum triglyceride levels.

[0053] "Non-familial hypercholesterolemia" means a condition characterized by elevated serum cholesterol that is not the result of a single gene mutation.

[0054] "Polygenic hypercholesterolemia" means a condition characterized by elevated cholesterol that results from the influence of a variety of genetic factors. In certain embodiments, polygenic hypercholesterolemia may be exacerbated by dietary intake of lipids.

[0055] "Familial hypercholesterolemia (FH)" means an autosomal dominant metabolic disorder characterized by a mutation in the LDL-receptor (LDL-R) gene, markedly elevated LDL-C and premature onset of atherosclerosis. A diagnosis of familial hypercholesterolemia is made when an individual meets one or more of the following criteria: genetic testing confirming 2 mutated LDL-receptor genes; genetic testing confirming one mutated LDL-receptor gene; document history of untreated serum LDL-cholesterol greater than 500 mg/dL; tendinous and/or cutaneous xanthoma prior to age 10 years; or, both parents have documented elevated serum LDL-cholesterol prior to lipid-lowering therapy consistent with heterozygous familial hypercholesterolemia.

[0056] "Homozygous familial hypercholesterolemia" or "HoFH" means a condition characterized by a mutation in both maternal and paternal LDL-R genes.

[0057] "Heterozygous familial hypercholesterolemia" or "HeFH" means a condition characterized by a mutation in either the maternal or paternal LDL-R gene.

[0058] "Mixed dyslipidemia" means a condition characterized by elevated serum cholesterol and elevated serum triglycerides.

[0059] "Diabetic dyslipidemia" or "Type II diabetes with dyslipidemia" means a condition characterized by Type II diabetes, reduced HDL-C, elevated serum triglycerides, and elevated small, dense LDL particles.

[0060] "CHD risk equivalents," means indicators of clinical atherosclerotic disease that confer a high risk for coronary heart disease, and include clinical coronary heart disease, symptomatic carotid artery disease, peripheral arterial disease, and/or abdominal aortic aneurysm.

[0061] "Metabolic syndrome" means a condition characterized by a clustering of lipid and non-lipid cardiovascular risk factors of metabolic origin. In certain embodiments, metabolic syndrome is identified by the presence of any 3 of the following factors: waist circumference of greater than 102 cm in men or greater than 88 cm in women; serum triglyceride of at least 150 mg/dL; HDL-C less than 40 mg/dL in men or less than 50 mg/dL in women; blood pressure of at least 130/85 mmHg; and fasting glucose of at least 110 mg/dL.

[0062] "Non-alcholic fatty liver disease (NAFLD)" means a condition characterized by fatty inflammation of the liver that is not due to excessive alcohol use (for example, alcohol consumption of over 20 g/day). In certain embodiments, NAFLD is related to insulin resistance and the metabolic syndrome.

[0063] "Non-alcoholic steatohepatitis (NASH)" means a condition characterized by inflammation and the accumulation of fat and fibrous tissue in the liver, that is not due to excessive alcohol use. NASH is an extreme form of NAFLD.

[0064] "Major risk factors" mean factors that contribute to a high risk for coronary heart disease, and include without limitation cigarette smoking, hypertension, low HDL-C, family history of coronary heart disease, and age.

[0065] "CHD risk factors" mean CHD risk equivalents and major risk factors.

[0066] "Coronary heart disease (CHD)" means a narrowing of the small blood vessels that supply blood and oxygen to the heart, which is often a result of atherosclerosis.

[0067] "Reduced coronary heart disease risk" means a reduction in the likelihood that an individual will develop coronary heart disease. In certain embodiments, a reduction in coronary heart disease risk is measured by an improvement in one or more CHD risk factors, for example, a decrease in LDL-C levels.

[0068] "Atherosclerosis" means a hardening of the arteries affecting large and medium-sized arteries and is characterized by the presence of fatty deposits. The fatty deposits are called "atheromas" or "plaques," which consist mainly of cholesterol and other fats, calcium and scar tissue, and damage the lining of arteries.

[0069] "History of coronary heart disease" means the occurrence of clinically evident coronary heart disease in the medical history of an individualor a individual's family member.

[0070] "Early onset coronary heart disease" means a diagnosis of coronary heart disease prior to age 50.

[0071] "Statin intolerant individual" means an individualwho as a result of statin therapy experiences one or more of creatine kinase increases, liver function test abnormalities, muscle aches, or central nervous system side effects.

[0072] "Efficacy" means the ability to produce a desired effect. For example, efficacy of a lipid-lowering therapy may be reduction in the concentration of one or more of LDL-C, VLDL-C, IDL-C, non-HDL-C, ApoB, lipoprotein(a), or triglycerides.

[0073] "Acceptable safety profile" means a pattern of side effects that is within clinically acceptable limits.

[0074] "Side effects" means physiological responses attributable to a treatment other than desired effects. In certain embodiments, side effects include, without limitation, injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, and myopathies. For example, increased aminotransferase levels in serum may indicate liver toxicity or liver function abnormality. For example, increased bilirubin may indicate liver toxicity or liver function abnormality.

[0075] "Injection site reaction" means inflammation or abnormal redness of skin at a site of injection in an individual.

[0076] "Individual compliance" means adherence to a recommended or prescribed therapy by an individual.

[0077] "Lipid-lowering therapy" means a therapeutic regimen provided to an individual to reduce one or more lipids in a individual. In certain embodiments, a lipid-lowering therapy is provide to reduce one or more of ApoB, total cholesterol, LDL-C, VLDL-C, IDL-C, non-HDL-C, triglycerides, small dense LDL particles, and Lp(a) in a individual.

[0078] "Lipid-lowering agent" means a pharmaceutical agent provided to an individual to achieve a lowering of lipids in the individual. For example, in certain embodiments, a lipid-lowering agent is provided to an individualto reduce one or more of ApoB, LDL-C, total cholesterol, and triglycerides.

[0079] "LDL-C target" means an LDL-C level that is desired following lipid-lowering therapy.

[0080] "Comply" means the adherence with a recommended therapy by a individual.

[0081] "Recommended therapy" means a therapeutic regimen recommended by a medical professional for the treatment, amelioration, or prevention of a disease.

[0082] "Low LDL-receptor activity" means LDL-receptor activity that is not sufficiently high to maintain clinically acceptable levels of LDL-C in the bloodstream.

[0083] "Cardiovascular outcome" means the occurrence of major adverse cardiovascular events.

[0084] "Improved cardiovascular outcome" means a reduction in the occurrence of major adverse cardiovascular events, or the risk thereof. Examples of major adverse cardiovascular events include, without limitation, death, reinfarction, stroke, cardiogenic shock, pulmonary edema, cardiac arrest, and atrial dysrhythmia.

[0085] "Surrogate markers of cardiovascular outcome" means indirect indicators of cardiovascular events, or the risk thereof. For example, surrogate markers of cardiovascular outcome include carotid intimal media thickness (CIMT). Another example of a surrogate marker of cardiovascular outcome includes atheroma size. Atheroma size may be determined by intravascular ultrasound (IVUS). Surrogate markers also include increased HDL-cholesterol, or any combination of the markers above.

[0086] "Increased HDL-C" means an increase in serum HDL-C in an individualover time.

[0087] "Lipid-lowering" means a reduction in one or more serum lipids in an individual over time.

[0088] "Co-administration" means administration of two or more pharmaceutical agents to a individual. The two or more pharmaceutical agents may be in a single pharmaceutical composition, or may be in separate pharmaceutical compositions. Each of the two or more pharmaceutical agents may be administered through the same or different routes of administration. Co-administration encompasses administration in parallel or sequentially.

[0089] "Administered concomitantly" refers to the administration of two agents at the same therapeutic time frame, in any manner in which the pharmacological effects of both are manifest in the patient at the same time. Concomitant administration does not require that both agents be administered in a single pharmaceutical composition, in the same dosage form, or by the same route of administration.

[0090] "Therapeutic lifestyle change" means dietary and lifestyle changes intended to lower cholesterol and reduce the risk of developing heart disease, and includes recommendations for dietary intake of total daily calories, total fat, saturated fat, polyunsaturated fat, monounsaturated fat, carbohydrate, protein, cholesterol, insoluble fiber, as well as recommendations for physical activity.

[0091] "Statin" means a pharmaceutical agent that inhibits the activity of HMG-CoA reductase.

[0092] "HMG-CoA reductase inhibitor" means a pharmaceutical agent that acts through the inhibition of the enzyme HMG-CoA reductase.

[0093] "Cholesterol absorption inhibitor" means a pharmaceutical agent that inhibits the absorption of exogenous cholesterol obtained from diet.

[0094] "LDL apheresis" means a form of apheresis by which LDL-C is removed from blood. Typically, a individual's blood is removed from a vein, and separated into red cells and plasma. LDL-C is filtered out of the plasma prior to return of the plasma and red blood cells to the individual.

[0095] "MTP inhibitor" means a pharmaceutical agent that inhibits the enzyme microsomal triglyceride transfer protein.

[0096] "Low density lipoprotein-cholesterol (LDL-C)" means cholesterol carried in low density lipoprotein particles. Concentration of LDL-C in serum (or plasma) is typically quantified in mg/dL or nmol/L. "Serum LDL-C" and "plasma LDL-C" mean LDL-C in the serum and plasma, respectively.

[0097] "Very low density lipoprotein-cholesterol (VLDL-C)" means cholesterol associated with very low density lipoprotein particles. Concentration of VLDL-C in serum (or plasma) is typically quantified in mg/dL or nmol/L. "Serum VLDL-C" and "plasma VLDL-C" mean VLDL-C in the serum or plasma, respectively.

[0098] "Intermediate low density lipoprotein-cholesterol (IDL-C)" means cholesterol associated with intermediate density lipoprotein. Concentration of IDL-C in serum (or plasma) is typically quantified in mg/dL or nmol/L. "Serum IDL-C" and "plasma IDL-C" mean IDL-C in the serum or plasma, respectively.

[0099] "Non-high density lipoprotein-cholesterol (Non-HDL-C)" means cholesterol associated with lipoproteins other than high density lipoproteins, and includes, without limitation, LDL-C, VLDL-C, and IDL-C.

[0100] "High density lipoprotein-C(HDL-C)" means cholesterol associated with high density lipoprotein particles. Concentration of HDL-C in serum (or plasma) is typically quantified in mg/dL or nmol/L. "Serum HDL-C" and "plasma HDL-C" mean HDL-C in the serum and plasma, respectively.

[0101] "Total cholesterol" means all types of cholesterol, including, but not limited to, LDL-C, HDL-C, IDL-C and VLDL-C. Concentration of total cholesterol in serum (or plasma) is typically quantified in mg/dL or nmol/L.

[0102] "Lipoprotein(a)" or "Lp(a)" means a lipoprotein particle that is comprised of LDL-C, an apolipoprotein(a) particle, and an apolipoproteinB-100 particle.

[0103] "ApoA1" means apolipoprotein-A1 protein in serum. Concentration of ApoA1 in serum is typically quantified in mg/dL or nmol/L.

[0104] "ApoB:ApoA1 ratio" means the ratio of ApoB concentration to ApoA1 concentration.

[0105] "ApoB-containing lipoprotein" means any lipoprotein that has apolipoprotein B as its protein component, and is understood to include LDL, VLDL, IDL, and lipoprotein (a).

[0106] "Small LDL particle" means a subclass of LDL particles characterized by a smaller, denser size compared to other LDL particles. In certain embodiments, large LDL particles are 23-27 nm in diameter. In certain embodiments, intermediate LDL particles are 21.2-23 nm in diameter. In certain embodiments, small LDL particles are 18-21.2 nm in diameter. In certain embodiments, particle size is measured by nuclear magnetic resonance analysis.

[0107] "Small VLDL particle" means a subclass of VLDL particles characterized by a smaller, denser size compared to other VLDL particles. In certain embodiments, large VLDL particles are greater than 60 nm in diameter. In certain embodiments, medium VLDL particles are 35-60 nm in diameter. In certain embodiments, small VLDL particles are 27-35 nm in diameter. In certain embodiments, particle size is measured by nuclear magnetic resonance analysis.

[0108] "Triglycerides" means lipids that are the triesters of glycerol. "Serum triglycerides" mean triglycerides present in serum. "Liver triglycerides" mean triglycerides present in liver tissue.

[0109] "Serum lipids" mean cholesterol and triglycerides in the serum.

[0110] "Elevated total cholesterol" means total cholesterol at a concentration in an individual at which lipid-lowering therapy is recommended, and includes, without limitation, elevated LDL-C", "elevated VLDL-C," "elevated IDL-C," and "elevated non-HDL-C." In certain embodiments, total cholesterol concentrations of less than 200 mg/dL, 200-239 mg/dL, and greater than 240 mg/dL are considered desirable, borderline high, and high, respectively. In certain embodiments, LDL-C concentrations of 100 mg/dL, 100-129 mg/dL, 130-159 mg/dL, 160-189 mg/dL, and greater than 190 mg/dL are considered optimal, near optimal/above optimal, borderline high, high, and very high, respectively.

[0111] "Elevated triglyceride" means concentrations of triglyceride in the serum or liver at which lipid-lowering therapy is recommended, and includes "elevated serum triglyceride" and "elevated liver triglyceride." In certain embodiments, serum triglyceride concentration of 150-199 mg/dL, 200-499 mg/dL, and greater than or equal to 500 mg/dL is considered borderline high, high, and very high, respectively.

[0112] "Elevated small LDL particles" means a concentration of small LDL particles in an individual at which lipid-lowering therapy is recommended.

[0113] "Elevated small VLDL particles" means a concentration of small VLDL particles in an individual at which lipid-lowering therapy is recommended.

[0114] "Elevated lipoprotein(a)" means a concentration of lipoprotein(a) in an individual at which lipid-lowering therapy is recommended.

[0115] "Low HDL-C" means a concentration of HDL-C in an individual at which lipid-lowering therapy is recommended. In certain embodiments lipid-lowering therapy is recommended when low HDL-C is accompanied by elevations in non-HDL-C and/or elevations in triglyceride. In certain embodiments, HDL-C concentrations of less than 40 mg/dL are considered low. In certain embodiments, HDL-C concentrations of less than 50 mg/dL are considered low.

[0116] "ApoB" means apolipoprotein B-100 protein. Concentration of ApoB in serum (or plasma) is typically quantified in mg/dL or nmol/L. "Serum ApoB" and "plasma ApoB" mean ApoB in the serum and plasma, respectively.

[0117] "LDL/HDL ratio" means the ratio of LDL-C to HDL-C.

[0118] "Oxidized-LDL" or "Ox-LDL-C" means LDL-C that is oxidized following exposure to free radicals.

[0119] "Individual having elevated LDL-C levels" means an individual who has been identified by a medical professional (e.g., a physician) as having LDL-C levels near or above the level at which therapeutic intervention is recommended, according to guidelines recognized by medical professionals. Such an individual may also be considered "in need of treatment" to decrease LDL-C levels.

[0120] "Individual having elevated apoB-100 levels" means an individual who has been identified as having apoB-100 levels near or above the level at which therapeutic intervention is recommended, according to guidelines recognized by medical professionals. Such an individual may also be considered "in need of treatment" to decrease apoB-100 levels.

[0121] "Treatment of elevated LDL-C levels" means administration of an antisense compound targeted to a PCSK9 nucleic acid to an individual having elevated LDL-C levels.

[0122] "Treatment of atherosclerosis" means administration of an antisense compound targeted to a PCSK9 nucleic acid to an individual who, based upon a physician's assessment, has or is likely to have atherosclerosis. "Prevention of atherosclerosis" means administration of an antisense compound targeted to a PCSK9 nucleic acid to an individual who, based upon a physician's assessment, is susceptible to atherosclerosis.

[0123] "Ameliorate" means to make better or improve a detrimental condition in an individual.

[0124] "Target nucleic acid," "target RNA," "target RNA transcript" and "nucleic acid target" all mean any nucleic acid capable of being targeted by antisense compounds.

[0125] "PCSK9 nucleic acid" means any nucleic acid encoding PCSK9. For example, in certain embodiments, a PCSK9 nucleic acid includes, without limitation, a DNA sequence encoding PCSK9, an RNA sequence transcribed from DNA encoding PCSK9, and an mRNA sequence encoding PCSK9. "PCSK9 mRNA" means an mRNA encoding a PCSK9 protein.

[0126] "Targeting" means the process of design and selection of an antisense compound that will specifically hybridize to a target nucleic acid and induce a desired effect.

[0127] "Targeted" means having a nucleobase sequence that will allow specific hybridization of an antisense compound to a target nucleic acid to induce a desired effect. In certain embodiments, a desired effect is reduction of a target nucleic acid. In certain such embodiments, a desired effect is reduction of PCSK9 mRNA.

[0128] "Antisense inhibition" means reduction of a target nucleic acid levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels in the absence of the antisense compound.

[0129] "Target region" means a fragment of a target nucleic acid to which one or more antisense compounds is targeted.

[0130] "Target segment" means the sequence of nucleotides of a target nucleic acid to which an antisense compound is targeted. "5' target site" refers to the 5'-most nucleotide of a target segment. "3' target site" refers to the 3'-most nucleotide of a target segment.

[0131] "Active target region" means a target region to which one or more active antisense compounds is targeted. "Active antisense compounds" means antisense compounds that reduce target nucleic acid levels.

[0132] "Hybridization" means the annealing of complementary nucleic acid molecules. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense compound and a nucleic acid target. In certain such embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense oligonucleotide and a nucleic acid target

[0133] "Stringent hybridization conditions" means conditions under which a nucleic acid molecule, such as an antisense compound, will hybridize to a target nucleic acid sequence, but to a minimal number of other sequences.

[0134] "Specifically hybridizable" means an antisense compound that hybridizes to a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids.

[0135] "Complementarity" means the capacity for pairing between nucleobases of a first nucleic acid and a second nucleic acid.

[0136] "Fully complementary" means each nucleobase of a first nucleic acid is capable of pairing with each nucleobase of a second nucleic acid. In certain embodiments, a first nucleic acid is an antisense compound and a target nucleic acid is a second nucleic acid. In certain such embodiments, an antisense oligonucleotide is a first nucleic acid and a target nucleic acid is a second nucleic acid.

[0137] "Non-complementary nucleobase" means a nucleobase of first nucleic acid that is not capable of pairing with the corresponding nucleobase of a target nucleic acid. "Mismatch" a nucleobase of first nucleic acid that is not capable of pairing with the corresponding nucleobase of a target nucleic acid.

[0138] "Portion" means a defined number of contiguous (i.e., linked) nucleobases. In certain embodiments, a portion is a defined number of contiguous nucleobases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an antisense compound.

[0139] "Oligomeric compound" means a polymer of linked monomeric subunits which is capable of hybridizing to at least a region of an RNA molecule.

[0140] "Antisense compound" means an oligomeric compound that is is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.

[0141] "Oligonucleotide" means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another.

[0142] "Oligonucleoside" means an oligonucleotide in which the internucleoside linkages do not contain a phosphorus atom.

[0143] "Unmodified nucleotide" means a nucleotide composed of naturally occurring nucleobases, sugar moieties and internucleoside linkages. In certain embodiments, an unmodified nucleotide is an RNA nucleotide (i.e., β-D-ribonucleosides) or a DNA nucleotide (i.e., β-D-deoxyribonucleoside).

[0144] "Antisense oligonucleotide" means a single-stranded oligonucleotide having a nucleobase sequence that permits hybridization to a corresponding region of a target nucleic acid.

[0145] "Motif" means the pattern of unmodified and modified nucleosides in an antisense compound.

[0146] "Chimeric antisense compound" means an antisense compound that has at least 2 chemically distinct regions, each region having a plurality of subunits.

[0147] A "gapmer" means an antisense compound in which an internal position having a plurality of nucleotides that supports RNaseH cleavage is positioned between external regions having one or more nucleotides that are chemically distinct from the nucleosides of the internal region.

[0148] A "gap segment" means the plurality of nucleotides that make up the internal region of a gapmer.

[0149] A "wing segment" means the external region of a gapmer.

[0150] "Gap-widened" means an antisense compound has a gap segment of 12 or more contiguous 2'-deoxyribonucleotides positioned between 5' and 3' wing segments having from one to six nucleotides having modified sugar moieties.

[0151] "Nucleoside" means a nucleobase linked to a sugar.

[0152] "Nucleobase" means a heterocyclic base moiety capable of pairing with a base of another nucleic acid.

[0153] "Nucleotide" means a nucleoside having a phosphate group covalently linked to the sugar portion of the nucleoside.

[0154] "Nucleobase sequence" means the order of contiguous nucleobases independent of any sugar, linkage, and/or nucleobase modification.

[0155] "Modified nucleotide" means a nucleotide having, independently, a modified sugar moiety, modified internucleoside linkage, or modified nucleobase.

[0156] "Modified nucleoside" means a nucleotide having, independently, a modified sugar moiety or modified nucleobase.

[0157] "Contiguous nucleobases" means nucleobases immediately adjacent to each other.

[0158] "Modified oligonucleotide" means an oligonucleotide comprising a modified internucleoside linkage, a modified sugar, and/or a modified nucleobase.

[0159] "Single-stranded modified oligonucleotide" means a modified oligonucleotide which is not hybridized to a complementary strand.

[0160] "Internucleoside linkage" refers to the chemical bond between nucleosides.

[0161] "Linked nucleoside" means adjacent nucleosides which are bonded together.

[0162] "Linked deoxynucleoside" means a nucleic acid base (A, G, C, T, U) substituted by deoxyribose linked by a phosphate ester to form a nucleotide.

[0163] "Naturally occurring internucleoside linkage" means a 3' to 5' phosphodiester linkage.

[0164] "Modified internucleoside linkage" means substitution and/or any change from a naturally occurring internucleoside bond. In certain instances, the modified internucleoside linkage refers to a phosphodiester internucleoside bond.

[0165] "Phosphorothioate linkage" means a linkage between nucleosides where the phosphodiester bond is modified by replacing one of the non-bridging oxygen atoms with a sulfur atom. A phosphorothioate linkage is a modified internucleoside linkage.

[0166] "Modified sugar moiety" means substitution and/or any change from a natural sugar moiety. For the purposes of this disclosure, a "natural sugar moiety" is a sugar moiety found in DNA (2'-H) or RNA (2'-OH).

[0167] "Modified sugar" refers to a substitution and/or any change from a natural sugar.

[0168] "Bicyclic sugar" means a furosyl ring modified by the bridging of the two non-geminal ring atoms. A bicyclic sugar is a modified sugar.

[0169] "Modified nucleobase" means any nucleobase other than adenine, cytosine, guanine, thymidine, or uracil. An "unmodified nucleobase" means the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U).

[0170] "Modified sugar moiety" means a sugar moiety having a substitution and/or any change from a natural sugar moiety.

[0171] "Natural sugar moiety" means a sugar moiety found in DNA (2'-H) or RNA (2'-OH).

[0172] "2'-O-methoxyethyl sugar moiety" means a 2'-substituted furosyl ring having a 2'-O(CH2)$_2$—OCH$_3$ (2'-O-methoxyethyl or 2'-MOE) substituent group.

[0173] "2'-O-methoxyethyl nucleotide" means a nucleotide comprising a 2'-O-methoxyethyl modified sugar moiety.

[0174] "2'-O-methoxyethyl" refers to an O-methoxy-ethyl modification of the 2' position of a furosyl ring. A 2'-O-methoxyethyl modified sugar is a modified sugar.

[0175] "Bicyclic nucleic acid sugar moiety" means a furosyl ring modified by the bridging of two non-geminal ring atoms.

[0176] "5-methylcytosine" means a cytosine modified with a methyl group attached to the 5' position. A 5-methylcytosine is a modified nucleobase.

[0177] "Prodrug" means a therapeutic agent that is prepared in an inactive form that is converted to an active form (i.e., drug) within the body or cells thereof by the action of endogenous enzymes or other chemicals and/or conditions.

[0178] "Pharmaceutically acceptable salts" means physiologically and pharmaceutically acceptable salts of antisense compounds, i.e., salts that retain the desired biological activity of the parent oligonucleotide and do not impart undesired toxicological effects thereto.

[0179] "Salts" refers to physiologically and pharmaceutically acceptable salts of antisense compounds, i.e., salts that retain the desired biological activity of the parent oligonucleotide and do not impart undesired toxicological effects thereto.

[0180] "Cures" means a method or course that restores health or a prescribed treatment for an illness.

[0181] "Slows progression" means decrease in the development of the said disease.

[0182] "Cap structure" or "terminal cap moiety" means chemical modifications, which have been incorporated at either terminus of an antisense compound.

[0183] "ISIS 301012" means a lipid-lowering agent that is an antisense oligonucleotide having the sequence "GCCTCAGTCTGCTTCGCACC" (SEQ ID NO: 457), where each internucleoside linkage is a phosphorothioate internucleoside linkage, each cytosine is a 5-methylcytosine, nucleotides 6-15 are 2'-deoxynucleotides, and nucleotides 1-5 and 16-20 are 2'-O-methoxyethyl nucleotides. ISIS 301012 does not target PCSK9. It is used herein as a non-PCSK9 control.

BRIEF DESCRIPTION OF THE FIGURES

[0184] FIG. **1** depicts the location of PCSK9 mRNA target sequences of exemplary antisense compounds.

[0185] FIG. **2** depicts the results of a five-point dose response experiment with exemplary antisense compounds in HEK-293 cells. The antisense compounds are identified by Isis number. The bars from left to right represent PCSK9 expression in relative units at 0 nM, 25 nM, 50 nM, 100 nM, 200 nM, and 300 nM antisense oligonucleotides.

[0186] FIG. **3** depicts the results of a five-point dose response experiment with exemplary antisense compounds in cyno primary hepatocytes. The antisense compounds are identified by Isis number. The bars from left to right represent PCSK9 expression in relative units at 300 nM, 150 nM, 50 nM, 25 nM, and 10 nM antisense oligonucleotides. Control values are shown in the rightmost two bars.

[0187] FIG. **4** depicts a reduction of murine PCSK9 mRNA expression (relative units) in the liver of h-apoB/CETP transgenic mice when administered Isis 394816. Isis 394816 was administered by intraperitoneal injection for 6 weeks at a dosing regimen of 20 mg/kg/wk or 50 mg/kg/wk. Isis 329993 was administered at 50 mg/kg/wk as a control.

[0188] FIG. **5**, Panel A, depicts inhibition of plasma LDL-C (mg/dL) in mice administered Isis 394816 at 20 mg/kg/wk or 50 mg/kg/wk. FIG. **5**, Panel B, depicts inhibition of liver TG (mg/gram) in mice administered Isis 394816 at 20 mg/kg/wk or 50 mg/kg/wk.

[0189] FIG. **6** depicts the dose-dependent inhibition (relative units) of liver PCSK9 mRNA, plasma LDL-C, and liver TG in response to the indicated doses of Isis 394816.

[0190] FIG. **7** depicts a correlation between LDL-C lowering and PCSK9 mRNA inhibition, using the data from FIG. **6**.

[0191] FIG. **8** depicts the accumulation of exemplary antisense compounds in rat tissues (μg/g tissue) following administration of antisense compounds. The antisense compounds are identified by Isis number. The bars from left to right represent antisense compound accumulation in kidney and liver.

[0192] FIG. **9** depicts the accumulation of the identified antisense compounds in tissues (μg/g tissue) following administration to a cyno monkey. Accumulation at each dosing concentration (15 mg/kg and 30 mg/kg) is shown in cyno kidney (left side bar) and cyno liver (right side bar).

[0193] FIG. **10** depicts PCSK9 mRNA levels (relative units) following administration of the identified antisense compounds to a cyno monkey. Results are shown at each dosing concentration (15 mg/kg and 30 mg/kg).

[0194] FIG. **11** depicts hepatic PCSK9 mRNA levels (relative units) detected by Northern analysis. Cyno monkeys were administered Isis 405879 ("BMS-844421") at a dose of 15 mg/kg or 30 mg/kg.

[0195] FIG. **12** depicts plasma LDL-C levels (mg/dL) following administration of Isis 405879 (labeled "BMS-844421") at a dose of 15 mg/kg or 30 mg/kg. Left side bars represent baseline levels of LDL-C in control animals; right side bars represent LDL-C following administration of antisense compounds.

[0196] FIG. **13** depicts the ratio of plasma HDL/LDL following administration of Isis 405879 (labeled "BMS-844421") at a dose of 15 mg/kg or 30 mg/kg. Left side bars represent baseline HDL/LDL in control animals; right side bars represent HDL/LDL following administration of antisense compounds.

[0197] FIG. **14** depicts the effect of BMS-844421 and control ASOs on liver human PCSK9 mRNA in transgenic mice (6 wk).

[0198] FIG. **15** depicts the effect of BMS-844421 and control ASOs on plasma levels of human PCSK9 protein in transgenic mice (6 wk).

[0199] FIG. **16** depicts plasma hPCSK9 at baseline (FIG. **16**A), 3 wk (FIG. **16**B), and 6 wk (FIG. **16**C) treatment.

[0200] FIG. **17** depicts the BMS-844421 dose response for plasma hPCSK9 levels in transgenic mice at 6 weeks (wk).

[0201] FIG. **18** depicts the BMS-844421 dose response for liver mRNA and plasma protein (hPCSK9).

[0202] FIG. **19** depicts the dose response for plasma hPCSK9 in male and female transgenic mice.

[0203] FIG. **20** depicts the dose response for liver hPCSK9 mRNA in male versus female transgenic mice.

[0204] FIG. **21** depicts the effect of the ASOs on liver endogenous PCSK9 mRNA in the transgenic mouse after 6 weeks of administration. The numbers following the hyphens provide the mg ASO/kg body weight dose of each ASO. For example, "844421-30" is the data for BMS-84421 administered to test animals at 30 mg/kg.

[0205] FIG. **22** depicts the effect of ASOs on liver endogenous HMGCoA reductase mRNA in the transgenic mouse after 6 weeks of administration.

[0206] FIG. **23** depicts LDLR protein concentration following a 6 wk ASO treatment, as assayed by a Western blot.

[0207] FIG. **24** depicts the quantification of the Western blot shown in FIG. **23**. The LDLR band is normalized to a transferring receptor (TR) band, as observed using anti-LDLR and anti-TR antibodies, respectively.

[0208] FIG. **25** depicts the Liver PCSK9 mRNA concentration in cyno monkeys after 13 weeks of treatment. The numbers in parentheses are administered doses in mg/kg.

[0209] FIG. **26** depicts the concentration of BMS-844421 in the liver of cyno monkeys after 13 weeks of treatment.

[0210] FIG. **27** depicts the average serum LDL-C levels in the cyno monkey study. The numbers in parentheses are administered doses in mg/kg.

[0211] FIG. **28**A depicts the pharmacodynamic responses in serum lipid assays for the cyno 2.5 and 5 wk ASO studies, as displayed in a scatterplot of individual animal data. FIG. **28**B depicts the average ratio of HDL/LDL in BMS-844421-treated cynos at 2.5 and 5 wk. The numbers in parentheses in FIG. **28**B are administered doses in mg/kg.

[0212] FIG. **29** depicts downregulation of PCSK9 target in cyno monkeys at 13 wk in the toxicology study of liver mRNA and plasma PCSK9.

DETAILED DESCRIPTION

Overview

[0213] Elevated levels of LDL-cholesterol (HDL-C) are recognized as a major independent risk factor for coronary heart disease (CHD). Even in individuals undergoing aggressive treatment with currently available cholesterol-lowering agents to reduce LDL-cholesterol (LDL-C) levels, coronary events still occur, and elevated LDL-C levels remain a major risk factor for coronary heart disease in these individuals. Furthermore, many individuals undergoing LDL-lowering therapy do not reach their target LDL-C levels, and thus remain at risk for CHD. Accordingly, there is a need for additional LDL-C lowering agents.

[0214] The antisense compounds and methods provided herein are useful for the treatment of hypercholesterolemia. Treatment of hypercholesterolemia encompasses a therapeutic regimen that results in a clinically desirable outcome. For example, the antisense compounds and methods provided herein are useful for the treatment of elevated cholesterol, such as elevated LDL-C. In addition, the antisense compounds and methods provided herein may be used to reduce the risk of CHD, in individuals exhibiting one or more risk factors for CHD. Furthermore, the antisense compounds and methods provided herein may be used to treat and/or prevent atherosclerosis.

[0215] As illustrated herein, administration of an antisense oligonucleotide targeted to PCSK9 to animals fed a high-fat diet (an experimental model of hyperlipidemia) resulted in antisense inhibition of PCSK9, upregulation of the LDL-R, reduction of LDL-C levels, and reduction of liver triglycerides. Thus, it is demonstrated that in an experimental model of hyperlipidemia, antisense inhibition of PCSK9 results in lowering LDL-C levels. Accordingly, provided herein are methods for the treatment of reducing LDL-C levels through the administration of an antisense compound targeted to a PCSK9 nucleic acid. Increased LDL-C levels are considered a risk factor for CHD, and are also linked to atherosclerosis. Accordingly, also provided herein are methods for the reduction of CHD risk, and for the prevention and/or treatment of atherosclerosis. Also provided herein are methods for the treatment of conditions characterized by elevated liver triglycerides, such as hepatic steatosis.

[0216] In a particular embodiment, methods comprise the use of antisense compounds targeted to particularly advantageous sequences within a PCSK9 nucleic acid. The PCSK9 target sequences are selected on the basis of superior results shown in one or more selection criteria. The presently disclosed antisense compounds all exhibit high in vitro efficacy, determined using a battery of cell models to measure antisense regulation of PCSK9 expression. Further, preferred antisense compounds exhibit relatively high in vivo efficacy, determined using a transgenic mouse model that expresses human PCSK9 or a cynomolgus ("cyno") monkey model. The antisense compound ISIS 405879 (SEQ ID NO: 248) displays superior results in suppressing PCSK9 mRNA expression in the liver of a host individual.

[0217] The present antisense compounds are tolerated to different extents when administered to a host individual. Increased tolerability can depend on a number of factors, including, but not limited to, the nucleotide sequence of the antisense compound, chemical modifications to the nucleotides, the particular motif of unmodified and modified nucleosides in the antisense compound, or combinations thereof. Antisense compounds that exhibit increased or superior tolerability are particularly preferred in the present methods. The present antisense compounds also can show different pharmacokinetic properties, depending on their nucleotide sequence, chemical modifications, motifs, or combinations thereof. Particularly preferred antisense compounds also exhibit favorable pharmacokinetics when administered to a host individual.

Certain Indications

[0218] In certain embodiments, the invention provides methods of treating an individual comprising administering one or more pharmaceutical compositions of the present invention. In certain embodiments, the individual has hypercholesterolemia, mixed dyslipidemia, atherosclerosis, a risk of developing atherosclerosis, coronary heart disease, a history of coronary heart disease, early onset coronary heart disease, acute coronary syndrome, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, dyslipidemia, hypertriglyceridemia, hyperlipidemia, hyperfattyacidemia, hepatic steatosis, non-alcoholic steatohepatitis, or non-alcoholic fatty liver disease.

[0219] Guidelines for lipid-lowering therapy were established in 2001 by Adult Treatment Panel III (ATP III) of the National Cholesterol Education Program (NCEP), and updated in 2004 (Grundy et al., Circulation, 2004, 110, 227-239). The guidelines include obtaining a complete lipoprotein profile, typically after a 9 to 12 hour fast, for determination of LDL-C, total cholesterol, and HDL-C levels. According to the most recently established guidelines, LDL-C levels of 130-159 mg/dL, 160-189 mg/dL, and greater than or equal to 190 mg/dL are considered borderline high, high, and very high, respectively. Total cholesterol levels of 200-239 and greater than or equal to 240 mg/dL are considered borderline high and high, respectively. HDL-C levels of less than 40 mg/dL are considered low.

[0220] In certain embodiments, the individual has been identified as in need of lipid-lowering therapy. In certain such embodiments, the individual has been identified as in need of lipid-lowering therapy according to the guidelines established in 2001 by Adult Treatment Panel III (ATP III) of the National Cholesterol Education Program (NCEP), and updated in 2004 (Grundy et al., Circulation, 2004, 110, 227-239). In certain such embodiments, the individual in need of lipid-lowering therapy has LDL-C above 190 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy has LDL-C above 160 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy has LDL-C above 130 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy has LDL-C above 100 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy should maintain LDL-C below 160 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy should maintain LDL-C below 130 mg/dL. In certain such embodiments, the individual in need of lipid-lowering therapy should maintain LDL-C below 100 mg/dL. In certain such embodiments, the individual should maintain LDL-C below 70 mg/dL or even below 50 mg/dL.

[0221] In certain embodiments, the invention provides methods for reducing ApoB in an individual. In certain embodiments, the invention provides methods for reducing ApoB-containing lipoprotein in an individual. In certain embodiments, the invention provides methods for reducing LDL-C in an individual. In certain embodiments, the invention provides methods for reducing VLDL-C in an individual. In certain embodiments, the invention provides methods for reducing IDL-C in an individual. In certain embodiments, the invention provides methods for reducing non-HDL-C in an individual. In certain embodiments the invention provides methods for reducing Lp(a) in an individual. In certain embodiments, the invention provides methods for reducing serum triglyceride in an individual. In certain embodiments, the invention provides methods for reducing liver triglyceride in an individual. In certain embodiments, the invention provides methods for reducing Ox-LDL-C in an individual. In certain embodiments, the invention provides methods for reducing small LDL particles in an individual. In certain embodiments, the invention provides methods for reducing small VLDL particles in an individual. In certain embodiments, the invention provides methods for reducing phospholipids in an individual. In certain embodiments, the invention provides methods for reducing oxidized phospholipids in an individual.

[0222] In certain embodiments, the methods provided by the present invention do not lower HDL-C. In certain embodiments, the methods provided by the present invention do not result in accumulation of lipids in the liver.

[0223] In one embodiment are methods for decreasing LDL-C levels, or alternatively methods for treating hypercholesterolemia, by administering to an individual suffering from elevated LDL-C levels a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid. In another embodiment, a method of decreasing LDL-C levels comprises selecting an individual in need of a decrease in LDL-C levels, and administering to the individual a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid. In a further embodiment, a method of reducing coronary heart disease risk includes selecting an individual having elevated LDL-C levels and one or more additional indicators of coronary heart disease risk, and administering to the individual a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid.

[0224] In other embodiments, the LDL-C level is from 100-129 mg/dL, from 130 to 159 mg/dL, from 160-189 mg/dL, or greater than or equal to 190 mg/dL.

[0225] In one embodiment, administration of a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid is accompanied by monitoring of LDL-C levels in the serum of an individual, to determine an individual's response to administration of the antisense compound. An individual's response to administration of the antisense compound is used by a physician to determine the amount and duration of therapeutic intervention.

[0226] In one embodiment, administration of an antisense compound targeted to a PCSK9 nucleic acid results in LDL-C levels below 190 mg/dL, below 160 mg/dL, below 130 mg/dL, below 100 mg/dL, below 70 mg/dL, or below 50 mg/dL. In another embodiment, administration of an antisense compound targeted to a PCSK9 nucleic acid decreases LDL-C by at least 15%, by at least 25%, by at least 50%, by at least 60%, by at least 70%, by at least 75%, by at least 80%, by at least 85%, by at least 90%, or by at least 95%.

[0227] An individual having elevated LDL-C levels may also exhibit reduced HDL-C levels and/or elevated total cholesterol levels. Accordingly, in one embodiment a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid is administered to an individual having elevated LDL-C levels, who also has reduced HDL-C levels and/or elevated total cholesterol levels.

[0228] Individuals having elevated LDL-C levels may also exhibit elevated triglyceride levels. Accordingly, in one embodiment a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid is administered to an individual having elevated LDL-C levels, and also having elevated triglyceride levels.

[0229] Atherosclerosis can lead to coronary heart disease, stroke, or peripheral vascular disease. Elevated LDL-C levels are considered a risk factor in the development and progression of atherosclerosis. Accordingly, in one embodiment, a therapeutically effective amount of an antisense compound targeted to a PCSK9 nucleic acid is administered to an individual having atherosclerosis. In a further embodiment, a therapeutically effective amount of antisense compound targeted to a PCSK9 nucleic acid is administered to an individual susceptible to atherosclerosis. Atherosclerosis is assessed directly through routine imaging techniques such as, for example, ultrasound imaging techniques that reveal carotid intimomedial thickness. Accordingly, treatment and/or prevention of atherosclerosis further include monitoring atherosclerosis through routine imaging techniques. In one embodiment, administration of an antisense compound targeted to a PCSK9 nucleic acid leads to a lessening of the severity of atherosclerosis, as indicated by, for example, a reduction of carotid intimomedial thickness in arteries.

[0230] Measurements of cholesterol, lipoproteins and triglycerides are obtained using serum or plasma collected from an individual. Methods of obtaining serum or plasma samples are routine, as are methods of preparation of the serum samples for analysis of cholesterol, triglycerides, and other serum markers.

[0231] A physician may determine the need for therapeutic intervention for individuals in cases where more or less aggressive LDL-lowering therapy is needed. The practice of the methods herein may be applied to any altered guidelines provided by the NCEP, or other entities that establish guidelines for physicians used in treating any of the diseases or conditions listed herein, for determining coronary heart disease risk and diagnosing metabolic syndrome.

[0232] In one embodiment, administration of an antisense compound targeted to a PCSK9 nucleic acid is parenteral administration. Parenteral administration may be intravenous or subcutaneous administration. Accordingly, in another embodiment, administration of an antisense compound targeted to a PCSK9 nucleic acid is intravenous or subcutaneous administration. Administration may include multiple doses of an antisense compound targeted to a PCSK9 nucleic acid.

[0233] In certain embodiments, a pharmaceutical composition comprising an antisense compound targeted to PCSK9 is for use in therapy. In certain embodiments, the therapy is the reduction of LDL-C, ApoB, VLDL-C, IDL-C, non-HDL-C, Lp(a), serum triglyceride, liver triglyceride, Ox-LDL-C, small LDL particles, small VLDL, phospholipids, or oxidized phospholipids in an individual. In certain embodiments, the therapy is the treatment of hypercholesterolemia, mixed dyslipidemia, atherosclerosis, a risk of developing atherosclerosis, coronary heart disease, acute coronary syndrome, a history of coronary heart disease, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, dyslipidemia, hypertriglyceridemia, hyperlipidemia, hyperfattyacidemia, hepatic steatosis, non-alcoholic steatohepatitis, or non-alcoholic fatty liver disease. In additional embodiments, the therapy is the reduction of CHD risk. In certain aspects, the therapy is prevention of atherosclerosis. In certain embodiments, the therapy is the prevention of coronary heart disease.

[0234] In certain embodiments, a pharmaceutical composition comprising an antisense compound targeted to PCSK9 is used for the preparation of a medicament for reducing LDL-C, ApoB, VLDL-C, IDL-C, non-HDL-C, Lp(a), serum triglyceride, liver triglyceride, Ox-LDL-C, small LDL particles, small VLDL, phospholipids, or oxidized phospholipids in an individual. In certain embodiments pharmaceutical composition comprising an antisense compound targeted to PCKS9 is used for the preparation of a medicament for reducing coronary heart disease risk. In certain embodiments, an antisense compound targeted to PCSK9 is used for the preparation of a medicament for the treatment of hypercholesterolemia, mixed dyslipidemia, atherosclerosis, a risk of developing atherosclerosis, coronary heart disease, a history of coronary heart disease, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, dyslipidemia, hypertriglyceridemia, hyperlipidemia, hyperfattyacidemia, hepatic steatosis, non-alcoholic steatohepatitis, or non-alcoholic fatty liver disease.

Certain Combination Therapies

[0235] In certain embodiments, one or more pharmaceutical compositions of the present invention are co-administered with one or more other pharmaceutical agents. In certain embodiments, such one or more other pharmaceutical agents are designed to treat the same disease or condition as the one or more pharmaceutical compositions of the present invention. In certain embodiments, such one or more other pharmaceutical agents are designed to treat a different disease or condition as the one or more pharmaceutical compositions of the present invention. In certain embodiments, such one or more other pharmaceutical agents are designed to treat an undesired effect of one or more pharmaceutical compositions of the present invention. In certain embodiments, one or more pharmaceutical compositions of the present invention are co-administered with another pharmaceutical agent to treat an undesired effect of that other pharmaceutical agent. In certain embodiments, one or more pharmaceutical compositions of the present invention and one or more other pharmaceutical agents are administered at the same time. In certain embodiments, one or more pharmaceutical compositions of the present invention and one or more other pharmaceutical agents are administered at different times. In certain embodiments, one or more pharmaceutical compositions of the present invention and one or more other pharmaceutical agents are prepared together in a single formulation. In certain embodiments, one or more pharmaceutical compositions of the present invention and one or more other pharmaceutical agents are prepared separately. For example, a composition may comprise a pharmaceutical agent for separate, sequential, or simultaneous administration with an antisense compound.

[0236] In certain embodiments, pharmaceutical agents that may be co-administered with a pharmaceutical composition of the present invention include lipid-lowering agents or LXR agonists. In certain such embodiments, pharmaceutical agents that may be co-administered with a pharmaceutical composition of the present invention include, but are not limited to atorvastatin, simvastatin, rosuvastatin, and ezetimibe. In certain such embodiments, the lipid-lowering agent is administered prior to administration of a pharmaceutical composition of the present invention. In certain such embodiments, the lipid-lowering agent is administered following administration of a pharmaceutical composition of the present invention. In certain such embodiments the lipid-lowering agent is administered at the same time as a pharmaceutical composition of the present invention. In certain such embodiments the dose of a co-administered lipid-lowering agent is the same as the dose that would be administered if the lipid-lowering agent was administered alone. In certain such embodiments the dose of a co-administered lipid-lowering agent is lower than the dose that would be administered if the lipid-lowering agent was administered alone. In certain such embodiments the dose of a co-administered lipid-lowering agent is greater than the dose that would be administered if the lipid-lowering agent was administered alone.

[0237] In certain embodiments, a co-administered lipid-lowering agent is a HMG-CoA reductase inhibitor. In certain such embodiments the HMG-CoA reductase inhibitor is a statin. In certain such embodiments, the statin is selected from, for example, atorvastatin, simvastatin, pravastatin, fluvastatin, and rosuvastatin.

[0238] In certain embodiments, a co-administered lipid-lowering agent is a cholesterol absorption inhibitor. In certain such embodiments, cholesterol absorption inhibitor is ezetimibe.

[0239] In certain embodiments, a co-administered lipid-lowering agent is a co-formulated HMG-CoA reductase inhibitor and cholesterol absorption inhibitor. In certain such embodiments the co-formulated lipid-lowering agent is ezetimibe/simvastatin.

[0240] In certain embodiments, a co-administered lipid-lowering agent is a microsomal triglyceride transfer protein inhibitor (MTP inhibitor).

[0241] In certain embodiments, a co-administered lipid-lowering agent is an oligonucleotide targeted to ApoB.

[0242] In certain embodiments, a co-administered pharmaceutical agent is a bile acid sequestrant. In certain such embodiments, the bile acid sequestrant is selected from cholestyramine, colestipol, and colesevelam.

[0243] In certain embodiments, a co-administered pharmaceutical agent is a nicotinic acid. In certain such embodiments, the nicotinic acid is selected from immediate release nicotinic acid, extended release nicotinic acid, and sustained release nicotinic acid.

[0244] In certain embodiments, a co-administered pharmaceutical agent is a fibric acid. In certain such embodiments, a fibric acid is selected from gemfibrozil, fenofibrate, clofibrate, bezafibrate, and ciprofibrate.

[0245] Further examples of pharmaceutical agents that may be co-administered with a pharmaceutical composition of the present invention include, but are not limited to, corticosteroids, including but not limited to prednisone; LXR agonists; immunoglobulins, including, but not limited to intravenous immunoglobulin (IVIg); analgesics (e.g., acetaminophen); anti-inflammatory agents, including, but not limited to non-steroidal anti-inflammatory drugs (e.g., ibuprofen, COX-1 inhibitors, and COX-2, inhibitors); salicylates; antibiotics; antivirals; antifungal agents; antidiabetic agents (e.g., biguanides, glucosidase inhibitors, insulins, sulfonylureas, and thiazolidenediones); adrenergic modifiers; diuretics; hormones (e.g., anabolic steroids, androgen, estrogen, calcitonin, progestin, somatostan, and thyroid hormones); immunomodulators; muscle relaxants;

antihistamines; osteoporosis agents (e.g., biphosphonates, calcitonin, and estrogens); prostaglandins, antineoplastic agents; psychotherapeutic agents; sedatives; poison oak or poison sumac products; antibodies; and vaccines.

[0246] In certain embodiments, the pharmaceutical compositions of the present invention may be administered in conjunction with a lipid-lowering therapy. In certain such embodiments, a lipid-lowering therapy is therapeutic lifestyle change. In certain such embodiments, a lipid-lowering therapy is LDL apheresis.

Antisense Compounds

[0247] Oligomeric compounds include, but are not limited to, oligonucleotides, oligonucleosides, oligonucleotide analogs, oligonucleotide mimetics, antisense compounds, antisense oligonucleotides, and siRNAs. An oligomeric compound may be “antisense” to a target nucleic acid, meaning that is is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.

[0248] In certain embodiments, an antisense compound has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted. In certain such embodiments, an antisense oligonucleotide has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

[0249] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid is 8 to 80, 12 to 50, 12 to 30 or 15 to 30 subunits in length. In other words, antisense compounds are from 8 to 80, 12 to 50, 12 to 30 or 15 to 30 linked subunits. In certain such embodiments, the antisense compounds are 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 subunits in length.

[0250] In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 12 to 30 nucleotides in length. In certain such embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length.

[0251] In certain embodiment, an antisense compound targeted to a PCSK9 nucleic acid is 15 to 30 subunits in length. In other words, antisense compounds are from 15 to 30 linked subunits. In certain such embodiments, the antisense compounds are 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 subunits in length.

[0252] In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 15 to 30 nucleotides in length. In certain such embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length.

[0253] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid is 18 to 24 subunits in length. In other words, antisense compounds are from 18 to 24 linked subunits. In one embodiment, the antisense compounds are 18, 19, 20, 21, 22, 23, or 24 subunits in length.

[0254] In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 18 to 24 nucleotides in length. In certain such embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 18, 19, 20, 21, 22, 23, or 24 nucleotides in length.

[0255] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid is 19 to 22 subunits in length. In other words, antisense compounds are from 19 to 22 linked subunits. This embodies antisense compounds of 19, 20, 21, or 22 subunits in length.

[0256] In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 19 to 22 nucleotides in length. In certain such embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 19, 20, 21, or 22 nucleotides in length.

[0257] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid is 20 subunits in length. In certain such embodiments, antisense compounds are 20 linked subunits in length.

[0258] In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid is 20 nucleotides in length. In certain such embodiments, an antisense oligonucleotide targeted to an PCSK9 nucleic acid is 20 linked nucleotides in length.

[0259] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence target a region identified herein.

[0260] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid may target the following nucleotide regions of SEQ ID NO: 1: 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569.

[0261] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 1: 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1021, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569.

[0262] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid may target the following nucleotide regions of SEQ ID NO: 2: 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376.

[0263] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 2: 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376.

[0264] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid may target the following nucleotide regions of SEQ ID NO: 3: 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824.

[0265] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 3: 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824.

[0266] In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid may target the following nucleotide regions of SEQ ID NO: 1: 320-405, 441-445, 527-544, 582-590, 744-781, 811-820, 918-922, 953-959, 1034-1036, 1152-1153, 1174-1199, 1251-1272, 1445-1464, 1603-1604, 1625-1627, 1707-1734, 1766-1811, 1832-1848, 1880-1904, 1956-1961, 1982-1939, 2030-2039, 2060-2099, 2140-2149, 2170-2186, 2207-2304, 2355-2409, 2435-2503, 2529-2581, 2626-2648, 2669-2749, 2770-2827, 2856-2876, 2891-2899, 2930-2982, 3014-3226, 3253-3436, 3457-3471, or 3497-3542.

[0267] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 1: 320-405, 441-445, 527-544, 582-590, 744-781, 811-820, 918-922, 953-959, 1034-1036, 1152-1153, 1174-1199, 1251-1272, 1445-1464, 1603-1604, 1625-1627, 1707-1734, 1766-1811, 1832-1848, 1880-1904, 1956-1961, 1982-1939, 2030-2039, 2060-2099, 2140-2149, 2170-2186, 2207-2304, 2355-2409, 2435-2503, 2529-2581, 2626-2648, 2669-2749, 2770-2827, 2856-2876, 2891-2899, 2930-2982, 3014-3226, 3253-3436, 3457-3471, or 3497-3542.

[0268] In certain embodiments, a shortened or truncated antisense compound targeted to a PCSK9 nucleic acid has a single subunit deleted from the 5' end (5' truncation), or alternatively from the 3' end (3' truncation). A shortened or truncated antisense compound targeted to a PCSK9 nucleic acid may have two subunits deleted from the 5' end, or alternatively may have two subunits deleted from the 3' end, of the antisense compound. Alternatively, the deleted nucleosides may be dispersed throughout the antisense compound, for example, in an antisense compound having one nucleoside deleted from the 5' end and one nucleoside deleted from the 3' end.

[0269] When a single additional subunit is present in a lengthened antisense compound, the additional subunit may be located at the 5' or 3' end of the antisense compound. When two are more additional subunits are present, the added subunits may be adjacent to each other, for example, in an antisense compound having two subunits added to the 5' end (5' addition), or alternatively to the 3' end (3' addition), of the antisense compound. Alternatively, the added subunits may be dispersed throughout the antisense compound, for example, in an antisense compound having one subunit added to the 5' end and one subunit added to the 3' end.

[0270] It is possible to increase or decrease the length of an antisense compound, such as an antisense oligonucleotide, and/or introduce mismatch bases without eliminating activity. For example, in Woolf et al. (Proc. Natl. Acad. Sci. USA 89:7305-7309, 1992), a series of antisense oligonucleotides 13-25 nucleobases in length were tested for their ability to induce cleavage of a target RNA in an oocyte injection model. Antisense oligonucleotides 25 nucleobases in length with 8 or 11 mismatch bases near the ends of the antisense oligonucleotides were able to direct specific cleavage of the target mRNA, albeit to a lesser extent than the antisense oligonucleotides that contained no mismatches. Similarly, target specific cleavage was achieved using 13 nucleobase antisense oligonucleotides, including those with 1 or 3 mismatches.

[0271] Gautschi et al (J. Natl. Cancer Inst. 93:463-471, March 2001) demonstrated the ability of an oligonucleotide having 100% complementarity to the bcl-2 mRNA and having 3 mismatches to the bcl-xL mRNA to reduce the expression of both bcl-2 and bcl-xL in vitro and in vivo. Furthermore, this oligonucleotide demonstrated potent antitumor activity in vivo.

[0272] Maher and Dolnick (Nuc. Acid. Res. 16:3341-3358, 1988) tested a series of tandem 14 nucleobase antisense oligonucleotides, and a 28 and 42 nucleobase antisense oligonucleotides comprised of the sequence of two or three of the tandem antisense oligonucleotides, respectively, for their ability to arrest translation of human DHFR in a rabbit reticulocyte assay. Each of the three 14 nucleobase antisense oligonucleotides alone was able to inhibit translation, albeit at a more modest level than the 28 or 42 nucleobase antisense oligonucleotides.

[0273] PCT/US2007/068404 describes incorporation of chemically-modified high-affinity nucleotides into short antisense compounds about 8-16 nucleobases in length and that such compounds are useful in the reduction of target RNAs in animals with increased potency and improved therapeutic index.

[0274] In certain embodiments, antisense compounds targeted to a PCSK9 nucleic acid are short antisense compounds. In certain embodiments, such short antisense compounds are oligonucleotide compounds. In certain embodiments, such short antisense compounds are about 8 to 16, preferably 9 to 15, more preferably 9 to 14, more preferably 10 to 14 nucleotides in length and comprises a gap region flanked on each side by a wing, wherein each wing independently consists of 1 to 3 nucleotides. Preferred motifs include but are not limited to wing—deoxy gap—wing motifs selected from 3-10-3, 2-10-3, 2-10-2, 1-10-1, 2-8-2, 1-8-1, 3-6-3 or 1-6-1.

[0275] Antisense compounds targeted to a PCSK9 nucleic acid are synthesized in vitro and do not include antisense compositions of biological origin, or genetic vector constructs designed to direct the in vivo synthesis of antisense molecules.

[0276] In certain embodiments, an antisense compound is targeted to a region of a PCSK9 nucleic acid that does not contain a single nucleotide polymorphism (SNPs). In certain embodiments, an antisense compound is targeted to a region of a PCSK9 nucleic acid that does contain a single nucleotide polymorph (SNPs). A single nucleotide polymorphism refers to polymorphisms that are the result of a single nucleotide alteration or the existence of two or more alternative sequences which can be, for example, different allelic forms of a gene. A polymorphism may comprise one or more base changes including, for example, an insertion, a repeat, or a deletion. In certain embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid overlaps with a SNP at the following positions: 428, 432, 449, 996, 1011, 1044, 1317, 1565, 1617, 1618, 1671, 1711, 1722, 1836, 1911. In certain embodiments, the compounds provided herein that target a region of a PCSK9 nucleic acid that contains one or more SNPs will contain the appropriate base substitution, insertion, repeat or deletion such that the compound is fully complementary to the altered PCSK9 nucleic acid sequence.

[0277] A subgroup of these antisense compounds were selected for further characterization, based on the ability of the selected antisense compounds to inhibit PCSK9 expression selectively and effectively in cell culture assays, set forth in the Examples. The antisense compounds selected for further evaluation are listed in Table 1. The 5' and 3' boundaries of the PCSK9 target sequence are shown for each antisense compound, with reference to the nucleotide positions of SEQ ID NO: 1.

[0278] "Motif" in Table 1 refers to the use of chemically modified nucleosides at the 5' and 3' ends of the antisense compounds. In a "5-10-5" motif, for example, five chemical modified nucleosides at both the 5' and 3' ends flank ten unmodified nucleosides in the center of the antisense compound. As disclosed in more detail below, modified nucleosides may make the antisense compounds more resistant to nucleases, among other things, which generally improves the pharmacodynamic and pharmacokinetic properties of the antisense compounds when administered to an individual.

[0279] Table 1 further discloses the nucleobase sequence of the antisense compounds. In certain embodiments, an antisense compound has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

TABLE 1

Isis No.	5' Target Site	3' Target Site	Sequence 5'-3'	Motif	SEQ ID NO
405881	1005	1024	CACCCTTGGCCACGCCGGCA	5-10-5	250
399819	2509	2528	CCCACTCAAGGGCCAGGCCA	5-10-5	65
395165	1004	1023	ACCCTTGGCCACGCCGGCAT	5-10-5	28
405879	1002	1021	CCTTGGCCACGCCGGCATCC	5-10-5	248
406008	602	621	CTTGGTGAGGTATCCCCGGC	5-10-5	188
405891	1567	1586	TCCTCAGGGAACCAGGCCTC	5-10-5	352
395186	2100	2119	CTTTGCATTCCAGACCTGGG	5-10-5	60
405988	1854	1873	GGCAGCACCTGGCAATGGCG	5-10-5	381
405994	1928	1947	GCAGTGGACACGGGTCCCCA	5-10-5	400
406023	787	806	TGGTATTCATCCGCCCGGTA	5-10-5	212
395187	2310	2329	GGCAGCAGATGGCAACGGCT	5-10-5	62
395185	1920	1939	CACGGGTCCCCATGCTGGCC	5-10-5	59
406033	967	986	CCTGCCAGGTGGGTGCCATG	5-10-5	237
405923	1295	1314	GGCATTGGTGGCCCCAACTG	5-10-5	288
399900	1569	1588	GGTCCTCAGGGAACCAGGCC	3-14-3	50
405995	1930	1949	TGGCAGTGGACACGGGTCCC	5-10-5	402
405991	1922	1941	GACACGGGTCCCCATGCTGG	5-10-5	394
406005	559	578	CGGGCAGTGCGCTCTGACTG	5-10-5	180

TABLE 1-continued

Isis No.	5' Target Site	3' Target Site	Sequence 5'-3'	Motif	SEQ ID NO
399793	417	436	CCTCGGAACGCAAGGCTAGC	5-10-5	8
395152	410	429	ACGCAAGGCTAGCACCAGCT	5-10-5	7

Antisense Compound Motifs

[0280] In certain embodiments, antisense compounds targeted to a PCSK9 nucleic acid have chemically modified subunits arranged in patterns, or motifs, to confer to the antisense compounds properties such as enhanced the inhibitory activity, increased binding affinity for a target nucleic acid, or resistance to degradation by in vivo nucleases.

[0281] Chimeric antisense compounds typically contain at least one region modified so as to confer increased resistance to nuclease degradation, increased cellular uptake, increased binding affinity for the target nucleic acid, and/or increased inhibitory activity. A second region of a chimeric antisense compound may serve as a substrate for the cellular endonuclease RNase H, which cleaves the RNA strand of an RNA:DNA duplex.

[0282] Antisense compounds having a gapmer motif are considered chimeric antisense compounds. In a gapmer an internal position having a plurality of nucleotides that supports RNaseH cleavage is positioned between external regions having a plurality of nucleotides that are chemically distinct from the nucleosides of the internal region. In the case of an antisense oligonucleotide having a gapmer motif, the gap segment generally serves as the substrate for endonuclease cleavage, while the wing segments comprise modified nucleosides. The regions of a gapmer are differentiated by the types of sugar moieties comprising each distinct region. The types of sugar moieties that are used to differentiate the regions of a gapmer may in some embodiments include β-D-ribonucleosides, β-D-deoxyribonucleosides, 2'-modified nucleosides (such 2'-modified nucleosides may include 2'-MOE, and 2'-O—CH_3, among others), and bicyclic sugar modified nucleosides (such bicyclic sugar modified nucleosides may include those having a 4'-(CH2)n-O-2' bridge, where n=1 or n=2). In general, each distinct region comprises uniform sugar moieties. The wing-gap-wing motif is frequently described as "X-Y-Z", where "X" represents the length of the 5' wing region, "Y" represents the length of the gap region, and "Z" represents the length of the 3' wing region.

[0283] In some embodiments, an antisense compound targeted to a PCSK9 nucleic acid has a gap-widened motif. In other embodiments, an antisense oligonucleotide targeted to a PCSK9 nucleic acid has a gap-widened motif.

[0284] The gap-widened antisense oligonucleotides described herein may have various wing-gap-wing motifs selected from: 1-16-1, 2-15-1, 1-15-2, 1-14-3, 3-14-1, 2-14-2, 1-13-4, 4-13-1, 2-13-3, 3-13-2, 1-12-5, 5-12-1, 2-12-4, 4-12-2, 3-12-3, 1-11-6, 6-11-1, 2-11-5, 5-11-2, 3-11-4, 4-11-3, 1-17-1, 2-16-1, 1-16-2, 1-15-3, 3-15-1, 2-15-2, 1-14-4, 4-14-1, 2-14-3, 3-14-2, 1-13-5, 5-13-1, 2- 13-4, 4-13-2, 3-13-3, 1-12-6, 6-12-1, 2-12-5, 5-12-2, 3-12-4, 4-12-3, 1-11-7, 7-11-1, 2-11-6, 6-11-2, 3-11-5, 5-11-3, 4-11-4, 1-18-1, 1-17-2, 2-17-1, 1-16-3, 1-16-3, 2-16-2, 1-15-4, 4-15-1, 2-15-3, 3-15-2, 1-14-5, 5-14-1, 2-14-4, 4-14-2, 3-14-3, 1-13-6, 6-13-1, 2-13-5, 5-13-2, 3-13-4, 4-13-3, 1-12-7, 7-12-1, 2-12-6, 6-12-2, 3-12-5, 5-12-3, 4-12-4, 1-11-8, 8-11-1, 2-11-7, 7-11-2, 3-11-6, 6-11-3, 4-11-5, 5-11-4, 1-18-1, 1-17-2, 2-17-1, 1-16-3, 3-16-1, 2-16-2, 1-15-4, 4-15-1, 2-15-3, 3-15-2, 1-14-5, 2-14-4, 4 2, 3-14-3, 1-13-6, 6-13-1, 2-13-5, 5-13-2, 3-13-4, 4-13-3, 1-12-7, 7-12-1, 2-12-6, 6-12-2, 3-12-5, 5-12-3, 4-12-4, 1-11-8, 8-11-1, 2-11-7, 7-11-2, 3-11-6, 6-11-3, 4-11-5, 5-11-4, 1-19-1, 1-18-2, 2-18-1, 1-17-3, 3-17-1, 2-17-2, 1-16-4, 4-16-1, 2-16-3, 3-16-2, 1-15-5, 2-15-4, 4-15-2, 3-15-3, 1-14-6, 6 1, 2-14-5, 5-14-2, 3-14-4, 4-14-3, 1-13-7, 7-13-1, 2-13-6, 6-13-2, 3-13-5, 5-13-3, 4-13-4, 1-12-8, 8-12-1, 2-12-7, 7-12-2, 3-12-6, 6-12-3, 4-12-5, 5-12-4, 2-11-8, 8-11-2, 3-11-7, 7-11-3, 4-11-6, 6-11-4, 5-11-5, 1-20-1, 1-19-2, 2-19-1, 1-18-3, 3-18-1, 2-18-2, 1-17-4, 4-17-1, 2-17-3, 3-17-2, 1-16-5, 2-16-4, 4-16-2, 3-16-3, 1-15-6, 6-15-1, 2-15-5, 5-15-2, 3-15-4, 4-15-3, 1-14-7, 7-14-1, 2-14-6, 6- 14-2, 3-14-5, 5-14-3, 4-14-4, 1-13-8, 8-13-1, 2-13-7, 7-13-2, 3-13-6, 6-13-3, 4-13-5, 5-13-4, 2-12-8, 8-12-2, 3-12-7, 7-12-3, 4-12-6, 6-12-4, 5-12-5, 3-11-8, 8-11-3, 4-11-7, 7-11-4, 5-11-6, 6-11-5, 1-21-1, 1-20-2, 2-20-1, 1-20-3, 3-19-1, 2-19-2, 1-18-4, 4-18-1, 2-18-3, 3-18-2, 1-17-5, 2-17-4, 4-17-2, 3- 17-3, 1-16-6, 6-16-1, 2-16-5, 5-16-2, 3-16-4, 4-16-3, 1-15-7, 7-15-1, 2-15-6, 6-15-2, 3-15-5, 5-15-3, 4-15-4, 1-14-8, 8-14-1, 2-14-7, 7-14-2, 3-14-6, 6-14-3, 4-14-5, 5-14-4, 2-13-8, 8-13-2, 3-13-7, 7-13-3, 4-13-6, 6-13-4, 5-13-5, 1-12-10, 10-12-1, 2-12-9, 9-12-2, 3-12-8, 8-12-3, 4-12-7, 7-12-4, 5-12-6, 6-12-5, 4-11-8, 8-11-4, 5-11-7, 7-11-5, 6-11-6, 1-22-1, 1-21-2, 2-21-1, 1-21-3, 3-20-1, 2-20-2, 1-19-4, 4-19-1, 2-19-3, 3-19-2, 1-18-5, 2-18-4, 4-18-2, 3-18-3, 1-17-6, 6-17-1, 2-17-5, 5-17-2, 3-17-4, 4- 17-3, 1-16-7, 7-16-1, 2-16-6, 6-16-2, 3-16-5, 5-16-3, 4-16-4, 1-15-8, 8-15-1, 2-15-7, 7-15-2, 3-15-6, 6-15-3, 4-15-5, 5-15-4, 2-14-8, 8-14-2, 3-14-7, 7-14-3, 4-14-6, 6-14-4, 5-14-5, 3-13-8, 8-13-3, 4-13-7, 7-13-4, 5-13-6, 6-13-5, 4-12-8, 8-12-4, 5-12-7, 7-12-5, 6-12-6, 5-11-8, 8-11-5, 6-11-7, or 7-11-6. In certain preferred embodiments, a gap-widened motif includes, but is not limited to, 2-13-5, 3-14-3, 3-14-4 gapmer motif.

[0285] In one embodiment, a gap-widened antisense oligonucleotide targeted to a PCSK9 nucleic acid has a gap segment of fourteen 2'-deoxyribonucleotides positioned between wing segments of three chemically modified nucleosides. In one embodiment, the chemical modification comprises a 2'-sugar modification. In another embodiment, the chemical modification comprises a 2'-MOE sugar modification.

[0286] In one embodiment, antisense compounds targeted to a PCSK9 nucleic acid possess a 5-10-5 gapmer motif.

Target Nucleic Acids, Target Regions and Nucleotide Sequences

[0287] Nucleotide sequences that encode PCSK9 include, without limitation, the following: GENBANK® Accession No. NM_174936.2, first deposited with GENBANK® on Jun. 1, 2003, and incorporated herein as SEQ ID NO: 1; nucleotides 25475000 to 25504000 of GENBANK® Accession No. NT_032977.8, first deposited with GENBANK® on Feb. 26, 2006, and incorporated herein as SEQ ID NO: 2; and GENBANK® Accession No. AK124635.1, first deposited with GENBANK® on Sep. 8, 2003, and incorporated herein as SEQ ID NO: 3.

[0288] It is noted that some portions of these nucleotide sequences share identical sequence. For example, portions of SEQ ID NO: 1 are identical to portions of SEQ ID NO:

2; portions of SEQ ID NO: 1 are identical to portions of SEQ ID NO: 3; and portions of SEQ ID NO: 2 are identical to portions of SEQ ID NO: 3. Accordingly, antisense compounds targeted to SEQ ID NO: 1 may also target SEQ ID NO: 2 and/or SEQ ID NO: 3; antisense compounds targeted to SEQ ID NO: 2 may also target SEQ ID NO: 1 and/or SEQ ID NO: 3; and antisense compounds targeted to SEQ ID NO: 3 may also target SEQ ID NO: 1 and/or SEQ ID NO: 2. Examples of such antisense compounds are shown in the following tables.

[0289] In certain embodiments, antisense compounds target a PCSK9 nucleic acid having the sequence of GENBANK® Accession No. NM_174936.2, first deposited with GENBANK® on Jun. 1, 2003, and incorporated herein as SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide targets SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 1 is at least 90% complementary to SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 1 is at least 95% complementary to SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 1 is 100% complementary to SEQ ID NO: 1. In certain embodiments, an antisense oligonucleotide targeted to SEQ ID NO: 1 comprises a nucleotide sequence selected from the nucleotide sequences set forth in Table 2.

TABLE 2

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
4	135	154	GCGCGGAATCCTGGCTGGGA
5	242	261	GAGGAGACCTAGAGGCCGTG
159	294	313	GCCTGGAGCTGACGGTGCCC
160	298	317	GACCGCCTGGAGCTGACGGT
6	300	319	AGGACCGCCTGGAGCTGACG
162	406	425	AAGGCTAGCACCAGCTCCTC
163	407	426	CAAGGCTAGCACCAGCTCCT
164	408	427	GCAAGGCTAGCACCAGCTCC
165	409	428	CGCAAGGCTAGCACCAGCTC
7	410	429	ACGCAAGGCTAGCACCAGCT
166	411	430	AACGCAAGGCTAGCACCAGC
167	412	431	GAACGCAAGGCTAGCACCAG
168	413	432	GGAACGCAAGGCTAGCACCA
169	414	433	CGGAACGCAAGGCTAGCACC
8	417	436	CCTCGGAACGCAAGGCTAGC
170	421	440	TCCTCCTCGGAACGCAAGGC
171	446	465	GTGCTCGGGTGCTTCGGCCA
172	466	485	TGGAAGGTGGCTGTGGTTCC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
9	480	499	CCTTGGCGCAGCGGTGGAAG
173	482	501	ATCCTTGGCGCAGCGGTGGA
174	484	503	GGATCCTTGGCGCAGCGGTG
175	488	507	CCACGGATCCTTGGCGCAGC
176	507	526	CGTAGGTGCCAGGCAACCTC
177	545	564	TGACTGCGAGAGGTGGGTCT
178	555	574	CAGTGCGCTCTGACTGCGAG
179	557	576	GGCAGTGCGCTCTGACTGCG
180	559	578	CGGGCAGTGCGCTCTGACTG
10	561	580	GGCGGGCAGTGCGCTCTGAC
181	562	581	CGGCGGGCAGTGCGCTCTGA
182	591	610	ATCCCCGGCGGGCAGCCTGG
183	595	614	AGGTATCCCCGGCGGGCAGC
184	597	616	TGAGGTATCCCCGGCGGGCA
185	598	617	GTGAGGTATCCCCGGCGGGC
186	599	618	GGTGAGGTATCCCCGGCGGG
11	600	619	TGGTGAGGTATCCCCGGCGG
187	601	620	TTGGTGAGGTATCCCCGGCG
188	602	621	CTTGGTGAGGTATCCCCGGC
189	603	622	TCTTGGTGAGGTATCCCCGG
190	604	623	ATCTTGGTGAGGTATCCCCG
191	605	624	GATCTTGGTGAGGTATCCCC
12	606	625	GGATCTTGGTGAGGTATCCC
192	607	626	AGGATCTTGGTGAGGTATCC
193	609	628	GCAGGATCTTGGTGAGGTAT
194	611	630	ATGCAGGATCTTGGTGAGGT
195	613	632	ACATGCAGGATCTTGGTGAG
13	615	634	AGACATGCAGGATCTTGGTG
196	617	636	GAAGACATGCAGGATCTTGG
14	620	639	ATGGAAGACATGCAGGATCT
197	628	647	AGAAGGCCATGGAAGACATG
198	638	657	GAAGCCAGGAAGAAGGCCAT
15	646	665	TTCACCAGGAAGCCAGGAAG
199	648	667	TCTTCACCAGGAAGCCAGGA
16	651	670	TCATCTTCACCAGGAAGCCA

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
200	653	672	ACTCATCTTCACCAGGAAGC
201	655	674	CCACTCATCTTCACCAGGAA
202	657	676	CGCCACTCATCTTCACCAGG
203	659	678	GTCGCCACTCATCTTCACCA
204	661	680	AGGTCGCCACTCATCTTCAC
205	663	682	GCAGGTCGCCACTCATCTTC
206	665	684	CAGCAGGTCGCCACTCATCT
207	667	686	TCCAGCAGGTCGCCACTCAT
208	685	704	GGCAACTTCAAGGCCAGCTC
17	705	724	CCTCGATGTAGTCGACATGG
209	724	743	GCAAAGACAGAGGAGTCCTC
210	782	801	TTCATCCGCCCGGTACCGTG
211	784	803	TATTCATCCGCCCGGTACCG
18	785	804	GTATTCATCCGCCCGGTACC
212	787	806	TGGTATTCATCCGCCCGGTA
213	789	808	GCTGGTATTCATCCGCCCGG
214	791	810	GGGCTGGTATTCATCCGCCC
215	821	840	ATACACCTCCACCAGGCTGC
216	832	851	GTGTCTAGGAGATACACCTC
19	835	854	CTGGTGTCTAGGAGATACAC
217	837	856	TGCTGGTGTCTAGGAGATAC
20	840	859	GTATGCTGGTGTCTAGGAGA
21	860	879	GATTTCCCGGTGGTCACTCT
218	862	881	TCGATTTCCCGGTGGTCACT
219	863	882	CTCGATTTCCCGGTGGTCAC
220	864	883	CCTCGATTTCCCGGTGGTCA
221	865	884	CCCTCGATTTCCCGGTGGTC
22	866	885	GCCCTCGATTTCCCGGTGGT
222	867	886	TGCCCTCGATTTCCCGGTGG
223	868	887	CTGCCCTCGATTTCCCGGTG
224	869	888	CCTGCCCTCGATTTCCCGGT
225	870	889	CCCTGCCCTCGATTTCCCGG
226	874	893	ATGACCCTGCCCTCGATTTC
227	876	895	CCATGACCCTGCCCTCGATT
228	878	897	GACCATGACCCTGCCCTCGA
23	880	899	GTGACCATGACCCTGCCCTC
229	882	901	CGGTGACCATGACCCTGCCC
230	884	903	GTCGGTGACCATGACCCTGC
231	886	905	AAGTCGGTGACCATGACCCT
232	888	907	CGAAGTCGGTGACCATGACC
24	890	909	CTCGAAGTCGGTGACCATGA
233	898	917	GGCACATTCTCGAAGTCGGT
25	923	942	GTGGAAGCGGGTCCCGTCCT
234	933	952	TGGCCTGTCTGTGGAAGCGG
235	960	979	GGTGGGTGCCATGACTGTCA
236	963	982	CCAGGTGGGTGCCATGACTG
237	967	986	CCTGCCAGGTGGGTGCCATG
26	970	989	ACCCCTGCCAGGTGGGTGCC
238	972	991	CCACCCCTGCCAGGTGGGTG
27	975	994	TGACCACCCCTGCCAGGTGG
239	977	996	GCTGACCACCCCTGCCAGGT
240	985	1004	TCCCGGCCGCTGACCACCCC
241	989	1008	GGCATCCCGGCCGCTGACCA
242	992	1011	GCCGGCATCCCGGCCGCTGA
243	997	1016	GCCACGCCGGCATCCCGGCC
244	998	1017	GGCCACGCCGGCATCCCGGC
245	999	1018	TGGCCACGCCGGCATCCCGG
246	1000	1019	TTGGCCACGCCGGCATCCCG
247	1001	1020	CTTGGCCACGCCGGCATCCC
248	1002	1021	CCTTGGCCACGCCGGCATCC
249	1003	1022	CCCTTGGCCACGCCGGCATC
28	1004	1023	ACCCTTGGCCACGCCGGCAT
447	1004	1023	ACCCTTGGTCACGCCGGCAT
250	1005	1024	CACCCTTGGCCACGCCGGCA
251	1006	1025	GCACCCTTGGCCACGCCGGC
252	1007	1026	GGCACCCTTGGCCACGCCGG
253	1008	1027	TGGCACCCTTGGCCACGCCG
254	1009	1028	CTGGCACCCTTGGCCACGCC
255	1010	1029	GCTGGCACCCTTGGCCACGC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
256	1015	1034	CGCATGCTGGCACCCTTGGC
257	1036	1055	CAGTTGAGCACGCGCAGGCT
258	1038	1057	GGCAGTTGAGCACGCGCAGG
29	1040	1059	TTGGCAGTTGAGCACGCGCA
259	1042	1061	CCTTGGCAGTTGAGCACGCG
30	1045	1064	TTCCCTTGGCAGTTGAGCAC
260	1047	1066	CCTTCCCTTGGCAGTTGAGC
261	1051	1070	GTGCCCTTCCCTTGGCAGTT
262	1053	1072	CCGTGCCCTTCCCTTGGCAG
263	1064	1083	GGTGCCGCTAACCGTGCCCT
264	1076	1095	CAGGCCTATGAGGGTGCCGC
31	1077	1096	CCAGGCCTATGAGGGTGCCG
458	1079	1092	GCCTATGAGGGTGC
459	1084	1097	TCCAGGCCTATGAG
265	1088	1107	CCGAATAAACTCCAGGCCTA
266	1096	1115	TGGCTTTTCCGAATAAACTC
32	1098	1117	GCTGGCTTTTCCGAATAAAC
267	1100	1119	CAGCTGGCTTTTCCGAATAA
268	1102	1121	ACCAGCTGGCTTTTCCGAAT
269	1104	1123	GGACCAGCTGGCTTTTCCGA
270	1108	1127	GGCTGGACCAGCTGGCTTTT
271	1119	1138	GTGGCCCCACAGGCTGGACC
272	1132	1151	AGCAGCACCACCAGTGGCCC
273	1154	1173	GCTGTACCCACCCGCCAGGG
274	1200	1219	CGACCCCAGCCCTCGCCAGG
33	1210	1229	GTGACCAGCACGACCCCAGC
275	1212	1231	CGGTGACCAGCACGACCCCA
276	1214	1233	AGCGGTGACCAGCACGACCC
277	1216	1235	GCAGCGGTGACCAGCACGAC
278	1218	1237	CGGCAGCGGTGACCAGCACG
279	1219	1238	CCGGCAGCGGTGACCAGCAC
280	1222	1241	TTGCCGGCAGCGGTGACCAG
281	1224	1243	AGTTGCCGGCAGCGGTGACC
282	1226	1245	GAAGTTGCCGGCAGCGGTGA

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
283	1228	1247	CGGAAGTTGCCGGCAGCGGT
284	1230	1249	CCCGGAAGTTGCCGGCAGCG
285	1232	1251	GTCCCGGAAGTTGCCGGCAG
286	1273	1292	ATGACCTCGGGAGCTGAGGC
287	1283	1302	CCCAACTGTGATGACCTCGG
288	1295	1314	GGCATTGGTGGCCCCAACTG
149	1297	1316	TGGGCATTGGTGGCCCCAAC
289	1305	1324	GCTGGTCTTGGGCATTGGTG
290	1318	1337	CCCAGGGTCACCGGCTGGTC
291	1320	1339	TCCCCAGGGTCACCGGCTGG
292	1322	1341	AGTCCCCAGGGTCACCGGCT
293	1324	1343	AAAGTCCCCAGGGTCACCGG
34	1326	1345	CCAAAGTCCCCAGGGTCACC
294	1328	1347	CCCCAAAGTCCCCAGGGTCA
128	1330	1349	GTCCCCAAAGTCCCCAGGGT
295	1333	1352	TTGGTCCCCAAAGTCCCCAG
35	1335	1354	AGTTGGTCCCCAAAGTCCCC
296	1337	1356	AAAGTTGGTCCCCAAAGTCC
36	1340	1359	GCCAAAGTTGGTCCCCAAAG
297	1342	1361	CGGCCAAAGTTGGTCCCCAA
298	1344	1363	AGCGGCCAAAGTTGGTCCCC
299	1346	1365	ACAGCGGCCAAAGTTGGTCC
300	1348	1367	ACACAGCGGCCAAAGTTGGT
301	1350	1369	CCACACAGCGGCCAAAGTTG
37	1352	1371	GTCCACACAGCGGCCAAAGT
302	1354	1373	AGGTCCACACAGCGGCCAAA
303	1356	1375	AGAGGTCCACACAGCGGCCA
304	1358	1377	AAAGAGGTCCACACAGCGGC
38	1361	1380	GGCAAAGAGGTCCACACAGC
305	1380	1399	CAATGATGTCCTCCCCTGGG
306	1387	1406	GAGGCACCAATGATGTCCTC
39	1389	1408	TGGAGGCACCAATGATGTCC
307	1391	1410	GCTGGAGGCACCAATGATGT
308	1393	1412	TCGCTGGAGGCACCAATGAT
309	1395	1414	AGTCGCTGGAGGCACCAATG

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
310	1397	1416	GCAGTCGCTGGAGGCACCAA
40	1400	1419	GCTGCAGTCGCTGGAGGCAC
311	1402	1421	GTGCTGCAGTCGCTGGAGGC
312	1404	1423	AGGTGCTGCAGTCGCTGGAG
313	1406	1425	GCAGGTGCTGCAGTCGCTGG
314	1407	1426	AGCAGGTGCTGCAGTCGCTG
315	1409	1428	AAAGCAGGTGCTGCAGTCGC
41	1411	1430	ACAAAGCAGGTGCTGCAGTC
316	1413	1432	ACACAAAGCAGGTGCTGCAG
317	1415	1434	TGACACAAAGCAGGTGCTGC
318	1425	1444	TCCCACTCTGTGACACAAAG
101	1465	1484	ATGGCTGCAATGCCAGCCAC
319	1467	1486	TCATGGCTGCAATGCCAGCC
42	1470	1489	GCATCATGGCTGCAATGCCA
320	1472	1491	CAGCATCATGGCTGCAATGC
321	1474	1493	GACAGCATCATGGCTGCAAT
322	1476	1495	CAGACAGCATCATGGCTGCA
43	1478	1497	GGCAGACAGCATCATGGCTG
323	1480	1499	TCGGCAGACAGCATCATGGC
324	1482	1501	GCTCGGCAGACAGCATCATG
325	1484	1503	CGGCTCGGCAGACAGCATCA
326	1486	1505	TCCGGCTCGGCAGACAGCAT
327	1500	1519	CGGCCAGGGTGAGCTCCGGC
328	1513	1532	CTCTGCCTCAACTCGGCCAG
329	1515	1534	GTCTCTGCCTCAACTCGGCC
330	1517	1536	CAGTCTCTGCCTCAACTCGG
331	1519	1538	ATCAGTCTCTGCCTCAACTC
332	1521	1540	GGATCAGTCTCTGCCTCAAC
333	1523	1542	GTGGATCAGTCTCTGCCTCA
334	1525	1544	AAGTGGATCAGTCTCTGCCT
44	1526	1545	GAAGTGGATCAGTCTCTGCC
335	1528	1547	GAGAAGTGGATCAGTCTCTG
336	1530	1549	CAGAGAAGTGGATCAGTCTC
337	1532	1551	GGCAGAGAAGTGGATCAGTC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
45	1534	1553	TTGGCAGAGAAGTGGATCAG
338	1536	1555	CTTTGGCAGAGAAGTGGATC
46	1539	1558	CATCTTTGGCAGAGAAGTGG
339	1541	1560	GACATCTTTGGCAGAGAAGT
340	1543	1562	ATGACATCTTTGGCAGAGAA
47	1545	1564	TGATGACATCTTTGGCAGAG
341	1547	1566	ATTGATGACATCTTTGGCAG
342	1549	1568	TCATTGATGACATCTTTGGC
48	1552	1571	GCCTCATTGATGACATCTTT
343	1554	1573	AGGCCTCATTGATGACATCT
344	1556	1575	CCAGGCCTCATTGATGACAT
345	1558	1577	AACCAGGCCTCATTGATGAC
346	1560	1579	GGAACCAGGCCTCATTGATG
347	1561	1580	GGGAACCAGGCCTCATTGAT
348	1562	1581	AGGGAACCAGGCCTCATTGA
349	1563	1582	CAGGGAACCAGGCCTCATTG
49	1564	1583	TCAGGGAACCAGGCCTCATT
49	1564	1583	TCAGGGAACCAGGCCTCATT
350	1565	1584	CTCAGGGAACCAGGCCTCAT
351	1566	1585	CCTCAGGGAACCAGGCCTCA
352	1567	1586	TCCTCAGGGAACCAGGCCTC
353	1568	1587	GTCCTCAGGGAACCAGGCCT
50	1569	1588	GGTCCTCAGGGAACCAGGCC
354	1570	1589	TGGTCCTCAGGGAACCAGGC
355	1571	1590	CTGGTCCTCAGGGAACCAGG
356	1572	1591	GCTGGTCCTCAGGGAACCAG
357	1573	1592	CGCTGGTCCTCAGGGAACCA
87	1576	1595	ACCCGCTGGTCCTCAGGGAA
358	1578	1597	GTACCCGCTGGTCCTCAGGG
359	1580	1599	CAGTACCCGCTGGTCCTCAG
51	1583	1602	GGTCAGTACCCGCTGGTCCT
119	1605	1624	GCAGGGCGGCCACCAGGTTG
360	1628	1647	ACCTGCCCCATGGGTGCTGG
52	1640	1659	AAACAGCTGCCAACCTGCCC
361	1642	1661	CAAAACAGCTGCCAACCTGC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
53	1645	1664	CTGCAAAACAGCTGCCAACC
362	1647	1666	TCCTGCAAAACAGCTGCCAA
363	1649	1668	AGTCCTGCAAAACAGCTGCC
364	1660	1679	GCTGACCATACAGTCCTGCA
365	1672	1691	GGCCCCGAGTGTGCTGACCA
54	1675	1694	GTAGGCCCCGAGTGTGCTGA
366	1677	1696	GTGTAGGCCCCGAGTGTGCT
367	1679	1698	CCGTGTAGGCCCCGAGTGTG
368	1681	1700	ATCCGTGTAGGCCCCGAGTG
369	1683	1702	CCATCCGTGTAGGCCCCGAG
370	1685	1704	GGCCATCCGTGTAGGCCCCG
371	1687	1706	GTGGCCATCCGTGTAGGCCC
372	1735	1754	CTGGAGCAGCTCAGCAGCTC
460	1735	1748	CAGCTCAGCAGCTC
373	1737	1756	AACTGGAGCAGCTCAGCAGC
55	1740	1759	AGAAACTGGAGCAGCTCAGC
374	1742	1761	GGAGAAACTGGAGCAGCTCA
375	1744	1763	CTGGAGAAACTGGAGCAGCT
56	1746	1765	TCCTGGAGAAACTGGAGCAG
57	1812	1831	CGTTGTGGGCCCGGCAGACC
376	1849	1868	CACCTGGCAATGGCGTAGAC
377	1850	1869	GCACCTGGCAATGGCGTAGA
378	1851	1870	AGCACCTGGCAATGGCGTAG
379	1852	1871	CAGCACCTGGCAATGGCGTA
380	1853	1872	GCAGCACCTGGCAATGGCGT
381	1854	1873	GGCAGCACCTGGCAATGGCG
382	1855	1874	AGGCAGCACCTGGCAATGGC
383	1856	1875	CAGGCAGCACCTGGCAATGG
384	1857	1876	GCAGGCAGCACCTGGCAATG
58	1858	1877	AGCAGGCAGCACCTGGCAAT
385	1859	1878	TAGCAGGCAGCACCTGGCAA
386	1860	1879	GTAGCAGGCAGCACCTGGCA
387	1905	1924	TGGCCTCAGCTGGTGGAGCT
388	1915	1934	GTCCCCATGCTGGCCTCAGC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)

SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
389	1916	1935	GGTCCCCATGCTGGCCTCAG
390	1917	1936	GGGTCCCCATGCTGGCCTCA
391	1918	1937	CGGGTCCCCATGCTGGCCTC
392	1919	1938	ACGGGTCCCCATGCTGGCCT
59	1920	1939	CACGGGTCCCCATGCTGGCC
59	1920	1939	CACGGGTCCCCATGCTGGCC
393	1921	1940	ACACGGGTCCCCATGCTGGC
394	1922	1941	GACACGGGTCCCCATGCTGG
395	1923	1942	GGACACGGGTCCCCATGCTG
396	1924	1943	TGGACACGGGTCCCCATGCT
397	1925	1944	GTGGACACGGGTCCCCATGC
398	1926	1945	AGTGGACACGGGTCCCCATG
399	1927	1946	CAGTGGACACGGGTCCCCAT
400	1928	1947	GCAGTGGACACGGGTCCCCA
401	1929	1948	GGCAGTGGACACGGGTCCCC
402	1930	1949	TGGCAGTGGACACGGGTCCC
403	1931	1950	GTGGCAGTGGACACGGGTCC
404	1932	1951	GGTGGCAGTGGACACGGGTC
405	1933	1952	TGGTGGCAGTGGACACGGGT
406	1936	1955	TGTTGGTGGCAGTGGACACG
407	1962	1981	AGCTGCAGCCTGTGAGGACG
408	1990	2009	GTGCCAAGGTCCTCCACCTC
409	2010	2029	TCAGCACAGGCGGCTTGTGG
410	2040	2059	CCACGCACTGGTTGGGCTGA
60	2100	2119	CTTTGCATTCCAGACCTGGG
411	2101	2120	ACTTTGCATTCCAGACCTGG
412	2102	2121	GACTTTGCATTCCAGACCTG
413	2103	2122	TGACTTTGCATTCCAGACCT
414	2104	2123	TTGACTTTGCATTCCAGACC
61	2105	2124	CTTGACTTTGCATTCCAGAC
415	2107	2126	TCCTTGACTTTGCATTCCAG
416	2120	2139	CGGGATTCCATGCTCCTTGA
417	2150	2169	GCAGGCCACGGTCACCTGCT
418	2187	2206	GGAGGGCACTGCAGCCAGTC
419	2305	2324	CAGATGGCAACGGCTGTCAC

TABLE 2-continued

Nucleotide sequences targeted to NM_174936.2 (SEQ ID NO: 1)			
SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Sequence (5'-3')
420	2306	2325	GCAGATGGCAACGGCTGTCA
421	2307	2326	AGCAGATGGCAACGGCTGTC
422	2308	2327	CAGCAGATGGCAACGGCTGT
423	2309	2328	GCAGCAGATGGCAACGGCTG
62	2310	2329	GGCAGCAGATGGCAACGGCT
424	2311	2330	CGGCAGCAGATGGCAACGGC
425	2312	2331	CCGGCAGCAGATGGCAACGG
426	2313	2332	TCCGGCAGCAGATGGCAACG
427	2314	2333	CTCCGGCAGCAGATGGCAAC
428	2315	2334	GCTCCGGCAGCAGATGGCAA
429	2325	2344	CCAGGTGCCGGCTCCGGCAG
430	2335	2354	GAGGCCTGCGCCAGGTGCCG
154	2410	2429	TTTTAAAGCTCAGCCCCAGC
63	2415	2434	AACCATTTTAAAGCTCAGCC
64	2504	2523	TCAAGGGCCAGGCCAGCAGC
65	2509	2528	CCCACTCAAGGGCCAGGCCA
122	2582	2601	GGAGGGAGCTTCCTGGCACC
66	2597	2616	ATGCCCCACAGTGAGGGAGG
67	2606	2625	AATGGTGAAATGCCCCACAG
153	2649	2668	TTGGGAGCAGCTGGCAGCAC
68	2750	2769	CATGGGAAGAATCCTGCCTC
431	2828	2847	ATGAGGGCCATCAGCACCTT
432	2829	2848	GATGAGGGCCATCAGCACCT
433	2830	2849	AGATGAGGGCCATCAGCACC
434	2831	2850	GAGATGAGGGCCATCAGCAC
69	2832	2851	GGAGATGAGGGCCATCAGCA
435	2833	2852	TGGAGATGAGGGCCATCAGC
436	2834	2853	CTGGAGATGAGGGCCATCAG
437	2835	2854	GCTGGAGATGAGGGCCATCA
438	2836	2855	AGCTGGAGATGAGGGCCATC
461	2877	2890	TTAATCAGGGAGCC
70	2900	2919	TAGATGCCATCCAGAAAGCT
439	2902	2921	GCTAGATGCCATCCAGAAAG
440	2903	2922	GGCTAGATGCCATCCAGAAA
441	2904	2923	TGGCTAGATGCCATCCAGAA
442	2905	2924	CTGGCTAGATGCCATCCAGA
71	2906	2925	TCTGGCTAGATGCCATCCAG
443	2907	2926	CTCTGGCTAGATGCCATCCA
444	2908	2927	CCTCTGGCTAGATGCCATCC
445	2909	2928	GCCTCTGGCTAGATGCCATC
446	2910	2929	AGCCTCTGGCTAGATGCCAT
72	2983	3002	GGCATAGAGCAGAGTAAAGG
73	2988	3007	AGCCTGGCATAGAGCAGAGT
135	2994	3013	TAGCACAGCCTGGCATAGAG
112	3227	3246	GAAGAGGCTTGGCTTCAGAG
74	3233	3252	AAGTAAGAAGAGGCTTGGCT
75	3437	3456	GCTCAAGGAGGGACAGTTGT
76	3472	3491	AAAGATAAATGTCTGCTTGC
77	3477	3496	ACCCAAAAGATAAATGTCTG
78	3543	3562	TCTTCAAGTTACAAAAGCAA
99	3550	3569	ATAAATATCTTCAAGTTACA

[0290] In certain embodiments, gapmer antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gapmer antisense compounds are targeted to SEQ ID NO: 1. In certain such embodiments, the nucleotide sequences illustrated in Table 2 have a 5-10-5 gapmer motif. Table 3 illustrates gapmer antisense compounds targeted to SEQ ID NO: 1, having a 5-10-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 3

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
395149	5-10-5	4	135	154	0	GCGCGGAATCCTGGCTGGGA
395150	5-10-5	5	242	261	0	GAGGAGACCTAGAGGCCGTG
395151	5-10-5	6	300	319	0	AGGACCGCCTGGAGCTGACG
395152	5-10-5	7	410	429	0	ACGCAAGGCTAGCACCAGCT
399793	5-10-5	8	417	436	0	CCTCGGAACGCAAGGCTAGC
395153	5-10-5	9	480	499	0	CCTTGGCGCAGCGGTGGAAG
395154	5-10-5	10	561	580	0	GGCGGGCAGTGCGCTCTGAC
395155	5-10-5	11	600	619	0	TGGTGAGGTATCCCCGGCGG
399794	5-10-5	12	606	625	0	GGATCTTGGTGAGGTATCCC
399795	5-10-5	13	615	634	0	AGACATGCAGGATCTTGGTG
395156	5-10-5	14	620	639	0	ATGGAAGACATGCAGGATCT
395157	5-10-5	15	646	665	0	TTCACCAGGAAGCCAGGAAG
399796	5-10-5	16	651	670	0	TCATCTTCACCAGGAAGCCA
395158	5-10-5	17	705	724	0	CCTCGATGTAGTCGACATGG
395159	5-10-5	18	785	804	0	GTATTCATCCGCCCGGTACC
395160	5-10-5	19	835	854	0	CTGGTGTCTAGGAGATACAC
399797	5-10-5	20	840	859	0	GTATGCTGGTGTCTAGGAGA
395161	5-10-5	21	860	879	0	GATTTCCCGGTGGTCACTCT
399798	5-10-5	22	866	885	0	GCCCTCGATTTCCCGGTGGT
399799	5-10-5	23	880	899	0	GTGACCATGACCCTGCCCTC
395162	5-10-5	24	890	909	0	CTCGAAGTCGGTGACCATGA
395163	5-10-5	25	923	942	0	GTGGAAGCGGGTCCCGTCCT
395164	5-10-5	26	970	989	0	ACCCCTGCCAGGTGGGTGCC
399800	5-10-5	27	975	994	0	TGACCACCCCTGCCAGGTGG
395165	5-10-5	28	1004	1023	0	ACCCTTGGCCACGCCGGCAT
395166	5-10-5	29	1040	1059	0	TTGGCAGTTGAGCACGCGCA
399801	5-10-5	30	1045	1064	0	TTCCCTTGGCAGTTGAGCAC
395167	5-10-5	31	1077	1096	0	CCAGGCCTATGAGGGTGCCG
395168	5-10-5	32	1098	1117	0	GCTGGCTTTTCCGAATAAAC
395169	5-10-5	33	1210	1229	0	GTGACCAGCACGACCCCAGC
395170	5-10-5	149	1297	1316	0	TGGGCATTGGTGGCCCCAAC
395171	5-10-5	34	1326	1345	0	CCAAAGTCCCCAGGGTCACC
395172	5-10-5	128	1330	1349	0	GTCCCCAAAGTCCCCAGGGT
399802	5-10-5	35	1335	1354	0	AGTTGGTCCCCAAAGTCCCC
395173	5-10-5	36	1340	1359	0	GCCAAAGTTGGTCCCCAAAG

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
399803	5-10-5	37	1352	1371	0	GTCCACACAGCGGCCAAAGT
395174	5-10-5	38	1361	1380	0	GGCAAAGAGGTCCACACAGC
395175	5-10-5	39	1389	1408	0	TGGAGGCACCAATGATGTCC
399804	5-10-5	40	1400	1419	0	GCTGCAGTCGCTGGAGGCAC
399805	5-10-5	41	1411	1430	0	ACAAAGCAGGTGCTGCAGTC
395176	5-10-5	101	1465	1484	0	ATGGCTGCAATGCCAGCCAC
399806	5-10-5	42	1470	1489	0	GCATCATGGCTGCAATGCCA
399807	5-10-5	43	1478	1497	0	GGCAGACAGCATCATGGCTG
399808	5-10-5	44	1526	1545	0	GAAGTGGATCAGTCTCTGCC
395177	5-10-5	45	1534	1553	0	TTGGCAGAGAAGTGGATCAG
399809	5-10-5	46	1539	1558	0	CATCTTTGGCAGAGAAGTGG
399810	5-10-5	47	1545	1564	0	TGATGACATCTTTGGCAGAG
399811	5-10-5	48	1552	1571	0	GCCTCATTGATGACATCTTT
399812	5-10-5	49	1564	1583	0	TCAGGGAACCAGGCCTCATT
395178	5-10-5	50	1569	1588	0	GGTCCTCAGGGAACCAGGCC
395179	5-10-5	87	1576	1595	0	ACCCGCTGGTCCTCAGGGAA
399813	5-10-5	51	1583	1602	0	GGTCAGTACCCGCTGGTCCT
395180	5-10-5	119	1605	1624	0	GCAGGGCGGCCACCAGGTTG
395181	5-10-5	52	1640	1659	0	AAACAGCTGCCAACCTGCCC
399814	5-10-5	53	1645	1664	0	CTGCAAAACAGCTGCCAACC
395182	5-10-5	54	1675	1694	0	GTAGGCCCCGAGTGTGCTGA
399815	5-10-5	55	1740	1759	0	AGAAACTGGAGCAGCTCAGC
399816	5-10-5	56	1746	1765	0	TCCTGGAGAAACTGGAGCAG
395183	5-10-5	57	1812	1831	0	CGTTGTGGGCCCGGCAGACC
395184	5-10-5	58	1858	1877	0	AGCAGGCAGCACCTGGCAAT
395185	5-10-5	59	1920	1939	0	CACGGGTCCCCATGCTGGCC
395186	5-10-5	60	2100	2119	0	CTTTGCATTCCAGACCTGGG
399817	5-10-5	61	2105	2124	0	CTTGACTTTGCATTCCAGAC
395187	5-10-5	62	2310	2329	0	GGCAGCAGATGGCAACGGCT
395188	5-10-5	154	2410	2429	0	TTTTAAAGCTCAGCCCCAGC
399818	5-10-5	63	2415	2434	0	AACCATTTTAAAGCTCAGCC
395189	5-10-5	64	2504	2523	0	TCAAGGGCCAGGCCAGCAGC
399819	5-10-5	65	2509	2528	0	CCCACTCAAGGGCCAGGCCA
399820	5-10-5	122	2582	2601	0	GGAGGGAGCTTCCTGGCACC

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
395190	5-10-5	66	2597	2616	0	ATGCCCCACAGTGAGGGAGG
395191	5-10-5	67	2606	2625	0	AATGGTGAAATGCCCCACAG
395192	5-10-5	153	2649	2668	0	TTGGGAGCAGCTGGCAGCAC
395193	5-10-5	68	2750	2769	0	CATGGGAAGAATCCTGCCTC
395194	5-10-5	69	2832	2851	0	GGAGATGAGGGCCATCAGCA
395195	5-10-5	70	2900	2919	0	TAGATGCCATCCAGAAAGCT
399821	5-10-5	71	2906	2925	0	TCTGGCTAGATGCCATCCAG
395196	5-10-5	72	2983	3002	0	GGCATAGAGCAGAGTAAAGG
399822	5-10-5	73	2988	3007	0	AGCCTGGCATAGAGCAGAGT
399823	5-10-5	135	2994	3013	0	TAGCACAGCCTGGCATAGAG
395197	5-10-5	112	3227	3246	0	GAAGAGGCTTGGCTTCAGAG
399824	5-10-5	74	3233	3252	0	AAGTAAGAAGAGGCTTGGCT
395198	5-10-5	75	3437	3456	0	GCTCAAGGAGGGACAGTTGT
395199	5-10-5	76	3472	3491	0	AAAGATAAATGTCTGCTTGC
399825	5-10-5	77	3477	3496	0	ACCCAAAAGATAAATGTCTG
395200	5-10-5	78	3543	3562	0	TCTTCAAGTTACAAAAGCAA
399826	5-10-5	99	3550	3569	0	ATAAATATCTTCAAGTTACA
405861	5-10-5	162	406	425	0	AAGGCTAGCACCAGCTCCTC
405862	5-10-5	163	407	426	0	CAAGGCTAGCACCAGCTCCT
405863	5-10-5	164	408	427	0	GCAAGGCTAGCACCAGCTCC
405864	5-10-5	165	409	428	0	CGCAAGGCTAGCACCAGCTC
405865	5-10-5	166	411	430	0	AACGCAAGGCTAGCACCAGC
405866	5-10-5	167	412	431	0	GAACGCAAGGCTAGCACCAG
405867	5-10-5	168	413	432	0	GGAACGCAAGGCTAGCACCA
405868	5-10-5	169	414	433	0	CGGAACGCAAGGCTAGCACC
405869	5-10-5	218	862	881	0	TCGATTTCCCGGTGGTCACT
405870	5-10-5	219	863	882	0	CTCGATTTCCCGGTGGTCAC
405871	5-10-5	220	864	883	0	CCTCGATTTCCCGGTGGTCA
405872	5-10-5	221	865	884	0	CCCTCGATTTCCCGGTGGTC
405873	5-10-5	222	867	886	0	TGCCCTCGATTTCCCGGTGG
405874	5-10-5	223	868	887	0	CTGCCCTCGATTTCCCGGTG
405875	5-10-5	224	869	888	0	CCTGCCCTCGATTTCCCGGT
405876	5-10-5	225	870	889	0	CCCTGCCCTCGATTTCCCGG
405877	5-10-5	246	1000	1019	0	TTGGCCACGCCGGCATCCCG
405878	5-10-5	247	1001	1020	0	CTTGGCCACGCCGGCATCCC

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
405879	5-10-5	248	1002	1021	0	CCTTGGCCACGCCGGCATCC
405880	5-10-5	249	1003	1022	0	CCCTTGGCCACGCCGGCATC
405881	5-10-5	250	1005	1024	0	CACCCTTGGCCACGCCGGCA
405882	5-10-5	251	1006	1025	0	GCACCCTTGGCCACGCCGGC
405883	5-10-5	252	1007	1026	0	GGCACCCTTGGCCACGCCGG
405884	5-10-5	253	1008	1027	0	TGGCACCCTTGGCCACGCCG
405885	5-10-5	346	1560	1579	0	GGAACCAGGCCTCATTGATG
405886	5-10-5	347	1561	1580	0	GGGAACCAGGCCTCATTGAT
405887	5-10-5	348	1562	1581	0	AGGGAACCAGGCCTCATTGA
405888	5-10-5	349	1563	1582	0	CAGGGAACCAGGCCTCATTG
405889	5-10-5	350	1565	1584	0	CTCAGGGAACCAGGCCTCAT
405890	5-10-5	351	1566	1585	0	CCTCAGGGAACCAGGCCTCA
405891	5-10-5	352	1567	1586	0	TCCTCAGGGAACCAGGCCTC
405892	5-10-5	353	1568	1587	0	GTCCTCAGGGAACCAGGCCT
405893	5-10-5	431	2828	2847	0	ATGAGGGCCATCAGCACCTT
405894	5-10-5	432	2829	2848	0	GATGAGGGCCATCAGCACCT
405895	5-10-5	433	2830	2849	0	AGATGAGGGCCATCAGCACC
405896	5-10-5	434	2831	2850	0	GAGATGAGGGCCATCAGCAC
405897	5-10-5	435	2833	2852	0	TGGAGATGAGGGCCATCAGC
405898	5-10-5	436	2834	2853	0	CTGGAGATGAGGGCCATCAG
405899	5-10-5	437	2835	2854	0	GCTGGAGATGAGGGCCATCA
405900	5-10-5	438	2836	2855	0	AGCTGGAGATGAGGGCCATC
405901	5-10-5	439	2902	2921	0	GCTAGATGCCATCCAGAAAG
405902	5-10-5	440	2903	2922	0	GGCTAGATGCCATCCAGAAA
405903	5-10-5	441	2904	2923	0	TGGCTAGATGCCATCCAGAA
405904	5-10-5	442	2905	2924	0	CTGGCTAGATGCCATCCAGA
405905	5-10-5	443	2907	2926	0	CTCTGGCTAGATGCCATCCA
405906	5-10-5	444	2908	2927	0	CCTCTGGCTAGATGCCATCC
405907	5-10-5	445	2909	2928	0	GCCTCTGGCTAGATGCCATC
405908	5-10-5	446	2910	2929	0	AGCCTCTGGCTAGATGCCAT
405909	5-10-5	267	1100	1119	0	CAGCTGGCTTTTCCGAATAA
405910	5-10-5	268	1102	1121	0	ACCAGCTGGCTTTTCCGAAT
405911	5-10-5	269	1104	1123	0	GGACCAGCTGGCTTTTCCGA
405912	5-10-5	270	1108	1127	0	GGCTGGACCAGCTGGCTTTT

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
405913	5-10-5	275	1212	1231	0	CGGTGACCAGCACGACCCCA
405914	5-10-5	276	1214	1233	0	AGCGGTGACCAGCACGACCC
405915	5-10-5	277	1216	1235	0	GCAGCGGTGACCAGCACGAC
405916	5-10-5	278	1218	1237	0	CGGCAGCGGTGACCAGCACG
405917	5-10-5	280	1222	1241	0	TTGCCGGCAGCGGTGACCAG
405918	5-10-5	281	1224	1243	0	AGTTGCCGGCAGCGGTGACC
405919	5-10-5	282	1226	1245	0	GAAGTTGCCGGCAGCGGTGA
405920	5-10-5	283	1228	1247	0	CGGAAGTTGCCGGCAGCGGT
405921	5-10-5	284	1230	1249	0	CCCGGAAGTTGCCGGCAGCG
405922	5-10-5	285	1232	1251	0	GTCCCGGAAGTTGCCGGCAG
405923	5-10-5	288	1295	1314	0	GGCATTGGTGGCCCCAACTG
405924	5-10-5	290	1318	1337	0	CCCAGGGTCACCGGCTGGTC
405925	5-10-5	292	1322	1341	0	AGTCCCCAGGGTCACCGGCT
405926	5-10-5	293	1324	1343	0	AAAGTCCCCAGGGTCACCGG
405927	5-10-5	294	1328	1347	0	CCCCAAAGTCCCCAGGGTCA
405928	5-10-5	295	1333	1352	0	TTGGTCCCCAAAGTCCCCAG
405929	5-10-5	296	1337	1356	0	AAAGTTGGTCCCCAAAGTCC
405930	5-10-5	297	1342	1361	0	CGGCCAAAGTTGGTCCCCAA
405931	5-10-5	298	1344	1363	0	AGCGGCCAAAGTTGGTCCCC
405932	5-10-5	299	1346	1365	0	ACAGCGGCCAAAGTTGGTCC
405933	5-10-5	300	1348	1367	0	ACACAGCGGCCAAAGTTGGT
405934	5-10-5	301	1350	1369	0	CCACACAGCGGCCAAAGTTG
405935	5-10-5	302	1354	1373	0	AGGTCCACACAGCGGCCAAA
405936	5-10-5	304	1358	1377	0	AAAGAGGTCCACACAGCGGC
405937	5-10-5	306	1387	1406	0	GAGGCACCAATGATGTCCTC
405938	5-10-5	307	1391	1410	0	GCTGGAGGCACCAATGATGT
405939	5-10-5	308	1393	1412	0	TCGCTGGAGGCACCAATGAT
405940	5-10-5	309	1395	1414	0	AGTCGCTGGAGGCACCAATG
405941	5-10-5	310	1397	1416	0	GCAGTCGCTGGAGGCACCAA
405942	5-10-5	311	1402	1421	0	GTGCTGCAGTCGCTGGAGGC
405943	5-10-5	312	1404	1423	0	AGGTGCTGCAGTCGCTGGAG
405944	5-10-5	314	1407	1426	0	AGCAGGTGCTGCAGTCGCTG
405945	5-10-5	315	1409	1428	0	AAAGCAGGTGCTGCAGTCGC
405946	5-10-5	316	1413	1432	0	ACACAAAGCAGGTGCTGCAG
405947	5-10-5	317	1415	1434	0	TGACACAAAGCAGGTGCTGC

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
405948	5-10-5	319	1467	1486	0	TCATGGCTGCAATGCCAGCC
405949	5-10-5	320	1472	1491	0	CAGCATCATGGCTGCAATGC
405950	5-10-5	321	1474	1493	0	GACAGCATCATGGCTGCAAT
405951	5-10-5	322	1476	1495	0	CAGACAGCATCATGGCTGCA
405952	5-10-5	323	1480	1499	0	TCGGCAGACAGCATCATGGC
405953	5-10-5	325	1484	1503	0	CGGCTCGGCAGACAGCATCA
405954	5-10-5	326	1486	1505	0	TCCGGCTCGGCAGACAGCAT
405955	5-10-5	328	1513	1532	0	CTCTGCCTCAACTCGGCCAG
405956	5-10-5	329	1515	1534	0	GTCTCTGCCTCAACTCGGCC
405957	5-10-5	330	1517	1536	0	CAGTCTCTGCCTCAACTCGG
405958	5-10-5	331	1519	1538	0	ATCAGTCTCTGCCTCAACTC
405959	5-10-5	332	1521	1540	0	GGATCAGTCTCTGCCTCAAC
405960	5-10-5	333	1523	1542	0	GTGGATCAGTCTCTGCCTCA
405961	5-10-5	334	1525	1544	0	AAGTGGATCAGTCTCTGCCT
405962	5-10-5	335	1528	1547	0	GAGAAGTGGATCAGTCTCTG
405963	5-10-5	337	1532	1551	0	GGCAGAGAAGTGGATCAGTC
405964	5-10-5	338	1536	1555	0	CTTTGGCAGAGAAGTGGATC
405965	5-10-5	339	1541	1560	0	GACATCTTTGGCAGAGAAGT
405966	5-10-5	340	1543	1562	0	ATGACATCTTTGGCAGAGAA
405967	5-10-5	341	1547	1566	0	ATTGATGACATCTTTGGCAG
405968	5-10-5	342	1549	1568	0	TCATTGATGACATCTTTGGC
405969	5-10-5	343	1554	1573	0	AGGCCTCATTGATGACATCT
405970	5-10-5	344	1556	1575	0	CCAGGCCTCATTGATGACAT
405971	5-10-5	355	1571	1590	0	CTGGTCCTCAGGGAACCAGG
405972	5-10-5	357	1573	1592	0	CGCTGGTCCTCAGGGAACCA
405973	5-10-5	358	1578	1597	0	GTACCCGCTGGTCCTCAGGG
405974	5-10-5	359	1580	1599	0	CAGTACCCGCTGGTCCTCAG
405975	5-10-5	361	1642	1661	0	CAAAACAGCTGCCAACCTGC
405976	5-10-5	362	1647	1666	0	TCCTGCAAAACAGCTGCCAA
405977	5-10-5	363	1649	1668	0	AGTCCTGCAAAACAGCTGCC
405978	5-10-5	365	1672	1691	0	GGCCCCGAGTGTGCTGACCA
405979	5-10-5	366	1677	1696	0	GTGTAGGCCCCGAGTGTGCT
405980	5-10-5	367	1679	1698	0	CCGTGTAGGCCCCGAGTGTG
405981	5-10-5	368	1681	1700	0	ATCCGTGTAGGCCCCGAGTG

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
405982	5-10-5	370	1685	1704	0	GGCCATCCGTGTAGGCCCCG
405983	5-10-5	371	1687	1706	0	GTGGCCATCCGTGTAGGCCC
405984	5-10-5	372	1735	1754	0	CTGGAGCAGCTCAGCAGCTC
405985	5-10-5	373	1737	1756	0	AACTGGAGCAGCTCAGCAGC
405986	5-10-5	374	1742	1761	0	GGAGAAACTGGAGCAGCTCA
405987	5-10-5	375	1744	1763	0	CTGGAGAAACTGGAGCAGCT
405988	5-10-5	381	1854	1873	0	GGCAGCACCTGGCAATGGCG
405989	5-10-5	383	1856	1875	0	CAGGCAGCACCTGGCAATGG
405990	5-10-5	386	1860	1879	0	GTAGCAGGCAGCACCTGGCA
405991	5-10-5	394	1922	1941	0	GACACGGGTCCCCATGCTGG
405992	5-10-5	396	1924	1943	0	TGGACACGGGTCCCCATGCT
405993	5-10-5	398	1926	1945	0	AGTGGACACGGGTCCCCATG
405994	5-10-5	400	1928	1947	0	GCAGTGGACACGGGTCCCCA
405995	5-10-5	402	1930	1949	0	TGGCAGTGGACACGGGTCCC
405996	5-10-5	412	2102	2121	0	GACTTTGCATTCCAGACCTG
405997	5-10-5	415	2107	2126	0	TCCTTGACTTTGCATTCCAG
405998	5-10-5	426	2313	2332	0	TCCGGCAGCAGATGGCAACG
405999	5-10-5	160	298	317	0	GACCGCCTGGAGCTGACGGT
406000	5-10-5	173	482	501	0	ATCCTTGGCGCAGCGGTGGA
406001	5-10-5	174	484	503	0	GGATCCTTGGCGCAGCGGTG
406002	5-10-5	175	488	507	0	CCACGGATCCTTGGCGCAGC
406003	5-10-5	178	555	574	0	CAGTGCGCTCTGACTGCGAG
406004	5-10-5	179	557	576	0	GGCAGTGCGCTCTGACTGCG
406005	5-10-5	180	559	578	0	CGGGCAGTGCGCTCTGACTG
406006	5-10-5	181	562	581	0	CGGCGGGCAGTGCGCTCTGA
406007	5-10-5	183	595	614	0	AGGTATCCCCGGCGGGCAGC
406008	5-10-5	188	602	621	0	CTTGGTGAGGTATCCCCGGC
406009	5-10-5	190	604	623	0	ATCTTGGTGAGGTATCCCCG
406010	5-10-5	193	609	628	0	GCAGGATCTTGGTGAGGTAT
406011	5-10-5	194	611	630	0	ATGCAGGATCTTGGTGAGGT
406012	5-10-5	195	613	632	0	ACATGCAGGATCTTGGTGAG
406013	5-10-5	196	617	636	0	GAAGACATGCAGGATCTTGG
406014	5-10-5	199	648	667	0	TCTTCACCAGGAAGCCAGGA
406015	5-10-5	200	653	672	0	ACTCATCTTCACCAGGAAGC
406016	5-10-5	201	655	674	0	CCACTCATCTTCACCAGGAA

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
406017	5-10-5	203	659	678	0	GTCGCCACTCATCTTCACCA
406018	5-10-5	204	661	680	0	AGGTCGCCACTCATCTTCAC
406019	5-10-5	205	663	682	0	GCAGGTCGCCACTCATCTTC
406020	5-10-5	206	665	684	0	CAGCAGGTCGCCACTCATCT
406021	5-10-5	210	782	801	0	TTCATCCGCCCGGTACCGTG
406022	5-10-5	211	784	803	0	TATTCATCCGCCCGGTACCG
406023	5-10-5	212	787	806	0	TGGTATTCATCCGCCCGGTA
406024	5-10-5	213	789	808	0	GCTGGTATTCATCCGCCCGG
406025	5-10-5	216	832	851	0	GTGTCTAGGAGATACACCTC
406026	5-10-5	217	837	856	0	TGCTGGTGTCTAGGAGATAC
406027	5-10-5	226	874	893	0	ATGACCCTGCCCTCGATTTC
406028	5-10-5	227	876	895	0	CCATGACCCTGCCCTCGATT
406029	5-10-5	228	878	897	0	GACCATGACCCTGCCCTCGA
406030	5-10-5	229	882	901	0	CGGTGACCATGACCCTGCCC
406031	5-10-5	230	884	903	0	GTCGGTGACCATGACCCTGC
406032	5-10-5	232	888	907	0	CGAAGTCGGTGACCATGACC
406033	5-10-5	237	967	986	0	CCTGCCAGGTGGGTGCCATG
406034	5-10-5	238	972	991	0	CCACCCCTGCCAGGTGGGTG
406035	5-10-5	239	977	996	0	GCTGACCACCCCTGCCAGGT
406036	5-10-5	241	989	1008	0	GGCATCCCGGCCGCTGACCA
406037	5-10-5	242	992	1011	0	GCCGGCATCCCGGCCGCTGA
406038	5-10-5	245	999	1018	0	TGGCCACGCCGGCATCCCGG
406039	5-10-5	257	1036	1055	0	CAGTTGAGCACGCGCAGGCT
406040	5-10-5	258	1038	1057	0	GGCAGTTGAGCACGCGCAGG
406041	5-10-5	259	1042	1061	0	CCTTGGCAGTTGAGCACGCG
406042	5-10-5	260	1047	1066	0	CCTTCCCTTGGCAGTTGAGC
406043	5-10-5	261	1051	1070	0	GTGCCCTTCCCTTGGCAGTT
406044	5-10-5	262	1053	1072	0	CCGTGCCCTTCCCTTGGCAG
406045	5-10-5	266	1096	1115	0	TGGCTTTTCCGAATAAACTC
408642	5-10-5	264	1076	1095	0	CAGGCCTATGAGGGTGCCGC
408653	5-10-5	354	1570	1589	0	TGGTCCTCAGGGAACCAGGC
409126	5-10-5	447	1004	1023	1	ACCCTTGGTCACGCCGGCAT
410529	5-10-5	184	597	616	0	TGAGGTATCCCCGGCGGGCA
410530	5-10-5	185	598	617	0	GTGAGGTATCCCCGGCGGGC

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410531	5-10-5	186	599	618	0	GGTGAGGTATCCCCGGCGGG
410532	5-10-5	187	601	620	0	TTGGTGAGGTATCCCCGGCG
410533	5-10-5	189	603	622	0	TCTTGGTGAGGTATCCCCGG
410534	5-10-5	191	605	624	0	GATCTTGGTGAGGTATCCCC
410535	5-10-5	192	607	626	0	AGGATCTTGGTGAGGTATCC
410536	5-10-5	243	997	1016	0	GCCACGCCGGCATCCCGGCC
410537	5-10-5	244	998	1017	0	GGCCACGCCGGCATCCCGGC
410538	5-10-5	254	1009	1028	0	CTGGCACCCTTGGCCACGCC
410539	5-10-5	255	1010	1029	0	GCTGGCACCCTTGGCCACGC
410540	5-10-5	356	1572	1591	0	GCTGGTCCTCAGGGAACCAG
410541	5-10-5	376	1849	1868	0	CACCTGGCAATGGCGTAGAC
410542	5-10-5	377	1850	1869	0	GCACCTGGCAATGGCGTAGA
410543	5-10-5	378	1851	1870	0	AGCACCTGGCAATGGCGTAG
410544	5-10-5	379	1852	1871	0	CAGCACCTGGCAATGGCGTA
410545	5-10-5	380	1853	1872	0	GCAGCACCTGGCAATGGCGT
410546	5-10-5	382	1855	1874	0	AGGCAGCACCTGGCAATGGC
410547	5-10-5	384	1857	1876	0	GCAGGCAGCACCTGGCAATG
410548	5-10-5	385	1859	1878	0	TAGCAGGCAGCACCTGGCAA
410549	5-10-5	388	1915	1934	0	GTCCCCATGCTGGCCTCAGC
410550	5-10-5	389	1916	1935	0	GGTCCCCATGCTGGCCTCAG
410551	5-10-5	390	1917	1936	0	GGGTCCCCATGCTGGCCTCA
410552	5-10-5	391	1918	1937	0	CGGGTCCCCATGCTGGCCTC
410553	5-10-5	392	1919	1938	0	ACGGGTCCCCATGCTGGCCT
410554	5-10-5	393	1921	1940	0	ACACGGGTCCCCATGCTGGC
410555	5-10-5	395	1923	1942	0	GGACACGGGTCCCCATGCTG
410556	5-10-5	397	1925	1944	0	GTGGACACGGGTCCCCATGC
410557	5-10-5	399	1927	1946	0	CAGTGGACACGGGTCCCCAT
410558	5-10-5	401	1929	1948	0	GGCAGTGGACACGGGTCCCC
410559	5-10-5	403	1931	1950	0	GTGGCAGTGGACACGGGTCC
410560	5-10-5	404	1932	1951	0	GGTGGCAGTGGACACGGGTC
410561	5-10-5	405	1933	1952	0	TGGTGGCAGTGGACACGGGT
410562	5-10-5	411	2101	2120	0	ACTTTGCATTCCAGACCTGG
410563	5-10-5	413	2103	2122	0	TGACTTTGCATTCCAGACCT
410564	5-10-5	414	2104	2123	0	TTGACTTTGCATTCCAGACC
410565	5-10-5	419	2305	2324	0	CAGATGGCAACGGCTGTCAC

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410566	5-10-5	420	2306	2325	0	GCAGATGGCAACGGCTGTCA
410567	5-10-5	421	2307	2326	0	AGCAGATGGCAACGGCTGTC
410568	5-10-5	422	2308	2327	0	CAGCAGATGGCAACGGCTGT
410569	5-10-5	423	2309	2328	0	GCAGCAGATGGCAACGGCTG
410570	5-10-5	424	2311	2330	0	CGGCAGCAGATGGCAACGGC
410571	5-10-5	425	2312	2331	0	CCGGCAGCAGATGGCAACGG
410572	5-10-5	427	2314	2333	0	CTCCGGCAGCAGATGGCAAC
410573	5-10-5	428	2315	2334	0	GCTCCGGCAGCAGATGGCAA
410730	5-10-5	202	657	676	0	CGCCACTCATCTTCACCAGG
410731	5-10-5	207	667	686	0	TCCAGCAGGTCGCCACTCAT
410732	5-10-5	214	791	810	0	GGGCTGGTATTCATCCGCCC
410733	5-10-5	231	886	905	0	AAGTCGGTGACCATGACCCT
410734	5-10-5	279	1219	1238	0	CCGGCAGCGGTGACCAGCAC
410735	5-10-5	291	1320	1339	0	TCCCCAGGGTCACCGGCTGG
410736	5-10-5	303	1356	1375	0	AGAGGTCCACACAGCGGCCA
410737	5-10-5	313	1406	1425	0	GCAGGTGCTGCAGTCGCTGG
410738	5-10-5	324	1482	1501	0	GCTCGGCAGACAGCATCATG
410739	5-10-5	336	1530	1549	0	CAGAGAAGTGGATCAGTCTC
410740	5-10-5	345	1558	1577	0	AACCAGGCCTCATTGATGAC
410741	5-10-5	369	1683	1702	0	CCATCCGTGTAGGCCCCGAG
410742	5-10-5	159	294	313	0	GCCTGGAGCTGACGGTGCCC
410743	5-10-5	170	421	440	0	TCCTCCTCGGAACGCAAGGC
410744	5-10-5	171	446	465	0	GTGCTCGGGTGCTTCGGCCA
410745	5-10-5	172	466	485	0	TGGAAGGTGGCTGTGGTTCC
410746	5-10-5	176	507	526	0	CGTAGGTGCCAGGCAACCTC
410747	5-10-5	177	545	564	0	TGACTGCGAGAGGTGGGTCT
410748	5-10-5	182	591	610	0	ATCCCCGGCGGGCAGCCTGG
410749	5-10-5	197	628	647	0	AGAAGGCCATGGAAGACATG
410750	5-10-5	198	638	657	0	GAAGCCAGGAAGAAGGCCAT
410751	5-10-5	208	685	704	0	GGCAACTTCAAGGCCAGCTC
410752	5-10-5	209	724	743	0	GCAAAGACAGAGGAGTCCTC
410753	5-10-5	215	821	840	0	ATACACCTCCACCAGGCTGC
410754	5-10-5	233	898	917	0	GGCACATTCTCGAAGTCGGT
410755	5-10-5	234	933	952	0	TGGCCTGTCTGTGGAAGCGG

TABLE 3-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410756	5-10-5	235	960	979	0	GGTGGGTGCCATGACTGTCA
410757	5-10-5	240	985	1004	0	TCCCGGCCGCTGACCACCCC
410758	5-10-5	256	1015	1034	0	CGCATGCTGGCACCCTTGGC
410759	5-10-5	263	1064	1083	0	GGTGCCGCTAACCGTGCCCT
410760	5-10-5	265	1088	1107	0	CCGAATAAACTCCAGGCCTA
410761	5-10-5	271	1119	1138	0	GTGGCCCCACAGGCTGGACC
410762	5-10-5	272	1132	1151	0	AGCAGCACCACCAGTGGCCC
410763	5-10-5	273	1154	1173	0	GCTGTACCCACCCGCCAGGG
410764	5-10-5	274	1200	1219	0	CGACCCCAGCCCTCGCCAGG
410765	5-10-5	286	1273	1292	0	ATGACCTCGGGAGCTGAGGC
410766	5-10-5	287	1283	1302	0	CCCAACTGTGATGACCTCGG
410767	5-10-5	289	1305	1324	0	GCTGGTCTTGGGCATTGGTG
410768	5-10-5	305	1380	1399	0	CAATGATGTCCTCCCCTGGG
410769	5-10-5	318	1425	1444	0	TCCCACTCTGTGACACAAAG
410770	5-10-5	327	1500	1519	0	CGGCCAGGGTGAGCTCCGGC
410771	5-10-5	360	1628	1647	0	ACCTGCCCCATGGGTGCTGG
410772	5-10-5	364	1660	1679	0	GCTGACCATACAGTCCTGCA
410773	5-10-5	387	1905	1924	0	TGGCCTCAGCTGGTGGAGCT
410774	5-10-5	406	1936	1955	0	TGTTGGTGGCAGTGGACACG
410775	5-10-5	407	1962	1981	0	AGCTGCAGCCTGTGAGGACG
410776	5-10-5	408	1990	2009	0	GTGCCAAGGTCCTCCACCTC
410777	5-10-5	409	2010	2029	0	TCAGCACAGGCGGCTTGTGG
410778	5-10-5	410	2040	2059	0	CCACGCACTGGTTGGGCTGA
410779	5-10-5	416	2120	2139	0	CGGGATTCCATGCTCCTTGA
410780	5-10-5	417	2150	2169	0	GCAGGCCACGGTCACCTGCT
410781	5-10-5	418	2187	2206	0	GGAGGGCACTGCAGCCAGTC
410782	5-10-5	429	2325	2344	0	CCAGGTGCCGGCTCCGGCAG
410783	5-10-5	430	2335	2354	0	GAGGCCTGCGCCAGGTGCCG

[0291] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 1. In certain such embodiments, the nucleotide sequences illustrated in Table 2 have a 3-14-3 gap-widened motif. Table 4 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 1, having a 3-14-3 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 4

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 1						
Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
399871	3-14-3	4	135	154	0	GCGCGGAATCCTGGCTGGGA
399872	3-14-3	5	242	261	0	GAGGAGACCTAGAGGCCGTG
399873	3-14-3	6	300	319	0	AGGACCGCCTGGAGCTGACG
399874	3-14-3	7	410	429	0	ACGCAAGGCTAGCACCAGCT
399949	3-14-3	8	417	436	0	CCTCGGAACGCAAGGCTAGC
399875	3-14-3	9	480	499	0	CCTTGGCGCAGCGGTGGAAG
399876	3-14-3	10	561	580	0	GGCGGGCAGTGCGCTCTGAC
399877	3-14-3	11	600	619	0	TGGTGAGGTATCCCCGGCGG
399950	3-14-3	12	606	625	0	GGATCTTGGTGAGGTATCCC
399951	3-14-3	13	615	634	0	AGACATGCAGGATCTTGGTG
399878	3-14-3	14	620	639	0	ATGGAAGACATGCAGGATCT
399879	3-14-3	15	646	665	0	TTCACCAGGAAGCCAGGAAG
399952	3-14-3	16	651	670	0	TCATCTTCACCAGGAAGCCA
399880	3-14-3	17	705	724	0	CCTCGATGTAGTCGACATGG
399881	3-14-3	18	785	804	0	GTATTCATCCGCCCGGTACC
399882	3-14-3	19	835	854	0	CTGGTGTCTAGGAGATACAC
399953	3-14-3	20	840	859	0	GTATGCTGGTGTCTAGGAGA
399883	3-14-3	21	860	879	0	GATTTCCCGGTGGTCACTCT
399954	3-14-3	22	866	885	0	GCCCTCGATTTCCCGGTGGT
399955	3-14-3	23	880	899	0	GTGACCATGACCCTGCCCTC
399884	3-14-3	24	890	909	0	CTCGAAGTCGGTGACCATGA
399885	3-14-3	25	923	942	0	GTGGAAGCGGGTCCCGTCCT
399886	3-14-3	26	970	989	0	ACCCCTGCCAGGTGGGTGCC
399956	3-14-3	27	975	994	0	TGACCACCCCTGCCAGGTGG
399887	3-14-3	28	1004	1023	0	ACCCTTGGCCACGCCGGCAT
399888	3-14-3	29	1040	1059	0	TTGGCAGTTGAGCACGCGCA
399957	3-14-3	30	1045	1064	0	TTCCCTTGGCAGTTGAGCAC
399889	3-14-3	31	1077	1096	0	CCAGGCCTATGAGGGTGCCG
399890	3-14-3	32	1098	1117	0	GCTGGCTTTTCCGAATAAAC
399891	3-14-3	33	1210	1229	0	GTGACCAGCACGACCCCAGC
399892	3-14-3	149	1297	1316	0	TGGGCATTGGTGGCCCCAAC
399893	3-14-3	34	1326	1345	0	CCAAAGTCCCCAGGGTCACC
399894	3-14-3	128	1330	1349	0	GTCCCCAAAGTCCCCAGGGT
399958	3-14-3	35	1335	1354	0	AGTTGGTCCCCAAAGTCCCC
399895	3-14-3	36	1340	1359	0	GCCAAAGTTGGTCCCCAAAG

TABLE 4-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
399959	3-14-3	37	1352	1371	0	GTCCACACAGCGGCCAAAGT
399896	3-14-3	38	1361	1380	0	GGCAAAGAGGTCCACACAGC
399897	3-14-3	39	1389	1408	0	TGGAGGCACCAATGATGTCC
399960	3-14-3	40	1400	1419	0	GCTGCAGTCGCTGGAGGCAC
399961	3-14-3	41	1411	1430	0	ACAAAGCAGGTGCTGCAGTC
399898	3-14-3	101	1465	1484	0	ATGGCTGCAATGCCAGCCAC
399962	3-14-3	42	1470	1489	0	GCATCATGGCTGCAATGCCA
399963	3-14-3	43	1478	1497	0	GGCAGACAGCATCATGGCTG
399964	3-14-3	44	1526	1545	0	GAAGTGGATCAGTCTCTGCC
399899	3-14-3	45	1534	1553	0	TTGGCAGAGAAGTGGATCAG
399965	3-14-3	46	1539	1558	0	CATCTTTGGCAGAGAAGTGG
399966	3-14-3	47	1545	1564	0	TGATGACATCTTTGGCAGAG
399967	3-14-3	48	1552	1571	0	GCCTCATTGATGACATCTTT
399968	3-14-3	49	1564	1583	0	TCAGGGAACCAGGCCTCATT
399900	3-14-3	50	1569	1588	0	GGTCCTCAGGGAACCAGGCC
399901	3-14-3	87	1576	1595	0	ACCCGCTGGTCCTCAGGGAA
399969	3-14-3	51	1583	1602	0	GGTCAGTACCCGCTGGTCCT
399902	3-14-3	119	1605	1624	0	GCAGGGCGGCCACCAGGTTG
399903	3-14-3	52	1640	1659	0	AAACAGCTGCCAACCTGCCC
399970	3-14-3	53	1645	1664	0	CTGCAAAACAGCTGCCAACC
399904	3-14-3	54	1675	1694	0	GTAGGCCCCGAGTGTGCTGA
399971	3-14-3	55	1740	1759	0	AGAAACTGGAGCAGCTCAGC
399972	3-14-3	56	1746	1765	0	TCCTGGAGAAACTGGAGCAG
399905	3-14-3	57	1812	1831	0	CGTTGTGGGCCCGGCAGACC
399906	3-14-3	58	1858	1877	0	AGCAGGCAGCACCTGGCAAT
399907	3-14-3	59	1920	1939	0	CACGGGTCCCCATGCTGGCC
399908	3-14-3	60	2100	2119	0	CTTTGCATTCCAGACCTGGG
399973	3-14-3	61	2105	2124	0	CTTGACTTTGCATTCCAGAC
399909	3-14-3	62	2310	2329	0	GGCAGCAGATGGCAACGGCT
399910	3-14-3	154	2410	2429	0	TTTTAAAGCTCAGCCCCAGC
399974	3-14-3	63	2415	2434	0	AACCATTTTAAAGCTCAGCC
399911	3-14-3	64	2504	2523	0	TCAAGGGCCAGGCCAGCAGC
399975	3-14-3	65	2509	2528	0	CCCACTCAAGGGCCAGGCCA
399976	3-14-3	122	2582	2601	0	GGAGGGAGCTTCCTGGCACC
399912	3-14-3	66	2597	2616	0	ATGCCCCACAGTGAGGGAGG

TABLE 4-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
399913	3-14-3	67	2606	2625	0	AATGGTGAAATGCCCCACAG
399914	3-14-3	153	2649	2668	0	TTGGGAGCAGCTGGCAGCAC
399915	3-14-3	68	2750	2769	0	CATGGGAAGAATCCTGCCTC
399916	3-14-3	69	2832	2851	0	GGAGATGAGGGCCATCAGCA
399917	3-14-3	70	2900	2919	0	TAGATGCCATCCAGAAAGCT
399977	3-14-3	71	2906	2925	0	TCTGGCTAGATGCCATCCAG
399918	3-14-3	72	2983	3002	0	GGCATAGAGCAGAGTAAAGG
399978	3-14-3	73	2988	3007	0	AGCCTGGCATAGAGCAGAGT
399979	3-14-3	135	2994	3013	0	TAGCACAGCCTGGCATAGAG
399919	3-14-3	112	3227	3246	0	GAAGAGGCTTGGCTTCAGAG
399980	3-14-3	74	3233	3252	0	AAGTAAGAAGAGGCTTGGCT
399920	3-14-3	75	3437	3456	0	GCTCAAGGAGGGACAGTTGT
399921	3-14-3	76	3472	3491	0	AAAGATAAATGTCTGCTTGC
399981	3-14-3	77	3477	3496	0	ACCCAAAAGATAAATGTCTG
399922	3-14-3	78	3543	3562	0	TCTTCAAGTTACAAAAGCAA
399982	3-14-3	99	3550	3569	0	ATAAATATCTTCAAGTTACA
410574	3-14-3	184	597	616	0	TGAGGTATCCCCGGCGGGCA
410575	3-14-3	185	598	617	0	GTGAGGTATCCCCGGCGGGC
410576	3-14-3	186	599	618	0	GGTGAGGTATCCCCGGCGGG
410577	3-14-3	187	601	620	0	TTGGTGAGGTATCCCCGGCG
410578	3-14-3	188	602	621	0	CTTGGTGAGGTATCCCCGGC
410579	3-14-3	189	603	622	0	TCTTGGTGAGGTATCCCCGG
410580	3-14-3	190	604	623	0	ATCTTGGTGAGGTATCCCCG
410581	3-14-3	191	605	624	0	GATCTTGGTGAGGTATCCCC
410582	3-14-3	192	607	626	0	AGGATCTTGGTGAGGTATCC
405604	3-14-3	236	963	982	0	CCAGGTGGGTGCCATGACTG
410583	3-14-3	243	997	1016	0	GCCACGCCGGCATCCCGGCC
410584	3-14-3	244	998	1017	0	GGCCACGCCGGCATCCCGGC
410585	3-14-3	245	999	1018	0	TGGCCACGCCGGCATCCCGG
410586	3-14-3	246	1000	1019	0	TTGGCCACGCCGGCATCCCG
410587	3-14-3	247	1001	1020	0	CTTGGCCACGCCGGCATCCC
410588	3-14-3	248	1002	1021	0	CCTTGGCCACGCCGGCATCC
410589	3-14-3	249	1003	1022	0	CCCTTGGCCACGCCGGCATC
410590	3-14-3	250	1005	1024	0	CACCCTTGGCCACGCCGGCA
410591	3-14-3	251	1006	1025	0	GCACCCTTGGCCACGCCGGC

TABLE 4-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410592	3-14-3	252	1007	1026	0	GGCACCCTTGGCCACGCCGG
410593	3-14-3	253	1008	1027	0	TGGCACCCTTGGCCACGCCG
410594	3-14-3	254	1009	1028	0	CTGGCACCCTTGGCCACGCC
410595	3-14-3	255	1010	1029	0	GCTGGCACCCTTGGCCACGC
410596	3-14-3	348	1562	1581	0	AGGGAACCAGGCCTCATTGA
410597	3-14-3	349	1563	1582	0	CAGGGAACCAGGCCTCATTG
410598	3-14-3	350	1565	1584	0	CTCAGGGAACCAGGCCTCAT
410599	3-14-3	351	1566	1585	0	CCTCAGGGAACCAGGCCTCA
410600	3-14-3	352	1567	1586	0	TCCTCAGGGAACCAGGCCTC
410601	3-14-3	353	1568	1587	0	GTCCTCAGGGAACCAGGCCT
410602	3-14-3	354	1570	1589	0	TGGTCCTCAGGGAACCAGGC
410603	3-14-3	355	1571	1590	0	CTGGTCCTCAGGGAACCAGG
410604	3-14-3	356	1572	1591	0	GCTGGTCCTCAGGGAACCAG
405641	3-14-3	373	1737	1756	0	AACTGGAGCAGCTCAGCAGC
410605	3-14-3	376	1849	1868	0	CACCTGGCAATGGCGTAGAC
410606	3-14-3	377	1850	1869	0	GCACCTGGCAATGGCGTAGA
410607	3-14-3	378	1851	1870	0	AGCACCTGGCAATGGCGTAG
410608	3-14-3	379	1852	1871	0	CAGCACCTGGCAATGGCGTA
410609	3-14-3	380	1853	1872	0	GCAGCACCTGGCAATGGCGT
410610	3-14-3	381	1854	1873	0	GGCAGCACCTGGCAATGGCG
410611	3-14-3	382	1855	1874	0	AGGCAGCACCTGGCAATGGC
410612	3-14-3	383	1856	1875	0	CAGGCAGCACCTGGCAATGG
410613	3-14-3	384	1857	1876	0	GCAGGCAGCACCTGGCAATG
410614	3-14-3	385	1859	1878	0	TAGCAGGCAGCACCTGGCAA
410615	3-14-3	388	1915	1934	0	GTCCCCATGCTGGCCTCAGC
410616	3-14-3	389	1916	1935	0	GGTCCCCATGCTGGCCTCAG
410617	3-14-3	390	1917	1936	0	GGGTCCCCATGCTGGCCTCA
410618	3-14-3	391	1918	1937	0	CGGGTCCCCATGCTGGCCTC
410619	3-14-3	392	1919	1938	0	ACGGGTCCCCATGCTGGCCT
410620	3-14-3	393	1921	1940	0	ACACGGGTCCCCATGCTGGC
410621	3-14-3	394	1922	1941	0	GACACGGGTCCCCATGCTGG
410622	3-14-3	395	1923	1942	0	GGACACGGGTCCCCATGCTG
410623	3-14-3	396	1924	1943	0	TGGACACGGGTCCCCATGCT
410624	3-14-3	397	1925	1944	0	GTGGACACGGGTCCCCATGC
410625	3-14-3	398	1926	1945	0	AGTGGACACGGGTCCCCATG

TABLE 4-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410626	3-14-3	399	1927	1946	0	CAGTGGACACGGGTCCCCAT
410627	3-14-3	400	1928	1947	0	GCAGTGGACACGGGTCCCCA
410628	3-14-3	401	1929	1948	0	GGCAGTGGACACGGGTCCCC
410629	3-14-3	402	1930	1949	0	TGGCAGTGGACACGGGTCCC
410630	3-14-3	403	1931	1950	0	GTGGCAGTGGACACGGGTCC
410631	3-14-3	404	1932	1951	0	GGTGGCAGTGGACACGGGTC
410632	3-14-3	405	1933	1952	0	TGGTGGCAGTGGACACGGGT
410633	3-14-3	411	2101	2120	0	ACTTTGCATTCCAGACCTGG
410634	3-14-3	412	2102	2121	0	GACTTTGCATTCCAGACCTG
410635	3-14-3	413	2103	2122	0	TGACTTTGCATTCCAGACCT
410636	3-14-3	414	2104	2123	0	TTGACTTTGCATTCCAGACC
410637	3-14-3	419	2305	2324	0	CAGATGGCAACGGCTGTCAC
410638	3-14-3	420	2306	2325	0	GCAGATGGCAACGGCTGTCA
410639	3-14-3	421	2307	2326	0	AGCAGATGGCAACGGCTGTC
410640	3-14-3	422	2308	2327	0	CAGCAGATGGCAACGGCTGT
410641	3-14-3	423	2309	2328	0	GCAGCAGATGGCAACGGCTG
410642	3-14-3	424	2311	2330	0	CGGCAGCAGATGGCAACGGC
410643	3-14-3	425	2312	2331	0	CCGGCAGCAGATGGCAACGG
410644	3-14-3	426	2313	2332	0	TCCGGCAGCAGATGGCAACG
410645	3-14-3	427	2314	2333	0	CTCCGGCAGCAGATGGCAAC
410646	3-14-3	428	2315	2334	0	GCTCCGGCAGCAGATGGCAA

[0292] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 1. In certain such embodiments, the nucleotide sequences illustrated in Table 2 have a 2-13-5 gap-widened motif. Table 5 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 1, having a 2-13-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 5

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410647	2-13-5	184	597	616	0	TGAGGTATCCCCGGCGGGCA
410648	2-13-5	185	598	617	0	GTGAGGTATCCCCGGCGGGC
410649	2-13-5	186	599	618	0	GGTGAGGTATCCCCGGCGGG
410650	2-13-5	11	600	619	0	TGGTGAGGTATCCCCGGCGG

TABLE 5-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410651	2-13-5	187	601	620	0	TTGGTGAGGTATCCCCGGCG
410652	2-13-5	188	602	621	0	CTTGGTGAGGTATCCCCGGC
410653	2-13-5	189	603	622	0	TCTTGGTGAGGTATCCCCGG
410654	2-13-5	190	604	623	0	ATCTTGGTGAGGTATCCCCG
410655	2-13-5	191	605	624	0	GATCTTGGTGAGGTATCCCC
410656	2-13-5	12	606	625	0	GGATCTTGGTGAGGTATCCC
410657	2-13-5	192	607	626	0	AGGATCTTGGTGAGGTATCC
410658	2-13-5	243	997	1016	0	GCCACGCCGGCATCCCGGCC
410659	2-13-5	244	998	1017	0	GGCCACGCCGGCATCCCGGC
410660	2-13-5	245	999	1018	0	TGGCCACGCCGGCATCCCGG
410661	2-13-5	246	1000	1019	0	TTGGCCACGCCGGCATCCCG
410662	2-13-5	247	1001	1020	0	CTTGGCCACGCCGGCATCCC
410663	2-13-5	248	1002	1021	0	CCTTGGCCACGCCGGCATCC
410664	2-13-5	249	1003	1022	0	CCCTTGGCCACGCCGGCATC
410665	2-13-5	28	1004	1023	0	ACCCTTGGCCACGCCGGCAT
410666	2-13-5	250	1005	1024	0	CACCCTTGGCCACGCCGGCA
410667	2-13-5	251	1006	1025	0	GCACCCTTGGCCACGCCGGC
410668	2-13-5	252	1007	1026	0	GGCACCCTTGGCCACGCCGG
410669	2-13-5	253	1008	1027	0	TGGCACCCTTGGCCACGCCG
410670	2-13-5	254	1009	1028	0	CTGGCACCCTTGGCCACGCC
410671	2-13-5	255	1010	1029	0	GCTGGCACCCTTGGCCACGC
410672	2-13-5	348	1562	1581	0	AGGGAACCAGGCCTCATTGA
410673	2-13-5	349	1563	1582	0	CAGGGAACCAGGCCTCATTG
410674	2-13-5	49	1564	1583	0	TCAGGGAACCAGGCCTCATT
410675	2-13-5	350	1565	1584	0	CTCAGGGAACCAGGCCTCAT
410676	2-13-5	351	1566	1585	0	CCTCAGGGAACCAGGCCTCA
410677	2-13-5	352	1567	1586	0	TCCTCAGGGAACCAGGCCTC
410678	2-13-5	353	1568	1587	0	GTCCTCAGGGAACCAGGCCT
410679	2-13-5	50	1569	1588	0	GGTCCTCAGGGAACCAGGCC
410680	2-13-5	354	1570	1589	0	TGGTCCTCAGGGAACCAGGC
410681	2-13-5	355	1571	1590	0	CTGGTCCTCAGGGAACCAGG
410682	2-13-5	356	1572	1591	0	GCTGGTCCTCAGGGAACCAG
410683	2-13-5	376	1849	1868	0	CACCTGGCAATGGCGTAGAC
410684	2-13-5	377	1850	1869	0	GCACCTGGCAATGGCGTAGA
410685	2-13-5	378	1851	1870	0	AGCACCTGGCAATGGCGTAG

TABLE 5-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410686	2-13-5	379	1852	1871	0	CAGCACCTGGCAATGGCGTA
410687	2-13-5	380	1853	1872	0	GCAGCACCTGGCAATGGCGT
410688	2-13-5	381	1854	1873	0	GGCAGCACCTGGCAATGGCG
410689	2-13-5	382	1855	1874	0	AGGCAGCACCTGGCAATGGC
410690	2-13-5	383	1856	1875	0	CAGGCAGCACCTGGCAATGG
410691	2-13-5	384	1857	1876	0	GCAGGCAGCACCTGGCAATG
410692	2-13-5	58	1858	1877	0	AGCAGGCAGCACCTGGCAAT
410693	2-13-5	385	1859	1878	0	TAGCAGGCAGCACCTGGCAA
410694	2-13-5	388	1915	1934	0	GTCCCCATGCTGGCCTCAGC
410695	2-13-5	389	1916	1935	0	GGTCCCCATGCTGGCCTCAG
410696	2-13-5	390	1917	1936	0	GGGTCCCCATGCTGGCCTCA
410697	2-13-5	391	1918	1937	0	CGGGTCCCCATGCTGGCCTC
410698	2-13-5	392	1919	1938	0	ACGGGTCCCCATGCTGGCCT
410699	2-13-5	59	1920	1939	0	CACGGGTCCCCATGCTGGCC
410700	2-13-5	393	1921	1940	0	ACACGGGTCCCCATGCTGGC
410701	2-13-5	394	1922	1941	0	GACACGGGTCCCCATGCTGG
410702	2-13-5	395	1923	1942	0	GGACACGGGTCCCCATGCTG
410703	2-13-5	396	1924	1943	0	TGGACACGGGTCCCCATGCT
410704	2-13-5	397	1925	1944	0	GTGGACACGGGTCCCCATGC
410705	2-13-5	398	1926	1945	0	AGTGGACACGGGTCCCCATG
410706	2-13-5	399	1927	1946	0	CAGTGGACACGGGTCCCCAT
410707	2-13-5	400	1928	1947	0	GCAGTGGACACGGGTCCCCA
410708	2-13-5	401	1929	1948	0	GGCAGTGGACACGGGTCCCC
410709	2-13-5	402	1930	1949	0	TGGCAGTGGACACGGGTCCC
410710	2-13-5	403	1931	1950	0	GTGGCAGTGGACACGGGTCC
410711	2-13-5	404	1932	1951	0	GGTGGCAGTGGACACGGGTC
410712	2-13-5	405	1933	1952	0	TGGTGGCAGTGGACACGGGT
410713	2-13-5	60	2100	2119	0	CTTTGCATTCCAGACCTGGG
410714	2-13-5	411	2101	2120	0	ACTTTGCATTCCAGACCTGG
410715	2-13-5	412	2102	2121	0	GACTTTGCATTCCAGACCTG
410716	2-13-5	413	2103	2122	0	TGACTTTGCATTCCAGACCT
410717	2-13-5	414	2104	2123	0	TTGACTTTGCATTCCAGACC
410718	2-13-5	61	2105	2124	0	CTTGACTTTGCATTCCAGAC
410719	2-13-5	419	2305	2324	0	CAGATGGCAACGGCTGTCAC
410720	2-13-5	420	2306	2325	0	GCAGATGGCAACGGCTGTCA

TABLE 5-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 1

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	Mismatches	Sequence (5'-3')
410721	2-13-5	421	2307	2326	0	AGCAGATGGCAACGGCTGTC
410722	2-13-5	422	2308	2327	0	CAGCAGATGGCAACGGCTGT
410723	2-13-5	423	2309	2328	0	GCAGCAGATGGCAACGGCTG
410724	2-13-5	62	2310	2329	0	GGCAGCAGATGGCAACGGCT
410725	2-13-5	424	2311	2330	0	CGGCAGCAGATGGCAACGGC
410726	2-13-5	425	2312	2331	0	CCGGCAGCAGATGGCAACGG
410727	2-13-5	426	2313	2332	0	TCCGGCAGCAGATGGCAACG
410728	2-13-5	427	2314	2333	0	CTCCGGCAGCAGATGGCAAC
410729	2-13-5	428	2315	2334	0	GCTCCGGCAGCAGATGGCAA

[0293] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 1. In certain such embodiments, the nucleotide sequences illustrated in Table 2 have a 3-13-4 gap-widened motif. Table 6 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 1, having a 3-13-4 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 6

Gapmer antisense compounds having a 3-13-4 motif targeted to SEQ ID NO: 1

Oligo	Motif	SEQ ID NO:	5' start site	3' Start Site	Mis-matches	OligoSeq
405526	3-13-4	236	963	982	0	CCAGGTGGGT GCCATGACTG
405557	3-13-4	50	1569	1588	0	GGTCCTCAGG GAACCAGGCC
405564	3-13-4	373	1737	1756	0	AACTGGAGCA GCTCAGCAGC

[0294] The following embodiments set forth target regions of PCSK9 nucleic acids. Also illustrated are examples of antisense compounds targeted to the target regions. It is understood that the sequence set forth in each SEQ ID NO is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by Isis Number (Isis No) indicate a combination of nucleobase sequence and motif.

[0295] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 1: 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1021, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569.

[0296] In certain embodiments, a target region is nucleotides 294-317 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 294-317 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 159 or 160.

In certain such embodiments, an antisense compound targeted to nucleotides 294-317 of SEQ ID NO: 1 is selected from ISIS NOs: 410742 or 405999.

[0297] In certain embodiments, a target region is nucleotides 406-440 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 406-440 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 7, 8, 162, 163, 164, 165, 166, 167, 168, 169 or 170. In certain such embodiments, an antisense compound targeted to nucleotides 406-440 of SEQ ID NO: 1 is selected from ISIS NOs: 405861, 405862, 405863, 405864, 395152, 399874, 405865, 405866, 405867, 405868, 399793, 399949, or 410743.

[0298] In certain embodiments, a target region is nucleotides 406-526 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 406-526 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 7, 8, 9, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, or 176. In certain such embodiments, an antisense compound targeted to nucleotides 406-526 of SEQ ID NO: 1 is selected from ISIS NOs: 405861, 405862, 405863, 405864, 395152, 399874, 405865, 405866, 405867, 405868, 399793, 399950, 410743, 410744, 410745, 395153, 399875, 406000, 406001, 406002 or 410746.

[0299] In certain embodiments, a target region is nucleotides 410-436 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 410-436 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 7, 8, 166, 167, 168, 169 or 169. In certain such embodiments, an antisense compound targeted to nucleotides 410-436 of SEQ ID NO: 1 is selected from ISIS NOs: 395152, 399874, 405865, 405866, 405867, 405868, or 399793.

[0300] In certain embodiments, a target region is nucleotides 410-499 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 410-499 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 7, 8, 9, 166, 167, 169, 170, 171 or 172. In certain such embodiments, an antisense compound targeted to nucleotides 410-499 of SEQ ID NO: 1 is selected from ISIS NOs: 395152, 399874, 405865, 405866, 405867, 405868, 399793, 399949, 410743, 410744, 410745, 395153, or 399875.

[0301] In certain embodiments, a target region is nucleotides 446-526 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 446-526 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 9, 171, 172, 173, 174, 175 or 176. In certain such embodiments, an antisense compound targeted to nucleotides 446-526 of SEQ ID NO: 1 is selected from ISIS NOs: 410744, 410745, 395153, 399875, 406000, 406001, 406002, or 410746.

[0302] In certain embodiments, a target region is nucleotides 545-581 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 545-581 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 10, 177, 178, 179, 180 or 181. In certain such embodiments, an antisense compound targeted to nucleotides 545-581 of SEQ ID NO: 1 is selected from ISIS NOs: 410747, 406003, 406004, 406005, 395154, 399876, or 406006.

[0303] In certain embodiments, a target region is nucleotides 591-619 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 591-619 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 182, 183, 184, 185 or 186. In certain such embodiments, an antisense compound targeted to nucleotides 591-619 of SEQ ID NO: 1 is selected from ISIS NOs: 410748, 406007, 410529, 410574, 410647, 410530, 410575, 410648, 410531, 410576, 410649, 395155, 399877, or 410650.

[0304] In certain embodiments, a target region is nucleotides 591-704 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 591-704 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 13, 14, 15, 16, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, or 208. In certain such embodiments, an antisense compound targeted to nucleotides 591-704 of SEQ ID NO: 1 is selected from ISIS NOs: 410748, 406007, 410529, 410574, 410647, 410530, 410575, 410648, 410531, 410576, 410649, 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, or 410751.

[0305] In certain embodiments, a target region is nucleotides 591-743 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 591-743 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 13, 14, 15, 16, 17, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, or 209. In certain such embodiments, an antisense compound targeted to nucleotides 591-743 of SEQ ID NO: 1 is selected from ISIS NOs: 410748, 406007, 410529, 410574, 410647, 410530, 410575, 410648, 410531, 410576, 410649, 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, or 410752.

[0306] In certain embodiments, a target region is nucleotides 595-622 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 595-622 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 183, 184, 185, 186, 187, 188, or 189. In certain such embodiments, an antisense compound targeted to nucleotides 595-622 of SEQ ID NO: 1 is selected from ISIS NOs: 406007, 410529, 410574, 410647, 410530, 410575, 410648, 410531, 410576, 410649, 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, or 410653.

[0307] In certain embodiments, a target region is nucleotides 600-626 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 600-626 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 87, 188, 189, 190, 191, or 192. In certain such embodiments, an antisense compound targeted to nucleotides 600-626 of SEQ ID NO: 1 is selected from ISIS NOs: 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, or 410657.

[0308] In certain embodiments, a target region is nucleotides 600-639 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 600-639 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 13, 14, 187, 188, 189, 190, 191, 192, 193, 194, 195, or 196. In certain such embodiments, an antisense compound targeted to nucleotides 600-639 of SEQ ID NO: 1 is selected from ISIS NOs: 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, 410657, 406010, 406011, 406012, 399795, 399951, 406013, 395156, or 399878.

[0309] In certain embodiments, a target region is nucleotides 600-670 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 600-670 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 13, 14, 15, 16, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, or 199. In certain such embodiments, an antisense compound targeted to nucleotides 600-670 of SEQ ID NO: 1 is selected from ISIS NOs: 395155, 399877, 410650, 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, 410657, 406010, 406011, 406012, 399795, 399951, 406013, 395156, 399878, or 399952.

[0310] In certain embodiments, a target region is nucleotides 601-628 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 601-628 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 187, 188, 189, 190, 191, 192 or 193. In certain such embodiments, an antisense compound targeted to nucleotides 601-628 of SEQ ID NO: 1 is selected from ISIS NOs: 410532, 410577, 410651, 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, 410657, or 406010.

[0311] In certain embodiments, a target region is nucleotides 602-628 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 602-628 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 188, 189, 190, 191, 192 or 193. In certain such embodiments, an antisense compound targeted to nucleotides 602-628 of SEQ ID NO: 1 is selected from ISIS NOs: 406008, 410578, 410652, 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, 410657, or 406010.

[0312] In certain embodiments, a target region is nucleotides 603-630 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 603-630 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 189, 190, 191, 192, 193 or 194. In certain such embodiments, an antisense compound targeted to nucleotides 603-630 of SEQ ID NO: 1 is selected from ISIS NOs: 410533, 410579, 410653, 406009, 410580, 410654, 410534, 410581, 410655, 399794, 399950, 410656, 410535, 410582, 410657, 406010, or 406011.

[0313] In certain embodiments, a target region is nucleotides 611-636 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 611-636 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 13, 194, 195 or 196. In certain such embodiments, an antisense compound targeted to nucleotides 611-636 of SEQ ID NO: 1 is selected from ISIS NOs: 406011, 406012, 399795, 399951, or 406013.

[0314] In certain embodiments, a target region is nucleotides 620-647 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 620-647 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 14 or 197. In certain such embodiments, an antisense compound targeted to nucleotides 620-647 of SEQ ID NO: 1 is selected from ISIS NOs: 395156, 399878, or 410749.

[0315] In certain embodiments, a target region is nucleotides 638-665 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 638-665 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 15 or 198. In certain such embodiments, an antisense compound targeted to nucleotides 638-665 of SEQ ID NO: 1 is selected from ISIS NOs: 410750, 395157, or 399879.

[0316] In certain embodiments, a target region is nucleotides 648-674 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 648-674 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 16, 199, 200, or 201. In certain such embodiments, an antisense compound targeted to nucleotides 648-674 of SEQ ID NO: 1 is selected from ISIS NOs: 406014, 399796, 399952, 406015, or 406016.

[0317] In certain embodiments, a target region is nucleotides 657-684 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 657-684 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 202, 203, 204, 205, or 206. In certain such embodiments, an antisense compound targeted to nucleotides 657-684 of SEQ ID NO: 1 is selected from ISIS NOs: 410730, 406017, 406018, 406019, or 406020.

[0318] In certain embodiments, a target region is nucleotides 705-743 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 705-743 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 17 or 209. In certain such embodiments, an antisense compound targeted to nucleotides 705-743 of SEQ ID NO: 1 is selected from ISIS NOs: 395158, 399880, or 410752.

[0319] In certain embodiments, a target region is nucleotides 782-810 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 782-810 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 18, 210, 211, 212, 213, or 214. In certain such embodiments, an antisense compound targeted to nucleotides 782-810 of SEQ ID NO: 1 is selected from ISIS NOs: 406021, 406022, 395159, 399881, 406023, 406024, or 410732.

[0320] In certain embodiments, a target region is nucleotides 821-859 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 821-859 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 215, 216 or 217. In certain such embodiments, an antisense compound targeted to nucleotides 821-859 of SEQ ID NO: 1 is selected from ISIS NOs: 410753, 406025, 395160, 399882, 406026, 399797, or 399953.

[0321] In certain embodiments, a target region is nucleotides 835-859 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 835-859 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20 or 217. In certain such embodiments, an antisense compound targeted to nucleotides 835-859 of SEQ ID NO: 1 is selected from ISIS NOs: 395160, 399882, 406026, 399797, or 399953.

[0322] In certain embodiments, a target region is nucleotides 835-917 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 835-917 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 21, 22, 23, 24, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232 or 233. In certain such embodiments, an antisense compound targeted to nucleotides 835-917 of SEQ ID NO: 1 is selected from ISIS NOs: 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, 399884, or 410754.

[0323] In certain embodiments, a target region is nucleotides 835-942 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 835-942 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 21, 22, 23, 24, 25, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, or 233. In certain such embodiments, an antisense compound targeted to nucleotides 835-942 of SEQ ID NO: 1 is selected from ISIS NOs: 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, 399884, 410754, 395163, or 399885.

[0324] In certain embodiments, a target region is nucleotides 860-887 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 860-887 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 218, 219, 220, 221, 222 or 223. In certain such embodiments, an antisense compound targeted to nucleotides 860-887 of SEQ ID NO: 1 is selected from ISIS NOs: 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, or 405874.

[0325] In certain embodiments, a target region is nucleotides 860-899 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 860-899 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227 or 228. In certain such embodiments, an antisense compound targeted to nucleotides 860-899 of SEQ ID NO: 1 is selected from ISIS NOs: 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, or 399955.

[0326] In certain embodiments, a target region is nucleotides 860-909 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 860-909 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 24, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231 or 232. In certain such embodiments, an antisense compound targeted to nucleotides 860-909 of SEQ ID NO: 1 is selected from ISIS NOs: 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, or 399884.

[0327] In certain embodiments, a target region is nucleotides 860-917 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 860-917 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 24, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, or 233. In certain such embodiments, an antisense compound targeted to nucleotides 860-917 of SEQ ID NO: 1 is selected from ISIS NOs: 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, 399884, or 410754.

[0328] In certain embodiments, a target region is nucleotides 869-895 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 869-895 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 224, 225, 226, or 227. In certain such embodiments, an antisense compound targeted to nucleotides 869-895 of SEQ ID NO: 1 is selected from ISIS NOs: 405875, 405876, 406027, or 406028.

[0329] In certain embodiments, a target region is nucleotides 878-905 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 878-905 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 23, 228, 229, 230, or 231. In certain such embodiments, an antisense compound targeted to nucleotides 878-905 of SEQ ID NO: 1 is selected from ISIS NOs: 406029, 399799, 399955, 406030, 406031, or 410733.

[0330] In certain embodiments, a target region is nucleotides 888-909 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 888-909 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 24 or 232. In certain such embodiments, an antisense compound targeted to nucleotides 888-909 of SEQ ID NO: 1 is selected from ISIS NOs: 406032, 395162, or 399884.

[0331] In certain embodiments, a target region is nucleotides 923-952 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 923-952 of SEQ ID NO: 1. In 5 certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 25 or 234. In certain such embodiments, an antisense compound targeted to nucleotides 923-952 of SEQ ID NO: 1 is selected from ISIS NOs: 395163, 399885, or 410755.

[0332] In certain embodiments, a target region is nucleotides 960-1034 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 960-1034 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, or 256. In certain such embodiments, an antisense compound targeted to nucleotides 960-1034 of SEQ ID NO: 1 is selected from ISIS NOs: 410756, 405526, 405604, 406033, 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, or 410758.

[0333] In certain embodiments, a target region is nucleotides 960-1173 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 960-1173 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, or 273. In certain such embodiments, an antisense compound targeted to nucleotides 960-1173 of SEQ ID NO: 1 is selected from ISIS NOs: 410756, 405526, 405604, 406033, 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, or 410763.

[0334] In certain embodiments, a target region is nucleotides 960-986 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 960-986 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 235, 236, or 237. In certain such embodiments, an antisense compound targeted to nucleotides 960-986 of SEQ ID NO: 1 is selected from ISIS NOs: 410756, 405526, 405604, or 406033.

[0335] In certain embodiments, a target region is nucleotides 967-991 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 967-991 of SEQ ID NO: 1. In 5 certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 237 or 238. In certain such embodiments, an antisense compound targeted to nucleotides 967-991 of SEQ ID NO: 1 is selected from ISIS NOs: 406033 or 406034.

[0336] In certain embodiments, a target region is nucleotides 970-1023 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 970-1023 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, or 249. In certain such embodiments, an antisense compound targeted to nucleotides 970-1023 of SEQ ID NO: 1 is selected from ISIS NOs: 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, or 410665.

[0337] In certain embodiments, a target region is nucleotides 970-1064 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 970-1064 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 29, 30, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 149, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, or 263. In certain such embodiments, an antisense compound targeted to nucleotides 970-1064 of SEQ ID NO: 1 is selected from ISIS NOs: 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, or 399957.

[0338] In certain embodiments, a target region is nucleotides 970-1117 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 970-1117 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 29, 30, 31, 32, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 149, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, or 266. In certain such embodiments, an antisense compound targeted to nucleotides 970-1117 of SEQ ID NO: 1 is selected from ISIS NOs: 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, or 399890.

[0339] In certain embodiments, a target region is nucleotides 970-996 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 970-996 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, or 238. In certain such embodiments, an antisense compound targeted to nucleotides 970-996 of SEQ ID NO: 1 is selected from ISIS NOs: 395164, 399886, 406034, 399800, 399956, or 406035.

[0340] In certain embodiments, a target region is nucleotides 977-1004 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 977-1004 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 239 or 240. In certain such embodiments, an antisense compound targeted to nucleotides 977-1004 of SEQ ID NO: 1 is selected from ISIS NOs: 406035 or 410757.

[0341] In certain embodiments, a target region is nucleotides 985-1011 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 985-1011 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 240, 241, or 242. In certain such embodiments, an antisense compound targeted to nucleotides 985-1011 of SEQ ID NO: 1 is selected from ISIS NOs: 410757, 406036, or 406037.

[0342] In certain embodiments, a target region is nucleotides 989-1016 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 989-1016 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 241, 242, or 243. In certain such embodiments, an antisense compound targeted to nucleotides 989-1016 of SEQ ID NO: 1 is selected from ISIS NOs: 406036, 406037, 410536, 410583, or 410658.

[0343] In certain embodiments, a target region is nucleotides 992-1019 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 992-1019 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 242, 243, 244, 245, or 246. In certain such embodiments, an antisense compound targeted to nucleotides 992-1019 of SEQ ID NO: 1 is selected from ISIS NOs: 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, or 410661.

[0344] In certain embodiments, a target region is nucleotides 997-1024 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 997-1024 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 243, 244, 245, 246, 247, 248, 249, or 250. In certain such embodiments, an antisense compound targeted to nucleotides 997-1024 of SEQ ID NO: 1 is selected from ISIS NOs: 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 409126, 410665, 405881, 410590, or 410666.

[0345] In certain embodiments, a target region is nucleotides 997-1024 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 997-1024 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 243, 244, 245, 246, 247, 248, 249, or 250. In certain such embodiments, an antisense compound targeted to nucleotides 997-1024 of SEQ ID NO: 1 is selected from ISIS NOs: 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 409126, 410665, 405881, 410590, or 410666.

[0346] In certain embodiments, a target region is nucleotides 998-1025 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 998-1025 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 244, 245, 246, 247, 248, 249, 250, or 251. In certain such embodiments, an antisense compound targeted to nucleotides 998-1025 of SEQ ID NO: 1 is selected from ISIS NOs: 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, or 410667.

[0347] In certain embodiments, a target region is nucleotides 999-1026 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 999-1026 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 245, 246, 247, 248, 249, 250, 251, or 252. In certain such embodiments, an antisense compound targeted to nucleotides 999-1026 of SEQ ID NO: 1 is selected from ISIS NOs: 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405833, 410592, or 410668.

[0348] In certain embodiments, a target region is nucleotides 1000-1027 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1000-1027 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 246, 247, 248, 249, 250, 251, 252, or 253. In certain such embodiments, an antisense compound targeted to nucleotides 1000-1027 of SEQ ID NO: 1 is selected from ISIS NOs: 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405833, 410592, 410668, 405884, 410593, or 410669.

[0349] In certain embodiments, a target region is nucleotides 1001-1028 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1001-1028 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 247, 248, 249, 250, 251, 252, 253, or 254. In certain such embodiments, an antisense compound targeted to nucleotides 1001-1028 of SEQ ID NO: 1 is selected from ISIS NOs: 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405833, 410592, 410668, 405884, 410593, 410669, 410538, 410594, or 410670.

[0350] In certain embodiments, a target region is nucleotides 1002-1021 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1002-1021 of SEQ ID NO: 1. In one such embodiment, an antisense compound targeted to nucleotides 1002-1021 of SEQ ID NO: 1 is ISIS NO: 405879.

[0351] In certain embodiments, a target region is nucleotides 1002-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1002-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 248, 249, 250, 251, 252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1002-1029 of SEQ ID NO: 1 is selected from ISIS NOs: 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405833, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0352] In certain embodiments, a target region is nucleotides 1003-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1003-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 249, 250, 251,252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1003-1029 of SEQ ID NO: 1 is selected from ISIS NOs: 405880, 410589, 410664, 395165, 399887, 409126, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0353] In certain embodiments, a target region is nucleotides 1004-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1004-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 250, 251,252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1004-1029 of SEQ ID NO: 1 is selected from ISIS NOs: 395165, 399887, 409126, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0354] In certain embodiments, a target region is nucleotides 1005-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1005-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 250, 251, 252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1005-1029 of SEQ ID NO: 1 is selected from ISIS NOs: 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0355] In certain embodiments, a target region is nucleotides 1006-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1006-1029 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 251, 252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1006-1029 of SEQ ID NO: 1 is selected from ISIS NOs: 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0356] In certain embodiments, a target region is nucleotides 1007-1034 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1007-1034 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 252, 253, 254, 255, or 256. In certain such embodiments, an antisense compound targeted to nucleotides 1007-1034 of SEQ ID NO: 1 is selected from ISIS NOs: 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, 410671, or 410758.

[0357] In certain embodiments, a target region is nucleotides 1036-1061 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1036-1061 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 29, 257, 258, or 259. In certain such embodiments, an antisense compound targeted to nucleotides 1036-1061 of SEQ ID NO: 1 is selected from ISIS NOs: 406039, 406040, 395166, 399888, or 406041.

[0358] In certain embodiments, a target region is nucleotides 1045-1072 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1045-1072 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 30, 260, 261, or 262. In certain such embodiments, an antisense compound targeted to nucleotides 1045-1072 of SEQ ID NO: 1 is selected from ISIS NOs: 399801, 399957, 406042, 406043, or 406044.

[0359] In certain embodiments, a target region is nucleotides 1076-1096 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1076-1096 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 31 or 264. In certain such embodiments, an antisense compound targeted to nucleotides 1076-1096 of SEQ ID NO: 1 is selected from ISIS NOs: 408642, 395167, or 399889.

[0360] In certain embodiments, a target region is nucleotides 1088-1115 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1088-1115 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 265 or 266. In certain such embodiments, an antisense compound targeted to nucleotides 1088-1115 of SEQ ID NO: 1 is selected from ISIS NOs: 410760 or 406045.

[0361] In certain embodiments, a target region is nucleotides 1098-1123 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1098-1123 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 32, 267, 268, or 269. In certain such embodiments, an antisense compound targeted to nucleotides 1098-1123 of SEQ ID NO: 1 is selected from ISIS NOs: 395168, 399890, 405909, 405910, or 405911.

[0362] In certain embodiments, a target region is nucleotides 1200-1251 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1200-1251 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, or 285. In certain such embodiments, an antisense compound targeted to nucleotides 1200-1251 of SEQ ID NO: 1 is selected from ISIS NOs: 410764, 395169, 399891, 405913, 405914, 405915, 405916, 410734, 405917, 405918, 405919, 405920, 405921, or 405922.

[0363] In certain embodiments, a target region is nucleotides 1210-1237 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1210-1237 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 275, 276, 277, or 278. In certain such embodiments, an antisense compound targeted to nucleotides 1210-1237 of SEQ ID NO: 1 is selected from ISIS NOs: 395169, 399891, 405913, 405914, 405915, or 405916.

[0364] In certain embodiments, a target region is nucleotides 1219-1245 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1219-1245 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 279, 280, 281 or 282. In certain such embodiments, an antisense compound targeted to nucleotides 1219-1245 of SEQ ID NO: 1 is selected from ISIS NOs: 410734, 405917, 405918, or 405919.

[0365] In certain embodiments, a target region is nucleotides 1228-1251 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1228-1251 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 283, 284, or 285. In certain such embodiments, an antisense compound targeted to nucleotides 1228-1251 of SEQ ID NO: 1 is selected from ISIS NOs: 405920, 405921, or 405922.

[0366] In certain embodiments, a target region is nucleotides 1273-1444 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1273-1444 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 34, 35, 36, 37, 38, 39, 40, 41, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307,308, 309, 310, 311, 312, 313, 314, 315, 316, 316, and 318. In certain such embodiments, an antisense compound targeted to nucleotides 1273-1444 of SEQ ID NO: 1 is selected from ISIS NOs: 410765, 410766, 405923, 395170, 399892, 410767, 405924, 410735, 405925, 405926, 395171, 399893, 405927, 395172, 399894, 405928, 399802, 399958, 405929, 395173, 399895, 405930, 405931, 405932, 405933, 405934, 399803, 399959, 405935, 410736, 405936, 395174, or 410769.

[0367] In certain embodiments, a target region is nucleotides 1295-1316 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1295-1316 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 149 or 288. In certain such embodiments, an antisense compound targeted to nucleotides 1295-1316 of SEQ ID NO: 1 is selected from ISIS NOs: 405923, 395170, or 399892.

[0368] In certain embodiments, a target region is nucleotides 1318-1345 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1318-1345 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 34, 290, 291, 292 or 293. In certain such embodiments, an antisense compound targeted to nucleotides 1318-1345 of SEQ ID NO: 1 is selected from ISIS NOs: 405924, 410735, 405925, 405926, 395171, or 399893.

[0369] In certain embodiments, a target region is nucleotides 1328-1354 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1328-1354 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 35, 128, 294 or 295. In certain such embodiments, an antisense compound targeted to nucleotides 1328-1354 of SEQ ID NO: 1 is selected from ISIS NOs: 405927, 395172, 399894, 405928, 399802, or 399958.

[0370] In certain embodiments, a target region is nucleotides 1337-1361 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1337-1361 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 36, 296, or 297. In certain such embodiments, an antisense compound targeted to nucleotides 1337-1361 of SEQ ID NO: 1 is selected from ISIS NOs: 405929, 395173, 399895, or 405930.

[0371] In certain embodiments, a target region is nucleotides 1344-1371 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1344-1371 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 37, 298, 299, 300, or 301. In certain such embodiments, an antisense compound targeted to nucleotides 1344-1371 of SEQ ID NO: 1 is selected from ISIS NOs: 405931, 405932, 405933, 405934, 399803, or 399959.

[0372] In certain embodiments, a target region is nucleotides 1354-1377 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1354-1377 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 302, 303, or 304. In certain such embodiments, an antisense compound targeted to nucleotides 1354-1377 of SEQ ID NO: 1 is selected from ISIS NOs: 405935, 410736, or 405936.

[0373] In certain embodiments, a target region is nucleotides 1380-1406 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1380-1406 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 305 or 306. In certain such embodiments, an antisense compound targeted to nucleotides 1380-1406 of SEQ ID NO: 1 is selected from ISIS NOs: 410768 or 405937.

[0374] In certain embodiments, a target region is nucleotides 1389-1416 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1389-1416 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 39, 307, 308, 309 or 310. In certain such embodiments, an antisense compound targeted to nucleotides 1389-1416 of SEQ ID NO: 1 is selected from ISIS NOs: 395175, 399897, 405938, 405939, 405940, or 405941.

[0375] In certain embodiments, a target region is nucleotides 1400-1426 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1400-1426 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 40, 311, 312, 313, or 314. In certain such embodiments, an antisense compound targeted to nucleotides 1400-1426 of SEQ ID NO: 1 is selected from ISIS NOs: 399804, 399960, 405942, 405943, 410737, or 405944.

[0376] In certain embodiments, a target region is nucleotides 1409-1434 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1409-1434 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 41, 315, 316, or 317. In certain such embodiments, an antisense compound targeted to nucleotides 1409-1434 of SEQ ID NO: 1 is selected from ISIS NOs: 405945, 399805, 399961, 405946, or 405947.

[0377] In certain embodiments, a target region is nucleotides 1465-1491 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1465-1491 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 42, 101, 319, or 320. In certain such embodiments, an antisense compound targeted to nucleotides 1465-1491 of SEQ ID NO: 1 is selected from ISIS NOs: 395176, 399898, 405948, 399806, 399962, or 405949.

[0378] In certain embodiments, a target region is nucleotides 1465-1602 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1465-1602 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 87, 101, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 1465-1602 of SEQ ID NO: 1 is selected from ISIS NOs: 395176, 399898, 405948, 399806, 399962, 405949, 405950, 405951, 399807, 399963, 405952, 410738, 405953, 405954, 410770, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, or 399969.

[0379] In certain embodiments, a target region is nucleotides 1474-1499 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1474-1499 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 321, 322, or 323. In certain such embodiments, an antisense compound targeted to nucleotides 1474-1499 of SEQ ID NO: 1 is selected from ISIS NOs: 405950, 405951, 399807, 399963, or 405952.

[0380] In certain embodiments, a target region is nucleotides 1482-1519 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1482-1519 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 324, 325, 326, or 327. In certain such embodiments, an antisense compound targeted to nucleotides 1482-1519 of SEQ ID NO: 1 is selected from ISIS NOs: 410738, 405953, 405954, or 410770.

[0381] In certain embodiments, a target region is nucleotides 1513-1540 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1513-1540 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 328, 329, 330, 331, or 332. In certain such embodiments, an antisense compound targeted to nucleotides 1513-1540 of SEQ ID NO: 1 is selected from ISIS NOs: 405955, 405956, 405957, 405958, or 405959.

[0382] In certain embodiments, a target region is nucleotides 1523-1549 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1523-1549 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 333, 334, 335, or 336. In certain such embodiments, an antisense compound targeted to nucleotides 1523-1549 of SEQ ID NO: 1 is selected from ISIS NOs: 405960, 405961, 399808, 399964, 405962, or 410739.

[0383] In certain embodiments, a target region is nucleotides 1526-1602 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1526-1602 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 45, 46, 47, 48, 49, 50, 51, 87, 335, 336, 337, 338, 339, 340, 341, 342, 343, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 1526-1602 of SEQ ID NO: 1 is selected from ISIS NOs: 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, 399966, 405967, 405968, 399811, 399967, 405969, 405970, 410740, 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, or 399969.

[0384] In certain embodiments, a target region is nucleotides 1526-1624 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1526-1624 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 45, 46, 47, 48, 49, 50, 51, 87, 119, 335, 336, 337, 338, 339, 340, 341, 342, 343, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 1526-1624 of SEQ ID NO: 1 is selected from ISIS NOs: 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, 399966, 405967, 405968, 399811, 399967, 405969, 405970, 410740, 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, or 399902.

[0385] In certain embodiments, a target region is nucleotides 1532-1558 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1532-1558 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 45, 46, 337, or 338. In certain such embodiments, an antisense compound targeted to nucleotides 1532-1558 of SEQ ID NO: 1 is selected from ISIS NOs: 405963, 395177, 399899, 405964, 399809, or 399965.

[0386] In certain embodiments, a target region is nucleotides 1541-1568 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1541-1568 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 47, 340, 341, or 342. In certain such embodiments, an antisense compound targeted to nucleotides 1541-1568 of SEQ ID NO: 1 is selected from ISIS NOs: 405965, 405966, 399810, 399966, 405967, or 405968.

[0387] In certain embodiments, a target region is nucleotides 1552-1579 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1552-1579 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 48, 343, 344, 345, or 346. In certain such embodiments, an antisense compound targeted to nucleotides 1552-1579 of SEQ ID NO: 1 is selected from ISIS NOs: 399811, 399967, 405969, 405970, 410740, or 405885.

[0388] In certain embodiments, a target region is nucleotides 1560-1587 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1560-1587 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 346, 347, 348, 349, 350, 351, 352, or 353. In certain such embodiments, an antisense compound targeted to nucleotides 1560-1587 of SEQ ID NO: 1 is selected from ISIS NOs: 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, or 410678.

[0389] In certain embodiments, a target region is nucleotides 1561-1589 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1561-1589 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 346, 347, 348, 349, 350, 351, 352, 353, or 354. In certain such embodiments, an antisense compound targeted to nucleotides 1561-1589 of SEQ ID NO: 1 is selected from ISIS NOs: 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 405557, 410679, 408653, 410602, or 410680.

[0390] In certain embodiments, a target region is nucleotides 1564-1591 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1564-1591 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 350, 351, 352, 353, 354, 355, or 356. In certain such embodiments, an antisense compound targeted to nucleotides 1564-1591 of SEQ ID NO: 1 is selected from ISIS NOs: 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 405557, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, or 410682.

[0391] In certain embodiments, a target region is nucleotides 1565-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1565-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 350, 351, 352, 353, 354, 355, 356, or 357. In certain such embodiments, an antisense compound targeted to nucleotides 1565-1592 of SEQ ID NO: 1 is selected from ISIS NOs: 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 405557, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0392] In certain embodiments, a target region is nucleotides 1566-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1566-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 351, 352, 353, 354, 355, 356, or 357. In certain such embodiments, an antisense compound targeted to nucleotides 1566-1592 of SEQ ID NO: 1 is selected from ISIS NOs: 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 405557, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0393] In certain embodiments, a target region is nucleotides 1567-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1567-1592 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 352, 353, 354, 355, 356, or 357. In certain such embodiments, an antisense compound targeted to nucleotides 1567-1592 of SEQ ID NO: 1 is selected from ISIS NOs: 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 405557, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0394] In certain embodiments, a target region is nucleotides 1570-1597 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1570-1597 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 354, 355, 356, 357 or 358. In certain such embodiments, an antisense compound targeted to nucleotides 1570-1597 of SEQ ID NO: 1 is selected from ISIS NOs: 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, 405972 or 405973.

[0395] In certain embodiments, a target region is nucleotides 1571-1599 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1571-1599 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 87, 356, 357, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 1571-1599 of SEQ ID NO: 1 is selected from ISIS NOs: 405971, 410603, 410681, 410540, 410604, 410682, 405972, 395179, 399901, 405973, or 405974.

[0396] In certain embodiments, a target region is nucleotides 1605-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1605-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 54, 119, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 1605-1706 of SEQ ID NO: 1 is selected from ISIS NOs: 395180, 399902, 410771, 395181, 399903, 405975, 399814, 399970, 405976, 405977, 410772, 405978, 395182, 399904, 405979, 405980, 405981, 410741, 405982, or 405983.

[0397] In certain embodiments, a target region is nucleotides 1628-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1628-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 54, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 1628-1706 of SEQ ID NO: 1 is selected from ISIS NOs: 410771, 395181, 399903, 405975, 399814, 399970, 405976, 405977, 410772, 405978, 395182, 399904, 405979, 405980, 405981, 410741, 405982, or 405983.

[0398] In certain embodiments, a target region is nucleotides 1640-1666 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1640-1666 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 361, or 362. In certain such embodiments, an antisense compound targeted to nucleotides 1640-1666 of SEQ ID NO: 1 is selected from ISIS NOs: 395181, 399903, 405975, 399814, 399970, or 405976.

[0399] In certain embodiments, a target region is nucleotides 1672-1698 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1672-1698 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 54, 365, 366, or 367. In certain such embodiments, an antisense compound targeted to nucleotides 1672-1698 of SEQ ID NO: 1 is selected from ISIS NOs: 405978, 395182, 399904, 405979, or 405980.

[0400] In certain embodiments, a target region is nucleotides 1681-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1681-1706 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 1681-1706 of SEQ ID NO: 1 is selected from ISIS NOs: 405981, 410741, 405982, or 405983.

[0401] In certain embodiments, a target region is nucleotides 1735-1761 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1735-1761 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 372, 373, or 374. In certain such embodiments, an antisense compound targeted to nucleotides 1735-1761 of SEQ ID NO: 1 is selected from ISIS NOs: 405984, 405564, 405641, 405985, 399815, 399971, or 405986.

[0402] In certain embodiments, a target region is nucleotides 1735-1765 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1735-1765 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 56, 372, 373, 374, or 375. In certain such embodiments, an antisense compound targeted to nucleotides 1735-1765 of SEQ ID NO: 1 is selected from ISIS NOs: 405984, 405564, 405641, 405985, 399815, 399971, 405986, 405987, 399816 or 399972.

[0403] In certain embodiments, a target region is nucleotides 1740-1765 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1740-1765 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 56, 374, or 375. In certain such embodiments, an antisense compound targeted to nucleotides 1740-1765 of SEQ ID NO: 1 is selected from ISIS NOs: 399815, 399971, 405986, 405987, 399816, or 399972.

[0404] In certain embodiments, a target region is nucleotides 1849-1876 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1849-1876 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 376, 377, 378, 379, 380, 381, 382, 383, or 384. In certain such embodiments, an antisense compound targeted to nucleotides 1849-1876 of SEQ ID NO: 1 is selected from ISIS NOs: 410541, 410605, 410683, 410542, 410606, 410684, 410543, 410607, 410685, 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, or 410691.

[0405] In certain embodiments, a target region is nucleotides 1849-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1849-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, or 386. In certain such embodiments, an antisense compound targeted to nucleotides 1849-1879 of SEQ ID NO: 1 is selected from ISIS NOs: 410541, 410605, 410683, 410542, 410606, 410684, 410543, 410607, 410685, 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, 410692, 410548, 410614, or 405990.

[0406] In certain embodiments, a target region is nucleotides 1850-1877 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1850-1877 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 377, 378, 379, 380, 381, 382, 383 or 384. In certain such embodiments, an antisense compound targeted to nucleotides 1850-1877 of SEQ ID NO: 1 is selected from ISIS NOs: 410542, 410606, 410684, 410543, 410607, 410685, 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, or 410692.

[0407] In certain embodiments, a target region is nucleotides 1851-1877 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1851-1877 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 378, 379, 380, 381, 382, 383, or 384. In certain such embodiments, an antisense compound targeted to nucleotides 1851-1877 of SEQ ID NO: 1 is selected from ISIS NOs: 410543, 410607, 410685, 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, or 410692.

[0408] In certain embodiments, a target region is nucleotides 1852-1878 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1852-1878 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 379, 380, 381, 382, 383, 384, or 385. In certain such embodiments, an antisense compound targeted to nucleotides 1852-1878 of SEQ ID NO: 1 is selected from ISIS NOs: 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, 410692, 410548, 410614, or 410693.

[0409] In certain embodiments, a target region is nucleotides 1852-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1852-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 379, 380, 381, 382, 383, 384, 385, or 386. In certain such embodiments, an antisense compound targeted to nucleotides 1852-1879 of SEQ ID NO: 1 is selected from ISIS NOs: 410544, 410608, 410686, 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, 410692, 410548, 410614, 410693, or 405990.

[0410] In certain embodiments, a target region is nucleotides 1853-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1853-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 380, 381, 382, 383, 384, 385 or 386. In certain such embodiments, an antisense compound targeted to nucleotides 1853-1879 of SEQ ID NO: 1 is selected from ISIS NOs: 410545, 410609, 410687, 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, 410692, 410548, 410614, 410693, or 405990.

[0411] In certain embodiments, a target region is nucleotides 1854-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1854-1879 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 381, 382, 383, 384, 385, or 386. In certain such embodiments, an antisense compound targeted to nucleotides 1854-1879 of SEQ ID NO: 1 is selected from ISIS NOs: 405988, 410610, 410688, 410546, 410611, 410689, 405989, 410612, 410690, 410547, 410613, 395184, 410691, 399906, 410692, 410548, 410614, 410693, or 405990.

[0412] In certain embodiments, a target region is nucleotides 1905-1955 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1905-1955 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, or 406. In certain such embodiments, an antisense compound targeted to nucleotides 1905-1955 of SEQ ID NO: 1 is selected from ISIS NOs: 410773, 410549, 410615, 410694, 410550, 410616, 410695, 410551, 410617, 410696, 410552, 410618, 410697, 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, or 410774.

[0413] In certain embodiments, a target region is nucleotides 1915-1942 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1915-1942 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 388, 389, 390, 391, 392, 393, 394, or 395. In certain such embodiments, an antisense compound targeted to nucleotides 1915-1942 of SEQ ID NO: 1 is selected from ISIS NOs: 410549, 410615, 410694, 410550, 410616, 410695, 410551, 410617, 410696, 410552, 410618, 410697, 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, or 410702.

[0414] In certain embodiments, a target region is nucleotides 1916-1943 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1916-1943 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 389, 390, 391, 392, 393, 394, 395, or 396. In certain such embodiments, an antisense compound targeted to nucleotides 1916-1943 of SEQ ID NO: 1 is selected from ISIS NOs: 410550, 410616, 410695, 410551, 410617, 410696, 410552, 410618, 410697, 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, or 410703.

[0415] In certain embodiments, a target region is nucleotides 1917-1944 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1917-1944 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 390, 391, 392, 393, 394, 395, 396, or 397. In certain such embodiments, an antisense compound targeted to nucleotides 1917-1944 of SEQ ID NO: 1 is selected from ISIS NOs: 410551, 410617, 410696, 410552, 410618, 410697, 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, or 410704.

[0416] In certain embodiments, a target region is nucleotides 1918-1945 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1918-1945 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 391, 392, 393, 394, 395, 396, 397, or 398. In certain such embodiments, an antisense compound targeted to nucleotides 1918-1945 of SEQ ID NO: 1 is selected from ISIS NOs: 410552, 410618, 410697, 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, or 410705.

[0417] In certain embodiments, a target region is nucleotides 1919-1946 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1919-1946 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 392, 393, 394, 395, 396, 397, or 399. In certain such embodiments, an antisense compound targeted to nucleotides 1919-1946 of SEQ ID NO: 1 is selected from ISIS NOs: 410553, 410619, 410698, 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, or 410706.

[0418] In certain embodiments, a target region is nucleotides 1920-1939 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1920-1939 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59. In certain such embodiments, an antisense compound targeted to nucleotides 1920-1939 of SEQ ID NO: 1 is selected from ISIS NOs: 395185, 399907, or 410699.

[0419] In certain embodiments, a target region is nucleotides 1920-1947 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1920-1947 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 393, 394, 395, 396, 397, 398, 399, or 400. In certain such embodiments, an antisense compound targeted to nucleotides 1920-1947 of SEQ ID NO: 1 is selected from ISIS NOs: 395185, 399907, 410699, 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405944, 410627, or 410707.

[0420] In certain embodiments, a target region is nucleotides 1921-1948 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1921-1948 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 393, 394, 395, 396, 397, 398, 399, 400, or 401. In certain such embodiments, an antisense compound targeted to nucleotides 1921-1948 of SEQ ID NO: 1 is selected from ISIS NOs: 410554, 410620, 410700, 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405944, 410627, 410707, 410558, 410628, or 410708.

[0421] In certain embodiments, a target region is nucleotides 1922-1949 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1922-1949 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 394, 395, 396, 397, 398, 399, 400, 401, or 402. In certain such embodiments, an antisense compound targeted to nucleotides 1922-1949 of SEQ ID NO: 1 is selected from ISIS NOs: 405991, 410621, 410701, 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405944, 410627, 410707, 410558, 410628, 410708, 405995, 410629, or 410709.

[0422] In certain embodiments, a target region is nucleotides 1923-1950 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1923-1950 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 395, 396, 397, 398, 399, 400, 401, 402, or 403. In certain such embodiments, an antisense compound targeted to nucleotides 1923-1950 of SEQ ID NO: 1 is selected from ISIS NOs: 410555, 410622, 410702, 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405944, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, or 410710.

[0423] In certain embodiments, a target region is nucleotides 1924-1951 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1924-1951 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 396, 397, 398, 399, 400, 401, 402, 403, or 404. In certain such embodiments, an antisense compound targeted to nucleotides 1924-1951 of SEQ ID NO: 1 is selected from ISIS NOs: 405992, 410623, 410703, 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405994, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, 410710, 410560, 410631, or 410711.

[0424] In certain embodiments, a target region is nucleotides 1925-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1925-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 397, 398, 399, 400, 401, 402, 403, 404, or 405. In certain such embodiments, an antisense compound targeted to nucleotides 1925-1952 of SEQ ID NO: 1 is selected from ISIS NOs: 410556, 410624, 410704, 405993, 410625, 410705, 410557, 410626, 410706, 405994, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, 410710, 410560, 410631, 410711, 410561, 410632, or 410712.

[0425] In certain embodiments, a target region is nucleotides 1926-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1926-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 398, 399, 400, 401, 402, 403, 404, or 405. In certain such embodiments, an antisense compound targeted to nucleotides 1926-

1952 of SEQ ID NO: 1 is selected from ISIS NOs: 405993, 410625, 410705, 410557, 410626, 410706, 405994, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, 410710, 410560, 410631, 410711, 410561, 410632, or 410712.

[0426] In certain embodiments, a target region is nucleotides 1927-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1927-1952 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 399, 400, 401, 402, 403, 404, or 405. In certain such embodiments, an antisense compound targeted to nucleotides 1927-1952 of SEQ ID NO: 1 is selected from ISIS NOs: 410557, 410626, 410706, 405994, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, 410710, 410560, 410631, 410711, 410561, 410632, or 410712.

[0427] In certain embodiments, a target region is nucleotides 1928-1955 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1928-1955 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 400, 401, 402, 403, 404, 405, or 406. In certain such embodiments, an antisense compound targeted to nucleotides 1928-1955 of SEQ ID NO: 1 is selected from ISIS NOs: 405994, 410627, 410707, 410558, 410628, 410708, 405995, 410629, 410709, 410559, 410630, 410710, 410560, 410631, 410711, 410561, 410632, 410712, or 410774.

[0428] In certain embodiments, a target region is nucleotides 1962-2059 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 1962-2059 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 407, 408, 409, or 410. In certain such embodiments, an antisense compound targeted to nucleotides 1962-2059 of SEQ ID NO: 1 is selected from ISIS NOs: 410775, 410776, 410777, or 410778.

[0429] In certain embodiments, a target region is nucleotides 2040-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2040-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 410, 411, 412, 413, 414, or 415. In certain such embodiments, an antisense compound targeted to nucleotides 2040-2126 of SEQ ID NO: 1 is selected from ISIS NOs: 410778, 395186, 399908, 410713, 410562, 410633, 410714, 405996, 410634, 410715, 410563, 410635, 410716, 410564, 410636, 410717, 399817, 399973, 410718, or 405997.

[0430] In certain embodiments, a target region is nucleotides 2100-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2100-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414, or 415. In certain such embodiments, an antisense compound targeted to nucleotides 2100-2126 of SEQ ID NO: 1 is selected from ISIS NOs: 395186, 399908, 410713, 410562, 410633, 410714, 405996, 410634, 410715, 410563, 410635, 410716, 410564, 410636, 410717, 399817, 399973, 410718 or 405997.

[0431] In certain embodiments, a target region is nucleotides 2100-2139 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2100-2139 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414, 415, or 416. In certain such embodiments, an antisense compound targeted to nucleotides 2100-2139 of SEQ ID NO: 1 is selected from ISIS NOs: 395186, 399908, 410713, 410562, 410633, 410714, 405996, 410634, 410715, 410563, 410635, 410716, 410564, 410636, 410717, 399817, 399973, 410718, 405997 or 410779.

[0432] In certain embodiments, a target region is nucleotides 2100-2206 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2100-2206 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414, 415, 416, 417, or 418. In certain such embodiments, an antisense compound targeted to nucleotides 2100-2206 of SEQ ID NO: 1 is selected from ISIS NOs: 395186, 399908, 410713, 410562, 410633, 410714, 405996, 410634, 410715, 410563, 410635, 410716, 410564, 410636, 410717, 399817, 399973, 410718, 405997, 405997, 410780, or 410781.

[0433] In certain embodiments, a target region is nucleotides 2101-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2101-2126 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 61, 411, 412, 413, 414, or 415. In certain such embodiments, an antisense compound targeted to nucleotides 2101-2126 of SEQ ID NO: 1 is selected from ISIS NOs: 410562, 410633, 410714, 405996, 410634, 410715, 410563, 410635, 410716, 410564, 410636, 410717, 399817, 399973, 410718, or 405997.

[0434] In certain embodiments, a target region is nucleotides 2305-2332 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2305-2332 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425, or 426. In certain such embodiments, an antisense compound targeted to nucleotides 2305-2332 of SEQ ID NO: 1 is selected from ISIS NOs: 410565, 410637, 410719, 410566, 410638, 410720, 410567, 410639, 410721, 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, or 410727.

[0435] In certain embodiments, a target region is nucleotides 2305-2354 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2305-2354 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, or 430. In certain such embodiments, an antisense compound targeted to nucleotides 2305-2354 of SEQ ID NO: 1 is selected from ISIS NOs: 410565, 410637, 410719, 410566, 410638, 410720, 410567, 410639, 410721, 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410783.

[0436] In certain embodiments, a target region is nucleotides 2306-2333 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2306-2333 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 420, 421, 422, 423, 424, 425, 426, or 427. In certain such embodiments, an antisense compound targeted to nucleotides 2306-2333 of SEQ ID NO: 1 is selected from ISIS NOs: 410566, 410638, 410720, 410567, 410639, 410721, 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, or 410728.

[0437] In certain embodiments, a target region is nucleotides 2307-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2307-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 421, 422, 423, 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2307-2334 of SEQ ID NO: 1 is selected from ISIS NOs: 410567, 410639, 410721, 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0438] In certain embodiments, a target region is nucleotides 2308-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2308-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 422, 423, 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2308-2334 of SEQ ID NO: 1 is selected from ISIS NOs: 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0439] In certain embodiments, a target region is nucleotides 2309-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2309-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 423, 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2309-2334 of SEQ ID NO: 1 is selected from ISIS NOs: 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0440] In certain embodiments, a target region is nucleotides 2310-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2310-2334 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2310-2334 of SEQ ID NO: 1 is selected from ISIS NOs: 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0441] In certain embodiments, a target region is nucleotides 2410-2434 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2410-2434 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 63 or 154. In certain such embodiments, an antisense compound targeted to nucleotides 2410-2434 of SEQ ID NO: 1 is selected from ISIS NOs: 395188, 399910, 399818, or 399974.

[0442] In certain embodiments, a target region is nucleotides 2504-2528 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2504-2528 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 64 or 65. In certain such embodiments, an antisense compound targeted to nucleotides 2504-2528 of SEQ ID NO: 1 is selected from ISIS NOs: 395189, 399911, 399819, or 399975.

[0443] In certain embodiments, a target region is nucleotides 2509-2528 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2509-2528 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 65. In certain such embodiments, an antisense compound targeted to nucleotides 2509-2528 of SEQ ID NO: 1 is selected from ISIS NOs: 399819 or 399975.

[0444] In certain embodiments, a target region is nucleotides 2582-2625 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2582-2625 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 66, 67 or 122. In certain such embodiments, an antisense compound targeted to nucleotides 2582-2625 of SEQ ID NO: 1 is selected from ISIS NOs: 399820, 399976, 395190, 399912, 395191, or 399913.

[0445] In certain embodiments, a target region is nucleotides 2606-2668 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2606-2668 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 67 or 153. In certain such embodiments, an antisense compound targeted to nucleotides 2606-2668 of SEQ ID NO: 1 is selected from ISIS NOs: 395191, 399913, 395192, or 399914.

[0446] In certain embodiments, a target region is nucleotides 2828-2855 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2828-2855 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 69, 431, 432, 433, 434, 435, 436, 437 or 438. In certain such embodiments, an antisense compound targeted to nucleotides 2828-2855 of SEQ ID NO: 1 is selected from ISIS NOs: 405893, 405894, 405895, 405896, 395194, 399916, 405897, 405898, 405899, or 405900.

[0447] In certain embodiments, a target region is nucleotides 2832-2851 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2832-2851 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 69. In certain such embodiments, an antisense compound targeted to nucleotides 2832-2851 of SEQ ID NO: 1 is selected from ISIS NOs: 395194, or 399916.

[0448] In certain embodiments, a target region is nucleotides 2900-2927 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2900-2927 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 70, 71, 439, 440, 441, 442, 443 or 444. In certain such embodiments, an antisense compound targeted to nucleotides 2900-2927 of SEQ ID NO: 1 is selected from ISIS NOs: 395195, 399917, 405901, 405902, 405903, 405904, 399821, 399977, 405905, or 405906.

[0449] In certain embodiments, a target region is nucleotides 2900-2929 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2900-2929 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 70, 71, 439, 440, 441, 442, 443, 444, 446 or 446. In certain such embodiments, an antisense compound targeted to nucleotides 2900-2929 of SEQ ID NO: 1 is selected from ISIS NOs: 395195, 399917, 405901, 405902, 405903, 405904, 399821, 399977, 405905, 405906, 405907, or 405908.

[0450] In certain embodiments, a target region is nucleotides 2902-2927 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2902-2927 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 71, 439, 440, 441, 442, 443 or 444. In certain such embodiments, an antisense compound targeted to nucleotides 2902-2927 of SEQ ID NO: 1 is selected from ISIS NOs: 405901, 405902, 405903, 405904, 399821, 399977, 405905, or 405906.

[0451] In certain embodiments, a target region is nucleotides 2983-3007 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2983-3007 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 72 or 73. In certain such embodiments, an antisense compound targeted to nucleotides 2983-3007 of SEQ ID NO: 1 is selected from ISIS NOs: 395196, 399918, 399822, or 399978.

[0452] In certain embodiments, a target region is nucleotides 2983-3013 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 2983-3013 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 72, 73 or 135. In certain such embodiments, an antisense compound targeted to nucleotides 2983-3013 of SEQ ID NO: 1 is selected from ISIS NOs: 395196, 399918, 399822, 399978, 399823, or 399979.

[0453] In certain embodiments, a target region is nucleotides 3227-3252 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 3227-3252 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 74 or 112. In certain such embodiments, an 5 antisense compound targeted to nucleotides 3227-3252 of SEQ ID NO: 1 is selected from ISIS NOs: 395197, 399919, 399824, or 399980.

[0454] In certain embodiments, a target region is nucleotides 3227-3456 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 3227-3456 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 74, 75 or 112. In certain such embodiments, an antisense compound targeted to nucleotides 3227-3456 of SEQ ID NO: 1 is selected from ISIS NOs: 395197, 399919, 399824, 399980, 395198, or 399920.

[0455] In certain embodiments, a target region is nucleotides 3472-3496 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 3472-3496 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 76 or 77. In certain such embodiments, an antisense compound targeted to nucleotides 3472-3496 of SEQ ID NO: 1 is selected from ISIS NOs: 395199, 399921, 399825 or 399981.

[0456] In certain embodiments, a target region is nucleotides 3543-3569 of SEQ ID NO: 1. In certain embodiments, an antisense compound is targeted to nucleotides 3543-3569 of SEQ ID NO: 1. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 78 or 99. In certain such embodiments, an antisense compound targeted to nucleotides 3543-3569 of SEQ ID NO: 1 is selected from ISIS NOs: 395200, 399922, 399826 or 399982.

[0457] In certain embodiments, antisense compound target a PCSK9 nucleic acid having the sequence of nucleotides 25475000 to 25504000 of GENBANK® Accession No. NT_032977.8, first deposited with GENBANK® on Feb. 26, 2006, and incorporated herein as SEQ ID NO: 2. In certain such embodiments, an antisense oligonucleotide is targeted to SEQ ID NO: 2. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 2 is at least 90% complementary to SEQ ID NO: 2. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 2 is at least 95% complementary to SEQ ID NO: 2. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 1 is 100% complementary to SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide comprises a nucleotide sequence selected from a nucleotide sequence set forth in Table 7.

TABLE 7

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
4	2274	2293	GCGCGGAATCCTGGCTGGGA
5	2381	2400	GAGGAGACCTAGAGGCCGTG
159	2433	2452	GCCTGGAGCTGACGGTGCCC
160	2437	2456	GACCGCCTGGAGCTGACGGT

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
6	2439	2458	AGGACCGCCTGGAGCTGACG
162	2545	2564	AAGGCTAGCACCAGCTCCTC
163	2546	2565	CAAGGCTAGCACCAGCTCCT
164	2547	2566	GCAAGGCTAGCACCAGCTCC
165	2548	2567	CGCAAGGCTAGCACCAGCTC
7	2549	2568	ACGCAAGGCTAGCACCAGCT
166	2550	2569	AACGCAAGGCTAGCACCAGC
167	2551	2570	GAACGCAAGGCTAGCACCAG
168	2552	2571	GGAACGCAAGGCTAGCACCA
169	2553	2572	CGGAACGCAAGGCTAGCACC
8	2556	2575	CCTCGGAACGCAAGGCTAGC
170	2560	2579	TCCTCCTCGGAACGCAAGGC
171	2585	2604	GTGCTCGGGTGCTTCGGCCA
172	2605	2624	TGGAAGGTGGCTGTGGTTCC
9	2619	2638	CCTTGGCGCAGCGGTGGAAG
107	3056	3075	CCCACTATAATGGCAAGCCC
80	4306	4325	AACCCAGTTCTAATGCACCT
106	5140	5159	CCAGTCAGAGTAGAACAGAG
102	5590	5609	ATGTGCAGAGATCAATCACA
121	5599	5618	GGAGCCTACATGTGCAGAGA
94	5667	5686	AGCATGGCACCAGCATCTGC
176	6444	6463	CGTAGGTGCCAGGCAACCTC
177	6482	6501	TGACTGCGAGAGGTGGGTCT
178	6492	6511	CAGTGCGCTCTGACTGCGAG
179	6494	6513	GGCAGTGCGCTCTGACTGCG
180	6496	6515	CGGGCAGTGCGCTCTGACTG
10	6498	6517	GGCGGGCAGTGCGCTCTGAC
181	6499	6518	CGGCGGGCAGTGCGCTCTGA
182	6528	6547	ATCCCCGGCGGGCAGCCTGG
183	6532	6551	AGGTATCCCCGGCGGGCAGC
184	6534	6553	TGAGGTATCCCCGGCGGGCA
185	6535	6554	GTGAGGTATCCCCGGCGGGC
186	6536	6555	GGTGAGGTATCCCCGGCGGG
11	6537	6556	TGGTGAGGTATCCCCGGCGG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
187	6538	6557	TTGGTGAGGTATCCCCGGCG
188	6539	6558	CTTGGTGAGGTATCCCCGGC
189	6540	6559	TCTTGGTGAGGTATCCCCGG
190	6541	6560	ATCTTGGTGAGGTATCCCCG
191	6542	6561	GATCTTGGTGAGGTATCCCC
12	6543	6562	GGATCTTGGTGAGGTATCCC
192	6544	6563	AGGATCTTGGTGAGGTATCC
193	6546	6565	GCAGGATCTTGGTGAGGTAT
194	6548	6567	ATGCAGGATCTTGGTGAGGT
195	6550	6569	ACATGCAGGATCTTGGTGAG
13	6552	6571	AGACATGCAGGATCTTGGTG
196	6554	6573	GAAGACATGCAGGATCTTGG
14	6557	6576	ATGGAAGACATGCAGGATCT
197	6565	6584	AGAAGGCCATGGAAGACATG
198	6575	6594	GAAGCCAGGAAGAAGGCCAT
15	6583	6602	TTCACCAGGAAGCCAGGAAG
199	6585	6604	TCTTCACCAGGAAGCCAGGA
16	6588	6607	TCATCTTCACCAGGAAGCCA
200	6590	6609	ACTCATCTTCACCAGGAAGC
201	6592	6611	CCACTCATCTTCACCAGGAA
202	6594	6613	CGCCACTCATCTTCACCAGG
203	6596	6615	GTCGCCACTCATCTTCACCA
204	6598	6617	AGGTCGCCACTCATCTTCAC
205	6600	6619	GCAGGTCGCCACTCATCTTC
206	6602	6621	CAGCAGGTCGCCACTCATCT
207	6604	6623	TCCAGCAGGTCGCCACTCAT
108	6652	6671	CCCAGCCCTATCAGGAAGTG
144	7099	7118	TGACATCCAGGAGGGAGGAG
91	7556	7575	AGACTGATGGAAGGCATTGA
131	7565	7584	GTGTTGAGCAGACTGATGGA
145	8836	8855	TGACATCTTGTCTGGGAGCC
90	8948	8967	AGACTAGGAGCCTGAGTTTT
125	9099	9118	GGCCTGCAGAAGCCAGAGAG
17	9130	9149	CCTCGATGTAGTCGACATGG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
209	9149	9168	GCAAAGACAGAGGAGTCCTC
210	9207	9226	TTCATCCGCCCGGTACCGTG
211	9209	9228	TATTCATCCGCCCGGTACCG
18	9210	9229	GTATTCATCCGCCCGGTACC
212	9212	9231	TGGTATTCATCCGCCCGGTA
213	9214	9233	GCTGGTATTCATCCGCCCGG
214	9216	9235	GGGCTGGTATTCATCCGCCC
148	10252	10271	TGGCAGCAACTCAGACATAT
127	10633	10652	GGTGGTAATTTGTCACAGCA
84	11308	11327	AAGGTCACACAGTTAAGAGT
79	11472	11491	AAATGCAGGGCTAAAATCAC
88	12715	12734	ACTGGATACATTGGCAGACA
111	12928	12947	CTAGAGGAACCACTAGATAT
85	13681	13700	ACAAATTCCCAGACTCAGCA
100	13746	13765	ATCTCAGGACAGGTGAGCAA
116	13760	13779	GAGTAGAGATTCTCATCTCA
129	13816	13835	GTGCCATCTGAACAGCACCT
117	13828	13847	GAGTCTTCTGAAGTGCCATC
81	13903	13922	AAGCAGGGCCTCAGGTGGAA
110	13926	13945	CCTGGAACCCCTGCAGCCAG
152	13977	13996	TTCAGGCAGGTTGCTGCTAG
83	13986	14005	AAGGAAGACTTCAGGCAGGT
140	13998	14017	TCAGCCAGGCCAAAGGAAGA
137	14112	14131	TAGGGAGAGCTCACAGATGC
136	14122	14141	TAGGAGAAAGTAGGGAGAGC
132	14179	14198	TAAAAGCTGCAAGAGACTCA
139	14267	14286	TCAGAGAAAACAGTCACCGA
92	14397	14416	AGAGACAGGAAGCTGCAGCT
142	14404	14423	TCATTTTAGAGACAGGAAGC
113	14441	14460	GAATAACAGTGATGTCTGGC
138	14494	14513	TCACAGCTCACCGAGTCTGC
98	14524	14543	AGTGTAAAATAAAGCCCCTA
96	14601	14620	AGGACCCAAGTCATCCTGCT
124	14631	14650	GGCCATCAGCTGGCAATGCT

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
82	14670	14689	AAGGAAAGGGAGGCCTAGAG
133	14675	14694	TAGACAAGGAAAGGGAGGCC
103	14681	14700	ATTTCATAGACAAGGAAAGG
155	14801	14820	CTTATAGTTAACACACAGAA
156	14809	14828	AAGTCAACCTTATAGTTAAC
215	14877	14896	ATACACCTCCACCAGGCTGC
216	14888	14907	GTGTCTAGGAGATACACCTC
19	14891	14910	CTGGTGTCTAGGAGATACAC
217	14893	14912	TGCTGGTGTCTAGGAGATAC
20	14896	14915	GTATGCTGGTGTCTAGGAGA
21	14916	14935	GATTTCCCGGTGGTCACTCT
218	14918	14937	TCGATTTCCCGGTGGTCACT
219	14919	14938	CTCGATTTCCCGGTGGTCAC
220	14920	14939	CCTCGATTTCCCGGTGGTCA
221	14921	14940	CCCTCGATTTCCCGGTGGTC
22	14922	14941	GCCCTCGATTTCCCGGTGGT
222	14923	14942	TGCCCTCGATTTCCCGGTGG
223	14924	14943	CTGCCCTCGATTTCCCGGTG
224	14925	14944	CCTGCCCTCGATTTCCCGGT
225	14926	14945	CCCTGCCCTCGATTTCCCGG
226	14930	14949	ATGACCCTGCCCTCGATTTC
227	14932	14951	CCATGACCCTGCCCTCGATT
228	14934	14953	GACCATGACCCTGCCCTCGA
23	14936	14955	GTGACCATGACCCTGCCCTC
229	14938	14957	CGGTGACCATGACCCTGCCC
230	14940	14959	GTCGGTGACCATGACCCTGC
231	14942	14961	AAGTCGGTGACCATGACCCT
232	14944	14963	CGAAGTCGGTGACCATGACC
24	14946	14965	CTCGAAGTCGGTGACCATGA
233	14954	14973	GGCACATTCTCGAAGTCGGT
25	14979	14998	GTGGAAGCGGGTCCCGTCCT
235	15254	15273	GGTGGGTGCCATGACTGTCA
236	15257	15276	CCAGGTGGGTGCCATGACTG
237	15261	15280	CCTGCCAGGTGGGTGCCATG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
26	15264	15283	ACCCCTGCCAGGTGGGTGCC
238	15266	15285	CCACCCCTGCCAGGTGGGTG
27	15269	15288	TGACCACCCCTGCCAGGTGG
239	15271	15290	GCTGACCACCCCTGCCAGGT
240	15279	15298	TCCCGGCCGCTGACCACCCC
241	15283	15302	GGCATCCCGGCCGCTGACCA
242	15286	15305	GCCGGCATCCCGGCCGCTGA
243	15291	15310	GCCACGCCGGCATCCCGGCC
244	15292	15311	GGCCACGCCGGCATCCCGGC
245	15293	15312	TGGCCACGCCGGCATCCCGG
246	15294	15313	TTGGCCACGCCGGCATCCCG
247	15295	15314	CTTGGCCACGCCGGCATCCC
248	15296	15315	CCTTGGCCACGCCGGCATCC
249	15297	15316	CCCTTGGCCACGCCGGCATC
28	15298	15317	ACCCTTGGCCACGCCGGCAT
447	15298	15317	ACCCTTGGTCACGCCGGCAT
250	15299	15318	CACCCTTGGCCACGCCGGCA
251	15300	15319	GCACCCTTGGCCACGCCGGC
252	15301	15320	GGCACCCTTGGCCACGCCGG
253	15302	15321	TGGCACCCTTGGCCACGCCG
254	15303	15322	CTGGCACCCTTGGCCACGCC
255	15304	15323	GCTGGCACCCTTGGCCACGC
256	15309	15328	CGCATGCTGGCACCCTTGGC
257	15330	15349	CAGTTGAGCACGCGCAGGCT
258	15332	15351	GGCAGTTGAGCACGCGCAGG
29	15334	15353	TTGGCAGTTGAGCACGCGCA
259	15336	15355	CCTTGGCAGTTGAGCACGCG
30	15339	15358	TTCCCTTGGCAGTTGAGCAC
260	15341	15360	CCTTCCCTTGGCAGTTGAGC
261	15345	15364	GTGCCCTTCCCTTGGCAGTT
262	15347	15366	CCGTGCCCTTCCCTTGGCAG
263	15358	15377	GGTGCCGCTAACCGTGCCCT
86	15471	15490	ACAGCATTCTTGGTTAGGAG
97	16134	16153	AGTCAAGCTGCTGCCCAGAG
120	16668	16687	GCTAGTTATTAAGCACCTGC
150	17267	17286	TGTGAGCTCTGGCCCAGTGG
115	18377	18396	GAGTAAGGCAGGTTACTCTC
134	18408	18427	TAGATGTGACTAACATTTAA
157	18561	18580	AGGAACAAAGCCAAGGTCAC
266	18591	18610	TGGCTTTTCCGAATAAACTC
32	18593	18612	GCTGGCTTTTCCGAATAAAC
267	18595	18614	CAGCTGGCTTTTCCGAATAA
268	18597	18616	ACCAGCTGGCTTTTCCGAAT
269	18599	18618	GGACCAGCTGGCTTTTCCGA
270	18603	18622	GGCTGGACCAGCTGGCTTTT
271	18614	18633	GTGGCCCCACAGGCTGGACC
272	18627	18646	AGCAGCACCACCAGTGGCCC
273	18649	18668	GCTGTACCCACCCGCCAGGG
274	18695	18714	CGACCCCAGCCCTCGCCAGG
33	18705	18724	GTGACCAGCACGACCCCAGC
275	18707	18726	CGGTGACCAGCACGACCCCA
276	18709	18728	AGCGGTGACCAGCACGACCC
277	18711	18730	GCAGCGGTGACCAGCACGAC
278	18713	18732	CGGCAGCGGTGACCAGCACG
279	18714	18733	CCGGCAGCGGTGACCAGCAC
280	18717	18736	TTGCCGGCAGCGGTGACCAG
281	18719	18738	AGTTGCCGGCAGCGGTGACC
282	18721	18740	GAAGTTGCCGGCAGCGGTGA
283	18723	18742	CGGAAGTTGCCGGCAGCGGT
284	18725	18744	CCCGGAAGTTGCCGGCAGCG
285	18727	18746	GTCCCGGAAGTTGCCGGCAG
105	19203	19222	CACATTAGCCTTGCTCAAGT
151	19913	19932	TGTGATGACCTGGAAAGGTG
288	19931	19950	GGCATTGGTGGCCCCAACTG
149	19933	19952	TGGGCATTGGTGGCCCCAAC
289	19941	19960	GCTGGTCTTGGGCATTGGTG
290	19954	19973	CCCAGGGTCACCGGCTGGTC
291	19956	19975	TCCCCAGGGTCACCGGCTGG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
292	19958	19977	AGTCCCCAGGGTCACCGGCT
293	19960	19979	AAAGTCCCCAGGGTCACCGG
34	19962	19981	CCAAAGTCCCCAGGGTCACC
294	19964	19983	CCCCAAAGTCCCCAGGGTCA
128	19966	19985	GTCCCCAAAGTCCCCAGGGT
295	19969	19988	TTGGTCCCCAAAGTCCCCAG
35	19971	19990	AGTTGGTCCCCAAAGTCCCC
296	19973	19992	AAAGTTGGTCCCCAAAGTCC
36	19976	19995	GCCAAAGTTGGTCCCCAAAG
297	19978	19997	CGGCCAAAGTTGGTCCCCAA
298	19980	19999	AGCGGCCAAAGTTGGTCCCC
299	19982	20001	ACAGCGGCCAAAGTTGGTCC
300	19984	20003	ACACAGCGGCCAAAGTTGGT
301	19986	20005	CCACACAGCGGCCAAAGTTG
37	19988	20007	GTCCACACAGCGGCCAAAGT
302	19990	20009	AGGTCCACACAGCGGCCAAA
303	19992	20011	AGAGGTCCACACAGCGGCCA
304	19994	20013	AAAGAGGTCCACACAGCGGC
38	19997	20016	GGCAAAGAGGTCCACACAGC
305	20016	20035	CAATGATGTCCTCCCCTGGG
306	20023	20042	GAGGCACCAATGATGTCCTC
39	20025	20044	TGGAGGCACCAATGATGTCC
307	20027	20046	GCTGGAGGCACCAATGATGT
308	20029	20048	TCGCTGGAGGCACCAATGAT
309	20031	20050	AGTCGCTGGAGGCACCAATG
310	20033	20052	GCAGTCGCTGGAGGCACCAA
40	20036	20055	GCTGCAGTCGCTGGAGGCAC
311	20038	20057	GTGCTGCAGTCGCTGGAGGC
312	20040	20059	AGGTGCTGCAGTCGCTGGAG
313	20042	20061	GCAGGTGCTGCAGTCGCTGG
314	20043	20062	AGCAGGTGCTGCAGTCGCTG
315	20045	20064	AAAGCAGGTGCTGCAGTCGC
41	20047	20066	ACAAAGCAGGTGCTGCAGTC
316	20049	20068	ACACAAAGCAGGTGCTGCAG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
317	20051	20070	TGACACAAAGCAGGTGCTGC
318	20061	20080	TCCCACTCTGTGACACAAAG
158	20100	20119	GTGGTGACTTACCAGCCACG
109	20188	20207	CCCCTGCACAGAGCCTGGCA
141	20624	20643	TCATGGCTGCAATGCCTGGT
320	20629	20648	CAGCATCATGGCTGCAATGC
321	20631	20650	GACAGCATCATGGCTGCAAT
322	20633	20652	CAGACAGCATCATGGCTGCA
43	20635	20654	GGCAGACAGCATCATGGCTG
323	20637	20656	TCGGCAGACAGCATCATGGC
324	20639	20658	GCTCGGCAGACAGCATCATG
325	20641	20660	CGGCTCGGCAGACAGCATCA
326	20643	20662	TCCGGCTCGGCAGACAGCAT
327	20657	20676	CGGCCAGGGTGAGCTCCGGC
328	20670	20689	CTCTGCCTCAACTCGGCCAG
329	20672	20691	GTCTCTGCCTCAACTCGGCC
330	20674	20693	CAGTCTCTGCCTCAACTCGG
331	20676	20695	ATCAGTCTCTGCCTCAACTC
332	20678	20697	GGATCAGTCTCTGCCTCAAC
333	20680	20699	GTGGATCAGTCTCTGCCTCA
334	20682	20701	AAGTGGATCAGTCTCTGCCT
44	20683	20702	GAAGTGGATCAGTCTCTGCC
335	20685	20704	GAGAAGTGGATCAGTCTCTG
336	20687	20706	CAGAGAAGTGGATCAGTCTC
337	20689	20708	GGCAGAGAAGTGGATCAGTC
45	20691	20710	TTGGCAGAGAAGTGGATCAG
338	20693	20712	CTTTGGCAGAGAAGTGGATC
46	20696	20715	CATCTTTGGCAGAGAAGTGG
339	20698	20717	GACATCTTTGGCAGAGAAGT
340	20700	20719	ATGACATCTTTGGCAGAGAA
47	20702	20721	TGATGACATCTTTGGCAGAG
341	20704	20723	ATTGATGACATCTTTGGCAG
342	20706	20725	TCATTGATGACATCTTTGGC
48	20709	20728	GCCTCATTGATGACATCTTT

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
343	20711	20730	AGGCCTCATTGATGACATCT
344	20713	20732	CCAGGCCTCATTGATGACAT
345	20715	20734	AACCAGGCCTCATTGATGAC
346	20717	20736	GGAACCAGGCCTCATTGATG
347	20718	20737	GGGAACCAGGCCTCATTGAT
348	20719	20738	AGGGAACCAGGCCTCATTGA
349	20720	20739	CAGGGAACCAGGCCTCATTG
49	20721	20740	TCAGGGAACCAGGCCTCATT
350	20722	20741	CTCAGGGAACCAGGCCTCAT
351	20723	20742	CCTCAGGGAACCAGGCCTCA
352	20724	20743	TCCTCAGGGAACCAGGCCTC
353	20725	20744	GTCCTCAGGGAACCAGGCCT
50	20726	20745	GGTCCTCAGGGAACCAGGCC
354	20727	20746	TGGTCCTCAGGGAACCAGGC
355	20728	20747	CTGGTCCTCAGGGAACCAGG
356	20729	20748	GCTGGTCCTCAGGGAACCAG
357	20730	20749	CGCTGGTCCTCAGGGAACCA
87	20733	20752	ACCCGCTGGTCCTCAGGGAA
358	20735	20754	GTACCCGCTGGTCCTCAGGG
359	20737	20756	CAGTACCCGCTGGTCCTCAG
51	20740	20759	GGTCAGTACCCGCTGGTCCT
119	20762	20781	GCAGGGCGGCCACCAGGTTG
360	20785	20804	ACCTGCCCCATGGGTGCTGG
93	20995	21014	AGAGAGGAGGGCTTAAAGAA
95	21082	21101	AGCTGCCAACCTGCAAAAAG
361	21088	21107	CAAAACAGCTGCCAACCTGC
53	21091	21110	CTGCAAAACAGCTGCCAACC
362	21093	21112	TCCTGCAAAACAGCTGCCAA
363	21095	21114	AGTCCTGCAAAACAGCTGCC
364	21106	21125	GCTGACCATACAGTCCTGCA
365	21118	21137	GGCCCCGAGTGTGCTGACCA
54	21121	21140	GTAGGCCCCGAGTGTGCTGA
366	21123	21142	GTGTAGGCCCCGAGTGTGCT
367	21125	21144	CCGTGTAGGCCCCGAGTGTG

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
368	21127	21146	ATCCGTGTAGGCCCCGAGTG
369	21129	21148	CCATCCGTGTAGGCCCCGAG
370	21131	21150	GGCCATCCGTGTAGGCCCCG
371	21133	21152	GTGGCCATCCGTGTAGGCCC
372	21181	21200	CTGGAGCAGCTCAGCAGCTC
460	21181	21194	CAGCTCAGCAGCTC
373	21183	21202	AACTGGAGCAGCTCAGCAGC
55	21186	21205	AGAAACTGGAGCAGCTCAGC
374	21188	21207	GGAGAAACTGGAGCAGCTCA
375	21190	21209	CTGGAGAAACTGGAGCAGCT
56	21192	21211	TCCTGGAGAAACTGGAGCAG
143	21481	21500	TGAAAATCCATCCAGCACTG
89	21589	21608	AGAACCATGGAGCACCTGAG
448	21692	21711	CTGCCCTTCCACCAAAATGC
449	21693	21712	ACTGCCCTTCCACCAAAATG
450	21694	21713	CACTGCCCTTCCACCAAAAT
451	21695	21714	GCACTGCCCTTCCACCAAAA
123	21696	21715	GGCACTGCCCTTCCACCAAA
452	21697	21716	GGGCACTGCCCTTCCACCAA
453	21698	21717	TGGGCACTGCCCTTCCACCA
454	21699	21718	CTGGGCACTGCCCTTCCACC
455	21700	21719	GCTGGGCACTGCCCTTCCAC
57	22096	22115	CGTTGTGGGCCCGGCAGACC
376	22133	22152	CACCTGGCAATGGCGTAGAC
377	22134	22153	GCACCTGGCAATGGCGTAGA
378	22135	22154	AGCACCTGGCAATGGCGTAG
379	22136	22155	CAGCACCTGGCAATGGCGTA
380	22137	22156	GCAGCACCTGGCAATGGCGT
381	22138	22157	GGCAGCACCTGGCAATGGCG
382	22139	22158	AGGCAGCACCTGGCAATGGC
383	22140	22159	CAGGCAGCACCTGGCAATGG
384	22141	22160	GCAGGCAGCACCTGGCAATG
58	22142	22161	AGCAGGCAGCACCTGGCAAT
385	22143	22162	TAGCAGGCAGCACCTGGCAA

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
386	22144	22163	GTAGCAGGCAGCACCTGGCA
387	22189	22208	TGGCCTCAGCTGGTGGAGCT
388	22199	22218	GTCCCCATGCTGGCCTCAGC
389	22200	22219	GGTCCCCATGCTGGCCTCAG
390	22201	22220	GGGTCCCCATGCTGGCCTCA
391	22202	22221	CGGGTCCCCATGCTGGCCTC
392	22203	22222	ACGGGTCCCCATGCTGGCCT
59	22204	22223	CACGGGTCCCCATGCTGGCC
393	22205	22224	ACACGGGTCCCCATGCTGGC
394	22206	22225	GACACGGGTCCCCATGCTGG
395	22207	22226	GGACACGGGTCCCCATGCTG
396	22208	22227	TGGACACGGGTCCCCATGCT
397	22209	22228	GTGGACACGGGTCCCCATGC
398	22210	22229	AGTGGACACGGGTCCCCATG
399	22211	22230	CAGTGGACACGGGTCCCCAT
400	22212	22231	GCAGTGGACACGGGTCCCCA
401	22213	22232	GGCAGTGGACACGGGTCCCC
402	22214	22233	TGGCAGTGGACACGGGTCCC
403	22215	22234	GTGGCAGTGGACACGGGTCC
404	22216	22235	GGTGGCAGTGGACACGGGTC
405	22217	22236	TGGTGGCAGTGGACACGGGT
406	22220	22239	TGTTGGTGGCAGTGGACACG
126	22292	22311	GGTGCATAAGGAGAAAGAGA
408	23985	24004	GTGCCAAGGTCCTCCACCTC
409	24005	24024	TCAGCACAGGCGGCTTGTGG
410	24035	24054	CCACGCACTGGTTGGGCTGA
60	24095	24114	CTTTGCATTCCAGACCTGGG
411	24096	24115	ACTTTGCATTCCAGACCTGG
412	24097	24116	GACTTTGCATTCCAGACCTG
413	24098	24117	TGACTTTGCATTCCAGACCT
414	24099	24118	TTGACTTTGCATTCCAGACC
61	24100	24119	CTTGACTTTGCATTCCAGAC
415	24102	24121	TCCTTGACTTTGCATTCCAG
416	24115	24134	CGGGATTCCATGCTCCTTGA

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)

SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
147	24858	24877	TGAGTTCATTTAAGAGTGGA
118	24907	24926	GCACCATCCAGACCAGAATC
114	25413	25432	GAGAGGTTCAGATCCAGGCC
418	25994	26013	GGAGGGCACTGCAGCCAGTC
419	26112	26131	CAGATGGCAACGGCTGTCAC
420	26113	26132	GCAGATGGCAACGGCTGTCA
421	26114	26133	AGCAGATGGCAACGGCTGTC
422	26115	26134	CAGCAGATGGCAACGGCTGT
423	26116	26135	GCAGCAGATGGCAACGGCTG
62	26117	26136	GGCAGCAGATGGCAACGGCT
424	26118	26137	CGGCAGCAGATGGCAACGGC
425	26119	26138	CCGGCAGCAGATGGCAACGG
426	26120	26139	TCCGGCAGCAGATGGCAACG
427	26121	26140	CTCCGGCAGCAGATGGCAAC
428	26122	26141	GCTCCGGCAGCAGATGGCAA
429	26132	26151	CCAGGTGCCGGCTCCGGCAG
430	26142	26161	GAGGCCTGCGCCAGGTGCCG
154	26217	26236	TTTTAAAGCTCAGCCCCAGC
63	26222	26241	AACCATTTTAAAGCTCAGCC
64	26311	26330	TCAAGGGCCAGGCCAGCAGC
65	26316	26335	CCCACTCAAGGGCCAGGCCA
122	26389	26408	GGAGGGAGCTTCCTGGCACC
66	26404	26423	ATGCCCCACAGTGAGGGAGG
67	26413	26432	AATGGTGAAATGCCCCACAG
153	26456	26475	TTGGGAGCAGCTGGCAGCAC
68	26557	26576	CATGGGAAGAATCCTGCCTC
431	26635	26654	ATGAGGGCCATCAGCACCTT
432	26636	26655	GATGAGGGCCATCAGCACCT
433	26637	26656	AGATGAGGGCCATCAGCACC
434	26638	26657	GAGATGAGGGCCATCAGCAC
69	26639	26658	GGAGATGAGGGCCATCAGCA
435	26640	26659	TGGAGATGAGGGCCATCAGC
436	26641	26660	CTGGAGATGAGGGCCATCAG
437	26642	26661	GCTGGAGATGAGGGCCATCA

TABLE 7-continued

Antisense sequences targeted to NT_032977.8 (SEQ ID NO: 2)			
SEQ ID NO	5' Start Site to SEQ ID NO: 2	3' Start Site to SEQ ID NO: 2	Sequence (5' to 3')
438	26643	26662	AGCTGGAGATGAGGGCCATC
461	26684	26697	TTAATCAGGGAGCC
70	26707	26726	TAGATGCCATCCAGAAAGCT
439	26709	26728	GCTAGATGCCATCCAGAAAG
440	26710	26729	GGCTAGATGCCATCCAGAAA
441	26711	26730	TGGCTAGATGCCATCCAGAA
442	26712	26731	CTGGCTAGATGCCATCCAGA
71	26713	26732	TCTGGCTAGATGCCATCCAG
443	26714	26733	CTCTGGCTAGATGCCATCCA
444	26715	26734	CCTCTGGCTAGATGCCATCC
445	26716	26735	GCCTCTGGCTAGATGCCATC
446	26717	26736	AGCCTCTGGCTAGATGCCAT
72	26790	26809	GGCATAGAGCAGAGTAAAGG
73	26795	26814	AGCCTGGCATAGAGCAGAGT
135	26801	26820	TAGCACAGCCTGGCATAGAG
112	27034	27053	GAAGAGGCTTGGCTTCAGAG
74	27040	27059	AAGTAAGAAGAGGCTTGGCT
75	27244	27263	GCTCAAGGAGGGACAGTTGT
76	27279	27298	AAAGATAAATGTCTGCTTGC
77	27284	27303	ACCCAAAAGATAAATGTCTG
78	27350	27369	TCTTCAAGTTACAAAAGCAA
99	27357	27376	ATAAATATCTTCAAGTTACA

[0458] In certain embodiments, gapmer antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gapmer antisense compounds are targeted to SEQ ID NO: 2. In certain such embodiments, the nucleotide sequences illustrated in Table 7 have a 5-10-5 gapmer motif. Table 8 illustrates gapmer antisense compounds targeted to SEQ ID NO: 2, having a 5-10-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 8

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2						
Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
395149	5-10-5	4	2274	2293	0	GCGCGGAATCCTGGCTGGGA
395150	5-10-5	5	2381	2400	0	GAGGAGACCTAGAGGCCGTG
395151	5-10-5	6	2439	2458	0	AGGACCGCCTGGAGCTGACG
395152	5-10-5	7	2549	2568	0	ACGCAAGGCTAGCACCAGCT
399793	5-10-5	8	2556	2575	0	CCTCGGAACGCAAGGCTAGC
395153	5-10-5	9	2619	2638	0	CCTTGGCGCAGCGGTGGAAG
399837	5-10-5	107	3056	3075	0	CCCACTATAATGGCAAGCCC
399838	5-10-5	80	4306	4325	0	AACCCAGTTCTAATGCACCT
399839	5-10-5	106	5140	5159	0	CCAGTCAGAGTAGAACAGAG
395221	5-10-5	102	5590	5609	0	ATGTGCAGAGATCAATCACA
399840	5-10-5	121	5599	5618	0	GGAGCCTACATGTGCAGAGA
399841	5-10-5	94	5667	5686	0	AGCATGGCACCAGCATCTGC
395154	5-10-5	10	6498	6517	0	GGCGGGCAGTGCGCTCTGAC
395155	5-10-5	11	6537	6556	0	TGGTGAGGTATCCCCGGCGG

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
399794	5-10-5	12	6543	6562	0	GGATCTTGGTGAGGTATCCC
399795	5-10-5	13	6552	6571	0	AGACATGCAGGATCTTGGTG
395156	5-10-5	14	6557	6576	0	ATGGAAGACATGCAGGATCT
395157	5-10-5	15	6583	6602	0	TTCACCAGGAAGCCAGGAAG
399796	5-10-5	16	6588	6607	0	TCATCTTCACCAGGAAGCCA
399842	5-10-5	108	6652	6671	0	CCCAGCCCTATCAGGAAGTG
399843	5-10-5	144	7099	7118	0	TGACATCCAGGAGGGAGGAG
399844	5-10-5	91	7556	7575	0	AGACTGATGGAAGGCATTGA
399845	5-10-5	131	7565	7584	0	GTGTTGAGCAGACTGATGGA
399846	5-10-5	145	8836	8855	0	TGACATCTTGTCTGGGAGCC
399847	5-10-5	90	8948	8967	0	AGACTAGGAGCCTGAGTTTT
399848	5-10-5	125	9099	9118	0	GGCCTGCAGAAGCCAGAGAG
395158	5-10-5	17	9130	9149	0	CCTCGATGTAGTCGACATGG
395159	5-10-5	18	9210	9229	0	GTATTCATCCGCCCGGTACC
399849	5-10-5	148	10252	10271	0	TGGCAGCAACTCAGACATAT
395222	5-10-5	127	10633	10652	0	GGTGGTAATTTGTCACAGCA
395223	5-10-5	84	11308	11327	0	AAGGTCACACAGTTAAGAGT
399850	5-10-5	79	11472	11491	0	AAATGCAGGGCTAAAATCAC
399851	5-10-5	88	12715	12734	0	ACTGGATACATTGGCAGACA
399852	5-10-5	111	12928	12947	0	CTAGAGGAACCACTAGATAT
395201	5-10-5	85	13681	13700	0	ACAAATTCCCAGACTCAGCA
399827	5-10-5	100	13746	13765	0	ATCTCAGGACAGGTGAGCAA
399828	5-10-5	116	13760	13779	0	GAGTAGAGATTCTCATCTCA
395202	5-10-5	129	13816	13835	0	GTGCCATCTGAACAGCACCT
399829	5-10-5	117	13828	13847	0	GAGTCTTCTGAAGTGCCATC
399830	5-10-5	81	13903	13922	0	AAGCAGGGCCTCAGGTGGAA
395203	5-10-5	110	13926	13945	0	CCTGGAACCCCTGCAGCCAG
395204	5-10-5	152	13977	13996	0	TTCAGGCAGGTTGCTGCTAG
399831	5-10-5	83	13986	14005	0	AAGGAAGACTTCAGGCAGGT
395205	5-10-5	140	13998	14017	0	TCAGCCAGGCCAAAGGAAGA
399832	5-10-5	137	14112	14131	0	TAGGGAGAGCTCACAGATGC
395206	5-10-5	136	14122	14141	0	TAGGAGAAAGTAGGGAGAGC
395207	5-10-5	132	14179	14198	0	TAAAAGCTGCAAGAGACTCA
395208	5-10-5	139	14267	14286	0	TCAGAGAAAACAGTCACCGA
399833	5-10-5	92	14397	14416	0	AGAGACAGGAAGCTGCAGCT

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
395209	5-10-5	142	14404	14423	0	TCATTTTAGAGACAGGAAGC
395210	5-10-5	113	14441	14460	0	GAATAACAGTGATGTCTGGC
395211	5-10-5	138	14494	14513	0	TCACAGCTCACCGAGTCTGC
395212	5-10-5	98	14524	14543	0	AGTGTAAAATAAAGCCCCTA
395213	5-10-5	96	14601	14620	0	AGGACCCAAGTCATCCTGCT
395214	5-10-5	124	14631	14650	0	GGCCATCAGCTGGCAATGCT
399834	5-10-5	82	14670	14689	0	AAGGAAAGGGAGGCCTAGAG
395215	5-10-5	133	14675	14694	0	TAGACAAGGAAAGGGAGGCC
395216	5-10-5	103	14681	14700	0	ATTTCATAGACAAGGAAAGG
395217	5-10-5	155	14801	14820	0	CTTATAGTTAACACACAGAA
399835	5-10-5	156	14809	14828	0	AAGTCAACCTTATAGTTAAC
395160	5-10-5	19	14891	14910	0	CTGGTGTCTAGGAGATACAC
399797	5-10-5	20	14896	14915	0	GTATGCTGGTGTCTAGGAGA
395161	5-10-5	21	14916	14935	0	GATTTCCCGGTGGTCACTCT
399798	5-10-5	22	14922	14941	0	GCCCTCGATTTCCCGGTGGT
399799	5-10-5	23	14936	14955	0	GTGACCATGACCCTGCCCTC
395162	5-10-5	24	14946	14965	0	CTCGAAGTCGGTGACCATGA
395163	5-10-5	25	14979	14998	0	GTGGAAGCGGGTCCCGTCCT
395164	5-10-5	26	15264	15283	0	ACCCCTGCCAGGTGGGTGCC
399800	5-10-5	27	15269	15288	0	TGACCACCCCTGCCAGGTGG
395165	5-10-5	28	15298	15317	0	ACCCTTGGCCACGCCGGCAT
395166	5-10-5	29	15334	15353	0	TTGGCAGTTGAGCACGCGCA
399801	5-10-5	30	15339	15358	0	TTCCCTTGGCAGTTGAGCAC
399853	5-10-5	86	15471	15490	0	ACAGCATTCTTGGTTAGGAG
399854	5-10-5	97	16134	16153	0	AGTCAAGCTGCTGCCCAGAG
399855	5-10-5	120	16668	16687	0	GCTAGTTATTAAGCACCTGC
399856	5-10-5	150	17267	17286	0	TGTGAGCTCTGGCCCAGTGG
399857	5-10-5	115	18377	18396	0	GAGTAAGGCAGGTTACTCTC
399858	5-10-5	134	18408	18427	0	TAGATGTGACTAACATTTAA
395224	5-10-5	157	18561	18580	0	AGGAACAAAGCCAAGGTCAC
395168	5-10-5	32	18593	18612	0	GCTGGCTTTTCCGAATAAAC
395169	5-10-5	33	18705	18724	0	GTGACCAGCACGACCCCAGC
399859	5-10-5	105	19203	19222	0	CACATTAGCCTTGCTCAAGT
399860	5-10-5	151	19913	19932	0	TGTGATGACCTGGAAAGGTG
395170	5-10-5	149	19933	19952	0	TGGGCATTGGTGGCCCCAAC

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
395171	5-10-5	34	19962	19981	0	CCAAAGTCCCCAGGGTCACC
395172	5-10-5	128	19966	19985	0	GTCCCCAAAGTCCCCAGGGT
399802	5-10-5	35	19971	19990	0	AGTTGGTCCCCAAAGTCCCC
395173	5-10-5	36	19976	19995	0	GCCAAAGTTGGTCCCCAAAG
399803	5-10-5	37	19988	20007	0	GTCCACACAGCGGCCAAAGT
395174	5-10-5	38	19997	20016	0	GGCAAAGAGGTCCACACAGC
395175	5-10-5	39	20025	20044	0	TGGAGGCACCAATGATGTCC
399804	5-10-5	40	20036	20055	0	GCTGCAGTCGCTGGAGGCAC
399805	5-10-5	41	20047	20066	0	ACAAAGCAGGTGCTGCAGTC
399861	5-10-5	158	20100	20119	0	GTGGTGACTTACCAGCCACG
399862	5-10-5	109	20188	20207	0	CCCCTGCACAGAGCCTGGCA
399863	5-10-5	141	20624	20643	0	TCATGGCTGCAATGCCTGGT
399807	5-10-5	43	20635	20654	0	GGCAGACAGCATCATGGCTG
399808	5-10-5	44	20683	20702	0	GAAGTGGATCAGTCTCTGCC
395177	5-10-5	45	20691	20710	0	TTGGCAGAGAAGTGGATCAG
399809	5-10-5	46	20696	20715	0	CATCTTTGGCAGAGAAGTGG
399810	5-10-5	47	20702	20721	0	TGATGACATCTTTGGCAGAG
399811	5-10-5	48	20709	20728	0	GCCTCATTGATGACATCTTT
399812	5-10-5	49	20721	20740	0	TCAGGGAACCAGGCCTCATT
395178	5-10-5	50	20726	20745	0	GGTCCTCAGGGAACCAGGCC
395179	5-10-5	87	20733	20752	0	ACCCGCTGGTCCTCAGGGAA
399813	5-10-5	51	20740	20759	0	GGTCAGTACCCGCTGGTCCT
395180	5-10-5	119	20762	20781	0	GCAGGGCGGCCACCAGGTTG
399864	5-10-5	93	20995	21014	0	AGAGAGGAGGGCTTAAAGAA
399865	5-10-5	95	21082	21101	0	AGCTGCCAACCTGCAAAAAG
399814	5-10-5	53	21091	21110	0	CTGCAAAACAGCTGCCAACC
395182	5-10-5	54	21121	21140	0	GTAGGCCCCGAGTGTGCTGA
399815	5-10-5	55	21186	21205	0	AGAAACTGGAGCAGCTCAGC
399816	5-10-5	56	21192	21211	0	TCCTGGAGAAACTGGAGCAG
399866	5-10-5	143	21481	21500	0	TGAAAATCCATCCAGCACTG
399867	5-10-5	89	21589	21608	0	AGAACCATGGAGCACCTGAG
399868	5-10-5	123	21696	21715	0	GGCACTGCCCTTCCACCAAA
395183	5-10-5	57	22096	22115	0	CGTTGTGGGCCCGGCAGACC
395184	5-10-5	58	22142	22161	0	AGCAGGCAGCACCTGGCAAT
395185	5-10-5	59	22204	22223	0	CACGGGTCCCCATGCTGGCC

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
395225	5-10-5	126	22292	22311	0	GGTGCATAAGGAGAAAGAGA
395186	5-10-5	60	24095	24114	0	CTTTGCATTCCAGACCTGGG
399817	5-10-5	61	24100	24119	0	CTTGACTTTGCATTCCAGAC
395226	5-10-5	147	24858	24877	0	TGAGTTCATTTAAGAGTGGA
399869	5-10-5	118	24907	24926	0	GCACCATCCAGACCAGAATC
399870	5-10-5	114	25413	25432	0	GAGAGGTTCAGATCCAGGCC
395187	5-10-5	62	26117	26136	0	GGCAGCAGATGGCAACGGCT
395188	5-10-5	154	26217	26236	0	TTTTAAAGCTCAGCCCCAGC
399818	5-10-5	63	26222	26241	0	AACCATTTTAAAGCTCAGCC
395189	5-10-5	64	26311	26330	0	TCAAGGGCCAGGCCAGCAGC
399819	5-10-5	65	26316	26335	0	CCCACTCAAGGGCCAGGCCA
399820	5-10-5	122	26389	26408	0	GGAGGGAGCTTCCTGGCACC
395190	5-10-5	66	26404	26423	0	ATGCCCCACAGTGAGGGAGG
395191	5-10-5	67	26413	26432	0	AATGGTGAAATGCCCCACAG
395192	5-10-5	153	26456	26475	0	TTGGGAGCAGCTGGCAGCAC
395193	5-10-5	68	26557	26576	0	CATGGGAAGAATCCTGCCTC
395194	5-10-5	69	26639	26658	0	GGAGATGAGGGCCATCAGCA
395195	5-10-5	70	26707	26726	0	TAGATGCCATCCAGAAAGCT
399821	5-10-5	71	26713	26732	0	TCTGGCTAGATGCCATCCAG
395196	5-10-5	72	26790	26809	0	GGCATAGAGCAGAGTAAAGG
399822	5-10-5	73	26795	26814	0	AGCCTGGCATAGAGCAGAGT
399823	5-10-5	135	26801	26820	0	TAGCACAGCCTGGCATAGAG
395197	5-10-5	112	27034	27053	0	GAAGAGGCTTGGCTTCAGAG
399824	5-10-5	74	27040	27059	0	AAGTAAGAAGAGGCTTGGCT
395198	5-10-5	75	27244	27263	0	GCTCAAGGAGGGACAGTTGT
395199	5-10-5	76	27279	27298	0	AAAGATAAATGTCTGCTTGC
399825	5-10-5	77	27284	27303	0	ACCCAAAAGATAAATGTCTG
395200	5-10-5	78	27350	27369	0	TCTTCAAGTTACAAAAGCAA
399826	5-10-5	99	27357	27376	0	ATAAATATCTTCAAGTTACA
405861	5-10-5	162	2545	2564	0	AAGGCTAGCACCAGCTCCTC
405862	5-10-5	163	2546	2565	0	CAAGGCTAGCACCAGCTCCT
405863	5-10-5	164	2547	2566	0	GCAAGGCTAGCACCAGCTCC
405864	5-10-5	165	2548	2567	0	CGCAAGGCTAGCACCAGCTC
405865	5-10-5	166	2550	2569	0	AACGCAAGGCTAGCACCAGC
405866	5-10-5	167	2551	2570	0	GAACGCAAGGCTAGCACCAG

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405867	5-10-5	168	2552	2571	0	GGAACGCAAGGCTAGCACCA
405868	5-10-5	169	2553	2572	0	CGGAACGCAAGGCTAGCACC
405869	5-10-5	218	14918	14937	0	TCGATTTCCCGGTGGTCACT
405870	5-10-5	219	14919	14938	0	CTCGATTTCCCGGTGGTCAC
405871	5-10-5	220	14920	14939	0	CCTCGATTTCCCGGTGGTCA
405872	5-10-5	221	14921	14940	0	CCCTCGATTTCCCGGTGGTC
405873	5-10-5	222	14923	14942	0	TGCCCTCGATTTCCCGGTGG
405874	5-10-5	223	14924	14943	0	CTGCCCTCGATTTCCCGGTG
405875	5-10-5	224	14925	14944	0	CCTGCCCTCGATTTCCCGGT
405876	5-10-5	225	14926	14945	0	CCCTGCCCTCGATTTCCCGG
405877	5-10-5	246	15294	15313	0	TTGGCCACGCCGGCATCCCG
405878	5-10-5	247	15295	15314	0	CTTGGCCACGCCGGCATCCC
405879	5-10-5	248	15296	15315	0	CCTTGGCCACGCCGGCATCC
405880	5-10-5	249	15297	15316	0	CCCTTGGCCACGCCGGCATC
405881	5-10-5	250	15299	15318	0	CACCCTTGGCCACGCCGGCA
405882	5-10-5	251	15300	15319	0	GCACCCTTGGCCACGCCGGC
405883	5-10-5	252	15301	15320	0	GGCACCCTTGGCCACGCCGG
405884	5-10-5	253	15302	15321	0	TGGCACCCTTGGCCACGCCG
405885	5-10-5	346	20717	20736	0	GGAACCAGGCCTCATTGATG
405886	5-10-5	347	20718	20737	0	GGGAACCAGGCCTCATTGAT
405887	5-10-5	348	20719	20738	0	AGGGAACCAGGCCTCATTGA
405888	5-10-5	349	20720	20739	0	CAGGGAACCAGGCCTCATTG
405889	5-10-5	350	20722	20741	0	CTCAGGGAACCAGGCCTCAT
405890	5-10-5	351	20723	20742	0	CCTCAGGGAACCAGGCCTCA
405891	5-10-5	352	20724	20743	0	TCCTCAGGGAACCAGGCCTC
405892	5-10-5	353	20725	20744	0	GTCCTCAGGGAACCAGGCCT
405893	5-10-5	431	26635	26654	0	ATGAGGGCCATCAGCACCTT
405894	5-10-5	432	26636	26655	0	GATGAGGGCCATCAGCACCT
405895	5-10-5	433	26637	26656	0	AGATGAGGGCCATCAGCACC
405896	5-10-5	434	26638	26657	0	GAGATGAGGGCCATCAGCAC
405897	5-10-5	435	26640	26659	0	TGGAGATGAGGGCCATCAGC
405898	5-10-5	436	26641	26660	0	CTGGAGATGAGGGCCATCAG
405899	5-10-5	437	26642	26661	0	GCTGGAGATGAGGGCCATCA
405900	5-10-5	438	26643	26662	0	AGCTGGAGATGAGGGCCATC
405901	5-10-5	439	26709	26728	0	GCTAGATGCCATCCAGAAAG

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405902	5-10-5	440	26710	26729	0	GGCTAGATGCCATCCAGAAA
405903	5-10-5	441	26711	26730	0	TGGCTAGATGCCATCCAGAA
405904	5-10-5	442	26712	26731	0	CTGGCTAGATGCCATCCAGA
405905	5-10-5	443	26714	26733	0	CTCTGGCTAGATGCCATCCA
405906	5-10-5	444	26715	26734	0	CCTCTGGCTAGATGCCATCC
405907	5-10-5	445	26716	26735	0	GCCTCTGGCTAGATGCCATC
405908	5-10-5	446	26717	26736	0	AGCCTCTGGCTAGATGCCAT
405909	5-10-5	267	18595	18614	0	CAGCTGGCTTTTCCGAATAA
405910	5-10-5	268	18597	18616	0	ACCAGCTGGCTTTTCCGAAT
405911	5-10-5	269	18599	18618	0	GGACCAGCTGGCTTTTCCGA
405912	5-10-5	270	18603	18622	0	GGCTGGACCAGCTGGCTTTT
405913	5-10-5	275	18707	18726	0	CGGTGACCAGCACGACCCCA
405914	5-10-5	276	18709	18728	0	AGCGGTGACCAGCACGACCC
405915	5-10-5	277	18711	18730	0	GCAGCGGTGACCAGCACGAC
405916	5-10-5	278	18713	18732	0	CGGCAGCGGTGACCAGCACG
405917	5-10-5	280	18717	18736	0	TTGCCGGCAGCGGTGACCAG
405918	5-10-5	281	18719	18738	0	AGTTGCCGGCAGCGGTGACC
405919	5-10-5	282	18721	18740	0	GAAGTTGCCGGCAGCGGTGA
405920	5-10-5	283	18723	18742	0	CGGAAGTTGCCGGCAGCGGT
405921	5-10-5	284	18725	18744	0	CCCGGAAGTTGCCGGCAGCG
405922	5-10-5	285	18727	18746	0	GTCCCGGAAGTTGCCGGCAG
405923	5-10-5	288	19931	19950	0	GGCATTGGTGGCCCCAACTG
405924	5-10-5	290	19954	19973	0	CCCAGGGTCACCGGCTGGTC
405925	5-10-5	292	19958	19977	0	AGTCCCCAGGGTCACCGGCT
405926	5-10-5	293	19960	19979	0	AAAGTCCCCAGGGTCACCGG
405927	5-10-5	294	19964	19983	0	CCCCAAAGTCCCCAGGGTCA
405928	5-10-5	295	19969	19988	0	TTGGTCCCCAAAGTCCCCAG
405929	5-10-5	296	19973	19992	0	AAAGTTGGTCCCCAAAGTCC
405930	5-10-5	297	19978	19997	0	CGGCCAAAGTTGGTCCCCAA
405931	5-10-5	298	19980	19999	0	AGCGGCCAAAGTTGGTCCCC
405932	5-10-5	299	19982	20001	0	ACAGCGGCCAAAGTTGGTCC
405933	5-10-5	300	19984	20003	0	ACACAGCGGCCAAAGTTGGT
405934	5-10-5	301	19986	20005	0	CCACACAGCGGCCAAAGTTG
405935	5-10-5	302	19990	20009	0	AGGTCCACACAGCGGCCAAA
405936	5-10-5	304	19994	20013	0	AAAGAGGTCCACACAGCGGC

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405937	5-10-5	306	20023	20042	0	GAGGCACCAATGATGTCCTC
405938	5-10-5	307	20027	20046	0	GCTGGAGGCACCAATGATGT
405939	5-10-5	308	20029	20048	0	TCGCTGGAGGCACCAATGAT
405940	5-10-5	309	20031	20050	0	AGTCGCTGGAGGCACCAATG
405941	5-10-5	310	20033	20052	0	GCAGTCGCTGGAGGCACCAA
405942	5-10-5	311	20038	20057	0	GTGCTGCAGTCGCTGGAGGC
405943	5-10-5	312	20040	20059	0	AGGTGCTGCAGTCGCTGGAG
405944	5-10-5	314	20043	20062	0	AGCAGGTGCTGCAGTCGCTG
405945	5-10-5	315	20045	20064	0	AAAGCAGGTGCTGCAGTCGC
405946	5-10-5	316	20049	20068	0	ACACAAAGCAGGTGCTGCAG
405947	5-10-5	317	20051	20070	0	TGACACAAAGCAGGTGCTGC
405949	5-10-5	320	20629	20648	0	CAGCATCATGGCTGCAATGC
405950	5-10-5	321	20631	20650	0	GACAGCATCATGGCTGCAAT
405951	5-10-5	322	20633	20652	0	CAGACAGCATCATGGCTGCA
405952	5-10-5	323	20637	20656	0	TCGGCAGACAGCATCATGGC
405953	5-10-5	325	20641	20660	0	CGGCTCGGCAGACAGCATCA
405954	5-10-5	326	20643	20662	0	TCCGGCTCGGCAGACAGCAT
405955	5-10-5	328	20670	20689	0	CTCTGCCTCAACTCGGCCAG
405956	5-10-5	329	20672	20691	0	GTCTCTGCCTCAACTCGGCC
405957	5-10-5	330	20674	20693	0	CAGTCTCTGCCTCAACTCGG
405958	5-10-5	331	20676	20695	0	ATCAGTCTCTGCCTCAACTC
405959	5-10-5	332	20678	20697	0	GGATCAGTCTCTGCCTCAAC
405960	5-10-5	333	20680	20699	0	GTGGATCAGTCTCTGCCTCA
405961	5-10-5	334	20682	20701	0	AAGTGGATCAGTCTCTGCCT
405962	5-10-5	335	20685	20704	0	GAGAAGTGGATCAGTCTCTG
405963	5-10-5	337	20689	20708	0	GGCAGAGAAGTGGATCAGTC
405964	5-10-5	338	20693	20712	0	CTTTGGCAGAGAAGTGGATC
405965	5-10-5	339	20698	20717	0	GACATCTTTGGCAGAGAAGT
405966	5-10-5	340	20700	20719	0	ATGACATCTTTGGCAGAGAA
405967	5-10-5	341	20704	20723	0	ATTGATGACATCTTTGGCAG
405968	5-10-5	342	20706	20725	0	TCATTGATGACATCTTTGGC
405969	5-10-5	343	20711	20730	0	AGGCCTCATTGATGACATCT
405970	5-10-5	344	20713	20732	0	CCAGGCCTCATTGATGACAT
405971	5-10-5	355	20728	20747	0	CTGGTCCTCAGGGAACCAGG
405972	5-10-5	357	20730	20749	0	CGCTGGTCCTCAGGGAACCA

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405973	5-10-5	358	20735	20754	0	GTACCCGCTGGTCCTCAGGG
405974	5-10-5	359	20737	20756	0	CAGTACCCGCTGGTCCTCAG
405975	5-10-5	361	21088	21107	0	CAAAACAGCTGCCAACCTGC
405976	5-10-5	362	21093	21112	0	TCCTGCAAAACAGCTGCCAA
405977	5-10-5	363	21095	21114	0	AGTCCTGCAAAACAGCTGCC
405978	5-10-5	365	21118	21137	0	GGCCCCGAGTGTGCTGACCA
405979	5-10-5	366	21123	21142	0	GTGTAGGCCCCGAGTGTGCT
405980	5-10-5	367	21125	21144	0	CCGTGTAGGCCCCGAGTGTG
405981	5-10-5	368	21127	21146	0	ATCCGTGTAGGCCCCGAGTG
405982	5-10-5	370	21131	21150	0	GGCCATCCGTGTAGGCCCCG
405983	5-10-5	371	21133	21152	0	GTGGCCATCCGTGTAGGCCC
405984	5-10-5	372	21181	21200	0	CTGGAGCAGCTCAGCAGCTC
405985	5-10-5	373	21183	21202	0	AACTGGAGCAGCTCAGCAGC
405986	5-10-5	374	21188	21207	0	GGAGAAACTGGAGCAGCTCA
405987	5-10-5	375	21190	21209	0	CTGGAGAAACTGGAGCAGCT
405988	5-10-5	381	22138	22157	0	GGCAGCACCTGGCAATGGCG
405989	5-10-5	383	22140	22159	0	CAGGCAGCACCTGGCAATGG
405990	5-10-5	386	22144	22163	0	GTAGCAGGCAGCACCTGGCA
405991	5-10-5	394	22206	22225	0	GACACGGGTCCCCATGCTGG
405992	5-10-5	396	22208	22227	0	TGGACACGGGTCCCCATGCT
405993	5-10-5	398	22210	22229	0	AGTGGACACGGGTCCCCATG
405994	5-10-5	400	22212	22231	0	GCAGTGGACACGGGTCCCCA
405995	5-10-5	402	22214	22233	0	TGGCAGTGGACACGGGTCCC
405996	5-10-5	412	24097	24116	0	GACTTTGCATTCCAGACCTG
405997	5-10-5	415	24102	24121	0	TCCTTGACTTTGCATTCCAG
405998	5-10-5	426	26120	26139	0	TCCGGCAGCAGATGGCAACG
405999	5-10-5	160	2437	2456	0	GACCGCCTGGAGCTGACGGT
406003	5-10-5	178	6492	6511	0	CAGTGCGCTCTGACTGCGAG
406004	5-10-5	179	6494	6513	0	GGCAGTGCGCTCTGACTGCG
406005	5-10-5	180	6496	6515	0	CGGGCAGTGCGCTCTGACTG
406006	5-10-5	181	6499	6518	0	CGGCGGGCAGTGCGCTCTGA
406007	5-10-5	183	6532	6551	0	AGGTATCCCCGGCGGGCAGC
406008	5-10-5	188	6539	6558	0	CTTGGTGAGGTATCCCCGGC
406009	5-10-5	190	6541	6560	0	ATCTTGGTGAGGTATCCCCG
406010	5-10-5	193	6546	6565	0	GCAGGATCTTGGTGAGGTAT

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
406011	5-10-5	194	6548	6567	0	ATGCAGGATCTTGGTGAGGT
406012	5-10-5	195	6550	6569	0	ACATGCAGGATCTTGGTGAG
406013	5-10-5	196	6554	6573	0	GAAGACATGCAGGATCTTGG
406014	5-10-5	199	6585	6604	0	TCTTCACCAGGAAGCCAGGA
406015	5-10-5	200	6590	6609	0	ACTCATCTTCACCAGGAAGC
406016	5-10-5	201	6592	6611	0	CCACTCATCTTCACCAGGAA
406017	5-10-5	203	6596	6615	0	GTCGCCACTCATCTTCACCA
406018	5-10-5	204	6598	6617	0	AGGTCGCCACTCATCTTCAC
406019	5-10-5	205	6600	6619	0	GCAGGTCGCCACTCATCTTC
406020	5-10-5	206	6602	6621	0	CAGCAGGTCGCCACTCATCT
406021	5-10-5	210	9207	9226	0	TTCATCCGCCCGGTACCGTG
406022	5-10-5	211	9209	9228	0	TATTCATCCGCCCGGTACCG
406023	5-10-5	212	9212	9231	0	TGGTATTCATCCGCCCGGTA
406024	5-10-5	213	9214	9233	0	GCTGGTATTCATCCGCCCGG
406025	5-10-5	216	14888	14907	0	GTGTCTAGGAGATACACCTC
406026	5-10-5	217	14893	14912	0	TGCTGGTGTCTAGGAGATAC
406027	5-10-5	226	14930	14949	0	ATGACCCTGCCCTCGATTTC
406028	5-10-5	227	14932	14951	0	CCATGACCCTGCCCTCGATT
406029	5-10-5	228	14934	14953	0	GACCATGACCCTGCCCTCGA
406030	5-10-5	229	14938	14957	0	CGGTGACCATGACCCTGCCC
406031	5-10-5	230	14940	14959	0	GTCGGTGACCATGACCCTGC
406032	5-10-5	232	14944	14963	0	CGAAGTCGGTGACCATGACC
406033	5-10-5	237	15261	15280	0	CCTGCCAGGTGGGTGCCATG
406034	5-10-5	238	15266	15285	0	CCACCCCTGCCAGGTGGGTG
406035	5-10-5	239	15271	15290	0	GCTGACCACCCCTGCCAGGT
406036	5-10-5	241	15283	15302	0	GGCATCCCGGCCGCTGACCA
406037	5-10-5	242	15286	15305	0	GCCGGCATCCCGGCCGCTGA
406038	5-10-5	245	15293	15312	0	TGGCCACGCCGGCATCCCGG
406039	5-10-5	257	15330	15349	0	CAGTTGAGCACGCGCAGGCT
406040	5-10-5	258	15332	15351	0	GGCAGTTGAGCACGCGCAGG
406041	5-10-5	259	15336	15355	0	CCTTGGCAGTTGAGCACGCG
406042	5-10-5	260	15341	15360	0	CCTTCCCTTGGCAGTTGAGC
406043	5-10-5	261	15345	15364	0	GTGCCCTTCCCTTGGCAGTT
406044	5-10-5	262	15347	15366	0	CCGTGCCCTTCCCTTGGCAG
406045	5-10-5	266	18591	18610	0	TGGCTTTTCCGAATAAACTC

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
406478	5-10-5	448	21692	21711	0	CTGCCCTTCCACCAAAATGC
406479	5-10-5	449	21693	21712	0	ACTGCCCTTCCACCAAAATG
406480	5-10-5	450	21694	21713	0	CACTGCCCTTCCACCAAAAT
406481	5-10-5	451	21695	21714	0	GCACTGCCCTTCCACCAAAA
406482	5-10-5	452	21697	21716	0	GGGCACTGCCCTTCCACCAA
406483	5-10-5	453	21698	21717	0	TGGGCACTGCCCTTCCACCA
406484	5-10-5	454	21699	21718	0	CTGGGCACTGCCCTTCCACC
406485	5-10-5	455	21700	21719	0	GCTGGGCACTGCCCTTCCAC
408653	5-10-5	354	20727	20746	0	TGGTCCTCAGGGAACCAGGC
409126	5-10-5	447	15298	15317	1	ACCCTTGGTCACGCCGGCAT
410529	5-10-5	184	6534	6553	0	TGAGGTATCCCCGGCGGGCA
410530	5-10-5	185	6535	6554	0	GTGAGGTATCCCCGGCGGGC
410531	5-10-5	186	6536	6555	0	GGTGAGGTATCCCCGGCGGG
410532	5-10-5	187	6538	6557	0	TTGGTGAGGTATCCCCGGCG
410533	5-10-5	189	6540	6559	0	TCTTGGTGAGGTATCCCCGG
410534	5-10-5	191	6542	6561	0	GATCTTGGTGAGGTATCCCC
410535	5-10-5	192	6544	6563	0	AGGATCTTGGTGAGGTATCC
410536	5-10-5	243	15291	15310	0	GCCACGCCGGCATCCCGGCC
410537	5-10-5	244	15292	15311	0	GGCCACGCCGGCATCCCGGC
410538	5-10-5	254	15303	15322	0	CTGGCACCCTTGGCCACGCC
410539	5-10-5	255	15304	15323	0	GCTGGCACCCTTGGCCACGC
410540	5-10-5	356	20729	20748	0	GCTGGTCCTCAGGGAACCAG
410541	5-10-5	376	22133	22152	0	CACCTGGCAATGGCGTAGAC
410542	5-10-5	377	22134	22153	0	GCACCTGGCAATGGCGTAGA
410543	5-10-5	378	22135	22154	0	AGCACCTGGCAATGGCGTAG
410544	5-10-5	379	22136	22155	0	CAGCACCTGGCAATGGCGTA
410545	5-10-5	380	22137	22156	0	GCAGCACCTGGCAATGGCGT
410546	5-10-5	382	22139	22158	0	AGGCAGCACCTGGCAATGGC
410547	5-10-5	384	22141	22160	0	GCAGGCAGCACCTGGCAATG
410548	5-10-5	385	22143	22162	0	TAGCAGGCAGCACCTGGCAA
410549	5-10-5	388	22199	22218	0	GTCCCCATGCTGGCCTCAGC
410550	5-10-5	389	22200	22219	0	GGTCCCCATGCTGGCCTCAG
410551	5-10-5	390	22201	22220	0	GGGTCCCCATGCTGGCCTCA
410552	5-10-5	391	22202	22221	0	CGGGTCCCCATGCTGGCCTC
410553	5-10-5	392	22203	22222	0	ACGGGTCCCCATGCTGGCCT

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410554	5-10-5	393	22205	22224	0	ACACGGGTCCCCATGCTGGC
410555	5-10-5	395	22207	22226	0	GGACACGGGTCCCCATGCTG
410556	5-10-5	397	22209	22228	0	GTGGACACGGGTCCCCATGC
410557	5-10-5	399	22211	22230	0	CAGTGGACACGGGTCCCCAT
410558	5-10-5	401	22213	22232	0	GGCAGTGGACACGGGTCCCC
410559	5-10-5	403	22215	22234	0	GTGGCAGTGGACACGGGTCC
410560	5-10-5	404	22216	22235	0	GGTGGCAGTGGACACGGGTC
410561	5-10-5	405	22217	22236	0	TGGTGGCAGTGGACACGGGT
410562	5-10-5	411	24096	24115	0	ACTTTGCATTCCAGACCTGG
410563	5-10-5	413	24098	24117	0	TGACTTTGCATTCCAGACCT
410564	5-10-5	414	24099	24118	0	TTGACTTTGCATTCCAGACC
410565	5-10-5	419	26112	26131	0	CAGATGGCAACGGCTGTCAC
410566	5-10-5	420	26113	26132	0	GCAGATGGCAACGGCTGTCA
410567	5-10-5	421	26114	26133	0	AGCAGATGGCAACGGCTGTC
410568	5-10-5	422	26115	26134	0	CAGCAGATGGCAACGGCTGT
410569	5-10-5	423	26116	26135	0	GCAGCAGATGGCAACGGCTG
410570	5-10-5	424	26118	26137	0	CGGCAGCAGATGGCAACGGC
410571	5-10-5	425	26119	26138	0	CCGGCAGCAGATGGCAACGG
410572	5-10-5	427	26121	26140	0	CTCCGGCAGCAGATGGCAAC
410573	5-10-5	428	26122	26141	0	GCTCCGGCAGCAGATGGCAA
410730	5-10-5	202	6594	6613	0	CGCCACTCATCTTCACCAGG
410731	5-10-5	207	6604	6623	0	TCCAGCAGGTCGCCACTCAT
410732	5-10-5	214	9216	9235	0	GGGCTGGTATTCATCCGCCC
410733	5-10-5	231	14942	14961	0	AAGTCGGTGACCATGACCCT
410734	5-10-5	279	18714	18733	0	CCGGCAGCGGTGACCAGCAC
410735	5-10-5	291	19956	19975	0	TCCCCAGGGTCACCGGCTGG
410736	5-10-5	303	19992	20011	0	AGAGGTCCACACAGCGGCCA
410737	5-10-5	313	20042	20061	0	GCAGGTGCTGCAGTCGCTGG
410738	5-10-5	324	20639	20658	0	GCTCGGCAGACAGCATCATG
410739	5-10-5	336	20687	20706	0	CAGAGAAGTGGATCAGTCTC
410740	5-10-5	345	20715	20734	0	AACCAGGCCTCATTGATGAC
410741	5-10-5	369	21129	21148	0	CCATCCGTGTAGGCCCCGAG
410742	5-10-5	159	2433	2452	0	GCCTGGAGCTGACGGTGCCC
410743	5-10-5	170	2560	2579	0	TCCTCCTCGGAACGCAAGGC
410744	5-10-5	171	2585	2604	0	GTGCTCGGGTGCTTCGGCCA

TABLE 8-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO:2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410745	5-10-5	172	2605	2624	0	TGGAAGGTGGCTGTGGTTCC
410746	5-10-5	176	6444	6463	0	CGTAGGTGCCAGGCAACCTC
410747	5-10-5	177	6482	6501	0	TGACTGCGAGAGGTGGGTCT
410748	5-10-5	182	6528	6547	0	ATCCCCGGCGGGCAGCCTGG
410749	5-10-5	197	6565	6584	0	AGAAGGCCATGGAAGACATG
410750	5-10-5	198	6575	6594	0	GAAGCCAGGAAGAAGGCCAT
410752	5-10-5	209	9149	9168	0	GCAAAGACAGAGGAGTCCTC
410753	5-10-5	215	14877	14896	0	ATACACCTCCACCAGGCTGC
410754	5-10-5	233	14954	14973	0	GGCACATTCTCGAAGTCGGT
410756	5-10-5	235	15254	15273	0	GGTGGGTGCCATGACTGTCA
410757	5-10-5	240	15279	15298	0	TCCCGGCCGCTGACCACCCC
410758	5-10-5	256	15309	15328	0	CGCATGCTGGCACCCTTGGC
410759	5-10-5	263	15358	15377	0	GGTGCCGCTAACCGTGCCCT
410761	5-10-5	271	18614	18633	0	GTGGCCCCACAGGCTGGACC
410762	5-10-5	272	18627	18646	0	AGCAGCACCACCAGTGGCCC
410763	5-10-5	273	18649	18668	0	GCTGTACCCACCCGCCAGGG
410764	5-10-5	274	18695	18714	0	CGACCCCAGCCCTCGCCAGG
410767	5-10-5	289	19941	19960	0	GCTGGTCTTGGGCATTGGTG
410768	5-10-5	305	20016	20035	0	CAATGATGTCCTCCCCTGGG
410769	5-10-5	318	20061	20080	0	TCCCACTCTGTGACACAAAG
410770	5-10-5	327	20657	20676	0	CGGCCAGGGTGAGCTCCGGC
410771	5-10-5	360	20785	20804	0	ACCTGCCCCATGGGTGCTGG
410772	5-10-5	364	21106	21125	0	GCTGACCATACAGTCCTGCA
410773	5-10-5	387	22189	22208	0	TGGCCTCAGCTGGTGGAGCT
410774	5-10-5	406	22220	22239	0	TGTTGGTGGCAGTGGACACG
410776	5-10-5	408	23985	24004	0	GTGCCAAGGTCCTCCACCTC
410777	5-10-5	409	24005	24024	0	TCAGCACAGGCGGCTTGTGG
410778	5-10-5	410	24035	24054	0	CCACGCACTGGTTGGGCTGA
410779	5-10-5	416	24115	24134	0	CGGGATTCCATGCTCCTTGA
410781	5-10-5	418	25994	26013	0	GGAGGGCACTGCAGCCAGTC
410782	5-10-5	429	26132	26151	0	CCAGGTGCCGGCTCCGGCAG
410783	5-10-5	430	26142	26161	0	GAGGCCTGCGCCAGGTGCCG

[0459] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 2. In certain such embodiments, the nucleotide sequences illustrated in Table 7 have a 3-14-3 gap-widened motif. Table 9 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 2, having a 3-14-3 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 9

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
399871	3-14-3	4	2274	2293	0	GCGCGGAATCCTGGCTGGGA
399872	3-14-3	5	2381	2400	0	GAGGAGACCTAGAGGCCGTG
399873	3-14-3	6	2439	2458	0	AGGACCGCCTGGAGCTGACG
399874	3-14-3	7	2549	2568	0	ACGCAAGGCTAGCACCAGCT
399875	3-14-3	9	2619	2638	0	CCTTGGCGCAGCGGTGGAAG
399876	3-14-3	10	6498	6517	0	GGCGGGCAGTGCGCTCTGAC
399877	3-14-3	11	6537	6556	0	TGGTGAGGTATCCCCGGCGG
399878	3-14-3	14	6557	6576	0	ATGGAAGACATGCAGGATCT
399879	3-14-3	15	6583	6602	0	TTCACCAGGAAGCCAGGAAG
399880	3-14-3	17	9130	9149	0	CCTCGATGTAGTCGACATGG
399881	3-14-3	18	9210	9229	0	GTATTCATCCGCCCGGTACC
399882	3-14-3	19	14891	14910	0	CTGGTGTCTAGGAGATACAC
399883	3-14-3	21	14916	14935	0	GATTTCCCGGTGGTCACTCT
399884	3-14-3	24	14946	14965	0	CTCGAAGTCGGTGACCATGA
399885	3-14-3	25	14979	14998	0	GTGGAAGCGGGTCCCGTCCT
399886	3-14-3	26	15264	15283	0	ACCCCTGCCAGGTGGGTGCC
399887	3-14-3	28	15298	15317	0	ACCCTTGGCCACGCCGGCAT
399888	3-14-3	29	15334	15353	0	TTGGCAGTTGAGCACGCGCA
399890	3-14-3	32	18593	18612	0	GCTGGCTTTTCCGAATAAAC
399891	3-14-3	33	18705	18724	0	GTGACCAGCACGACCCCAGC
399892	3-14-3	149	19933	19952	0	TGGGCATTGGTGGCCCCAAC
399893	3-14-3	34	19962	19981	0	CCAAAGTCCCCAGGGTCACC
399894	3-14-3	128	19966	19985	0	GTCCCCAAAGTCCCCAGGGT
399895	3-14-3	36	19976	19995	0	GCCAAAGTTGGTCCCCAAAG
399896	3-14-3	38	19997	20016	0	GGCAAAGAGGTCCACACAGC
399897	3-14-3	39	20025	20044	0	TGGAGGCACCAATGATGTCC
399899	3-14-3	45	20691	20710	0	TTGGCAGAGAAGTGGATCAG
399900	3-14-3	50	20726	20745	0	GGTCCTCAGGGAACCAGGCC
399901	3-14-3	87	20733	20752	0	ACCCGCTGGTCCTCAGGGAA
399902	3-14-3	119	20762	20781	0	GCAGGGCGGCCACCAGGTTG
399904	3-14-3	54	21121	21140	0	GTAGGCCCCGAGTGTGCTGA
399905	3-14-3	57	22096	22115	0	CGTTGTGGGCCCGGCAGACC
399906	3-14-3	58	22142	22161	0	AGCAGGCAGCACCTGGCAAT

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
399907	3-14-3	59	22204	22223	0	CACGGGTCCCCATGCTGGCC
399908	3-14-3	60	24095	24114	0	CTTTGCATTCCAGACCTGGG
399909	3-14-3	62	26117	26136	0	GGCAGCAGATGGCAACGGCT
399910	3-14-3	154	26217	26236	0	TTTTAAAGCTCAGCCCCAGC
399911	3-14-3	64	26311	26330	0	TCAAGGGCCAGGCCAGCAGC
399912	3-14-3	66	26404	26423	0	ATGCCCCACAGTGAGGGAGG
399913	3-14-3	67	26413	26432	0	AATGGTGAAATGCCCCACAG
399914	3-14-3	153	26456	26475	0	TTGGGAGCAGCTGGCAGCAC
399915	3-14-3	68	26557	26576	0	CATGGGAAGAATCCTGCCTC
399916	3-14-3	69	26639	26658	0	GGAGATGAGGGCCATCAGCA
399917	3-14-3	70	26707	26726	0	TAGATGCCATCCAGAAAGCT
399918	3-14-3	72	26790	26809	0	GGCATAGAGCAGAGTAAAGG
399919	3-14-3	112	27034	27053	0	GAAGAGGCTTGGCTTCAGAG
399920	3-14-3	75	27244	27263	0	GCTCAAGGAGGGACAGTTGT
399921	3-14-3	76	27279	27298	0	AAAGATAAATGTCTGCTTGC
399922	3-14-3	78	27350	27369	0	TCTTCAAGTTACAAAAGCAA
399923	3-14-3	85	13681	13700	0	ACAAATTCCCAGACTCAGCA
399924	3-14-3	129	13816	13835	0	GTGCCATCTGAACAGCACCT
399925	3-14-3	110	13926	13945	0	CCTGGAACCCCTGCAGCCAG
399926	3-14-3	152	13977	13996	0	TTCAGGCAGGTTGCTGCTAG
399927	3-14-3	140	13998	14017	0	TCAGCCAGGCCAAAGGAAGA
399928	3-14-3	136	14122	14141	0	TAGGAGAAAGTAGGGAGAGC
399929	3-14-3	132	14179	14198	0	TAAAAGCTGCAAGAGACTCA
399930	3-14-3	139	14267	14286	0	TCAGAGAAAACAGTCACCGA
399931	3-14-3	142	14404	14423	0	TCATTTTAGAGACAGGAAGC
399932	3-14-3	113	14441	14460	0	GAATAACAGTGATGTCTGGC
399933	3-14-3	138	14494	14513	0	TCACAGCTCACCGAGTCTGC
399934	3-14-3	98	14524	14543	0	AGTGTAAAATAAAGCCCCTA
399935	3-14-3	96	14601	14620	0	AGGACCCAAGTCATCCTGCT
399936	3-14-3	124	14631	14650	0	GGCCATCAGCTGGCAATGCT
399937	3-14-3	133	14675	14694	0	TAGACAAGGAAAGGGAGGCC
399938	3-14-3	103	14681	14700	0	ATTTCATAGACAAGGAAAGG
399939	3-14-3	155	14801	14820	0	CTTATAGTTAACACACAGAA
399943	3-14-3	102	5590	5609	0	ATGTGCAGAGATCAATCACA
399944	3-14-3	127	10633	10652	0	GGTGGTAATTTGTCACAGCA

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
399945	3-14-3	84	11308	11327	0	AAGGTCACACAGTTAAGAGT
399946	3-14-3	157	18561	18580	0	AGGAACAAAGCCAAGGTCAC
399947	3-14-3	126	22292	22311	0	GGTGCATAAGGAGAAAGAGA
399948	3-14-3	147	24858	24877	0	TGAGTTCATTTAAGAGTGGA
399949	3-14-3	8	2556	2575	0	CCTCGGAACGCAAGGCTAGC
399950	3-14-3	12	6543	6562	0	GGATCTTGGTGAGGTATCCC
399951	3-14-3	13	6552	6571	0	AGACATGCAGGATCTTGGTG
399952	3-14-3	16	6588	6607	0	TCATCTTCACCAGGAAGCCA
399953	3-14-3	20	14896	14915	0	GTATGCTGGTGTCTAGGAGA
399954	3-14-3	22	14922	14941	0	GCCCTCGATTTCCCGGTGGT
399955	3-14-3	23	14936	14955	0	GTGACCATGACCCTGCCCTC
399956	3-14-3	27	15269	15288	0	TGACCACCCCTGCCAGGTGG
399957	3-14-3	30	15339	15358	0	TTCCCTTGGCAGTTGAGCAC
399958	3-14-3	35	19971	19990	0	AGTTGGTCCCCAAAGTCCCC
399959	3-14-3	37	19988	20007	0	GTCCACACAGCGGCCAAAGT
399960	3-14-3	40	20036	20055	0	GCTGCAGTCGCTGGAGGCAC
399961	3-14-3	41	20047	20066	0	ACAAAGCAGGTGCTGCAGTC
399963	3-14-3	43	20635	20654	0	GGCAGACAGCATCATGGCTG
399964	3-14-3	44	20683	20702	0	GAAGTGGATCAGTCTCTGCC
399965	3-14-3	46	20696	20715	0	CATCTTTGGCAGAGAAGTGG
399966	3-14-3	47	20702	20721	0	TGATGACATCTTTGGCAGAG
399967	3-14-3	48	20709	20728	0	GCCTCATTGATGACATCTTT
399968	3-14-3	49	20721	20740	0	TCAGGGAACCAGGCCTCATT
399969	3-14-3	51	20740	20759	0	GGTCAGTACCCGCTGGTCCT
399970	3-14-3	53	21091	21110	0	CTGCAAAACAGCTGCCAACC
399971	3-14-3	55	21186	21205	0	AGAAACTGGAGCAGCTCAGC
399972	3-14-3	56	21192	21211	0	TCCTGGAGAAACTGGAGCAG
399973	3-14-3	61	24100	24119	0	CTTGACTTTGCATTCCAGAC
399974	3-14-3	63	26222	26241	0	AACCATTTTAAAGCTCAGCC
399975	3-14-3	65	26316	26335	0	CCCACTCAAGGGCCAGGCCA
399976	3-14-3	122	26389	26408	0	GGAGGGAGCTTCCTGGCACC
399977	3-14-3	71	26713	26732	0	TCTGGCTAGATGCCATCCAG
399978	3-14-3	73	26795	26814	0	AGCCTGGCATAGAGCAGAGT
399979	3-14-3	135	26801	26820	0	TAGCACAGCCTGGCATAGAG
399980	3-14-3	74	27040	27059	0	AAGTAAGAAGAGGCTTGGCT

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
399981	3-14-3	77	27284	27303	0	ACCCAAAAGATAAATGTCTG
399982	3-14-3	99	27357	27376	0	ATAAATATCTTCAAGTTACA
399983	3-14-3	100	13746	13765	0	ATCTCAGGACAGGTGAGCAA
399984	3-14-3	116	13760	13779	0	GAGTAGAGATTCTCATCTCA
399985	3-14-3	117	13828	13847	0	GAGTCTTCTGAAGTGCCATC
399986	3-14-3	81	13903	13922	0	AAGCAGGGCCTCAGGTGGAA
399987	3-14-3	83	13986	14005	0	AAGGAAGACTTCAGGCAGGT
399988	3-14-3	137	14112	14131	0	TAGGGAGAGCTCACAGATGC
399989	3-14-3	92	14397	14416	0	AGAGACAGGAAGCTGCAGCT
399990	3-14-3	82	14670	14689	0	AAGGAAAGGGAGGCCTAGAG
399991	3-14-3	156	14809	14828	0	AAGTCAACCTTATAGTTAAC
399993	3-14-3	107	3056	3075	0	CCCACTATAATGGCAAGCCC
399994	3-14-3	80	4306	4325	0	AACCCAGTTCTAATGCACCT
399995	3-14-3	106	5140	5159	0	CCAGTCAGAGTAGAACAGAG
399996	3-14-3	121	5599	5618	0	GGAGCCTACATGTGCAGAGA
399997	3-14-3	94	5667	5686	0	AGCATGGCACCAGCATCTGC
399998	3-14-3	108	6652	6671	0	CCCAGCCCTATCAGGAAGTG
399999	3-14-3	144	7099	7118	0	TGACATCCAGGAGGGAGGAG
400000	3-14-3	91	7556	7575	0	AGACTGATGGAAGGCATTGA
400001	3-14-3	131	7565	7584	0	GTGTTGAGCAGACTGATGGA
400002	3-14-3	145	8836	8855	0	TGACATCTTGTCTGGGAGCC
400003	3-14-3	90	8948	8967	0	AGACTAGGAGCCTGAGTTTT
400004	3-14-3	125	9099	9118	0	GGCCTGCAGAAGCCAGAGAG
400005	3-14-3	148	10252	10271	0	TGGCAGCAACTCAGACATAT
400006	3-14-3	79	11472	11491	0	AAATGCAGGGCTAAAATCAC
400007	3-14-3	88	12715	12734	0	ACTGGATACATTGGCAGACA
400008	3-14-3	111	12928	12947	0	CTAGAGGAACCACTAGATAT
400009	3-14-3	86	15471	15490	0	ACAGCATTCTTGGTTAGGAG
400010	3-14-3	97	16134	16153	0	AGTCAAGCTGCTGCCCAGAG
400011	3-14-3	120	16668	16687	0	GCTAGTTATTAAGCACCTGC
400012	3-14-3	150	17267	17286	0	TGTGAGCTCTGGCCCAGTGG
400013	3-14-3	115	18377	18396	0	GAGTAAGGCAGGTTACTCTC
400014	3-14-3	134	18408	18427	0	TAGATGTGACTAACATTTAA
400015	3-14-3	105	19203	19222	0	CACATTAGCCTTGCTCAAGT
400016	3-14-3	151	19913	19932	0	TGTGATGACCTGGAAAGGTG

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
400017	3-14-3	158	20100	20119	0	GTGGTGACTTACCAGCCACG
400018	3-14-3	109	20188	20207	0	CCCCTGCACAGAGCCTGGCA
400019	3-14-3	141	20624	20643	0	TCATGGCTGCAATGCCTGGT
400020	3-14-3	93	20995	21014	0	AGAGAGGAGGGCTTAAAGAA
400021	3-14-3	95	21082	21101	0	AGCTGCCAACCTGCAAAAAG
400022	3-14-3	143	21481	21500	0	TGAAAATCCATCCAGCACTG
400023	3-14-3	89	21589	21608	0	AGAACCATGGAGCACCTGAG
400024	3-14-3	123	21696	21715	0	GGCACTGCCCTTCCACCAAA
400025	3-14-3	118	24907	24926	0	GCACCATCCAGACCAGAATC
400026	3-14-3	114	25413	25432	0	GAGAGGTTCAGATCCAGGCC
405604	3-14-3	236	15257	15276	0	CCAGGTGGGTGCCATGACTG
405641	3-14-3	373	21183	21202	0	AACTGGAGCAGCTCAGCAGC
410574	3-14-3	184	6534	6553	0	TGAGGTATCCCCGGCGGGCA
410575	3-14-3	185	6535	6554	0	GTGAGGTATCCCCGGCGGGC
410576	3-14-3	186	6536	6555	0	GGTGAGGTATCCCCGGCGGG
410577	3-14-3	187	6538	6557	0	TTGGTGAGGTATCCCCGGCG
410578	3-14-3	188	6539	6558	0	CTTGGTGAGGTATCCCCGGC
410579	3-14-3	189	6540	6559	0	TCTTGGTGAGGTATCCCCGG
410580	3-14-3	190	6541	6560	0	ATCTTGGTGAGGTATCCCCG
410581	3-14-3	191	6542	6561	0	GATCTTGGTGAGGTATCCCC
410582	3-14-3	192	6544	6563	0	AGGATCTTGGTGAGGTATCC
410583	3-14-3	243	15291	15310	0	GCCACGCCGGCATCCCGGCC
410584	3-14-3	244	15292	15311	0	GGCCACGCCGGCATCCCGGC
410585	3-14-3	245	15293	15312	0	TGGCCACGCCGGCATCCCGG
410586	3-14-3	246	15294	15313	0	TTGGCCACGCCGGCATCCCG
410587	3-14-3	247	15295	15314	0	CTTGGCCACGCCGGCATCCC
410588	3-14-3	248	15296	15315	0	CCTTGGCCACGCCGGCATCC
410589	3-14-3	249	15297	15316	0	CCCTTGGCCACGCCGGCATC
410590	3-14-3	250	15299	15318	0	CACCCTTGGCCACGCCGGCA
410591	3-14-3	251	15300	15319	0	GCACCCTTGGCCACGCCGGC
410592	3-14-3	252	15301	15320	0	GGCACCCTTGGCCACGCCGG
410593	3-14-3	253	15302	15321	0	TGGCACCCTTGGCCACGCCG
410594	3-14-3	254	15303	15322	0	CTGGCACCCTTGGCCACGCC
410595	3-14-3	255	15304	15323	0	GCTGGCACCCTTGGCCACGC
410596	3-14-3	348	20719	20738	0	AGGGAACCAGGCCTCATTGA

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410597	3-14-3	349	20720	20739	0	CAGGGAACCAGGCCTCATTG
410598	3-14-3	350	20722	20741	0	CTCAGGGAACCAGGCCTCAT
410599	3-14-3	351	20723	20742	0	CCTCAGGGAACCAGGCCTCA
410600	3-14-3	352	20724	20743	0	TCCTCAGGGAACCAGGCCTC
410601	3-14-3	353	20725	20744	0	GTCCTCAGGGAACCAGGCCT
410602	3-14-3	354	20727	20746	0	TGGTCCTCAGGGAACCAGGC
410603	3-14-3	355	20728	20747	0	CTGGTCCTCAGGGAACCAGG
410604	3-14-3	356	20729	20748	0	GCTGGTCCTCAGGGAACCAG
410605	3-14-3	376	22133	22152	0	CACCTGGCAATGGCGTAGAC
410606	3-14-3	377	22134	22153	0	GCACCTGGCAATGGCGTAGA
410607	3-14-3	378	22135	22154	0	AGCACCTGGCAATGGCGTAG
410608	3-14-3	379	22136	22155	0	CAGCACCTGGCAATGGCGTA
410609	3-14-3	380	22137	22156	0	GCAGCACCTGGCAATGGCGT
410610	3-14-3	381	22138	22157	0	GGCAGCACCTGGCAATGGCG
410611	3-14-3	382	22139	22158	0	AGGCAGCACCTGGCAATGGC
410612	3-14-3	383	22140	22159	0	CAGGCAGCACCTGGCAATGG
410613	3-14-3	384	22141	22160	0	GCAGGCAGCACCTGGCAATG
410614	3-14-3	385	22143	22162	0	TAGCAGGCAGCACCTGGCAA
410615	3-14-3	388	22199	22218	0	GTCCCCATGCTGGCCTCAGC
410616	3-14-3	389	22200	22219	0	GGTCCCCATGCTGGCCTCAG
410617	3-14-3	390	22201	22220	0	GGGTCCCCATGCTGGCCTCA
410618	3-14-3	391	22202	22221	0	CGGGTCCCCATGCTGGCCTC
410619	3-14-3	392	22203	22222	0	ACGGGTCCCCATGCTGGCCT
410620	3-14-3	393	22205	22224	0	ACACGGGTCCCCATGCTGGC
410621	3-14-3	394	22206	22225	0	GACACGGGTCCCCATGCTGG
410622	3-14-3	395	22207	22226	0	GGACACGGGTCCCCATGCTG
410623	3-14-3	396	22208	22227	0	TGGACACGGGTCCCCATGCT
410624	3-14-3	397	22209	22228	0	GTGGACACGGGTCCCCATGC
410625	3-14-3	398	22210	22229	0	AGTGGACACGGGTCCCCATG
410626	3-14-3	399	22211	22230	0	CAGTGGACACGGGTCCCCAT
410627	3-14-3	400	22212	22231	0	GCAGTGGACACGGGTCCCCA
410628	3-14-3	401	22213	22232	0	GGCAGTGGACACGGGTCCCC
410629	3-14-3	402	22214	22233	0	TGGCAGTGGACACGGGTCCC
410630	3-14-3	403	22215	22234	0	GTGGCAGTGGACACGGGTCC
410631	3-14-3	404	22216	22235	0	GGTGGCAGTGGACACGGGTC

TABLE 9-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410632	3-14-3	405	22217	22236	0	TGGTGGCAGTGGACACGGGT
410633	3-14-3	411	24096	24115	0	ACTTTGCATTCCAGACCTGG
410634	3-14-3	412	24097	24116	0	GACTTTGCATTCCAGACCTG
410635	3-14-3	413	24098	24117	0	TGACTTTGCATTCCAGACCT
410636	3-14-3	414	24099	24118	0	TTGACTTTGCATTCCAGACC
410637	3-14-3	419	26112	26131	0	CAGATGGCAACGGCTGTCAC
410638	3-14-3	420	26113	26132	0	GCAGATGGCAACGGCTGTCA
410639	3-14-3	421	26114	26133	0	AGCAGATGGCAACGGCTGTC
410640	3-14-3	422	26115	26134	0	CAGCAGATGGCAACGGCTGT
410641	3-14-3	423	26116	26135	0	GCAGCAGATGGCAACGGCTG
410642	3-14-3	424	26118	26137	0	CGGCAGCAGATGGCAACGGC
410643	3-14-3	425	26119	26138	0	CCGGCAGCAGATGGCAACGG
410644	3-14-3	426	26120	26139	0	TCCGGCAGCAGATGGCAACG
410645	3-14-3	427	26121	26140	0	CTCCGGCAGCAGATGGCAAC
410646	3-14-3	428	26122	26141	0	GCTCCGGCAGCAGATGGCAA

[0460] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 2. In certain such embodiments, the nucleotide sequences illustrated in Table 7 have a 2-13-5 gap-widened motif. Table 10 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 2, having a 2-13-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 10

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410647	2-13-5	184	6534	6553	0	TGAGGTATCCCCGGCGGGCA
410648	2-13-5	185	6535	6554	0	GTGAGGTATCCCCGGCGGGC
410649	2-13-5	186	6536	6555	0	GGTGAGGTATCCCCGGCGGG
410650	2-13-5	11	6537	6556	0	TGGTGAGGTATCCCCGGCGG
410651	2-13-5	187	6538	6557	0	TTGGTGAGGTATCCCCGGCG

TABLE 10-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410652	2-13-5	188	6539	6558	0	CTTGGTGAGGTATCCCCGGC
410653	2-13-5	189	6540	6559	0	TCTTGGTGAGGTATCCCCGG
410654	2-13-5	190	6541	6560	0	ATCTTGGTGAGGTATCCCCG
410655	2-13-5	191	6542	6561	0	GATCTTGGTGAGGTATCCCC
410656	2-13-5	12	6543	6562	0	GGATCTTGGTGAGGTATCCC
410657	2-13-5	192	6544	6563	0	AGGATCTTGGTGAGGTATCC
410658	2-13-5	243	15291	15310	0	GCCACGCCGGCATCCCGGCC
410659	2-13-5	244	15292	15311	0	GGCCACGCCGGCATCCCGGC
410660	2-13-5	245	15293	15312	0	TGGCCACGCCGGCATCCCGG
410661	2-13-5	246	15294	15313	0	TTGGCCACGCCGGCATCCCG
410662	2-13-5	247	15295	15314	0	CTTGGCCACGCCGGCATCCC
410663	2-13-5	248	15296	15315	0	CCTTGGCCACGCCGGCATCC
410664	2-13-5	249	15297	15316	0	CCCTTGGCCACGCCGGCATC
410665	2-13-5	28	15298	15317	0	ACCCTTGGCCACGCCGGCAT
410666	2-13-5	250	15299	15318	0	CACCCTTGGCCACGCCGGCA
410667	2-13-5	251	15300	15319	0	GCACCCTTGGCCACGCCGGC
410668	2-13-5	252	15301	15320	0	GGCACCCTTGGCCACGCCGG
410669	2-13-5	253	15302	15321	0	TGGCACCCTTGGCCACGCCG
410670	2-13-5	254	15303	15322	0	CTGGCACCCTTGGCCACGCC
410671	2-13-5	255	15304	15323	0	GCTGGCACCCTTGGCCACGC
410672	2-13-5	348	20719	20738	0	AGGGAACCAGGCCTCATTGA

TABLE 10-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410673	2-13-5	349	20720	20739	0	CAGGGAACCAGGCCTCATTG
410674	2-13-5	49	20721	20740	0	TCAGGGAACCAGGCCTCATT
410675	2-13-5	350	20722	20741	0	CTCAGGGAACCAGGCCTCAT
410676	2-13-5	351	20723	20742	0	CCTCAGGGAACCAGGCCTCA
410677	2-13-5	352	20724	20743	0	TCCTCAGGGAACCAGGCCTC
410678	2-13-5	353	20725	20744	0	GTCCTCAGGGAACCAGGCCT
410679	2-13-5	50	20726	20745	0	GGTCCTCAGGGAACCAGGCC
410680	2-13-5	354	20727	20746	0	TGGTCCTCAGGGAACCAGGC
410681	2-13-5	355	20728	20747	0	CTGGTCCTCAGGGAACCAGG
410682	2-13-5	356	20729	20748	0	GCTGGTCCTCAGGGAACCAG
410683	2-13-5	376	22133	22152	0	CACCTGGCAATGGCGTAGAC
410684	2-13-5	377	22134	22153	0	GCACCTGGCAATGGCGTAGA
410685	2-13-5	378	22135	22154	0	AGCACCTGGCAATGGCGTAG
410686	2-13-5	379	22136	22155	0	CAGCACCTGGCAATGGCGTA
410687	2-13-5	380	22137	22156	0	GCAGCACCTGGCAATGGCGT
410688	2-13-5	381	22138	22157	0	GGCAGCACCTGGCAATGGCG
410689	2-13-5	382	22139	22158	0	AGGCAGCACCTGGCAATGGC
410690	2-13-5	383	22140	22159	0	CAGGCAGCACCTGGCAATGG
410691	2-13-5	384	22141	22160	0	GCAGGCAGCACCTGGCAATG
410692	2-13-5	58	22142	22161	0	AGCAGGCAGCACCTGGCAAT
410693	2-13-5	385	22143	22162	0	TAGCAGGCAGCACCTGGCAA
410694	2-13-5	388	22199	22218	0	GTCCCCATGCTGGCCTCAGC

TABLE 10-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 2						
Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410695	2-13-5	389	22200	22219	0	GGTCCCCATGCTGGCCTCAG
410696	2-13-5	390	22201	22220	0	GGGTCCCCATGCTGGCCTCA
410697	2-13-5	391	22202	22221	0	CGGGTCCCCATGCTGGCCTC
410698	2-13-5	392	22203	22222	0	ACGGGTCCCCATGCTGGCCT
410699	2-13-5	59	22204	22223	0	CACGGGTCCCCATGCTGGCC
410700	2-13-5	393	22205	22224	0	ACACGGGTCCCCATGCTGGC
410701	2-13-5	394	22206	22225	0	GACACGGGTCCCCATGCTGG
410702	2-13-5	395	22207	22226	0	GGACACGGGTCCCCATGCTG
410703	2-13-5	396	22208	22227	0	TGGACACGGGTCCCCATGCT
410704	2-13-5	397	22209	22228	0	GTGGACACGGGTCCCCATGC
410705	2-13-5	398	22210	22229	0	AGTGGACACGGGTCCCCATG
410706	2-13-5	399	22211	22230	0	CAGTGGACACGGGTCCCCAT
410707	2-13-5	400	22212	22231	0	GCAGTGGACACGGGTCCCCA
410708	2-13-5	401	22213	22232	0	GGCAGTGGACACGGGTCCCC
410709	2-13-5	402	22214	22233	0	TGGCAGTGGACACGGGTCCC
410710	2-13-5	403	22215	22234	0	GTGGCAGTGGACACGGGTCC
410711	2-13-5	404	22216	22235	0	GGTGGCAGTGGACACGGGTC
410712	2-13-5	405	22217	22236	0	TGGTGGCAGTGGACACGGGT
410713	2-13-5	60	24095	24114	0	CTTTGCATTCCAGACCTGGG
410714	2-13-5	411	24096	24115	0	ACTTTGCATTCCAGACCTGG
410715	2-13-5	412	24097	24116	0	GACTTTGCATTCCAGACCTG
410716	2-13-5	413	24098	24117	0	TGACTTTGCATTCCAGACCT

TABLE 10-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
410717	2-13-5	414	24099	24118	0	TTGACTTTGCATTCCAGACC
410718	2-13-5	61	24100	24119	0	CTTGACTTTGCATTCCAGAC
410719	2-13-5	419	26112	26131	0	CAGATGGCAACGGCTGTCAC
410720	2-13-5	420	26113	26132	0	GCAGATGGCAACGGCTGTCA
410721	2-13-5	421	26114	26133	0	AGCAGATGGCAACGGCTGTC
410722	2-13-5	422	26115	26134	0	CAGCAGATGGCAACGGCTGT
410723	2-13-5	423	26116	26135	0	GCAGCAGATGGCAACGGCTG
410724	2-13-5	62	26117	26136	0	GGCAGCAGATGGCAACGGCT
410725	2-13-5	424	26118	26137	0	CGGCAGCAGATGGCAACGGC
410726	2-13-5	425	26119	26138	0	CCGGCAGCAGATGGCAACGG
410727	2-13-5	426	26120	26139	0	TCCGGCAGCAGATGGCAACG
410728	2-13-5	427	26121	26140	0	CTCCGGCAGCAGATGGCAAC
410729	2-13-5	428	26122	26141	0	GCTCCGGCAGCAGATGGCAA

[0461] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 2. In certain such embodiments, the nucleotide sequences illustrated in Table 7 have a 3-13-4 gap-widened motif. Table 11 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 2, having a 3-13-4 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 11

Gapmer antisense compounds having a 3-13-4 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405526	3-13-4	236	15257	15276	0	CCAGGTGGGTGCCATGACTG

TABLE 11-continued

Gapmer antisense compounds having a 3-13-4 motif targeted to SEQ ID NO: 2

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 2	3' Target Site on SEQ ID NO: 2	Mismatches	Sequence (5'-3')
405557	3-13-4	5	20726	20745	0	GGTCCTCAGGGAACCAGGCC
405564	3-13-4	373	21183	21202	0	AACTGGAGCAGCTCAGCAGC

[0462] The following embodiments set forth target regions of PCSK9 nucleic acids. Also illustrated are examples of antisense compounds targeted to the target regions. It is understood that the sequence set forth in each SEQ ID NO is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by Isis Number (Isis No) indicate a combination of nucleobase sequence and motif.

[0463] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 2: 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15315, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376.

[0464] In certain embodiments, a target region is nucleotides 2274-2400 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2274-2400 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 4 or 5. In certain such embodiments, an antisense compound targeted to nucleotides 2274-2400 of SEQ ID NO: 2 is selected from ISIS NOs: 395149, 399871, 395150 or 399872.

[0465] In certain embodiments, a target region is nucleotides 2274-2575 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2274-2575 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 4, 5, 6, 7, 8, 159, 160, 162, 163, 164, 165, 166, 167, 168 or 169. In certain such embodiments, an antisense compound targeted to nucleotides 2274-2575 of SEQ ID NO: 2 is selected from ISIS NOs: 395149, 399871, 395150, 399872, 410742, 405999, 395151, 399873, 405861, 405862, 405863, 405864, 395152, 399874, 405865, 405866, 405867, 405868, 399793, or 399949.

[0466] In certain embodiments, a target region is nucleotides 2433-2570 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2433-2570 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 6, 7, 159, 160, 162, 163, 164, 165, 166, or 167. In certain such embodiments, an antisense compound targeted to nucleotides 2433-2570 of SEQ ID NO: 2 is selected from ISIS NOs: 395151, 395152, 399873, 399874, 405861, 405862, 405863, 405864, 405865, 405866, 405999, or 410742.

[0467] In certain embodiments, a target region is nucleotides 2433-2579 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2433-2579 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 6, 7, 8, 159, 160, 162, 163, 164, 165, 166, 167, 168, 169, or 170. In certain such embodiments, an antisense compound targeted to nucleotides 2433-2579 of SEQ ID NO: 2 is selected from ISIS NOs: 395151, 395152, 399793, 399873, 399874, 399949, 405861, 405862, 405863, 405864, 405865, 405866, 405867, 405868, 405999, 410742, or 410743.

[0468] In certain embodiments, a target region is nucleotides 2549-2575 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2549-2575 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 7, 8, 166, 167, 168 or 169. In certain such embodiments, an antisense compound targeted to nucleotides 2549-2575 of SEQ ID NO: 2 is selected from ISIS NOs: 395152, 399793, 399874, 399949, 405865, 405866, 405867, or 405868.

[0469] In certain embodiments, a target region is nucleotides 2552-2579 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2552-2579 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 8, 168, 169 or 170. In certain such embodiments, an antisense compound targeted to nucleotides 2552-2579 of SEQ ID NO: 2 is selected from ISIS NOs: 399793, 399949, 405867, 405868, or 410743.

[0470] In certain embodiments, a target region is nucleotides 2585-2638 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2585-2638 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 9, 171 or 172. In certain such embodiments, an antisense compound targeted to nucleotides 2585-2638 of SEQ ID NO: 2 is selected from ISIS NOs: 395153, 399875, 410744, or 410745.

[0471] In certain embodiments, a target region is nucleotides 2605-2638 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 2605-2638 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 8, 168 or 169. In certain such embodiments, an antisense compound targeted to nucleotides 2605-2638 of SEQ ID NO: 2 is selected from ISIS NOs: 410745, 395153, or 399875.

[0472] In certain embodiments, a target region is nucleotides 3056-3075 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 3056-3075 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 107. In certain such embodiments, an antisense compound targeted to nucleotides 3056-3075 of SEQ ID NO: 2 is selected from ISIS NOs: 399837 or 399993.

[0473] In certain embodiments, a target region is nucleotides 4150-5159 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 4150-5159 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 106. In certain such embodiments, an antisense compound targeted to nucleotides 4150-5159 of SEQ ID NO: 2 is selected from ISIS NOs: 399839 or 399995.

[0474] In certain embodiments, a target region is nucleotides 4306-4325 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 4306-4325 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 80. In certain such embodiments, an antisense compound targeted to nucleotides 4306-4325 of SEQ ID NO: 2 is selected from ISIS NOs: 399838 or 399994.

[0475] In certain embodiments, a target region is nucleotides 5590-5618 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 5590-5618 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 102 or 121. In certain such embodiments, an antisense compound targeted to nucleotides 5590-5618 of SEQ ID NO: 2 is selected from ISIS NOs: 395221, 399840, 399943, or 399996.

[0476] In certain embodiments, a target region is nucleotides 5667-5686 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 5667-5686 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 94. In certain such embodiments, an antisense compound targeted to nucleotides 5667-5686 of SEQ ID NO: 2 is selected from ISIS NOs: 399841 or 399997.

[0477] In certain embodiments, a target region is nucleotides 6444-6463 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6444-6463 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 176. In certain such embodiments, an antisense compound targeted to nucleotides 6444-6463 of SEQ ID NO: 2 is selected from ISIS NOs: 410746.

[0478] In certain embodiments, a target region is nucleotides 6482-6518 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6482-6518 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 10, 177, 178, 179, 180 or 181. In certain such embodiments, an antisense compound targeted to nucleotides 6482-6518 of SEQ ID NO: 2 is selected from ISIS NOs: 395154, 399876, 406003, 406004, 406005, 406006 or 410747.

[0479] In certain embodiments, a target region is nucleotides 6492-6518 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6492-6518 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 10, 178, 179, 180 or 181. In certain such embodiments, an antisense compound targeted to nucleotides 6492-6518 of SEQ ID NO: 2 is selected from ISIS NOs: 395154, 399876, 406003, 406004, 406005 or 406006.

[0480] In certain embodiments, a target region is nucleotides 6528-6555 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6528-6555 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 182, 183, 184, 185 or 186. In certain such embodiments, an antisense compound targeted to nucleotides 6528-6555 of SEQ ID NO: 2 is selected from ISIS NOs: 406007, 410529, 410530, 410531, 410574, 410575, 410576, 410647, 410648, 410649 or 410748.

[0481] In certain embodiments, a target region is nucleotides 6528-6623 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6528-6623 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 13, 13, 14, 15, 16, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, or 207. In certain such embodiments, an antisense compound targeted to nucleotides 6528-6623 of SEQ ID NO: 2 is selected from ISIS NOs: 395155, 395156, 395157, 399794, 399795, 399796, 399877, 399878, 399879, 399950, 399951, 399952, 406007, 406008, 406009, 406010, 406011, 406012, 406013, 406014, 406015, 406016, 406017, 406018, 406019, 406020, 410529, 410530, 410531, 410532, 410533, 410534, 410535, 410574, 410575, 410576, 410577, 410578, 410579, 410580, 410581, 410582, 410647, 410648, 410649, 410650, 410651, 410652, 410653, 410654, 410655, 410656, 410657, 410730, 410731, 410748, 410749, or 410750.

[0482] In certain embodiments, a target region is nucleotides 6534-6561 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6534-6561 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 184, 185, 186, 187, 188, 189, 190 or 191. In certain such embodiments, an antisense compound targeted to nucleotides 6534-6561 of SEQ ID NO: 2 is selected from ISIS NOs: 395155, 399877, 406008, 406009, 410529, 410530, 410531, 410532, 410533, 410534, 410574, 410575, 410576, 410577, 410578, 410579, 410580, 410581, 410647, 410648, 410649, 410650, 410651, 410652, 410653, 410654, or 410655.

[0483] In certain embodiments, a target region is nucleotides 6535-6562 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6535-6562 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 185, 186, 187, 188, 189, 190 or 191. In certain such embodiments, an antisense compound targeted to nucleotides 6535-6562 of SEQ ID NO: 2 is selected from ISIS NOs: 395155, 399794, 399877, 399950, 406008, 406009, 410530, 410531, 410532, 410533, 410534, 410575, 410576, 410577, 410578, 410579, 410580, 410581, 410648, 410649, 410650, 410651, 410652, 410653, 410654, 410655 or 410656.

[0484] In certain embodiments, a target region is nucleotides 6536-6563 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6536-6563 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 186, 187, 188, 189, 190, 191 or 192. In certain such embodiments, an antisense compound targeted to nucleotides 6536-6563 of SEQ ID NO: 2 is selected from ISIS NOs: 395155, 399794, 399877, 399950, 406008, 406009, 410531, 410532, 410533, 410534, 410535, 410576, 410577, 410578, 410579, 410580, 410581, 410582, 410649, 410650, 410651, 410652, 410653, 410654, 410655, 410656 or 410657.

[0485] In certain embodiments, a target region is nucleotides 6537-6563 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6537-6563 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 11, 12, 187, 188, 189, 190, 191 or 192. In certain such embodiments, an antisense compound targeted to nucleotides 6537-6563 of SEQ ID NO: 2 is selected from ISIS NOs: 395155, 399794, 399877, 399950, 406008, 406009, 410532, 410533, 410534, 410535, 410577, 410578, 410579, 410580, 410581, 410582, 410650, 410651, 410652, 410653, 410654, 410655, 410656 or 410657.

[0486] In certain embodiments, a target region is nucleotides 6538-6565 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6538-6565 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 187, 188, 189, 190, 191, 192 or 193. In certain such embodiments, an antisense compound targeted to nucleotides 6538-6565 of SEQ ID NO: 2 is selected from ISIS NOs: 399794, 399950, 406008, 406009, 406010, 410532, 410533, 410534, 410535, 410577, 410578, 410579, 410580, 410581, 410582, 410651, 410652, 410653, 410654, 410655, 410656 or 410657.

[0487] In certain embodiments, a target region is nucleotides 6539-6565 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6539-6565 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 188, 189, 190, 191, 192 or 193. In certain such embodiments, an antisense compound targeted to nucleotides 6539-6565 of SEQ ID NO: 2 is selected from ISIS NOs: 399794, 399950, 406008, 406009, 406010, 410533, 410534, 410535, 410578, 410579, 410580, 410581, 410582, 410652, 410653, 410654, 410655, 410656 or 410657.

[0488] In certain embodiments, a target region is nucleotides 6540-6567 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6540-6567 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 189, 190, 191, 192, 193 or 194. In certain such embodiments, an antisense compound targeted to nucleotides 6540-6567 of SEQ ID NO: 2 is selected from ISIS NOs: 399794, 399950, 406009, 406010, 406011, 410533, 410534, 410535, 410579, 410580, 410581, 410582, 410653, 410654, 410655, 410656 or 410657.

[0489] In certain embodiments, a target region is nucleotides 6541-6567 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6541-6567 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 190, 191, 192, 193 or 194. In certain such embodiments, an antisense compound targeted to nucleotides 6541-6567 of SEQ ID NO: 2 is selected from ISIS NOs: 399794, 399950, 406009, 406010, 406011, 410534, 410535, 410580, 410581, 410582, 410654, 410655, 410656 or 410657.

[0490] In certain embodiments, a target region is nucleotides 6542-6569 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6542-6569 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 12, 191, 192, 193, 194 or 195. In certain such embodiments, an antisense compound targeted to nucleotides 6542-6569 of SEQ ID NO: 2 is selected from ISIS NOs: 399794, 399950, 406010, 406011, 406012, 410534, 410535, 410581, 410582, 410655, 410656 or 410657.

[0491] In certain embodiments, a target region is nucleotides 6546-6573 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6546-6573 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 13, 193, 194, 195 or 196. In certain such embodiments, an antisense compound targeted to nucleotides 6546-6573 of SEQ ID NO: 2 is selected from ISIS NOs: 399795, 399951, 406010, 406011, 406012 or 406013.

[0492] In certain embodiments, a target region is nucleotides 6557-6584 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6557-6584 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 14, 197 or 198. In certain such embodiments, an antisense compound targeted to nucleotides 6557-6584 of SEQ ID NO: 2 is selected from ISIS NOs: 395156, 399878, 410749 or 410750.

[0493] In certain embodiments, a target region is nucleotides 6575-6602 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6575-6602 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 15 or 198. In certain such embodiments, an antisense compound targeted to nucleotides 6575-6602 of SEQ ID NO: 2 is selected from ISIS NOs: 395157, 399879 or 410750.

[0494] In certain embodiments, a target region is nucleotides 6585-6611 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6585-6611 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 16, 199, 200 or 201. In certain such embodiments, an antisense compound targeted to nucleotides 6585-6611 of SEQ ID NO: 2 is selected from ISIS NOs: 399796, 399952, 406014, 406015 or 406016.

[0495] In certain embodiments, a target region is nucleotides 6594-6621 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6594-6621 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 202, 203, 204, 205 or 206. In certain such embodiments, an antisense compound targeted to nucleotides 6594-6621 of SEQ ID NO: 2 is selected from ISIS NOs: 406017, 406018, 406019, 406020 or 410730.

[0496] In certain embodiments, a target region is nucleotides 6596-6623 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6596-6623 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 203, 204, 205, 206 or 207. In certain such embodiments, an antisense compound targeted to nucleotides 6596-6623 of SEQ ID NO: 2 is selected from ISIS NOs: 406017, 406018, 406019, 406020 or 410731.

[0497] In certain embodiments, a target region is nucleotides 6652-6671 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 6652-6671 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 108. In certain such embodiments, an antisense compound targeted to nucleotides 6652-6671 of SEQ ID NO: 2 is selected from ISIS NOs: 399842 or 399998.

[0498] In certain embodiments, a target region is nucleotides 7099-7118 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 7099-7118 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 114. In certain such embodiments, an antisense compound targeted to nucleotides 7099-7118 of SEQ ID NO: 2 is selected from ISIS NOs: 399843 or 399999.

[0499] In certain embodiments, a target region is nucleotides 7556-7584 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 7556-7584 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 91 or 131. In certain such embodiments, an antisense compound targeted to nucleotides 7556-7584 of SEQ ID NO: 2 is selected from ISIS NOs: 399844, 399845, 400000 or 400001.

[0500] In certain embodiments, a target region is nucleotides 8836-8855 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 8836-8855 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 145. In certain such embodiments, an antisense compound targeted to nucleotides 8836-8855 of SEQ ID NO: 2 is selected from ISIS NOs: 399846 or 400002.

[0501] In certain embodiments, a target region is nucleotides 8948-8967 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 8948-8967 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 90. In certain such embodiments, an antisense compound targeted to nucleotides 8948-8967 of SEQ ID NO: 2 is selected from ISIS NOs: 399847 or 400003.

[0502] In certain embodiments, a target region is nucleotides 9099-9118 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9099-9118 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 125. In certain such embodiments, an antisense compound targeted to nucleotides 9099-9118 of SEQ ID NO: 2 is selected from ISIS NOs: 399848 or 400004.

[0503] In certain embodiments, a target region is nucleotides 9099-9168 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9099-9168 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 17, 125 or 209. In certain such embodiments, an antisense compound targeted to nucleotides 9099-9168 of SEQ ID NO: 2 is selected from ISIS NOs: 395158, 399848, 399880, 400004 or 410752.

[0504] In certain embodiments, a target region is nucleotides 9130-9168 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9130-9168 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 17. In certain such embodiments, an antisense compound targeted to nucleotides 9130-9168 of SEQ ID NO: 2 is selected from ISIS NOs: 395158 or 399880.

[0505] In certain embodiments, a target region is nucleotides 9207-9233 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9207-9233 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 18, 210, 211, 212 or 213. In certain such embodiments, an antisense compound targeted to nucleotides 9207-9233 of SEQ ID NO: 2 is selected from ISIS NOs: 395159, 399881, 406021, 406022, 406023 or 406024.

[0506] In certain embodiments, a target region is nucleotides 9207-9235 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9207-9235 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 18, 210, 211, 212, 213 or 214. In certain such embodiments, an antisense compound targeted to nucleotides 9207-9235 of SEQ ID NO: 2 is selected from ISIS NOs: 395159, 399881, 406021, 406022, 406023, 406024 or 410732.

[0507] In certain embodiments, a target region is nucleotides 9209-9235 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 9209-9235 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 18, 211, 212, 213 or 214. In certain such embodiments, an antisense compound targeted to nucleotides 9209-9235 of SEQ ID NO: 2 is selected from ISIS NOs: 395159, 399881, 406022, 406023, 406024 or 410732.

[0508] In certain embodiments, a target region is nucleotides 10252-10271 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 10252-10271 of SEQ ID 5 NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 148. In certain such embodiments, an antisense compound targeted to nucleotides 10252-10271 of SEQ ID NO: 2 is selected from ISIS NOs: 399849 or 400005.

[0509] In certain embodiments, a target region is nucleotides 10633-10652 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 10633-10652 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 127. In certain such embodiments, an antisense compound targeted to nucleotides 10633-10652 of SEQ ID NO: 2 is selected from ISIS NOs: 395222 or 399944.

[0510] In certain embodiments, a target region is nucleotides 11308-11491 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 11308-11491 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 79 or 84. In certain such embodiments, an antisense compound targeted to nucleotides 11308-11491 of SEQ ID NO: 2 is selected from ISIS NOs: 395223, 399850, 399945 or 400006.

[0511] In certain embodiments, a target region is nucleotides 12715-12734 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 12715-12734 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 88. In certain such embodiments, an antisense compound targeted to nucleotides 12715-12734 of SEQ ID NO: 2 is selected from ISIS NOs: 399851 or 400007.

[0512] In certain embodiments, a target region is nucleotides 12928-12947 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 12928-12947 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 111. In certain such embodiments, an antisense compound targeted to nucleotides 12928-12947 of SEQ ID NO: 2 is selected from ISIS NOs: 399852 or 400008.

[0513] In certain embodiments, a target region is nucleotides 13681-13700 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 13681-13700 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 85. In certain such embodiments, an antisense compound targeted to nucleotides 13681-13700 of SEQ ID NO: 2 is selected from ISIS NOs: 395201 or 399923.

[0514] In certain embodiments, a target region is nucleotides 13746-13779 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 13746-13779 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 100 or 116. In certain such embodiments, an antisense compound targeted to nucleotides 13746-13779 of SEQ ID NO: 2 is selected from ISIS NOs: 399827, 399828, 399983 or 399984.

[0515] In certain embodiments, a target region is nucleotides 13816-13847 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 13816-13847 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 117 or 129. In certain such embodiments, an antisense compound targeted to nucleotides 13816-13847 of SEQ ID NO: 2 is selected from ISIS NOs: 395202, 399829, 399924 or 399985.

[0516] In certain embodiments, a target region is nucleotides 13903-13945 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 13903-13945 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 81 or 110. In certain such embodiments, an antisense compound targeted to nucleotides 13903-13945 of SEQ ID NO: 2 is selected from ISIS NOs: 395203, 399830, 399925 or 399986.

[0517] In certain embodiments, a target region is nucleotides 13977-14141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 13977-14141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 83, 136, 137, 140 or 152. In certain such embodiments, an antisense compound targeted to nucleotides 13977-14141 of SEQ ID NO: 2 is selected from ISIS NOs: 395204, 395205, 395206, 399831, 399832, 399926, 399927, 399928, 399987 or 399988.

[0518] In certain embodiments, a target region is nucleotides 14179-14198 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14179-14198 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 132. In certain such embodiments, an antisense compound targeted to nucleotides 14179-14198 of SEQ ID NO: 2 is selected from ISIS NOs: 395207 or 399929.

[0519] In certain embodiments, a target region is nucleotides 14267-14286 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14267-14286 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 139. In certain such embodiments, an antisense compound targeted to nucleotides 14267-14286 of SEQ ID NO: 2 is selected from ISIS NOs: 395208 or 399930.

[0520] In certain embodiments, a target region is nucleotides 14397-14423 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14397-14423 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 92 or 142. In certain such embodiments, an antisense compound targeted to nucleotides 14397-14423 of SEQ ID NO: 2 is selected from ISIS NOs: 395209, 399833, 399931 or 399989.

[0521] In certain embodiments, a target region is nucleotides 14441-14460 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14441-14460 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 113. In certain such embodiments, an antisense compound targeted to nucleotides 14441-14460 of SEQ ID NO: 2 is selected from ISIS NOs: 395210 or 399932.

[0522] In certain embodiments, a target region is nucleotides 14494-14513 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14494-14513 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 138. In certain such embodiments, an antisense compound targeted to nucleotides 14494-14513 of SEQ ID NO: 2 is selected from ISIS NOs: 395211 or 399933.

[0523] In certain embodiments, a target region is nucleotides 14494-14543 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14494-14543 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 98 or 138. In certain such embodiments, an antisense compound targeted to nucleotides 14494-14543 of SEQ ID NO: 2 is selected from ISIS NOs: 395211, 395212, 399933 or 399934.

[0524] In certain embodiments, a target region is nucleotides 14524-14543 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14524-14543 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 98. In certain such embodiments, an antisense compound targeted to nucleotides 14524-14543 of SEQ ID NO: 2 is selected from ISIS NOs: 395212 or 399934.

[0525] In certain embodiments, a target region is nucleotides 14601-14650 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14601-14650 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 96 or 124. In certain such embodiments, an antisense compound targeted to nucleotides 14601-14650 of SEQ ID NO: 2 is selected from ISIS NOs: 395213, 395214, 399935 or 399936.

[0526] In certain embodiments, a target region is nucleotides 14670-14700 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14670-14700 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 82, 103 or 133. In certain such embodiments, an antisense compound targeted to nucleotides 14670-14700 of SEQ ID NO: 2 is selected from ISIS NOs: 395215, 395216, 399834, 399937, 399938 or 399990.

[0527] In certain embodiments, a target region is nucleotides 14675-14700 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14675-14700 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 103 or 133. In certain such embodiments, an antisense compound targeted to nucleotides 14675-14700 of SEQ ID NO: 2 is selected from ISIS NOs: 395215, 395216, 399937 or 399938.

[0528] In certain embodiments, a target region is nucleotides 14801-14828 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14801-14828 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 155 or 156. In certain such embodiments, an antisense compound targeted to nucleotides 14801-14828 of SEQ ID NO: 2 is selected from ISIS NOs: 395217, 399835, 399939 or 399991.

[0529] In certain embodiments, a target region is nucleotides 14877-14912 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14877-14912 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 215, 216 or 217. In certain such embodiments, an antisense compound targeted to nucleotides 14877-14912 of SEQ ID NO: 2 is selected from ISIS NOs: 395160, 399882, 406025, 406026 or 410753.

[0530] In certain embodiments, a target region is nucleotides 14877-14915 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14877-14915 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 215, 216 or 217. In certain such embodiments, an antisense compound targeted to nucleotides 14877-14915 of SEQ ID NO: 2 is selected from ISIS NOs: 395160, 399797, 399882, 399953, 406025, 406026 or 410753.

[0531] In certain embodiments, a target region is nucleotides 14877-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14877-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 21, 22, 23, 24, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, or 233. In certain such embodiments, an antisense compound targeted to nucleotides 14877-14973 of SEQ ID NO: 2 is selected from ISIS NOs: 395160, 395161, 395162, 399797, 399798, 399799, 399882, 399883, 399884, 399953, 399954, 399955, 405869, 405870, 405871, 405872, 405873, 405874, 405875, 405876, 406025, 406026, 406027, 406028, 406029, 406030, 406031, 406032, 410733, 410753, or 410754.

[0532] In certain embodiments, a target region is nucleotides 14916-14943 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14916-14943 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 218, 219, 220, 221, 222 or 223. In certain such embodiments, an antisense compound targeted to nucleotides 14916-14943 of SEQ ID NO: 2 is selected from ISIS NOs: 395161, 399798, 399883, 399954, 405869, 405870, 405871, 405872, 405873 or 405874.

[0533] In certain embodiments, a target region is nucleotides 14916-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14916-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 24, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, or 223. In certain such embodiments, an antisense compound targeted to nucleotides 14916-14973 of SEQ ID NO: 2 is selected from ISIS NOs: 395161, 395162, 399798, 399799, 399883, 399884, 399954, 399955, 405869, 405870, 405871, 405872, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 406030, 406031, 406032, 410733, or 410754.

[0534] In certain embodiments, a target region is nucleotides 14925-14951 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14925-14951 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 224, 225, 226 or 227. In certain such embodiments, an antisense compound targeted to nucleotides 14925-14951 of SEQ ID NO: 2 is selected from ISIS NOs: 405875, 405876, 406027 or 406028.

[0535] In certain embodiments, a target region is nucleotides 14934-14963 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14934-14963 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 23, 228, 229, 230, 231 or 232. In certain such embodiments, an antisense compound targeted to nucleotides 14934-14963 of SEQ ID NO: 2 is selected from ISIS NOs: 399799, 399955, 406029, 406030, 406031, 406032 or 410733.

[0536] In certain embodiments, a target region is nucleotides 14946-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14946-14973 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 24 or 233. In certain such embodiments, an antisense compound targeted to nucleotides 14946-14973 of SEQ ID NO: 2 is selected from ISIS NOs: 395162, 399884 or 410754.

[0537] In certain embodiments, a target region is nucleotides 14979-14998 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 14979-14998 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 25. In certain such embodiments, an antisense compound targeted to nucleotides 14979-14998 of SEQ ID NO: 2 is selected from ISIS NOs: 395163 or 399885.

[0538] In certain embodiments, a target region is nucleotides 15254-15280 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15254-15280 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 235, 236 or 237. In certain such embodiments, an antisense compound targeted to nucleotides 15254-15280 of SEQ ID NO: 2 is selected from ISIS NOs: 405526, 405604, 406033 or 410756.

[0539] In certain embodiments, a target region is nucleotides 15254-15328 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15254-15328 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 28, 235, 236, 237, 238, 239, 240, 241, 242, 243, 243, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15254-15328 of SEQ ID NO: 2 is selected from ISIS NOs: 395164, 395165, 399800, 399886, 399887, 399956, 405526, 405604, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 406033, 406034, 406035, 406036, 406037, 406038, 409126, 410536, 410537, 410538, 410539, 410583, 410584, 410585, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410594, 410595, 410658, 410659, 410660, 410661, 410662, 410663, 410664, 410665, 410666, 410667, 410668, 410669, 410670, 410671, 410756, 410757, or 410758.

[0540] In certain embodiments, a target region is nucleotides 15264-15290 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15264-15290 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 27, 238 or 239. In certain such embodiments, an antisense compound targeted to nucleotides 15264-15290 of SEQ ID NO: 2 is selected from ISIS NOs: 395164, 399800, 399886, 399956, 406034, or 406035.

[0541] In certain embodiments, a target region is nucleotides 15279-15305 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15279-15305 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 240, 241 or 242. In certain such embodiments, an antisense compound targeted to nucleotides 15279-15305 of SEQ ID NO: 2 is selected from ISIS NOs: 406036, 406037, or 410757.

[0542] In certain embodiments, a target region is nucleotides 15291-15318 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15291-15318 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 243, 244, 245, 246, 247, 248, 249, 250, or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15291-15318 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 406038, 409126, 410536, 410537, 410583, 410584, 410585, 410586, 410587, 410588, 410589, 410590, 410658, 410659, 410660, 410661, 410662, 410663, 410664, 410665, or 410666.

[0543] In certain embodiments, a target region is nucleotides 15292-15319 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15292-15319 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 244, 245, 246, 247, 248, 249, 250 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15292-15319 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 406038, 409126, 410537, 410584, 410585, 410586, 410587, 410588, 410589, 410590, 410659, 410660, 410661, 410662, 410663, 410664, 410665, or 410666.

[0544] In certain embodiments, a target region is nucleotides 15293-15320 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15293-15320 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 245, 246, 247, 248, 249, 250, 251, 252 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15293-15320 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 406038, 409126, 410585, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410660, 410661, 410662, 410663, 410664, 410665, 410666, 410667, or 410668.

[0545] In certain embodiments, a target region is nucleotides 15294-15321 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15294-15321 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 246, 247, 248, 249, 250, 251, 252, 253 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15294-15321 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410661, 410662, 410663, 410664, 410665, 410666, 410667, 410668 or 410669.

[0546] In certain embodiments, a target region is nucleotides 15294-15321 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15294-15321 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 246, 247, 248, 249, 250, 251, 252, 253 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15294-15321 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410661, 410662, 410663, 410664, 410665, 410666, 410667, 410668 or 410669.

[0547] In certain embodiments, a target region is nucleotides 15295-15322 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15295-15322 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 247, 248, 249, 250, 250, 251, 252, 253, 254 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15295-15322 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410538, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410594, 410662, 410663, 410664, 410665, 410666, 410667, 410668, 410669 or 410670.

[0548] In certain embodiments, a target region is nucleotides 15296-15315 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15296-15315 of SEQ ID NO: 2. In one such embodiment, an antisense compound targeted to nucleotides 15296-15315 of SEQ ID NO: 2 is ISIS NO: 405879.

[0549] In certain embodiments, a target region is nucleotides 15296-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15296-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 248, 249, 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15296-15323 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410538, 410539, 410588, 410589, 410590, 410591, 410592, 410593, 410594, 410595, 410663, 410664, 410665, 410666, 410667, 410668, 410669, 410670 or 410671.

[0550] In certain embodiments, a target region is nucleotides 15297-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15297-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 249, 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15297-15323 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405880, 405881, 405882, 405883, 405884, 409126, 410538, 410539, 410589, 410590, 410591, 410592, 410593, 410594, 410595, 410664, 410665, 410666, 410667, 410668, 410669, 410670 or 410671.

[0551] In certain embodiments, a target region is nucleotides 15298-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15298-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 15298-15323 of SEQ ID NO: 2 is selected from ISIS NOs: 395165, 399887, 405881, 405882, 405883, 405884, 409126, 410538, 410539, 410590, 410591, 410592, 410593, 410594, 410595, 410665, 410666, 410667, 410668, 410669, 410670 or 410671.

[0552] In certain embodiments, a target region is nucleotides 15299-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15299-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 250, 251, 252, 253, 254 or 255. In certain such embodiments, an antisense compound targeted to nucleotides 15299-15323 of SEQ ID NO: 2 is selected from ISIS NOs: 405881, 405882, 405883, 405884, 410538, 410539, 410590, 410591, 410592, 410593, 410594, 410595, 410666, 410667, 410668, 410669, 410670 or 410671.

[0553] In certain embodiments, a target region is nucleotides 15300-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15300-15323 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 251, 252, 253, 254 or 255. In certain such embodiments, an antisense compound targeted to nucleotides 15300-15323 of SEQ ID NO: 2 is selected from ISIS NOs: 405882, 405883, 405884, 410538, 410539, 410591, 410592, 410593, 410594, 410595, 410667, 410668, 410669, 410670 or 410671.

[0554] In certain embodiments, a target region is nucleotides 15301-15328 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15301-15328 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 252, 253, 254, 255 or 256. In certain such embodiments, an antisense compound targeted to nucleotides 15301-15328 of SEQ ID NO: 2 is selected from ISIS NOs: 405883, 405884, 410538, 410539, 410592, 410593, 410594, 410595, 410668, 410669, 410670, 410671 or 410758.

[0555] In certain embodiments, a target region is nucleotides 15330-15355 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15330-15355 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 29, 257, 258 or 259. In certain such embodiments, an antisense compound targeted to nucleotides 15330-15355 of SEQ ID NO: 2 is selected from ISIS NOs: 395166, 399888, 406039, 406040 or 406041.

[0556] In certain embodiments, a target region is nucleotides 15330-15490 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15330-15490 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 29, 30, 86, 257, 258, 259, 260, 261, 262 or 263. In certain such embodiments, an antisense compound targeted to nucleotides 15330-15490 of SEQ ID NO: 2 is selected from ISIS NOs: 395166, 399801, 399853, 399888, 399957, 400009, 406039, 406040, 406041, 406042, 406043, 406044 or 410759.

[0557] In certain embodiments, a target region is nucleotides 15339-15366 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15339-15366 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 30, 260, 261 or 262. In certain such embodiments, an antisense compound targeted to nucleotides 15339-15366 of SEQ ID NO: 2 is selected from ISIS NOs: 399801, 399957, 406042, 406043 or 406044.

[0558] In certain embodiments, a target region is nucleotides 15358-15490 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 15358-15490 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 86 or 263. In certain such embodiments, an antisense compound targeted to nucleotides 15358-15490 of SEQ ID NO: 2 is selected from ISIS NOs: 399853, 400009 or 410759.

[0559] In certain embodiments, a target region is nucleotides 16134-16153 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 16134-16153 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 97. In certain such embodiments, an antisense compound targeted to nucleotides 16134-16153 of SEQ ID NO: 2 is selected from ISIS NOs: 399854 or 400010.

[0560] In certain embodiments, a target region is nucleotides 16668-16687 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 16668-16687 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 120. In certain such embodiments, an antisense compound targeted to nucleotides 16668-16687 of SEQ ID NO: 2 is selected from ISIS NOs: 399855 or 400011.

[0561] In certain embodiments, a target region is nucleotides 17267-17286 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 17267-17286 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 150. In certain such embodiments, an antisense compound targeted to nucleotides 17267-17286 of SEQ ID NO: 2 is selected from ISIS NOs: 399856 or 400012.

[0562] In certain embodiments, a target region is nucleotides 18377-18427 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18377-18427 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 115 or 134. In certain such embodiments, an antisense compound targeted to nucleotides 18377-18427 of SEQ ID NO: 2 is selected from ISIS NOs: 399857, 399858, 400013 or 400014.

[0563] In certain embodiments, a target region is nucleotides 18561-18580 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18561-18580 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 157. In certain such embodiments, an antisense compound targeted to nucleotides 18561-18580 of SEQ ID NO: 2 is selected from ISIS NOs: 395224 or 399946.

[0564] In certain embodiments, a target region is nucleotides 18591-18618 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18591-18618 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 32, 266, 267, 268 or 269. In certain such embodiments, an antisense compound targeted to nucleotides 18591-18618 of SEQ ID NO: 2 is selected from ISIS NOs: 395168, 399890, 405909, 405910, 405911 or 406045.

[0565] In certain embodiments, a target region is nucleotides 18591-18646 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18591-18646 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 32, 32, 266, 267, 268, 269, 270, 271 or 272. In certain such embodiments, an antisense compound targeted to nucleotides 18591-18646 of SEQ ID NO: 2 is selected from ISIS NOs: 395168, 399890, 405909, 405910, 405911, 405912, 406045, 410761 or 410762.

[0566] In certain embodiments, a target region is nucleotides 18591-18668 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18591-18668 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 32, 266, 267, 268, 269, 270, 271, 272 or 273. In certain such embodiments, an antisense compound targeted to nucleotides 18591-18668 of SEQ ID NO: 2 is selected from ISIS NOs: 395168, 399890, 405909, 405910, 405911, 405912, 406045, 410761, 410762 or 410763.

[0567] In certain embodiments, a target region is nucleotides 18695-18746 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18695-18746 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284 or 285. In certain such embodiments, an antisense compound targeted to nucleotides 18695-18746 of SEQ ID NO: 2 is selected from ISIS NOs: 395169, 399891, 405913, 405914, 405915, 405916, 405917, 405918, 405919, 405920, 405921, 405922, 410734 or 410764.

[0568] In certain embodiments, a target region is nucleotides 18705-18730 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18705-18730 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 275, 276 or 277. In certain such embodiments, an antisense compound targeted to nucleotides 18705-18730 of SEQ ID NO: 2 is selected from ISIS NOs: 395169, 399891, 405913, 405914 or 405915.

[0569] In certain embodiments, a target region is nucleotides 18709-18736 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18709-18736 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 276, 277, 278, 279 or 280. In certain such embodiments, an antisense compound targeted to nucleotides 18709-18736 of SEQ ID NO: 2 is selected from ISIS NOs: 405914, 405915, 405916, 405917 or 410734.

[0570] In certain embodiments, a target region is nucleotides 18719-18746 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 18719-18746 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 281, 282, 283, 284 or 285. In certain such embodiments, an antisense compound targeted to nucleotides 18719-18746 of SEQ ID NO: 2 is selected from ISIS NOs: 405918, 405919, 405920, 405921 or 405922.

[0571] In certain embodiments, a target region is nucleotides 19203-20080 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19203-20080 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 34, 35, 36, 37, 38, 39, 40, 41, 105, 128, 149, 151, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317 or 318. In certain such embodiments, an antisense compound targeted to nucleotides 19203-20080 of SEQ ID NO: 2 is selected from ISIS NOs: 395170, 395171, 395172, 395173, 395174, 395175, 399802, 399803, 399804, 399805, 399859, 399860, 399892, 399893, 399894, 399895, 399896, 399897, 399958, 399959, 399960, 399961, 400015, 400016, 405923, 405924, 405925, 405926, 405927, 405928, 405929, 405930, 405931, 405932, 405933, 405934, 405935, 405936, 405937, 405938, 405939, 405940, 405941, 405942, 405943, 405944, 405945, 405946, 405947, 410735, 410736, 410737, 410767, 410768 or 410769.

[0572] In certain embodiments, a target region is nucleotides 19931-19952 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19931-19952 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 149 or 188. In certain such embodiments, an antisense compound targeted to nucleotides 19931-19952 of SEQ ID NO: 2 is selected from ISIS NOs: 395170, 399892 or 405923.

[0573] In certain embodiments, a target region is nucleotides 19954-19981 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19954-19981 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 34, 290, 291, 292 or 293. In certain such embodiments, an antisense compound targeted to nucleotides 19954-19981 of SEQ ID NO: 2 is selected from ISIS NOs: 395171, 399893, 405924, 405925, 405926 or 410735.

[0574] In certain embodiments, a target region is nucleotides 19964-19990 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19964-19990 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 35, 128, 128, 294 or 295. In certain such embodiments, an antisense compound targeted to nucleotides 19964-19990 of SEQ ID NO: 2 is selected from ISIS NOs: 395172, 399802, 399894, 399958, 405927 or 405928.

[0575] In certain embodiments, a target region is nucleotides 19973-19999 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19973-19999 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 36, 296, 297 or 298. In certain such embodiments, an antisense compound targeted to nucleotides 19973-19999 of SEQ ID NO: 2 is selected from ISIS NOs: 395173, 399895, 405929, 405930 or 405931.

[0576] In certain embodiments, a target region is nucleotides 19982-20009 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19982-20009 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 37, 299, 300, 301 or 302. In certain such embodiments, an antisense compound targeted to nucleotides 19982-20009 of SEQ ID NO: 2 is selected from ISIS NOs: 399803, 399959, 405932, 405933, 405934 or 405935.

[0577] In certain embodiments, a target region is nucleotides 19992-20016 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 19992-20016 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 38, 303 or 304. In certain such embodiments, an antisense compound targeted to nucleotides 19992-20016 of SEQ ID NO: 2 is selected from ISIS NOs: 395174, 399896, 405936 or 410736.

[0578] In certain embodiments, a target region is nucleotides 20016-20042 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20016-20042 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 305 or 306. In certain such embodiments, an antisense compound targeted to nucleotides 20016-20042 of SEQ ID NO: 2 is selected from ISIS NOs: 405937 or 410768.

[0579] In certain embodiments, a target region is nucleotides 20025-20052 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20025-20052 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 39, 307, 308, 309 or 310. In certain such embodiments, an antisense compound targeted to nucleotides 20025-20052 of SEQ ID NO: 2 is selected from ISIS NOs: 395175, 399897, 405938, 405939, 405940 or 405941.

[0580] In certain embodiments, a target region is nucleotides 20036-20062 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20036-20062 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 40, 311, 312, 313 or 314. In certain such embodiments, an antisense compound targeted to nucleotides 20036-20062 of SEQ ID NO: 2 is selected from ISIS NOs: 399804, 399960, 405942, 405943, 405944 or 410737.

[0581] In certain embodiments, a target region is nucleotides 20045-20070 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20045-20070 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 41, 315, 316 or 317. In certain such embodiments, an antisense compound targeted to nucleotides 20045-20070 of SEQ ID NO: 2 is selected from ISIS NOs: 399805, 399961, 405945, 405946 or 405947.

[0582] In certain embodiments, a target region is nucleotides 20100-20119 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20100-20119 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 158. In certain such embodiments, an antisense compound targeted to nucleotides 20100-20119 of SEQ ID NO: 2 is selected from ISIS NOs: 399861 or 400017.

[0583] In certain embodiments, a target region is nucleotides 20188-20207 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20188-20207 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 109. In certain such embodiments, an antisense compound targeted to nucleotides 20188-20207 of SEQ ID NO: 2 is selected from ISIS NOs: 399862 or 400018.

[0584] In certain embodiments, a target region is nucleotides 20624-20650 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20624-20650 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 141, 320 or 321. In certain such embodiments, an antisense compound targeted to nucleotides 20624-20650 of SEQ ID NO: 2 is selected from ISIS NOs: 399863, 400019, 405949 or 405950.

[0585] In certain embodiments, a target region is nucleotides 20624-20759 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20624-20759 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 44, 45, 46, 47, 48, 49, 50, 51, 87, 141, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358 or 359. In certain such embodiments, an antisense compound targeted to nucleotides 20624-20759 of SEQ ID NO: 2 is selected from ISIS NOs: 395177, 395178, 395179, 399807, 399808, 399809, 399810, 399811, 399812, 399813, 399863, 399899, 399900, 399901, 399963, 399964, 399965, 399966, 399967, 399968, 399969, 400019, 405557, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 405949, 405950, 405951, 405952, 405953, 405954, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 405962, 405963, 405964, 405965, 405966, 405967, 405968, 405969, 405970, 405971, 405972, 405973, 405974, 408653, 410540, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681, 410682, 410738, 410739, 410740 or 410770.

[0586] In certain embodiments, a target region is nucleotides 20629-20804 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20629-20804 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 44, 45, 46, 47, 48, 49, 50, 51, 87, 119, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 356, 357, 358, 359 or 360. In certain such embodiments, an antisense compound targeted to nucleotides 20629-20804 of SEQ ID NO: 2 is selected from ISIS NOs: 395177, 395178, 395179, 395180, 399807, 399808, 399809, 399810, 399811, 399812, 399813, 399899, 399900, 399901, 399902, 399963, 399964, 399965, 399966, 399967, 399968, 399969, 405557, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 405949, 405950, 405951, 405952, 405953, 405954, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 405962, 405963, 405964, 405965, 405966, 405967, 405968, 405969, 405970, 405971, 405972, 405973, 405974, 408653, 410540, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681, 410682, 410738, 410739, 410740, 410770 or 410771.

[0587] In certain embodiments, a target region is nucleotides 20633-20660 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20633-20660 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 322, 323, 324 or 325. In certain such embodiments, an antisense compound targeted to nucleotides 20633-20660 of SEQ ID NO: 2 is selected from ISIS NOs: 399807, 399963, 405951, 405952, 405953 or 410738.

[0588] In certain embodiments, a target region is nucleotides 20635-20781 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20635-20781 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 44, 45, 46, 47, 48, 49, 50, 51, 87, 119, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358 or 359. In certain such embodiments, an antisense compound targeted to nucleotides 20635-20781 of SEQ ID NO: 2 is selected from ISIS NOs: 395177, 395178, 395179, 395180, 399807, 399808, 399809, 399810, 399811, 399812, 399813, 399899, 399900, 399901, 399902, 399963, 399964, 399965, 399966, 399967, 399968, 399969, 405557, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 405952, 405953, 405954, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 405962, 405963, 405964, 405965, 405966, 405967, 405968, 405969, 405970, 405971, 405972, 405973, 405974, 408653, 410540, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681, 410682, 410738, 410739, 410740 or 410770.

[0589] In certain embodiments, a target region is nucleotides 20643-20662 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20643-20662 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 326. In certain such embodiments, an antisense compound targeted to nucleotides 20643-20662 of SEQ ID NO: 2 is selected from ISIS NOs: 405954.

[0590] In certain embodiments, a target region is nucleotides 20657-20676 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20657-20676 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 327. In certain such embodiments, an antisense compound targeted to nucleotides 20657-20676 of SEQ ID NO: 2 is selected from ISIS NOs: 410770.

[0591] In certain embodiments, a target region is nucleotides 20670-20697 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20670-20697 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 328, 329, 330, 331 or 332. In certain such embodiments, an antisense compound targeted to nucleotides 20670-20697 of SEQ ID NO: 2 is selected from ISIS NOs: 405955, 405956, 405957, 405958 or 405959.

[0592] In certain embodiments, a target region is nucleotides 20680-20706 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20680-20706 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 333, 334, 335 or 336. In certain such embodiments, an antisense compound targeted to nucleotides 20680-20706 of SEQ ID NO: 2 is selected from ISIS NOs: 399808, 399964, 405960, 405961, 405962 or 410739.

[0593] In certain embodiments, a target region is nucleotides 20683-20781 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20683-20781 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 45, 46, 47, 48, 49, 50, 51, 87, 119, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358 or 359. In certain such embodiments, an antisense compound targeted to nucleotides 20683-20781 of SEQ ID NO: 2 is selected from ISIS NOs: 395177, 395178, 395179, 395180, 399808, 399809, 399810, 399811, 399812, 399813, 399899, 399900, 399901, 399902, 399964, 399965, 399966, 399967, 399968, 399969, 405557, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 405962, 405963, 405964, 405965, 405966, 405967, 405968, 405969, 405970, 405971, 405972, 405973, 405974, 408653, 410540, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681, 410682, 410739 or 410740.

[0594] In certain embodiments, a target region is nucleotides 20689-20715 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20689-20715 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 45, 46, 337 or 338. In certain such embodiments, an antisense compound targeted to nucleotides 20689-20715 of SEQ ID NO: 2 is selected from ISIS NOs: 395177, 399809, 399899, 399965, 405963 or 405964.

[0595] In certain embodiments, a target region is nucleotides 20698-20725 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20698-20725 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 47, 47, 339, 340, 341 or 342. In certain such embodiments, an antisense compound targeted to nucleotides 20698-20725 of SEQ ID NO: 2 is selected from ISIS NOs: 399810, 399966, 405965, 405966, 405967 or 405968.

[0596] In certain embodiments, a target region is nucleotides 20709-20736 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20709-20736 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 48, 48, 343, 344, 345 or 346. In certain such embodiments, an antisense compound targeted to nucleotides 20709-20736 of SEQ ID NO: 2 is selected from ISIS NOs: 399811, 399967, 405885, 405969, 405970 or 410740.

[0597] In certain embodiments, a target region is nucleotides 20717-20744 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20717-20744 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 346, 347, 348, 349, 350, 351, 352 or 353. In certain such embodiments, an antisense compound targeted to nucleotides 20717-20744 of SEQ ID NO: 2 is selected from ISIS NOs: 399812, 399968, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 410596, 410597, 410598, 410599, 410600, 410601, 410672, 410673, 410674, 410675, 410676, 410677 or 410678.

[0598] In certain embodiments, a target region is nucleotides 20718-20745 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20718-20745 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 347, 348, 349, 350, 351, 352 or 353. In certain such embodiments, an antisense compound targeted to nucleotides 20718-20745 of SEQ ID NO: 2 is selected from ISIS NOs: 395178, 399812, 399900, 399968, 405557, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 410596, 410597, 410598, 410599, 410600, 410601, 410672, 410673, 410674, 410675, 410676, 410677, 410678 or 410679.

[0599] In certain embodiments, a target region is nucleotides 20719-20746 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20719-20746 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 348, 349, 350, 351, 352, 353 or 354. In certain such embodiments, an antisense compound targeted to nucleotides 20719-20746 of SEQ ID NO: 2 is selected from ISIS NOs: 395178, 399812, 399900, 399968, 405557, 405887, 405888, 405889, 405890, 405891, 405892, 408653, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679 or 410680.

[0600] In certain embodiments, a target region is nucleotides 20720-20747 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20720-20747 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 349, 350, 351, 352, 353, 354 or 355. In certain such embodiments, an antisense compound targeted to nucleotides 20720-20747 of SEQ ID NO: 2 is selected from ISIS NOs: 395178, 399812, 399900, 399968, 405557, 405888, 405889, 405890, 405891, 405892, 405971, 408653, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680 or 410681.

[0601] In certain embodiments, a target region is nucleotides 20721-20748 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20721-20748 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 350, 351, 352, 353, 354, 355 or 356. In certain such embodiments, an antisense compound targeted to nucleotides 20721-20748 of SEQ ID NO: 2 is selected from ISIS NOs: 395178, 399812, 399900, 399968, 405557, 405889, 405890, 405891, 405892, 405971, 408653, 410540, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681 or 410682.

[0602] In certain embodiments, a target region is nucleotides 20722-20749 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20722-20749 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 350, 351, 352, 353, 354, 355, 356 or 357. In certain such embodiments, an antisense compound targeted to nucleotides 20722-20749 of SEQ ID NO: 2 is selected from ISIS NOs: 395178, 399900, 405557, 405889, 405890, 405891, 405892, 405971, 405972, 408653, 410540, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410675, 410676, 410677, 410678, 410679, 410680, 410681 or 410682.

[0603] In certain embodiments, a target region is nucleotides 20727-20752 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20727-20752 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 87, 354, 355, 356 or 357. In certain such embodiments, an antisense compound targeted to nucleotides 20727-20752 of SEQ ID NO: 2 is selected from ISIS NOs: 395179, 399901, 405971, 405972, 408653, 410540, 410602, 410603, 410604, 410680, 410681 or 410682.

[0604] In certain embodiments, a target region is nucleotides 20735-20759 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20735-20759 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 51, 358 or 359. In certain such embodiments, an antisense compound targeted to nucleotides 20735-20759 of SEQ ID NO: 2 is selected from ISIS NOs: 399813, 399969, 405973 or 405974.

[0605] In certain embodiments, a target region is nucleotides 20762-21014 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20762-21014 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 93, 119 or 360. In certain such embodiments, an antisense compound targeted to nucleotides 20762-21014 of SEQ ID NO: 2 is selected from ISIS NOs: 395180, 399864, 399902, 400020 or 410771.

[0606] In certain embodiments, a target region is nucleotides 20785-21014 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 20785-21014 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 93 or 360. In certain such embodiments, an antisense compound targeted to nucleotides 20785-21014 of SEQ ID NO: 2 is selected from ISIS NOs: 399864, 400020 or 410771.

[0607] In certain embodiments, a target region is nucleotides 21082-21107 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21082-21107 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 95 or 361. In certain such embodiments, an antisense compound targeted to nucleotides 21082-21107 of SEQ ID NO: 2 is selected from ISIS NOs: 399865, 400021 or 405975.

[0608] In certain embodiments, a target region is nucleotides 21082-21152 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21082-21152 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 53, 54, 95, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370 or 371. In certain such embodiments, an antisense compound targeted to nucleotides 21082-21152 of SEQ ID NO: 2 is selected from ISIS NOs: 395182, 399814, 399865, 399904, 399970, 400021, 405975, 405976, 405977, 405978, 405979, 405980, 405981, 405982, 405983, 410741 or 410772.

[0609] In certain embodiments, a target region is nucleotides 21091-21114 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21091-21114 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 53, 362 or 363. In certain such embodiments, an antisense compound targeted to nucleotides 21091-21114 of SEQ ID NO: 2 is selected from ISIS NOs: 399814, 399970, 405976 or 405977.

[0610] In certain embodiments, a target region is nucleotides 21118-21144 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21118-21144 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 54, 365, 366 or 367. In certain such embodiments, an antisense compound targeted to nucleotides 21118-21144 of SEQ ID NO: 2 is selected from ISIS NOs: 395182, 399904, 405978, 405979 or 405980.

[0611] In certain embodiments, a target region is nucleotides 21127-21152 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21127-21152 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 368, 369, 370 or 371. In certain such embodiments, an antisense compound targeted to nucleotides 21127-21152 of SEQ ID NO: 2 is selected from ISIS NOs: 405981, 405982, 405983 or 410741.

[0612] In certain embodiments, a target region is nucleotides 21181-21209 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21181-21209 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 372, 373, 374 or 375. In certain such embodiments, an antisense compound targeted to nucleotides 21181-21209 of SEQ ID NO: 2 is selected from ISIS NOs: 399815, 399971, 405564, 405641, 405984, 405985, 405986 or 405987.

[0613] In certain embodiments, a target region is nucleotides 21181-21211 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21181-21211 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 56, 372, 373, 374 or 375. In certain such embodiments, an antisense compound targeted to nucleotides 21181-21211 of SEQ ID NO: 2 is selected from ISIS NOs: 399815, 399816, 399971, 399972, 405564, 405641, 405984, 405985, 405986 or 405987.

[0614] In certain embodiments, a target region is nucleotides 21183-21211 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21183-21211 of SEQ ID 5 NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 56, 373, 374 or 375. In certain such embodiments, an antisense compound targeted to nucleotides 21183-21211 of SEQ ID NO: 2 is selected from ISIS NOs: 399815, 399816, 399971, 399972, 405564, 405641, 405985, 405986 or 405987.

[0615] In certain embodiments, a target region is nucleotides 21481-21500 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21481-21500 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 143. In certain such embodiments, an antisense compound targeted to nucleotides 21481-21500 of SEQ ID NO: 2 is selected from ISIS NOs: 399866 or 400022.

[0616] In certain embodiments, a target region is nucleotides 21589-21608 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 21589-21608 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 89. In certain such embodiments, an antisense compound targeted to nucleotides 21589-21608 of SEQ ID NO: 2 is selected from ISIS NOs: 399867 or 400023.

[0617] In certain embodiments, a target region is nucleotides 21692-21719 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides

21692-21719 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 123, 448, 449, 450, 451, 452, 453, 454 or 455. In certain such embodiments, an antisense compound targeted to nucleotides 21692-21719 of SEQ ID NO: 2 is selected from ISIS NOs: 399868, 400024, 406478, 406479, 406480, 406481, 406482, 406483, 406484 or 406485.

[0618] In certain embodiments, a target region is nucleotides 22000-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22000-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 389, 390, 391, 392, 393, 394, 395 or 396. In certain such embodiments, an antisense compound targeted to nucleotides 22000-22227 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 410550, 410551, 410552, 410553, 410554, 410555, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410695, 410696, 410697, 410698, 410699, 410700, 410701, 410702 or 410703.

[0619] In certain embodiments, a target region is nucleotides 22096-22115 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22096-22115 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 57. In certain such embodiments, an antisense compound targeted to nucleotides 22096-22115 of SEQ ID NO: 2 is selected from ISIS NOs: 395183 or 399905.

[0620] In certain embodiments, a target region is nucleotides 22096-22223 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22096-22223 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 57, 58, 59, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391 or 392. In certain such embodiments, an antisense compound targeted to nucleotides 22096-22223 of SEQ ID NO: 2 is selected from ISIS NOs: 395183, 395184, 395185, 399905, 399906, 399907, 405988, 405989, 405990, 410541, 410542, 410543, 410544, 410545, 410546, 410547, 410548, 410549, 410550, 410551, 410552, 410553, 410605, 410606, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410614, 410615, 410616, 410617, 410618, 410619, 410683, 410684, 410685, 410686, 410687, 410688, 410689, 410690, 410691, 410692, 410693, 410694, 410695, 410696, 410697, 410698, 410699 or 410773.

[0621] In certain embodiments, a target region is nucleotides 22096-22311 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22096-22311 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 57, 58, 59, 126, 126, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 401, 401, 402, 402, 402, 403, 403, 403, 404, 404, 404, 405, 405, 405 or 406. In certain such embodiments, an antisense compound targeted to nucleotides 22096-22311 of SEQ ID NO: 2 is selected from ISIS NOs: 395183, 395184, 395185, 395225, 399905, 399906, 399907, 399947, 405988, 405989, 405990, 405991, 405992, 405993, 405994, 405995, 410541, 410542, 410543, 410544, 410545, 410546, 410547, 410548, 410549, 410550, 410551, 410552, 410553, 410554, 410555, 410556, 410557, 410558, 410559, 410560, 410561, 410605, 410606, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410614, 410615, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410624, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410683, 410684, 410685, 410686, 410687, 410688, 410689, 410690, 410691, 410692, 410693, 410694, 410695, 410696, 410697, 410698, 410699, 410700, 410701, 410702, 410703, 410704, 410705, 410706, 410707, 410708, 410709, 410710, 410711, 410712, 410773 or 410774.

[0622] In certain embodiments, a target region is nucleotides 22133-22160 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22133-22160 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 376, 377, 378, 379, 380, 381, 382, 383 or 384. In certain such embodiments, an antisense compound targeted to nucleotides 22133-22160 of SEQ ID NO: 2 is selected from ISIS NOs: 405988, 405989, 410541, 410542, 410543, 410544, 410545, 410546, 410547, 410605, 410606, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410683, 410684, 410685, 410686, 410687, 410688, 410689, 410690 or 410691.

[0623] In certain embodiments, a target region is nucleotides 22133-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22133-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385 or 386. In certain such embodiments, an antisense compound targeted to nucleotides 22133-22163 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 405990, 410541, 410542, 410543, 410544, 410545, 410546, 410547, 410548, 410605, 410606, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410614, 410683, 410684, 410685, 410686, 410687, 410688, 410689, 410690, 410691, 410692 or 410693.

[0624] In certain embodiments, a target region is nucleotides 22134-22161 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22134-22161 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 377, 378, 379, 380, 381, 382, 383 or 384. In certain such embodiments, an antisense compound targeted to nucleotides 22134-22161 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 410542, 410543, 410544, 410545, 410546, 410547, 410606, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410684, 410685, 410686, 410687, 410688, 410689, 410690, 410691 or 410692.

[0625] In certain embodiments, a target region is nucleotides 22135-22162 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22135-22162 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 378, 379, 380, 381, 382, 383, 384 or 385. In certain such embodiments, an antisense compound targeted to nucleotides 22135-22162 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 410543, 410544, 410545, 410546, 410547, 410548, 410607, 410608, 410609, 410610, 410611, 410612, 410613, 410614, 410685, 410686, 410687, 410688, 410689, 410690, 410691, 410692 or 410693.

[0626] In certain embodiments, a target region is nucleotides 22136-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22136-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 379, 380, 381, 382, 383, 384, 385 or 386. In certain such embodiments, an antisense compound targeted to nucleotides 22136-22163 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 405990, 410544, 410545, 410546, 410547, 410548, 410608, 410609, 410610, 410611, 410612, 410613, 410614, 410686, 410687, 410688, 410689, 410690, 410691, 410692 or 410693.

[0627] In certain embodiments, a target region is nucleotides 22137-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22137-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 380, 381, 382, 383, 384, 385 or 386. In certain such embodiments, an antisense compound targeted to nucleotides 22137-22163 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 405990, 410545, 410546, 410547, 410548, 410609, 410610, 410611, 410612, 410613, 410614, 410687, 410688, 410689, 410690, 410691, 410692 or 410693.

[0628] In certain embodiments, a target region is nucleotides 22138-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22138-22163 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 58, 381, 382, 383, 384, 385 or 386. In certain such embodiments, an antisense compound targeted to nucleotides 22138-22163 of SEQ ID NO: 2 is selected from ISIS NOs: 395184, 399906, 405988, 405989, 405990, 410546, 410547, 410548, 410610, 410611, 410612, 410613, 410614, 410688, 410689, 410690, 410691, 410692 or 410693.

[0629] In certain embodiments, a target region is nucleotides 22189-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22189-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405 or 406. In certain such embodiments, an antisense compound targeted to nucleotides 22189-22239 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 405993, 405994, 405995, 410549, 410550, 410551, 410552, 410553, 410554, 410555, 410556, 410557, 410558, 410559, 410560, 410561, 410615, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410624, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410694, 410695, 410696, 410697, 410698, 410699, 410700, 410701, 410702, 410703, 410704, 410705, 410706, 410707, 410708, 410709, 410710, 410711, 410712, 410773 or 410774.

[0630] In certain embodiments, a target region is nucleotides 22199-22226 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22199-22226 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 388, 389, 390, 391, 392, 393, 394 or 395. In certain such embodiments, an antisense compound targeted to nucleotides 22199-22226 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 410549, 410550, 410551, 410552, 410553, 410554, 410555, 410615, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410694, 410695, 410696, 410697, 410698, 410699, 410700, 410701 or 410702.

[0631] In certain embodiments, a target region is nucleotides 22199-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22199-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 388, 389, 390, 391, 392, 393, 394, 395 or 396. In certain such embodiments, an antisense compound targeted to nucleotides 22199-22227 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 410549, 410550, 410551, 410552, 410553, 410554, 410555, 410615, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410694, 410695, 410696, 410697, 410698, 410699, 410700, 410701, 410702 or 410703.

[0632] In certain embodiments, a target region is nucleotides 22200-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22200-22227 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 389, 390, 391, 392, 393, 394, 395 or 396. In certain such embodiments, an antisense compound targeted to nucleotides 22200-22227 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 410550, 410551, 410552, 410553, 410554, 410555, 410616, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410695, 410696, 410697, 410698, 410699, 410700, 410701, 410702 or 410703.

[0633] In certain embodiments, a target region is nucleotides 22201-22228 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22201-22228 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 390, 391, 392, 393, 394, 395, 396 or 397. In certain such embodiments, an antisense compound targeted to nucleotides 22201-22228 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 410551, 410552, 410553, 410554, 410555, 410556, 410617, 410618, 410619, 410620, 410621, 410622, 410623, 410624, 410696, 410697, 410698, 410699, 410700, 410701, 410702, 410703 or 410704.

[0634] In certain embodiments, a target region is nucleotides 22202-22229 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22202-22229 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 391, 392, 393, 394, 395, 396, 397 or 398. In certain such embodiments, an antisense compound targeted to nucleotides 22202-22229 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 405993, 410552, 410553, 410554, 410555, 410556, 410618, 410619, 410620, 410621, 410622, 410623, 410624, 410625, 410697, 410698, 410699, 410700, 410701, 410702, 410703, 410704 or 410705.

[0635] In certain embodiments, a target region is nucleotides 22203-22230 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22203-22230 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 392, 393, 394, 395, 396, 397, 398 or 399. In certain such embodiments, an antisense compound targeted to nucleotides 22203-22230 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 405993, 410553, 410554, 410555, 410556, 410557, 410619, 410620, 410621, 410622, 410623, 410624, 410625, 410626, 410698, 410699, 410700, 410701, 410702, 410703, 410704, 410705 or 410706.

[0636] In certain embodiments, a target region is nucleotides 22204-22231 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22204-22231 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 59, 393, 394, 395, 396, 397, 398, 399 or 400. In certain such embodiments, an antisense compound targeted to nucleotides 22204-22231 of SEQ ID NO: 2 is selected from ISIS NOs: 395185, 399907, 405991, 405992, 405993, 405994, 410554, 410555, 410556, 410557, 410620, 410621, 410622, 410623, 410624, 410625, 410626, 410627, 410699, 410700, 410701, 410702, 410703, 410704, 410705, 410706 or 410707.

[0637] In certain embodiments, a target region is nucleotides 22205-22232 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22205-22232 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 393, 394, 395, 396, 397, 398, 399, 400 or 401. In certain such embodiments, an antisense compound targeted to nucleotides 22205-22232 of SEQ ID NO: 2 is selected from ISIS NOs: 405991, 405992, 405993, 405994, 410554, 410555, 410556, 410557, 410558, 410620, 410621, 410622, 410623, 410624, 410625, 410626, 410627, 410628, 410700, 410701, 410702, 410703, 410704, 410705, 410706, 410707 or 410708.

[0638] In certain embodiments, a target region is nucleotides 22206-22233 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22206-22233 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 394, 395, 396, 397, 398, 399, 400, 401 or 402. In certain such embodiments, an antisense compound targeted to nucleotides 22206-22233 of SEQ ID NO: 2 is selected from ISIS NOs: 405991, 405992, 405993, 405994, 405995, 410555, 410556, 410557, 410558, 410621, 410622, 410623, 410624, 410625, 410626, 410627, 410628, 410629, 410701, 410702, 410703, 410704, 410705, 410706, 410707, 410708 or 410709.

[0639] In certain embodiments, a target region is nucleotides 22207-22234 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22207-22234 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 395, 396, 397, 398, 399, 400, 401, 402 or 403. In certain such embodiments, an antisense compound targeted to nucleotides 22207-22234 of SEQ ID NO: 2 is selected from ISIS NOs: 405992, 405993, 405994, 405995, 410555, 410556, 410557, 410558, 410559, 410622, 410623, 410624, 410625, 410626, 410627, 410628, 410629, 410630, 410702, 410703, 410704, 410705, 410706, 410707, 410708, 410709 or 410710.

[0640] In certain embodiments, a target region is nucleotides 22208-22235 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22208-22235 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 396, 397, 398, 399, 400, 401, 402, 403 or 404. In certain such embodiments, an antisense compound targeted to nucleotides 22208-22235 of SEQ ID NO: 2 is selected from ISIS NOs: 405992, 405993, 405994, 405995, 410556, 410557, 410558, 410559, 410560, 410623, 410624, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410703, 410704, 410705, 410706, 410707, 410708, 410709, 410710 or 410711.

[0641] In certain embodiments, a target region is nucleotides 22209-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22209-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 397, 398, 399, 400, 401, 402, 403, 404 or 405. In certain such embodiments, an antisense compound targeted to nucleotides 22209-22236 of SEQ ID NO: 2 is selected from ISIS NOs: 405993, 405994, 405995, 410556, 410557, 410558, 410559, 410560, 410561, 410624, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410704, 410705, 410706, 410707, 410708, 410709, 410710, 410711 or 410712.

[0642] In certain embodiments, a target region is nucleotides 22210-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22210-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 398, 399, 400, 401, 402, 403, 404 or 405. In certain such embodiments, an antisense compound targeted to nucleotides 22210-22236 of SEQ ID NO: 2 is selected from ISIS NOs: 405993, 405994, 405995, 410557, 410558, 410559, 410560, 410561, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410705, 410706, 410707, 410708, 410709, 410710, 410711 or 410712.

[0643] In certain embodiments, a target region is nucleotides 22210-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22210-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 398, 399, 400, 401, 402, 403, 404, 405 or 405. In certain such embodiments, an antisense compound targeted to nucleotides 22210-22239 of SEQ ID NO: 2 is selected from ISIS NOs: 405993, 405994, 405995, 410557, 410558, 410559, 410560, 410561, 410625, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410705, 410706, 410707, 410708, 410709, 410710, 410711, 410712 or 410774.

[0644] In certain embodiments, a target region is nucleotides 22211-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22211-22236 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 399, 400, 401, 402, 403, 404 or 405. In certain such embodiments, an antisense compound targeted to nucleotides 22211-22236 of SEQ ID NO: 2 is selected from ISIS NOs: 405994, 405995, 410557, 410558, 410559, 410560, 410561, 410626, 410627, 410628, 410629, 410630, 410631, 410632, 410706, 410707, 410708, 410709, 410710, 410711 or 410712.

[0645] In certain embodiments, a target region is nucleotides 22212-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22212-22239 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 400, 401, 402, 403, 404, 405 or 406. In certain such embodiments, an antisense compound targeted to nucleotides 22212-22239 of SEQ ID NO: 2 is selected from ISIS NOs: 405994, 405995, 410558, 410559, 410560, 410561, 410627, 410628, 410629, 410630, 410631, 410632, 410707, 410708, 410709, 410710, 410711, 410712 or 410774.

[0646] In certain embodiments, a target region is nucleotides 22292-22311 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 22292-22311 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 126. In certain such embodiments, an antisense compound targeted to nucleotides 22292-22311 of SEQ ID NO: 2 is selected from ISIS NOs: 395225 or 399947.

[0647] In certain embodiments, a target region is nucleotides 23985-24054 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 23985-24054 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 408, 409 or 410. In certain such embodiments, an antisense compound targeted to nucleotides 23985-24054 of SEQ ID NO: 2 is selected from ISIS NOs: 410776, 410777 or 410778.

[0648] In certain embodiments, a target region is nucleotides 24035-24134 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 24035-24134 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 410, 411, 412, 413, 414, 415 or 416. In certain such embodiments, an antisense compound targeted to nucleotides 24035-24134 of SEQ ID NO: 2 is selected from ISIS NOs: 395186, 399817, 399908, 399973, 405996, 405997, 410562, 410563, 410564, 410633, 410634, 410635, 410636, 410713, 410714, 410715, 410716, 410717, 410718, 410778 or 410779.

[0649] In certain embodiments, a target region is nucleotides 24095-24121 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 24095-24121 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414 or 415. In certain such embodiments, an antisense compound targeted to nucleotides 24095-24121 of SEQ ID NO: 2 is selected from ISIS NOs: 395186, 399817, 399908, 399973, 405996, 405997, 410562, 410563, 410564, 410633, 410634, 410635, 410636, 410713, 410714, 410715, 410716, 410717 or 410718.

[0650] In certain embodiments, a target region is nucleotides 24858-24877 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 24858-24877 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 147. In certain such embodiments, an antisense compound targeted to nucleotides 24858-24877 of SEQ ID NO: 2 is selected from ISIS NOs: 395226 or 399948.

[0651] In certain embodiments, a target region is nucleotides 24907-24926 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 24907-24926 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 118. In certain such embodiments, an antisense compound targeted to nucleotides 24907-24926 of SEQ ID NO: 2 is selected from ISIS NOs: 399869 or 400025.

[0652] In certain embodiments, a target region is nucleotides 25413-25432 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 25413-25432 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 114. In certain such embodiments, an antisense compound targeted to nucleotides 25413-25432 of SEQ ID NO: 2 is selected from ISIS NOs: 399870 or 400026.

[0653] In certain embodiments, a target region is nucleotides 25994-26013 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 25994-26013 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 418. In certain such embodiments, an antisense compound targeted to nucleotides 25994-26013 of SEQ ID NO: 2 is selected from ISIS NOs: 410781.

[0654] In certain embodiments, a target region is nucleotides 26112-26139 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26112-26139 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425 or 426. In certain such embodiments, an antisense compound targeted to nucleotides 26112-26139 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410565, 410566, 410567, 410568, 410569, 410570, 410571, 410637, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410719, 410720, 410721, 410722, 410723, 410724, 410725, 410726 or 410727.

[0655] In certain embodiments, a target region is nucleotides 26112-26161 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26112-26161 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429 or 430. In certain such embodiments, an antisense compound targeted to nucleotides 26112-26161 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410565, 410566, 410567, 410568, 410569, 410570, 410571, 410572, 410573, 410637, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410719, 410720, 410721, 410722, 410723, 410724, 410725, 410726, 410727, 410728, 410729, 410782 or 410783.

[0656] In certain embodiments, a target region is nucleotides 26112-27303 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26112-27303 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 112, 122, 135, 153, 154, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445 or 446. In certain such embodiments, an antisense compound targeted to nucleotides 26112-27303 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 395188, 395189, 395190, 395191, 395192, 395193, 395194, 395195, 395196, 395197, 395198, 395199, 399818, 399819, 399820, 399821, 399822, 399823, 399824, 399825, 399909, 399910, 399911, 399912, 399913, 399914, 399915, 399916, 399917, 399918, 399919, 399920, 399921, 399974, 399975, 399976, 399977, 399978, 399979, 399980, 399981, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405900, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405908, 405998, 410565, 410566, 410567, 410568, 410569, 410570, 410571, 410572, 410573, 410637, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410719, 410720, 410721, 410722, 410723, 410724, 410725, 410726, 410727, 410728, 410729, 410782 or 410783.

[0657] In certain embodiments, a target region is nucleotides 26113-26140 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26113-26140 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 420, 421, 422, 423, 424, 425, 426 or 427. In certain such embodiments, an antisense compound targeted to nucleotides 26113-26140 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410566, 410567, 410568, 410569, 410570, 410571, 410572, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410720, 410721, 410722, 410723, 410724, 410725, 410726, 410727 or 410728.

[0658] In certain embodiments, a target region is nucleotides 26114-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26114-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 421, 422, 423, 424, 425, 426, 427 or 248. In certain such embodiments, an antisense compound targeted to nucleotides 26114-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410567, 410568, 410569, 410570, 410571, 410572, 410573, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410721, 410722, 410723, 410724, 410725, 410726, 410727, 410728 or 410729.

[0659] In certain embodiments, a target region is nucleotides 26115-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26115-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 422, 423, 424, 425, 426, 427 or 248. In certain such embodiments, an antisense compound targeted to nucleotides 26115-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410568, 410569, 410570, 410571, 410572, 410573, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410722, 410723, 410724, 410725, 410726, 410727, 410728 or 410729.

[0660] In certain embodiments, a target region is nucleotides 26116-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26116-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 423, 424, 425, 426, 427 or 428. In certain such embodiments, an antisense compound targeted to nucleotides 26116-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410569, 410570, 410571, 410572, 410573, 410641, 410642, 410643, 410644, 410645, 410646, 410723, 410724, 410725, 410726, 410727 or 410728.

[0661] In certain embodiments, a target region is nucleotides 26117-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26117-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 424, 425, 426, 427 or 428. In certain such embodiments, an antisense compound targeted to nucleotides 26117-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 399909, 405998, 410570, 410571, 410572, 410573, 410642, 410643, 410644, 410645, 410646, 410724, 410725, 410726, 410727, 410728 or 410729.

[0662] In certain embodiments, a target region is nucleotides 26117-26475 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26117-26475 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 63, 64, 65, 66, 67, 122, 153, 154, 424, 425, 426, 427, 428, 429 or 430. In certain such embodiments, an antisense compound targeted to nucleotides 26117-26475 of SEQ ID NO: 2 is selected from ISIS NOs: 395187, 395188, 395189, 395190, 395191, 395192, 399818, 399819, 399820, 399909, 399910, 399911, 399912, 399913, 399914, 399974, 399975, 399976, 405998, 410570, 410571, 410572, 410573, 410642, 410643, 410644, 410645, 410646, 410724, 410725, 410726, 410727, 410728, 410729, 410782 or 410783.

[0663] In certain embodiments, a target region is nucleotides 26118-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26118-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 424, 425, 426, 427 or 428. In certain such embodiments, an antisense compound targeted to nucleotides 26118-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 405998, 410570, 410571, 410572, 410573, 410642, 410643, 410644, 410645, 410646, 410725, 410726, 410727, 410728 or 410729.

[0664] In certain embodiments, a target region is nucleotides 26120-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26120-26141 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 426, 427 or 428. In certain such embodiments, an antisense compound targeted to nucleotides 26120-26141 of SEQ ID NO: 2 is selected from ISIS NOs: 405998, 410572, 410573, 410644, 410645, 410646, 410727, 410728 or 410729.

[0665] In certain embodiments, a target region is nucleotides 26132-26151 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26132-26151 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 429. In certain such embodiments, an antisense compound targeted to nucleotides 26132-26151 of SEQ ID NO: 2 is selected from ISIS NOs: 410782.

[0666] In certain embodiments, a target region is nucleotides 26142-26161 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26142-26161 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 430. In certain such embodiments, an antisense compound targeted to nucleotides 26142-26161 of SEQ ID NO: 2 is selected from ISIS NOs: 410783.

[0667] In certain embodiments, a target region is nucleotides 26217-26241 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26217-26241 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 63 or 154. In certain such embodiments, an antisense compound targeted to nucleotides 26217-26241 of SEQ ID NO: 2 is selected from ISIS NOs: 395188, 399818 or 399910.

[0668] In certain embodiments, a target region is nucleotides 26311-26335 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26311-26335 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 64 or 65. In certain such embodiments, an antisense compound targeted to nucleotides 26311-26335 of SEQ ID NO: 2 is selected from ISIS NOs: 395189, 399819, 399911 or 399975.

[0669] In certain embodiments, a target region is nucleotides 26389-26432 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26389-26432 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 66, 67 or 122. In certain such embodiments, an antisense compound targeted to nucleotides 26389-26432 of SEQ ID NO: 2 is selected from ISIS NOs: 395190, 395191, 399820, 399912, 399913 or 399976.

[0670] In certain embodiments, a target region is nucleotides 26456-26576 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26456-26576 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 68 or 153. In certain such embodiments, an antisense compound targeted to nucleotides 26456-26576 of SEQ ID NO: 2 is selected from ISIS NOs: 395192, 395193, 399914 or 399915.

[0671] In certain embodiments, a target region is nucleotides 26635-26662 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26635-26662 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 69, 431, 432, 433, 434, 435, 436, 437 or 438. In certain such embodiments, an antisense compound targeted to nucleotides 26635-26662 of SEQ ID NO: 2 is selected from ISIS NOs: 395194, 399916, 405893, 405894, 405895, 405896, 405897, 405898, 405899 or 405900.

[0672] In certain embodiments, a target region is nucleotides 26707-26734 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26707-26734 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 70, 71, 439, 440, 441, 442, 443 or 444. In certain such embodiments, an antisense compound targeted to nucleotides 26707-26734 of SEQ ID NO: 2 is selected from ISIS NOs: 395195, 399821, 399917, 399977, 405901, 405902, 405903, 405904, 405905 or 405906.

[0673] In certain embodiments, a target region is nucleotides 26707-26736 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26707-26736 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 70, 71, 439, 440, 441, 442, 443, 444, 445 or 446. In certain such embodiments, an antisense compound targeted to nucleotides 26707-26736 of SEQ ID NO: 2 is selected from ISIS NOs: 395195, 399821, 399917, 399977, 405901, 405902, 405903, 405904, 405905, 405906, 405907 or 405908.

[0674] In certain embodiments, a target region is nucleotides 26790-26820 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 26790-26820 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 72, 73 or 135. In certain such embodiments, an antisense compound targeted to nucleotides 26790-26820 of SEQ ID NO: 2 is selected from ISIS NOs: 395196, 399822, 399823, 399918, 399978 or 399979.

[0675] In certain embodiments, a target region is nucleotides 27034-27263 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 27034-27263 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 74, 75 or 112. In certain such embodiments, an antisense compound targeted to nucleotides 27034-27263 of SEQ ID NO: 2 is selected from ISIS NOs: 395197, 395198, 399824, 399919, 399920 or 399980.

[0676] In certain embodiments, a target region is nucleotides 27279-27303 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 27279-27303 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 76 or 77. In certain such embodiments, an antisense compound targeted to nucleotides 27279-27303 of SEQ ID NO: 2 is selected from ISIS NOs: 395199, 399825, 399921 or 399981.

[0677] In certain embodiments, a target region is nucleotides 27350-27376 of SEQ ID NO: 2. In certain embodiments, an antisense compound is targeted to nucleotides 27350-27376 of SEQ ID NO: 2. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 78 or 99. In certain such embodiments, an antisense compound targeted to nucleotides 27350-27376 of SEQ ID NO: 2 is selected from ISIS NOs: 395200, 399826, 399922 or 399982.

[0678] In certain embodiments, antisense compounds target a PCSK9 nucleic acid having the sequence of GENBANK® Accession No. AK124635.1, first deposited with GENBANK® on Sep. 8, 2003, and incorporated herein as SEQ ID NO: 3. In certain such embodiments, an antisense oligonucleotide is targeted to SEQ ID NO: 3. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 3 is at least 90% complementary to SEQ ID NO: 1. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 3 is at least 95% complementary to SEQ ID NO: 3. In certain such embodiments, an antisense oligonucleotide that is targeted to SEQ ID NO: 3 is 100% complementary to SEQ ID NO: 3. In certain such embodiments, an antisense oligonucleotide comprises a nucleotide sequence selected from a nucleotide sequence set forth in Table 12.

TABLE 12

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
85	155	174	ACAAATTCCCAGACTCAGCA
100	220	239	ATCTCAGGACAGGTGAGCAA
116	234	253	GAGTAGAGATTCTCATCTCA
129	290	309	GTGCCATCTGAACAGCACCT
117	302	321	GAGTCTTCTGAAGTGCCATC
81	377	396	AAGCAGGGCCTCAGGTGGAA
110	400	419	CCTGGAACCCCTGCAGCCAG
152	451	470	TTCAGGCAGGTTGCTGCTAG
83	460	479	AAGGAAGACTTCAGGCAGGT
140	472	491	TCAGCCAGGCCAAAGGAAGA
137	586	605	TAGGGAGAGCTCACAGATGC
136	596	615	TAGGAGAAAGTAGGGAGAGC
132	653	672	TAAAAGCTGCAAGAGACTCA
139	741	760	TCAGAGAAAACAGTCACCGA
92	871	890	AGAGACAGGAAGCTGCAGCT

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
142	878	897	TCATTTTAGAGACAGGAAGC
113	915	934	GAATAACAGTGATGTCTGGC
138	968	987	TCACAGCTCACCGAGTCTGC
98	998	1017	AGTGTAAAATAAAGCCCCTA
96	1075	1094	AGGACCCAAGTCATCCTGCT
124	1105	1124	GGCCATCAGCTGGCAATGCT
82	1144	1163	AAGGAAAGGGAGGCCTAGAG
133	1149	1168	TAGACAAGGAAAGGGAGGCC
103	1155	1174	ATTTCATAGACAAGGAAAGG
155	1275	1294	CTTATAGTTAACACACAGAA
156	1283	1302	AAGTCAACCTTATAGTTAAC
146	1315	1334	TGACATTTGTGGGAGAGGAG
161	1322	1341	TCCAAGGTGACATTTGTGGG
215	1351	1370	ATACACCTCCACCAGGCTGC
216	1362	1381	GTGTCTAGGAGATACACCTC
19	1365	1384	CTGGTGTCTAGGAGATACAC
217	1367	1386	TGCTGGTGTCTAGGAGATAC
20	1370	1389	GTATGCTGGTGTCTAGGAGA
21	1390	1409	GATTTCCCGGTGGTCACTCT
218	1392	1411	TCGATTTCCCGGTGGTCACT
219	1393	1412	CTCGATTTCCCGGTGGTCAC
220	1394	1413	CCTCGATTTCCCGGTGGTCA
221	1395	1414	CCCTCGATTTCCCGGTGGTC
22	1396	1415	GCCCTCGATTTCCCGGTGGT
222	1397	1416	TGCCCTCGATTTCCCGGTGG
223	1398	1417	CTGCCCTCGATTTCCCGGTG
224	1399	1418	CCTGCCCTCGATTTCCCGGT
225	1400	1419	CCCTGCCCTCGATTTCCCGG
226	1404	1423	ATGACCCTGCCCTCGATTTC
227	1406	1425	CCATGACCCTGCCCTCGATT
228	1408	1427	GACCATGACCCTGCCCTCGA
23	1410	1429	GTGACCATGACCCTGCCCTC
229	1412	1431	CGGTGACCATGACCCTGCCC
230	1414	1433	GTCGGTGACCATGACCCTGC

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
231	1416	1435	AAGTCGGTGACCATGACCCT
232	1418	1437	CGAAGTCGGTGACCATGACC
24	1420	1439	CTCGAAGTCGGTGACCATGA
233	1428	1447	GGCACATTCTCGAAGTCGGT
25	1453	1472	GTGGAAGCGGGTCCCGTCCT
234	1463	1482	TGGCCTGTCTGTGGAAGCGG
235	1490	1509	GGTGGGTGCCATGACTGTCA
236	1493	1512	CCAGGTGGGTGCCATGACTG
237	1497	1516	CCTGCCAGGTGGGTGCCATG
26	1500	1519	ACCCCTGCCAGGTGGGTGCC
238	1502	1521	CCACCCCTGCCAGGTGGGTG
27	1505	1524	TGACCACCCCTGCCAGGTGG
239	1507	1526	GCTGACCACCCCTGCCAGGT
240	1515	1534	TCCCGGCCGCTGACCACCCC
241	1519	1538	GGCATCCCGGCCGCTGACCA
242	1522	1541	GCCGGCATCCCGGCCGCTGA
243	1527	1546	GCCACGCCGGCATCCCGGCC
244	1528	1547	GGCCACGCCGGCATCCCGGC
245	1529	1548	TGGCCACGCCGGCATCCCGG
246	1530	1549	TTGGCCACGCCGGCATCCCG
247	1531	1550	CTTGGCCACGCCGGCATCCC
248	1532	1551	CCTTGGCCACGCCGGCATCC
249	1533	1552	CCCTTGGCCACGCCGGCATC
28	1534	1553	ACCCTTGGCCACGCCGGCAT
447	1534	1553	ACCCTTGGTCACGCCGGCAT
250	1535	1554	CACCCTTGGCCACGCCGGCA
251	1536	1555	GCACCCTTGGCCACGCCGGC
252	1537	1556	GGCACCCTTGGCCACGCCGG
253	1538	1557	TGGCACCCTTGGCCACGCCG
254	1539	1558	CTGGCACCCTTGGCCACGCC
255	1540	1559	GCTGGCACCCTTGGCCACGC
256	1545	1564	CGCATGCTGGCACCCTTGGC
257	1566	1585	CAGTTGAGCACGCGCAGGCT
258	1568	1587	GGCAGTTGAGCACGCGCAGG
29	1570	1589	TTGGCAGTTGAGCACGCGCA

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
259	1572	1591	CCTTGGCAGTTGAGCACGCG
30	1575	1594	TTCCCTTGGCAGTTGAGCAC
260	1577	1596	CCTTCCCTTGGCAGTTGAGC
261	1581	1600	GTGCCCTTCCCTTGGCAGTT
262	1583	1602	CCGTGCCCTTCCCTTGGCAG
263	1594	1613	GGTGCCGCTAACCGTGCCCT
264	1606	1625	CAGGCCTATGAGGGTGCCGC
31	1607	1626	CCAGGCCTATGAGGGTGCCG
458	1609	1622	GCCTATGAGGGTGC
459	1614	1627	TCCAGGCCTATGAG
265	1618	1637	CCGAATAAACTCCAGGCCTA
266	1626	1645	TGGCTTTTCCGAATAAACTC
32	1628	1647	GCTGGCTTTTCCGAATAAAC
267	1630	1649	CAGCTGGCTTTTCCGAATAA
268	1632	1651	ACCAGCTGGCTTTTCCGAAT
269	1634	1653	GGACCAGCTGGCTTTTCCGA
270	1638	1657	GGCTGGACCAGCTGGCTTTT
271	1649	1668	GTGGCCCCACAGGCTGGACC
272	1662	1681	AGCAGCACCACCAGTGGCCC
273	1684	1703	GCTGTACCCACCCGCCAGGG
274	1730	1749	CGACCCCAGCCCTCGCCAGG
33	1740	1759	GTGACCAGCACGACCCCAGC
275	1742	1761	CGGTGACCAGCACGACCCCA
276	1744	1763	AGCGGTGACCAGCACGACCC
277	1746	1765	GCAGCGGTGACCAGCACGAC
278	1748	1767	CGGCAGCGGTGACCAGCACG
279	1749	1768	CCGGCAGCGGTGACCAGCAC
280	1752	1771	TTGCCGGCAGCGGTGACCAG
281	1754	1773	AGTTGCCGGCAGCGGTGACC
282	1756	1775	GAAGTTGCCGGCAGCGGTGA
283	1758	1777	CGGAAGTTGCCGGCAGCGGT
284	1760	1779	CCCGGAAGTTGCCGGCAGCG
285	1762	1781	GTCCCGGAAGTTGCCGGCAG
306	1820	1839	GAGGCACCAATGATGTCCTC

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
39	1822	1841	TGGAGGCACCAATGATGTCC
307	1824	1843	GCTGGAGGCACCAATGATGT
308	1826	1845	TCGCTGGAGGCACCAATGAT
309	1828	1847	AGTCGCTGGAGGCACCAATG
310	1830	1849	GCAGTCGCTGGAGGCACCAA
40	1833	1852	GCTGCAGTCGCTGGAGGCAC
311	1835	1854	GTGCTGCAGTCGCTGGAGGC
312	1837	1856	AGGTGCTGCAGTCGCTGGAG
313	1839	1858	GCAGGTGCTGCAGTCGCTGG
314	1840	1859	AGCAGGTGCTGCAGTCGCTG
315	1842	1861	AAAGCAGGTGCTGCAGTCGC
41	1844	1863	ACAAAGCAGGTGCTGCAGTC
316	1846	1865	ACACAAAGCAGGTGCTGCAG
317	1848	1867	TGACACAAAGCAGGTGCTGC
318	1858	1877	TCCCACTCTGTGACACAAAG
101	1898	1917	ATGGCTGCAATGCCAGCCAC
319	1900	1919	TCATGGCTGCAATGCCAGCC
42	1903	1922	GCATCATGGCTGCAATGCCA
320	1905	1924	CAGCATCATGGCTGCAATGC
321	1907	1926	GACAGCATCATGGCTGCAAT
322	1909	1928	CAGACAGCATCATGGCTGCA
43	1911	1930	GGCAGACAGCATCATGGCTG
323	1913	1932	TCGGCAGACAGCATCATGGC
324	1915	1934	GCTCGGCAGACAGCATCATG
325	1917	1936	CGGCTCGGCAGACAGCATCA
326	1919	1938	TCCGGCTCGGCAGACAGCAT
327	1933	1952	CGGCCAGGGTGAGCTCCGGC
328	1946	1965	CTCTGCCTCAACTCGGCCAG
329	1948	1967	GTCTCTGCCTCAACTCGGCC
330	1950	1969	CAGTCTCTGCCTCAACTCGG
331	1952	1971	ATCAGTCTCTGCCTCAACTC
332	1954	1973	GGATCAGTCTCTGCCTCAAC
333	1956	1975	GTGGATCAGTCTCTGCCTCA
334	1958	1977	AAGTGGATCAGTCTCTGCCT
44	1959	1978	GAAGTGGATCAGTCTCTGCC
335	1961	1980	GAGAAGTGGATCAGTCTCTG
336	1963	1982	CAGAGAAGTGGATCAGTCTC
337	1965	1984	GGCAGAGAAGTGGATCAGTC
45	1967	1986	TTGGCAGAGAAGTGGATCAG
338	1969	1988	CTTTGGCAGAGAAGTGGATC
46	1972	1991	CATCTTTGGCAGAGAAGTGG
339	1974	1993	GACATCTTTGGCAGAGAAGT
340	1976	1995	ATGACATCTTTGGCAGAGAA
47	1978	1997	TGATGACATCTTTGGCAGAG
341	1980	1999	ATTGATGACATCTTTGGCAG
342	1982	2001	TCATTGATGACATCTTTGGC
48	1985	2004	GCCTCATTGATGACATCTTT
343	1987	2006	AGGCCTCATTGATGACATCT
344	1989	2008	CCAGGCCTCATTGATGACAT
345	1991	2010	AACCAGGCCTCATTGATGAC
346	1993	2012	GGAACCAGGCCTCATTGATG
347	1994	2013	GGGAACCAGGCCTCATTGAT
348	1995	2014	AGGGAACCAGGCCTCATTGA
349	1996	2015	CAGGGAACCAGGCCTCATTG
49	1997	2016	TCAGGGAACCAGGCCTCATT
350	1998	2017	CTCAGGGAACCAGGCCTCAT
351	1999	2018	CCTCAGGGAACCAGGCCTCA
352	2000	2019	TCCTCAGGGAACCAGGCCTC
353	2001	2020	GTCCTCAGGGAACCAGGCCT
50	2002	2021	GGTCCTCAGGGAACCAGGCC
354	2003	2022	TGGTCCTCAGGGAACCAGGC
355	2004	2023	CTGGTCCTCAGGGAACCAGG
356	2005	2024	GCTGGTCCTCAGGGAACCAG
357	2006	2025	CGCTGGTCCTCAGGGAACCA
87	2009	2028	ACCCGCTGGTCCTCAGGGAA
358	2011	2030	GTACCCGCTGGTCCTCAGGG
359	2013	2032	CAGTACCCGCTGGTCCTCAG
51	2016	2035	GGTCAGTACCCGCTGGTCCT
119	2038	2057	GCAGGGCGGCCACCAGGTTG

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
360	2061	2080	ACCTGCCCCATGGGTGCTGG
52	2073	2092	AAACAGCTGCCAACCTGCCC
361	2075	2094	CAAAACAGCTGCCAACCTGC
53	2078	2097	CTGCAAAACAGCTGCCAACC
362	2080	2099	TCCTGCAAAACAGCTGCCAA
363	2082	2101	AGTCCTGCAAAACAGCTGCC
104	2085	2104	CACAGTCCTGCAAAACAGCT
130	2095	2114	GTGCTGACCACACAGTCCTG
365	2105	2124	GGCCCCGAGTGTGCTGACCA
54	2108	2127	GTAGGCCCCGAGTGTGCTGA
366	2110	2129	GTGTAGGCCCCGAGTGTGCT
367	2112	2131	CCGTGTAGGCCCCGAGTGTG
368	2114	2133	ATCCGTGTAGGCCCCGAGTG
369	2116	2135	CCATCCGTGTAGGCCCCGAG
370	2118	2137	GGCCATCCGTGTAGGCCCCG
371	2120	2139	GTGGCCATCCGTGTAGGCCC
372	2168	2187	CTGGAGCAGCTCAGCAGCTC
460	2168	2181	CAGCTCAGCAGCTC
373	2170	2189	AACTGGAGCAGCTCAGCAGC
55	2173	2192	AGAAACTGGAGCAGCTCAGC
374	2175	2194	GGAGAAACTGGAGCAGCTCA
375	2177	2196	CTGGAGAAACTGGAGCAGCT
56	2179	2198	TCCTGGAGAAACTGGAGCAG
408	2245	2264	GTGCCAAGGTCCTCCACCTC
409	2265	2284	TCAGCACAGGCGGCTTGTGG
410	2295	2314	CCACGCACTGGTTGGGCTGA
60	2355	2374	CTTTGCATTCCAGACCTGGG
411	2356	2375	ACTTTGCATTCCAGACCTGG
412	2357	2376	GACTTTGCATTCCAGACCTG
413	2358	2377	TGACTTTGCATTCCAGACCT
414	2359	2378	TTGACTTTGCATTCCAGACC
61	2360	2379	CTTGACTTTGCATTCCAGAC
415	2362	2381	TCCTTGACTTTGCATTCCAG
416	2375	2394	CGGGATTCCATGCTCCTTGA
417	2405	2424	GCAGGCCACGGTCACCTGCT

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
418	2442	2461	GGAGGGCACTGCAGCCAGTC
419	2560	2579	CAGATGGCAACGGCTGTCAC
420	2561	2580	GCAGATGGCAACGGCTGTCA
421	2562	2581	AGCAGATGGCAACGGCTGTC
422	2563	2582	CAGCAGATGGCAACGGCTGT
423	2564	2583	GCAGCAGATGGCAACGGCTG
62	2565	2584	GGCAGCAGATGGCAACGGCT
424	2566	2585	CGGCAGCAGATGGCAACGGC
425	2567	2586	CCGGCAGCAGATGGCAACGG
426	2568	2587	TCCGGCAGCAGATGGCAACG
427	2569	2588	CTCCGGCAGCAGATGGCAAC
428	2570	2589	GCTCCGGCAGCAGATGGCAA
429	2580	2599	CCAGGTGCCGGCTCCGGCAG
430	2590	2609	GAGGCCTGCGCCAGGTGCCG
154	2665	2684	TTTTAAAGCTCAGCCCCAGC
63	2670	2689	AACCATTTTAAAGCTCAGCC
64	2759	2778	TCAAGGGCCAGGCCAGCAGC
65	2764	2783	CCCACTCAAGGGCCAGGCCA
122	2837	2856	GGAGGGAGCTTCCTGGCACC
66	2852	2871	ATGCCCCACAGTGAGGGAGG
67	2861	2880	AATGGTGAAATGCCCCACAG
153	2904	2923	TTGGGAGCAGCTGGCAGCAC
68	3005	3024	CATGGGAAGAATCCTGCCTC
431	3083	3102	ATGAGGGCCATCAGCACCTT
432	3084	3103	GATGAGGGCCATCAGCACCT
433	3085	3104	AGATGAGGGCCATCAGCACC
434	3086	3105	GAGATGAGGGCCATCAGCAC
69	3087	3106	GGAGATGAGGGCCATCAGCA
435	3088	3107	TGGAGATGAGGGCCATCAGC
436	3089	3108	CTGGAGATGAGGGCCATCAG
437	3090	3109	GCTGGAGATGAGGGCCATCA
438	3091	3110	AGCTGGAGATGAGGGCCATC
461	3132	3145	TTAATCAGGGAGCC
70	3155	3174	TAGATGCCATCCAGAAAGCT

TABLE 12-continued

Nucleotide sequences targeted to AK124635.1 (SEQ ID NO: 3)

SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Sequences (5'-3')
439	3157	3176	GCTAGATGCCATCCAGAAAG
440	3158	3177	GGCTAGATGCCATCCAGAAA
441	3159	3178	TGGCTAGATGCCATCCAGAA
442	3160	3179	CTGGCTAGATGCCATCCAGA
71	3161	3180	TCTGGCTAGATGCCATCCAG
443	3162	3181	CTCTGGCTAGATGCCATCCA
444	3163	3182	CCTCTGGCTAGATGCCATCC
445	3164	3183	GCCTCTGGCTAGATGCCATC
446	3165	3184	AGCCTCTGGCTAGATGCCAT
72	3238	3257	GGCATAGAGCAGAGTAAAGG
73	3243	3262	AGCCTGGCATAGAGCAGAGT
135	3249	3268	TAGCACAGCCTGGCATAGAG
112	3482	3501	GAAGAGGCTTGGCTTCAGAG
74	3488	3507	AAGTAAGAAGAGGCTTGGCT
75	3692	3711	GCTCAAGGAGGGACAGTTGT
76	3727	3746	AAAGATAAATGTCTGCTTGC
77	3732	3751	ACCCAAAAGATAAATGTCTG
78	3798	3817	TCTTCAAGTTACAAAAGCAA
99	3805	3824	ATAAATATCTTCAAGTTACA

[0679] In certain embodiments, gapmer antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gapmer antisense compounds are targeted to SEQ ID NO: 3. In certain such embodiments, the nucleotide sequences illustrated in Table 12 have a 5-10-5 gapmer motif. Table 13 illustrates gapmer antisense compounds targeted to SEQ ID NO: 3, having a 5-10-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 13

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
395201	5-10-5	85	155	174	0	ACAAATTCCCAGACTCAGCA
399827	5-10-5	100	220	239	0	ATCTCAGGACAGGTGAGCAA
399828	5-10-5	116	234	253	0	GAGTAGAGATTCTCATCTCA
395202	5-10-5	129	290	309	0	GTGCCATCTGAACAGCACCT
399829	5-10-5	117	302	321	0	GAGTCTTCTGAAGTGCCATC
399830	5-10-5	81	377	396	0	AAGCAGGGCCTCAGGTGGAA
395203	5-10-5	110	400	419	0	CCTGGAACCCCTGCAGCCAG
395204	5-10-5	152	451	470	0	TTCAGGCAGGTTGCTGCTAG
399831	5-10-5	83	460	479	0	AAGGAAGACTTCAGGCAGGT
395205	5-10-5	140	472	491	0	TCAGCCAGGCCAAAGGAAGA
399832	5-10-5	137	586	605	0	TAGGGAGAGCTCACAGATGC
395206	5-10-5	136	596	615	0	TAGGAGAAAGTAGGGAGAGC
395207	5-10-5	132	653	672	0	TAAAAGCTGCAAGAGACTCA
395208	5-10-5	139	741	760	0	TCAGAGAAAACAGTCACCGA
399833	5-10-5	92	871	890	0	AGAGACAGGAAGCTGCAGCT
395209	5-10-5	142	878	897	0	TCATTTTAGAGACAGGAAGC

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
395210	5-10-5	113	915	934	0	GAATAACAGTGATGTCTGGC
395211	5-10-5	138	968	987	0	TCACAGCTCACCGAGTCTGC
395212	5-10-5	98	998	1017	0	AGTGTAAAATAAAGCCCCTA
395213	5-10-5	96	1075	1094	0	AGGACCCAAGTCATCCTGCT
395214	5-10-5	124	1105	1124	0	GGCCATCAGCTGGCAATGCT
399834	5-10-5	82	1144	1163	0	AAGGAAAGGGAGGCCTAGAG
395215	5-10-5	133	1149	1168	0	TAGACAAGGAAAGGGAGGCC
395216	5-10-5	103	1155	1174	0	ATTTCATAGACAAGGAAAGG
395217	5-10-5	155	1275	1294	0	CTTATAGTTAACACACAGAA
399835	5-10-5	156	1283	1302	0	AAGTCAACCTTATAGTTAAC
395218	5-10-5	146	1315	1334	0	TGACATTTGTGGGAGAGGAG
395219	5-10-5	161	1322	1341	0	TCCAAGGTGACATTTGTGGG
395160	5-10-5	19	1365	1384	0	CTGGTGTCTAGGAGATACAC
399797	5-10-5	20	1370	1389	0	GTATGCTGGTGTCTAGGAGA
395161	5-10-5	21	1390	1409	0	GATTTCCCGGTGGTCACTCT
399798	5-10-5	22	1396	1415	0	GCCCTCGATTTCCCGGTGGT
399799	5-10-5	23	1410	1429	0	GTGACCATGACCCTGCCCTC
395162	5-10-5	24	1420	1439	0	CTCGAAGTCGGTGACCATGA
395163	5-10-5	25	1453	1472	0	GTGGAAGCGGGTCCCGTCCT
395164	5-10-5	26	1500	1519	0	ACCCCTGCCAGGTGGGTGCC
399800	5-10-5	27	1505	1524	0	TGACCACCCCTGCCAGGTGG
395165	5-10-5	28	1534	1553	0	ACCCTTGGCCACGCCGGCAT
395166	5-10-5	29	1570	1589	0	TTGGCAGTTGAGCACGCGCA
399801	5-10-5	30	1575	1594	0	TTCCCTTGGCAGTTGAGCAC
395167	5-10-5	31	1607	1626	0	CCAGGCCTATGAGGGTGCCG
395168	5-10-5	32	1628	1647	0	GCTGGCTTTTCCGAATAAAC
395169	5-10-5	33	1740	1759	0	GTGACCAGCACGACCCCAGC
395175	5-10-5	39	1822	1841	0	TGGAGGCACCAATGATGTCC
399804	5-10-5	40	1833	1852	0	GCTGCAGTCGCTGGAGGCAC
399805	5-10-5	41	1844	1863	0	ACAAAGCAGGTGCTGCAGTC
395176	5-10-5	101	1898	1917	0	ATGGCTGCAATGCCAGCCAC
399806	5-10-5	42	1903	1922	0	GCATCATGGCTGCAATGCCA
399807	5-10-5	43	1911	1930	0	GGCAGACAGCATCATGGCTG
399808	5-10-5	44	1959	1978	0	GAAGTGGATCAGTCTCTGCC
395177	5-10-5	45	1967	1986	0	TTGGCAGAGAAGTGGATCAG

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
399809	5-10-5	46	1972	1991	0	CATCTTTGGCAGAGAAGTGG
399810	5-10-5	47	1978	1997	0	TGATGACATCTTTGGCAGAG
399811	5-10-5	48	1985	2004	0	GCCTCATTGATGACATCTTT
399812	5-10-5	49	1997	2016	0	TCAGGGAACCAGGCCTCATT
395178	5-10-5	50	2002	2021	0	GGTCCTCAGGGAACCAGGCC
395179	5-10-5	87	2009	2028	0	ACCCGCTGGTCCTCAGGGAA
399813	5-10-5	51	2016	2035	0	GGTCAGTACCCGCTGGTCCT
395180	5-10-5	119	2038	2057	0	GCAGGGCGGCCACCAGGTTG
395181	5-10-5	52	2073	2092	0	AAACAGCTGCCAACCTGCCC
399814	5-10-5	53	2078	2097	0	CTGCAAAACAGCTGCCAACC
395220	5-10-5	104	2085	2104	0	CACAGTCCTGCAAAACAGCT
399836	5-10-5	130	2095	2114	0	GTGCTGACCACACAGTCCTG
395182	5-10-5	54	2108	2127	0	GTAGGCCCCGAGTGTGCTGA
399815	5-10-5	55	2173	2192	0	AGAAACTGGAGCAGCTCAGC
399816	5-10-5	56	2179	2198	0	TCCTGGAGAAACTGGAGCAG
395186	5-10-5	60	2355	2374	0	CTTTGCATTCCAGACCTGGG
399817	5-10-5	61	2360	2379	0	CTTGACTTTGCATTCCAGAC
395187	5-10-5	62	2565	2584	0	GGCAGCAGATGGCAACGGCT
395188	5-10-5	154	2665	2684	0	TTTTAAAGCTCAGCCCCAGC
399818	5-10-5	63	2670	2689	0	AACCATTTTAAAGCTCAGCC
395189	5-10-5	64	2759	2778	0	TCAAGGGCCAGGCCAGCAGC
399819	5-10-5	65	2764	2783	0	CCCACTCAAGGGCCAGGCCA
399820	5-10-5	122	2837	2856	0	GGAGGGAGCTTCCTGGCACC
395190	5-10-5	66	2852	2871	0	ATGCCCCACAGTGAGGGAGG
395191	5-10-5	67	2861	2880	0	AATGGTGAAATGCCCCACAG
395192	5-10-5	153	2904	2923	0	TTGGGAGCAGCTGGCAGCAC
395193	5-10-5	68	3005	3024	0	CATGGGAAGAATCCTGCCTC
395194	5-10-5	69	3087	3106	0	GGAGATGAGGGCCATCAGCA
395195	5-10-5	70	3155	3174	0	TAGATGCCATCCAGAAAGCT
399821	5-10-5	71	3161	3180	0	TCTGGCTAGATGCCATCCAG
395196	5-10-5	72	3238	3257	0	GGCATAGAGCAGAGTAAAGG
399822	5-10-5	73	3243	3262	0	AGCCTGGCATAGAGCAGAGT
399823	5-10-5	135	3249	3268	0	TAGCACAGCCTGGCATAGAG
395197	5-10-5	112	3482	3501	0	GAAGAGGCTTGGCTTCAGAG
399824	5-10-5	74	3488	3507	0	AAGTAAGAAGAGGCTTGGCT

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
395198	5-10-5	75	3692	3711	0	GCTCAAGGAGGGACAGTTGT
395199	5-10-5	76	3727	3746	0	AAAGATAAATGTCTGCTTGC
399825	5-10-5	77	3732	3751	0	ACCCAAAAGATAAATGTCTG
395200	5-10-5	78	3798	3817	0	TCTTCAAGTTACAAAAGCAA
399826	5-10-5	99	3805	3824	0	ATAAATATCTTCAAGTTACA
405869	5-10-5	218	1392	1411	0	TCGATTTCCCGGTGGTCACT
405870	5-10-5	219	1393	1412	0	CTCGATTTCCCGGTGGTCAC
405871	5-10-5	220	1394	1413	0	CCTCGATTTCCCGGTGGTCA
405872	5-10-5	221	1395	1414	0	CCCTCGATTTCCCGGTGGTC
405873	5-10-5	222	1397	1416	0	TGCCCTCGATTTCCCGGTGG
405874	5-10-5	223	1398	1417	0	CTGCCCTCGATTTCCCGGTG
405875	5-10-5	224	1399	1418	0	CCTGCCCTCGATTTCCCGGT
405876	5-10-5	225	1400	1419	0	CCCTGCCCTCGATTTCCCGG
405877	5-10-5	246	1530	1549	0	TTGGCCACGCCGGCATCCCG
405878	5-10-5	247	1531	1550	0	CTTGGCCACGCCGGCATCCC
405879	5-10-5	248	1532	1551	0	CCTTGGCCACGCCGGCATCC
405880	5-10-5	249	1533	1552	0	CCCTTGGCCACGCCGGCATC
405881	5-10-5	250	1535	1554	0	CACCCTTGGCCACGCCGGCA
405882	5-10-5	251	1536	1555	0	GCACCCTTGGCCACGCCGGC
405883	5-10-5	252	1537	1556	0	GGCACCCTTGGCCACGCCGG
405884	5-10-5	253	1538	1557	0	TGGCACCCTTGGCCACGCCG
405885	5-10-5	346	1993	2012	0	GGAACCAGGCCTCATTGATG
405886	5-10-5	347	1994	2013	0	GGGAACCAGGCCTCATTGAT
405887	5-10-5	348	1995	2014	0	AGGGAACCAGGCCTCATTGA
405888	5-10-5	349	1996	2015	0	CAGGGAACCAGGCCTCATTG
405889	5-10-5	350	1998	2017	0	CTCAGGGAACCAGGCCTCAT
405890	5-10-5	351	1999	2018	0	CCTCAGGGAACCAGGCCTCA
405891	5-10-5	352	2000	2019	0	TCCTCAGGGAACCAGGCCTC
405892	5-10-5	353	2001	2020	0	GTCCTCAGGGAACCAGGCCT
405893	5-10-5	431	3083	3102	0	ATGAGGGCCATCAGCACCTT
405894	5-10-5	432	3084	3103	0	GATGAGGGCCATCAGCACCT
405895	5-10-5	433	3085	3104	0	AGATGAGGGCCATCAGCACC
405896	5-10-5	434	3086	3105	0	GAGATGAGGGCCATCAGCAC
405897	5-10-5	435	3088	3107	0	TGGAGATGAGGGCCATCAGC
405898	5-10-5	436	3089	3108	0	CTGGAGATGAGGGCCATCAG

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
405899	5-10-5	437	3090	3109	0	GCTGGAGATGAGGGCCATCA
405900	5-10-5	438	3091	3110	0	AGCTGGAGATGAGGGCCATC
405901	5-10-5	439	3157	3176	0	GCTAGATGCCATCCAGAAAG
405902	5-10-5	440	3158	3177	0	GGCTAGATGCCATCCAGAAA
405903	5-10-5	441	3159	3178	0	TGGCTAGATGCCATCCAGAA
405904	5-10-5	442	3160	3179	0	CTGGCTAGATGCCATCCAGA
405905	5-10-5	443	3162	3181	0	CTCTGGCTAGATGCCATCCA
405906	5-10-5	444	3163	3182	0	CCTCTGGCTAGATGCCATCC
405907	5-10-5	445	3164	3183	0	GCCTCTGGCTAGATGCCATC
405908	5-10-5	446	3165	3184	0	AGCCTCTGGCTAGATGCCAT
405909	5-10-5	267	1630	1649	0	CAGCTGGCTTTTCCGAATAA
405910	5-10-5	268	1632	1651	0	ACCAGCTGGCTTTTCCGAAT
405911	5-10-5	269	1634	1653	0	GGACCAGCTGGCTTTTCCGA
405912	5-10-5	270	1638	1657	0	GGCTGGACCAGCTGGCTTTT
405913	5-10-5	275	1742	1761	0	CGGTGACCAGCACGACCCCA
405914	5-10-5	276	1744	1763	0	AGCGGTGACCAGCACGACCC
405915	5-10-5	277	1746	1765	0	GCAGCGGTGACCAGCACGAC
405916	5-10-5	278	1748	1767	0	CGGCAGCGGTGACCAGCACG
405917	5-10-5	280	1752	1771	0	TTGCCGGCAGCGGTGACCAG
405918	5-10-5	281	1754	1773	0	AGTTGCCGGCAGCGGTGACC
405919	5-10-5	282	1756	1775	0	GAAGTTGCCGGCAGCGGTGA
405920	5-10-5	283	1758	1777	0	CGGAAGTTGCCGGCAGCGGT
405921	5-10-5	284	1760	1779	0	CCCGGAAGTTGCCGGCAGCG
405922	5-10-5	285	1762	1781	0	GTCCCGGAAGTTGCCGGCAG
405937	5-10-5	306	1820	1839	0	GAGGCACCAATGATGTCCTC
405938	5-10-5	307	1824	1843	0	GCTGGAGGCACCAATGATGT
405939	5-10-5	308	1826	1845	0	TCGCTGGAGGCACCAATGAT
405940	5-10-5	309	1828	1847	0	AGTCGCTGGAGGCACCAATG
405941	5-10-5	310	1830	1849	0	GCAGTCGCTGGAGGCACCAA
405942	5-10-5	311	1835	1854	0	GTGCTGCAGTCGCTGGAGGC
405943	5-10-5	312	1837	1856	0	AGGTGCTGCAGTCGCTGGAG
405944	5-10-5	314	1840	1859	0	AGCAGGTGCTGCAGTCGCTG
405945	5-10-5	315	1842	1861	0	AAAGCAGGTGCTGCAGTCGC
405946	5-10-5	316	1846	1865	0	ACACAAAGCAGGTGCTGCAG
405947	5-10-5	317	1848	1867	0	TGACACAAAGCAGGTGCTGC

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3						
Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
405948	5-10-5	319	1900	1919	0	TCATGGCTGCAATGCCAGCC
405949	5-10-5	320	1905	1924	0	CAGCATCATGGCTGCAATGC
405950	5-10-5	321	1907	1926	0	GACAGCATCATGGCTGCAAT
405951	5-10-5	322	1909	1928	0	CAGACAGCATCATGGCTGCA
405952	5-10-5	323	1913	1932	0	TCGGCAGACAGCATCATGGC
405953	5-10-5	325	1917	1936	0	CGGCTCGGCAGACAGCATCA
405954	5-10-5	326	1919	1938	0	TCCGGCTCGGCAGACAGCAT
405955	5-10-5	328	1946	1965	0	CTCTGCCTCAACTCGGCCAG
405956	5-10-5	329	1948	1967	0	GTCTCTGCCTCAACTCGGCC
405957	5-10-5	330	1950	1969	0	CAGTCTCTGCCTCAACTCGG
405958	5-10-5	331	1952	1971	0	ATCAGTCTCTGCCTCAACTC
405959	5-10-5	332	1954	1973	0	GGATCAGTCTCTGCCTCAAC
405960	5-10-5	333	1956	1975	0	GTGGATCAGTCTCTGCCTCA
405961	5-10-5	334	1958	1977	0	AAGTGGATCAGTCTCTGCCT
405962	5-10-5	335	1961	1980	0	GAGAAGTGGATCAGTCTCTG
405963	5-10-5	337	1965	1984	0	GGCAGAGAAGTGGATCAGTC
405964	5-10-5	338	1969	1988	0	CTTTGGCAGAGAAGTGGATC
405965	5-10-5	339	1974	1993	0	GACATCTTTGGCAGAGAAGT
405966	5-10-5	340	1976	1995	0	ATGACATCTTTGGCAGAGAA
405967	5-10-5	341	1980	1999	0	ATTGATGACATCTTTGGCAG
405968	5-10-5	342	1982	2001	0	TCATTGATGACATCTTTGGC
405969	5-10-5	343	1987	2006	0	AGGCCTCATTGATGACATCT
405970	5-10-5	344	1989	2008	0	CCAGGCCTCATTGATGACAT
405971	5-10-5	355	2004	2023	0	CTGGTCCTCAGGGAACCAGG
405972	5-10-5	357	2006	2025	0	CGCTGGTCCTCAGGGAACCA
405973	5-10-5	358	2011	2030	0	GTACCCGCTGGTCCTCAGGG
405974	5-10-5	359	2013	2032	0	CAGTACCCGCTGGTCCTCAG
405975	5-10-5	361	2075	2094	0	CAAAACAGCTGCCAACCTGC
405976	5-10-5	362	2080	2099	0	TCCTGCAAAACAGCTGCCAA
405977	5-10-5	363	2082	2101	0	AGTCCTGCAAAACAGCTGCC
405978	5-10-5	365	2105	2124	0	GGCCCCGAGTGTGCTGACCA
405979	5-10-5	366	2110	2129	0	GTGTAGGCCCCGAGTGTGCT
405980	5-10-5	367	2112	2131	0	CCGTGTAGGCCCCGAGTGTG
405981	5-10-5	368	2114	2133	0	ATCCGTGTAGGCCCCGAGTG
405982	5-10-5	370	2118	2137	0	GGCCATCCGTGTAGGCCCCG

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
405983	5-10-5	371	2120	2139	0	GTGGCCATCCGTGTAGGCCC
405984	5-10-5	372	2168	2187	0	CTGGAGCAGCTCAGCAGCTC
405985	5-10-5	373	2170	2189	0	AACTGGAGCAGCTCAGCAGC
405986	5-10-5	374	2175	2194	0	GGAGAAACTGGAGCAGCTCA
405987	5-10-5	375	2177	2196	0	CTGGAGAAACTGGAGCAGCT
405996	5-10-5	412	2357	2376	0	GACTTTGCATTCCAGACCTG
405997	5-10-5	415	2362	2381	0	TCCTTGACTTTGCATTCCAG
405998	5-10-5	426	2568	2587	0	TCCGGCAGCAGATGGCAACG
406025	5-10-5	216	1362	1381	0	GTGTCTAGGAGATACACCTC
406026	5-10-5	217	1367	1386	0	TGCTGGTGTCTAGGAGATAC
406027	5-10-5	226	1404	1423	0	ATGACCCTGCCCTCGATTTC
406028	5-10-5	227	1406	1425	0	CCATGACCCTGCCCTCGATT
406029	5-10-5	228	1408	1427	0	GACCATGACCCTGCCCTCGA
406030	5-10-5	229	1412	1431	0	CGGTGACCATGACCCTGCCC
406031	5-10-5	230	1414	1433	0	GTCGGTGACCATGACCCTGC
406032	5-10-5	232	1418	1437	0	CGAAGTCGGTGACCATGACC
406033	5-10-5	237	1497	1516	0	CCTGCCAGGTGGGTGCCATG
406034	5-10-5	238	1502	1521	0	CCACCCCTGCCAGGTGGGTG
406035	5-10-5	239	1507	1526	0	GCTGACCACCCCTGCCAGGT
406036	5-10-5	241	1519	1538	0	GGCATCCCGGCCGCTGACCA
406037	5-10-5	242	1522	1541	0	GCCGGCATCCCGGCCGCTGA
406038	5-10-5	245	1529	1548	0	TGGCCACGCCGGCATCCCGG
406039	5-10-5	257	1566	1585	0	CAGTTGAGCACGCGCAGGCT
406040	5-10-5	258	1568	1587	0	GGCAGTTGAGCACGCGCAGG
406041	5-10-5	259	1572	1591	0	CCTTGGCAGTTGAGCACGCG
406042	5-10-5	260	1577	1596	0	CCTTCCCTTGGCAGTTGAGC
406043	5-10-5	261	1581	1600	0	GTGCCCTTCCCTTGGCAGTT
406044	5-10-5	262	1583	1602	0	CCGTGCCCTTCCCTTGGCAG
406045	5-10-5	266	1626	1645	0	TGGCTTTTCCGAATAAACTC
408642	5-10-5	264	1606	1625	0	CAGGCCTATGAGGGTGCCGC
408653	5-10-5	354	2003	2022	0	TGGTCCTCAGGGAACCAGGC
409126	5-10-5	447	1534	1553	1	ACCCTTGGTCACGCCGGCAT
410536	5-10-5	243	1527	1546	0	GCCACGCCGGCATCCCGGCC
410537	5-10-5	244	1528	1547	0	GGCCACGCCGGCATCCCGGC
410538	5-10-5	254	1539	1558	0	CTGGCACCCTTGGCCACGCC

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
410539	5-10-5	255	1540	1559	0	GCTGGCACCCTTGGCCACGC
410540	5-10-5	356	2005	2024	0	GCTGGTCCTCAGGGAACCAG
410562	5-10-5	411	2356	2375	0	ACTTTGCATTCCAGACCTGG
410563	5-10-5	413	2358	2377	0	TGACTTTGCATTCCAGACCT
410564	5-10-5	414	2359	2378	0	TTGACTTTGCATTCCAGACC
410565	5-10-5	419	2560	2579	0	CAGATGGCAACGGCTGTCAC
410566	5-10-5	420	2561	2580	0	GCAGATGGCAACGGCTGTCA
410567	5-10-5	421	2562	2581	0	AGCAGATGGCAACGGCTGTC
410568	5-10-5	422	2563	2582	0	CAGCAGATGGCAACGGCTGT
410569	5-10-5	423	2564	2583	0	GCAGCAGATGGCAACGGCTG
410570	5-10-5	424	2566	2585	0	CGGCAGCAGATGGCAACGGC
410571	5-10-5	425	2567	2586	0	CCGGCAGCAGATGGCAACGG
410572	5-10-5	427	2569	2588	0	CTCCGGCAGCAGATGGCAAC
410573	5-10-5	428	2570	2589	0	GCTCCGGCAGCAGATGGCAA
410733	5-10-5	231	1416	1435	0	AAGTCGGTGACCATGACCCT
410734	5-10-5	279	1749	1768	0	CCGGCAGCGGTGACCAGCAC
410737	5-10-5	313	1839	1858	0	GCAGGTGCTGCAGTCGCTGG
410738	5-10-5	324	1915	1934	0	GCTCGGCAGACAGCATCATG
410739	5-10-5	336	1963	1982	0	CAGAGAAGTGGATCAGTCTC
410740	5-10-5	345	1991	2010	0	AACCAGGCCTCATTGATGAC
410741	5-10-5	369	2116	2135	0	CCATCCGTGTAGGCCCCGAG
410753	5-10-5	215	1351	1370	0	ATACACCTCCACCAGGCTGC
410754	5-10-5	233	1428	1447	0	GGCACATTCTCGAAGTCGGT
410755	5-10-5	234	1463	1482	0	TGGCCTGTCTGTGGAAGCGG
410756	5-10-5	235	1490	1509	0	GGTGGGTGCCATGACTGTCA
410757	5-10-5	240	1515	1534	0	TCCCGGCCGCTGACCACCCC
410758	5-10-5	256	1545	1564	0	CGCATGCTGGCACCCTTGGC
410759	5-10-5	263	1594	1613	0	GGTGCCGCTAACCGTGCCCT
410760	5-10-5	265	1618	1637	0	CCGAATAAACTCCAGGCCTA
410761	5-10-5	271	1649	1668	0	GTGGCCCCACAGGCTGGACC
410762	5-10-5	272	1662	1681	0	AGCAGCACCACCAGTGGCCC
410763	5-10-5	273	1684	1703	0	GCTGTACCCACCCGCCAGGG
410764	5-10-5	274	1730	1749	0	CGACCCCAGCCCTCGCCAGG
410769	5-10-5	318	1858	1877	0	TCCCACTCTGTGACACAAAG
410770	5-10-5	327	1933	1952	0	CGGCCAGGGTGAGCTCCGGC

TABLE 13-continued

Gapmer antisense compounds having a 5-10-5 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
410771	5-10-5	360	2061	2080	0	ACCTGCCCCATGGGTGCTGG
410776	5-10-5	408	2245	2264	0	GTGCCAAGGTCCTCCACCTC
410777	5-10-5	409	2265	2284	0	TCAGCACAGGCGGCTTGTGG
410778	5-10-5	410	2295	2314	0	CCACGCACTGGTTGGGCTGA
410779	5-10-5	416	2375	2394	0	CGGGATTCCATGCTCCTTGA
410780	5-10-5	417	2405	2424	0	GCAGGCCACGGTCACCTGCT
410781	5-10-5	418	2442	2461	0	GGAGGGCACTGCAGCCAGTC
410782	5-10-5	429	2580	2599	0	CCAGGTGCCGGCTCCGGCAG
410783	5-10-5	430	2590	2609	0	GAGGCCTGCGCCAGGTGCCG

[0680] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 3. In certain such embodiments, the nucleotide sequences illustrated in Table 12 have a 3-14-3 gap-widened motif. Table 14 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 3, having a 3-14-3 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 14

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
399923	3-14-3	85	155	174	0	ACAAATTCCCAGACTCAGCA
399983	3-14-3	100	220	239	0	ATCTCAGGACAGGTGAGCAA
399984	3-14-3	116	234	253	0	GAGTAGAGATTCTCATCTCA
399924	3-14-3	129	290	309	0	GTGCCATCTGAACAGCACCT
399985	3-14-3	117	302	321	0	GAGTCTTCTGAAGTGCCATC
399986	3-14-3	81	377	396	0	AAGCAGGGCCTCAGGTGGAA
399925	3-14-3	110	400	419	0	CCTGGAACCCCTGCAGCCAG
399926	3-14-3	152	451	470	0	TTCAGGCAGGTTGCTGCTAG
399987	3-14-3	83	460	479	0	AAGGAAGACTTCAGGCAGGT
399927	3-14-3	140	472	491	0	TCAGCCAGGCCAAAGGAAGA
399988	3-14-3	137	586	605	0	TAGGGAGAGCTCACAGATGC
399928	3-14-3	136	596	615	0	TAGGAGAAAGTAGGGAGAGC
399929	3-14-3	132	653	672	0	TAAAAGCTGCAAGAGACTCA
399930	3-14-3	139	741	760	0	TCAGAGAAAACAGTCACCGA
399989	3-14-3	92	871	890	0	AGAGACAGGAAGCTGCAGCT
399931	3-14-3	142	878	897	0	TCATTTTAGAGACAGGAAGC

TABLE 14-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
399932	3-14-3	113	915	934	0	GAATAACAGTGATGTCTGGC
399933	3-14-3	138	968	987	0	TCACAGCTCACCGAGTCTGC
399934	3-14-3	98	998	1017	0	AGTGTAAAATAAAGCCCCTA
399935	3-14-3	96	1075	1094	0	AGGACCCAAGTCATCCTGCT
399936	3-14-3	124	1105	1124	0	GGCCATCAGCTGGCAATGCT
399990	3-14-3	82	1144	1163	0	AAGGAAAGGGAGGCCTAGAG
399937	3-14-3	133	1149	1168	0	TAGACAAGGAAAGGGAGGCC
399938	3-14-3	103	1155	1174	0	ATTTCATAGACAAGGAAAGG
399939	3-14-3	155	1275	1294	0	CTTATAGTTAACACACAGAA
399991	3-14-3	156	1283	1302	0	AAGTCAACCTTATAGTTAAC
399940	3-14-3	146	1315	1334	0	TGACATTTGTGGGAGAGGAG
399941	3-14-3	161	1322	1341	0	TCCAAGGTGACATTTGTGGG
399882	3-14-3	19	1365	1384	0	CTGGTGTCTAGGAGATACAC
399953	3-14-3	20	1370	1389	0	GTATGCTGGTGTCTAGGAGA
399883	3-14-3	21	1390	1409	0	GATTTCCCGGTGGTCACTCT
399954	3-14-3	22	1396	1415	0	GCCCTCGATTTCCCGGTGGT
399955	3-14-3	23	1410	1429	0	GTGACCATGACCCTGCCCTC
399884	3-14-3	24	1420	1439	0	CTCGAAGTCGGTGACCATGA
399885	3-14-3	25	1453	1472	0	GTGGAAGCGGGTCCCGTCCT
399886	3-14-3	26	1500	1519	0	ACCCCTGCCAGGTGGGTGCC
399956	3-14-3	27	1505	1524	0	TGACCACCCCTGCCAGGTGG
399887	3-14-3	28	1534	1553	0	ACCCTTGGCCACGCCGGCAT
399888	3-14-3	29	1570	1589	0	TTGGCAGTTGAGCACGCGCA
399957	3-14-3	30	1575	1594	0	TTCCCTTGGCAGTTGAGCAC
399889	3-14-3	31	1607	1626	0	CCAGGCCTATGAGGGTGCCG
399890	3-14-3	32	1628	1647	0	GCTGGCTTTTCCGAATAAAC
399891	3-14-3	33	1740	1759	0	GTGACCAGCACGACCCCAGC
399897	3-14-3	39	1822	1841	0	TGGAGGCACCAATGATGTCC
399960	3-14-3	40	1833	1852	0	GCTGCAGTCGCTGGAGGCAC
399961	3-14-3	41	1844	1863	0	ACAAAGCAGGTGCTGCAGTC
399898	3-14-3	101	1898	1917	0	ATGGCTGCAATGCCAGCCAC
399962	3-14-3	42	1903	1922	0	GCATCATGGCTGCAATGCCA
399963	3-14-3	43	1911	1930	0	GGCAGACAGCATCATGGCTG
399964	3-14-3	44	1959	1978	0	GAAGTGGATCAGTCTCTGCC
399899	3-14-3	45	1967	1986	0	TTGGCAGAGAAGTGGATCAG

TABLE 14-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
399965	3-14-3	46	1972	1991	0	CATCTTTGGCAGAGAAGTGG
399966	3-14-3	47	1978	1997	0	TGATGACATCTTTGGCAGAG
399967	3-14-3	48	1985	2004	0	GCCTCATTGATGACATCTTT
399968	3-14-3	49	1997	2016	0	TCAGGGAACCAGGCCTCATT
399900	3-14-3	50	2002	2021	0	GGTCCTCAGGGAACCAGGCC
399901	3-14-3	87	2009	2028	0	ACCCGCTGGTCCTCAGGGAA
399969	3-14-3	51	2016	2035	0	GGTCAGTACCCGCTGGTCCT
399902	3-14-3	119	2038	2057	0	GCAGGGCGGCCACCAGGTTG
399903	3-14-3	52	2073	2092	0	AAACAGCTGCCAACCTGCCC
399970	3-14-3	53	2078	2097	0	CTGCAAAACAGCTGCCAACC
399942	3-14-3	104	2085	2104	0	CACAGTCCTGCAAAACAGCT
399992	3-14-3	130	2095	2114	0	GTGCTGACCACACAGTCCTG
399904	3-14-3	54	2108	2127	0	GTAGGCCCCGAGTGTGCTGA
399971	3-14-3	55	2173	2192	0	AGAAACTGGAGCAGCTCAGC
399972	3-14-3	56	2179	2198	0	TCCTGGAGAAACTGGAGCAG
399908	3-14-3	60	2355	2374	0	CTTTGCATTCCAGACCTGGG
399973	3-14-3	61	2360	2379	0	CTTGACTTTGCATTCCAGAC
399909	3-14-3	62	2565	2584	0	GGCAGCAGATGGCAACGGCT
399910	3-14-3	154	2665	2684	0	TTTTAAAGCTCAGCCCCAGC
399974	3-14-3	63	2670	2689	0	AACCATTTTAAAGCTCAGCC
399911	3-14-3	64	2759	2778	0	TCAAGGGCCAGGCCAGCAGC
399975	3-14-3	65	2764	2783	0	CCCACTCAAGGGCCAGGCCA
399976	3-14-3	122	2837	2856	0	GGAGGGAGCTTCCTGGCACC
399912	3-14-3	66	2852	2871	0	ATGCCCCACAGTGAGGGAGG
399913	3-14-3	67	2861	2880	0	AATGGTGAAATGCCCCACAG
399914	3-14-3	153	2904	2923	0	TTGGGAGCAGCTGGCAGCAC
399915	3-14-3	68	3005	3024	0	CATGGGAAGAATCCTGCCTC
399916	3-14-3	69	3087	3106	0	GGAGATGAGGGCCATCAGCA
399917	3-14-3	70	3155	3174	0	TAGATGCCATCCAGAAAGCT
399977	3-14-3	71	3161	3180	0	TCTGGCTAGATGCCATCCAG
399918	3-14-3	72	3238	3257	0	GGCATAGAGCAGAGTAAAGG
399978	3-14-3	73	3243	3262	0	AGCCTGGCATAGAGCAGAGT
399979	3-14-3	135	3249	3268	0	TAGCACAGCCTGGCATAGAG
399919	3-14-3	112	3482	3501	0	GAAGAGGCTTGGCTTCAGAG
399980	3-14-3	74	3488	3507	0	AAGTAAGAAGAGGCTTGGCT

TABLE 14-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 3

Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
399920	3-14-3	75	3692	3711	0	GCTCAAGGAGGGACAGTTGT
399921	3-14-3	76	3727	3746	0	AAAGATAAATGTCTGCTTGC
399981	3-14-3	77	3732	3751	0	ACCCAAAAGATAAATGTCTG
399922	3-14-3	78	3798	3817	0	TCTTCAAGTTACAAAAGCAA
399982	3-14-3	99	3805	3824	0	ATAAATATCTTCAAGTTACA
405604	3-14-3	236	1493	1512	0	CCAGGTGGGTGCCATGACTG
405641	3-14-3	373	2170	2189	0	AACTGGAGCAGCTCAGCAGC
410583	3-14-3	243	1527	1546	0	GCCACGCCGGCATCCCGGCC
410584	3-14-3	244	1528	1547	0	GGCCACGCCGGCATCCCGGC
410585	3-14-3	245	1529	1548	0	TGGCCACGCCGGCATCCCGG
410586	3-14-3	246	1530	1549	0	TTGGCCACGCCGGCATCCCG
410587	3-14-3	247	1531	1550	0	CTTGGCCACGCCGGCATCCC
410588	3-14-3	248	1532	1551	0	CCTTGGCCACGCCGGCATCC
410589	3-14-3	249	1533	1552	0	CCCTTGGCCACGCCGGCATC
410590	3-14-3	250	1535	1554	0	CACCCTTGGCCACGCCGGCA
410591	3-14-3	251	1536	1555	0	GCACCCTTGGCCACGCCGGC
410592	3-14-3	252	1537	1556	0	GGCACCCTTGGCCACGCCGG
410593	3-14-3	253	1538	1557	0	TGGCACCCTTGGCCACGCCG
410594	3-14-3	254	1539	1558	0	CTGGCACCCTTGGCCACGCC
410595	3-14-3	255	1540	1559	0	GCTGGCACCCTTGGCCACGC
410596	3-14-3	348	1995	2014	0	AGGGAACCAGGCCTCATTGA
410597	3-14-3	349	1996	2015	0	CAGGGAACCAGGCCTCATTG
410598	3-14-3	350	1998	2017	0	CTCAGGGAACCAGGCCTCAT
410599	3-14-3	351	1999	2018	0	CCTCAGGGAACCAGGCCTCA
410600	3-14-3	352	2000	2019	0	TCCTCAGGGAACCAGGCCTC
410601	3-14-3	353	2001	2020	0	GTCCTCAGGGAACCAGGCCT
410602	3-14-3	354	2003	2022	0	TGGTCCTCAGGGAACCAGGC
410603	3-14-3	355	2004	2023	0	CTGGTCCTCAGGGAACCAGG
410604	3-14-3	356	2005	2024	0	GCTGGTCCTCAGGGAACCAG
410633	3-14-3	411	2356	2375	0	ACTTTGCATTCCAGACCTGG
410634	3-14-3	412	2357	2376	0	GACTTTGCATTCCAGACCTG
410635	3-14-3	413	2358	2377	0	TGACTTTGCATTCCAGACCT
410636	3-14-3	414	2359	2378	0	TTGACTTTGCATTCCAGACC
410637	3-14-3	419	2560	2579	0	CAGATGGCAACGGCTGTCAC
410638	3-14-3	420	2561	2580	0	GCAGATGGCAACGGCTGTCA

TABLE 14-continued

Gapmer antisense compounds having a 3-14-3 motif targeted to SEQ ID NO: 3						
Isis No	Motif	SEQ ID NO	5' Target Site on SEQ ID NO: 3	3' Target Site on SEQ ID NO: 3	Mismatches	Sequence (5'-3')
410639	3-14-3	421	2562	2581	0	AGCAGATGGCAACGGCTGTC
410640	3-14-3	422	2563	2582	0	CAGCAGATGGCAACGGCTGT
410641	3-14-3	423	2564	2583	0	GCAGCAGATGGCAACGGCTG
410642	3-14-3	424	2566	2585	0	CGGCAGCAGATGGCAACGGC
410643	3-14-3	425	2567	2586	0	CCGGCAGCAGATGGCAACGG
410644	3-14-3	426	2568	2587	0	TCCGGCAGCAGATGGCAACG
410645	3-14-3	427	2569	2588	0	CTCCGGCAGCAGATGGCAAC
410646	3-14-3	428	2570	2589	0	GCTCCGGCAGCAGATGGCAA

[0681] In certain embodiments, gap-widened antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gap-widened antisense compounds are targeted to SEQ ID NO: 3. In certain such embodiments, the nucleotide sequences illustrated in Table 12 have a 2-13-5 gap-widened motif. Table 15 illustrates gap-widened antisense compounds targeted to SEQ ID NO: 3, having a 2-13-5 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 15

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 3						
ISIS No	Motiff	SEQ ID NO	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Mismatches	Sequences (5'-3')
410658	2-13-5	243	1527	1546	0	GCCACGCCGGCATCCCGGCC
410659	2-13-5	244	1528	1547	0	GGCCACGCCGGCATCCCGGC
410660	2-13-5	245	1529	1548	0	TGGCCACGCCGGCATCCCGG
410661	2-13-5	246	1530	1549	0	TTGGCCACGCCGGCATCCCG
410662	2-13-5	247	1531	1550	0	CTTGGCCACGCCGGCATCCC
410663	2-13-5	248	1532	1551	0	CCTTGGCCACGCCGGCATCC
410664	2-13-5	249	1533	1552	0	CCCTTGGCCACGCCGGCATC

TABLE 15-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 3						
ISIS No	Motiff	SEQ ID NO	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Mismatches	Sequences (5'-3')
410665	2-13-5	28	1534	1553	0	ACCCTTGGCCACGCCGGCAT
410666	2-13-5	250	1535	1554	0	CACCCTTGGCCACGCCGGCA
410667	2-13-5	251	1536	1555	0	GCACCCTTGGCCACGCCGGC
410668	2-13-5	252	1537	1556	0	GGCACCCTTGGCCACGCCGG
410669	2-13-5	253	1538	1557	0	TGGCACCCTTGGCCACGCCG
410670	2-13-5	254	1539	1558	0	CTGGCACCCTTGGCCACGCC
410671	2-13-5	255	1540	1559	0	GCTGGCACCCTTGGCCACGC
410672	2-13-5	348	1995	2014	0	AGGGAACCAGGCCTCATTGA
410673	2-13-5	349	1996	2015	0	CAGGGAACCAGGCCTCATTG
410674	2-13-5	49	1997	2016	0	TCAGGGAACCAGGCCTCATT
410675	2-13-5	350	1998	2017	0	CTCAGGGAACCAGGCCTCAT
410676	2-13-5	351	1999	2018	0	CCTCAGGGAACCAGGCCTCA

TABLE 15-continued

Gapmer antisense compounds having a 2-13-5 motif targeted to SEQ ID NO: 3

ISIS No	Motiff	SEQ ID NO	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Mismatches	Sequences (5'-3')
410677	2-13-5	352	2000	2019	0	TCCTCAGGGAACCAGGCCTC
410678	2-13-5	353	2001	2020	0	GTCCTCAGGGAACCAGGCCT
410679	2-13-5	50	2002	2021	0	GGTCCTCAGGGAACCAGGCC
410680	2-13-5	354	2003	2022	0	TGGTCCTCAGGGAACCAGGC
410681	2-13-5	355	2004	2023	0	CTGGTCCTCAGGGAACCAGG
410682	2-13-5	356	2005	2024	0	GCTGGTCCTCAGGGAACCAG
410713	2-13-5	60	2355	2374	0	CTTTGCATTCCAGACCTGGG
410714	2-13-5	411	2356	2375	0	ACTTTGCATTCCAGACCTGG
410715	2-13-5	412	2357	2376	0	GACTTTGCATTCCAGACCTG
410716	2-13-5	413	2358	2377	0	TGACTTTGCATTCCAGACCT
410717	2-13-5	414	2359	2378	0	TTGACTTTGCATTCCAGACC
410718	2-13-5	61	2360	2379	0	CTTGACTTTGCATTCCAGAC
410719	2-13-5	419	2560	2579	0	CAGATGGCAACGGCTGTCAC
410720	2-13-5	420	2561	2580	0	GCAGATGGCAACGGCTGTCA
410721	2-13-5	421	2562	2581	0	AGCAGATGGCAACGGCTGTC
410722	2-13-5	422	2563	2582	0	CAGCAGATGGCAACGGCTGT
410723	2-13-5	423	2564	2583	0	GCAGCAGATGGCAACGGCTG
410724	2-13-5	62	2565	2584	0	GGCAGCAGATGGCAACGGCT
410725	2-13-5	424	2566	2585	0	CGGCAGCAGATGGCAACGGC
410726	2-13-5	425	2567	2586	0	CCGGCAGCAGATGGCAACGG
410727	2-13-5	426	2568	2587	0	TCCGGCAGCAGATGGCAACG
410728	2-13-5	427	2569	2588	0	CTCCGGCAGCAGATGGCAAC
410729	2-13-5	428	2570	2589	0	GCTCCGGCAGCAGATGGCAA

[0682] The following embodiments set forth target regions of PCSK9 nucleic acids. Also illustrated are examples of antisense compounds targeted to the target regions. It is understood that the sequence set forth in each SEQ ID NO is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by Isis Number (Isis No) indicate a combination of nucleobase sequence and motif.

[0683] In certain embodiments, antisense compounds target a range of a PCSK9 nucleic acid. In certain embodiment, such compounds contain at least an 8 nucleotide core sequence in common. In certain embodiments, such compounds sharing at least an 8 nucleotide core sequence targets the following nucleotide regions of SEQ ID NO: 3: 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1551, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, or 3727-3751, or 3798-3824.

[0684] In certain embodiments, a target region is nucleotides 290-321 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 290-321 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 117 or 129.

In certain such embodiments, an antisense compound targeted to nucleotides 290-321 of SEQ ID NO: 3 is selected from ISIS NOs: 395202, 399924, 399829 or 399985.

[0685] In certain embodiments, a target region is nucleotides 220-253 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 220-253 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 100 or 116. In certain such embodiments, an antisense compound targeted to nucleotides 220-253 of SEQ ID NO: 3 is selected from ISIS NOs: 399827, 399983, 399828 or 399984.

[0686] In certain embodiments, a target region is nucleotides 377-419 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 377-419 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 81 or 110. In certain such embodiments, an antisense compound targeted to nucleotides 377-419 of SEQ ID NO: 3 is selected from ISIS NOs: 399830, 399986, 395203 or 399925.

[0687] In certain embodiments, a target region is nucleotides 451-615 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 451-615 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 152, 140, 137 or 136. In certain such embodiments, an antisense compound targeted to nucleotides 451-615 of SEQ ID NO: 3 is selected from ISIS NOs: 395204, 399926, 399831, 399987, 395205, 399927, 399832, 399988, 395206 or 399928.

[0688] In certain embodiments, a target region is nucleotides 653-672 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 653-672 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 132. In certain such embodiments, an antisense compound targeted to nucleotides 653-672 of SEQ ID NO: 3 is selected from ISIS NOs: 395207 or 399929.

[0689] In certain embodiments, a target region is nucleotides 741-760 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 741-760 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 139. In certain such embodiments, an antisense compound targeted to nucleotides 741-760 of SEQ ID NO: 3 is selected from ISIS NOs: 395208 or 399930.

[0690] In certain embodiments, a target region is nucleotides 871-897 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 871-897 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 92 or 142. In certain such embodiments, an antisense compound targeted to nucleotides 871-897 of SEQ ID NO: 3 is selected from ISIS NOs: 399833, 399989, 395209 or 399931.

[0691] In certain embodiments, a target region is nucleotides 915-934 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 915-934 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 113. In certain such embodiments, an antisense compound targeted to nucleotides 915-934 of SEQ ID NO: 3 is selected from ISIS NOs: 395210 or 399932.

[0692] In certain embodiments, a target region is nucleotides 968-1017 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 968-1017 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 98 or 138. In certain such embodiments, an antisense compound targeted to nucleotides 968-1017 of SEQ ID NO: 3 is selected from ISIS NOs: 395211, 399933, 395212, or 399934.

[0693] In certain embodiments, a target region is nucleotides 1075-1124 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1075-1124 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 96 or 124. In certain such embodiments, an antisense compound targeted to nucleotides 1075-1124 of SEQ ID NO: 3 is selected from ISIS NOs: 395213, 399935, 395214, or 399936.

[0694] In certain embodiments, a target region is nucleotides 1075-1174 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1075-1174 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 82, 96, 103, 124, or 133. In certain such embodiments, an antisense compound targeted to nucleotides 1075-1174 of SEQ ID NO: 3 is selected from ISIS NOs: 395213, 399935, 395214, 399936, 399834, 399990, 395215, 399937, 395216, or 399938.

[0695] In certain embodiments, a target region is nucleotides 1144-1174 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1144-1174 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 82, 133 or 103. In certain such embodiments, an antisense compound targeted to nucleotides 1144-1174 of SEQ ID NO: 3 is selected from ISIS NOs: 399834, 399990, 395215, 399937, 395216 or 399938.

[0696] In certain embodiments, a target region is nucleotides 1275-1302 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1275-1302 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 155 or 156. In certain such embodiments, an antisense compound targeted to nucleotides 1275-1302 of SEQ ID NO: 3 is selected from ISIS NOs: 395217, 399939, 399835 or 399991.

[0697] In certain embodiments, a target region is nucleotides 1315-1341 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1315-1341 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 146 or 161. In certain such embodiments, an antisense compound targeted to nucleotides 1315-1341 of SEQ ID NO: 3 is selected from ISIS NOs: 395218, 399940, 395219 or 399941.

[0698] In certain embodiments, a target region is nucleotides 1351-1389 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1351-1389 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 215, 216, 19, 217 or 20. In certain such embodiments, an antisense compound targeted to nucleotides 1351-1389 of SEQ ID NO: 3 is selected from ISIS NOs: 410753, 406025, 395160, 399882, 406026, 399797 or 399953.

[0699] In certain embodiments, a target region is nucleotides 1351-1447 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1351-1447 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 215, 216, 19, 217, 20, 21, 218, 219, 220, 221, 22,222, 223, 224, 225, 226, 227, 228, 23, 229, 230,231, 232, 24, or 233. In certain such embodiments, an antisense compound targeted to nucleotides 1351-1447 of SEQ ID NO: 3 is selected from ISIS NOs: 410753, 406025, 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, 399884 or 410754.

[0700] In certain embodiments, a target region is nucleotides 1365-1439 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1365-1439 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 19, 20, 21, 22, 23, 24, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, or 232. In certain such embodiments, an antisense compound targeted to nucleotides 1365-1439 of SEQ ID NO: 3 is selected from ISIS NOs: 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, or 399884.

[0701] In certain embodiments, a target region is nucleotides 1390-1417 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1390-1417 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 218, 219, 220, 221, 22, 222 or 223. In certain such embodiments, an antisense compound targeted to nucleotides 1390-1417 of SEQ ID NO: 3 is selected from ISIS NOs: 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873 or 405874.

[0702] In certain embodiments, a target region is nucleotides 1390-1429 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1390-1429 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, or 228. In certain such embodiments, an antisense compound targeted to nucleotides 1390-1429 of SEQ ID NO: 3 is selected from ISIS NOs: 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, or 399955.

[0703] In certain embodiments, a target region is nucleotides 1390-1439 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1390-1439 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 21, 22, 23, 24, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, or 232. In certain such embodiments, an antisense compound targeted to nucleotides 1390-1439 of SEQ ID NO: 3 is selected from ISIS NOs: 395160, 399882, 406026, 399797, 399953, 395161, 399883, 405869, 405870, 405871, 405872, 399798, 399954, 405873, 405874, 405875, 405876, 406027, 406028, 406029, 399799, 399955, 406030, 406031, 410733, 406032, 395162, or 399884.

[0704] In certain embodiments, a target region is nucleotides 1399-1425 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1399-1425 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 224, 225, 226 or 227. In certain such embodiments, an antisense compound targeted to nucleotides 1399-1425 of SEQ ID NO: 3 is selected from ISIS NOs: 405875, 405876, 406027 or 406028.

[0705] In certain embodiments, a target region is nucleotides 1408-1435 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1408-1435 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 228, 23, 229, 230 or 231. In certain such embodiments, an antisense compound targeted to nucleotides 1408-1435 of SEQ ID NO: 3 is selected from ISIS NOs: 406029, 399799, 399955, 406030, 406031 or 410733.

[0706] In certain embodiments, a target region is nucleotides 1420-1447 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1420-1447 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 24 or 233. In certain such embodiments, an antisense compound targeted to nucleotides 1420-1447 of SEQ ID NO: 3 is selected from ISIS NOs: 395162, 399884 or 410754.

[0707] In certain embodiments, a target region is nucleotides 1453-1482 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1453-1482 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 25 or 234. In certain such embodiments, an antisense compound targeted to nucleotides 1453-1482 of SEQ ID NO: 3 is selected from ISIS NOs: 395163, 399885 or 410755.

[0708] In certain embodiments, a target region is nucleotides 1490-1516 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1490-1516 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 235, 236, or 237. In certain such embodiments, an antisense compound targeted to nucleotides 1490-1516 of SEQ ID NO: 3 is selected from ISIS NOs: 410756, 405604, or 406033.

[0709] In certain embodiments, a target region is nucleotides 1490-1564 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1490-1564 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 235, 236, 237, 26, 238, 27, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 447, 28, 250, 251, 252, 253, 254, 255, or 256. In certain such embodiments, an antisense compound targeted to nucleotides 1490-1564 of SEQ ID NO: 3 is selected from ISIS NOs: 410756, 405604, 406033, 395164, 399886, 406034, 399800, 399956, 406035, 410757, 406036, 406037, 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, or 410758.

[0710] In certain embodiments, a target region is nucleotides 1500-1526 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1500-1526 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 26, 238, 27, or 239. In certain such embodiments, an antisense compound targeted to nucleotides 1500-1526 of SEQ ID NO: 3 is selected from ISIS NOs: 395164, 399886, 406034, 399800, 399956, or 406035.

[0711] In certain embodiments, a target region is nucleotides 1515-1541 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1515-1541 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 240, 241, or 242. In certain such embodiments, an antisense compound targeted to nucleotides 1515-1541 of SEQ ID NO: 3 is selected from ISIS NOs: 399886, 406034, 399800, 399956, or 406035.

[0712] In certain embodiments, a target region is nucleotides 1527-1553 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1527-1553 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 243, 244, 245, 246, 247, 248, 447, 28 or 249. In certain such embodiments, an antisense compound targeted to nucleotides 1527-1553 of SEQ ID NO: 3 is selected from ISIS NOs: 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, or 410665.

[0713] In certain embodiments, a target region is nucleotides 1527-1554 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1527-1554 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 243, 244, 245, 246, 247, 248, 249, 447, 28, or 250. In certain such embodiments, an antisense compound targeted to nucleotides 1527-1554 of SEQ ID NO: 3 is selected from ISIS NOs: 410536, 410583, 410658, 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 405881, 410590, 410666, or 410665.

[0714] In certain embodiments, a target region is nucleotides 1528-1554 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1528-1554 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 244, 245, 246, 247, 248, 249, 447, 28, or 250. In certain such embodiments, an antisense compound targeted to nucleotides 1528-1554 of SEQ ID NO: 3 is selected from ISIS NOs: 410537, 410584, 410659, 406038, 410585, 410660, 405877, 410586, 410661, 405878, 410587, 410662, 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 409126, 410665, or 405881.

[0715] In certain embodiments, a target region is nucleotides 1529-1555 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1529-1555 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 245, 246, 247, 248, 249, 447, 28, 250 or 251. In certain such embodiments, an antisense compound targeted to nucleotides 1529-1555 of SEQ ID NO: 3 is selected from ISIS NOs: 406038, 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 409126, 410585, 410586, 410587, 410588, 410589, 410590, 410591, 410660, 410661, 410662, 410663, 410664, 410665, 410666, 410667, or 410667.

[0716] In certain embodiments, a target region is nucleotides 1529-1556 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1529-1556 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 245, 246, 247, 248, 249, 447, 28, 250, 251, or 252. In certain such embodiments, an antisense compound targeted to nucleotides 1529-1556 of SEQ ID NO: 3 is selected from ISIS NOs: 406038, 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 409126, 410585, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410660, 410661, 410662, 410663, 410664, 410665, 410666, 410667, or 410668.

[0717] In certain embodiments, a target region is nucleotides 1530-1556 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1530-1556 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 246, 247, 248, 249, 447, 28, 250, 251, or 252. In certain such embodiments, an antisense compound targeted to nucleotides 1530-1556 of SEQ ID NO: 3 is selected from ISIS NOs: 406038, 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 409126, 410585, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410660, 410661, 410662, 410663, 410664, 410665, 410666, 410667, 405884, 410593, 410669 or 410668.

[0718] In certain embodiments, a target region is nucleotides 1530-1557 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1530-1557 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 246, 247, 248, 249, 250, 251, 252, 253, or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1530-1557 of SEQ ID NO: 3 is selected from ISIS NOs: 395165, 399887, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410586, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410661, 410662, 410663, 410664, 410665, 410666, 410667, 410668, or 410669.

[0719] In certain embodiments, a target region is nucleotides 1531-1557 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1531-1557 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 247, 248, 249, 250, 251, 252, 253, or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1531-1557 of SEQ ID NO: 3 is selected from ISIS NOs: 395165, 399887, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 409126, 410587, 410588, 410589, 410590, 410591, 410592, 410593, 410662, 410663, 410664, 410665, 410666, 410667, 410668, or 410669.

[0720] In certain embodiments, a target region is nucleotides 1532-1551 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1532-1551 of SEQ ID NO: 3. In one such embodiment, an antisense compound targeted to nucleotides 1532-1551 of SEQ ID NO: 3 is ISIS NO: 405879.

[0721] In certain embodiments, a target region is nucleotides 1532-1558 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1532-1558 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 248, 249, 250, 251, 252, 253, 254 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1532-1558 of SEQ ID NO: 3 is selected from ISIS NOs: 405879, 410588, 410663, 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, or 410670.

[0722] In certain embodiments, a target region is nucleotides 1533-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1533-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 249, 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1533-1559 of SEQ ID NO: 3 is selected from ISIS NOs: 405880, 410589, 410664, 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0723] In certain embodiments, a target region is nucleotides 1534-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1534-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 28, 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1534-1559 of SEQ ID NO: 3 is selected from ISIS NOs: 395165, 399887, 410665, 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, or 410671.

[0724] In certain embodiments, a target region is nucleotides 1535-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1535-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 250, 251, 252, 253, 254, 255 or 447. In certain such embodiments, an antisense compound targeted to nucleotides 1535-1559 of SEQ ID NO: 3 is selected from ISIS NOs: 405881, 410590, 410666, 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594 or 410671.

[0725] In certain embodiments, a target region is nucleotides 1536-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1536-1559 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 251, 252, 253, 254, or 255. In certain such embodiments, an antisense compound targeted to nucleotides 1536-1559 of SEQ ID NO: 3 is selected from ISIS NOs: 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, or 410671.

[0726] In certain embodiments, a target region is nucleotides 1537-1564 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1537-1564 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 251, 252, 253, 254, 255, or 256. In certain such embodiments, an antisense compound targeted to nucleotides 1537-1564 of SEQ ID NO: 3 is selected from ISIS NOs: 405882, 410591, 410667, 405883, 410592, 410668, 405884, 410593, 410669, 410538, 410594, 410670, 410539, 410595, 410671, or 410758.

[0727] In certain embodiments, a target region is nucleotides 1566-1602 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1566-1602 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 29, 30, 31, 32, 257, 258, 259, 260, 261, or 262. In certain such embodiments, an antisense compound targeted to nucleotides 1566-1602 of SEQ ID NO: 3 is selected from ISIS NOs: 395166, 395167, 395168, 399801, 399888, 399889, 399890, 399957, 405909, 405910, 405911, 405912, 406039, 406040, 406041, 406042, 406043, or 406044.

[0728] In certain embodiments, a target region is nucleotides 1566-1681 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1566-1681 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 29, 30, 31, 32, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, or 272. In certain such embodiments, an antisense compound targeted to nucleotides 1566-1681 of SEQ ID NO: 3 is selected from ISIS NOs: 395166, 395167, 395168, 399801, 399888, 399889, 399890, 399957, 405909, 405910, 405911, 405912, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 408642, 410759, 410760, 410761, or 410762.

[0729] In certain embodiments, a target region is nucleotides 1606-1626 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1606-1626 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 264 or 31. In certain such embodiments, an antisense compound targeted to nucleotides 1606-1626 of SEQ ID NO: 3 is selected from ISIS NOs: 408642, 395167, or 399889.

[0730] In certain embodiments, a target region is nucleotides 1618-1645 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1618-1645 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 265 or 266.

In certain such embodiments, an antisense compound targeted to nucleotides 1618-1645 of SEQ ID NO: 3 is selected from ISIS NOs: 410760 or 406045.

[0731] In certain embodiments, a target region is nucleotides 1626-1653 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1626-1653 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 32, 266, 267, 268, or 269. In certain such embodiments, an antisense compound targeted to nucleotides 1626-1653 of SEQ ID NO: 3 is selected from ISIS NOs: 395168, 399890, 405909, 405910, 405911, or 406045.

[0732] In certain embodiments, a target region is nucleotides 1684-1703 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1684-1703 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 273. In certain such embodiments, an antisense compound targeted to nucleotides 1684-1703 of SEQ ID NO: 3 is selected from ISIS NOs: 410763.

[0733] In certain embodiments, a target region is nucleotides 1730-1781 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1730-1781 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 33, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, or 285. In certain such embodiments, an antisense compound targeted to nucleotides 1730-1781 of SEQ ID NO: 3 is selected from ISIS NOs: 395169, 399891, 405913, 405914, 405915, 405916, 405917, 405918, 405919, 405920, 405921, 405922, 410734, or 410764.

[0734] In certain embodiments, a target region is nucleotides 1740-1767 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1740-1767 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 33, 275, 276, 277, or 278. In certain such embodiments, an antisense compound targeted to nucleotides 1740-1767 of SEQ ID NO: 3 is selected from ISIS NOs: 395169, 399891, 405913, 405914, 405915, or 405916.

[0735] In certain embodiments, a target region is nucleotides 1749-1775 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1749-1775 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 279, 280, 281 or 282. In certain such embodiments, an antisense compound targeted to nucleotides 1749-1775 of SEQ ID NO: 3 is selected from ISIS NOs: 410734, 405917, 405918 or 405919.

[0736] In certain embodiments, a target region is nucleotides 1758-1781 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1758-1781 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 283, 284, or 285. In certain such embodiments, an antisense compound targeted to nucleotides 1758-1781 of SEQ ID NO: 3 is selected from ISIS NOs: 405920, 405921, or 405922.

[0737] In certain embodiments, a target region is nucleotides 1820-1847 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1820-1847 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 39, 306, 307, 308, or 309. In certain such embodiments, an antisense compound targeted to nucleotides 1820-1847 of SEQ ID NO: 3 is selected from ISIS NOs: 395175, 399897, 405937, 405938, 405939, or 405940.

[0738] In certain embodiments, a target region is nucleotides 1820-1877 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1820-1877 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 39, 40, 41, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, or 318. In certain such embodiments, an antisense compound targeted to nucleotides 1820-1877 of SEQ ID NO: 3 is selected from ISIS NOs: 395175, 399804, 399805, 399897, 399960, 399961, 405937, 405938, 405939, 405940, 405941, 405942, 405943, 405944, 405945, 405946, 405947, 410737, or 410769.

[0739] In certain embodiments, a target region is nucleotides 1822-2198 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1822-2198 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 39, 307, 308, 309, 310, 40, 311, 312, 313, 314, 315, 41, 41, 316, 317, 318, 101, 319, 42, 320, 321, 322, 43, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 44, 335, 336, 337, 45, 338, 46, 339, 340, 47, 341, 342, 48, 343, 344, 345, 346, 347, 3, or 366. In certain such embodiments, an antisense compound targeted to nucleotides 1822-2198 of SEQ ID NO: 3 is selected from ISIS NOs: 395175, 399897, 405938, 405939, 405940, 405941, 399804, 399960, 405942, 405943, 410737, 405944, 405945, 399805, 399961, 405946, 405947, 410769, 395176, 399898, 405948, 399806, 399962, 405949, 405950, 405951, 399807, 399963, 405952, 410738, 405953, 405954, or 405979.

[0740] In certain embodiments, a target region is nucleotides 1830-1856 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1830-1856 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 40, 310, 311, or 312. In certain such embodiments, an antisense compound targeted to nucleotides 1830-1856 of SEQ ID NO: 3 is selected from ISIS NOs: 399804, 399960, 405941, 405942, or 405943.

[0741] In certain embodiments, a target region is nucleotides 1839-1865 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1839-1865 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 41, 313, 314, 315, or 316. In certain such embodiments, an antisense compound targeted to nucleotides 1839-1865 of SEQ ID NO: 3 is selected from ISIS NOs: 399805, 399961, 405944, 405945, 405946, or 410737.

[0742] In certain embodiments, a target region is nucleotides 1840-1867 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1840-1867 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 314, 315, 316, or 317. In certain such embodiments, an antisense compound targeted to nucleotides 1840-1867 of SEQ ID NO: 3 is selected from ISIS NOs: 399961, 405944, 405945, 405946, 410737, or 405947.

[0743] In certain embodiments, a target region is nucleotides 1898-1924 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1898-1924 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 42, 101, 319, or 320. In certain such embodiments, an antisense compound targeted to nucleotides 1898-1924 of SEQ ID NO: 3 is selected from ISIS NOs: 395176, 399898, 405948, 399806, 399962, 405949, or 405949.

[0744] In certain embodiments, a target region is nucleotides 1898-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1898-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 87, 101, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 1898-2035 of SEQ ID NO: 3 is selected from ISIS NOs: 395176, 395177, 395178, 395179, 399806, 399807, 399808, 399809, 399810, 399811, 399812, 399813, 399898, 399899, 399900, 399901, 399962, 399963, 399964, 399965, 399966, 399967, 399968, 399969, 405885, 405886, 405887, 405888, 405889, 405890, 405891, 405892, 405948, 405949, 405950, 405951, 405952, 405953, 405954, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 405962, 405963, 405964, 405965, 405966, 405967, 405968, 405969, 405970, 405971, 405972, 405973, 405974, 408653, 410540, 410596, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410604, 410672, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, 410681, 410682, 410738, 410739, 410740, or 410770.

[0745] In certain embodiments, a target region is nucleotides 1903-2127 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1903-2127 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 42, 320, 321, 322, 43, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 44, 335, 336, 337, 45, 338, 46, 339, 340, 47, 341, 342, 48, 343, 344, 345, 346, 347, 348, 349, 49, 350, 351, 352, 353, 50, 354, 355, 356, 357, 87, 358, 359, 51, 119, 360, or 54. In certain such embodiments, an antisense compound targeted to nucleotides 1903-2127 of SEQ ID NO: 3 is selected from ISIS NOs: 399806, 399962, 405949, 405950, 405951, 399807, 399963, 405952, 410738, 405953, 405954, 410770, 405955, 405956, 405957, 405958, 405959, 405960, 405961, 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, or 399967.

[0746] In certain embodiments, a target region is nucleotides 1907-1934 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1907-1934 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 321, 322, 323, or 324. In certain such embodiments, an antisense compound targeted to nucleotides 1907-1934 of SEQ ID NO: 3 is selected from ISIS NOs: 405950, 405951, 399807, 399963, 405952, or 410738.

[0747] In certain embodiments, a target region is nucleotides 1911-1938 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1911-1938 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 43, 323, 324, 325, or 326. In certain such embodiments, an antisense compound targeted to nucleotides 1911-1938 of SEQ ID NO: 3 is selected from ISIS NOs: 399807, 399963, 405952, 410738, 405953, or 405954.

[0748] In certain embodiments, a target region is nucleotides 1946-1971 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1946-1971 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 328, 329, 330, or 331. In certain such embodiments, an antisense compound targeted to nucleotides 1946-1971 of SEQ ID NO: 3 is selected from ISIS NOs: 405955, 405956, 405957, or 405958.

[0749] In certain embodiments, a target region is nucleotides 1954-1980 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1954-1980 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 332, 333, 334, or 335. In certain such embodiments, an antisense compound targeted to nucleotides 1954-1980 of SEQ ID NO: 3 is selected from ISIS NOs: 405959, 405960, 405961, 399808, 399964, or 405962.

[0750] In certain embodiments, a target region is nucleotides 1959-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1959-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 335, 336, 337, 45, 338, 46, 339, 340, 47, 341, 342, 48, 343, 344, 345, 346, 347, 348, 349, 49, 350, 351, 352, 353, 50, 354, 355, 356, 357, 87, 358, 359, or 51. In certain such embodiments, an antisense compound targeted to nucleotides 1959-2035 of SEQ ID NO: 3 is selected from ISIS NOs: 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, 399966, 405967, 405968, 399811, 399967, 405969, 405970, 410740, 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, or 399969.

[0751] In certain embodiments, a target region is nucleotides 1959-2057 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1959-2057 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 44, 335, 336, 337, 45, 338, 46, 339, 340, 47, 341, 342, 48, 343, 344, 345, 346, 347, 348, 349, 49, 350, 351, 352, 353, 50, 354, 355, 356, 357, 87, 358, 359, 51, or 119. In certain such embodiments, an antisense compound targeted to nucleotides 1959-2057 of SEQ ID NO: 3 is selected from ISIS NOs: 399808, 399964, 405962, 410739, 405963, 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, 399966, 405967, 405968, 399811, 399967, 405969, 405970, 410740, 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, or 399902.

[0752] In certain embodiments, a target region is nucleotides 1963-1988 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1963-1988 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 45, 336, 337 or 338. In certain such embodiments, an antisense compound targeted to nucleotides 1963-1988 of SEQ ID NO: 3 is selected from ISIS NOs: 410739, 405963, 395177, 399899, or 405964.

[0753] In certain embodiments, a target region is nucleotides 1967-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1967-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 45, 338, 46, 339, 340, 47, 341, 342, 48, 343, 344, 345, 346, 347, 348, 349, 49, 350, 351, 352, 353, 50, 354, 355, 356, 357, 87, 358, 359 or 51. In certain such embodiments, an antisense compound targeted to nucleotides 1967-2035 of SEQ ID NO: 3 is selected from ISIS NOs: 395177, 399899, 405964, 399809, 399965, 405965, 405966, 399810, 399966, 405967, 405968, 399811, 399967, 405969, 405970, 410740, 405885, 405886, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, or 399969.

[0754] In certain embodiments, a target region is nucleotides 1972-1999 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1972-1999 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 46, 47, 339, 340, or 341. In certain such embodiments, an antisense compound targeted to nucleotides 1972-1999 of SEQ ID NO: 3 is selected from ISIS NOs: 399809, 399965, 405965, 405966, 399810, 399966, or 405967.

[0755] In certain embodiments, a target region is nucleotides 1982-2008 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1982-2008 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 48, 342, 343, or 344. In certain such embodiments, an antisense compound targeted to nucleotides 1982-2008 of SEQ ID NO: 3 is selected from ISIS NOs: 405968, 399811, 399967, 405969, or 405970.

[0756] In certain embodiments, a target region is nucleotides 1991-2018 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1991-2018 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 345, 346, 347, 348, 349, 350, or 351. In certain such embodiments, an antisense compound targeted to nucleotides 1991-2018 of SEQ ID NO: 3 is selected from ISIS NOs: 399812, 399968, 405885, 405886, 405887, 405888, 405889, 405890, 410596, 410597, 410598, 410599, 410672, 410673, 410674, 410675, 410676, or 410740.

[0757] In certain embodiments, a target region is nucleotides 1993-2019 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1993-2019 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 346, 347, 348, 349, 350, or 354. In certain such embodiments, an antisense compound targeted to nucleotides 1993-2019 of SEQ ID NO: 3 is selected from ISIS NOs: 405885, 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, or 410677.

[0758] In certain embodiments, a target region is nucleotides 1995-2022 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1995-2022 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 348, 349, 350, 351, 352, 353, 50, or 351 In certain such embodiments, an antisense compound targeted to nucleotides 1995-2022 of SEQ ID NO: 3 is selected from ISIS NOs: 405887, 410596, 410672, 405888, 410597, 410673, 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 405892, 410601, 410678, 395178, 399900, 410679, 408653, 410602, or 410680.

[0759] In certain embodiments, a target region is nucleotides 1996-2023 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1996-2023 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 349, 350, 351, 352, 353, 354, or 355. In certain such embodiments, an antisense compound targeted to nucleotides 1996-2023 of SEQ ID NO: 3 is selected from ISIS NOs: 395178, 399812, 399900, 399968, 405888, 405889, 405890, 405891, 405892, 405971, 408653, 410597, 410598, 410599, 410600, 410601, 410602, 410603, 410673, 410674, 410675, 410676, 410677, 410678, 410679, 410680, or 410681.

[0760] In certain embodiments, a target region is nucleotides 1997-2024 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1997-2024 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 49, 50, 350, 351, 352, 353, 354, 355, or 356. In certain such embodiments, an antisense compound targeted to nucleotides 1997-2024 of SEQ ID NO: 3 is selected from ISIS NOs: 399812, 399968, 410674, 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, or 410682.

[0761] In certain embodiments, a target region is nucleotides 1998-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1998-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 350, 351, 352, 353, 354, 355, 356 or 357. In certain such embodiments, an antisense compound targeted to nucleotides 1998-2025 of SEQ ID NO: 3 is selected from ISIS NOs: 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0762] In certain embodiments, a target region is nucleotides 1999-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 1999-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 50, 351, 352, 353, 354, 355, 356, or 357. In certain such embodiments, an antisense compound targeted to nucleotides 1999-2025 of SEQ ID NO: 3 is selected from ISIS NOs: 405889, 410598, 410675, 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0763] In certain embodiments, a target region is nucleotides 2000-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2000-2025 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 351, 352, 353, 354, 355, 356, or 357. In certain such embodiments, an antisense compound targeted to nucleotides 2000-2025 of SEQ ID NO: 3 is selected from ISIS NOs: 405890, 410599, 410676, 405891, 410600, 410677, 405892, 410601, 410678, 395178, 399900, 410679, 408653, 410602, 410680, 405971, 410603, 410681, 410540, 410604, 410682, or 405972.

[0764] In certain embodiments, a target region is nucleotides 2009-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2009-2035 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 51, 87, 358, or 359. In certain such embodiments, an antisense compound targeted to nucleotides 2009-2035 of SEQ ID NO: 3 is selected from ISIS NOs: 395179, 399813, 399901, 399969, 405973, or 405974.

[0765] In certain embodiments, a target region is nucleotides 2038-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2038-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 54, 104, 119, 130, 360, 361, 362, 363, 365, 366, 367, 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 2038-2139 of SEQ ID NO: 3 is selected from ISIS NOs: 395180, 395181, 395182, 395220, 399814, 399836, 399902, 399903, 399904, 399942, 399970, 399992, 405975, 405976, 405977, 405978, 405979, 405980, 405981, 405982, 405983, 410741, or 410771.

[0766] In certain embodiments, a target region is nucleotides 2061-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2061-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 54, 104, 130, 360, 361, 362, 363, 365, 366, 367, 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 2061-2139 of SEQ ID NO: 3 is selected from ISIS NOs: 395181, 395182, 395220, 399814, 399836, 399903, 399904, 399942, 399970, 399992, 405975, 405976, 405977, 405978, 405979, 405980, 405981, 405982, 405983, 410741, or 410771.

[0767] In certain embodiments, a target region is nucleotides 2073-2099 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2073-2099 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 52, 53, 361, or 362. In certain such embodiments, an antisense compound targeted to nucleotides 2073-2099 of SEQ ID NO: 3 is selected from ISIS NOs: 395181, 399814, 399903, 399970, 405975, or 405976.

[0768] In certain embodiments, a target region is nucleotides 2078-2104 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2078-2104 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 53, 104, 362, or 363. In certain such embodiments, an antisense compound targeted to nucleotides 2078-2104 of SEQ ID NO: 3 is selected from ISIS NOs: 395220, 399814, 399942, 399970, 405976, or 405977.

[0769] In certain embodiments, a target region is nucleotides 2105-2131 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2105-2131 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 54, 365, 366, or 367. In certain such embodiments, an antisense compound targeted to nucleotides 2105-2131 of SEQ ID NO: 3 is selected from ISIS NOs: 395182, 399904, 405978, 405979, or 405980.

[0770] In certain embodiments, a target region is nucleotides 2112-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2112-2139 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 367, 368, 369, 370, or 371. In certain such embodiments, an antisense compound targeted to nucleotides 2112-2139 of SEQ ID NO: 3 is selected from ISIS NOs: 405980, 405981, 410741, 405982, or 405983.

[0771] In certain embodiments, a target region is nucleotides 2168-2198 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2168-2198 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 56, 372, 373, 374, or 375. In certain such embodiments, an antisense compound targeted to nucleotides 2168-2198 of SEQ ID NO: 3 is selected from ISIS NOs: 399815, 399816, 399971, 399972, 405641, 405984, 405985, 405986, or 405987.

[0772] In certain embodiments, a target region is nucleotides 2170-2177 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2170-2177 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 55, 373, 374, or 375. In certain such embodiments, an antisense compound targeted to nucleotides 2170-2177 of SEQ ID NO: 3 is selected from ISIS NOs: 399815, 399971, 405641, 405985, 405986, or 405987.

[0773] In certain embodiments, a target region is nucleotides 2245-2284 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2245-2284 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 408 or 409.

In certain such embodiments, an antisense compound targeted to nucleotides 2245-2284 of SEQ ID NO: 3 is selected from ISIS NOs: 410776 or 410777.

[0774] In certain embodiments, a target region is nucleotides 2295-2394 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2295-2394 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 410, 411, 412, 413, 414, 415, or 416. In certain such embodiments, an antisense compound targeted to nucleotides 2295-2394 of SEQ ID NO: 3 is selected from ISIS NOs: 395186, 399817, 399908, 399973, 405996, 405997, 410562, 410563, 410564, 410633, 410634, 410635, 410636, 410713, 410714, 410715, 410716, 410717, 410718, 410778, or 410779.

[0775] In certain embodiments, a target region is nucleotides 2355-2381 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2355-2381 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414, or 415. In certain such embodiments, an antisense compound targeted to nucleotides 2355-2381 of SEQ ID NO: 3 is selected from ISIS NOs: 395186, 399817, 399908, 399973, 405996, 405997, 410562, 410563, 410564, 410633, 410634, 410635, 410636, 410713, 410714, 410715, 410716, 410717, or 410718.

[0776] In certain embodiments, a target region is nucleotides 2355-2394 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2355-2394 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 60, 61, 411, 412, 413, 414, 415, or 416. In certain such embodiments, an antisense compound targeted to nucleotides 2355-2394 of SEQ ID NO: 3 is selected from ISIS NOs: 395186, 399817, 399908, 399973, 405996, 405997, 410562, 410563, 410564, 410633, 410634, 410635, 410636, 410713, 410714, 410715, 410716, 410717, 410718, or 410779.

[0777] In certain embodiments, a target region is nucleotides 2405-2461 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2405-2461 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 417 or 418. In certain such embodiments, an antisense compound targeted to nucleotides 2405-2461 of SEQ ID NO: 3 is selected from ISIS NOs: 410780 or 410781.

[0778] In certain embodiments, a target region is nucleotides 2560-2587 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2560-2587 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425, or 426. In certain such embodiments, an antisense compound targeted to nucleotides 2560-2587 of SEQ ID NO: 3 is selected from ISIS NOs: 395187, 399909, 405998, 410565, 410566, 410567, 410568, 410569, 410570, 410571, 410637, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410719, 410720, 410721, 410722, 410723, 410724, 410725, 410726, or 410727.

[0779] In certain embodiments, a target region is nucleotides 2560-2609 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2560-2609 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, or 430. In certain such embodiments, an antisense compound targeted to nucleotides 2560-2609 of SEQ ID NO: 3 is selected from ISIS NOs: 395187, 399909, 405998, 410565, 410566, 410567, 410568, 410569, 410570, 410571, 410572, 410573, 410637, 410638, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410719, 410720, 410721, 410722, 410723, 410724, 410725, 410726, 410727, 410728, or 410783.

[0780] In certain embodiments, a target region is nucleotides 2561-2588 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2561-2588 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 420, 421, 422, 423, 424, 425, 426, 427, or 62. In certain such embodiments, an antisense compound targeted to nucleotides 2561-2588 of SEQ ID NO: 3 is selected from ISIS NOs: 410566, 410638, 410720, 410567, 410639, 410721, 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, or 410728.

[0781] In certain embodiments, a target region is nucleotides 2562-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2562-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 62, 421, 422, 423, 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2562-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 395187, 399909, 405998, 410567, 410568, 410569, 410570, 410571, 410572, 410573, 410639, 410640, 410641, 410642, 410643, 410644, 410645, 410646, 410721, 410722, 410723, 410724, 410725, 410726, 410727, 410728, or 410729.

[0782] In certain embodiments, a target region is nucleotides 2563-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2563-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 422, 423, 424, 425, 426, 427, 428 or 62. In certain such embodiments, an antisense compound targeted to nucleotides 2563-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 410568, 410640, 410722, 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0783] In certain embodiments, a target region is nucleotides 2564-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2564-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 423, 424, 425, 426, 427, 428 or 62. In certain such embodiments, an antisense compound targeted to nucleotides 2564-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 410569, 410641, 410723, 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0784] In certain embodiments, a target region is nucleotides 2565-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2565-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 424, 425, 426, 427, 428 or 62. In certain such embodiments, an antisense compound targeted to nucleotides 2565-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 395187, 399909, 410724, 410570, 410642, 410725, 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0785] In certain embodiments, a target region is nucleotides 2566-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2566-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 424, 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2566-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 410570, 410642, 410725, 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0786] In certain embodiments, a target region is nucleotides 2567-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2567-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 425, 426, 427, or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2567-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 410571, 410643, 410726, 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0787] In certain embodiments, a target region is nucleotides 2568-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2568-2589 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 426, 427 or 428. In certain such embodiments, an antisense compound targeted to nucleotides 2568-2589 of SEQ ID NO: 3 is selected from ISIS NOs: 405988, 405998, 410644, 410727, 410572, 410645, 410728, 410573, 410646, or 410729.

[0788] In certain embodiments, a target region is nucleotides 2665-2689 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2665-2689 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 63 or 154. In certain such embodiments, an antisense compound targeted to nucleotides 2665-2689 of SEQ ID NO: 3 is selected from ISIS NOs: 395188, 399910, 399818, or 399974.

[0789] In certain embodiments, a target region is nucleotides 2759-2783 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2759-2783 SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 64 or 65. In certain such embodiments, an antisense compound targeted to nucleotides 2759-2783 of SEQ ID NO: 3 is selected from ISIS NOs: 395189, 399911, 399819, or 399975.

[0790] In certain embodiments, a target region is nucleotides 2837-2880 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2837-2880 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 66, 67, or 122. In certain such embodiments, an antisense compound targeted to nucleotides 2837-2880 of SEQ ID NO: 3 is selected from ISIS NOs: 399820, 399976, 395190, 399912, 395191, or 399913.

[0791] In certain embodiments, a target region is nucleotides 2904-2923 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 2904-2923 SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 153. In certain such embodiments, an antisense compound targeted to nucleotides 2904-2923 of SEQ ID NO: 3 is selected from ISIS NOs: 395192 or 399914.

[0792] In certain embodiments, a target region is nucleotides 3005-3024 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3005-3024 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 68. In certain such embodiments, an antisense compound targeted to nucleotides 3005-3024 of SEQ ID NO: 3 is selected from ISIS NOs: 395193 or 399915.

[0793] In certain embodiments, a target region is nucleotides 3005-3174 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3005-3174 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 68, 431, 432, 433, 434, 69, 435, 436, 437, 438 or 70. In certain such embodiments, an antisense compound targeted to nucleotides 3005-3174 of SEQ ID NO: 3 is selected from ISIS NOs: 395193, 399915, 405893, 405894, 405895, 405896, 395194, 399916, 405897, 405898, 405899, 405900, 395195, or 399917.

[0794] In certain embodiments, a target region is nucleotides 3083-3110 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3083-3110 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 69, 69, 431, 432, 433, 434, 435, 436, 437, or 438. In certain such embodiments, an antisense compound targeted to nucleotides 3083-3110 of SEQ ID NO: 3 is selected from ISIS NOs: 395194, 399916, 405893, 405894, 405895, 405896, 405897, 405898, 405899, or 405900.

[0795] In certain embodiments, a target region is nucleotides 3155-3184 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3155-3184 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 70, 71, 439, 440, 441, 442, 443, 444, 445, or 446. In certain such embodiments, an antisense compound targeted to nucleotides 3155-3184 of SEQ ID NO: 3 is selected from ISIS NOs: 395195, 399821, 399917, 399977, 405901, 405902, 405903, 405904, 405905, 405906, 405907, or 405908.

[0796] In certain embodiments, a target region is nucleotides 3238-3268 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3238-3268 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 72, 73, or 135. In certain such embodiments, an antisense compound targeted to nucleotides 3238-3268 of SEQ ID NO: 3 is selected from ISIS NOs: 395196, 399822, 399823, 399918, 399978, or 399979.

[0797] In certain embodiments, a target region is nucleotides 3482-3711 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3482-3711 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 74, 75, or 112. In certain such embodiments, an antisense compound targeted to nucleotides 3482-3711 of SEQ ID NO: 3 is selected from ISIS NOs: 395197, 395198, 399824, 399919, 399920, or 399980.

[0798] In certain embodiments, a target region is nucleotides 3727-3751 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3727-3751 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 76 or 77. In certain such embodiments, an antisense compound targeted to nucleotides 3727-3751 of SEQ ID NO: 3 is selected from ISIS NOs: 395199, 399921, 399825, or 399981.

[0799] In certain embodiments, a target region is nucleotides 3798-3824 of SEQ ID NO: 3. In certain embodiments, an antisense compound is targeted to nucleotides 3798-3824 of SEQ ID NO: 3. In certain embodiments, an antisense compound targeted to a PCSK9 nucleic acid comprises a nucleotide sequence selected from SEQ ID NOs: 78 or 99. In certain such embodiments, an antisense compound targeted to nucleotides 3798-3824 of SEQ ID NO: 3 is selected from ISIS NOs: 395200, 399922, 399826, or 399982.

[0800] In certain embodiments, short gapmer antisense compounds are targeted to a PCSK9 nucleic acid. In certain such embodiments, gapmer antisense compounds are targeted to SEQ ID NO: 1, SEQ ID NO: 2 and/or SEQ ID NO: 3. In certain such embodiments, the nucleotide sequences illustrated in Table 15.1 have a 2-10-2 gapmer motif. Table 15.1 illustrates gapmer antisense compounds targeted to SEQ ID NO: 1, SEQ ID NO: 2 and/or SEQ ID NO: 3, having a 2-10-2 motif, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises nucleotides comprising a 2'-O-methoxyethyl sugar modification. Internucleoside linkages are phosphorthioate, and cytidines are 5-methylcytidines.

TABLE 15.1

Gapmer antisense compounds having a 2-10-2 motif targeted to SEQ ID NO: 1, SEQ ID NO: 2 and/or SEQ ID NO: 3

Target SEQ ID NO	ISIS NO	Motif	OligoSeq	5' Target Site on SEQ ID NO: 1	3' Target Site on SEQ ID NO: 1	SEQ ID NO
1	406539	2-10-2	GCCTATGAGGGTGC	1079	1092	458
1	406540	2-10-2	TCCAGGCCTATGAG	1084	1097	459
1	406551	2-10-2	CAGCTCAGCAGCTC	1735	1748	460
1	406595	2-10-2	TTAATCAGGGAGCC	2877	2890	461
2	406551	2-10-2	CAGCTCAGCAGCTC	21181	21194	460
2	406595	2-10-2	TTAATCAGGGAGCC	26684	26697	461
3	406539	2-10-2	GCCTATGAGGGTGC	1609	1622	458
3	406540	2-10-2	TCCAGGCCTATGAG	1614	1627	459
3	406551	2-10-2	CAGCTCAGCAGCTC	2168	2181	460
3	406595	2-10-2	TTAATCAGGGAGCC	3132	3145	461

[0801] In a certain embodiment, a short antisense compound has increased potency compared to a 20 nucleotide compound. In a certain embodiment such short antisense compound and 20 nucleotide compound target the same region. In a certain embodiment, a short antisense compound has increased potency compared to a 20 nucleotide coumpound containing the sequence of the short antisense compound.

[0802] In certain embodiments, the desired effect is a reduction in mRNA target nucleic acid levels. In other embodiments, the desired effect is reduction of levels of protein encoded by the target nucleic acid or a phenotypic change associated with the target nucleic acid.

[0803] In one embodiment, a target region is a structurally defined region of the nucleic acid. For example, a target region may encompass a 3' UTR, a 5' UTR, an exon, an intron, a coding region, a translation initiation region, translation termination region, or other defined nucleic acid region. In other embodiments, a target region may encompass the sequence from a 5' target site of one target segment within the target region to a 3' target site of another target segment within the target region.

[0804] Targeting includes determination of at least one target segment to which an antisense compound hybridizes, such that a desired effect occurs. A target region may contain one or more target segments. Multiple target segments within a target region may be overlapping. Alternatively, they may be non-overlapping. In one embodiment, target segments within a target region are separated by no more than about 10 nucleotides on the target nucleic acid. In another embodiment, target segments within a target region are separated by no more than about 5 nucleotides on the target nucleic acid. In additional embodiments, target segments are contiguous.

[0805] Suitable target segments may be found within a 5' UTR, a coding region, a 3' UTR, an intron, or an exon. Target segments containing a start codon or a stop codon are also suitable target segments.

[0806] The determination of suitable target segments may include a comparison of the sequence of a target nucleic acid to other sequences throughout the genome. For example, the BLAST algorithm may be used to identify regions of similarity amongst different nucleic acids. This comparison can prevent the selection of antisense compound sequences that may hybridize in a non-specific manner to sequences other than a selected target nucleic acid (i.e., non-target or off-target sequences).

[0807] There may be variation in activity (e.g., as defined by percent reduction of target nucleic acid levels) of the antisense compounds within an active target region. In one embodiment, reductions in PCSK9 mRNA levels are indicative of inhibition of PCSK9 expression. Reductions in levels of a PCSK9 protein are also indicative of inhibition of target mRNA expression. Further, phenotypic changes are indicative of inhibition of PCSK9 expression. For example, a decrease in LDL-C levels is indicative of inhibition of PCSK9 expression.

Hybridization

[0808] For example, hybridization may occur between an antisense compound disclosed herein and a PCSK9 nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.

[0809] Hybridization can occur under varying conditions. Stringent conditions are sequence-dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.

[0810] Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in the art. In one embodiment, the antisense compounds provided herein are specifically hybridizable with a PCSK9 nucleic acid.

[0811] The PCSK9 sequences targeted by some of the preferred antisense compounds disclosed in Table 1 are shown in FIG. **1**. The antisense compounds are labeled with the corresponding Isis number. The bar in FIG. **1** labeled "NM_174936.2" displays the relative position of the target sequences within GENBANK® Accession No. NM_174936.2, first deposited with GENBANK® on Jun. 1, 2003 (SEQ ID NO: 1). The bar below that shows the corresponding structural elements of the PCSK9 mRNA, where "CDS" is the protein-encoding region of the mRNA and "UTRs" are the 5' and 3' untranslated regions. As shown in FIG. **1**, the sequences targeted by the present antisense compounds are found in the protein-encoding region of the PCSK9 mRNA.

Complementarity

[0812] An antisense compound and a target nucleic acid are complementary to each other when a sufficient number of nucleobases of the antisense compound can hydrogen bond with the corresponding nucleobases of the target nucleic acid, such that a desired effect will occur (e.g., antisense inhibition of a target nucleic acid, such as a PCSK9 nucleic acid).

[0813] Non-complementary nucleobases between an antisense compound and a PCSK9 nucleic acid may be tolerated provided that the antisense compound remains able to specifically hybridize to a target nucleic acid. Moreover, an antisense compound may hybridize over one or more segments of a PCSK9 nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).

[0814] In some embodiments, the antisense compounds provided herein are at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% complementary to a PCSK9 nucleic acid. Percent complementarity of an antisense compound with a target nucleic acid can be determined using routine methods.

[0815] In other embodiments, the antisense compounds provided herein are fully complementary (i.e, 100% complementary) to a target nucleic acid. For example, antisense compound may be fully complementary to a PCSK9 nucleic acid. As used herein, "fully complementary" means each nucleobase of an antisense compound is capable of precise base pairing with the corresponding nucleobases of a target nucleic acid.

[0816] The location of a non-complementary nucleobase may be at the 5' end or 3' end of the antisense compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the antisense compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e., linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer antisense oligonucleotide.

[0817] In one embodiment, antisense compounds up to 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2 or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a PCSK9 nucleic acid.

[0818] In another embodiment, antisense compounds up to 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2 or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a PCSK9 nucleic acid.

[0819] The antisense compounds provided herein also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e., linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of an antisense compound. In one embodiment, the antisense compounds are complementary to at least an 8 nucleobase portion of a target segment. In another embodiment, the antisense compounds are complementary to at least a 12 nucleobase portion of a target segment. In yet another embodiment, the antisense compounds are complementary to at least a 15 nucleobase portion of a target segment.

Identity

[0820] The antisense compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific Isis number. As used herein, an antisense compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the antisense compounds described herein as well as compounds having non-identical bases relative to the antisense compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the antisense compound. Percent identity of an antisense compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

[0821] In one embodiment, the antisense compounds are at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% or 100% identical to one or more of the antisense compounds disclosed herein.

Modifications

[0822] A nucleoside is a base-sugar combination. The nucleobase (also known as base) portion of the nucleoside is normally a heterocyclic base moiety. Nucleotides are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2', 3' or 5' hydroxyl moiety of the sugar. Oligonucleotides are formed through the covalent linkage of adjacent nucleosides to one another, to form a linear polymeric oligonucleotide. Within the oligonucleotide structure, the phosphate groups are commonly referred to as forming the internucleoside linkages of the oligonucleotide.

[0823] Modifications to antisense compounds encompass substitutions or changes to internucleoside linkages, sugar moieties, or nucleobases. Modified antisense compounds are often preferred over native forms because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for nucleic acid target, increased stability in the presence of nucleases, or increased inhibitory activity.

[0824] Chemically modified nucleosides may also be employed to increase the binding affinity of a shortened or truncated antisense oligonucleotide for its target nucleic acid. Consequently, comparable results can often be obtained with shorter antisense compounds that have such chemically modified nucleosides.

Modified Internucleoside Linkages

[0825] The naturally occurring internucleoside linkage of RNA and DNA is a 3' to 5' phosphodiester linkage. Antisense compounds having one or more modified, i.e., non-naturally occurring, internucleoside linkages are often selected over antisense compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

[0826] Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiesters, phosphotriesters, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

[0827] In one embodiment, antisense compounds targeted to a PCSK9 nucleic acid comprise one or more modified internucleoside linkages. In some embodiments, the modified internucleoside linkages are phosphorothioate linkages. In other embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage.

Modified Sugar Moieties

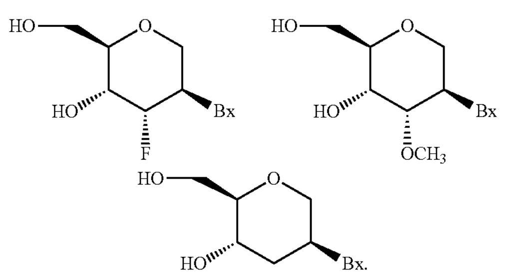
[0828] Antisense compounds targeted to a PCSK9 nucleic acid may contain one or more nucleotides having modified sugar moieties. Sugar modifications may impart nuclease stability, binding affinity or some other beneficial biological property to the antisense compounds. The furanosyl sugar ring of a nucleoside can be modified in a number of ways including, but not limited to: addition of a substituent group, particularly at the 2' position; bridging of two non-geminal ring atoms to form a bicyclic nucleic acid (BNA); and substitution of an atom or group such as —S—, —N(R)— or —C(R_1)(R_2) (R═H, C1-C12 alkyl or a protecting group) for the ring oxygen at the 4'-position and combinations of these such as for example a 2'-F-5'-methyl substituted nucleoside (see WO 2008/101157, published Aug. 21, 2008, for other disclosed 5',2'-bis substituted nucleosides) or replacement of the ribosyl ring oxygen atom with S with further substitution at the 2'-position (see U.S. Published Application No. 2005/0130923, published on Jun. 16, 2005) or alternatively 5'-substitution of a BNA (see WO 2007/134181, published Nov. 22, 2007, wherein LNA is substituted with for example a 5'-methyl or a 5'-vinyl group).

[0829] Modified sugars include, but are not limited to: substituted sugars, especially 2'-substituted sugars having a 2'-F, 2'-OCH_3 (2'-OMe) or a 2'-O$(CH_2)_2$—OCH_3 (2'-O-methoxyethyl or 2'-MOE) substituent group; and bicyclic modified sugars (BNAs), having a 4'-$(CH_2)_n$—O-2' bridge, where n=1 or n=2, including α-L-methyleneoxy (4'-CH2-O-2' or 4'-α-L-(CH2)-O-2' (α-L-LNA)) BNA, β-D-methyleneoxy (4'-CH2-O-2' or 4'-β-D-(CH2)-O-2' (β-D-LNA)) BNA and ethyleneoxy (4'-$(CH_2)_2$—O-2' (ENA)) BNA. BNAs also include 4'-(CH2)-S-2' and 4'-C(CH3)2-O-2' (see International Application No. PCT/US2008/068922); 4'-CH(CH3)-O-2' and 4'-CH(CH2OCH3)-O-2' (see U.S. Pat. No. 7,399,845, issued Jul. 15, 2008); 4'-CH2-N(OCH3)-2' (see International Application No. PCT/US2008/064591); 4'-CH2-O—N(CH3)-2' (see U.S. Published Application No. 2004/0171570, published Sep. 2, 2004); 4'-CH2-N(R)—O-2' (see U.S. Pat. No. 7,427,672, issued Sep. 23, 2008); 4'-CH2-C(CH3)-2' and 4'-CH2-C(═CH2)-2' (see International Application No. PCT/US2008/066154); and wherein R is, independently, H, C1-C12 alkyl, or a protecting group. Bicyclic modified sugars also include (6'S)-6' methyl BNA, aminooxy (4'-CH2-O—N(R)-2') BNA, oxyamino (4'-CH2-N(R)—O-2') BNA wherein R is, independently, H, a protecting group or C1-C12 alkyl. Examples of nucleosides having modified sugar moieties also include without limitation nucleosides comprising 5'-vinyl, 5'-methyl (R or S) and 4'-S.

[0830] The substituent at the 2' position can also be selected from allyl, amino, azido, thio, O-allyl, O—C1-C10 alkyl, OCF_3, O$(CH_2)_2SCH_3$, O$(CH_2)_2$—O—N(Rm)(Rn), and O—CH_2—C(═O)—N(Rm)(Rn), where each Rm and Rn is, independently, H or substituted or unsubstituted C1-C10 alkyl.

[0831] In certain embodiments, such BNA modified nucleotides are high-affinity nucleotides and their incorporation into antisense compounds allows for increased potency and improved therapeutic index. Methods for the preparations of modified sugars are well known to those skilled in the art.

[0832] In certain embodiments, nucleosides are modified by substitution of the ribosyl ring with a sugar surrogate. Such modification includes without limitation, replacement of the ribosyl ring with a surrogate ring system (sometimes referred to as DNA analogs) such as a morpholino ring, a cyclohexenyl ring, a cyclohexyl ring or a tetrahydropyranyl ring such as one having one of the formula:



[0833] Many other bicyclo and tricyclo sugar surrogate ring systems are also know in the art that can be used to modify nucleosides for incorporation into antisense compounds. Such ring systems can undergo various additional substitutions to enhance activity.

[0834] Methods for the preparations of modified sugars are well known to those skilled in the art.

[0835] In nucleotides having modified sugar moieties (including substitution of a sugar surrogate), the nucleobase moieties (natural, modified or a combination thereof) are maintained for hybridization with an appropriate nucleic acid target.

[0836] In one embodiment, antisense compounds targeted to a PCSK9 nucleic acid comprise one or more nucleotides having modified sugar moieties. In a suitable embodiment, the modified sugar moiety is 2'-MOE. In other embodiments, the 2'-MOE modified nucleotides are arranged in a gapmer motif.

Modified Nucleobases

[0837] Nucleobase (or base) modifications or substitutions are structurally distinguishable from, yet functionally interchangeable with, naturally occurring or synthetic unmodified nucleobases. Both natural and modified nucleobases are capable of participating in hydrogen bonding. Such nucleobase modifications may impart nuclease stability, binding affinity or some other beneficial biological property to antisense compounds. Modified nucleobases include synthetic and natural nucleobases such as, for example, 5-methylcytosine (5-me-C). Certain nucleobase substitutions, including 5-methylcytosine substitutions, are particularly useful for increasing the binding affinity of an antisense compound for a target nucleic acid. For example, 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., eds., *Antisense Research and Applications*, CRC Press, Boca Raton, 1993, pp. 276-278).

[0838] Additional unmodified nucleobases include 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl (—C≡C—CH_3) uracil and cytosine and other alkynyl derivatives of pyrimidine bases, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 2-F-adenine, 2-amino-adenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-deazaadenine and 3-deazaguanine and 3-deazaadenine.

[0839] Heterocyclic base moieties may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Nucleobases that are particularly useful for increasing the binding affinity of antisense compounds include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2 aminopropyladenine, 5-propynyluracil and 5-propynylcytosine.

[0840] In one embodiment, antisense compounds targeted to a PCSK9 nucleic acid comprise one or more modified nucleobases. In an additional embodiment, gap-widened antisense oligonucleotides targeted to a PCSK9 nucleic acid comprise one or more modified nucleobases. In some embodiments, the modified nucleobase is 5-methylcytosine. In further embodiments, each cytosine is a 5-methylcytosine.

Compositions and Methods for Formulating Pharmaceutical Compositions

[0841] Antisense oligonucleotides may be admixed with pharmaceutically acceptable active and/or inert substances for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

[0842] Antisense compound targeted to a PCSK9 nucleic acid can be utilized in pharmaceutical compositions by combining the antisense compound with a suitable pharmaceutically acceptable diluent or carrier. A pharmaceutically acceptable diluent includes phosphate-buffered saline (PBS). PBS is a diluent suitable for use in compositions to be delivered parenterally. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising an antisense compound targeted to a PCSK9 nucleic acid and a pharmaceutically acceptable diluent. In one embodiment, the pharmaceutically acceptable diluent is PBS. In other embodiments, the antisense compound is an antisense oligonucleotide.

[0843] Pharmaceutical compositions comprising antisense compounds encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. Accordingly, for example, the disclosure is also drawn to pharmaceutically acceptable salts of antisense compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

[0844] A prodrug can include the incorporation of additional nucleosides at one or both ends of an antisense compound which are cleaved by endogenous nucleases within the body, to form the active antisense compound.

[0845] In certain embodiments, pharmaceutical compositions of the present invention comprise one or more oligonucleotides and one or more excipients. In certain such embodiments, excipients are selected from water, salt solutions, alcohol, polyethylene glycols, gelatin, lactose, amylase, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulosem and polyvinylpyrrolidone.

[0846] In certain embodiments, a pharmaceutical composition of the present invention is prepared using known techniques, including, but not limited to mixing, dissolving, granulating, dragee-making, levigating, emulsifying, encapsulating, entrapping or tabletting processes.

[0847] In certain embodiments, a pharmaceutical composition of the present invention is a liquid (e.g., a suspension, elixir and/or solution). In certain of such embodiments, a liquid pharmaceutical composition is prepared using ingredients known in the art, including, but not limited to, water, glycols, oils, alcohols, flavoring agents, preservatives, and coloring agents.

[0848] In certain embodiments, a pharmaceutical composition of the present invention is a solid (e.g., a powder, tablet, and/or capsule). In certain of such embodiments, a solid pharmaceutical composition comprising one or more oligonucleotides is prepared using ingredients known in the art, including, but not limited to, starches, sugars, diluents, granulating agents, lubricants, binders, and disintegrating agents.

[0849] In certain embodiments, a pharmaceutical composition of the present invention is formulated as a depot preparation. Certain such depot preparations are typically longer acting than non-depot preparations. In certain embodiments, such preparations are administered by implantation (for example subcutaneously or intramuscularly) or by intramuscular injection. In certain embodiments, depot preparations are prepared using suitable polymeric or hydrophobic materials (for example an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives, for example, as a sparingly soluble salt.

[0850] In certain embodiments, a pharmaceutical composition of the present invention comprises a delivery system. Examples of delivery systems include, but are not limited to, liposomes and emulsions. Certain delivery systems are useful for preparing certain pharmaceutical compositions including those comprising hydrophobic compounds. In certain embodiments, certain organic solvents such as dimethylsulfoxide are used.

[0851] In certain embodiments, a pharmaceutical composition of the present invention comprises one or more tissue-specific delivery molecules designed to deliver the one or more pharmaceutical agents of the present invention to specific tissues or cell types. For example, in certain embodiments, pharmaceutical compositions include liposomes coated with a tissue-specific antibody.

[0852] In certain embodiments, a pharmaceutical composition of the present invention comprises a co-solvent system. Certain of such co-solvent systems comprise, for example, benzyl alcohol, a nonpolar surfactant, a water-miscible organic polymer, and an aqueous phase. In certain embodiments, such co-solvent systems are used for hydrophobic compounds. A non-limiting example of such a co-solvent system is the VPD co-solvent system, which is a solution of absolute ethanol comprising 3% w/v benzyl alcohol, 8% w/v of the nonpolar surfactant Polysorbate 80™, and 65% w/v polyethylene glycol 300. The proportions of such co-solvent systems may be varied considerably without significantly altering their solubility and toxicity characteristics. Furthermore, the identity of co-solvent components may be varied: for example, other surfactants may be used instead of Polysorbate 80™; the fraction size of polyethylene glycol may be varied; other biocompatible polymers may replace polyethylene glycol, e.g., polyvinyl pyrrolidone; and other sugars or polysaccharides may substitute for dextrose.

[0853] In certain embodiments, a pharmaceutical composition of the present invention comprises a sustained-release system. A non-limiting example of such a sustained-release system is a semi-permeable matrix of solid hydrophobic polymers. In certain embodiments, sustained-release systems may, depending on their chemical nature, release pharmaceutical agents over a period of hours, days, weeks or months.

[0854] In certain embodiments, a pharmaceutical composition of the present invention is prepared for oral administration. In certain of such embodiments, a pharmaceutical composition is formulated by combining one or more oligonucleotides with one or more pharmaceutically acceptable carriers. Certain of such carriers enable pharmaceutical compositions to be formulated as tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspensions and the like, for oral ingestion by a subject. In certain embodiments, pharmaceutical compositions for oral use are obtained by mixing oligonucleotide and one or more solid excipient. Suitable excipients include, but are not limited to, fillers, such as sugars, including lactose, sucrose, mannitol, or sorbitol; cellulose preparations such as, for example, maize starch, wheat starch, rice starch, potato starch, gelatin, gum tragacanth, methyl cellulose, hydroxypropylmethyl-cellulose, sodium carboxymethylcellulose, and/or polyvinylpyrrolidone (PVP). In certain embodiments, such a mixture is optionally ground and auxiliaries are optionally added. In certain embodiments, pharmaceutical compositions are formed to obtain tablets or dragee cores. In certain embodiments, disintegrating agents (e.g., cross-linked polyvinyl pyrrolidone, agar, or alginic acid or a salt thereof, such as sodium alginate) are added.

[0855] In certain embodiments, dragee cores are provided with coatings. In certain such embodiments, concentrated sugar solutions may be used, which may optionally contain gum arabic, talc, polyvinyl pyrrolidone, carbopol gel, polyethylene glycol, and/or titanium dioxide, lacquer solutions, and suitable organic solvents or solvent mixtures. Dyestuffs or pigments may be added to tablets or dragee coatings.

[0856] In certain embodiments, pharmaceutical compositions for oral administration are push-fit capsules made of gelatin. Certain of such push-fit capsules comprise one or more pharmaceutical agents of the present invention in admixture with one or more filler such as lactose, binders such as starches, and/or lubricants such as talc or magnesium stearate and, optionally, stabilizers. In certain embodiments, pharmaceutical compositions for oral administration are soft, sealed capsules made of gelatin and a plasticizer, such as glycerol or sorbitol. In certain soft capsules, one or more pharmaceutical agents of the present invention are be dissolved or suspended in suitable liquids, such as fatty oils, liquid paraffin, or liquid polyethylene glycols. In addition, stabilizers may be added.

[0857] In certain embodiments, pharmaceutical compositions are prepared for buccal administration. Certain of such pharmaceutical compositions are tablets or lozenges formulated in conventional manner.

[0858] In certain embodiments, a pharmaceutical composition is prepared for administration by injection (e.g., intravenous, subcutaneous, intramuscular, etc.). In certain of such embodiments, a pharmaceutical composition comprises a carrier and is formulated in aqueous solution, such as water or physiologically compatible buffers such as Hanks's solution, Ringer's solution, or physiological saline buffer. In certain embodiments, other ingredients are included (e.g., ingredients that aid in solubility or serve as preservatives). In certain embodiments, injectable suspensions are prepared using appropriate liquid carriers, suspending agents and the like. Certain pharmaceutical compositions for injection are presented in unit dosage form, e.g., in ampoules or in multi-dose containers. Certain pharmaceutical compositions for injection are suspensions, solutions or emulsions in oily or aqueous vehicles, and may contain formulatory agents such as suspending, stabilizing and/or dispersing agents. Certain solvents suitable for use in pharmaceutical compositions for injection include, but are not limited to, lipophilic solvents and fatty oils, such as sesame oil, synthetic fatty acid esters, such as ethyl oleate or triglycerides, and liposomes. Aqueous injection suspensions may contain substances that increase the viscosity of the suspension, such as sodium carboxymethyl cellulose, sorbitol, or dextran. Optionally, such suspensions may also contain suitable stabilizers or agents that increase the solubility of the pharmaceutical agents to allow for the preparation of highly concentrated solutions.

[0859] In certain embodiments, a pharmaceutical composition is prepared for transmucosal administration. In certain of such embodiments penetrants appropriate to the barrier to be permeated are used in the formulation. Such penetrants are generally known in the art.

[0860] In certain embodiments, a pharmaceutical composition is prepared for administration by inhalation. Certain of such pharmaceutical compositions for inhalation are prepared in the form of an aerosol spray in a pressurized pack or a nebulizer. Certain of such pharmaceutical compositions comprise a propellant, e.g., dichlorodifluoromethane, trichlorofluoromethane, dichlorotetrafluoroethane, carbon dioxide or other suitable gas. In certain embodiments using a pressurized aerosol, the dosage unit may be determined with a valve that delivers a metered amount. In certain embodiments, capsules and cartridges for use in an inhaler or insufflator may be formulated. Certain of such formulations comprise a powder mixture of a pharmaceutical agent of the invention and a suitable powder base such as lactose or starch.

[0861] In certain embodiments, a pharmaceutical composition is prepared for rectal administration, such as a suppositories or retention enema. Certain of such pharmaceutical compositions comprise known ingredients, such as cocoa butter and/or other glycerides.

[0862] In certain embodiments, a pharmaceutical composition is prepared for topical administration. Certain of such pharmaceutical compositions comprise bland moisturizing bases, such as ointments or creams. Exemplary suitable ointment bases include, but are not limited to, petrolatum, petrolatum plus volatile silicones, lanolin and water in oil emulsions such as Eucerin™, available from Beiersdorf (Cincinnati, Ohio). Exemplary suitable cream bases include, but are not limited to, Nivea™ Cream, available from Beiersdorf (Cincinnati, Ohio), cold cream (USP), Purpose Cream™, available from Johnson & Johnson (New Brunswick, N.J.), hydrophilic ointment (USP) and Lubriderm™, available from Pfizer (Morris Plains, N.J.).

[0863] In certain embodiments, a pharmaceutical composition of the present invention comprises an oligonucleotide in a therapeutically effective amount. In certain embodiments, the therapeutically effective amount is sufficient to prevent, alleviate or ameliorate symptoms of a disease or to prolong the survival of the subject being treated. Determination of a therapeutically effective amount is well within the capability of those skilled in the art.

[0864] In certain embodiments, one or more oligonucleotides of the present invention is formulated as a prodrug. In certain embodiments, upon in vivo administration, a prodrug is chemically converted to the biologically, pharmaceutically or therapeutically more active form of the oligonucleotide. In certain embodiments, prodrugs are useful because they are easier to administer than the corresponding active form. For example, in certain instances, a prodrug may be more bioavailable (e.g., through oral administration) than is the corresponding active form. In certain instances, a prodrug may have improved solubility compared to the corresponding active form. In certain embodiments, prodrugs are less water soluble than the corresponding active form. In certain instances, such prodrugs possess superior transmittal across cell membranes, where water solubility is detrimental to mobility. In certain embodiments, a prodrug is an ester. In certain such embodiments, the ester is metabolically hydrolyzed to carboxylic acid upon administration. In certain instances the carboxylic acid containing compound is the corresponding active form. In certain embodiments, a prodrug comprises a short peptide (polyaminoacid) bound to an acid group. In certain of such embodiments, the peptide is cleaved upon administration to form the corresponding active form.

[0865] In certain embodiments, a prodrug is produced by modifying a pharmaceutically active compound such that the active compound will be regenerated upon in vivo administration. The prodrug can be designed to alter the metabolic stability or the transport characteristics of a drug, to mask side effects or toxicity, to improve the flavor of a drug or to alter other characteristics or properties of a drug. By virtue of knowledge of pharmacodynamic processes and drug metabolism in vivo, those of skill in this art, once a pharmaceutically active compound is known, can design prodrugs of the compound (see, e.g., Nogrady (1985) Medicinal Chemistry A Biochemical Approach, Oxford University Press, New York, pages 388-392).

[0866] In certain embodiments, a pharmaceutical composition comprising one or more pharmaceutical agents of the present invention is useful for treating a conditions or disorders in a mammalian, and particularly in a human, subject. Suitable administration routes include, but are not limited to, oral, rectal, transmucosal, intestinal, enteral, topical, suppository, through inhalation, intrathecal, intraventricular, intraperitoneal, intranasal, intraocular and parenteral (e.g., intravenous, intramuscular, intramedullary, and subcutaneous). In certain embodiments, pharmaceutical intrathecals are administered to achieve local rather than systemic exposures. For example, pharmaceutical compositions may be injected directly in the area of desired effect (e.g., in the renal or cardiac area).

[0867] In certain embodiments, a pharmaceutical composition of the present invention is administered in the form of a dosage unit (e.g., tablet, capsule, bolus, etc.). In certain embodiments, such pharmaceutical compositions comprise an oligonucleotide in a dose selected from 25 mg, 30 mg, 35 mg, 40 mg, 45 mg, 50 mg, 55 mg, 60 mg, 65 mg, 70 mg, 75 mg, 80 mg, 85 mg, 90 mg, 95 mg, 100 mg, 105 mg, 110 mg, 115 mg, 120 mg, 125 mg, 130 mg, 135 mg, 140 mg, 145 mg, 150 mg, 155 mg, 160 mg, 165 mg, 170 mg, 175 mg, 180 mg, 185 mg, 190 mg, 195 mg, 200 mg, 205 mg, 210 mg, 215 mg, 220 mg, 225 mg, 230 mg, 235 mg, 240 mg, 245 mg, 250 mg, 255 mg, 260 mg, 265 mg, 270 mg, 270 mg, 280 mg, 285 mg, 290 mg, 295 mg, 300 mg, 305 mg, 310 mg, 315 mg, 320 mg, 325 mg, 330 mg, 335 mg, 340 mg, 345 mg, 350 mg, 355 mg, 360 mg, 365 mg, 370 mg, 375 mg, 380 mg, 385 mg, 390 mg, 395 mg, 400 mg, 405 mg, 410 mg, 415 mg, 420 mg, 425 mg, 430 mg, 435 mg, 440 mg, 445 mg, 450 mg, 455 mg, 460 mg, 465 mg, 470 mg, 475 mg, 480 mg, 485 mg, 490 mg, 495 mg, 500 mg, 505 mg, 510 mg, 515 mg, 520 mg, 525 mg, 530 mg, 535 mg, 540 mg, 545 mg, 550 mg, 555 mg, 560 mg, 565 mg, 570 mg, 575 mg, 580 mg, 585 mg, 590 mg, 595 mg, 600 mg, 605 mg, 610 mg, 615 mg, 620 mg, 625 mg, 630 mg, 635 mg, 640 mg, 645 mg, 650 mg, 655 mg, 660 mg, 665 mg, 670 mg, 675 mg, 680 mg, 685 mg, 690 mg, 695 mg, 700 mg, 705 mg, 710 mg, 715 mg, 720 mg, 725 mg, 730 mg, 735 mg, 740 mg, 745 mg, 750 mg, 755 mg, 760 mg, 765 mg, 770 mg, 775 mg, 780 mg, 785 mg, 790 mg, 795 mg, and 800 mg. In certain such embodiments, a pharmaceutical composition of the present invention comprises a dose of oligonucleotide selected from 25 mg, 50 mg, 75 mg, 100 mg, 150 mg, 200 mg, 250 mg, 300 mg, 350 mg, 400 mg, 500 mg, 600 mg, 700 mg, and 800 mg. In certain embodiments, a pharmaceutical composition is comprises a dose of oligonucleotide selected from 50 mg, 100 mg, 150 mg, 200 mg, 250 mg, 300 mg, and 400 mg. In certain embodiments the dose is administered at intervals ranging from more than once per day, once per day, once per week, twice per week, three times per week, four times per week, five times per week, 6 times per week, once per month to once per three months, for as long as needed to sustain the desired effect.

[0868] In a further aspect, a pharmaceutical agent is sterile lyophilized oligonucleotide that is reconstituted with a suitable diluent, e.g., sterile water for injection. The reconstituted product is administered as a subcutaneous injection or as an intravenous infusion after dilution into saline. The lyophilized drug product consists of the oligonucleotide which has been prepared in water for injection, adjusted to pH 7.0-9.0 with acid or base during preparation, and then lyophilized. The lyophilized oligonucleotide may be 25-800 mg of the oligonucleotide. It is understood that this encompasses 25, 50, 75, 100, 125, 150, 175, 200, 225, 250, 275, 300, 325, 350, 375, 425, 450, 475, 500, 525, 550, 575, 600, 625, 650, 675, 700, 725, 750, 775, and 800 mg of lyophilized oligonucleotide. The lyophilized drug product may be packaged in a 2 mL Type I, clear glass vial (ammonium sulfate-treated), stoppered with a bromobutyl rubber closure and sealed with an aluminum FLIP-OFF® overseal. In one embodiment, the lyophilized pharmaceutical agent comprises ISIS 405879.

[0869] The compositions of the present invention may additionally contain other adjunct components conventionally found in pharmaceutical compositions, at their art-established usage levels. Thus, for example, the compositions may contain additional, compatible, pharmaceutically-active materials such as, for example, antipruritics, astringents, local anesthetics or anti-inflammatory agents, or may contain additional materials useful in physically formulating various dosage forms of the compositions of the present invention, such as dyes, flavoring agents, preservatives, antioxidants, opacifiers, thickening agents and stabilizers. However, such materials, when added, should not unduly interfere with the biological activities of the components of the compositions of the present invention. The formulations can be sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, colorings, flavorings and/or aromatic substances and the like which do not deleteriously interact with the oligonucleotide(s) of the formulation.

Conjugated Antisense Compounds

[0870] Antisense compounds may be covalently linked to one or more moieties or conjugates which enhance the activity, visualization of cellular distribution, or cellular uptake of the resulting antisense oligonucleotides. Typical conjugate groups include cholesterol moieties and lipid moieties. Additional conjugate groups include carbohydrates, phospholipids, biotin, phenazine, folate, phenanthridine, anthraquinone, acridine, fluoresceins, rhodamines, coumarins, and dyes.

[0871] Antisense compounds can also be modified to have one or more stabilizing groups that are generally attached to one or both termini of antisense compounds to enhance properties such as, for example, nuclease stability. Included in stabilizing groups are cap structures. These terminal modifications protect the antisense compound having terminal nucleic acid from exonuclease degradation, and can help in delivery and/or localization within a cell. The cap can be present at the 5'-terminus (5'-cap), or at the 3'-terminus (3'-cap), or can be present on both termini. Cap structures are well known in the art and include, for example, inverted deoxy abasic caps. Further 3' and 5'-stabilizing groups that can be used to cap one or both ends of an antisense compound to impart nuclease stability include those disclosed in WO 03/004602 published on Jan. 16, 2003.

Cell Culture and Antisense Compounds Treatment

[0872] The effects of antisense compounds on the level, activity or expression of PCSK9 nucleic acids can be tested in vitro in a variety of cell types. Cell types used for such analyses are available from commercial vendors (e.g., American Type Culture Collection, Manassas, Va.; Zen-Bio, Inc., Research Triangle Park, N.C.; Clonetics Corporation, Walkersville, Md.) and cells are cultured according to the vendor's instructions using commercially available reagents (e.g., Invitrogen Life Technologies, Carlsbad, Calif.). Illustrative cell types include, but are not limited to, HepG2 cells, Hep3B cells, and primary hepatocytes.

In Vitro Testing of Antisense Oligonucleotides

[0873] Described herein are methods for treatment of cells with antisense oligonucleotides, which can be modified appropriately for treatment with other antisense compounds.

[0874] In general, cells are treated with antisense oligonucleotides when they cells reach approximately 60-80% confluency in culture.

[0875] One reagent commonly used to introduce antisense oligonucleotides into cultured cells includes the cationic lipid transfection reagent LIPOFECTIN® (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotides are mixed with LIPOFECTIN® in OPTI-MEM® 1 (Invitrogen, Carlsbad, Calif.) to achieve the desired final concentration of antisense oligonucleotide and a LIPOFECTIN® concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0876] Another reagent used to introduce antisense oligonucleotides into cultured cells includes LIPOFECTAMINE® (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotide is mixed with LIPOFECTAMINE® in OPTI-MEM® 1 reduced serum medium (Invitrogen, Carlsbad, Calif.) to achieve the desired concentration of antisense oligonucleotide and a LIPOFECTAMINE® concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0877] Cells are treated with antisense oligonucleotides by routine methods. Cells are typically harvested 16-24 hours after antisense oligonucleotide treatment, at which time RNA or protein levels of target nucleic acids are measured by methods known in the art and described herein. In general, when treatments are performed in multiple replicates, the data are presented as the average of the replicate treatments.

[0878] The concentration of antisense oligonucleotide used varies from cell line to cell line. Methods to determine the optimal antisense oligonucleotide concentration for a particular cell line are well known in the art. Antisense oligonucleotides are typically used at concentrations ranging from 1 nM to 300 nM.

RNA Isolation

[0879] RNA analysis can be performed on total cellular RNA or poly(A)+ mRNA. Methods of RNA isolation are well known in the art. RNA is prepared using methods well known in the art, for example, using the TRIZOL® Reagent (Invitrogen, Carlsbad, Calif.) according to the manufacturer's recommended protocols.

Analysis of Inhibition of Target Levels or Expression

[0880] Inhibition of levels or expression of a PCSK9 nucleic acid can be assayed in a variety of ways known in the art. For example, target nucleic acid levels can be quantitated by, e.g., Northern blot analysis, competitive polymerase chain reaction (PCR), or quantitaive real-time PCR. RNA analysis can be performed on total cellular RNA or poly(A)+ mRNA. Methods of RNA isolation are well known in the art. Northern blot analysis is also routine in the art. Quantitative real-time PCR can be conveniently accomplished using the commercially available ABI PRISM® 7600, 7700, or 7900 Sequence Detection System, available from PE-Applied Biosystems, Foster City, Calif. and used according to manufacturer's instructions.

Quantitative Real-Time PCR Analysis of Target RNA Levels

[0881] Quantitation of target RNA levels may be accomplished by quantitative real-time PCR using the ABI PRISM® 7600, 7700, or 7900 Sequence Detection System (PE-Applied Biosystems, Foster City, Calif.) according to manufacturer's instructions. Methods of quantitative real-time PCR are well known in the art.

[0882] Prior to real-time PCR, the isolated RNA is subjected to a reverse transcriptase (RT) reaction, which produces complementary DNA (cDNA) that is then used as the substrate for the real-time PCR amplification. The RT and real-time PCR reactions are performed sequentially in the same sample well. RT and real-time PCR reagents are obtained from Invitrogen (Carlsbad, Calif.). RT, real-time-PCR reactions are carried out by methods well known to those skilled in the art.

[0883] Gene (or RNA) target quantities obtained by real time PCR are normalized using either the expression level of a gene whose expression is constant, such as GAPDH, or by quantifying total RNA using RIBOGREEN® (Invitrogen, Inc. Carlsbad, Calif.). GAPDH expression is quantified by real time PCR, by being run simultaneously with the target, multiplexing, or separately. Total RNA is quantified using RIBOGREEN® RNA quantification reagent (Invitrogen, Inc. Carlsbad, Calif.). Methods of RNA quantification by RIBOGREEN® are taught in Jones, L. J., et al, (Analytical Biochemistry, 1998, 265, 368-374). A CYTOFLUOR® 4000 instrument (PE Applied Biosystems) is used to measure RIBOGREEN® fluorescence.

[0884] Probes and primers are designed to hybridize to a PCSK9 nucleic acid. Methods for designing real-time PCR probes and primers are well known in the art, and may include the use of software such as PRIMER EXPRESS® Software (Applied Biosystems, Foster City, Calif.).

Analysis of Protein Levels

[0885] Antisense inhibition of PCSK9 nucleic acids can be assessed by measuring PCSK9 protein levels. Protein levels of PCSK9 can be evaluated or quantitated in a variety of ways well known in the art, such as immunoprecipitation, Western blot analysis (immunoblotting), enzyme-linked immunosorbent assay (ELISA), quantitative protein assays, protein activity assays (for example, caspase activity assays), immunohistochemistry, immunocytochemistry or fluorescence-activated cell sorting (FACS). Antibodies directed to a target can be identified and obtained from a variety of sources, such as the MSRS catalog of antibodies (Aerie Corporation, Birmingham, Mich.), or can be prepared via conventional monoclonal or polyclonal antibody generation methods well known in the art. Antibodies useful for the detection of human and rat PCSK9 are commercially available.

Serum Lipid Analysis

[0886] Plasma concentrations of total cholesterol, LDL-C, HDL-C, free cholesterol, triglycerides, glucose, ketones, transaminases, and phospholipids can be measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). Serum lipoprotein and cholesterol profiling can be performed as described by Crooke et al., *Lipid Res.* (2005) 46: 872-884, using a Beckman System Gold 126 HPLC system, a 507e refrigerated autosampler, a 126 photodiode array detector (Beckman Instruments, Fullerton, Calif.), and a Superose 6 HR 10/30 column (Pfizer, Chicago, Ill.). HDL, LDL and VLDL fractions are measured at a wavelength of 505 nm and validated with a cholesterol calibration kit. (Sigma).

Liver Triglyceride Analysis

[0887] Liver triglyceride content can be measured according to routine experimental procedures, for example, per procedures described by Desai et al., *Diabetes* (2001) 50: 2287-2295. Liver triglyceride content may be measured after antisense inhibition of PCSK9 for 6 weeks.

In Vivo Testing of Antisense Compounds

[0888] Antisense compounds, for example, antisense oligonucleotides, are tested in animals to assess their ability to inhibit expression of PCSK9 and produce phenotypic changes, such as decreases in LDL-C. Testing may be performed in normal animals, or in experimental disease models. For administration to animals, antisense oligonucleotides are formulated in a pharmaceutically acceptable diluent, such as phosphate-buffered saline. Administration includes parenteral routes of administration, such as intraperitoneal, intravenous, and subcutaneous. Calculation of antisense oligonucleotide dosage and dosing frequency is within the abilities of those skilled in the art, and depends upon factors such as route of administration and animal body weight. Following a period of treatment with antisense oligonucleotides, RNA is isolated from various tissues and changes in PCSK9 nucleic acid expression are measured. Changes in PCSK9 protein levels may also be measured.

Nonlimiting Disclosure and Incorporation by Reference

[0889] The following examples serve only to illustrate the methods, antisense oligonucleotides, and compositions provided and are not intended to limit the same. Each of the references, GENBANK® accession numbers, and the like recited in the present application is incorporated herein by reference in its entirety.

EXAMPLES

Example 1: Antisense Inhibition of Human PCSK9: A549 Cells

[0890] Antisense oligonucleotides targeted to a PCSK9 nucleic acid were tested for their effects on PCSK9 mRNA in vitro. Cultured A549 cells at a density of 4000 cells per well in a 96-well plate were treated with 60 nM of antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. PCSK9 mRNA levels were adjusted according to total RNA content as measured by RIBOGREEN®. Results are presented as percent inhibition of PCSK9, relative to untreated control cells. Antisense oligonucleotides that exhibited at least 30% inhibition of PCSK9 expression are shown in Table 16.

[0891] The motif column indicates the wing-gap-wing motif of each antisense oligonucleotide. Antisense oligonucleotides were designed as 5-10-5 gapmers, or alternatively as 3-14-3 gapmers, where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises 2'-MOE nucleotides. As illustrated in Table 16, a single nucleobase sequence may be represented by a 5-10-5 motif as well as a 3-14-3 motif "5' target site" indicates the 5'-most nucleotide which the antisense oligonucleotide is targeted on the indicated GENBANK Accession No.

TABLE 16

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
395150	5	242	NM_174936.2	GAGGAGACCTAGAGGCCGTG	5-10-5	57
395151	6	300	NM_174936.2	AGGACCGCCTGGAGCTGACG	5-10-5	77
395152	7	410	NM_174936.2	ACGCAAGGCTAGCACCAGCT	5-10-5	81
395153	9	480	NM_174936.2	CCTTGGCGCAGCGGTGGAAG	5-10-5	51
395154	10	561	NM_174936.2	GGCGGGCAGTGCGCTCTGAC	5-10-5	40
395155	11	600	NM_174936.2	TGGTGAGGTATCCCCGGCGG	5-10-5	79
395156	14	620	NM_174936.2	ATGGAAGACATGCAGGATCT	5-10-5	75
395157	15	646	NM_174936.2	TTCACCAGGAAGCCAGGAAG	5-10-5	72
395158	17	705	NM_174936.2	CCTCGATGTAGTCGACATGG	5-10-5	61
395159	18	785	NM_174936.2	GTATTCATCCGCCCGGTACC	5-10-5	35
395160	19	835	NM_174936.2	CTGGTGTCTAGGAGATACAC	5-10-5	47
395161	21	860	NM_174936.2	GATTTCCCGGTGGTCACTCT	5-10-5	70
395162	24	890	NM_174936.2	CTCGAAGTCGGTGACCATGA	5-10-5	52
395163	25	923	NM_174936.2	GTGGAAGCGGGTCCCGTCCT	5-10-5	73

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
395164	26	970	NM_174936.2	ACCCCTGCCAGGTGGGTGCC	5-10-5	69
395165	28	1004	NM_174936.2	ACCCTTGGCCACGCCGGCAT	5-10-5	81
395166	29	1040	NM_174936.2	TTGGCAGTTGAGCACGCGCA	5-10-5	65
395167	31	1077	NM_174936.2	CCAGGCCTATGAGGGTGCCG	5-10-5	34
395168	32	1098	NM_174936.2	GCTGGCTTTTCCGAATAAAC	5-10-5	69
395169	33	1210	NM_174936.2	GTGACCAGCACGACCCCAGC	5-10-5	61
395171	34	1326	NM_174936.2	CCAAAGTCCCCAGGGTCACC	5-10-5	38
395173	36	1340	NM_174936.2	GCCAAAGTTGGTCCCCAAAG	5-10-5	64
395174	38	1361	NM_174936.2	GGCAAAGAGGTCCACACAGC	5-10-5	72
395175	39	1389	NM_174936.2	TGGAGGCACCAATGATGTCC	5-10-5	41
395177	45	1534	NM_174936.2	TTGGCAGAGAAGTGGATCAG	5-10-5	41
395178	50	1569	NM_174936.2	GGTCCTCAGGGAACCAGGCC	5-10-5	67
395181	52	1640	NM_174936.2	AAACAGCTGCCAACCTGCCC	5-10-5	37
395182	54	1675	NM_174936.2	GTAGGCCCCGAGTGTGCTGA	5-10-5	51
395183	57	1812	NM_174936.2	CGTTGTGGGCCCGGCAGACC	5-10-5	74
395184	58	1858	NM_174936.2	AGCAGGCAGCACCTGGCAAT	5-10-5	61
395185	59	1920	NM_174936.2	CACGGGTCCCCATGCTGGCC	5-10-5	73
395186	60	2100	NM_174936.2	CTTTGCATTCCAGACCTGGG	5-10-5	83
395187	62	2310	NM_174936.2	GGCAGCAGATGGCAACGGCT	5-10-5	80
395189	64	2504	NM_174936.2	TCAAGGGCCAGGCCAGCAGC	5-10-5	65
395190	66	2597	NM_174936.2	ATGCCCCACAGTGAGGGAGG	5-10-5	60
395191	67	2606	NM_174936.2	AATGGTGAAATGCCCCACAG	5-10-5	76
395193	68	2750	NM_174936.2	CATGGGAAGAATCCTGCCTC	5-10-5	48
395194	69	2832	NM_174936.2	GGAGATGAGGGCCATCAGCA	5-10-5	64
395195	70	2900	NM_174936.2	TAGATGCCATCCAGAAAGCT	5-10-5	44
395196	72	2983	NM_174936.2	GGCATAGAGCAGAGTAAAGG	5-10-5	39
395198	75	3437	NM_174936.2	GCTCAAGGAGGGACAGTTGT	5-10-5	73
395199	76	3472	NM_174936.2	AAAGATAAATGTCTGCTTGC	5-10-5	83
395200	78	3543	NM_174936.2	TCTTCAAGTTACAAAAGCAA	5-10-5	40
395202	129	13816	NT_032977.8	GTGCCATCTGAACAGCACCT	5-10-5	36
395203	110	13926	NT_032977.8	CCTGGAACCCCTGCAGCCAG	5-10-5	64
395204	152	13977	NT_032977.8	TTCAGGCAGGTTGCTGCTAG	5-10-5	55
395206	136	14122	NT_032977.8	TAGGAGAAAGTAGGGAGAGC	5-10-5	45
395207	132	14179	NT_032977.8	TAAAAGCTGCAAGAGACTCA	5-10-5	49
395208	139	14267	NT_032977.8	TCAGAGAAAACAGTCACCGA	5-10-5	56
395209	142	14404	NT_032977.8	TCATTTTAGAGACAGGAAGC	5-10-5	73

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
395210	113	14441	NT_032977.8	GAATAACAGTGATGTCTGGC	5-10-5	82
395211	138	14494	NT_032977.8	TCACAGCTCACCGAGTCTGC	5-10-5	60
395212	98	14524	NT_032977.8	AGTGTAAAATAAAGCCCCTA	5-10-5	50
395213	96	14601	NT_032977.8	AGGACCCAAGTCATCCTGCT	5-10-5	57
395214	124	14631	NT_032977.8	GGCCATCAGCTGGCAATGCT	5-10-5	50
395215	133	14675	NT_032977.8	TAGACAAGGAAAGGGAGGCC	5-10-5	60
395216	103	14681	NT_032977.8	ATTTCATAGACAAGGAAAGG	5-10-5	51
395218	146	1315	AK124635.1	TGACATTTGTGGGAGAGGAG	5-10-5	35
395220	104	2085	AK124635.1	CACAGTCCTGCAAAACAGCT	5-10-5	54
395221	102	5590	NT_032977.8	ATGTGCAGAGATCAATCACA	5-10-5	69
395222	127	10633	NT_032977.8	GGTGGTAATTTGTCACAGCA	5-10-5	64
395223	84	11308	NT_032977.8	AAGGTCACACAGTTAAGAGT	5-10-5	55
395226	147	24858	NT_032977.8	TGAGTTCATTTAAGAGTGGA	5-10-5	42
399793	8	417	NM_174936.2	CCTCGGAACGCAAGGCTAGC	5-10-5	71
399794	12	606	NM_174936.2	GGATCTTGGTGAGGTATCCC	5-10-5	42
399795	13	615	NM_174936.2	AGACATGCAGGATCTTGGTG	5-10-5	54
399796	16	651	NM_174936.2	TCATCTTCACCAGGAAGCCA	5-10-5	50
399797	20	840	NM_174936.2	GTATGCTGGTGTCTAGGAGA	5-10-5	38
399798	22	866	NM_174936.2	GCCCTCGATTTCCCGGTGGT	5-10-5	54
399799	23	880	NM_174936.2	GTGACCATGACCCTGCCCTC	5-10-5	75
399800	27	975	NM_174936.2	TGACCACCCCTGCCAGGTGG	5-10-5	67
399801	30	1045	NM_174936.2	TTCCCTTGGCAGTTGAGCAC	5-10-5	79
399802	35	1335	NM_174936.2	AGTTGGTCCCCAAAGTCCCC	5-10-5	71
399803	37	1352	NM_174936.2	GTCCACACAGCGGCCAAAGT	5-10-5	63
399804	40	1400	NM_174936.2	GCTGCAGTCGCTGGAGGCAC	5-10-5	46
399805	41	1411	NM_174936.2	ACAAAGCAGGTGCTGCAGTC	5-10-5	46
399806	42	1470	NM_174936.2	GCATCATGGCTGCAATGCCA	5-10-5	85
399807	43	1478	NM_174936.2	GGCAGACAGCATCATGGCTG	5-10-5	71
399808	44	1526	NM_174936.2	GAAGTGGATCAGTCTCTGCC	5-10-5	38
399809	46	1539	NM_174936.2	CATCTTTGGCAGAGAAGTGG	5-10-5	53
399810	47	1545	NM_174936.2	TGATGACATCTTTGGCAGAG	5-10-5	63
399811	48	1552	NM_174936.2	GCCTCATTGATGACATCTTT	5-10-5	61
399812	49	1564	NM_174936.2	TCAGGGAACCAGGCCTCATT	5-10-5	46
399813	51	1583	NM_174936.2	GGTCAGTACCCGCTGGTCCT	5-10-5	72
399814	53	1645	NM_174936.2	CTGCAAAACAGCTGCCAACC	5-10-5	88

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
399815	55	1740	NM_174936.2	AGAAACTGGAGCAGCTCAGC	5-10-5	38
399816	56	1746	NM_174936.2	TCCTGGAGAAACTGGAGCAG	5-10-5	51
399817	61	2105	NM_174936.2	CTTGACTTTGCATTCCAGAC	5-10-5	84
399818	63	2415	NM_174936.2	AACCATTTTAAAGCTCAGCC	5-10-5	68
399819	65	2509	NM_174936.2	CCCACTCAAGGGCCAGGCCA	5-10-5	87
399821	71	2906	NM_174936.2	TCTGGCTAGATGCCATCCAG	5-10-5	83
399822	73	2988	NM_174936.2	AGCCTGGCATAGAGCAGAGT	5-10-5	70
399824	74	3233	NM_174936.2	AAGTAAGAAGAGGCTTGGCT	5-10-5	43
399825	77	3477	NM_174936.2	ACCCAAAAGATAAATGTCTG	5-10-5	61
399827	100	13746	NT_032977.8	ATCTCAGGACAGGTGAGCAA	5-10-5	81
399828	116	13760	NT_032977.8	GAGTAGAGATTCTCATCTCA	5-10-5	73
399829	117	13828	NT_032977.8	GAGTCTTCTGAAGTGCCATC	5-10-5	65
399831	83	13986	NT_032977.8	AAGGAAGACTTCAGGCAGGT	5-10-5	52
399833	92	14397	NT_032977.8	AGAGACAGGAAGCTGCAGCT	5-10-5	54
399834	82	14670	NT_032977.8	AAGGAAAGGGAGGCCTAGAG	5-10-5	44
399836	130	2095	AK124635.1	GTGCTGACCACACAGTCCTG	5-10-5	96
399837	107	3056	NT_032977.8	CCCACTATAATGGCAAGCCC	5-10-5	83
399838	80	4306	NT_032977.8	AACCCAGTTCTAATGCACCT	5-10-5	81
399839	106	5140	NT_032977.8	CCAGTCAGAGTAGAACAGAG	5-10-5	68
399840	121	5599	NT_032977.8	GGAGCCTACATGTGCAGAGA	5-10-5	46
399841	94	5667	NT_032977.8	AGCATGGCACCAGCATCTGC	5-10-5	59
399842	108	6652	NT_032977.8	CCCAGCCCTATCAGGAAGTG	5-10-5	67
399843	144	7099	NT_032977.8	TGACATCCAGGAGGGAGGAG	5-10-5	33
399844	91	7556	NT_032977.8	AGACTGATGGAAGGCATTGA	5-10-5	56
399845	131	7565	NT_032977.8	GTGTTGAGCAGACTGATGGA	5-10-5	48
399846	145	8836	NT_032977.8	TGACATCTTGTCTGGGAGCC	5-10-5	58
399847	90	8948	NT_032977.8	AGACTAGGAGCCTGAGTTTT	5-10-5	38
399849	148	10252	NT_032977.8	TGGCAGCAACTCAGACATAT	5-10-5	74
399851	88	12715	NT_032977.8	ACTGGATACATTGGCAGACA	5-10-5	62
399852	111	12928	NT_032977.8	CTAGAGGAACCACTAGATAT	5-10-5	66
399854	97	16134	NT_032977.8	AGTCAAGCTGCTGCCCAGAG	5-10-5	82
399855	120	16668	NT_032977.8	GCTAGTTATTAAGCACCTGC	5-10-5	71
399856	150	17267	NT_032977.8	TGTGAGCTCTGGCCCAGTGG	5-10-5	64
399857	115	18377	NT_032977.8	GAGTAAGGCAGGTTACTCTC	5-10-5	79
399858	134	18408	NT_032977.8	TAGATGTGACTAACATTTAA	5-10-5	57
399859	105	19203	NT_032977.8	CACATTAGCCTTGCTCAAGT	5-10-5	75

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
399860	151	19913	NT_032977.8	TGTGATGACCTGGAAAGGTG	5-10-5	36
399862	109	20188	NT_032977.8	CCCCTGCACAGAGCCTGGCA	5-10-5	59
399863	141	20624	NT_032977.8	TCATGGCTGCAATGCCTGGT	5-10-5	46
399864	93	20995	NT_032977.8	AGAGAGGAGGGCTTAAAGAA	5-10-5	30
399865	95	21082	NT_032977.8	AGCTGCCAACCTGCAAAAAG	5-10-5	75
399866	143	21481	NT_032977.8	TGAAAATCCATCCAGCACTG	5-10-5	74
399867	89	21589	NT_032977.8	AGAACCATGGAGCACCTGAG	5-10-5	84
399868	123	21696	NT_032977.8	GGCACTGCCCTTCCACCAAA	5-10-5	53
399869	118	24907	NT_032977.8	GCACCATCCAGACCAGAATC	5-10-5	49
399870	114	25413	NT_032977.8	GAGAGGTTCAGATCCAGGCC	5-10-5	66
399871	4	135	NM_174936.2	GCGCGGAATCCTGGCTGGGA	3-14-3	40
399872	5	242	NM_174936.2	GAGGAGACCTAGAGGCCGTG	3-14-3	64
399873	6	300	NM_174936.2	AGGACCGCCTGGAGCTGACG	3-14-3	65
399874	7	410	NM_174936.2	ACGCAAGGCTAGCACCAGCT	3-14-3	80
399875	9	480	NM_174936.2	CCTTGGCGCAGCGGTGGAAG	3-14-3	67
399876	10	561	NM_174936.2	GGCGGGCAGTGCGCTCTGAC	3-14-3	62
399877	11	600	NM_174936.2	TGGTGAGGTATCCCCGGCGG	3-14-3	73
399878	14	620	NM_174936.2	ATGGAAGACATGCAGGATCT	3-14-3	54
399879	15	646	NM_174936.2	TTCACCAGGAAGCCAGGAAG	3-14-3	75
399880	17	705	NM_174936.2	CCTCGATGTAGTCGACATGG	3-14-3	80
399881	18	785	NM_174936.2	GTATTCATCCGCCCGGTACC	3-14-3	63
399882	19	835	NM_174936.2	CTGGTGTCTAGGAGATACAC	3-14-3	66
399883	21	860	NM_174936.2	GATTTCCCGGTGGTCACTCT	3-14-3	59
399884	24	890	NM_174936.2	CTCGAAGTCGGTGACCATGA	3-14-3	71
399885	25	923	NM_174936.2	GTGGAAGCGGGTCCCGTCCT	3-14-3	73
399886	26	970	NM_174936.2	ACCCCTGCCAGGTGGGTGCC	3-14-3	76
399887	28	1004	NM_174936.2	ACCCTTGGCCACGCCGGCAT	3-14-3	54
399888	29	1040	NM_174936.2	TTGGCAGTTGAGCACGCGCA	3-14-3	75
399889	31	1077	NM_174936.2	CCAGGCCTATGAGGGTGCCG	3-14-3	84
399890	32	1098	NM_174936.2	GCTGGCTTTTCCGAATAAAC	3-14-3	73
399891	33	1210	NM_174936.2	GTGACCAGCACGACCCCAGC	3-14-3	99
399892	149	1297	NM_174936.2	TGGGCATTGGTGGCCCCAAC	3-14-3	61
399893	34	1326	NM_174936.2	CCAAAGTCCCCAGGGTCACC	3-14-3	54
399894	128	1330	NM_174936.2	GTCCCCAAAGTCCCCAGGGT	3-14-3	80
399895	36	1340	NM_174936.2	GCCAAAGTTGGTCCCCAAAG	3-14-3	82

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
399896	38	1361	NM_174936.2	GGCAAAGAGGTCCACACAGC	3-14-3	83
399897	39	1389	NM_174936.2	TGGAGGCACCAATGATGTCC	3-14-3	70
399898	101	1465	NM_174936.2	ATGGCTGCAATGCCAGCCAC	3-14-3	37
399899	45	1534	NM_174936.2	TTGGCAGAGAAGTGGATCAG	3-14-3	64
399900	50	1569	NM_174936.2	GGTCCTCAGGGAACCAGGCC	3-14-3	82
399901	87	1576	NM_174936.2	ACCCGCTGGTCCTCAGGGAA	3-14-3	76
399902	119	1605	NM_174936.2	GCAGGGCGGCCACCAGGTTG	3-14-3	39
399903	52	1640	NM_174936.2	AAACAGCTGCCAACCTGCCC	3-14-3	73
399904	54	1675	NM_174936.2	GTAGGCCCCGAGTGTGCTGA	3-14-3	81
399905	57	1812	NM_174936.2	CGTTGTGGGCCCGGCAGACC	3-14-3	80
399906	58	1858	NM_174936.2	AGCAGGCAGCACCTGGCAAT	3-14-3	92
399907	59	1920	NM_174936.2	CACGGGTCCCCATGCTGGCC	3-14-3	95
399908	60	2100	NM_174936.2	CTTTGCATTCCAGACCTGGG	3-14-3	84
399909	62	2310	NM_174936.2	GGCAGCAGATGGCAACGGCT	3-14-3	73
399910	154	2410	NM_174936.2	TTTTAAAGCTCAGCCCCAGC	3-14-3	57
399911	64	2504	NM_174936.2	TCAAGGGCCAGGCCAGCAGC	3-14-3	82
399912	66	2597	NM_174936.2	ATGCCCCACAGTGAGGGAGG	3-14-3	68
399913	67	2606	NM_174936.2	AATGGTGAAATGCCCCACAG	3-14-3	92
399914	153	2649	NM_174936.2	TTGGGAGCAGCTGGCAGCAC	3-14-3	59
399915	68	2750	NM_174936.2	CATGGGAAGAATCCTGCCTC	3-14-3	78
399916	69	2832	NM_174936.2	GGAGATGAGGGCCATCAGCA	3-14-3	71
399917	70	2900	NM_174936.2	TAGATGCCATCCAGAAAGCT	3-14-3	81
399918	72	2983	NM_174936.2	GGCATAGAGCAGAGTAAAGG	3-14-3	82
399919	112	3227	NM_174936.2	GAAGAGGCTTGGCTTCAGAG	3-14-3	57
399920	75	3437	NM_174936.2	GCTCAAGGAGGGACAGTTGT	3-14-3	84
399921	76	3472	NM_174936.2	AAAGATAAATGTCTGCTTGC	3-14-3	78
399922	78	3543	NM_174936.2	TCTTCAAGTTACAAAAGCAA	3-14-3	61
399923	85	13681	NT_032977.8	ACAAATTCCCAGACTCAGCA	3-14-3	82
399924	129	13816	NT_032977.8	GTGCCATCTGAACAGCACCT	3-14-3	84
399925	110	13926	NT_032977.8	CCTGGAACCCCTGCAGCCAG	3-14-3	73
399926	152	13977	NT_032977.8	TTCAGGCAGGTTGCTGCTAG	3-14-3	47
399927	140	13998	NT_032977.8	TCAGCCAGGCCAAAGGAAGA	3-14-3	78
399928	136	14122	NT_032977.8	TAGGAGAAAGTAGGGAGAGC	3-14-3	47
399929	132	14179	NT_032977.8	TAAAAGCTGCAAGAGACTCA	3-14-3	69
399930	139	14267	NT_032977.8	TCAGAGAAAACAGTCACCGA	3-14-3	82
399931	142	14404	NT_032977.8	TCATTTTAGAGACAGGAAGC	3-14-3	79

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
399932	113	14441	NT_032977.8	GAATAACAGTGATGTCTGGC	3-14-3	83
399933	138	14494	NT_032977.8	TCACAGCTCACCGAGTCTGC	3-14-3	78
399934	98	14524	NT_032977.8	AGTGTAAAATAAAGCCCCTA	3-14-3	79
399935	96	14601	NT_032977.8	AGGACCCAAGTCATCCTGCT	3-14-3	89
399936	124	14631	NT_032977.8	GGCCATCAGCTGGCAATGCT	3-14-3	70
399937	133	14675	NT_032977.8	TAGACAAGGAAAGGGAGGCC	3-14-3	78
399938	103	14681	NT_032977.8	ATTTCATAGACAAGGAAAGG	3-14-3	67
399940	146	1315	AK124635.1	TGACATTTGTGGGAGAGGAG	3-14-3	35
399942	104	2085	AK124635.1	CACAGTCCTGCAAAACAGCT	3-14-3	75
399943	102	5590	NT_032977.8	ATGTGCAGAGATCAATCACA	3-14-3	64
399944	127	10633	NT_032977.8	GGTGGTAATTTGTCACAGCA	3-14-3	73
399945	84	11308	NT_032977.8	AAGGTCACACAGTTAAGAGT	3-14-3	89
399947	126	22292	NT_032977.8	GGTGCATAAGGAGAAAGAGA	3-14-3	70
399948	147	24858	NT_032977.8	TGAGTTCATTTAAGAGTGGA	3-14-3	46
399949	8	417	NM_174936.2	CCTCGGAACGCAAGGCTAGC	3-14-3	74
399950	12	606	NM_174936.2	GGATCTTGGTGAGGTATCCC	3-14-3	68
399951	13	615	NM_174936.2	AGACATGCAGGATCTTGGTG	3-14-3	67
399952	16	651	NM_174936.2	TCATCTTCACCAGGAAGCCA	3-14-3	74
399953	20	840	NM_174936.2	GTATGCTGGTGTCTAGGAGA	3-14-3	59
399954	22	866	NM_174936.2	GCCCTCGATTTCCCGGTGGT	3-14-3	83
399955	23	880	NM_174936.2	GTGACCATGACCCTGCCCTC	3-14-3	62
399956	27	975	NM_174936.2	TGACCACCCCTGCCAGGTGG	3-14-3	78
399957	30	1045	NM_174936.2	TTCCCTTGGCAGTTGAGCAC	3-14-3	82
399958	35	1335	NM_174936.2	AGTTGGTCCCCAAAGTCCCC	3-14-3	78
399959	37	1352	NM_174936.2	GTCCACACAGCGGCCAAAGT	3-14-3	89
399960	40	1400	NM_174936.2	GCTGCAGTCGCTGGAGGCAC	3-14-3	64
399961	41	1411	NM_174936.2	ACAAAGCAGGTGCTGCAGTC	3-14-3	63
399962	42	1470	NM_174936.2	GCATCATGGCTGCAATGCCA	3-14-3	68
399963	43	1478	NM_174936.2	GGCAGACAGCATCATGGCTG	3-14-3	70
399964	44	1526	NM_174936.2	GAAGTGGATCAGTCTCTGCC	3-14-3	70
399965	46	1539	NM_174936.2	CATCTTTGGCAGAGAAGTGG	3-14-3	49
399966	47	1545	NM_174936.2	TGATGACATCTTTGGCAGAG	3-14-3	78
399967	48	1552	NM_174936.2	GCCTCATTGATGACATCTTT	3-14-3	82
399968	49	1564	NM_174936.2	TCAGGGAACCAGGCCTCATT	3-14-3	71
399969	51	1583	NM_174936.2	GGTCAGTACCCGCTGGTCCT	3-14-3	82

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
399970	53	1645	NM_174936.2	CTGCAAAACAGCTGCCAACC	3-14-3	74
399971	55	1740	NM_174936.2	AGAAACTGGAGCAGCTCAGC	3-14-3	48
399972	56	1746	NM_174936.2	TCCTGGAGAAACTGGAGCAG	3-14-3	63
399973	61	2105	NM_174936.2	CTTGACTTTGCATTCCAGAC	3-14-3	74
399974	63	2415	NM_174936.2	AACCATTTTAAAGCTCAGCC	3-14-3	71
399975	65	2509	NM_174936.2	CCCACTCAAGGGCCAGGCCA	3-14-3	91
399976	122	2582	NM_174936.2	GGAGGGAGCTTCCTGGCACC	3-14-3	67
399977	71	2906	NM_174936.2	TCTGGCTAGATGCCATCCAG	3-14-3	89
399978	73	2988	NM_174936.2	AGCCTGGCATAGAGCAGAGT	3-14-3	80
399979	135	2994	NM_174936.2	TAGCACAGCCTGGCATAGAG	3-14-3	83
399980	74	3233	NM_174936.2	AAGTAAGAAGAGGCTTGGCT	3-14-3	50
399981	77	3477	NM_174936.2	ACCCAAAAGATAAATGTCTG	3-14-3	55
399982	99	3550	NM_174936.2	ATAAATATCTTCAAGTTACA	3-14-3	36
399983	100	13746	NT_032977.8	ATCTCAGGACAGGTGAGCAA	3-14-3	96
399984	116	13760	NT_032977.8	GAGTAGAGATTCTCATCTCA	3-14-3	71
399985	117	13828	NT_032977.8	GAGTCTTCTGAAGTGCCATC	3-14-3	76
399986	81	13903	NT_032977.8	AAGCAGGGCCTCAGGTGGAA	3-14-3	68
399987	83	13986	NT_032977.8	AAGGAAGACTTCAGGCAGGT	3-14-3	59
399988	137	14112	NT_032977.8	TAGGGAGAGCTCACAGATGC	3-14-3	31
399989	92	14397	NT_032977.8	AGAGACAGGAAGCTGCAGCT	3-14-3	65
399990	82	14670	NT_032977.8	AAGGAAAGGGAGGCCTAGAG	3-14-3	59
399992	130	2095	AK124635.1	GTGCTGACCACACAGTCCTG	3-14-3	94
399993	107	3056	NT_032977.8	CCCACTATAATGGCAAGCCC	3-14-3	76
399994	80	4306	NT_032977.8	AACCCAGTTCTAATGCACCT	3-14-3	82
399995	106	5140	NT_032977.8	CCAGTCAGAGTAGAACAGAG	3-14-3	74
399996	121	5599	NT_032977.8	GGAGCCTACATGTGCAGAGA	3-14-3	65
399997	94	5667	NT_032977.8	AGCATGGCACCAGCATCTGC	3-14-3	77
399998	108	6652	NT_032977.8	CCCAGCCCTATCAGGAAGTG	3-14-3	78
399999	144	7099	NT_032977.8	TGACATCCAGGAGGGAGGAG	3-14-3	43
400000	91	7556	NT_032977.8	AGACTGATGGAAGGCATTGA	3-14-3	84
400001	131	7565	NT_032977.8	GTGTTGAGCAGACTGATGGA	3-14-3	73
400002	145	8836	NT_032977.8	TGACATCTTGTCTGGGAGCC	3-14-3	82
400003	90	8948	NT_032977.8	AGACTAGGAGCCTGAGTTTT	3-14-3	48
400004	125	9099	NT_032977.8	GGCCTGCAGAAGCCAGAGAG	3-14-3	45
400005	148	10252	NT_032977.8	TGGCAGCAACTCAGACATAT	3-14-3	73
400006	79	11472	NT_032977.8	AAATGCAGGGCTAAAATCAC	3-14-3	78

TABLE 16-continued

Antisense inhibition of PCSK9 in human cells (Cell type A549)

IsisNo	SEQ ID NO	5' Target Site	Target Nucleic Acid	Sequence	Motif	% Inhibition
400007	88	12715	NT_032977.8	ACTGGATACATTGGCAGACA	3-14-3	77
400008	111	12928	NT_032977.8	CTAGAGGAACCACTAGATAT	3-14-3	78
400009	86	15471	NT_032977.8	ACAGCATTCTTGGTTAGGAG	3-14-3	73
400010	97	16134	NT_032977.8	AGTCAAGCTGCTGCCCAGAG	3-14-3	74
400011	120	16668	NT_032977.8	GCTAGTTATTAAGCACCTGC	3-14-3	75
400012	150	17267	NT_032977.8	TGTGAGCTCTGGCCCAGTGG	3-14-3	64
400013	115	18377	NT_032977.8	GAGTAAGGCAGGTTACTCTC	3-14-3	88
400014	134	18408	NT_032977.8	TAGATGTGACTAACATTTAA	3-14-3	58
400015	105	19203	NT_032977.8	CACATTAGCCTTGCTCAAGT	3-14-3	89
400018	109	20188	NT_032977.8	CCCCTGCACAGAGCCTGGCA	3-14-3	62
400019	141	20624	NT_032977.8	TCATGGCTGCAATGCCTGGT	3-14-3	47
400020	93	20995	NT_032977.8	AGAGAGGAGGGCTTAAAGAA	3-14-3	66
400021	95	21082	NT_032977.8	AGCTGCCAACCTGCAAAAAG	3-14-3	61
400022	143	21481	NT_032977.8	TGAAAATCCATCCAGCACTG	3-14-3	90
400023	89	21589	NT_032977.8	AGAACCATGGAGCACCTGAG	3-14-3	83
400024	123	21696	NT_032977.8	GGCACTGCCCTTCCACCAAA	3-14-3	79
400025	118	24907	NT_032977.8	GCACCATCCAGACCAGAATC	3-14-3	78
400026	114	25413	NT_032977.8	GAGAGGTTCAGATCCAGGCC	3-14-3	81

[0892] Antisense oligonucleotides with the following ISIS Nos exhibited at least 80% inhibition of PCSK9 mRNA levels: 399806, 399814, 399817, 399819, 399821, 399827, 399836, 399837, 399854, 399867, 399889, 399891, 399906, 399907, 399908, 399913, 399920, 399924, 399935, 399945, 399959, 399975, 399977, 399983, 399992, 400000, 400013, 400015, 400022, and 400023. The target segments to which these antisense oligonucleotides are targeted are active target segments. The target regions to which these antisense oligonucleotides are targeted are active target regions.

[0893] ISIS Nos 399814, 399819, 399836, 399891, 399906, 399907, 399913, 399935, 399945, 399959, 399975, 399977, 399983, 399992, 400013, 400015, and 400022 each exhibited at least 85% inhibition of PCSK9 mRNA levels.

[0894] ISIS Nos 399836, 399891, 399906, 399907, 399913, 399975, 399983, 399992, and 400022 each exhibited at least 90% inhibition of PCSK9 mRNA levels.

Example 2: Antisense Inhibition of Human PCSK9 in HepG2 Cells

[0895] Antisense oligonucleotides were tested for their ability to inhibit the expression of PCSK9 mRNA in cultured HepG2 cells. HepG2 cells at a density of 10000 cells per well in a 96-well plate were treated with 1500 nM of antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. PCSK9 mRNA levels were adjusted according to total RNA content as measured by RIBOGREEN®. Results are presented as percent inhibition of PCSK9, relative to untreated control cells.

[0896] Isis Nos. 399819, 395149, 395150, 395151, 395152, 395153, 395154, 395155, 395156, 395157, 395158, 395161, 395162, 395163, 395164, 395165, 395166, 395167, 395168, 395172, 395173, 395174, 395178, 395179, 395182, 395183, 395185, 395186, 395187, 395189, 395190, 395191, 395193, 395194, 395195, 395198, 395199, 395203, 395207, 395210, 395211, 395213, 395214, 395221, 395222, 399793, 399794, 399795, 399798, 399801, 399803, 399804, 399806, 399807, 399812, 399813, 399814, 399816, 399817, 399820, 399821, 399822, 399823, 399827, 399828, 399829, 399833, 399834, 399836, 399837, 399841, 399844, 399846, 399848, 399849, 399851, 399852, 399854, 399855, 399856, 399857, 399862, 399865, 399867, 399868, 399872, 399873, 399874, 399875, 399876, 399877, 399879, 399880, 399883, 399884, 399885, 399886, 399887, 399888, 399889, 399890, 399891, 399892, 399894, 399895, 399896, 399900, 399901, 399902, 399904, 399905, 399906, 399907, 399908, 399909, 399911, 399912, 399913, 399915, 399916, 399918, 399920, 399921, 399924, 399925, 399929, 399931, 399935, 399936, 399937, 399943, 399944, 399945, 399949, 399950, 399954, 399957, 399959, 399960, 399962, 399963, 399966, 399967, 399968, 399969, 399970, 399972, 399973, 399975, 399976, 399977, 399978, 399979, 399983, 399984, 399985, 399986, 399989, 399990, 399992, 399996, 399997, 399998, 400000, 400001, 400002, 400004, 400005, 400007, 400008, 400009, 400010, 400012, 400013, 400015, 400018, 400021, 400023, 400024, 400025, and 400026 each inhibited PCSK9 by at least 70% in this experiment. The target segments to which these antisense oligonucleotides are targeted are active target segments. The target regions to which these antisense oligonucleotides are targeted are active target regions.

[0897] Isis Nos 399819, 395150, 395152, 395153, 395154, 395155, 395156, 395158, 395163, 395164, 395165, 395166, 395167, 395168, 395173, 395178, 395183, 395185, 395186, 395187, 395189, 395190, 395193, 395194, 395199, 395203, 395210, 395213, 395214, 399793, 399798, 399801, 399804, 399806, 399813, 399816, 399820, 399821, 399822, 399836, 399849, 399856, 399857, 399862, 399868, 399875, 399877, 399880, 399885, 399886, 399887, 399888, 399889, 399890, 399894, 399896, 399900, 399901, 399904, 399905, 399906, 399907, 399911, 399912, 399916, 399920, 399925, 399935, 399936, 399937, 399945, 399954, 399957, 399959, 399960, 399962, 399968, 399969, 399973, 399976, 399977, 399978, 399989, 399992, 400002, 400004, 400007, 400012, 400013, 400023, and 400024 each inhibited PCSK9 by at least 80% in this experiment.

[0898] Isis Nos 399819, 395152, 395153, 395155, 395156, 395158, 395163, 395164, 395165, 395166, 395168, 395178, 395183, 395185, 395187, 395189, 395190, 395194, 395199, 395203, 395214, 399793, 399804, 399813, 399821, 399822, 399836, 399856, 399868, 399877, 399887, 399888, 399890, 399900, 399904, 399905, 399907, 399911, 399912, 399916, 399925, 399936, 399954, 399960, 399962, 399969, 399976, 399977, 399978, 399989, 399992, 400012, 400013, 400023, and 400024 each inhibited PCSK9 by at least 85% in this experiment.

[0899] Isis Nos 399819, 395165, 395183, 395185, 399868, 399900, 399907, 399911, 399912, 399916, 399936, 399969, and 400024 each inhibited PCSK9 by at least 90% in this experiment.

Example 3: Antisense Inhibition of Human PCSK9 (Hep3B Cells)

[0900] Antisense oligonucleotides targeted to a PCSK9 nucleic acid were tested for their effects on PCSK9 mRNA in vitro. Cultured Hep3B cells at a density of 4000 cells per well in a 96-well plate were treated with 75 nM of antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. PCSK9 mRNA levels were adjusted according to total RNA content as measured by RIBOGREEN®. Results are presented as percent inhibition of PCSK9, relative to untreated control cells. Antisense oligonucleotides that exhibited at least 30% inhibition of PCSK9 expression are shown in Tables 17, 18, and 19.

[0901] The motif column indicates the wing-gap-wing motif of each antisense oligonucleotide. Antisense oligonucleotides were designed as 5-10-5 gapmers, or alternatively as 3-14-3 gapmers, or alternatively as 2-13-5 gapmers where the gap segment comprises 2'-deoxynucleotides and each wing segment comprises 2'-MOE nucleotides. As illustrated in Table 17, 18 and 19, a single nucleobase sequence may be represented by a 5-10-5 motif as well as a 3-14-3 motif as well as a 2-13-5 motif. "5' target site" indicates the 5'-most nucleotide which the antisense oligonucleotide is targeted on the indicated GENBANK Accession No.

TABLE 17

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410742	294	313	GCCTGGAGCTGACGGTGCCC	5-10-5	75.1	159
405999	298	317	GACCGCCTGGAGCTGACGGT	5-10-5	70.3	160
395151	300	319	AGGACCGCCTGGAGCTGACG	5-10-5	57.3	6
405861	406	425	AAGGCTAGCACCAGCTCCTC	5-10-5	90.5	162
405862	407	426	CAAGGCTAGCACCAGCTCCT	5-10-5	92.3	163
405863	408	427	GCAAGGCTAGCACCAGCTCC	5-10-5	82.6	164
405864	409	428	CGCAAGGCTAGCACCAGCTC	5-10-5	91.8	165
395152	410	429	ACGCAAGGCTAGCACCAGCT	5-10-5	93.3	7
405865	411	430	AACGCAAGGCTAGCACCAGC	5-10-5	84.6	166
405866	412	431	GAACGCAAGGCTAGCACCAG	5-10-5	81.8	167
405867	413	432	GGAACGCAAGGCTAGCACCA	5-10-5	74.0	168
405868	414	433	CGGAACGCAAGGCTAGCACC	5-10-5	76.8	169
399793	417	436	CCTCGGAACGCAAGGCTAGC	5-10-5	77.9	8
410743	421	440	TCCTCCTCGGAACGCAAGGC	5-10-5	76.7	170

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410744	446	465	GTGCTCGGGTGCTTCGGCCA	5-10-5	76.9	171
410745	466	485	TGGAAGGTGGCTGTGGTTCC	5-10-5	41.7	172
395153	480	499	CCTTGGCGCAGCGGTGGAAG	5-10-5	81.1	9
406000	482	501	ATCCTTGGCGCAGCGGTGGA	5-10-5	79.1	173
406001	484	503	GGATCCTTGGCGCAGCGGTG	5-10-5	63.1	174
406002	488	507	CCACGGATCCTTGGCGCAGC	5-10-5	80.8	175
410746	507	526	CGTAGGTGCCAGGCAACCTC	5-10-5	44.8	176
410747	545	564	TGACTGCGAGAGGTGGGTCT	5-10-5	NA	177
406003	555	574	CAGTGCGCTCTGACTGCGAG	5-10-5	84.2	178
406004	557	576	GGCAGTGCGCTCTGACTGCG	5-10-5	56.4	179
406005	559	578	CGGGCAGTGCGCTCTGACTG	5-10-5	83.4	180
406006	562	581	CGGCGGGCAGTGCGCTCTGA	5-10-5	71.0	181
410748	591	610	ATCCCCGGCGGGCAGCCTGG	5-10-5	56.0	182
406007	595	614	AGGTATCCCCGGCGGGCAGC	5-10-5	76.3	183
410574	597	616	TGAGGTATCCCCGGCGGGCA	3-14-3	69.1	184
410529	597	616	TGAGGTATCCCCGGCGGGCA	5-10-5	77.4	184
410647	597	616	TGAGGTATCCCCGGCGGGCA	2-13-5	90.7	184
410575	598	617	GTGAGGTATCCCCGGCGGGC	3-14-3	71.5	185
410648	598	617	GTGAGGTATCCCCGGCGGGC	2-13-5	82.8	185
410530	598	617	GTGAGGTATCCCCGGCGGGC	5-10-5	83.2	185
410649	599	618	GGTGAGGTATCCCCGGCGGG	2-13-5	53.5	186
410576	599	618	GGTGAGGTATCCCCGGCGGG	3-14-3	56.2	186
410531	599	618	GGTGAGGTATCCCCGGCGGG	5-10-5	63.6	186
410650	600	619	TGGTGAGGTATCCCCGGCGG	2-13-5	86.1	11
399877	600	619	TGGTGAGGTATCCCCGGCGG	3-14-3	92.6	11
395155	600	619	TGGTGAGGTATCCCCGGCGG	5-10-5	92.7	11
410577	601	620	TTGGTGAGGTATCCCCGGCG	3-14-3	62.4	187
410532	601	620	TTGGTGAGGTATCCCCGGCG	5-10-5	63.5	187
410651	601	620	TTGGTGAGGTATCCCCGGCG	2-13-5	75.5	187
410652	602	621	CTTGGTGAGGTATCCCCGGC	2-13-5	81.2	188
410578	602	621	CTTGGTGAGGTATCCCCGGC	3-14-3	83.3	188
406008	602	621	CTTGGTGAGGTATCCCCGGC	5-10-5	95.2	188
410653	603	622	TCTTGGTGAGGTATCCCCGG	2-13-5	68.9	189
410579	603	622	TCTTGGTGAGGTATCCCCGG	3-14-3	74.0	189
410533	603	622	TCTTGGTGAGGTATCCCCGG	5-10-5	79.0	189

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410580	604	623	ATCTTGGTGAGGTATCCCCG	3-14-3	62.3	190
410654	604	623	ATCTTGGTGAGGTATCCCCG	2-13-5	65.2	190
406009	604	623	ATCTTGGTGAGGTATCCCCG	5-10-5	90.7	190
410534	605	624	GATCTTGGTGAGGTATCCCC	5-10-5	59.8	191
410581	605	624	GATCTTGGTGAGGTATCCCC	3-14-3	72.9	191
410655	605	624	GATCTTGGTGAGGTATCCCC	2-13-5	77.4	191
399794	606	625	GGATCTTGGTGAGGTATCCC	5-10-5	56.5	12
410656	606	625	GGATCTTGGTGAGGTATCCC	2-13-5	68.0	12
399950	606	625	GGATCTTGGTGAGGTATCCC	3-14-3	74.4	12
410535	607	626	AGGATCTTGGTGAGGTATCC	5-10-5	65.7	192
410582	607	626	AGGATCTTGGTGAGGTATCC	3-14-3	69.5	192
410657	607	626	AGGATCTTGGTGAGGTATCC	2-13-5	73.1	192
406010	609	628	GCAGGATCTTGGTGAGGTAT	5-10-5	56.9	193
406011	611	630	ATGCAGGATCTTGGTGAGGT	5-10-5	70.7	194
406012	613	632	ACATGCAGGATCTTGGTGAG	5-10-5	69.1	195
406013	617	636	GAAGACATGCAGGATCTTGG	5-10-5	77.4	196
395156	620	639	ATGGAAGACATGCAGGATCT	5-10-5	76.7	14
410749	628	647	AGAAGGCCATGGAAGACATG	5-10-5	59.4	197
410750	638	657	GAAGCCAGGAAGAAGGCCAT	5-10-5	57.5	198
399879	646	665	TTCACCAGGAAGCCAGGAAG	3-14-3	91.9	15
406014	648	667	TCTTCACCAGGAAGCCAGGA	5-10-5	94.0	199
406015	653	672	ACTCATCTTCACCAGGAAGC	5-10-5	78.8	200
406016	655	674	CCACTCATCTTCACCAGGAA	5-10-5	87.2	201
410730	657	676	CGCCACTCATCTTCACCAGG	5-10-5	85.4	202
406017	659	678	GTCGCCACTCATCTTCACCA	5-10-5	76.3	203
406018	661	680	AGGTCGCCACTCATCTTCAC	5-10-5	63.2	204
406019	663	682	GCAGGTCGCCACTCATCTTC	5-10-5	68.1	205
406020	665	684	CAGCAGGTCGCCACTCATCT	5-10-5	83.7	206
410731	667	686	TCCAGCAGGTCGCCACTCAT	5-10-5	72.9	207
410751	685	704	GGCAACTTCAAGGCCAGCTC	5-10-5	84.6	208
395158	705	724	CCTCGATGTAGTCGACATGG	5-10-5	93.9	17
410752	724	743	GCAAAGACAGAGGAGTCCTC	5-10-5	76.3	209
406021	782	801	TTCATCCGCCCGGTACCGTG	5-10-5	79.6	210
406022	784	803	TATTCATCCGCCCGGTACCG	5-10-5	78.7	211
406023	787	806	TGGTATTCATCCGCCCGGTA	5-10-5	92.5	212

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
406024	789	808	GCTGGTATTCATCCGCCCGG	5-10-5	69.7	213
410732	791	810	GGGCTGGTATTCATCCGCCC	5-10-5	30.4	214
410753	821	840	ATACACCTCCACCAGGCTGC	5-10-5	72.8	215
406025	832	851	GTGTCTAGGAGATACACCTC	5-10-5	74.2	216
406026	837	856	TGCTGGTGTCTAGGAGATAC	5-10-5	80.5	217
405869	862	881	TCGATTTCCCGGTGGTCACT	5-10-5	87.6	218
405870	863	882	CTCGATTTCCCGGTGGTCAC	5-10-5	83.3	219
405871	864	883	CCTCGATTTCCCGGTGGTCA	5-10-5	87.0	220
405872	865	884	CCCTCGATTTCCCGGTGGTC	5-10-5	88.2	221
399798	866	885	GCCCTCGATTTCCCGGTGGT	5-10-5	84.3	22
399954	866	885	GCCCTCGATTTCCCGGTGGT	3-14-3	90.3	22
405873	867	886	TGCCCTCGATTTCCCGGTGG	5-10-5	90.0	222
405874	868	887	CTGCCCTCGATTTCCCGGTG	5-10-5	91.4	223
405875	869	888	CCTGCCCTCGATTTCCCGGT	5-10-5	93.5	224
405876	870	889	CCCTGCCCTCGATTTCCCGG	5-10-5	90.1	225
406027	874	893	ATGACCCTGCCCTCGATTTC	5-10-5	73.9	226
406028	876	895	CCATGACCCTGCCCTCGATT	5-10-5	92.3	227
406029	878	897	GACCATGACCCTGCCCTCGA	5-10-5	91.4	228
406030	882	901	CGGTGACCATGACCCTGCCC	5-10-5	95.6	229
406031	884	903	GTCGGTGACCATGACCCTGC	5-10-5	89.6	230
410733	886	905	AAGTCGGTGACCATGACCCT	5-10-5	65.7	231
406032	888	907	CGAAGTCGGTGACCATGACC	5-10-5	88.5	232
410754	898	917	GGCACATTCTCGAAGTCGGT	5-10-5	80.1	233
395163	923	942	GTGGAAGCGGGTCCCGTCCT	5-10-5	83.9	25
410755	933	952	TGGCCTGTCTGTGGAAGCGG	5-10-5	NA	234
410756	960	979	GGTGGGTGCCATGACTGTCA	5-10-5	67.1	235
406033	967	986	CCTGCCAGGTGGGTGCCATG	5-10-5	91.2	237
406034	972	991	CCACCCCTGCCAGGTGGGTG	5-10-5	59.3	238
406035	977	996	GCTGACCACCCCTGCCAGGT	5-10-5	91.7	239
410757	985	1004	TCCCGGCCGCTGACCACCCC	5-10-5	88.9	240
406036	989	1008	GGCATCCCGGCCGCTGACCA	5-10-5	84.0	241
406037	992	1011	GCCGGCATCCCGGCCGCTGA	5-10-5	55.9	242
410536	997	1016	GCCACGCCGGCATCCCGGCC	5-10-5	63.7	243
410658	997	1016	GCCACGCCGGCATCCCGGCC	2-13-5	82.0	243
410583	997	1016	GCCACGCCGGCATCCCGGCC	3-14-3	87.3	243

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410537	998	1017	GGCCACGCCGGCATCCCGGC	5-10-5	58.5	244
410659	998	1017	GGCCACGCCGGCATCCCGGC	2-13-5	73.7	244
410584	998	1017	GGCCACGCCGGCATCCCGGC	3-14-3	79.5	244
410660	999	1018	TGGCCACGCCGGCATCCCGG	2-13-5	63.0	245
410585	999	1018	TGGCCACGCCGGCATCCCGG	3-14-3	73.2	245
406038	999	1018	TGGCCACGCCGGCATCCCGG	5-10-5	86.3	245
410661	1000	1019	TTGGCCACGCCGGCATCCCG	2-13-5	53.3	246
410586	1000	1019	TTGGCCACGCCGGCATCCCG	3-14-3	60.3	246
405877	1000	1019	TTGGCCACGCCGGCATCCCG	5-10-5	83.2	246
410587	1001	1020	CTTGGCCACGCCGGCATCCC	3-14-3	63.2	247
410662	1001	1020	CTTGGCCACGCCGGCATCCC	2-13-5	67.3	247
405878	1001	1020	CTTGGCCACGCCGGCATCCC	5-10-5	91.4	247
410588	1002	1021	CCTTGGCCACGCCGGCATCC	3-14-3	65.3	248
410663	1002	1021	CCTTGGCCACGCCGGCATCC	2-13-5	67.9	248
405879	1002	1021	CCTTGGCCACGCCGGCATCC	5-10-5	94.1	248
410589	1003	1022	CCCTTGGCCACGCCGGCATC	3-14-3	80.1	249
410664	1003	1022	CCCTTGGCCACGCCGGCATC	2-13-5	81.9	249
405880	1003	1022	CCCTTGGCCACGCCGGCATC	5-10-5	93.5	249
410665	1004	1023	ACCCTTGGCCACGCCGGCAT	2-13-5	78.7	28
399887	1004	1023	ACCCTTGGCCACGCCGGCAT	3-14-3	91.9	28
395165	1004	1023	ACCCTTGGCCACGCCGGCAT	5-10-5	96.7	28
410590	1005	1024	CACCCTTGGCCACGCCGGCA	3-14-3	82.2	250
410666	1005	1024	CACCCTTGGCCACGCCGGCA	2-13-5	82.7	250
405881	1005	1024	CACCCTTGGCCACGCCGGCA	5-10-5	98.3	250
410667	1006	1025	GCACCCTTGGCCACGCCGGC	2-13-5	84.7	251
410591	1006	1025	GCACCCTTGGCCACGCCGGC	3-14-3	86.3	251
405882	1006	1025	GCACCCTTGGCCACGCCGGC	5-10-5	91.6	251
410668	1007	1026	GGCACCCTTGGCCACGCCGG	2-13-5	91.4	252
405883	1007	1026	GGCACCCTTGGCCACGCCGG	5-10-5	92.5	252
410592	1007	1026	GGCACCCTTGGCCACGCCGG	3-14-3	93.4	252
410669	1008	1027	TGGCACCCTTGGCCACGCCG	2-13-5	69.0	253
410593	1008	1027	TGGCACCCTTGGCCACGCCG	3-14-3	88.8	253
405884	1008	1027	TGGCACCCTTGGCCACGCCG	5-10-5	90.8	253
410670	1009	1028	CTGGCACCCTTGGCCACGCC	2-13-5	74.9	254
410594	1009	1028	CTGGCACCCTTGGCCACGCC	3-14-3	75.2	254

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1						
Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410538	1009	1028	CTGGCACCCTTGGCCACGCC	5-10-5	78.2	254
410539	1010	1029	GCTGGCACCCTTGGCCACGC	5-10-5	61.9	255
410595	1010	1029	GCTGGCACCCTTGGCCACGC	3-14-3	83.1	255
410671	1010	1029	GCTGGCACCCTTGGCCACGC	2-13-5	83.3	255
410758	1015	1034	CGCATGCTGGCACCCTTGGC	5-10-5	73.6	256
406039	1036	1055	CAGTTGAGCACGCGCAGGCT	5-10-5	91.2	257
406040	1038	1057	GGCAGTTGAGCACGCGCAGG	5-10-5	96.5	258
399888	1040	1059	TTGGCAGTTGAGCACGCGCA	3-14-3	90.9	29
395166	1040	1059	TTGGCAGTTGAGCACGCGCA	5-10-5	96.2	29
406041	1042	1061	CCTTGGCAGTTGAGCACGCG	5-10-5	96.6	259
399801	1045	1064	TTCCCTTGGCAGTTGAGCAC	5-10-5	95.6	30
406042	1047	1066	CCTTCCCTTGGCAGTTGAGC	5-10-5	92.0	260
406043	1051	1070	GTGCCCTTCCCTTGGCAGTT	5-10-5	91.9	261
406044	1053	1072	CCGTGCCCTTCCCTTGGCAG	5-10-5	95.4	262
410759	1064	1083	GGTGCCGCTAACCGTGCCCT	5-10-5	83.7	263
410760	1088	1107	CCGAATAAACTCCAGGCCTA	5-10-5	96.7	265
406045	1096	1115	TGGCTTTTCCGAATAAACTC	5-10-5	88.4	266
395168	1098	1117	GCTGGCTTTTCCGAATAAAC	5-10-5	91.9	32
405909	1100	1119	CAGCTGGCTTTTCCGAATAA	5-10-5	80.3	267
405910	1102	1121	ACCAGCTGGCTTTTCCGAAT	5-10-5	90.9	268
405911	1104	1123	GGACCAGCTGGCTTTTCCGA	5-10-5	88.0	269
405912	1108	1127	GGCTGGACCAGCTGGCTTTT	5-10-5	78.2	270
410761	1119	1138	GTGGCCCCACAGGCTGGACC	5-10-5	53.7	271
410762	1132	1151	AGCAGCACCACCAGTGGCCC	5-10-5	57.6	272
410763	1154	1173	GCTGTACCCACCCGCCAGGG	5-10-5	68.8	273
410764	1200	1219	CGACCCCAGCCCTCGCCAGG	5-10-5	66.3	274
399891	1210	1229	GTGACCAGCACGACCCCAGC	3-14-3	91.1	33
405913	1212	1231	CGGTGACCAGCACGACCCCA	5-10-5	93.8	275
405914	1214	1233	AGCGGTGACCAGCACGACCC	5-10-5	90.4	276
405915	1216	1235	GCAGCGGTGACCAGCACGAC	5-10-5	86.8	277
405916	1218	1237	CGGCAGCGGTGACCAGCACG	5-10-5	91.3	278
410734	1219	1238	CCGGCAGCGGTGACCAGCAC	5-10-5	65.8	279
405917	1222	1241	TTGCCGGCAGCGGTGACCAG	5-10-5	62.0	280
405918	1224	1243	AGTTGCCGGCAGCGGTGACC	5-10-5	33.5	281
405919	1226	1245	GAAGTTGCCGGCAGCGGTGA	5-10-5	68.2	282

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
405920	1228	1247	CGGAAGTTGCCGGCAGCGGT	5-10-5	86.2	283
405921	1230	1249	CCCGGAAGTTGCCGGCAGCG	5-10-5	85.5	284
405922	1232	1251	GTCCCGGAAGTTGCCGGCAG	5-10-5	86.0	285
410765	1273	1292	ATGACCTCGGGAGCTGAGGC	5-10-5	50.4	286
410766	1283	1302	CCCAACTGTGATGACCTCGG	5-10-5	70.6	287
405923	1295	1314	GGCATTGGTGGCCCCAACTG	5-10-5	94.3	288
410767	1305	1324	GCTGGTCTTGGGCATTGGTG	5-10-5	65.1	289
405924	1318	1337	CCCAGGGTCACCGGCTGGTC	5-10-5	93.1	290
410735	1320	1339	TCCCCAGGGTCACCGGCTGG	5-10-5	78.1	291
405925	1322	1341	AGTCCCCAGGGTCACCGGCT	5-10-5	93.5	292
405926	1324	1343	AAAGTCCCCAGGGTCACCGG	5-10-5	94.6	293
405927	1328	1347	CCCCAAAGTCCCCAGGGTCA	5-10-5	94.5	294
405928	1333	1352	TTGGTCCCCAAAGTCCCCAG	5-10-5	90.0	295
405929	1337	1356	AAAGTTGGTCCCCAAAGTCC	5-10-5	85.5	296
399895	1340	1359	GCCAAAGTTGGTCCCCAAAG	3-14-3	84.6	36
395173	1340	1359	GCCAAAGTTGGTCCCCAAAG	5-10-5	94.8	36
405930	1342	1361	CGGCCAAAGTTGGTCCCCAA	5-10-5	92.9	297
405931	1344	1363	AGCGGCCAAAGTTGGTCCCC	5-10-5	88.8	298
405932	1346	1365	ACAGCGGCCAAAGTTGGTCC	5-10-5	90.6	299
405933	1348	1367	ACACAGCGGCCAAAGTTGGT	5-10-5	NA	300
405934	1350	1369	CCACACAGCGGCCAAAGTTG	5-10-5	92.7	301
405935	1354	1373	AGGTCCACACAGCGGCCAAA	5-10-5	86.0	302
410736	1356	1375	AGAGGTCCACACAGCGGCCA	5-10-5	73.8	303
405936	1358	1377	AAAGAGGTCCACACAGCGGC	5-10-5	93.9	304
410768	1380	1399	CAATGATGTCCTCCCCTGGG	5-10-5	79.7	305
405937	1387	1406	GAGGCACCAATGATGTCCTC	5-10-5	77.7	306
405938	1391	1410	GCTGGAGGCACCAATGATGT	5-10-5	75.5	307
405939	1393	1412	TCGCTGGAGGCACCAATGAT	5-10-5	70.6	308
405940	1395	1414	AGTCGCTGGAGGCACCAATG	5-10-5	84.4	309
405941	1397	1416	GCAGTCGCTGGAGGCACCAA	5-10-5	90.1	310
399804	1400	1419	GCTGCAGTCGCTGGAGGCAC	5-10-5	80.8	40
399960	1400	1419	GCTGCAGTCGCTGGAGGCAC	3-14-3	82.4	40
405942	1402	1421	GTGCTGCAGTCGCTGGAGGC	5-10-5	69.8	311
405943	1404	1423	AGGTGCTGCAGTCGCTGGAG	5-10-5	83.6	312
410737	1406	1425	GCAGGTGCTGCAGTCGCTGG	5-10-5	49.8	313

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
405944	1407	1426	AGCAGGTGCTGCAGTCGCTG	5-10-5	80.7	314
405945	1409	1428	AAAGCAGGTGCTGCAGTCGC	5-10-5	41.1	315
405946	1413	1432	ACACAAAGCAGGTGCTGCAG	5-10-5	70.4	316
405947	1415	1434	TGACACAAAGCAGGTGCTGC	5-10-5	72.5	317
410769	1425	1444	TCCCACTCTGTGACACAAAG	5-10-5	84.9	318
405948	1467	1486	TCATGGCTGCAATGCCAGCC	5-10-5	45.9	319
399806	1470	1489	GCATCATGGCTGCAATGCCA	5-10-5	82.8	42
399962	1470	1489	GCATCATGGCTGCAATGCCA	3-14-3	86.1	42
405949	1472	1491	CAGCATCATGGCTGCAATGC	5-10-5	84.9	320
405950	1474	1493	GACAGCATCATGGCTGCAAT	5-10-5	76.9	321
405951	1476	1495	CAGACAGCATCATGGCTGCA	5-10-5	75.4	322
405952	1480	1499	TCGGCAGACAGCATCATGGC	5-10-5	85.1	323
410738	1482	1501	GCTCGGCAGACAGCATCATG	5-10-5	68.4	324
405953	1484	1503	CGGCTCGGCAGACAGCATCA	5-10-5	93.6	325
405954	1486	1505	TCCGGCTCGGCAGACAGCAT	5-10-5	87.5	326
410770	1500	1519	CGGCCAGGGTGAGCTCCGGC	5-10-5	68.5	327
405955	1513	1532	CTCTGCCTCAACTCGGCCAG	5-10-5	94.2	328
405956	1515	1534	GTCTCTGCCTCAACTCGGCC	5-10-5	94.3	329
405958	1519	1538	ATCAGTCTCTGCCTCAACTC	5-10-5	80.3	331
405959	1521	1540	GGATCAGTCTCTGCCTCAAC	5-10-5	95.4	332
405960	1523	1542	GTGGATCAGTCTCTGCCTCA	5-10-5	90.1	333
405961	1525	1544	AAGTGGATCAGTCTCTGCCT	5-10-5	88.8	334
405962	1528	1547	GAGAAGTGGATCAGTCTCTG	5-10-5	56.1	335
410739	1530	1549	CAGAGAAGTGGATCAGTCTC	5-10-5	NA	336
405963	1532	1551	GGCAGAGAAGTGGATCAGTC	5-10-5	59.7	337
405964	1536	1555	CTTTGGCAGAGAAGTGGATC	5-10-5	45.3	338
405965	1541	1560	GACATCTTTGGCAGAGAAGT	5-10-5	55.9	339
405966	1543	1562	ATGACATCTTTGGCAGAGAA	5-10-5	51.3	340
405967	1547	1566	ATTGATGACATCTTTGGCAG	5-10-5	64.4	341
405968	1549	1568	TCATTGATGACATCTTTGGC	5-10-5	74.6	342
399967	1552	1571	GCCTCATTGATGACATCTTT	3-14-3	80.1	48
405969	1554	1573	AGGCCTCATTGATGACATCT	5-10-5	63.0	343
405970	1556	1575	CCAGGCCTCATTGATGACAT	5-10-5	66.8	344
410740	1558	1577	AACCAGGCCTCATTGATGAC	5-10-5	46.8	345
405885	1560	1579	GGAACCAGGCCTCATTGATG	5-10-5	73.4	346

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410596	1562	1581	AGGGAACCAGGCCTCATTGA	3-14-3	NA	348
410672	1562	1581	AGGGAACCAGGCCTCATTGA	2-13-5	NA	348
405887	1562	1581	AGGGAACCAGGCCTCATTGA	5-10-5	NA	348
410597	1563	1582	CAGGGAACCAGGCCTCATTG	3-14-3	NA	349
405888	1563	1582	CAGGGAACCAGGCCTCATTG	5-10-5	NA	349
410673	1563	1582	CAGGGAACCAGGCCTCATTG	2-13-5	NA	349
410674	1564	1583	TCAGGGAACCAGGCCTCATT	2-13-5	60.7	49
399812	1564	1583	TCAGGGAACCAGGCCTCATT	5-10-5	68.7	49
399968	1564	1583	TCAGGGAACCAGGCCTCATT	3-14-3	84.5	49
410598	1565	1584	CTCAGGGAACCAGGCCTCAT	3-14-3	74.5	350
410675	1565	1584	CTCAGGGAACCAGGCCTCAT	2-13-5	76.9	350
405889	1565	1584	CTCAGGGAACCAGGCCTCAT	5-10-5	80.2	350
410599	1566	1585	CCTCAGGGAACCAGGCCTCA	3-14-3	70.6	351
410676	1566	1585	CCTCAGGGAACCAGGCCTCA	2-13-5	77.8	351
405890	1566	1585	CCTCAGGGAACCAGGCCTCA	5-10-5	87.7	351
410600	1567	1586	TCCTCAGGGAACCAGGCCTC	3-14-3	76.5	352
410677	1567	1586	TCCTCAGGGAACCAGGCCTC	2-13-5	88.4	352
405891	1567	1586	TCCTCAGGGAACCAGGCCTC	5-10-5	96.1	352
405892	1568	1587	GTCCTCAGGGAACCAGGCCT	5-10-5	71.4	353
410601	1568	1587	GTCCTCAGGGAACCAGGCCT	3-14-3	72.7	353
410678	1568	1587	GTCCTCAGGGAACCAGGCCT	2-13-5	75.0	353
395178	1569	1588	GGTCCTCAGGGAACCAGGCC	5-10-5	76.7	50
410679	1569	1588	GGTCCTCAGGGAACCAGGCC	2-13-5	77.6	50
399900	1569	1588	GGTCCTCAGGGAACCAGGCC	3-14-3	92.0	50
408653	1570	1589	TGGTCCTCAGGGAACCAGGC	5-10-5	42.5	354
410602	1570	1589	TGGTCCTCAGGGAACCAGGC	3-14-3	66.5	354
410680	1570	1589	TGGTCCTCAGGGAACCAGGC	2-13-5	71.6	354
410603	1571	1590	CTGGTCCTCAGGGAACCAGG	3-14-3	40.8	355
410681	1571	1590	CTGGTCCTCAGGGAACCAGG	2-13-5	44.4	355
405971	1571	1590	CTGGTCCTCAGGGAACCAGG	5-10-5	65.5	355
410540	1572	1591	GCTGGTCCTCAGGGAACCAG	5-10-5	50.9	356
410682	1572	1591	GCTGGTCCTCAGGGAACCAG	2-13-5	54.1	356
410604	1572	1591	GCTGGTCCTCAGGGAACCAG	3-14-3	62.5	356
405972	1573	1592	CGCTGGTCCTCAGGGAACCA	5-10-5	77.6	357
405973	1578	1597	GTACCCGCTGGTCCTCAGGG	5-10-5	83.8	358

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
405974	1580	1599	CAGTACCCGCTGGTCCTCAG	5-10-5	89.0	359
399813	1583	1602	GGTCAGTACCCGCTGGTCCT	5-10-5	72.4	51
399969	1583	1602	GGTCAGTACCCGCTGGTCCT	3-14-3	93.4	51
410771	1628	1647	ACCTGCCCCATGGGTGCTGG	5-10-5	NA	360
395181	1640	1659	AAACAGCTGCCAACCTGCCC	5-10-5	87.5	52
405975	1642	1661	CAAAACAGCTGCCAACCTGC	5-10-5	77.4	361
405976	1647	1666	TCCTGCAAAACAGCTGCCAA	5-10-5	81.2	362
405977	1649	1668	AGTCCTGCAAAACAGCTGCC	5-10-5	89.5	363
410772	1660	1679	GCTGACCATACAGTCCTGCA	5-10-5	91.2	364
405978	1672	1691	GGCCCCGAGTGTGCTGACCA	5-10-5	95.3	365
399904	1675	1694	GTAGGCCCCGAGTGTGCTGA	3-14-3	92.6	54
405979	1677	1696	GTGTAGGCCCCGAGTGTGCT	5-10-5	85.5	366
405980	1679	1698	CCGTGTAGGCCCCGAGTGTG	5-10-5	86.5	367
405981	1681	1700	ATCCGTGTAGGCCCCGAGTG	5-10-5	77.8	368
410741	1683	1702	CCATCCGTGTAGGCCCCGAG	5-10-5	78.4	369
405982	1685	1704	GGCCATCCGTGTAGGCCCCG	5-10-5	86.7	370
405983	1687	1706	GTGGCCATCCGTGTAGGCCC	5-10-5	73.1	371
405984	1735	1754	CTGGAGCAGCTCAGCAGCTC	5-10-5	83.4	372
405985	1737	1756	AACTGGAGCAGCTCAGCAGC	5-10-5	50.7	373
405986	1742	1761	GGAGAAACTGGAGCAGCTCA	5-10-5	75.6	374
405987	1744	1763	CTGGAGAAACTGGAGCAGCT	5-10-5	88.0	375
399905	1812	1831	CGTTGTGGGCCCGGCAGACC	3-14-3	91.2	57
395183	1812	1831	CGTTGTGGGCCCGGCAGACC	5-10-5	93.1	57
410541	1849	1868	CACCTGGCAATGGCGTAGAC	5-10-5	46.1	376
410605	1849	1868	CACCTGGCAATGGCGTAGAC	3-14-3	67.4	376
410683	1849	1868	CACCTGGCAATGGCGTAGAC	2-13-5	71.8	376
410606	1850	1869	GCACCTGGCAATGGCGTAGA	3-14-3	74.3	377
410542	1850	1869	GCACCTGGCAATGGCGTAGA	5-10-5	75.5	377
410684	1850	1869	GCACCTGGCAATGGCGTAGA	2-13-5	78.7	377
410543	1851	1870	AGCACCTGGCAATGGCGTAG	5-10-5	76.4	378
410685	1851	1870	AGCACCTGGCAATGGCGTAG	2-13-5	76.9	378
410607	1851	1870	AGCACCTGGCAATGGCGTAG	3-14-3	77.1	378
410544	1852	1871	CAGCACCTGGCAATGGCGTA	5-10-5	62.7	379
410608	1852	1871	CAGCACCTGGCAATGGCGTA	3-14-3	69.6	379
410686	1852	1871	CAGCACCTGGCAATGGCGTA	2-13-5	81.0	379

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410545	1853	1872	GCAGCACCTGGCAATGGCGT	5-10-5	75.5	380
410687	1853	1872	GCAGCACCTGGCAATGGCGT	2-13-5	79.2	380
410609	1853	1872	GCAGCACCTGGCAATGGCGT	3-14-3	83.2	380
410610	1854	1873	GGCAGCACCTGGCAATGGCG	3-14-3	67.5	381
410688	1854	1873	GGCAGCACCTGGCAATGGCG	2-13-5	89.3	381
405988	1854	1873	GGCAGCACCTGGCAATGGCG	5-10-5	95.9	381
410546	1855	1874	AGGCAGCACCTGGCAATGGC	5-10-5	74.8	382
410689	1855	1874	AGGCAGCACCTGGCAATGGC	2-13-5	83.9	382
410611	1855	1874	AGGCAGCACCTGGCAATGGC	3-14-3	88.2	382
410612	1856	1875	CAGGCAGCACCTGGCAATGG	3-14-3	72.8	383
410690	1856	1875	CAGGCAGCACCTGGCAATGG	2-13-5	74.8	383
405989	1856	1875	CAGGCAGCACCTGGCAATGG	5-10-5	88.1	383
410547	1857	1876	GCAGGCAGCACCTGGCAATG	5-10-5	63.0	384
410691	1857	1876	GCAGGCAGCACCTGGCAATG	2-13-5	70.2	384
410613	1857	1876	GCAGGCAGCACCTGGCAATG	3-14-3	76.7	384
395184	1858	1877	AGCAGGCAGCACCTGGCAAT	5-10-5	68.5	58
410692	1858	1877	AGCAGGCAGCACCTGGCAAT	2-13-5	69.1	58
399906	1858	1877	AGCAGGCAGCACCTGGCAAT	3-14-3	94.2	58
410614	1859	1878	TAGCAGGCAGCACCTGGCAA	3-14-3	36.6	385
410548	1859	1878	TAGCAGGCAGCACCTGGCAA	5-10-5	75.0	385
410693	1859	1878	TAGCAGGCAGCACCTGGCAA	2-13-5	82.0	385
405990	1860	1879	GTAGCAGGCAGCACCTGGCA	5-10-5	96.8	386
410773	1905	1924	TGGCCTCAGCTGGTGGAGCT	5-10-5	61.3	387
410549	1915	1934	GTCCCCATGCTGGCCTCAGC	5-10-5	63.0	388
410615	1915	1934	GTCCCCATGCTGGCCTCAGC	3-14-3	72.5	388
410694	1915	1934	GTCCCCATGCTGGCCTCAGC	2-13-5	74.8	388
410550	1916	1935	GGTCCCCATGCTGGCCTCAG	5-10-5	69.4	389
410616	1916	1935	GGTCCCCATGCTGGCCTCAG	3-14-3	80.7	389
410695	1916	1935	GGTCCCCATGCTGGCCTCAG	2-13-5	85.5	389
410551	1917	1936	GGGTCCCCATGCTGGCCTCA	5-10-5	68.7	390
410696	1917	1936	GGGTCCCCATGCTGGCCTCA	2-13-5	86.3	390
410617	1917	1936	GGGTCCCCATGCTGGCCTCA	3-14-3	86.5	390
410552	1918	1937	CGGGTCCCCATGCTGGCCTC	5-10-5	77.9	391
410618	1918	1937	CGGGTCCCCATGCTGGCCTC	3-14-3	86.7	391
410697	1918	1937	CGGGTCCCCATGCTGGCCTC	2-13-5	87.0	391

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410553	1919	1938	ACGGGTCCCCATGCTGGCCT	5-10-5	72.8	392
410619	1919	1938	ACGGGTCCCCATGCTGGCCT	3-14-3	84.1	392
410698	1919	1938	ACGGGTCCCCATGCTGGCCT	2-13-5	87.0	392
410699	1920	1939	CACGGGTCCCCATGCTGGCC	2-13-5	79.5	59
395185	1920	1939	CACGGGTCCCCATGCTGGCC	5-10-5	92.7	59
399907	1920	1939	CACGGGTCCCCATGCTGGCC	3-14-3	93.7	59
410620	1921	1940	ACACGGGTCCCCATGCTGGC	3-14-3	74.4	393
410554	1921	1940	ACACGGGTCCCCATGCTGGC	5-10-5	79.7	393
410700	1921	1940	ACACGGGTCCCCATGCTGGC	2-13-5	84.4	393
410621	1922	1941	GACACGGGTCCCCATGCTGG	3-14-3	76.0	394
410701	1922	1941	GACACGGGTCCCCATGCTGG	2-13-5	83.4	394
405991	1922	1941	GACACGGGTCCCCATGCTGG	5-10-5	91.9	394
410555	1923	1942	GGACACGGGTCCCCATGCTG	5-10-5	77.0	395
410622	1923	1942	GGACACGGGTCCCCATGCTG	3-14-3	79.8	395
410702	1923	1942	GGACACGGGTCCCCATGCTG	2-13-5	85.0	395
410703	1924	1943	TGGACACGGGTCCCCATGCT	2-13-5	70.4	396
410623	1924	1943	TGGACACGGGTCCCCATGCT	3-14-3	78.9	396
405992	1924	1943	TGGACACGGGTCCCCATGCT	5-10-5	89.3	396
410704	1925	1944	GTGGACACGGGTCCCCATGC	2-13-5	78.6	397
410556	1925	1944	GTGGACACGGGTCCCCATGC	5-10-5	81.4	397
410624	1925	1944	GTGGACACGGGTCCCCATGC	3-14-3	82.8	397
410625	1926	1945	AGTGGACACGGGTCCCCATG	3-14-3	64.4	398
410705	1926	1945	AGTGGACACGGGTCCCCATG	2-13-5	74.7	398
405993	1926	1945	AGTGGACACGGGTCCCCATG	5-10-5	85.5	398
410706	1927	1946	CAGTGGACACGGGTCCCCAT	2-13-5	70.8	399
410557	1927	1946	CAGTGGACACGGGTCCCCAT	5-10-5	74.2	399
410626	1927	1946	CAGTGGACACGGGTCCCCAT	3-14-3	76.8	399
410627	1928	1947	GCAGTGGACACGGGTCCCCA	3-14-3	71.4	400
410707	1928	1947	GCAGTGGACACGGGTCCCCA	2-13-5	72.8	400
405994	1928	1947	GCAGTGGACACGGGTCCCCA	5-10-5	95.2	400
410708	1929	1948	GGCAGTGGACACGGGTCCCC	2-13-5	79.0	401
410628	1929	1948	GGCAGTGGACACGGGTCCCC	3-14-3	88.1	401
410558	1929	1948	GGCAGTGGACACGGGTCCCC	5-10-5	88.9	401
410629	1930	1949	TGGCAGTGGACACGGGTCCC	3-14-3	66.5	402
410709	1930	1949	TGGCAGTGGACACGGGTCCC	2-13-5	66.9	402

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
405995	1930	1949	TGGCAGTGGACACGGGTCCC	5-10-5	92.4	402
410630	1931	1950	GTGGCAGTGGACACGGGTCC	3-14-3	54.1	403
410710	1931	1950	GTGGCAGTGGACACGGGTCC	2-13-5	61.4	403
410559	1931	1950	GTGGCAGTGGACACGGGTCC	5-10-5	77.5	403
410711	1932	1951	GGTGGCAGTGGACACGGGTC	2-13-5	NA	404
410631	1932	1951	GGTGGCAGTGGACACGGGTC	3-14-3	NA	404
410560	1932	1951	GGTGGCAGTGGACACGGGTC	5-10-5	42.5	404
410712	1933	1952	TGGTGGCAGTGGACACGGGT	2-13-5	NA	405
410632	1933	1952	TGGTGGCAGTGGACACGGGT	3-14-3	NA	405
410561	1933	1952	TGGTGGCAGTGGACACGGGT	5-10-5	NA	405
410774	1936	1955	TGTTGGTGGCAGTGGACACG	5-10-5	47.8	406
410775	1962	1981	AGCTGCAGCCTGTGAGGACG	5-10-5	36.3	407
410776	1990	2009	GTGCCAAGGTCCTCCACCTC	5-10-5	77.3	408
410777	2010	2029	TCAGCACAGGCGGCTTGTGG	5-10-5	40.0	409
410778	2040	2059	CCACGCACTGGTTGGGCTGA	5-10-5	52.7	410
399908	2100	2119	CTTTGCATTCCAGACCTGGG	3-14-3	74.9	60
410713	2100	2119	CTTTGCATTCCAGACCTGGG	2-13-5	84.9	60
395186	2100	2119	CTTTGCATTCCAGACCTGGG	5-10-5	93.7	60
410714	2101	2120	ACTTTGCATTCCAGACCTGG	2-13-5	82.4	411
410633	2101	2120	ACTTTGCATTCCAGACCTGG	3-14-3	83.8	411
410562	2101	2120	ACTTTGCATTCCAGACCTGG	5-10-5	87.3	411
410715	2102	2121	GACTTTGCATTCCAGACCTG	2-13-5	80.5	412
410634	2102	2121	GACTTTGCATTCCAGACCTG	3-14-3	81.8	412
405996	2102	2121	GACTTTGCATTCCAGACCTG	5-10-5	91.0	412
410563	2103	2122	TGACTTTGCATTCCAGACCT	5-10-5	75.4	413
410635	2103	2122	TGACTTTGCATTCCAGACCT	3-14-3	75.9	413
410716	2103	2122	TGACTTTGCATTCCAGACCT	2-13-5	88.3	413
410636	2104	2123	TTGACTTTGCATTCCAGACC	3-14-3	61.3	414
410564	2104	2123	TTGACTTTGCATTCCAGACC	5-10-5	71.3	414
410717	2104	2123	TTGACTTTGCATTCCAGACC	2-13-5	76.7	414
410718	2105	2124	CTTGACTTTGCATTCCAGAC	2-13-5	55.8	61
399973	2105	2124	CTTGACTTTGCATTCCAGAC	3-14-3	82.1	61
399817	2105	2124	CTTGACTTTGCATTCCAGAC	5-10-5	93.6	61
405997	2107	2126	TCCTTGACTTTGCATTCCAG	5-10-5	95.6	415
410779	2120	2139	CGGGATTCCATGCTCCTTGA	5-10-5	79.7	416

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410780	2150	2169	GCAGGCCACGGTCACCTGCT	5-10-5	60.9	417
410781	2187	2206	GGAGGGCACTGCAGCCAGTC	5-10-5	66.0	418
410719	2305	2324	CAGATGGCAACGGCTGTCAC	2-13-5	61.0	419
410637	2305	2324	CAGATGGCAACGGCTGTCAC	3-14-3	66.0	419
410565	2305	2324	CAGATGGCAACGGCTGTCAC	5-10-5	71.6	419
410638	2306	2325	GCAGATGGCAACGGCTGTCA	3-14-3	71.9	420
410720	2306	2325	GCAGATGGCAACGGCTGTCA	2-13-5	75.1	420
410566	2306	2325	GCAGATGGCAACGGCTGTCA	5-10-5	77.7	420
410639	2307	2326	AGCAGATGGCAACGGCTGTC	3-14-3	67.7	421
410721	2307	2326	AGCAGATGGCAACGGCTGTC	2-13-5	68.8	421
410567	2307	2326	AGCAGATGGCAACGGCTGTC	5-10-5	72.1	421
410568	2308	2327	CAGCAGATGGCAACGGCTGT	5-10-5	54.2	422
410640	2308	2327	CAGCAGATGGCAACGGCTGT	3-14-3	57.7	422
410722	2308	2327	CAGCAGATGGCAACGGCTGT	2-13-5	59.2	422
410569	2309	2328	GCAGCAGATGGCAACGGCTG	5-10-5	57.7	423
410641	2309	2328	GCAGCAGATGGCAACGGCTG	3-14-3	66.1	423
410723	2309	2328	GCAGCAGATGGCAACGGCTG	2-13-5	81.1	423
399909	2310	2329	GGCAGCAGATGGCAACGGCT	3-14-3	69.9	62
410724	2310	2329	GGCAGCAGATGGCAACGGCT	2-13-5	87.4	62
395187	2310	2329	GGCAGCAGATGGCAACGGCT	5-10-5	92.7	62
410642	2311	2330	CGGCAGCAGATGGCAACGGC	3-14-3	70.4	424
410725	2311	2330	CGGCAGCAGATGGCAACGGC	2-13-5	75.6	424
410570	2311	2330	CGGCAGCAGATGGCAACGGC	5-10-5	78.3	424
410571	2312	2331	CCGGCAGCAGATGGCAACGG	5-10-5	52.8	425
410643	2312	2331	CCGGCAGCAGATGGCAACGG	3-14-3	59.8	425
410726	2312	2331	CCGGCAGCAGATGGCAACGG	2-13-5	69.8	425
410644	2313	2332	TCCGGCAGCAGATGGCAACG	3-14-3	59.0	426
410727	2313	2332	TCCGGCAGCAGATGGCAACG	2-13-5	68.3	426
405998	2313	2332	TCCGGCAGCAGATGGCAACG	5-10-5	75.6	426
410572	2314	2333	CTCCGGCAGCAGATGGCAAC	5-10-5	40.9	427
410728	2314	2333	CTCCGGCAGCAGATGGCAAC	2-13-5	56.4	427
410645	2314	2333	CTCCGGCAGCAGATGGCAAC	3-14-3	70.2	427
410729	2315	2334	GCTCCGGCAGCAGATGGCAA	2-13-5	67.7	428
410573	2315	2334	GCTCCGGCAGCAGATGGCAA	5-10-5	71.8	428
410646	2315	2334	GCTCCGGCAGCAGATGGCAA	3-14-3	72.8	428

TABLE 17-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 1

Isis No.	5' Target Site to SEQ ID NO: 1	3' Target Site to SEQ ID NO: 1	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
410782	2325	2344	CCAGGTGCCGGCTCCGGCAG	5-10-5	43.2	429
410783	2335	2354	GAGGCCTGCGCCAGGTGCCG	5-10-5	63.7	430
399819	2509	2528	CCCACTCAAGGGCCAGGCCA	5-10-5	93.4	65
399912	2597	2616	ATGCCCCACAGTGAGGGAGG	3-14-3	84.3	66
405893	2828	2847	ATGAGGGCCATCAGCACCTT	5-10-5	93.9	431
405894	2829	2848	GATGAGGGCCATCAGCACCT	5-10-5	93.9	432
405895	2830	2849	AGATGAGGGCCATCAGCACC	5-10-5	91.8	433
405896	2831	2850	GAGATGAGGGCCATCAGCAC	5-10-5	88.5	434
395194	2832	2851	GGAGATGAGGGCCATCAGCA	5-10-5	90.0	69
399916	2832	2851	GGAGATGAGGGCCATCAGCA	3-14-3	94.5	69
405897	2833	2852	TGGAGATGAGGGCCATCAGC	5-10-5	85.1	435
405898	2834	2853	CTGGAGATGAGGGCCATCAG	5-10-5	86.9	436
405899	2835	2854	GCTGGAGATGAGGGCCATCA	5-10-5	91.5	437
405900	2836	2855	AGCTGGAGATGAGGGCCATC	5-10-5	83.7	438
405901	2902	2921	GCTAGATGCCATCCAGAAAG	5-10-5	88.4	439
405902	2903	2922	GGCTAGATGCCATCCAGAAA	5-10-5	89.3	440
405903	2904	2923	TGGCTAGATGCCATCCAGAA	5-10-5	92.6	441
405904	2905	2924	CTGGCTAGATGCCATCCAGA	5-10-5	90.4	442
399821	2906	2925	TCTGGCTAGATGCCATCCAG	5-10-5	91.7	71
405905	2907	2926	CTCTGGCTAGATGCCATCCA	5-10-5	90.7	443
405906	2908	2927	CCTCTGGCTAGATGCCATCC	5-10-5	93.3	444
405907	2909	2928	GCCTCTGGCTAGATGCCATC	5-10-5	90.2	445
405908	2910	2929	AGCCTCTGGCTAGATGCCAT	5-10-5	83.3	446
399978	2988	3007	AGCCTGGCATAGAGCAGAGT	3-14-3	96.4	73

[0902] Antisense oligonucleotides that exhibited less than 30% inhibition of PCSK9 mRNA levels were marked with "NA".

[0903] Antisense oligonucleotides with the following ISIS Nos exhibited at least 80% inhibition of PCSK9 mRNA levels: 395152, 395153, 395155, 395158, 395163, 395165, 395166, 395168, 395173, 395181, 395183, 395185, 395186, 395187, 395194, 399798, 399801, 399804, 399806, 399817, 399819, 399821, 399877, 399879, 399887, 399888, 399891, 399895, 399900, 399904, 399905, 399906, 399907, 399912, 399916, 399954, 399960, 399962, 399967, 399968, 399969, 399973, 399978, 405861, 405862, 405863, 405864, 405865, 405866, 405869, 405870, 405871, 405872, 405873, 405874, 405875, 405876, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405889, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405900, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405908, 405909, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405923, 405924, 405925, 405926, 405927, 405928, 405929, 405930, 405931, 405932, 405934, 405935, 405936, 405940, 405941, 405943, 405944, 405949, 405952, 405953, 405954, 405955, 405956, 405958, 405959, 405960, 405961, 405973, 405974, 405976, 405977, 405978, 405979, 405980, 405982, 405984, 405987, 405988, 405989, 405990, 405991, 405992, 405993, 405994, 405995, 405996, 405997, 406002, 406003, 406005, 406008, 406009, 406014, 406016, 406020, 406023, 406026, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406036, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410530, 410556, 410558, 410562, 410578, 410583, 410589, 410590, 410591, 410592, 410593, 410595, 410609, 410611, 410616, 410617, 410618, 410619, 410624, 410628, 410633, 410634, 410647, 410648, 410650, 410652, 410658, 410664, 410666, 410667, 410668, 410671, 410677, 410686, 410688, 410689, 410693, 410695, 410696, 410697, 410698, 410700, 410701, 410702, 410713, 410714, 410715, 410716, 410723, 410724, 410730, 410751, 410754, 410757, 410759, 410760, 410769, and 410772. The target segments to which these antisense oligonucleotides are targeted are active target segments. The target regions to which these antisense oligonucleotides are targeted are active target regions.

[0904] ISIS Nos 395152, 395155, 395158, 395165, 395166, 395168, 395173, 395181, 395183, 395185, 395186, 395187, 395194, 399801, 399817, 399819, 399821, 399877, 399879, 399887, 399888, 399891, 399900, 399904, 399905, 399906, 399907, 399916, 399954, 399962, 399969, 399978, 405861, 405862, 405864, 405869, 405871, 405872, 405873, 405874, 405875, 405876, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405923, 405924, 405925, 405926, 405927, 405928, 405929, 405930, 405931, 405932, 405934, 405935, 405936, 405941, 405952, 405953, 405954, 405955, 405956, 405959, 405960, 405961, 405974, 405977, 405978, 405979, 405980, 405982, 405987, 405988, 405989, 405990, 405991, 405992, 405993, 405994, 405995, 405996, 405997, 406008, 406009, 406014, 406016, 406023, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410558, 410562, 410583, 410591, 410592, 410593, 410611, 410617, 410618, 410628, 410647, 410650, 410668, 410677, 410688, 410695, 410696, 410697, 410698, 410702, 410716, 410724, 410730, 410757, 410760, and 410772 each exhibited at least 85% inhibition of PCSK9 mRNA levels.

[0905] ISIS Nos 395152, 395155, 395158, 395165, 395166, 395168, 395173, 395183, 395185, 395186, 395187, 395194, 399801, 399817, 399819, 399821, 399877, 399879, 399887, 399888, 399891, 399900, 399904, 399905, 399906, 399907, 399916, 399954, 399969, 399978, 405861, 405862, 405864, 405873, 405874, 405875, 405876, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405891, 405893, 405894, 405895, 405899, 405903, 405904, 405905, 405906, 405907, 405910, 405913, 405914, 405916, 405923, 405924, 405925, 405926, 405927, 405928, 405930, 405932, 405934, 405936, 405941, 405953, 405955, 405956, 405959, 405960, 405978, 405988, 405990, 405991, 405994, 405995, 405996, 405997, 406008, 406009, 406014, 406023, 406028, 406029, 406030, 406033, 406035, 406039, 406040, 406041, 406042, 406043, 406044, 410592, 410647, 410668, 410760, and 410772 each exhibited at least 90% inhibition of PCSK9 mRNA levels.

[0906] ISIS Nos 395165, 395166, 399801, 399978, 405881, 405891, 405959, 405978, 405988, 405990, 405994, 405997, 406008, 406030, 406040, 406041, 406044, and 410760 each exhibited at least 95% inhibition of PCSK9 mRNA levels.

TABLE 18

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410742	2433	2452	GCCTGGAGCTGACGGTGCCC	5-10-5	75.1	159
405999	2437	2456	GACCGCCTGGAGCTGACGGT	5-10-5	70.3	160
395151	2439	2458	AGGACCGCCTGGAGCTGACG	5-10-5	57.3	6
405861	2545	2564	AAGGCTAGCACCAGCTCCTC	5-10-5	90.5	162
405862	2546	2565	CAAGGCTAGCACCAGCTCCT	5-10-5	92.3	163
405863	2547	2566	GCAAGGCTAGCACCAGCTCC	5-10-5	82.6	164
405864	2548	2567	CGCAAGGCTAGCACCAGCTC	5-10-5	91.8	165
395152	2549	2568	ACGCAAGGCTAGCACCAGCT	5-10-5	93.3	7
405865	2550	2569	AACGCAAGGCTAGCACCAGC	5-10-5	84.6	166
405866	2551	2570	GAACGCAAGGCTAGCACCAG	5-10-5	81.8	167
405867	2552	2571	GGAACGCAAGGCTAGCACCA	5-10-5	74.0	168
405868	2553	2572	CGGAACGCAAGGCTAGCACC	5-10-5	76.8	169
399793	2556	2575	CCTCGGAACGCAAGGCTAGC	5-10-5	77.9	8
410743	2560	2579	TCCTCCTCGGAACGCAAGGC	5-10-5	76.7	170
410744	2585	2604	GTGCTCGGGTGCTTCGGCCA	5-10-5	76.9	171
410745	2605	2624	TGGAAGGTGGCTGTGGTTCC	5-10-5	41.7	172
395153	2619	2638	CCTTGGCGCAGCGGTGGAAG	5-10-5	81.1	9

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410746	6444	6463	CGTAGGTGCCAGGCAACCTC	5-10-5	44.8	176
410747	6482	6501	TGACTGCGAGAGGTGGGTCT	5-10-5	NA	177
406003	6492	6511	CAGTGCGCTCTGACTGCGAG	5-10-5	84.2	178
406004	6494	6513	GGCAGTGCGCTCTGACTGCG	5-10-5	56.4	179
406005	6496	6515	CGGGCAGTGCGCTCTGACTG	5-10-5	83.4	180
406006	6499	6518	CGGCGGGCAGTGCGCTCTGA	5-10-5	71.0	181
410748	6528	6547	ATCCCCGGCGGGCAGCCTGG	5-10-5	56.0	182
406007	6532	6551	AGGTATCCCCGGCGGGCAGC	5-10-5	76.3	183
410574	6534	6553	TGAGGTATCCCCGGCGGGCA	3-14-3	69.1	184
410529	6534	6553	TGAGGTATCCCCGGCGGGCA	5-10-5	77.4	184
410647	6534	6553	TGAGGTATCCCCGGCGGGCA	2-13-5	90.7	184
410575	6535	6554	GTGAGGTATCCCCGGCGGGC	3-14-3	71.5	185
410648	6535	6554	GTGAGGTATCCCCGGCGGGC	2-13-5	82.8	185
410530	6535	6554	GTGAGGTATCCCCGGCGGGC	5-10-5	83.2	185
410649	6536	6555	GGTGAGGTATCCCCGGCGGG	2-13-5	53.5	186
410576	6536	6555	GGTGAGGTATCCCCGGCGGG	3-14-3	56.2	186
410531	6536	6555	GGTGAGGTATCCCCGGCGGG	5-10-5	63.6	186
410650	6537	6556	TGGTGAGGTATCCCCGGCGG	2-13-5	86.1	11
399877	6537	6556	TGGTGAGGTATCCCCGGCGG	3-14-3	92.6	11
395155	6537	6556	TGGTGAGGTATCCCCGGCGG	5-10-5	92.7	11
410577	6538	6557	TTGGTGAGGTATCCCCGGCG	3-14-3	62.4	187
410532	6538	6557	TTGGTGAGGTATCCCCGGCG	5-10-5	63.5	187
410651	6538	6557	TTGGTGAGGTATCCCCGGCG	2-13-5	75.5	187
410652	6539	6558	CTTGGTGAGGTATCCCCGGC	2-13-5	81.2	188
410578	6539	6558	CTTGGTGAGGTATCCCCGGC	3-14-3	83.3	188
406008	6539	6558	CTTGGTGAGGTATCCCCGGC	5-10-5	95.2	188
410653	6540	6559	TCTTGGTGAGGTATCCCCGG	2-13-5	68.9	189
410579	6540	6559	TCTTGGTGAGGTATCCCCGG	3-14-3	74.0	189
410533	6540	6559	TCTTGGTGAGGTATCCCCGG	5-10-5	79.0	189
410580	6541	6560	ATCTTGGTGAGGTATCCCCG	3-14-3	62.3	190
410654	6541	6560	ATCTTGGTGAGGTATCCCCG	2-13-5	65.2	190
406009	6541	6560	ATCTTGGTGAGGTATCCCCG	5-10-5	90.7	190
410534	6542	6561	GATCTTGGTGAGGTATCCCC	5-10-5	59.8	191
410581	6542	6561	GATCTTGGTGAGGTATCCCC	3-14-3	72.9	191
410655	6542	6561	GATCTTGGTGAGGTATCCCC	2-13-5	77.4	191

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
399794	6543	6562	GGATCTTGGTGAGGTATCCC	5-10-5	56.5	12
410656	6543	6562	GGATCTTGGTGAGGTATCCC	2-13-5	68.0	12
399950	6543	6562	GGATCTTGGTGAGGTATCCC	3-14-3	74.4	12
410535	6544	6563	AGGATCTTGGTGAGGTATCC	5-10-5	65.7	192
410582	6544	6563	AGGATCTTGGTGAGGTATCC	3-14-3	69.5	192
410657	6544	6563	AGGATCTTGGTGAGGTATCC	2-13-5	73.1	192
406010	6546	6565	GCAGGATCTTGGTGAGGTAT	5-10-5	56.9	193
406011	6548	6567	ATGCAGGATCTTGGTGAGGT	5-10-5	70.7	194
406012	6550	6569	ACATGCAGGATCTTGGTGAG	5-10-5	69.1	195
406013	6554	6573	GAAGACATGCAGGATCTTGG	5-10-5	77.4	196
395156	6557	6576	ATGGAAGACATGCAGGATCT	5-10-5	76.7	14
410749	6565	6584	AGAAGGCCATGGAAGACATG	5-10-5	59.4	197
410750	6575	6594	GAAGCCAGGAAGAAGGCCAT	5-10-5	57.5	198
399879	6583	6602	TTCACCAGGAAGCCAGGAAG	3-14-3	91.9	15
406014	6585	6604	TCTTCACCAGGAAGCCAGGA	5-10-5	94.0	199
406015	6590	6609	ACTCATCTTCACCAGGAAGC	5-10-5	78.8	200
406016	6592	6611	CCACTCATCTTCACCAGGAA	5-10-5	87.2	201
410730	6594	6613	CGCCACTCATCTTCACCAGG	5-10-5	85.4	202
406017	6596	6615	GTCGCCACTCATCTTCACCA	5-10-5	76.3	203
406018	6598	6617	AGGTCGCCACTCATCTTCAC	5-10-5	63.2	204
406019	6600	6619	GCAGGTCGCCACTCATCTTC	5-10-5	68.1	205
406020	6602	6621	CAGCAGGTCGCCACTCATCT	5-10-5	83.7	206
410731	6604	6623	TCCAGCAGGTCGCCACTCAT	5-10-5	72.9	207
395158	9130	9149	CCTCGATGTAGTCGACATGG	5-10-5	93.9	17
410752	9149	9168	GCAAAGACAGAGGAGTCCTC	5-10-5	76.3	209
406021	9207	9226	TTCATCCGCCCGGTACCGTG	5-10-5	79.6	210
406022	9209	9228	TATTCATCCGCCCGGTACCG	5-10-5	78.7	211
406023	9212	9231	TGGTATTCATCCGCCCGGTA	5-10-5	92.5	212
406024	9214	9233	GCTGGTATTCATCCGCCCGG	5-10-5	69.7	213
410732	9216	9235	GGGCTGGTATTCATCCGCCC	5-10-5	30.4	214
399935	14601	14620	AGGACCCAAGTCATCCTGCT	3-14-3	81.1	96
399936	14631	14650	GGCCATCAGCTGGCAATGCT	3-14-3	89.0	124
410753	14877	14896	ATACACCTCCACCAGGCTGC	5-10-5	72.8	315
406025	14888	14907	GTGTCTAGGAGATACACCTC	5-10-5	74.2	216

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
406026	14893	14912	TGCTGGTGTCTAGGAGATAC	5-10-5	80.5	217
405869	14918	14937	TCGATTTCCCGGTGGTCACT	5-10-5	87.6	218
405870	14919	14938	CTCGATTTCCCGGTGGTCAC	5-10-5	83.3	219
405871	14920	14939	CCTCGATTTCCCGGTGGTCA	5-10-5	87.0	220
405872	14921	14940	CCCTCGATTTCCCGGTGGTC	5-10-5	88.2	221
399798	14922	14941	GCCCTCGATTTCCCGGTGGT	5-10-5	84.3	22
399954	14922	14941	GCCCTCGATTTCCCGGTGGT	3-14-3	90.3	22
405873	14923	14942	TGCCCTCGATTTCCCGGTGG	5-10-5	90.0	222
405874	14924	14943	CTGCCCTCGATTTCCCGGTG	5-10-5	91.4	223
405875	14925	14944	CCTGCCCTCGATTTCCCGGT	5-10-5	93.5	224
405876	14926	14945	CCCTGCCCTCGATTTCCCGG	5-10-5	90.1	225
406027	14930	14949	ATGACCCTGCCCTCGATTTC	5-10-5	73.9	226
406028	14932	14951	CCATGACCCTGCCCTCGATT	5-10-5	92.3	227
406029	14934	14953	GACCATGACCCTGCCCTCGA	5-10-5	91.4	228
406030	14938	14957	CGGTGACCATGACCCTGCCC	5-10-5	95.6	229
406031	14940	14959	GTCGGTGACCATGACCCTGC	5-10-5	89.6	230
410733	14942	14961	AAGTCGGTGACCATGACCCT	5-10-5	65.7	231
406032	14944	14963	CGAAGTCGGTGACCATGACC	5-10-5	88.5	232
410754	14954	14973	GGCACATTCTCGAAGTCGGT	5-10-5	80.1	233
395163	14979	14998	GTGGAAGCGGGTCCCGTCCT	5-10-5	83.9	25
410756	15254	15273	GGTGGGTGCCATGACTGTCA	5-10-5	67.1	235
406033	15261	15280	CCTGCCAGGTGGGTGCCATG	5-10-5	91.2	237
406034	15266	15285	CCACCCCTGCCAGGTGGGTG	5-10-5	59.3	238
406035	15271	15290	GCTGACCACCCCTGCCAGGT	5-10-5	91.7	239
410757	15279	15298	TCCCGGCCGCTGACCACCCC	5-10-5	88.9	240
406036	15283	15302	GGCATCCCGGCCGCTGACCA	5-10-5	84.0	241
406037	15286	15305	GCCGGCATCCCGGCCGCTGA	5-10-5	55.9	242
410536	15291	15310	GCCACGCCGGCATCCCGGCC	5-10-5	63.7	243
410658	15291	15310	GCCACGCCGGCATCCCGGCC	2-13-5	82.0	243
410583	15291	15310	GCCACGCCGGCATCCCGGCC	3-14-3	87.3	243
410537	15292	15311	GGCCACGCCGGCATCCCGGC	5-10-5	58.5	244
410659	15292	15311	GGCCACGCCGGCATCCCGGC	2-13-5	73.7	244
410584	15292	15311	GGCCACGCCGGCATCCCGGC	3-14-3	79.5	244
410660	15293	15312	TGGCCACGCCGGCATCCCGG	2-13-5	63.0	245
410585	15293	15312	TGGCCACGCCGGCATCCCGG	3-14-3	73.2	245

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
406038	15293	15312	TGGCCACGCCGGCATCCCGG	5-10-5	86.3	245
410661	15294	15313	TTGGCCACGCCGGCATCCCG	2-13-5	53.3	246
410586	15294	15313	TTGGCCACGCCGGCATCCCG	3-14-3	60.3	246
405877	15294	15313	TTGGCCACGCCGGCATCCCG	5-10-5	83.2	246
410587	15295	15314	CTTGGCCACGCCGGCATCCC	3-14-3	63.2	247
410662	15295	15314	CTTGGCCACGCCGGCATCCC	2-13-5	67.3	247
405878	15295	15314	CTTGGCCACGCCGGCATCCC	5-10-5	91.4	247
410588	15296	15315	CCTTGGCCACGCCGGCATCC	3-14-3	65.3	248
410663	15296	15315	CCTTGGCCACGCCGGCATCC	2-13-5	67.9	248
405879	15296	15315	CCTTGGCCACGCCGGCATCC	5-10-5	94.1	248
410589	15297	15316	CCCTTGGCCACGCCGGCATC	3-14-3	80.1	249
410664	15297	15316	CCCTTGGCCACGCCGGCATC	2-13-5	81.9	249
405880	15297	15316	CCCTTGGCCACGCCGGCATC	5-10-5	93.5	249
410665	15298	15317	ACCCTTGGCCACGCCGGCAT	2-13-5	78.7	28
399887	15298	15317	ACCCTTGGCCACGCCGGCAT	3-14-3	91.9	28
395165	15298	15317	ACCCTTGGCCACGCCGGCAT	5-10-5	96.7	28
410590	15299	15318	CACCCTTGGCCACGCCGGCA	3-14-3	82.2	250
410666	15299	15318	CACCCTTGGCCACGCCGGCA	2-13-5	82.7	250
405881	15299	15318	CACCCTTGGCCACGCCGGCA	5-10-5	98.3	250
410667	15300	15319	GCACCCTTGGCCACGCCGGC	2-13-5	84.7	251
410591	15300	15319	GCACCCTTGGCCACGCCGGC	3-14-3	86.3	251
405882	15300	15319	GCACCCTTGGCCACGCCGGC	5-10-5	91.6	251
410668	15301	15320	GGCACCCTTGGCCACGCCGG	2-13-5	91.4	252
405883	15301	15320	GGCACCCTTGGCCACGCCGG	5-10-5	92.5	252
410592	15301	15320	GGCACCCTTGGCCACGCCGG	3-14-3	93.4	252
410669	15302	15321	TGGCACCCTTGGCCACGCCG	2-13-5	69.0	253
410593	15302	15321	TGGCACCCTTGGCCACGCCG	3-14-3	88.8	253
405884	15302	15321	TGGCACCCTTGGCCACGCCG	5-10-5	90.8	253
410670	15303	15322	CTGGCACCCTTGGCCACGCC	2-13-5	74.9	254
410594	15303	15322	CTGGCACCCTTGGCCACGCC	3-14-3	75.2	254
410538	15303	15322	CTGGCACCCTTGGCCACGCC	5-10-5	78.2	254
410539	15304	15323	GCTGGCACCCTTGGCCACGC	5-10-5	61.9	255
410595	15304	15323	GCTGGCACCCTTGGCCACGC	3-14-3	83.1	255
410671	15304	15323	GCTGGCACCCTTGGCCACGC	2-13-5	83.3	255

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410758	15309	15328	CGCATGCTGGCACCCTTGGC	5-10-5	73.6	256
406039	15330	15349	CAGTTGAGCACGCGCAGGCT	5-10-5	91.2	257
406040	15332	15351	GGCAGTTGAGCACGCGCAGG	5-10-5	96.5	258
399888	15334	15353	TTGGCAGTTGAGCACGCGCA	3-14-3	90.9	29
395166	15334	15353	TTGGCAGTTGAGCACGCGCA	5-10-5	96.2	29
406041	15336	15355	CCTTGGCAGTTGAGCACGCG	5-10-5	96.6	259
399801	15339	15358	TTCCCTTGGCAGTTGAGCAC	5-10-5	95.6	30
406042	15341	15360	CCTTCCCTTGGCAGTTGAGC	5-10-5	92.0	260
406043	15345	15364	GTGCCCTTCCCTTGGCAGTT	5-10-5	91.9	261
406044	15347	15366	CCGTGCCCTTCCCTTGGCAG	5-10-5	95.4	262
410759	15358	15377	GGTGCCGCTAACCGTGCCCT	5-10-5	83.7	263
406045	18591	18610	TGGCTTTTCCGAATAAACTC	5-10-5	88.4	266
395168	18593	18612	GCTGGCTTTTCCGAATAAAC	5-10-5	91.9	32
405909	18595	18614	CAGCTGGCTTTTCCGAATAA	5-10-5	80.3	267
405910	18597	18616	ACCAGCTGGCTTTTCCGAAT	5-10-5	90.9	268
405911	18599	18618	GGACCAGCTGGCTTTTCCGA	5-10-5	88.0	269
405912	18603	18622	GGCTGGACCAGCTGGCTTTT	5-10-5	78.2	270
410761	18614	18633	GTGGCCCCACAGGCTGGACC	5-10-5	53.7	271
410762	18627	18646	AGCAGCACCACCAGTGGCCC	5-10-5	57.6	272
410763	18649	18668	GCTGTACCCACCCGCCAGGG	5-10-5	68.8	273
410764	18695	18714	CGACCCCAGCCCTCGCCAGG	5-10-5	66.3	274
399891	18705	18724	GTGACCAGCACGACCCCAGC	3-14-3	91.1	33
405913	18707	18726	CGGTGACCAGCACGACCCCA	5-10-5	93.8	275
405914	18709	18728	AGCGGTGACCAGCACGACCC	5-10-5	90.4	276
405915	18711	18730	GCAGCGGTGACCAGCACGAC	5-10-5	86.8	277
405916	18713	18732	CGGCAGCGGTGACCAGCACG	5-10-5	91.3	278
410734	18714	18733	CCGGCAGCGGTGACCAGCAC	5-10-5	65.8	279
405917	18717	18736	TTGCCGGCAGCGGTGACCAG	5-10-5	62.0	280
405918	18719	18738	AGTTGCCGGCAGCGGTGACC	5-10-5	33.5	281
405919	18721	18740	GAAGTTGCCGGCAGCGGTGA	5-10-5	68.2	282
405920	18723	18742	CGGAAGTTGCCGGCAGCGGT	5-10-5	86.2	283
405921	18725	18744	CCCGGAAGTTGCCGGCAGCG	5-10-5	85.5	284
405922	18727	18746	GTCCCGGAAGTTGCCGGCAG	5-10-5	86.0	285
405923	19931	19950	GGCATTGGTGGCCCCAACTG	5-10-5	94.3	288
410767	19941	19960	GCTGGTCTTGGGCATTGGTG	5-10-5	65.1	289

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2						
Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
405924	19954	19973	CCCAGGGTCACCGGCTGGTC	5-10-5	93.1	290
410735	19956	19975	TCCCCAGGGTCACCGGCTGG	5-10-5	78.1	291
405925	19958	19977	AGTCCCCAGGGTCACCGGCT	5-10-5	93.5	292
405926	19960	19979	AAAGTCCCCAGGGTCACCGG	5-10-5	94.6	293
405927	19964	19983	CCCCAAAGTCCCCAGGGTCA	5-10-5	94.5	294
405928	19969	19988	TTGGTCCCCAAAGTCCCCAG	5-10-5	90.0	295
405929	19973	19992	AAAGTTGGTCCCCAAAGTCC	5-10-5	85.5	296
399895	19976	19995	GCCAAAGTTGGTCCCCAAAG	3-14-3	84.6	36
395173	19976	19995	GCCAAAGTTGGTCCCCAAAG	5-10-5	94.8	36
405930	19978	19997	CGGCCAAAGTTGGTCCCCAA	5-10-5	92.9	297
405931	19980	19999	AGCGGCCAAAGTTGGTCCCC	5-10-5	88.8	298
405932	19982	20001	ACAGCGGCCAAAGTTGGTCC	5-10-5	90.6	299
405933	19984	20003	ACACAGCGGCCAAAGTTGGT	5-10-5	NA	300
405934	19986	20005	CCACACAGCGGCCAAAGTTG	5-10-5	92.7	301
405935	19990	20009	AGGTCCACACAGCGGCCAAA	5-10-5	86.0	302
410736	19992	20011	AGAGGTCCACACAGCGGCCA	5-10-5	73.8	303
405936	19994	20013	AAAGAGGTCCACACAGCGGC	5-10-5	93.9	304
410768	20016	20035	CAATGATGTCCTCCCCTGGG	5-10-5	79.7	305
405937	20023	20042	GAGGCACCAATGATGTCCTC	5-10-5	77.7	306
405938	20027	20046	GCTGGAGGCACCAATGATGT	5-10-5	75.5	307
405939	20029	20048	TCGCTGGAGGCACCAATGAT	5-10-5	70.6	308
405940	20031	20050	AGTCGCTGGAGGCACCAATG	5-10-5	84.4	309
405941	20033	20052	GCAGTCGCTGGAGGCACCAA	5-10-5	90.1	310
399804	20036	20055	GCTGCAGTCGCTGGAGGCAC	5-10-5	80.8	40
399960	20036	20055	GCTGCAGTCGCTGGAGGCAC	3-14-3	82.4	40
405942	20038	20057	GTGCTGCAGTCGCTGGAGGC	5-10-5	69.8	311
405943	20040	20059	AGGTGCTGCAGTCGCTGGAG	5-10-5	83.6	312
410737	20042	20061	GCAGGTGCTGCAGTCGCTGG	5-10-5	49.8	313
405944	20043	20062	AGCAGGTGCTGCAGTCGCTG	5-10-5	80.7	314
405945	20045	20064	AAAGCAGGTGCTGCAGTCGC	5-10-5	41.1	315
405946	20049	20068	ACACAAAGCAGGTGCTGCAG	5-10-5	70.4	316
405947	20051	20070	TGACACAAAGCAGGTGCTGC	5-10-5	72.5	317
410769	20061	20080	TCCCACTCTGTGACACAAAG	5-10-5	84.9	318
405949	20629	20648	CAGCATCATGGCTGCAATGC	5-10-5	84.9	320

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
405950	20631	20650	GACAGCATCATGGCTGCAAT	5-10-5	76.9	321
405951	20633	20652	CAGACAGCATCATGGCTGCA	5-10-5	75.4	322
405952	20637	20656	TCGGCAGACAGCATCATGGC	5-10-5	85.1	323
410738	20639	20658	GCTCGGCAGACAGCATCATG	5-10-5	68.4	324
405953	20641	20660	CGGCTCGGCAGACAGCATCA	5-10-5	93.6	325
405954	20643	20662	TCCGGCTCGGCAGACAGCAT	5-10-5	87.5	326
410770	20657	20676	CGGCCAGGGTGAGCTCCGGC	5-10-5	68.5	327
405955	20670	20689	CTCTGCCTCAACTCGGCCAG	5-10-5	94.2	328
405956	20672	20691	GTCTCTGCCTCAACTCGGCC	5-10-5	94.3	329
405958	20676	20695	ATCAGTCTCTGCCTCAACTC	5-10-5	80.3	331
405959	20678	20697	GGATCAGTCTCTGCCTCAAC	5-10-5	95.4	332
405960	20680	20699	GTGGATCAGTCTCTGCCTCA	5-10-5	90.1	333
405961	20682	20701	AAGTGGATCAGTCTCTGCCT	5-10-5	88.8	334
405962	20685	20704	GAGAAGTGGATCAGTCTCTG	5-10-5	56.1	335
410739	20687	20706	CAGAGAAGTGGATCAGTCTC	5-10-5	NA	336
405963	20689	20708	GGCAGAGAAGTGGATCAGTC	5-10-5	59.7	337
405964	20693	20712	CTTTGGCAGAGAAGTGGATC	5-10-5	45.3	338
405965	20698	20717	GACATCTTTGGCAGAGAAGT	5-10-5	55.9	339
405966	20700	20719	ATGACATCTTTGGCAGAGAA	5-10-5	51.3	340
405967	20704	20723	ATTGATGACATCTTTGGCAG	5-10-5	64.4	341
405968	20706	20725	TCATTGATGACATCTTTGGC	5-10-5	74.6	342
399967	20709	20728	GCCTCATTGATGACATCTTT	3-14-3	80.1	48
405969	20711	20730	AGGCCTCATTGATGACATCT	5-10-5	63.0	343
405970	20713	20732	CCAGGCCTCATTGATGACAT	5-10-5	66.8	344
410740	20715	20734	AACCAGGCCTCATTGATGAC	5-10-5	46.8	345
405885	20717	20736	GGAACCAGGCCTCATTGATG	5-10-5	73.4	346
410596	20719	20738	AGGGAACCAGGCCTCATTGA	3-14-3	NA	348
410672	20719	20738	AGGGAACCAGGCCTCATTGA	2-13-5	NA	348
405887	20719	20738	AGGGAACCAGGCCTCATTGA	5-10-5	NA	348
410597	20720	20739	CAGGGAACCAGGCCTCATTG	3-14-3	NA	349
405888	20720	20739	CAGGGAACCAGGCCTCATTG	5-10-5	NA	349
410673	20720	20739	CAGGGAACCAGGCCTCATTG	2-13-5	NA	349
410674	20721	20740	TCAGGGAACCAGGCCTCATT	2-13-5	60.7	49
399812	20721	20740	TCAGGGAACCAGGCCTCATT	5-10-5	68.7	49
399968	20721	20740	TCAGGGAACCAGGCCTCATT	3-14-3	84.5	49

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410598	20722	20741	CTCAGGGAACCAGGCCTCAT	3-14-3	74.5	350
410675	20722	20741	CTCAGGGAACCAGGCCTCAT	2-13-5	76.9	350
405889	20722	20741	CTCAGGGAACCAGGCCTCAT	5-10-5	80.2	350
410599	20723	20742	CCTCAGGGAACCAGGCCTCA	3-14-3	70.6	351
410676	20723	20742	CCTCAGGGAACCAGGCCTCA	2-13-5	77.8	351
405890	20723	20742	CCTCAGGGAACCAGGCCTCA	5-10-5	87.7	351
410600	20724	20743	TCCTCAGGGAACCAGGCCTC	3-14-3	76.5	352
410677	20724	20743	TCCTCAGGGAACCAGGCCTC	2-13-5	88.4	352
405891	20724	20743	TCCTCAGGGAACCAGGCCTC	5-10-5	96.1	352
405892	20725	20744	GTCCTCAGGGAACCAGGCCT	5-10-5	71.4	353
410601	20725	20744	GTCCTCAGGGAACCAGGCCT	3-14-3	72.7	353
410678	20725	20744	GTCCTCAGGGAACCAGGCCT	2-13-5	75.0	353
395178	20726	20745	GGTCCTCAGGGAACCAGGCC	5-10-5	76.7	50
410679	20726	20745	GGTCCTCAGGGAACCAGGCC	2-13-5	77.6	50
399900	20726	20745	GGTCCTCAGGGAACCAGGCC	3-14-3	92.0	50
408653	20727	20746	TGGTCCTCAGGGAACCAGGC	5-10-5	42.5	354
410602	20727	20746	TGGTCCTCAGGGAACCAGGC	3-14-3	66.5	354
410680	20727	20746	TGGTCCTCAGGGAACCAGGC	2-13-5	71.6	354
410603	20728	20747	CTGGTCCTCAGGGAACCAGG	3-14-3	40.8	355
410681	20728	20747	CTGGTCCTCAGGGAACCAGG	2-13-5	44.4	355
405971	20728	20747	CTGGTCCTCAGGGAACCAGG	5-10-5	65.5	355
410540	20729	20748	GCTGGTCCTCAGGGAACCAG	5-10-5	50.9	356
410682	20729	20748	GCTGGTCCTCAGGGAACCAG	2-13-5	54.1	356
410604	20729	20748	GCTGGTCCTCAGGGAACCAG	3-14-3	62.5	356
405972	20730	20749	CGCTGGTCCTCAGGGAACCA	5-10-5	77.6	357
405973	20735	20754	GTACCCGCTGGTCCTCAGGG	5-10-5	83.8	358
405974	20737	20756	CAGTACCCGCTGGTCCTCAG	5-10-5	89.0	359
399813	20740	20759	GGTCAGTACCCGCTGGTCCT	5-10-5	72.4	51
399969	20740	20759	GGTCAGTACCCGCTGGTCCT	3-14-3	93.4	51
410771	20785	20804	ACCTGCCCCATGGGTGCTGG	5-10-5	NA	360
405975	21088	21107	CAAAACAGCTGCCAACCTGC	5-10-5	77.4	361
405976	21093	21112	TCCTGCAAAACAGCTGCCAA	5-10-5	81.2	362
405977	21095	21114	AGTCCTGCAAAACAGCTGCC	5-10-5	89.5	363
410772	21106	21125	GCTGACCATACAGTCCTGCA	5-10-5	91.2	264

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
405978	21118	21137	GGCCCCGAGTGTGCTGACCA	5-10-5	95.3	365
399904	21121	21140	GTAGGCCCCGAGTGTGCTGA	3-14-3	92.6	54
405979	21123	21142	GTGTAGGCCCCGAGTGTGCT	5-10-5	85.5	366
405980	21125	21144	CCGTGTAGGCCCCGAGTGTG	5-10-5	86.5	367
405981	21127	21146	ATCCGTGTAGGCCCCGAGTG	5-10-5	77.8	368
410741	21129	21148	CCATCCGTGTAGGCCCCGAG	5-10-5	78.4	369
405982	21131	21150	GGCCATCCGTGTAGGCCCCG	5-10-5	86.7	370
405983	21133	21152	GTGGCCATCCGTGTAGGCCC	5-10-5	73.1	371
405984	21181	21200	CTGGAGCAGCTCAGCAGCTC	5-10-5	83.4	372
405985	21183	21202	AACTGGAGCAGCTCAGCAGC	5-10-5	50.7	373
405986	21188	21207	GGAGAAACTGGAGCAGCTCA	5-10-5	75.6	374
405987	21190	21209	CTGGAGAAACTGGAGCAGCT	5-10-5	88.0	375
399868	21696	21715	GGCACTGCCCTTCCACCAAA	5-10-5	85.8	123
399905	22096	22115	CGTTGTGGGCCCGGCAGACC	3-14-3	91.2	57
395183	22096	22115	CGTTGTGGGCCCGGCAGACC	5-10-5	93.1	57
410541	22133	22152	CACCTGGCAATGGCGTAGAC	5-10-5	46.1	376
410605	22133	22152	CACCTGGCAATGGCGTAGAC	3-14-3	67.4	376
410683	22133	22152	CACCTGGCAATGGCGTAGAC	2-13-5	71.8	376
410606	22134	22153	GCACCTGGCAATGGCGTAGA	3-14-3	74.3	377
410542	22134	22153	GCACCTGGCAATGGCGTAGA	5-10-5	75.5	377
410684	22134	22153	GCACCTGGCAATGGCGTAGA	2-13-5	78.7	377
410543	22135	22154	AGCACCTGGCAATGGCGTAG	5-10-5	76.4	378
410685	22135	22154	AGCACCTGGCAATGGCGTAG	2-13-5	76.9	378
410607	22135	22154	AGCACCTGGCAATGGCGTAG	3-14-3	77.1	378
410544	22136	22155	CAGCACCTGGCAATGGCGTA	5-10-5	62.7	379
410608	22136	22155	CAGCACCTGGCAATGGCGTA	3-14-3	69.6	379
410686	22136	22155	CAGCACCTGGCAATGGCGTA	2-13-5	81.0	379
410545	22137	22156	GCAGCACCTGGCAATGGCGT	5-10-5	75.5	380
410687	22137	22156	GCAGCACCTGGCAATGGCGT	2-13-5	79.2	380
410609	22137	22156	GCAGCACCTGGCAATGGCGT	3-14-3	83.2	380
410610	22138	22157	GGCAGCACCTGGCAATGGCG	3-14-3	67.5	381
410688	22138	22157	GGCAGCACCTGGCAATGGCG	2-13-5	89.3	381
405988	22138	22157	GGCAGCACCTGGCAATGGCG	5-10-5	95.9	381
410546	22139	22158	AGGCAGCACCTGGCAATGGC	5-10-5	74.8	382
410689	22139	22158	AGGCAGCACCTGGCAATGGC	2-13-5	83.9	382

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2						
Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410611	22139	22158	AGGCAGCACCTGGCAATGGC	3-14-3	88.2	382
410612	22140	22159	CAGGCAGCACCTGGCAATGG	3-14-3	72.8	383
410690	22140	22159	CAGGCAGCACCTGGCAATGG	2-13-5	74.8	383
405989	22140	22159	CAGGCAGCACCTGGCAATGG	5-10-5	88.1	383
410547	22141	22160	GCAGGCAGCACCTGGCAATG	5-10-5	63.0	384
410691	22141	22160	GCAGGCAGCACCTGGCAATG	2-13-5	70.2	384
410613	22141	22160	GCAGGCAGCACCTGGCAATG	3-14-3	76.7	384
395184	22142	22161	AGCAGGCAGCACCTGGCAAT	5-10-5	68.5	58
410692	22142	22161	AGCAGGCAGCACCTGGCAAT	2-13-5	69.1	58
399906	22142	22161	AGCAGGCAGCACCTGGCAAT	3-14-3	94.2	58
410614	22143	22162	TAGCAGGCAGCACCTGGCAA	3-14-3	36.6	385
410548	22143	22162	TAGCAGGCAGCACCTGGCAA	5-10-5	75.0	385
410693	22143	22162	TAGCAGGCAGCACCTGGCAA	2-13-5	82.0	385
405990	22144	22163	GTAGCAGGCAGCACCTGGCA	5-10-5	96.8	386
410773	22189	22208	TGGCCTCAGCTGGTGGAGCT	5-10-5	61.3	387
410549	22199	22218	GTCCCCATGCTGGCCTCAGC	5-10-5	63.0	388
410615	22199	22218	GTCCCCATGCTGGCCTCAGC	3-14-3	72.5	388
410694	22199	22218	GTCCCCATGCTGGCCTCAGC	2-13-5	74.8	388
410550	22200	22219	GGTCCCCATGCTGGCCTCAG	5-10-5	69.4	389
410616	22200	22219	GGTCCCCATGCTGGCCTCAG	3-14-3	80.7	389
410695	22200	22219	GGTCCCCATGCTGGCCTCAG	2-13-5	85.5	389
410551	22201	22220	GGGTCCCCATGCTGGCCTCA	5-10-5	68.7	390
410696	22201	22220	GGGTCCCCATGCTGGCCTCA	2-13-5	86.3	390
410617	22201	22220	GGGTCCCCATGCTGGCCTCA	3-14-3	86.5	390
410552	22202	22221	CGGGTCCCCATGCTGGCCTC	5-10-5	77.9	391
410618	22202	22221	CGGGTCCCCATGCTGGCCTC	3-14-3	86.7	391
410697	22202	22221	CGGGTCCCCATGCTGGCCTC	2-13-5	87.0	391
410553	22203	22222	ACGGGTCCCCATGCTGGCCT	5-10-5	72.8	392
410619	22203	22222	ACGGGTCCCCATGCTGGCCT	3-14-3	84.1	392
410698	22203	22222	ACGGGTCCCCATGCTGGCCT	2-13-5	87.0	392
410699	22204	22223	CACGGGTCCCCATGCTGGCC	2-13-5	79.5	59
395185	22204	22223	CACGGGTCCCCATGCTGGCC	5-10-5	92.7	59
399907	22204	22223	CACGGGTCCCCATGCTGGCC	3-14-3	93.7	59
410620	22205	22224	ACACGGGTCCCCATGCTGGC	3-14-3	74.4	393

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410554	22205	22224	ACACGGGTCCCCATGCTGGC	5-10-5	79.7	393
410700	22205	22224	ACACGGGTCCCCATGCTGGC	2-13-5	84.4	393
410621	22206	22225	GACACGGGTCCCCATGCTGG	3-14-3	76.0	394
410701	22206	22225	GACACGGGTCCCCATGCTGG	2-13-5	83.4	394
405991	22206	22225	GACACGGGTCCCCATGCTGG	5-10-5	91.9	394
410555	22207	22226	GGACACGGGTCCCCATGCTG	5-10-5	77.0	395
410622	22207	22226	GGACACGGGTCCCCATGCTG	3-14-3	79.8	395
410702	22207	22226	GGACACGGGTCCCCATGCTG	2-13-5	85.0	395
410703	22208	22227	TGGACACGGGTCCCCATGCT	2-13-5	70.4	396
410623	22208	22227	TGGACACGGGTCCCCATGCT	3-14-3	78.9	396
405992	22208	22227	TGGACACGGGTCCCCATGCT	5-10-5	89.3	396
410704	22209	22228	GTGGACACGGGTCCCCATGC	2-13-5	78.6	397
410556	22209	22228	GTGGACACGGGTCCCCATGC	5-10-5	81.4	397
410624	22209	22228	GTGGACACGGGTCCCCATGC	3-14-3	82.8	397
410625	22210	22229	AGTGGACACGGGTCCCCATG	3-14-3	64.4	398
410705	22210	22229	AGTGGACACGGGTCCCCATG	2-13-5	74.7	398
405993	22210	22229	AGTGGACACGGGTCCCCATG	5-10-5	85.5	398
410706	22211	22230	CAGTGGACACGGGTCCCCAT	2-13-5	70.8	399
410557	22211	22230	CAGTGGACACGGGTCCCCAT	5-10-5	74.2	399
410626	22211	22230	CAGTGGACACGGGTCCCCAT	3-14-3	76.8	399
410627	22212	22231	GCAGTGGACACGGGTCCCCA	3-14-3	71.4	400
410707	22212	22231	GCAGTGGACACGGGTCCCCA	2-13-5	72.8	400
405994	22212	22231	GCAGTGGACACGGGTCCCCA	5-10-5	95.2	400
410708	22213	22232	GGCAGTGGACACGGGTCCCC	2-13-5	79.0	401
410628	22213	22232	GGCAGTGGACACGGGTCCCC	3-14-3	88.1	401
410558	22213	22232	GGCAGTGGACACGGGTCCCC	5-10-5	88.9	401
410629	22214	22233	TGGCAGTGGACACGGGTCCC	3-14-3	66.5	402
410709	22214	22233	TGGCAGTGGACACGGGTCCC	2-13-5	66.9	402
405995	22214	22233	TGGCAGTGGACACGGGTCCC	5-10-5	92.4	402
410630	22215	22234	GTGGCAGTGGACACGGGTCC	3-14-3	54.1	403
410710	22215	22234	GTGGCAGTGGACACGGGTCC	2-13-5	61.4	403
410559	22215	22234	GTGGCAGTGGACACGGGTCC	5-10-5	77.5	403
410711	22216	22235	GGTGGCAGTGGACACGGGTC	2-13-5	NA	404
410631	22216	22235	GGTGGCAGTGGACACGGGTC	3-14-3	NA	404
410560	22216	22235	GGTGGCAGTGGACACGGGTC	5-10-5	42.5	404

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410712	22217	22236	TGGTGGCAGTGGACACGGGT	2-13-5	NA	405
410632	22217	22236	TGGTGGCAGTGGACACGGGT	3-14-3	NA	405
410561	22217	22236	TGGTGGCAGTGGACACGGGT	5-10-5	NA	405
410774	22220	22239	TGTTGGTGGCAGTGGACACG	5-10-5	47.8	406
410776	23985	24004	GTGCCAAGGTCCTCCACCTC	5-10-5	77.3	408
410777	24005	24024	TCAGCACAGGCGGCTTGTGG	5-10-5	40.0	409
410778	24035	24054	CCACGCACTGGTTGGGCTGA	5-10-5	52.7	410
399908	24095	24114	CTTTGCATTCCAGACCTGGG	3-14-3	74.9	60
410713	24095	24114	CTTTGCATTCCAGACCTGGG	2-13-5	84.9	60
395186	24095	24114	CTTTGCATTCCAGACCTGGG	5-10-5	93.7	60
410714	24096	24115	ACTTTGCATTCCAGACCTGG	2-13-5	82.4	411
410633	24096	24115	ACTTTGCATTCCAGACCTGG	3-14-3	83.8	411
410562	24096	24115	ACTTTGCATTCCAGACCTGG	5-10-5	87.3	411
410715	24097	24116	GACTTTGCATTCCAGACCTG	2-13-5	80.5	412
410634	24097	24116	GACTTTGCATTCCAGACCTG	3-14-3	81.8	412
405996	24097	24116	GACTTTGCATTCCAGACCTG	5-10-5	91.0	412
410563	24098	24117	TGACTTTGCATTCCAGACCT	5-10-5	75.4	413
410635	24098	24117	TGACTTTGCATTCCAGACCT	3-14-3	75.9	413
410716	24098	24117	TGACTTTGCATTCCAGACCT	2-13-5	88.3	413
410636	24099	24118	TTGACTTTGCATTCCAGACC	3-14-3	61.3	414
410564	24099	24118	TTGACTTTGCATTCCAGACC	5-10-5	71.3	414
410717	24099	24118	TTGACTTTGCATTCCAGACC	2-13-5	76.7	414
410718	24100	24119	CTTGACTTTGCATTCCAGAC	2-13-5	55.8	61
399973	24100	24119	CTTGACTTTGCATTCCAGAC	3-14-3	82.1	61
399817	24100	24119	CTTGACTTTGCATTCCAGAC	5-10-5	93.6	61
405997	24102	24121	TCCTTGACTTTGCATTCCAG	5-10-5	95.6	415
410779	24115	24134	CGGGATTCCATGCTCCTTGA	5-10-5	79.7	416
410781	25994	26013	GGAGGGCACTGCAGCCAGTC	5-10-5	66.0	418
410719	26112	26131	CAGATGGCAACGGCTGTCAC	2-13-5	61.0	419
410637	26112	26131	CAGATGGCAACGGCTGTCAC	3-14-3	66.0	419
410565	26112	26131	CAGATGGCAACGGCTGTCAC	5-10-5	71.6	419
410638	26113	26132	GCAGATGGCAACGGCTGTCA	3-14-3	71.9	420
410720	26113	26132	GCAGATGGCAACGGCTGTCA	2-13-5	75.1	420
410566	26113	26132	GCAGATGGCAACGGCTGTCA	5-10-5	77.7	420

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410639	26114	26133	AGCAGATGGCAACGGCTGTC	3-14-3	67.7	421
410721	26114	26133	AGCAGATGGCAACGGCTGTC	2-13-5	68.8	421
410567	26114	26133	AGCAGATGGCAACGGCTGTC	5-10-5	72.1	421
410568	26115	26134	CAGCAGATGGCAACGGCTGT	5-10-5	54.2	422
410640	26115	26134	CAGCAGATGGCAACGGCTGT	3-14-3	57.7	422
410722	26115	26134	CAGCAGATGGCAACGGCTGT	2-13-5	59.2	422
410569	26116	26135	GCAGCAGATGGCAACGGCTG	5-10-5	57.7	423
410641	26116	26135	GCAGCAGATGGCAACGGCTG	3-14-3	66.1	423
410723	26116	26135	GCAGCAGATGGCAACGGCTG	2-13-5	81.1	423
399909	26117	26136	GGCAGCAGATGGCAACGGCT	3-14-3	69.9	62
410724	26117	26136	GGCAGCAGATGGCAACGGCT	2-13-5	87.4	62
395187	26117	26136	GGCAGCAGATGGCAACGGCT	5-10-5	92.7	62
410642	26118	26137	CGGCAGCAGATGGCAACGGC	3-14-3	70.4	424
410725	26118	26137	CGGCAGCAGATGGCAACGGC	2-13-5	75.6	424
410570	26118	26137	CGGCAGCAGATGGCAACGGC	5-10-5	78.3	424
410571	26119	26138	CCGGCAGCAGATGGCAACGG	5-10-5	52.8	425
410643	26119	26138	CCGGCAGCAGATGGCAACGG	3-14-3	59.8	425
410726	26119	26138	CCGGCAGCAGATGGCAACGG	2-13-5	69.8	425
410644	26120	26139	TCCGGCAGCAGATGGCAACG	3-14-3	59.0	426
410727	26120	26139	TCCGGCAGCAGATGGCAACG	2-13-5	68.3	426
405998	26120	26139	TCCGGCAGCAGATGGCAACG	5-10-5	75.6	426
410572	26121	26140	CTCCGGCAGCAGATGGCAAC	5-10-5	40.9	427
410728	26121	26140	CTCCGGCAGCAGATGGCAAC	2-13-5	56.4	427
410645	26121	26140	CTCCGGCAGCAGATGGCAAC	3-14-3	70.2	427
410729	26122	26141	GCTCCGGCAGCAGATGGCAA	2-13-5	67.7	428
410573	26122	26141	GCTCCGGCAGCAGATGGCAA	5-10-5	71.8	428
410646	26122	26141	GCTCCGGCAGCAGATGGCAA	3-14-3	72.8	428
410782	26132	26151	CCAGGTGCCGGCTCCGGCAG	5-10-5	43.2	429
410783	26142	26161	GAGGCCTGCGCCAGGTGCCG	5-10-5	63.7	430
399819	26316	26335	CCCACTCAAGGGCCAGGCCA	5-10-5	93.4	65
399912	26404	26423	ATGCCCCACAGTGAGGGAGG	3-14-3	84.3	66
405893	26635	26654	ATGAGGGCCATCAGCACCTT	5-10-5	93.9	431
405894	26636	26655	GATGAGGGCCATCAGCACCT	5-10-5	93.9	432
405895	26637	26656	AGATGAGGGCCATCAGCACC	5-10-5	91.8	433
405896	26638	26657	GAGATGAGGGCCATCAGCAC	5-10-5	88.5	434

TABLE 18-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 2

Isis No.	5' Target Site to SEQ ID NO: 2	3' Target Site to SEQ ID NO: 2	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
395194	26639	26658	GGAGATGAGGGCCATCAGCA	5-10-5	90.0	69
399916	26639	26658	GGAGATGAGGGCCATCAGCA	3-14-3	94.5	69
405897	26640	26659	TGGAGATGAGGGCCATCAGC	5-10-5	85.1	435
405898	26641	26660	CTGGAGATGAGGGCCATCAG	5-10-5	86.9	436
405899	26642	26661	GCTGGAGATGAGGGCCATCA	5-10-5	91.5	437
405900	26643	26662	AGCTGGAGATGAGGGCCATC	5-10-5	83.7	438
405901	26709	26728	GCTAGATGCCATCCAGAAAG	5-10-5	88.4	439
405902	26710	26729	GGCTAGATGCCATCCAGAAA	5-10-5	89.3	440
405903	26711	26730	TGGCTAGATGCCATCCAGAA	5-10-5	92.6	441
405904	26712	26731	CTGGCTAGATGCCATCCAGA	5-10-5	90.4	442
399821	26713	26732	TCTGGCTAGATGCCATCCAG	5-10-5	91.7	71
405905	26714	26733	CTCTGGCTAGATGCCATCCA	5-10-5	90.7	443
405906	26715	26734	CCTCTGGCTAGATGCCATCC	5-10-5	93.3	444
405907	26716	26735	GCCTCTGGCTAGATGCCATC	5-10-5	90.2	445
405908	26717	26736	AGCCTCTGGCTAGATGCCAT	5-10-5	83.3	446
399978	26795	26814	AGCCTGGCATAGAGCAGAGT	3-14-3	96.4	73

[0907] Antisense oligonucleotides that exhibited less than 30% inhibition of PCSK9 mRNA levels were marked with "NA".

[0908] Antisense oligonucleotides with the following ISIS Nos exhibited at least 80% inhibition of PCSK9 mRNA levels: 395152, 395153, 395155, 395158, 395163, 395165, 395166, 395168, 395173, 395183, 395185, 395186, 395187, 395194, 399798, 399801, 399804, 399817, 399819, 399821, 399868, 399877, 399879, 399887, 399888, 399891, 399895, 399900, 399904, 399905, 399906, 399907, 399912, 399916, 399935, 399936, 399954, 399960, 399967, 399968, 399969, 399973, 399978, 405861, 405862, 405863, 405864, 405865, 405866, 405869, 405870, 405871, 405872, 405873, 405874, 405875, 405876, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405889, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405900, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405908, 405909, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405923, 405924, 405925, 405926, 405927, 405928, 405929, 405930, 405931, 405932, 405934, 405935, 405936, 405940, 405941, 405943, 405944, 405949, 405952, 405953, 405954, 405955, 405956, 405958, 405959, 405960, 405961, 405973, 405974, 405976, 405977, 405978, 405979, 405980, 405982, 405984, 405987, 405988, 405989, 405990, 405991, 405992, 405993, 405994, 405995, 405996, 405997, 406003, 406005, 406008, 406009, 406014, 406016, 406020, 406023, 406026, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406036, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410530, 410556, 410558, 410562, 410578, 410583, 410589, 410590, 410591, 410592, 410593, 410595, 410609, 410611, 410616, 410617, 410618, 410619, 410624, 410628, 410633, 410634, 410647, 410648, 410650, 410652, 410658, 410664, 410666, 410667, 410668, 410671, 410677, 410686, 410688, 410689, 410693, 410695, 410696, 410697, 410698, 410700, 410701, 410702, 410713, 410714, 410715, 410716, 410723, 410724, 410730, 410754, 410757, 410759, 410769, and 410772. The target segments to which these antisense oligonucleotides are targeted are active target segments. The target regions to which these antisense oligonucleotides are targeted are active target regions.

[0909] ISIS Nos 395152, 395155, 395158, 395165, 395166, 395168, 395173, 395183, 395185, 395186, 395187, 395194, 399801, 399817, 399819, 399821, 399868, 399877, 399879, 399887, 399888, 399891, 399900, 399904, 399905, 399906, 399907, 399916, 399936, 399954, 399969, 399978, 405861, 405862, 405864, 405869, 405871, 405872, 405873, 405874, 405875, 405876, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405923, 405924, 405925, 405926, 405927, 405928, 405929, 405930, 405931, 405932, 405934, 405935, 405936, 405941, 405952, 405953, 405954, 405955, 405956, 405959, 405960, 405961, 405974, 405977, 405978, 405979, 405980, 405982, 405987, 405988, 405989, 405990, 405991, 405992, 405993, 405994, 405995, 405996, 405997, 406008, 406009, 406014, 406016, 406023, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410558, 410562, 410583, 410591, 410592, 410593, 410611, 410617, 410618, 410628, 410647, 410650, 410668, 410677, 410688, 410695, 410696, 410697, 410698, 410702, 410716, 410724, 410730, 410757, and 410772 each exhibited at least 85% inhibition of PCSK9 mRNA levels.

[0910] ISIS Nos 395152, 395155, 395158, 395165, 395166, 395168, 395173, 395183, 395185, 395186, 395187, 399801, 399817, 399819, 399821, 399877, 399879, 399887, 399888, 399891, 399900, 399904, 399905, 399906, 399907, 399916, 399969, 399978, 405862, 405864, 405874, 405875, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405891, 405893, 405894, 405895, 405899, 405903, 405906, 405910, 405913, 405916, 405923, 405924, 405925, 405926, 405927, 405930, 405934, 405936, 405953, 405955, 405956, 405959, 405978, 405988, 405990, 405991, 405994, 405995, 405996, 405997, 406008, 406014, 406023, 406028, 406029, 406030, 406033, 406035, 406039, 406040, 406041, 406042, 406043, 406044, 410592, 410647, 410668, and 410772 each exhibited at least 90% inhibition of PCSK9 mRNA levels.

[0911] ISIS Nos 395165, 395166, 399801, 399978, 405881, 405891, 405959, 405978, 405988, 405990, 405994, 405997, 406008, 406030, 406040, 406041, and 406044 each exhibited at least 95% inhibition of PCSK9 mRNA levels.

TABLE 19

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
399935	1075	1094	AGGACCCAAGTCATCCTGCT	3-14-3	81.1	96
399936	1105	1124	GGCCATCAGCTGGCAATGCT	3-14-3	89.0	124
410753	1351	1370	ATACACCTCCACCAGGCTGC	5-10-5	72.8	215
406025	1362	1381	GTGTCTAGGAGATACACCTC	5-10-5	74.2	216
406026	1367	1386	TGCTGGTGTCTAGGAGATAC	5-10-5	80.5	217
405869	1392	1411	TCGATTTCCCGGTGGTCACT	5-10-5	87.6	218
405870	1393	1412	CTCGATTTCCCGGTGGTCAC	5-10-5	83.3	219
405871	1394	1413	CCTCGATTTCCCGGTGGTCA	5-10-5	87.0	220
405872	1395	1414	CCCTCGATTTCCCGGTGGTC	5-10-5	88.2	221
399798	1396	1415	GCCCTCGATTTCCCGGTGGT	5-10-5	84.3	22
399954	1396	1415	GCCCTCGATTTCCCGGTGGT	3-14-3	90.3	22
405873	1397	1416	TGCCCTCGATTTCCCGGTGG	5-10-5	90.0	222
405874	1398	1417	CTGCCCTCGATTTCCCGGTG	5-10-5	91.4	223
405875	1399	1418	CCTGCCCTCGATTTCCCGGT	5-10-5	93.5	224
405876	1400	1419	CCCTGCCCTCGATTTCCCGG	5-10-5	90.1	225
406027	1404	1423	ATGACCCTGCCCTCGATTTC	5-10-5	73.9	226
406028	1406	1425	CCATGACCCTGCCCTCGATT	5-10-5	92.3	227
406029	1408	1427	GACCATGACCCTGCCCTCGA	5-10-5	91.4	228
406030	1412	1431	CGGTGACCATGACCCTGCCC	5-10-5	95.6	229
406031	1414	1433	GTCGGTGACCATGACCCTGC	5-10-5	89.6	230
410733	1416	1435	AAGTCGGTGACCATGACCCT	5-10-5	65.7	231
406032	1418	1437	CGAAGTCGGTGACCATGACC	5-10-5	88.5	232
410754	1428	1447	GGCACATTCTCGAAGTCGGT	5-10-5	80.1	233
395163	1453	1472	GTGGAAGCGGGTCCCGTCCT	5-10-5	83.9	25
410755	1463	1482	TGGCCTGTCTGTGGAAGCGG	5-10-5	NA	234
410756	1490	1509	GGTGGGTGCCATGACTGTCA	5-10-5	67.1	235

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
406033	1497	1516	CCTGCCAGGTGGGTGCCATG	5-10-5	91.2	237
406034	1502	1521	CCACCCCTGCCAGGTGGGTG	5-10-5	59.3	238
406035	1507	1526	GCTGACCACCCCTGCCAGGT	5-10-5	91.7	239
410757	1515	1534	TCCCGGCCGCTGACCACCCC	5-10-5	88.9	240
406036	1519	1538	GGCATCCCGGCCGCTGACCA	5-10-5	84.0	241
406037	1522	1541	GCCGGCATCCCGGCCGCTGA	5-10-5	55.9	242
410536	1527	1546	GCCACGCCGGCATCCCGGCC	5-10-5	63.7	243
410658	1527	1546	GCCACGCCGGCATCCCGGCC	2-13-5	82.0	243
410583	1527	1546	GCCACGCCGGCATCCCGGCC	3-14-3	87.3	243
410537	1528	1547	GGCCACGCCGGCATCCCGGC	5-10-5	58.5	244
410659	1528	1547	GGCCACGCCGGCATCCCGGC	2-13-5	73.7	244
410584	1528	1547	GGCCACGCCGGCATCCCGGC	3-14-3	79.5	244
410660	1529	1548	TGGCCACGCCGGCATCCCGG	2-13-5	63.0	245
410585	1529	1548	TGGCCACGCCGGCATCCCGG	3-14-3	73.2	245
406038	1529	1548	TGGCCACGCCGGCATCCCGG	5-10-5	86.3	245
410661	1530	1549	TTGGCCACGCCGGCATCCCG	2-13-5	53.3	246
410586	1530	1549	TTGGCCACGCCGGCATCCCG	3-14-3	60.3	246
405877	1530	1549	TTGGCCACGCCGGCATCCCG	5-10-5	83.2	246
410587	1531	1550	CTTGGCCACGCCGGCATCCC	3-14-3	63.2	247
410662	1531	1550	CTTGGCCACGCCGGCATCCC	2-13-5	67.3	247
405878	1531	1550	CTTGGCCACGCCGGCATCCC	5-10-5	91.4	247
410588	1532	1551	CCTTGGCCACGCCGGCATCC	3-14-3	65.3	248
410663	1532	1551	CCTTGGCCACGCCGGCATCC	2-13-5	67.9	248
405879	1532	1551	CCTTGGCCACGCCGGCATCC	5-10-5	94.1	248
410589	1533	1552	CCCTTGGCCACGCCGGCATC	3-14-3	80.1	249
410664	1533	1552	CCCTTGGCCACGCCGGCATC	2-13-5	81.9	249
405880	1533	1552	CCCTTGGCCACGCCGGCATC	5-10-5	93.5	249
410665	1534	1553	ACCCTTGGCCACGCCGGCAT	2-13-5	78.7	28
399887	1534	1553	ACCCTTGGCCACGCCGGCAT	3-14-3	91.9	28
395165	1534	1553	ACCCTTGGCCACGCCGGCAT	5-10-5	96.7	28
410590	1535	1554	CACCCTTGGCCACGCCGGCA	3-14-3	82.2	250
410666	1535	1554	CACCCTTGGCCACGCCGGCA	2-13-5	82.7	250
405881	1535	1554	CACCCTTGGCCACGCCGGCA	5-10-5	98.3	250
410667	1536	1555	GCACCCTTGGCCACGCCGGC	2-13-5	84.7	251

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410591	1536	1555	GCACCCTTGGCCACGCCGGC	3-14-3	86.3	251
405882	1536	1555	GCACCCTTGGCCACGCCGGC	5-10-5	91.6	251
410668	1537	1556	GGCACCCTTGGCCACGCCGG	2-13-5	91.4	252
405883	1537	1556	GGCACCCTTGGCCACGCCGG	5-10-5	92.5	252
410592	1537	1556	GGCACCCTTGGCCACGCCGG	3-14-3	93.4	252
410669	1538	1557	TGGCACCCTTGGCCACGCCG	2-13-5	69.0	253
410593	1538	1557	TGGCACCCTTGGCCACGCCG	3-14-3	88.8	253
405884	1538	1557	TGGCACCCTTGGCCACGCCG	5-10-5	90.8	253
410670	1539	1558	CTGGCACCCTTGGCCACGCC	2-13-5	74.9	254
410594	1539	1558	CTGGCACCCTTGGCCACGCC	3-14-3	75.2	254
410538	1539	1558	CTGGCACCCTTGGCCACGCC	5-10-5	78.2	254
410539	1540	1559	GCTGGCACCCTTGGCCACGC	5-10-5	61.9	255
410595	1540	1559	GCTGGCACCCTTGGCCACGC	3-14-3	83.1	255
410671	1540	1559	GCTGGCACCCTTGGCCACGC	2-13-5	83.3	255
410758	1545	1564	CGCATGCTGGCACCCTTGGC	5-10-5	73.6	256
406039	1566	1585	CAGTTGAGCACGCGCAGGCT	5-10-5	91.2	257
406040	1568	1587	GGCAGTTGAGCACGCGCAGG	5-10-5	96.5	258
399888	1570	1589	TTGGCAGTTGAGCACGCGCA	3-14-3	90.9	29
395166	1570	1589	TTGGCAGTTGAGCACGCGCA	5-10-5	96.2	29
406041	1572	1591	CCTTGGCAGTTGAGCACGCG	5-10-5	96.6	259
399801	1575	1594	TTCCCTTGGCAGTTGAGCAC	5-10-5	95.6	30
406042	1577	1596	CCTTCCCTTGGCAGTTGAGC	5-10-5	92.0	260
406043	1581	1600	GTGCCCTTCCCTTGGCAGTT	5-10-5	91.9	261
406044	1583	1602	CCGTGCCCTTCCCTTGGCAG	5-10-5	95.4	262
410759	1594	1613	GGTGCCGCTAACCGTGCCCT	5-10-5	83.7	263
410760	1618	1637	CCGAATAAACTCCAGGCCTA	5-10-5	96.7	265
406045	1626	1645	TGGCTTTTCCGAATAAACTC	5-10-5	88.4	266
395168	1628	1647	GCTGGCTTTTCCGAATAAAC	5-10-5	91.9	32
405909	1630	1649	CAGCTGGCTTTTCCGAATAA	5-10-5	80.3	267
405910	1632	1651	ACCAGCTGGCTTTTCCGAAT	5-10-5	90.9	268
405911	1634	1653	GGACCAGCTGGCTTTTCCGA	5-10-5	88.0	269
405912	1638	1657	GGCTGGACCAGCTGGCTTTT	5-10-5	78.2	270
410761	1649	1668	GTGGCCCCACAGGCTGGACC	5-10-5	53.7	271
410762	1662	1681	AGCAGCACCACCAGTGGCCC	5-10-5	57.6	272
410763	1684	1703	GCTGTACCCACCCGCCAGGG	5-10-5	68.8	273

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410764	1730	1749	CGACCCCAGCCCTCGCCAGG	5-10-5	66.3	274
399891	1740	1759	GTGACCAGCACGACCCCAGC	3-14-3	91.1	33
405913	1742	1761	CGGTGACCAGCACGACCCCA	5-10-5	93.8	275
405914	1744	1763	AGCGGTGACCAGCACGACCC	5-10-5	90.4	276
405915	1746	1765	GCAGCGGTGACCAGCACGAC	5-10-5	86.8	277
405916	1748	1767	CGGCAGCGGTGACCAGCACG	5-10-5	91.3	278
410734	1749	1768	CCGGCAGCGGTGACCAGCAC	5-10-5	65.8	249
405917	1752	1771	TTGCCGGCAGCGGTGACCAG	5-10-5	62.0	280
405918	1754	1773	AGTTGCCGGCAGCGGTGACC	5-10-5	33.5	281
405919	1756	1775	GAAGTTGCCGGCAGCGGTGA	5-10-5	68.2	282
405920	1758	1777	CGGAAGTTGCCGGCAGCGGT	5-10-5	86.2	283
405921	1760	1779	CCCGGAAGTTGCCGGCAGCG	5-10-5	85.5	284
405922	1762	1781	GTCCCGGAAGTTGCCGGCAG	5-10-5	86.0	285
405937	1820	1839	GAGGCACCAATGATGTCCTC	5-10-5	77.7	306
405938	1824	1843	GCTGGAGGCACCAATGATGT	5-10-5	75.5	307
405939	1826	1845	TCGCTGGAGGCACCAATGAT	5-10-5	70.6	308
405940	1828	1847	AGTCGCTGGAGGCACCAATG	5-10-5	84.4	309
405941	1830	1849	GCAGTCGCTGGAGGCACCAA	5-10-5	90.1	310
399804	1833	1852	GCTGCAGTCGCTGGAGGCAC	5-10-5	80.8	40
399960	1833	1852	GCTGCAGTCGCTGGAGGCAC	3-14-3	82.4	40
405942	1835	1854	GTGCTGCAGTCGCTGGAGGC	5-10-5	69.8	311
405943	1837	1856	AGGTGCTGCAGTCGCTGGAG	5-10-5	83.6	312
410737	1839	1858	GCAGGTGCTGCAGTCGCTGG	5-10-5	49.8	313
405944	1840	1859	AGCAGGTGCTGCAGTCGCTG	5-10-5	80.7	314
405945	1842	1861	AAAGCAGGTGCTGCAGTCGC	5-10-5	41.1	315
405946	1846	1865	ACACAAAGCAGGTGCTGCAG	5-10-5	70.4	316
405947	1848	1867	TGACACAAAGCAGGTGCTGC	5-10-5	72.5	317
410769	1858	1877	TCCCACTCTGTGACACAAAG	5-10-5	84.9	318
405948	1900	1919	TCATGGCTGCAATGCCAGCC	5-10-5	45.9	319
399806	1903	1922	GCATCATGGCTGCAATGCCA	5-10-5	82.8	42
399962	1903	1922	GCATCATGGCTGCAATGCCA	3-14-3	86.1	42
405949	1905	1924	CAGCATCATGGCTGCAATGC	5-10-5	84.9	320
405950	1907	1926	GACAGCATCATGGCTGCAAT	5-10-5	76.9	321
405951	1909	1928	CAGACAGCATCATGGCTGCA	5-10-5	75.4	322

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
405952	1913	1932	TCGGCAGACAGCATCATGGC	5-10-5	85.1	323
410738	1915	1934	GCTCGGCAGACAGCATCATG	5-10-5	68.4	324
405953	1917	1936	CGGCTCGGCAGACAGCATCA	5-10-5	93.6	325
405954	1919	1938	TCCGGCTCGGCAGACAGCAT	5-10-5	87.5	326
410770	1933	1952	CGGCCAGGGTGAGCTCCGGC	5-10-5	68.5	327
405955	1946	1965	CTCTGCCTCAACTCGGCCAG	5-10-5	94.2	328
405956	1948	1967	GTCTCTGCCTCAACTCGGCC	5-10-5	94.3	329
405958	1952	1971	ATCAGTCTCTGCCTCAACTC	5-10-5	80.3	331
405959	1954	1973	GGATCAGTCTCTGCCTCAAC	5-10-5	95.4	332
405960	1956	1975	GTGGATCAGTCTCTGCCTCA	5-10-5	90.1	333
405961	1958	1977	AAGTGGATCAGTCTCTGCCT	5-10-5	88.8	334
405962	1961	1980	GAGAAGTGGATCAGTCTCTG	5-10-5	56.1	335
410739	1963	1982	CAGAGAAGTGGATCAGTCTC	5-10-5	NA	336
405963	1965	1984	GGCAGAGAAGTGGATCAGTC	5-10-5	59.7	337
405964	1969	1988	CTTTGGCAGAGAAGTGGATC	5-10-5	45.3	338
405965	1974	1993	GACATCTTTGGCAGAGAAGT	5-10-5	55.9	339
405966	1976	1995	ATGACATCTTTGGCAGAGAA	5-10-5	51.3	340
405967	1980	1999	ATTGATGACATCTTTGGCAG	5-10-5	64.4	341
405968	1982	2001	TCATTGATGACATCTTTGGC	5-10-5	74.6	342
399967	1985	2004	GCCTCATTGATGACATCTTT	3-14-3	80.1	48
405969	1987	2006	AGGCCTCATTGATGACATCT	5-10-5	63.0	343
405970	1989	2008	CCAGGCCTCATTGATGACAT	5-10-5	66.8	344
410740	1991	2010	AACCAGGCCTCATTGATGAC	5-10-5	46.8	345
405885	1993	2012	GGAACCAGGCCTCATTGATG	5-10-5	73.4	346
410596	1995	2014	AGGGAACCAGGCCTCATTGA	3-14-3	NA	348
410672	1995	2014	AGGGAACCAGGCCTCATTGA	2-13-5	NA	348
405887	1995	2014	AGGGAACCAGGCCTCATTGA	5-10-5	NA	348
410597	1996	2015	CAGGGAACCAGGCCTCATTG	3-14-3	NA	349
405888	1996	2015	CAGGGAACCAGGCCTCATTG	5-10-5	NA	349
410673	1996	2015	CAGGGAACCAGGCCTCATTG	2-13-5	NA	349
410674	1997	2016	TCAGGGAACCAGGCCTCATT	2-13-5	60.7	49
399812	1997	2016	TCAGGGAACCAGGCCTCATT	5-10-5	68.7	49
399968	1997	2016	TCAGGGAACCAGGCCTCATT	3-14-3	84.5	49
410598	1998	2017	CTCAGGGAACCAGGCCTCAT	3-14-3	74.5	350
410675	1998	2017	CTCAGGGAACCAGGCCTCAT	2-13-5	76.9	350

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
405889	1998	2017	CTCAGGGAACCAGGCCTCAT	5-10-5	80.2	350
410599	1999	2018	CCTCAGGGAACCAGGCCTCA	3-14-3	70.6	351
410676	1999	2018	CCTCAGGGAACCAGGCCTCA	2-13-5	77.8	351
405890	1999	2018	CCTCAGGGAACCAGGCCTCA	5-10-5	87.7	351
410600	2000	2019	TCCTCAGGGAACCAGGCCTC	3-14-3	76.5	352
410677	2000	2019	TCCTCAGGGAACCAGGCCTC	2-13-5	88.4	352
405891	2000	2019	TCCTCAGGGAACCAGGCCTC	5-10-5	96.1	352
405892	2001	2020	GTCCTCAGGGAACCAGGCCT	5-10-5	71.4	353
410601	2001	2020	GTCCTCAGGGAACCAGGCCT	3-14-3	72.7	353
410678	2001	2020	GTCCTCAGGGAACCAGGCCT	2-13-5	75.0	353
395178	2002	2021	GGTCCTCAGGGAACCAGGCC	5-10-5	76.7	50
410679	2002	2021	GGTCCTCAGGGAACCAGGCC	2-13-5	77.6	50
399900	2002	2021	GGTCCTCAGGGAACCAGGCC	3-14-3	92.0	50
408653	2003	2022	TGGTCCTCAGGGAACCAGGC	5-10-5	42.5	354
410602	2003	2022	TGGTCCTCAGGGAACCAGGC	3-14-3	66.5	354
410680	2003	2022	TGGTCCTCAGGGAACCAGGC	2-13-5	71.6	354
410603	2004	2023	CTGGTCCTCAGGGAACCAGG	3-14-3	40.8	355
410681	2004	2023	CTGGTCCTCAGGGAACCAGG	2-13-5	44.4	355
405971	2004	2023	CTGGTCCTCAGGGAACCAGG	5-10-5	65.5	355
410540	2005	2024	GCTGGTCCTCAGGGAACCAG	5-10-5	50.9	356
410682	2005	2024	GCTGGTCCTCAGGGAACCAG	2-13-5	54.1	356
410604	2005	2024	GCTGGTCCTCAGGGAACCAG	3-14-3	62.5	356
405972	2006	2025	CGCTGGTCCTCAGGGAACCA	5-10-5	77.6	357
405973	2011	2030	GTACCCGCTGGTCCTCAGGG	5-10-5	83.8	358
405974	2013	2032	CAGTACCCGCTGGTCCTCAG	5-10-5	89.0	359
399813	2016	2035	GGTCAGTACCCGCTGGTCCT	5-10-5	72.4	51
399969	2016	2035	GGTCAGTACCCGCTGGTCCT	3-14-3	93.4	51
410771	2061	2080	ACCTGCCCCATGGGTGCTGG	5-10-5	NA	360
395181	2073	2092	AAACAGCTGCCAACCTGCCC	5-10-5	87.5	52
405975	2075	2094	CAAAACAGCTGCCAACCTGC	5-10-5	77.4	361
405976	2080	2099	TCCTGCAAAACAGCTGCCAA	5-10-5	81.2	362
405977	2082	2101	AGTCCTGCAAAACAGCTGCC	5-10-5	89.5	363
399992	2095	2114	GTGCTGACCACACAGTCCTG	3-14-3	92.6	130
405978	2105	2124	GGCCCCGAGTGTGCTGACCA	5-10-5	95.3	365

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
399904	2108	2127	GTAGGCCCCGAGTGTGCTGA	3-14-3	92.6	54
405979	2110	2129	GTGTAGGCCCCGAGTGTGCT	5-10-5	85.5	366
405980	2112	2131	CCGTGTAGGCCCCGAGTGTG	5-10-5	86.5	367
405981	2114	2133	ATCCGTGTAGGCCCCGAGTG	5-10-5	77.8	368
410741	2116	2135	CCATCCGTGTAGGCCCCGAG	5-10-5	78.4	369
405982	2118	2137	GGCCATCCGTGTAGGCCCCG	5-10-5	86.7	370
405983	2120	2139	GTGGCCATCCGTGTAGGCCC	5-10-5	73.1	371
405984	2168	2187	CTGGAGCAGCTCAGCAGCTC	5-10-5	83.4	372
405985	2170	2189	AACTGGAGCAGCTCAGCAGC	5-10-5	50.7	373
405986	2175	2194	GGAGAAACTGGAGCAGCTCA	5-10-5	75.6	374
405987	2177	2196	CTGGAGAAACTGGAGCAGCT	5-10-5	88.0	375
410776	2245	2264	GTGCCAAGGTCCTCCACCTC	5-10-5	77.3	408
410777	2265	2284	TCAGCACAGGCGGCTTGTGG	5-10-5	40.0	409
410778	2295	2314	CCACGCACTGGTTGGGCTGA	5-10-5	52.7	410
399908	2355	2374	CTTTGCATTCCAGACCTGGG	3-14-3	74.9	60
410713	2355	2374	CTTTGCATTCCAGACCTGGG	2-13-5	84.9	60
395186	2355	2374	CTTTGCATTCCAGACCTGGG	5-10-5	93.7	60
410714	2356	2375	ACTTTGCATTCCAGACCTGG	2-13-5	82.4	411
410633	2356	2375	ACTTTGCATTCCAGACCTGG	3-14-3	83.8	411
410562	2356	2375	ACTTTGCATTCCAGACCTGG	5-10-5	87.3	411
410715	2357	2376	GACTTTGCATTCCAGACCTG	2-13-5	80.5	412
410634	2357	2376	GACTTTGCATTCCAGACCTG	3-14-3	81.8	412
405996	2357	2376	GACTTTGCATTCCAGACCTG	5-10-5	91.0	412
410563	2358	2377	TGACTTTGCATTCCAGACCT	5-10-5	75.4	413
410635	2358	2377	TGACTTTGCATTCCAGACCT	3-14-3	75.9	413
410716	2358	2377	TGACTTTGCATTCCAGACCT	2-13-5	88.3	413
410636	2359	2378	TTGACTTTGCATTCCAGACC	3-14-3	61.3	414
410564	2359	2378	TTGACTTTGCATTCCAGACC	5-10-5	71.3	414
410717	2359	2378	TTGACTTTGCATTCCAGACC	2-13-5	76.7	414
410718	2360	2379	CTTGACTTTGCATTCCAGAC	2-13-5	55.8	61
399973	2360	2379	CTTGACTTTGCATTCCAGAC	3-14-3	82.1	61
399817	2360	2379	CTTGACTTTGCATTCCAGAC	5-10-5	93.6	61
405997	2362	2381	TCCTTGACTTTGCATTCCAG	5-10-5	95.6	415
410779	2375	2394	CGGGATTCCATGCTCCTTGA	5-10-5	79.7	416
410780	2405	2424	GCAGGCCACGGTCACCTGCT	5-10-5	60.9	417

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3

Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410781	2442	2461	GGAGGGCACTGCAGCCAGTC	5-10-5	66.0	418
410719	2560	2579	CAGATGGCAACGGCTGTCAC	2-13-5	61.0	419
410637	2560	2579	CAGATGGCAACGGCTGTCAC	3-14-3	66.0	419
410565	2560	2579	CAGATGGCAACGGCTGTCAC	5-10-5	71.6	419
410638	2561	2580	GCAGATGGCAACGGCTGTCA	3-14-3	71.9	420
410720	2561	2580	GCAGATGGCAACGGCTGTCA	2-13-5	75.1	420
410566	2561	2580	GCAGATGGCAACGGCTGTCA	5-10-5	77.7	420
410639	2562	2581	AGCAGATGGCAACGGCTGTC	3-14-3	67.7	421
410721	2562	2581	AGCAGATGGCAACGGCTGTC	2-13-5	68.8	421
410567	2562	2581	AGCAGATGGCAACGGCTGTC	5-10-5	72.1	421
410568	2563	2582	CAGCAGATGGCAACGGCTGT	5-10-5	54.2	422
410640	2563	2582	CAGCAGATGGCAACGGCTGT	3-14-3	57.7	422
410722	2563	2582	CAGCAGATGGCAACGGCTGT	2-13-5	59.2	422
410569	2564	2583	GCAGCAGATGGCAACGGCTG	5-10-5	57.7	423
410641	2564	2583	GCAGCAGATGGCAACGGCTG	3-14-3	66.1	423
410723	2564	2583	GCAGCAGATGGCAACGGCTG	2-13-5	81.1	423
399909	2565	2584	GGCAGCAGATGGCAACGGCT	3-14-3	69.9	62
410724	2565	2584	GGCAGCAGATGGCAACGGCT	2-13-5	87.4	62
395187	2565	2584	GGCAGCAGATGGCAACGGCT	5-10-5	92.7	62
410642	2566	2585	CGGCAGCAGATGGCAACGGC	3-14-3	70.4	424
410725	2566	2585	CGGCAGCAGATGGCAACGGC	2-13-5	75.6	424
410570	2566	2585	CGGCAGCAGATGGCAACGGC	5-10-5	78.3	424
410571	2567	2586	CCGGCAGCAGATGGCAACGG	5-10-5	52.8	425
410643	2567	2586	CCGGCAGCAGATGGCAACGG	3-14-3	59.8	425
410726	2567	2586	CCGGCAGCAGATGGCAACGG	2-13-5	69.8	425
410644	2568	2587	TCCGGCAGCAGATGGCAACG	3-14-3	59.0	426
410727	2568	2587	TCCGGCAGCAGATGGCAACG	2-13-5	68.3	426
405998	2568	2587	TCCGGCAGCAGATGGCAACG	5-10-5	75.6	426
405988	2568	2587	TCCGGCAGCAGATGGCAACG	5-10-5	84.5	426
410572	2569	2588	CTCCGGCAGCAGATGGCAAC	5-10-5	40.9	427
410728	2569	2588	CTCCGGCAGCAGATGGCAAC	2-13-5	56.4	427
410645	2569	2588	CTCCGGCAGCAGATGGCAAC	3-14-3	70.2	427
410729	2570	2589	GCTCCGGCAGCAGATGGCAA	2-13-5	67.7	428
410573	2570	2589	GCTCCGGCAGCAGATGGCAA	5-10-5	71.8	428

TABLE 19-continued

Antisense inhibition of PCSK9 in human cells (Hep3B) targeting SEQ ID NO: 3						
Isis No.	5' Target Site to SEQ ID NO: 3	3' Target Site to SEQ ID NO: 3	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO:
410646	2570	2589	GCTCCGGCAGCAGATGGCAA	3-14-3	72.8	428
410782	2580	2599	CCAGGTGCCGGCTCCGGCAG	5-10-5	43.2	429
410783	2590	2609	GAGGCCTGCGCCAGGTGCCG	5-10-5	63.7	430
399819	2764	2783	CCCACTCAAGGGCCAGGCCA	5-10-5	93.4	65
399912	2852	2871	ATGCCCCACAGTGAGGGAGG	3-14-3	84.3	66
405893	3083	3102	ATGAGGGCCATCAGCACCTT	5-10-5	93.9	431
405894	3084	3103	GATGAGGGCCATCAGCACCT	5-10-5	93.9	432
405895	3085	3104	AGATGAGGGCCATCAGCACC	5-10-5	91.8	433
405896	3086	3105	GAGATGAGGGCCATCAGCAC	5-10-5	88.5	434
395194	3087	3106	GGAGATGAGGGCCATCAGCA	5-10-5	90.0	69
399916	3087	3106	GGAGATGAGGGCCATCAGCA	3-14-3	94.5	69
405897	3088	3107	TGGAGATGAGGGCCATCAGC	5-10-5	85.1	435
405898	3089	3108	CTGGAGATGAGGGCCATCAG	5-10-5	86.9	436
405899	3090	3109	GCTGGAGATGAGGGCCATCA	5-10-5	91.5	437
405900	3091	3110	AGCTGGAGATGAGGGCCATC	5-10-5	83.7	438
405901	3157	3176	GCTAGATGCCATCCAGAAAG	5-10-5	88.4	439
405902	3158	3177	GGCTAGATGCCATCCAGAAA	5-10-5	89.3	440
405903	3159	3178	TGGCTAGATGCCATCCAGAA	5-10-5	92.6	441
405904	3160	3179	CTGGCTAGATGCCATCCAGA	5-10-5	90.4	442
399821	3161	3180	TCTGGCTAGATGCCATCCAG	5-10-5	91.7	71
405905	3162	3181	CTCTGGCTAGATGCCATCCA	5-10-5	90.7	443
405906	3163	3182	CCTCTGGCTAGATGCCATCC	5-10-5	93.3	444
405907	3164	3183	GCCTCTGGCTAGATGCCATC	5-10-5	90.2	445
405908	3165	3184	AGCCTCTGGCTAGATGCCAT	5-10-5	83.3	446
399978	3243	3262	AGCCTGGCATAGAGCAGAGT	3-14-3	96.4	73

[0912] Antisense oligonucleotides that exhibited less than 30% inhibition of PCSK9 mRNA levels were marked with "NA".

[0913] Antisense oligonucleotides with the following ISIS Nos exhibited at least 80% inhibition of PCSK9 mRNA levels: 395163, 395165, 395166, 395168, 395181, 395186, 395187, 395194, 399798, 399801, 399804, 399806, 399817, 399819, 399821, 399887, 399888, 399891, 399900, 399904, 399912, 399916, 399935, 399936, 399954, 399960, 399962, 399967, 399968, 399969, 399973, 399978, 399992, 405869, 405870, 405871, 405872, 405873, 405874, 405875, 405876, 405877, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405889, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405900, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405908, 405909, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405940, 405941, 405943, 405944, 405949, 405952, 405953, 405954, 405955, 405956, 405958, 405959, 405960, 405961, 405973, 405974, 405976, 405977, 405978, 405979, 405980, 405982, 405984, 405987, 405988, 405996, 405997, 406026, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406036, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410562, 410583, 410589, 410590, 410591, 410592, 410593, 410595, 410633, 410634, 410658, 410664, 410666, 410667, 410668, 410671, 410677, 410713, 410714, 410715, 410716, 410723, 410724, 410754, 410757, 410759, 410760, and 410769. The target segments to which these antisense oligonucleotides are targeted are active target segments. The target regions to which these antisense oligonucleotides are targeted are active target regions.

[0914] ISIS Nos 395165, 395166, 395168, 395181, 395186, 395187, 395194, 399801, 399817, 399819, 399821, 399887, 399888, 399891, 399900, 399904, 399916, 399936, 399954, 399962, 399969, 399978, 399992, 405869, 405871, 405872, 405873, 405874, 405875, 405876, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405890, 405891, 405893, 405894, 405895, 405896, 405897, 405898, 405899, 405901, 405902, 405903, 405904, 405905, 405906, 405907, 405910, 405911, 405913, 405914, 405915, 405916, 405920, 405921, 405922, 405941, 405952, 405953, 405954, 405955, 405956, 405959, 405960, 405961, 405974, 405977, 405978, 405979, 405980, 405982, 405987, 405996, 405997, 406028, 406029, 406030, 406031, 406032, 406033, 406035, 406038, 406039, 406040, 406041, 406042, 406043, 406044, 406045, 410562, 410583, 410591, 410592, 410593, 410668, 410677, 410716, 410724, 410757, and 410760 each exhibited at least 85% inhibition of PCSK9 mRNA levels.

[0915] ISIS Nos 395165, 395166, 395168, 395186, 395187, 395194, 399801, 399817, 399819, 399821, 399887, 399888, 399891, 399900, 399904, 399916, 399954, 399969, 399978, 399992, 405873, 405874, 405875, 405876, 405878, 405879, 405880, 405881, 405882, 405883, 405884, 405891, 405893, 405894, 405895, 405899, 405903, 405904, 405905, 405906, 405907, 405910, 405913, 405914, 405916, 405941, 405953, 405955, 405956, 405959, 405960, 405978, 405996, 405997, 406028, 406029, 406030, 406033, 406035, 406039, 406040, 406041, 406042, 406043, 406044, 410592, 410668, and 410760 each exhibited at least 90% inhibition of PCSK9 mRNA levels.

[0916] ISIS Nos 395165, 395166, 399801, 399978, 405881, 405891, 405959, 405978, 405997, 406030, 406040, 406041, 406044, and 410760 each exhibited at least 95% inhibition of PCSK9 mRNA levels.

Example 4: Antisense Reduction of Human PCSK9 mRNA in HepG2 Cells: Dose Response Experiment

[0917] Antisense oligonucleotides targeted to PCSK9 were tested at various doses in HepG2 cells. Cells were plated at densities of 10,000 cells per well and treated with nM concentrations of antisense oligonucleotide as indicated in Tables 20 and 21. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. Two different human PCSK9 primer probe sets were used to measure mRNA levels. Results with Primer Probe Set 2740 are shown in Table 20, and mRNA levels measured using primer probe set 2823 are shown in Table 21. Results are presented as percent PCSK9 mRNA levels relative to a control, as adjusted according to total RNA content measured by RIBOGREEN®. As illustrated in Tables 20 and 21, PCSK9 mRNA levels were reduced in a dose-dependent manner.

TABLE 20

Antisense Reduction of PCSK9 mRNA in HepG2 cells, Primer Probe Set 2740

Isis No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM
399819	93	113	84	71	26	12
395185	81	64	50	35	24	13
399916	65	56	41	18	20	25
399907	68	66	42	26	30	25
399954	71	63	38	22	16	15
395165	71	67	47	29	21	10
399936	65	51	31	20	30	28
399793	61	57	44	31	39	15
399969	57	54	40	27	25	19
395152	75	58	58	49	24	28

TABLE 21

Antisense Reduction of PCSK9 mRNA in HepG2 cells, Primer Probe Set 2823

Isis No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM
399819	80	264	87	85	34	15
395185	86	66	57	45	29	22
399916	75	47	45	20	30	45
399907	67	72	47	29	31	32
399954	59	50	35	17	22	27
395165	74	54	41	37	26	22
399936	62	41	33	32	28	42
399793	62	51	38	36	44	27
399969	73	40	58	28	29	35
395152	78	53	64	49	22	35

Example 5: Antisense Reduction of Human PCSK9 mRNA in Hep3B Cells: Dose Response Experiment

[0918] Antisense oligonucleotides targeted to PCSK9 were tested at various doses in Hep3B cells. Cells were plated at densities of 4,500 cells per well and treated with nM concentrations of antisense oligonucleotide as indicated in Table 22. ISIS 141923 is not complementary to any known gene sequence and is used in this and the following experiments as a negative control. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. Two different human PCSK9 primer probe sets were used to measure mRNA levels. Results with Primer Probe Set 2740 are shown in Table 22. Results are presented as percent PCSK9 mRNA levels relative to a control, as adjusted according to total RNA content measured by RIBOGREEN®. As illustrated in Table 22, PCSK9 mRNA levels were reduced in a dose-dependent manner.

TABLE 22

Antisense Reduction of PCSK9 mRNA in Hep3B cells, Primer Probe Set 2740

Isis No.	2.78 nM	8.33 nM	25.0 nM	75.0 nM
399819	127.5	85.0	36.1	21.9
405891	133.7	111.4	40.2	27.2
406008	144.6	108.3	49.9	17.7
395186	124.4	106.4	42.2	30.7

TABLE 22-continued

Antisense Reduction of PCSK9 mRNA in Hep3B cells, Primer Probe Set 2740				
Isis No.	2.78 nM	8.33 nM	25.0 nM	75.0 nM
395185	150.7	106.6	53.6	38.1
405994	141.9	101.2	49.3	30.8
405988	148.1	117.4	47.8	25.2
406033	138.1	109.9	62.4	35.3
395187	133.0	115.6	59.9	31.6
405995	124.1	109.2	57.8	42.9
406023	130.0	103.5	67.1	28.0
399900	131.2	95.7	81.1	30.9
301012	121.2	100.1	63.8	49.6
395165	113.2	86.9	41.6	13.3
405879	124.6	88.8	44.1	13.1
405991	100.6	101.4	95.9	41.0
405923	130.4	115.0	68.9	35.1
395152	143.8	98.6	61.3	59.0
405881	100.7	77.4	50.2	4.5
141923	131.4	131.9	144.0	167.3

Example 6: Antisense Reduction of Human PCSK9 mRNA in HeLa Cells: Dose Response Experiment

[0919] Antisense oligonucleotides targeted to PCSK9 were tested at various doses in HeLa cells. Cells were plated at densities of 5,000 cells per well and treated with nM concentrations of antisense oligonucleotide as indicated in Table 23. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. Two different human PCSK9 primer probe sets were used to measure mRNA levels. Results with Primer Probe Set 2740 are shown in Table 23. Results are presented as percent PCSK9 mRNA levels relative to a control, as adjusted according to total RNA content measured by RIBOGREEN®. As illustrated in Table 23, PCSK9 mRNA levels were reduced in a dose-dependent manner.

TABLE 23

Antisense Reduction of PCSK9 mRNA in HeLa cells, Primer Probe Set 2740				
Isis No.	2.963 nM	8.8889 nM	26.6667 nM	80.0 nM
399819	57	32	16	14
405891	57	46	17	18
406008	45	27	11	5.3
395186	40	28	11	6.6
395185	53	32	21	17
405994	48	34	20	13
405988	44	32	12	7.3
406033	61	42	25	12
395187	42	35	12	6.6
405995	54	39	21	15
406023	76	46	25	6.2
399900	85	48	23	6.4
301012	121	109	76	65
395165	61	30	10	2
405879	64	32	9.7	1
405991	64	35	24	10
405923	71	45	22	5
395152	73	34	13	3.1
405881	58	33	19	2.8
141923	122	118	131	140

Example 7: Antisense Reduction of Human PCSK9 mRNA in Primary Hepatocytes

[0920] Antisense oligonucleotides targeted to PCSK9 were tested for antisense inhibition of PCSK9 in human primary hepatocytes. Human primary hepatocytes were purchased from a commercial supplier, and cultured according to routine culture procedures. Cells were treated with 10, 25, 50, 150, or 300 nM of antisense oligonucleotide for a period of 24 hours, after which RNA was isolated and PCSK9 mRNA was measured by real-time PCR as described herein.

[0921] The antisense oligonucleotides tested in human primary hepatocytes were Isis 395152, 395155, 395165, 395185, 399819, 399821, 399916, 399954, 399978, and 399992. The antisense oligonucleotides inhibited PCSK9 in a dose-dependent manner in human primary hepatocytes.

[0922] Monkey primary hepatocytes were similarly treated with 10, 25, 50, 150, or 300 nM concentrations of antisense oligonucleotide targeted to PCSK9. The antisense oligonucleotides tested in monkey primary hepatocytes were Isis 395152, 395155, 395165, 395185, 399819, 399821, 399916, 399954, 399978, and 399992. Each of these antisense oligonucleotides is fully complementary to a monkey PCSK9 nucleic acid. The antisense oligonucleotides inhibited PCSK9 in a dose-dependent manner in monkey primary hepatocytes.

Example 8: Additional Antisense Reduction of Human PCSK9 mRNA in Primary Hepatocytes

[0923] Antisense oligonucleotides targeted to PCSK9 were tested for antisense inhibition of PCSK9 in human primary hepatocytes. Human primary hepatocytes were purchased from a commercial supplier, and cultured according to routine culture procedures. Cells were treated with 10, 25, 50, 150, or 300 nM of antisense oligonucleotide for a period of 24 hours, after which RNA was isolated and PCSK9 mRNA was measured by real-time PCR as described herein.

[0924] The antisense oligonucleotides tested in human primary hepatocytes were Isis 395165, 395185, 395186, 395187, 405879, 405881, 405891, 405988, 405994, and 406008. The antisense oligonucleotides inhibited PCSK9 in a dose-dependent manner in human primary hepatocytes as shown in Table 24.

TABLE 24

Dose-dependent Reduction in Human PCSK9 mRNA Expression in Human Primary Hepatocytes					
ISIS No.	10 nM	25 nM	50 nM	150 nM	300 nM
395165	76	48.7	22.5	20.2	16.2
395185	74.5	42.9	52.6	25.5	15.7
395186	123.8	58.3	34.4	21	19.9
395187	98.7	43.5	29.7	17.4	21.1
405879	119.2	176.6	64.9	48.4	47.7
405881	222.4	115.3	49.5	22.3	79.3
405891	169.9	138.7	73.9	86.9	48.7
405988	85.2	105.4	67.8	43.9	43.8
405994	95.7	83.3	43.5	24.6	10.5
406008	155	101.7	116	41.7	32.4

Example 9: Antisense Reduction of PCSK9 mRNA in Cyno Primary Hepatocytes

[0925] Antisense oligonucleotides targeted to PCSK9 were tested for antisense inhibition of PCSK9 in cyno primary hepatocytes. Cyno primary hepatocytes were purchased from a commercial supplier, and cultured according to routine culture procedures. Cells were treated with 10, 25, 50, 150, or 300 nM of antisense oligonucleotide for a period of 24 hours, after which RNA was isolated and PCSK9 mRNA was measured by real-time PCR as described herein.

[0926] The antisense oligonucleotides tested in cyno primary hepatocytes were Isis 395165, 395185, 395186, 395187, 405879, 405881, 405891, 405988, 405994, and 406008. The antisense oligonucleotides inhibited PCSK9 in a dose-dependent manner in cyno primary hepatocytes as shown in Table 25.

TABLE 25

Dose-dependent PCSK9 mRNA Inhibition in Cyno Primary Hepatocytes

ISIS No.	10 nM	25 nM	50 nM	150 nM	300 nM
395165	36.3	29.9	14.9	8.6	3.2
395185	65.0	31.9	35.1	16.6	12.0
395186	65.8	42.3	21.6	23.3	16.7
395187	61.3	32.9	13.2	7.7	9.02
405879	24.3	21.9	7.7	7.1	3.0
405881	75.0	33.5	26.5	11.2	4.9
405891	52.4	24.8	11.2	10.8	6.7
405988	59.4	23.7	22.5	10.7	8.2
405994	64.8	52.7	22.2	7.9	6.5
406008	56.3	28.2	26.5	13.4	13.3

Example 10: Antisense Reduction of Mouse PCSK9 mRNA In Vitro

[0927] Antisense oligonucleotides were designed to target murine PCSK9, and were evaluated for their ability to reduce PCSK9 mRNA in primary mouse hepatocytes.

[0928] Primary mouse hepatocytes were treated with antisense oligonucleotides at doses of 50, 150, or 300 nM, for a period of 24 hours. RNA was isolated using QIAGEN® RNeasy isolation kits and subjected to quantitative real-time PCR using commercially available reagents (Invitrogen, Carlsbad, Calif.). PCSK9 mRNA levels were measured using a mouse PCSK9 primer probe set, and normalized to G3PDH mRNA levels and/or RIBOGREEN® levels.

[0929] ISIS 394814 (GAGCAACTTCGGAGGCAGC, SEQ ID NO: 456) was identified as a potent inhibitor of mouse PCSK9. PCSK9 mRNA levels ISIS 394814 yielded a 50% reduction in PCSK9 mRNA at a concentration of 25 nM (i.e., an IC50 of 25 nM) after a 24 hour incubation with primary mouse hepatocytes. No effect on cultured cell viability was observed.

Example 11: Antisense Reduction of PCSK9 mRNA in an Animal Model of Hyperlipidemia: High Fat Fed Mice

Treatment

[0930] C57BL/6 mice fed a high-fat diet are routinely employed as an animal model of hyperlipidemia, as well as an animal model of atherosclerosis. Accordingly, to evaluate the effects of PCSK9 antisense inhibition on serum lipid levels in vivo, ISIS 394814 was evaluated in C57BL/6 mice fed a high-fat diet. C57BL/6 mice 4-5 weeks of age were obtained from the Jackson Laboratory and maintained on a diet consisting of 60% fat. Treatment groups of 5 mice each were as follows: a group treated with ISIS 394814; a control group treated with saline; and control group treated with an oligonucleotide not complementary to any known gene sequence (ISIS 141923). Oligonucleotide or saline was administered intraperitoneally twice weekly, for a period of 6 weeks; oligonucleotide doses were 50 mg/kg. After the treatment period, whole blood was collected for analysis of serum parameters, and whole liver was collected for RNA analysis, protein analysis, and histological evaluation. Statistical analyses included a nonparametric, two-tailed t-test comparison of experimental samples (ISIS 394814) to control samples (saline or control oligonucleotide).

[0931] RNA Analysis

[0932] Liver RNA was isolated for real-time PCR analysis of PCSK9, LDL-receptor and apolipoprotein B mRNA levels. Treatment with ISIS 394814 resulted in a 92% reduction in PCSK9 mRNA levels, while no significant reduction in PCSK9 mRNA levels was observed in ISIS 141923-treated or saline-treated mice. Antisense inhibition of PCSK9 expression did not significantly affect liver LDL-receptor or liver apolipoprotein B mRNA levels.

Protein Analysis

[0933] Immunoblotting was performed to assess the effects of PCSK9 antisense inhibition on LDL-receptor protein and apolipoprotein B levels. Protein isolated from mouse liver was subjected to electrophoresis, transferred to a polyvinylidene-fluoride membrane, and subsequently probed with an anti-mouse LDL-receptor antibody and a scavenger receptor B1 (SR-B1) antibody. Mouse plasma was similarly subjected to electrophoresis and immunoblotting using an anti-mouse apolipoprotein B antibody. Secondary antibodies conjugated to peroxidase were used to detect primary antibodies. Protein bands were visualized using the ECL plus Western blot detection kit (Amersham Biosciences, UK) and quantified using ImageQuant™ analysis software (MolecularDynamics, Santa Clara, Calif.).

[0934] While LDL-receptor mRNA levels were not significantly affected, antisense inhibition of PCSK9 resulted in an approximate 2-fold increase in the level of hepatic LDL-receptor protein relative to 141923-treated controls. No changes in SR-B1 protein were observed. Furthermore, serum apoB-100 levels were significantly reduced by 50%, relative to 141923-treated controls. Serum apoB-48 levels were significantly increased by approximately 3-fold relative to 141923-treated controls. No significant changes in apoA-I protein were observed, relative to 141923-treated controls.

[0935] The mRNA levels of the RNA editing enzyme apobec-1 were also measured. Apobec-1 is responsible for the RNA editing that produces apoB-48 in murine liver and intestine. Human apobec-1 is expressed only in intestinal cells, thus these cells are the source of apoB-48 in humans. Antisense inhibition of PCSK9 resulted in an increase in murine hepatic apobec-1 mRNA levels by approximately 2.7 fold compared to saline-treated controls

Serum Lipid Analysis

[0936] Plasma concentrations of total cholesterol, LDL-C, HDL-C, free cholesterol, triglycerides, glucose, ketones, transaminases, and phospholipids were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). Serum lipoprotein and cholesterol profiling was performed as described by Crooke et al. (Lipid Res., 2005, 46, 872-884) using a Beckman System Gold 126 HPLC system, a 507e refrigerated autosampler, a 126 photodiode array detector (Beckman Instruments, Fullerton, Calif.), and a Superose 6 HR 10/30 column (Pfizer, Chicago, Ill.). HDL, LDL and VLDL fractions were measured at a wavelength of 505 nm and validated with a cholesterol calibration kit. (Sigma).

[0937] Administration of ISIS 394814 resulted in a 52% reduction in total cholesterol (saline, 183 mg/dL±18; ISIS 141923, 194 mg/dL±14; ISIS 394814, 87 mg/dL±19, p<0.005) and a 36% reduction in LDL-C (saline 22 mg/dL±4; ISIS 141923, 25 mg/dL±2; ISIS 394814, 14 mg/dL±2, p<0.005). Serum free cholesterol was reduced by 25% (saline 41 mg/dL±10; ISIS 141923, 58 mg/dL±3; ISIS 394814, 31 mg/dL±5, p<0.005) and phospholipids were reduced 54% (saline, 354 mg/dL±29; ISIS 141923, 383 mg/dL±21; ISIS 394814, 169 mg/dL±35, p<0.0001). HDL-C was also reduced by approximately 54% (saline, 183 mg/dL±18; ISIS 141923, 194 mg/dL±14; ISIS 394814, 87 mg/dL±19; p<0.0001). HPLC profiling confirmed the reduction of LDL and HDL lipid classes.

Liver Triglyceride Analysis

[0938] Liver triglyceride content was measured according to routine experimental procedures, for example, per procedures described by Desai et al., Diabetes, 2001, 50:2287-2295.

[0939] Antisense inhibition of PCSK9 for 6 weeks reduced liver triglyceride content by approximately 65% (p=0.01) relative to saline controls. No statistically significant changes in liver triglyceride content following 141923 treatment were observed.

[0940] Accordingly, one embodiment is a method of lowering LDL-C levels through the administration of an antisense oligonucleotide targeted to a PCSK9 nucleic acid. An additional embodiment includes a method of lowering total cholesterol through the administration of an antisense oligonucleotide targeted to a PCSK9 nucleic acid. An addition embodiment is lowering liver triglycerides by administering an antisense oligonucleotide targeted to a PCSK9 target nucleic acid.

Example 12: Antisense Reduction of PCSK9 mRNA in LDL-R Deficient/apoB-100 Animals

[0941] LDL-receptor (LDL-R)-deficient/apoB-100 animals do not express the LDL-R gene, and express only the apoB-100 form of apoB. PCSK9 antisense oligonucleotide was administered to LDL-R deficient/apoB-100 mice, to evaluate the effects of antisense inhibition of PCSK9 in the absence of the LDL-R. Mice 4-5 weeks of age were maintained on a standard mouse diet. Treatment groups of 5 mice each were as follows: a group treated with ISIS 394814; a control group treated with saline; and control group treated with an oligonucleotide not complementary to any known gene sequence (ISIS 141923). Oligonucleotide or saline was administered intraperitoneally twice weekly, for a period of 6 weeks; oligonucleotide doses were 50 mg/kg (100 mg/kg/wk total). Real-time PCR of liver mRNA and serum analyses were performed as described for the C57Bl/6 study.

[0942] Antisense inhibition of PCSK9 reduced liver PCSK9 mRNA by approximately 90%, relative to saline controls. However, no reduction in serum cholesterol was observed. Liver triglyceride levels were unaffected by antisense inhibition of PCKS9 in these mice. An 80% increase in liver apobec1 mRNA levels was observed, relative to saline controls. These data suggest that a functioning LDL-R is required for cholesterol and liver triglyceride reduction via PCSK9 antisense inhibition.

Example 13: Antisense Reduction of PCSK9 mRNA in Mouse Primary Hepatocytes

[0943] Antisense oligonucleotides targeted to PCSK9 were tested for antisense inhibition of PCSK9 in mouse primary hepatocytes. Mouse primary hepatocytes were purchased from a commercial supplier, and cultured according to routine culture procedures. Cells were treated with 6.25, 12.5, 25, 50, 100, or 200 nM of antisense oligonucleotide for a period of 24 hours, after which RNA was isolated and PCSK9 mRNA was measured by real-time PCR as described herein.

[0944] The antisense oligonucleotides tested in human primary hepatocytes were Isis 395165, 395185, 395186, 395187, 405879, 405881, 405891, 405988, 405994, and 406008. The antisense oligonucleotides inhibited PCSK9 in a dose-dependent manner in human primary hepatocytes as shown in Table 26.

TABLE 26

Dose-Dependent Reduction PCSK9 mRNA in Mouse Primary Hepatocytes

ISIS No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM
395165	95.2	94.9	81.7	66.3	57.5	29.9
395185	109.2	100	90.5	81.2	64.7	35.8
395186	101.9	91.5	77.8	59.1	39.5	13.7
395187	102.4	96.1	83.9	66.4	40	15.3
405879	107.3	97.5	87.8	76.6	63	43.1
405881	104.8	102.1	98.1	84.9	68.7	48.9
405891	104.9	103.3	101.5	88.9	79.6	49.5
405988	97.3	101.1	91.7	82.3	61.2	34.8
405994	110.2	101	111.6	94.4	80.1	57.7
406008	99.5	91.3	83.9	73.7	61.8	39.4
157700	97.8	98.3	96.2	94.3	83.2	76.4
141923	84.3	80	79.9	77.3	98.8	64.2

Example 14: Antisense Reduction of Human PCSK9 mRNA in Hep3B and HeLa Cells: Dose Response Experiment

[0945] A subset of the antisense oligonucleotides that inhibited PCSK9 expression at high levels in the cell culture systems described above was selected for further screening. The group of selected antisense compounds is shown in Table 27. Table 27 shows the Isis number, nucleotide sequence, and percent inhibition obtained in Hep3B cells using the protocol disclosed in Example 3 (see Table 17) for the various antisense compounds.

TABLE 27

Isis No.	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
405881	CACCCTTGGCCACGCCGGCA	5-10-5	98.3	250
399819	CCCACTCAAGGGCCAGGCCA	5-10-5	93.4	65

TABLE 27-continued

Isis No.	Sequence 5'-3'	Motif	% inhibition	SEQ ID NO
395165	ACCCTTGGCCACGCCGGCAT	5-10-5	96.7	28
405879	CCTTGGCCACGCCGGCATCC	5-10-5	94.1	248
406008	CTTGGTGAGGTATCCCCGGC	5-10-5	95.2	188
405891	TCCTCAGGGAACCAGGCCTC	5-10-5	96.1	352
395186	CTTTGCATTCCAGACCTGGG	5-10-5	93.7	60
405988	GGCAGCACCTGGCAATGGCG	5-10-5	95.9	381
405994	GCAGTGGACACGGGTCCCCA	5-10-5	95.2	400
406023	TGGTATTCATCCGCCCGGTA	5-10-5	92.5	212
395187	GGCAGCAGATGGCAACGGCT	5-10-5	92.7	62
395185	CACGGGTCCCCATGCTGGCC	5-10-5	92.7	59
406033	CCTGCCAGGTGGGTGCCATG	5-10-5	91.2	237
405923	GGCATTGGTGGCCCCAACTG	5-10-5	94.3	288
399900	GGTCCTCAGGGAACCAGGCC	3-14-3	92.0	50
405995	TGGCAGTGGACACGGGTCCC	5-10-5	92.4	402
405991	GACACGGGTCCCCATGCTGG	5-10-5	91.9	394
406005	CGGGCAGTGCGCTCTGACTG	5-10-5	83.4	180
399793	CCTCGGAACGCAAGGCTAGC	5-10-5	77.9	8
395152	ACGCAAGGCTAGCACCAGCT	5-10-5	93.3	7

[0946] The IC_{50} of the various antisense compounds was determined using different cell culture systems. Antisense oligonucleotides targeted to PCSK9 were tested at various doses in Hep3B or HeLa cells, for example. Cells were plated at densities of 4,500 cells or 5,000 per well, respectively, and treated with of antisense oligonucleotides at various concentrations. After a treatment period of approximately 24 hours, RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein, relative to untreated control cells. Two different human PCSK9 primer probe sets were used to measure mRNA levels. Results with Primer Probe Set 2740 are shown in Table 28. PCSK9 mRNA levels were adjusted according to total RNA content as measured by RIBOGREEN®. Table 28 shows IC_{50} values (nM) obtained from two experiments with Hep3B cells and from one experiment with HeLa cells.

TABLE 28

ISIS #	Hep3B IC_{50} (nM)	Hep3B IC_{50} (nM)	HeLa IC_{50} (nM)	PCSK9 Target Sequence	SEQ ID NO
	IC_{50} values				
405881	21	26	13	CACCCTTGGCCACGCCGGCA	250
399819	21	28	12	CCCACTCAAGGGCCAGGCCA	65
395165	23	28	13	ACCCTTGGCCACGCCGGCAT	28
405879	24	28	14	CCTTGGCCACGCCGGCATCC	248
406008	31	35	8	CTTGGTGAGGTATCCCCGGC	188
405891	32	38	14	TCCTCAGGGAACCAGGCCTC	352
395186	33	33	7	CTTTGCATTCCAGACCTGGG	60
405988	34	42	8	GGCAGCACCTGGCAATGGCG	381
405994	35	39	9	GCAGTGGACACGGGTCCCCA	400
406023	40	50	18	TGGTATTCATCCGCCCGGTA	212
395187	41	44	8	GGCAGCAGATGGCAACGGCT	62
395185	42	43	11	CACGGGTCCCCATGCTGGCC	59
406033	44	61	15	CCTGCCAGGTGGGTGCCATG	237
405923	47	41	20	GGCATTGGTGGCCCCAACTG	288
399900	48	44	21	GGTCCTCAGGGAACCAGGCC	50
405995	48	63	12	TGGCAGTGGACACGGGTCCC	402
405991	63	52	15	GACACGGGTCCCCATGCTGG	394
406005	63			CGGGCAGTGCGCTCTGACTG	180
399793	>75			CCTCGGAACGCAAGGCTAGC	8
395152		46	16	ACGCAAGGCTAGCACCAGCT	7

[0947] Several of the antisense compounds examined above were selected for further study. These antisense compounds exhibited at least about 93% inhibition of human PCSK9 mRNA expression in the Hep3B cell culture system, as shown in Tables 17-19, for example. As shown in Table 28, the selected antisense compounds also displayed IC_{50} parameters of less than about 45 nM when tested with Hep3B cells and less than about 15 nM when tested with HeLa cells. The selected antisense compounds, listed in Table 29, are Isis 405881, Isis 395165, Isis 405879, Isis 406008, Isis 405891, Isis 395186, Isis 405988, Isis 405994, Isis 395187, and Isis 395185. Isis 405879, for example, showed superior results in both Hep3B and HeLa cells, with IC_{50} values of about 26 nM and 14 nM, respectively.

Example 15: Dose-Dependent Antisense Reduction of PCSK9 mRNA in Human Primary Hepatocytes; Cynomolgus Monkey ("Cyno") Primary Hepatocytes; and Human HEK-293 Cells

[0948] The selected antisense oligonucleotides were further tested for dose dependent antisense inhibition in human primary hepatocytes, cyno primary hepatocytes, and HEK-293 cells expressing human PCSK9. Cells were purchased from commercial suppliers and cultured according to routine culture procedures. Cells were treated with 10, 25, 50, 150, or 300 nM antisense oligonucleotide (i.e., a five-point dose response test) for a period of 24 hours, after which RNA was isolated. PCSK9 mRNA was measured by real-time PCR, as described herein.

[0949] Results of the five-point dose response tests using HEK-293 cells are shown in FIG. **2**. Results of the five-point dose response tests using cyno primary hepatocytes are shown in FIG. **3**. The IC_{50} results, expressed in nM units, for the five-point dose response test using human primary hepatocytes, cyno primary hepatocytes, HEK-293 cells, Hep3B cells, and HeLa cells are summarized in Table 29.

TABLE 29

IC_{50} (nM) for Selected Antisense Compounds

	Isis No.									
	395165	395185	395186	395187	405879	405881	405891	405988	405994	406008
	SEQ ID NO:									
	28	59	60	62	248	250	352	381	400	188
Human Primary Hepatocytes	24	33	39	30	ND	50	297	166	56	163
Cyno Primary Hepatocytes	5	17	18	14	2	19	10	12	21	11
HEK-293	40	116	6	11	35	86	21	11	17	27
Hep3B	28	43	33	44	28	26	38	42	39	35
HeLa	13	11	7	8	14	13	14	8	9	8

[0950] ISIS 405879 was tested at various doses in human primary hepatocytes. RNA was isolated from the cells and PCSK9 mRNA levels were measured by quantitative real-time PCR, as described herein. Human PCSK9 primer probe set 2740 was used to measure mRNA levels. Results are shown in Table 29.1.

TABLE 29.1

Dose-dependent Reduction in Human PCSK9 mRNA Expression in Human Primary Hepatocytes

ISIS NO.	300 nM	200 nM	150 nM	100 nM	50 nM	25 nM	10 nM
405879	13	18	26	30	27	54	75

[0951] Isis 405879 has an IC_{50} value of 29 nM in human primary hepatocytes. Overall Isis 405879 displays superior activity, particularly in cyno primary hepatocytes. Isis 405879 was thus further tested in the various cell models, and the results of the additional testing are shown in Table 30. In these experiments, Isis 405879 was administered to the cells using lipofectin transfection reagents. PCSK9 mRNA was assayed by PCR following 24 hr incubation after transfection. These results confirm that Isis 405879 potently inhibits PCSK9 mRNA expression at low concentrations.

TABLE 30

Isis 405879 activity in cell culture models

	Hep3B Cells (human)	HepG2 Cells (human)	HeLa Cells (human)	Hepatocyte Transgenic Mouse (h-PCSK9)	Primary Hepatocyte (human)	Primary Hepatocyte (cyno)	Transgenic HEK293 cells (h-PCSK9)
Mean IC_{50} (nM)	17	5	14	140	22	7.4	18

[0952] These experiments confirm that the presently disclosed antisense oligonucleotides are capable of potently inhibiting PCSK9 mRNA expression in a variety of cell model systems, including a human primary cell line. Exemplary antisense compounds inhibit PCSK9 mRNA expression with IC_{50} values in the nM range.

[0953] The transgenic PCSK9 cDNA construct expressed in mouse hepatocytes is thought to be rapidly exported to the cytoplasm, because it does not require post-transcriptional editing. The antisense oligonucleotides provided herein act by recruiting the RNase H enzyme, which is restricted to the nucleus. Therefore, inhibition by antisense oligonucleotides in this cell type is significantly diminished.

Example 16: In Vivo Testing Procedures

16.1. High Fat (HF)-Fed Hyperlipidemic Mouse Model

[0954] Antisense compounds may be tested in a high fat-fed, hyperlipidemic mouse model to test efficacy of the antisense compounds in reducing PCSK9 mRNA expression, increasing LDL-R expression, reducing total cholesterol and LDL, and affecting other metrics of antisense compound efficacy. A representative hyperlipidemic mouse model is described in Graham et al., *J. Lipid Res.* (2007) 48: 763-767.

[0955] All animal experiments were conducted in accordance with Institutional American Association for the Accreditation of Laboratory Animal Care guidelines. C57BL/6 mice were obtained from Jackson Laboratory (http://www.jax.org). A majority of the mice were male, and studies were initiated when animals were 4-5 weeks of age. The mice were maintained on a 12 h light/12 h dark cycle and fed ad libitum. C57BL/6 mice were fed a diet consisting of 60% lard (Research Diets, New Brunswick, N.J.), whereas experimental Ldlr-deficient/apoB-100 mice were fed regular chow.

[0956] Antisense compounds were administered twice weekly (50 mg/kg) for 6 weeks by intraperitoneal injection (10 mg/mL dosing solution formulated in saline; saline control). After the treatment period, whole blood was collected for analysis of serum parameters, and whole liver was collected for RNA analysis, protein analysis, and histological evaluation.

16.2. H-ApoB/CETP Transgenic Mouse Model

[0957] A transgenic mouse model that expresses both human apolipoprotein (apo) B and human cholesteryl ester transfer protein (CETP) provides another useful animal model system to test the efficacy of the presently disclosed antisense compounds. The lipoprotein cholesterol distribution in the serum of a chow-fed h-apoB/CETP mouse resembles a human profile. Specifically, the percentages of total cholesterol within the HDL, LDL, and VLDL fractions of apoB/CETP animals are approximately 30%, 65%, and 5%, respectively, similar to the distribution of cholesterol in the plasma of normolipidemic humans. The h-apoB/CETP mouse model is described further in Grass et al., *J. Lipid Res.* (1995) 36: 1082-1091.

16.3. H-PCSK9 Transgenic Mouse Model

[0958] A transgenic mouse expressing human PCSK9 provides another useful animal model to study in vivo repression of PCSK9 mRNA by the present antisense compounds. H-PCSK9 transgenic mice overexpress human PCSK9 in liver and secrete large amounts of the protein into plasma, which increases plasma LDL-C concentrations to levels similar to those of LDLR-knockout mice. H-PCSK9 transgenic mice were constructed by inserting a cDNA encoding human PCSK9 into a pLiv-11 vector that contained the constitutive human apoE promoter and its hepatic control region, as described in Simonet et al., *J. Biol. Chem.* (1993) 268: 8221-8229. Transgenic mice were generated by injecting linearized pLiv-11-hPCSK9 into the fertilized eggs as described in Shimano et al., *J. Clin. Invest.* (1996) 98: 1575-1584. H-PCSK9 transgenic mice are further described in Lagace et al., *J. Clin. Invest.* (2006) 116: 2995-3005.

16.4. Mouse and Rat Models for In Vivo Tolerability, Half-Life and Tissue Distribution

[0959] Mouse (CD1) and rat (Sprague-Dawley) models were used to test the in vivo tolerability of various antisense compounds. In the mouse CD1 model, antisense compounds were overdosed at 100 mg/kg/wk for six weeks. Possible proinflammatory effects and hepatotoxcity were assayed following administration of the antisense compounds. Serum levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), blood urea nitrogen (BUN), creatinine, albumin, and bilirubin were assayed. Weights of livers, kidneys, and spleens of animals were determined. Histopathologic screening was conducted to detect hepatocellular swelling, multifocal apoptosis, inflammatory cell infiltrates in the liver or lungs, or follicular hyperplasia. Additional studies were conducted in the rat model to screen for renal toxicity or high renal accumulation and proinflammatroy effects. In these experiments, antisense compounds were administered at 60 or 100 mg/kg/wk for six weeks. In addition to the markers examined above, the urine protein/creatine ratio and proteinuria were assayed.

[0960] The half-life of administered antisense compounds also was examined in the target organ, i.e., liver, in the mouse model. For these experiments, antisense compounds were administered twice a week for two weeks at 50 mg/kg, and target organs were then harvested and assayed for the presence of the antisense compounds, as described below.

[0961] The distribution of administered antisense compounds in various organs also was examined in the rat model. Rats in these studies were administered 20 mg/kg antisense compounds twice weekly for three weeks. Livers and kidneys of test animals were then harvested and assayed for the presence of the antisense compounds, as described below.

16.5. Cynomolgus Monkey (*M. fascicularis*) Model

[0962] Potencies and pharmacodynamics of antisense compounds were compared in a cyno model to identify the compounds exhibiting the best combination of high potency and low renal accumulation, when given subcutaneously to animals for a 5-week period. In another study, dosing was conducted over a 13-week period. The cyno monkey is widely used in toxicity testing, and a substantial historic toxicology database is available for this model.

[0963] The PCSK9 target sequence for an antisense compound may differ in humans versus cyno monkey. As a result, an antisense compound for use in humans may not be fully complementary to a monkey PCSK9 nucleic acid. Antisense compounds having identical target sequences in the two species are preferred. Isis 405881 and Isis 395165 each have a one nucleotide mismatch compared to the cyno sequence, meaning that these antisense compounds are less preferred in the cyno model.

[0964] Cyno monkeys (Charles River BRF, Inc.) were approximately 26 to 32 months old and 2.5 to 4 kg in weight on the first day of dosing. Monkeys were offered water ad libitum and fed a daily ration of 12 biscuits approximately 2 hours post dose. Antisense compounds were administered subcutaneously at 3 doses every other day during the first week, then by weekly injections. A low dose of 15 mg/kg was used to assess the potency of the antisense compounds. A high dose of 8 subcutaneous injections of 30 mg/kg over 5 weeks or 13 weeks was used to assay renal accumulation and pharmacodynamics. Antisense compounds were administered at a concentration of 60 or 120 mg/mL, respectively, in a vehicle of sterile phosphate buffered saline. Plasma concentrations of antisense compounds were determined at each dose level to estimate systemic exposure in monkeys following subcutaneous administration and to assess whether systemic exposure was altered after repeat dosing.

[0965] To measure the host reaction to the administered antisense compounds, 2 mL blood samples were collected from the femoral vein of host individuals in tubes of appropriate size containing K_2EDTA. Plasma was obtained by centrifugation and samples were stored frozen. The analysis of the plasma samples included a determination of the level of plasma PCSK9 protein. To measure an effect on pharmacodynamic end points, LDL, HDL, and apoB levels also were analyzed. To determine whether the host individuals suffered side effects, blood samples were further analyzed for an effect on the hematological parameters listed in Table 31 and the serum chemistry parameters listed in Table 32:

TABLE 31

Hematology markers	
erythrocyte count	red cell distribution width
hemoglobin	platelet count
hematocrit	mean platelet volume
mean corpuscular volume	absolute total and differential leukocyte counts
mean corpuscular hemoglobin concentration	evaluation of cell morphology
absolute reticulocyte count	

TABLE 32

Serum chemistry markers	
aspartate aminotransferase	triglycerides
alanine aminotransferase	glucose
gamma glutamyltransferase	urea nitrogen
alkaline phosphatase	creatinine
total bilirubin	calcium
total cholesterol	phosphorus
total protein	sodium
albumin	potassium
globulins	chloride
albumin/globulin ratio	

[0966] Possible additional side effects were determined by screening individuals administered antisense compounds for gross changes in organs or microscopic changes in tissues. A limited necropsy conducted on all test animals included gross examination of various organs and tissues. Terminal body weights were recorded, and kidneys and spleen were weighed for all test animals. For microscopic evaluation, liver, kidney, spleen and injection sites from all animals were fixed in 10% neutral buffered formalin and were processed, embedded in paraffin, sectioned, and stained with hematoxylin and eosin.

[0967] Representative tissue samples of liver and kidney cortex from each scheduled-necropsy monkey from the high-dose and control groups were collected for ultrastructural examination using electron microscopy. Thin slices (2-4 mm in thickness) of liver and kidney cortex were cut and transferred to a piece of dental wax containing a small volume of McDowell-Trump fixative. Thin slices were further minced into approximately 1×1-2 mm cubes in McDowell-Trump fixative and analyzed by electron microscopy.

[0968] To evaluate PCSK9 mRNA levels, samples from liver (approximately 1 g) were homogenized using a Polytron tissue disruptor in 10 mL of RLT solution (Qiagen) and snap-frozen and stored at −70° C. or colder. Additionally, samples (approximately 500 mg each) from kidney were collected and snap-frozen for mRNA analysis.

[0969] Tissue samples from liver and kidney cortex were also analyzed for the presence of the administered antisense compounds. Antisense compounds present in the tissues can be detected and quantified using capillary gel electrophoresis (CGE), as described in Leeds et al, *Drug Metab Dispos*. (1998) 26: 670-675, for example. In this procedure, tissues are weighed, mixed with an internal standard (e.g., a 27-mer (T) oligonucleotide) in proteinase K digestion buffer (2.0 mg/ml proteinase K in 20 mM Tris-HCl, pH 8.0, 20 mM EDTA, 100 mM NaCl, 0.5% Nonidet P-40), and homogenized. Overnight incubation at 37° C. is followed by phenol/chloroform extraction. Tissue extracts are purified by sequential anion-exchange solid-phase extraction (SPE) followed by reverse-phase SPE, described in Leeds et al., *Anal Biochem* (1996) 235: 36-43. Samples are further desalted by membrane dialysis before analysis using gel-filled capillaries, as described in Leeds et al. (1996).

[0970] Capillary gel electrophoresis separations can be performed as described by Leeds et al. (1998), using a Beckman P/ACE capillary electrophoresis instrument (model 5010) with a 27-cm column (effective length, 20 cm) containing 12% polyacrylamide, with 8.3 M urea in 100 mM Tris-borate, pH 8.5, as the running buffer. Separation is achieved at 50° C. and 550 V/cm. Oligomers eluting from the column are detected by UV absorption at 260 nm. Quantification of antisense oligonucleotides is based on the tissue weight extracted and the initial T27 concentration.

Example 17: In Vivo Testing Results

[0971] 17.1. Reducing PCSK9 mRNA In Vivo Using a Murine Antisense Compound Correlates with In Vivo Plasma LDL-C and Liver TG Reduction

[0972] In vivo pharmacodynamic studies were performed with the murine antisense compound Isis 394816 to validate the pharmacodynamic endpoint of PCSK9 mRNA reduction. Isis 394816 has the sequence 5' GGTAAGGTGCGGTAAGTCCT 3' (SEQ ID NO: 462), which targets position 3100 (5') of the coding sequence of murine PCSK9 mRNA (NCBI Accession No. NM_153565.1; SEQ ID NO: 463). Isis 394816 has a 3-14-3 motif.

[0973] Isis 394816 was administered by intraperitoneal injection to an h-apoB/CETP transgenic mouse for 6 weeks at a dosing regimen of 20 mg/kg/wk or 50 mg/kg/wk. See Example 16.2. As shown in FIG. **4**, Isis 394816 administered at either dose caused an approximate 80% reduction in PCSK9 mRNA expression in livers of test animals, compared to control animals. As shown in FIG. **5**, panel A, Isis 394816 produced a significant ($p<0.0001$) decrease in the level of plasma LDL-C at the end of six weeks. Additionally, as shown in FIG. **5**, panel B, Isis 394816 produced a significant ($p<0.02$) decrease in the level of liver TG at the end of six weeks.

[0974] Additionally, Isis 394816 reduced PCSK9 mRNA expression, plasma LDL-C, and liver TG in a dose dependent manner, as shown in FIG. **6**. The potency of Isis 394816 was much higher in a hyperlipidemic mouse model (see Example 16.1.) than in a lean mouse model (data not shown). The reduction of plasma LDL-C positively correlated with the inhibition of PCSK9 mRNA levels in liver, shown in FIG. **7**. These studies demonstrate that the reduction of liver PCSK9 mRNA correlates with the desired endpoint of reducing plasma LDL-C and liver TG. Accordingly, the reduction of liver PCSK9 mRNA provides a valid pharmacodynamic endpoint.

17.2. Differences in the Half-Life ($t_{1/2}$) of Antisense Compounds in Liver

[0975] The half-lives of selected antisense compounds were assayed in mouse CD1 livers, according to the procedures set forth in Example 16.4. Half-lives varied among the tested antisense compounds, which again is attributable to differences in structural features of the antisense compounds, such as nucleotide sequences and/or motifs. For example, the half-life of Isis 405879 ($t_{1/2}2$=27.7 days) was significantly higher than the half-life of Isis 395165 (20.4 days), although the target sequences of these two antisense compounds are shifted by only two nucleotides:

```
Isis 405879:
                                   (SEQ ID NO: 248)
CCTTGGCCACGCCGGCATCC

Isis 395165:
                                    (SEQ ID NO: 28)
ACCCTTGGCCACGCCGGCAT
```

Half-life data for exemplary antisense compounds is summarized in Table 33. Isis 405879 displayed a particularly high half-life in liver.

TABLE 33

Liver $t_{1/2}$ (days)	
Antisense Compound	Liver $t_{1/2}$ (d)
395165	20.4
395185	20.2
395186	19.9
395187	13.7
405879	27.7
405881	14.1
405891	17.1
405988	14.6
405994	20.2
406008	19.7

17.3. Differences in the Liver and Kidney Distributions of Antisense Compounds

[0976] The antisense compounds listed in Table 29 were administered to rats to determine the distribution of antisense compounds in liver and kidney. Accumulation of high levels of antisense compounds in the kidneys may cause adverse side effects. Antisense compounds were dosed and administered as described in Example 16.2. FIG. **8** shows the distribution of the antisense compounds shown in Table 29 in rat liver and kidneys. Isis 395185, Isis 395186, Isis 395187, Isis 405879, Isis 405881, Isis 405891, Isis 405988, and Isis 405994 show a particularly advantageous ratio of distribution between liver and kidney.

[0977] When tissue distribution was tested in a cyno model, further differences among antisense compounds were observed. Accumulation of Isis 395186, Isis 405879, and Isis 405891 in liver and kidney was examined after dosing each antisense compound at two concentrations. See Example 16.5. The results are shown in FIG. **9**. Accumulation at each dosing concentration (15 mg/kg and 30 mg/kg) is shown for kidney (left side bar) and liver (right side bar). Isis 405879 advantageously accumulates at a high concentration in liver, even at a dose of 15 mg/kg, and accumulates at a low level in kidney. Specifically, Isis 405879 accumulated in kidneys at a concentration of 1300 μg/g (15 mg/kg dose) to 1800 μg/g (30 mg/kg dose). The average accumulation of antisense compounds in kidneys is 2500 μg/g (15 mg/kg dose) to 4000 μg/g (30 mg/kg dose). Variation in kidney and liver accumulation is attributable to structural differences, such as nucleotide sequences and/or motifs, between the antisense compounds.

17.4. Differences in Host Tolerance of Antisense Compounds

[0978] Rodent tolerability studies, conducted as described in Example 16.4, showed that antisense compounds were well tolerated. Following administration of the exemplary antisense compounds listing in Table 29, serum chemistry markers were normal, with some variably high ALT and AST levels observed for Isis 405988 and Isis 405994. Whole body weight and major organ weights in CD-1 mice also were not significantly affected by administration of the exemplary antisense compounds. Histological findings revealed differences between the various antisense compounds, but were consistent with the overall good tolerability for the antisense compounds. The results of the histological screens are shown in Table 34.

17.5. Results of the 5 wk Cyno Toxicology Study

[0979] The methods used to conduct the 5 wk cyno toxicology study are described in Example 16.5. Isis 405879 is referred to as BMS-844421 below and in the corresponding figures. The PCSK9 mRNA levels assayed by q-RT-PCR in the cyno liver tissue samples were decreased by BMS-844421 treatment by an average of 65% and 70% (average of males and females) versus saline controls at the low and high doses, respectively (FIG. **25**). The effect was seen in both females and male monkeys, despite relatively high variability of mRNA level between animals. No dose-dependency was seen between the 2 doses. The data are mean+/− for male and female animals combined.

[0980] The levels of BMS-844421 ASO were measured in liver at the end of the 5 wk study. As shown in FIG. **26**, liver full-length BMS-844421 levels reached ~1 mg/gram tissue in both dose groups. This level is consistent with values from previous experiences with other liver-targeting ASOs. Most of the compound was detected as intact full-length BMS-844421. No dose-dependency was seen in the ASO liver accumulation data, in agreement with the lack of dose-dependent effects on liver PCSK9 target mRNA.

[0981] Pharmacodynamic responses were assessed in serum lipid assays for the cyno 5 wk ASO study. As seen in FIG. **27**, a modest trend (not statistically significant) toward lowering LDL-C was observed for BMS-844421 treated cyno monkeys at 2.5 and 5 wk of dosing. This trend at both time points for BMS-844421 relative to baseline was also seen in the scatterplot of individual animal data in FIG. **28**A. HDL/LDL ratios were sometimes employed as a sensitive indicator of plasma lipid status. A chart of the average ratio of HDL/LDL in the cyno study showed a trend for increase in the ratio for BMS-844421 treated cynos at 2.5 and 5 wk (FIG. **28**B).

[0982] Gene expression profiling by Affymetrix microarray was conducted on the RNA samples from this study. The analysis included liver RNA from animals administered vehicle alone, BMS-844421 at 15 mg/kg and 30 mg/kg and BMS-844419 and BMS-844423 at 30 mg/kg only. By microarray, there were significant ($p<0.01$) transcriptional changes in approximately 1 to 2% of genes with approximately 1% expected to change at this significance level by chance. For drug-dosed groups relative to control there were alterations in inflammation-, immune system-, and lipid homeostasis-related transcripts with statistically significant animal to animal variability. Other drug-related hepatic transcriptional changes common to multiple dose groups were associated with several or a few genes in various biological processes and are of unknown significance.

17.6. Results of the 13 wk Cyno Toxicology Study

[0983] An additional opportunity to monitor pharmacodynamic responses to BMS-844421 in monkeys was the 13 wk toxicology study, described in Example 16.5. Lean cyno monkeys with low LDL levels were again the subject for this study, as for the 5 wk cyno monkey study above. At 13 wks subcutaneous dosing at 4 different dose levels, BMS-844421 suppressed PCSK9 expression as measured by both liver PCSK9 mRNA and plasma PCSK9 protein levels. Despite the inter-animal variability in these values, it is apparent that BMS-844421 strongly suppressed both liver mRNA and plasma PCSK9 protein (FIG. **29**). 40 mg/kg suppressed PCSK9 protein levels very robustly with less variation between animals (FIG. **29**). No statistically significant differences were seen in plasma LDL-C or other lipid parameters in this study.

TABLE 34

Histological Findings in CD-I Mice

Antisense compound	Liver	Kidney	Lung	Heart	Spleen	Lymph node	Intestine	Thyroid	Sternum	Skeletal Muscle
Saline	–	–	–	–	–	–	–	–	–	–
ISIS 395165	+/++	–			–/+	–/+	–	–	–	–
ISIS 395185	+	–	–	–	–/+	–	–	–	–	–
ISIS 395186	–	–	–	–	–	–	–	–	–	–
ISIS 395187	–	–	–	–	–/+	–	–	–	–	–
ISIS 405879	–	–	–	–	–	–	–	–	–	–
ISIS 405881	inflam	–	inflam	–	–/+	–/+	–	–	–	–
ISIS 405891	–	–	–	–	–/+	–/+	–	–	–	–
ISIS 405988	–	–	–	–	–	–	–	–	–	–
ISIS 405994	++	–	–	–	–/+	–	–	–	–	–
ISIS 406008	+/++	–	–	–	–	–	–	–	–	–

Normal: –
Minimal changes: –/+
Mild changes: +
Moderate changes: ++
Severe changes: +++

[0984] Microscopic evaluation of mouse organs from animals treated with 10 different human PCSK-9 antisense compounds revealed that all compounds generally were well tolerated, but that some antisense compounds were particularly well tolerated. The following results were seen in the various examined organs.

[0985] Liver: Mild (+) to moderate (++) liver injuries (cytoplasmic swelling to multifocal apoptosis) were seen in Isis 395165 (+/++), Isis 395185 (+), Isis 405988 (+/++), Isis 405994 (++), and Isis 406008 (+/++) treated livers. Mild inflammatory changes were seen in Isis 405881 treated livers.

[0986] Lung: No significant abnormality was seen in treated animals except for mild histiocytosis noticed in Isis 405881 treated animals.

[0987] Spleen: Mild follicular hyperplasia was seen in Isis 395165, Isis 395185, Isis 395187, Isis 405881, Isis 405891, and Isis 405994 treated spleens.

[0988] Lymph node: Minimal histiocytosis was observed in Isis 395165, Isis 405881 and Isis 405891 treated animals.

[0989] Heart: Minimal levels of macrophage infiltration were seen in Isis 395165, Isis 395186, Isis 405879, Isis 405881, Isis 405891, Isis 405988, and Isis 406008 treated hearts.

[0990] Kidney, Thyroid, small intestine, skeletal muscle and sternum: No abnormality was visualized.

[0991] Differences in the histological findings were seen even between Isis 395165 and Isis 405879. As noted above, the target sequences of these two antisense compounds are only shifted by two nucleotide positions. Nevertheless, Isis 405879 advantageously displays no adverse reactions, whereas Isis 395165 elicits a mild inflammatory response in liver, spleen, and lymph nodes. In summary, Isis 405879, Isis 395186, and Isis 405988 display superior tolerance, compared to the other antisense compounds tested in this study.

17.7. Differences in Achieving Pharmacodynamic Endpoints in the Cyno Model.

[0992] Isis 395186 ("844419"), Isis 405879 ("844421"), and Isis 405891 ("844423") were selected for further study in the cyno model, using the procedures described in Example 16.6. As set forth above, these antisense compounds displayed high activity in cell culture models, were well tolerated, and displayed advantageous tissue distribution and half-life properties. Further, none of these antisense compounds targeted a PCSK9 sequence known to have differences in monkeys and humans. In addition, the target sequences in human PCSK9 were known not to contain SNPs having a wide distribution in the population.

[0993] The antisense compounds were tested for their ability to reduce PCSK9 mRNA in cyno liver. See Example 16.6. The results of this experiment are shown in FIG. **10**. Despite variability between the tested animals, Isis 405879 (labeled "844421") displayed a superior ability to reduce PCSK9 mRNA levels significantly. When administered at a low dose (15 mg/kg, shown in the left side bar), Isis 405879 significantly ($p<0.023$) reduced PCSK9 mRNA levels. When administered at a high dose (30 mg/kg, shown in the right side bar), Isis 405879 also significantly ($p<0.011$) inhibited PCSK9 mRNA expression. The results obtained with Isis 405879 were superior to those obtained with Isis 395186 (labeled "844419") and Isis 405891 (labeled "844423"). As shown in FIG. **11**, Northern analysis of liver PCSK9 mRNA confirms that 15 mg/kg and 30 mg/kg Isis 405879 substantially reduces PCSK9 mRNA levels.

[0994] As described in Example 17.1, the reduction of PCSK9 mRNA is expected to correlate with a reduction in plasma LDL-C and liver TG. Consistent with this expectation, Isis 405879 produced a significant ($p<0.06$) decrease in cyno serum LDL-C after five weeks of administration at 30 mg/kg, as shown in FIG. **12**. Isis 405879 additionally produced a significant ($p<0.05$) increase in the ratio of HDL to LDL following five weeks of administration at 30 mg/kg, as shown in FIG. **13**.

[0995] Generally confirming earlier data obtained in rodents, Isis 405879 elicited no differences in serum chemistry, hematology, or organ weights, except for a slight increase in spleen weight. Histology revealed indicia of only mild inflammation. The advantageous distribution of Isis 405879 in liver and kidney in the cyno model is described above and shown in FIG. **9**.

Example 18: Dose Response Study of Isis 405879 ("844421") in a Human PCSK9 Transgenic Mouse Model

[0996] BMS-844421 is a human specific PCSK9 antisense oligonucleotide (ASO) designed to downregulate hepatic PCSK9 expression, increase LDL receptor activity, and promote clearance of LDL and reduction of plasma LDL-C levels in humans. A human genomic PCSK9 transgenic mouse model was recently established. This mouse line expresses hemizygous transgene in the presence of endogenous mouse PCSK9 (i.e., both genes are simultaneously expressed). It was found that the transgenic PCSK9 mice responded to chronic dosing with BMS-844421 and related human PCSK9 ASOs with decreased liver mRNA expression and decreased circulating human PCSK9 protein levels.

[0997] The present study examined the dose response to BMS-844421 in the hemizygous genomic transgenic mouse model on a normal chow diet. Experiments were conducted comparing the effects of BMS-844421 (human specific PCSK9 ASO) to those of ISIS 394816 (murine specific PCSK9 ASO) in this model alone and in combination. BMS-844421 exhibited dose-dependent suppression of liver target mRNA, reflected in dose-dependent decreases in plasma human PCSK9 levels. Both liver mRNA and plasma protein gave ED50 values of approximately 15 mg/kg (given twice weekly) for BMS-844421, while the suppression of plasma PCSK9 protein was greater than liver PCSK9 mRNA at each dose in this model. Trends for increased LDLR protein in liver total membrane fraction, and a feedback effect on liver HMGCoA reductase mRNA expression were observed, but no changes in plasma lipids were seen for BMS-844421 treated hemizygous mice. The combination of ISIS 394816+BMS-844421 (each at 15 mg/kg) elicted a greater increase in liver LDLR protein than either ASO alone, consistent with the proposed mechanism of action and the dual expression of both human and murine PCSK9 in this model.

18.1. Materials and Methods

[0998] The sequences of BMS-844421 and other ASOs tested in this report are given in Table 35. The negative control ISIS 141923 is not complementary to any known gene. The design of the dose response study is described in the beginning of the Results section (section 18.2. below).

TABLE 35

Nucleotide sequences of ASOs

ASO	Sequence	SEQ ID NO
BMS 844421, human PCSK9 specific*	5'-CCTTGGCCACGCCGGCATCC-3'	248
ISIS 141923, non-targeting	5'-CCTTCCCTGAAGGTTCCTCC-3'	465
ISIS 394816, murine PCSK9 specific	5'-GGGCTCATAGCACATTATCC-3'	466

*BMS-844421 is identical to ISIS 405879

[0999] Parameters evaluated at study termination included hepatic mRNA, hepatic LDLR protein levels, plasma concentrations of total cholesterol, LDL, HDL, triglycerides, transaminases, and organ and body weights. Whole liver was processed for RNA, and protein, and held for possible histological examination and hepatic triglyceride levels. Blood and liver samples were obtained 24 hr after the prior dose. Blood samples were obtained from tail vein or retro orbital plexus and EDTA, chilled on ice, and plasma samples were obtained by standard procedures. For mRNA assays, liver samples (30-50 mg each) were placed in eppendorf tubes containing 1000 uL of TRIzol (Invitrogen #15596-018) and homogenized using stainless steel beads on a Tissue Lyser instrument (Qiagen, Valencia, Calif.). Chloroform (200 μL) was added to each tube, the contents were mixed vigorously by hand and tubes were centrifuged at 12,000×g for 20 minutes at 4° C. About 300 μL of each supernatant was removed and placed in 1.5 mL microcentrifuge tubes. One volume of 70% EtOH was added, the mixtures were vortexed thoroughly and transferred to RNeasy Mini Spin Columns (QIAgen Cat. No. 74106). RNA was purified, treated with DNase and eluted into microcentrifuge tubes following QIAgen's manufacturer's instructions.

[1000] Plasma human PCSK9 levels were assayed using the human specific ELISA with mAb 4H5 as capture antibody.

18.2. Results

[1001] A dose-response study was run in human PCSK9 genomic transgenic mouse line 66 mice on a normal chow diet. Groups of 6 mice (3 males, 3 females) received the following treatments for 6 weeks. BMS-844421 was administered p.o. at 5, 10, 15, and 30 mg/kg per dose given twice weekly (on Tuesdays and Fridays) for 6 weeks. Controls received saline, and an additional group received the non-targeting negative control ASO ISIS 141923 (30 mg/kg twice weekly). The murine PCSK9 ASO ISIS 394816 was administered to another group at 15 mg/kg per dose. The final group received BMS-844421 (15 mg/kg) plus ISIS 394816 (15 mg/kg) combined.

[1002] Liver mRNA assayed by qRT-PCR showed that BMS-844421 decreased liver PCSK9 mRNA in a dose dependent manner, with greatest effect at the top dose studied (FIG. **14**). The negative control ASO, ISIS 141923, had no significant effect on target mRNA, while the murine targeted ASO, ISIS 394816, decreased human PCSK9 mRNA by 33%, suggesting partial cross-hybridization to human PCSK9 mRNA.

[1003] The ASO effects on plasma human PCSK9 followed a similar pattern as mRNA, while the effects of the higher doses of BMS-844421 resulted in greater effects, up to 70% suppression at 30 mg/kg/dose (FIG. **15**). ISIS 394816 suppressed approximately 30% (similar to mRNA effects), while ISIS 141923 had no significant effect.

[1004] Plasma human PCSK9 levels exhibited time dependent suppression with effects at 6 weeks, greater than at 3 wks for all doses of BMS-844421 (FIG. **16**). At 3 wks dosing, only the highest dose of BMS-844421 showed significant suppression. As seen in FIG. **17**, data for 6 wks treatment with BMS-844421 showed a sigmoidal dose-response curve, with ED50 approximately 15 mg/kg/dose (given twice weekly). Plotting the mRNA and protein data together showed that the two endpoints generally, though imperfectly, reflect each other, with protein showing greater sensitivity to suppression (FIG. **18**).

[1005] Male transgenic mice responded more sensitively to BMS-844421 than the female transgenic mice in this study (FIG. **19**). The data was not the result of different plasma baseline levels of human PCSK9 between males and females, which is human PCSK9=67+/−31 ng/mL in males and 60+/−27 ng/mL in females in the predosing samples. The difference in PCSK9 protein response for males vs. females was less apparent for liver target mRNA (FIG. **20**).

[1006] Levels of endogenous mouse PCSK9 mRNA in liver were decreased by ISIS 394816 (and the combination of ISIS 394816 with BMS-844421), but were not decreased by BMS-844421 only (FIG. **21**).

[1007] A key cholesterol regulatory gene, HMGCoA reductase (HMGR), is known to respond to sterol feedback in the liver. An assay of endogenous mouse liver HMGR mRNA revealed 45% suppression of HMGR with BMS-844421 as well as ISIS 394816 (FIG. **22**). These findings are consistent with downregulation of PCSK9 promoting higher levels of LDLR, in turn leading to increased cholesterol in the regulatory pool in liver cells, resulting in downregulation of HMGR. 5 Similar, though quantitatively lesser effects, were also seen for the sterol pathway genes LDLR mRNA and for SREBP2 mRNA.

[1008] LDLR protein levels were assayed in liver samples from the same study. Total hepatic membranes were prepared, and western blots of SDS-PAGE gels were assayed with anti-LDLR antibodies. As a control, transferrin receptor (TR) protein levels were measured with anti-TR antibodies. Liver samples from the ASO study were pooled from n=3 for the preparation of the membrane fraction. Pairs of pools for each group were assayed. Treatment with BMS-844421, and especially the combination of BMS-844421 with the murine ASO ISIS 394816, led to increases up to >2-fold in levels of LDLR protein, compared to the level of TR protein (FIG. **23** and FIG. **24**).

[1009] The ELISA assay uses a monoclonal capture antibody and a polyclonal detector antibody developed against human recombinant PCSK9 produced in baculovirus. The recombinant human PCSK9 protein also serves as standard antigen for the ELISA assay. Cynomolgus monkey plasma PCSK9 was readily detectable using this assay. Human PCSK9 protein in plasma of human PCSK9 transgenic mouse lines was specifically detected with no cross-reactivity to murine PCSK9.

[1010] Despite these pharmacodynamic responses to BMS-844421, hemizygous transgenic mice did not exhibit changes in plasma LDL-C or total cholesterol following ASO treatment, including BMS-844421 (Table 36). However, normal chow fed mice have low levels of LDL-C (Table 36). It is plausible that these low LDL-C levels are not readily decreased by increases in LDLR protein. Another possible explanation is that the hemizygous expression of human PCSK9, i.e., the gene specifically targeted by BMS-844421, contributes only about ⅓ to ½ of the total murine+human PCSK9 in this hemizygous transgenic mouse model.

TABLE 36

Plasma lipids at 6 wk treatment with ASOs in transgenic mice

Both Sexes	Triglycerides	Total Cholesterol	HDL	LDL
Saline	72 ± 14	108 ± 8.2	57 ± 6.4	14 ± 1
Missense	64 ± 9.2	110 ± 8.6	57 ± 7.1	15 ± 1
BMS-844421 (30 mg/kg)	54 ± 8.6	129 ± 7.3	67 ± 4.6	18 ± 2
BMS-844421 (15 mg/kg)	54 ± 9.0	126 ± 9.2	65 ± 5.5	18 ± 1
BMS-844421 (10 mg/kg)	53 ± 9.4	118 ± 9.3	61 ± 6.9	17 ± 2
BMS-844421 (5 mg/kg)	56 ± 9.7	117 ± 8.5	61 ± 6.1	17 ± 1
Mouse ASO I-394816 (15 mg/kg)	50 ± 7.0	84.2 ± 2.9	45 ± 3.0	11 ± 2
BMS-844421 + I-394816 (15 mg/kg each)	55 ± 9.3	92 ± 5.3	48 ± 4.6	12 ± 2
WT Littermates	60 ± 8.1	111.7 ± 7.2	57 ± 5.3	15 ± 1

[1011] There were no changes in ALT or AST (not shown), suggesting that no liver abnormalities were present after 6 wks treatment.

18.3. Conclusions

[1012] The recently established human genomic PCSK9 transgenic mouse model responded to human PCSK9 ASO treatment with robust downregulation of human PCSK9 at both the mRNA and protein levels in liver. Plasma human PCSK9 was also decreased with the magnitude of decrease somewhat greater than liver mRNA suppression, suggesting that plasma PCSK9 levels are a good readout for suppression of the target mRNA. The ED50 for both endpoints was approximately 15 mg/kg (per dose given twice weekly). The transgenic mice did not exhibit significant changes in plasma LDL-C or total cholesterol following BMS-844421 or ISIS 394816 treatment, nor for the combination of the two ASOs. A trend for increasing liver LDLR protein was observed following treatment with BMS-844421, suggesting a functional effect consistent with the downregulation of PCSK9 mRNA and protein. Consistent with this was the downregulation of the sterol response gene HMGR in liver.

[1013] The limited pharmacodynamic responses in this model appear to be related both to the small LDL window in normal chow diet fed mice, and to the co-expression of the human PCSK9 gene with the endogenous murine PCSK9 gene. In the hemizygous mice, human PCSK9 appears to represent only about ⅓ of the total human+murine PCSK9 expressed. Consistent with this, in a previous study the murine specific PCSK9 ASO ISIS 394816 administered to human apoB-CETP double transgenic mice produced up to 90% suppression of liver PCSK9 mRNA and a corresponding decrease of up to 80% in plasma LDL-C levels.

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tgcccctggc gggtgggtac agccgcgtcc tcaacgccgc ctgccagcgc ctggcgaggg  1740
ctggggtcgt gctggtcacc gctgccggca acttccggga cgatgcctgc ctctactccc  1800
cagcctcagc tcccgagggg aggacatcat tggtgcctcc agcgactgca gcacctgctt  1860
tgtgtcacag agtgggacat cacaggctgc tgcccacgtg gctggcattg cagccatgat  1920
gctgtctgcc gagccggagc tcaccctggc cgagttgagg cagagactga tccacttctc  1980
tgccaaagat gtcatcaatg aggcctggtt ccctgaggac cagcgggtac tgacccccaa  2040
cctggtggcc gccctgcccc ccagcaccca tggggcaggt tggcagctgt tttgcaggac  2100
tgtgtggtca gcacactcgg ggcctacacg gatggccaca gccatcgccc gctgcgcccc  2160
agatgaggag ctgctgagct gctccagttt ctccaggagt gggaagcggc ggggcgagcg  2220
catggaggct gcagctccca ctgggaggtg gaggaccttg gcacccacaa gccgcctgtg  2280
ctgaggccac gaggtcagcc caaccagtgc gtgggccaca gggaggccag catccacgct  2340
tcctgctgcc atgccccagg tctggaatgc aaagtcaagg agcatggaat cccggcccct  2400
caggagcagg tgaccgtggc ctgcgaggag ggctggaccc tgactggctg cagtgccctc  2460
```

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cctgggacct cccacgtcct gggggcctac gccgtagaca acacgtgtgt agtcaggagc  2520
cgggacgtca gcactacagg cagcaccagc gaagaggccg tgacagccgt tgccatctgc  2580
tgccggagcc ggcacctggc gcaggcctcc caggagctcc agtgacagcc ccatcccagg  2640
atgggtgtct ggggagggtc aagggctggg gctgagcttt aaaatggttc cgacttgtcc  2700
ctctctcagc cctccatggc ctggcacgag gggatgggga tgcttccgcc tttccggggc  2760
tgctggcctg gcccttgagt ggggcagcct ccttgcctgg aactcactca ctctgggtgc  2820
ctcctcccca ggtggaggtg ccaggaagct ccctccctca ctgtggggca tttcaccatt  2880
caaacaggtc gagctgtgct cgggtgctgc cagctgctcc caatgtgccg atgtccgtgg  2940
gcagaatgac ttttattgag ctcttgttcc gtgccaggca ttcaatcctc aggtctccac  3000
caaggaggca ggattcttcc catggatagg ggagggggcg gtaggggctg cagggacaaa  3060
catcgttggg gggtgagtgt gaaaggtgct gatggccctc atctccagct aactgtggag  3120
aagcccctgg gggctccctg attaatggag gcttagcttt ctggatggca tctagccaga  3180
ggctggagac aggtgtgccc ctggtggtca caggctgtgc cttggtttcc tgagccacct  3240
ttactctgct ctatgccagg ctgtgctagc aacacccaaa ggtggcctgc ggggagccat  3300
cacctaggac tgactcggca gtgtgcagtg gtgcatgcac tgtctcagcc aacccgctcc  3360
actacccggc agggtacaca ttcgcacccc tacttcacag aggaagaaac ctggaaccag  3420
agggggcgtg cctgccaagc tcacacagca ggaactgagc cagaaacgca gattgggctg  3480
gctctgaagc caagcctctt cttacttcac ccggctgggc tcctcatttt tacgggtaac  3540
agtgaggctg ggaaggggaa cacagaccag gaagctcggt gagtgatggc agaacgatgc  3600
ctgcaggcat ggaacttttt ccgttatcac ccaggcctga ttcactggcc tggcggagat  3660
gcttctaagg catggtcggg ggagagggcc aacaactgtc cctccttgag caccagcccc  3720
acccaagcaa gcagacattt atcttttggg tctgtcctct ctgttgcctt tttacagcca  3780
acttttctag acctgtttg ctttgtaac ttgaagatat ttattctggg ttttgtagca  3840
tttttattaa tatggtgact ttttaaaata aaaacaaaca aacgttgtcc t           3891
```

```
SEQ ID NO: 4            moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 4
gcgcggaatc ctggctggga                                          20

SEQ ID NO: 5            moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 5
gaggagacct agaggccgtg                                          20

SEQ ID NO: 6            moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 6
aggaccgcct ggagctgacg                                          20

SEQ ID NO: 7            moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 7
acgcaaggct agcaccagct                                          20

SEQ ID NO: 8            moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 8
cctcggaacg caaggctagc                                          20

SEQ ID NO: 9            moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 9
ccttggcgca gcggtggaag                                               20

SEQ ID NO: 10           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 10
ggcgggcagt gcgctctgac                                               20

SEQ ID NO: 11           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 11
tggtgaggta tccccggcgg                                               20

SEQ ID NO: 12           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 12
ggatcttggt gaggtatccc                                               20

SEQ ID NO: 13           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 13
agacatgcag gatcttggtg                                               20

SEQ ID NO: 14           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 14
atggaagaca tgcaggatct                                               20

SEQ ID NO: 15           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 15
ttcaccagga agccaggaag                                               20

SEQ ID NO: 16           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 16
tcatcttcac caggaagcca                                                20

SEQ ID NO: 17           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 17
cctcgatgta gtcgacatgg                                                20

SEQ ID NO: 18           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 18
gtattcatcc gcccggtacc                                                20

SEQ ID NO: 19           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 19
ctggtgtcta ggagatacac                                                20

SEQ ID NO: 20           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 20
gtatgctggt gtctaggaga                                                20

SEQ ID NO: 21           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 21
gatttcccgg tggtcactct                                                20

SEQ ID NO: 22           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 22
gccctcgatt tcccggtggt                                                20

SEQ ID NO: 23           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 23
gtgaccatga ccctgccctc                                                20

SEQ ID NO: 24           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 24
ctcgaagtcg gtgaccatga                                          20

SEQ ID NO: 25           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 25
gtggaagcgg gtcccgtcct                                          20

SEQ ID NO: 26           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 26
acccctgcca ggtgggtgcc                                          20

SEQ ID NO: 27           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 27
tgaccacccc tgccaggtgg                                          20

SEQ ID NO: 28           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 28
acccttggcc acgccggcat                                          20

SEQ ID NO: 29           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 29
ttggcagttg agcacgcgca                                          20

SEQ ID NO: 30           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 30
ttcccttggc agttgagcac                                          20

SEQ ID NO: 31           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 31
ccaggcctat gagggtgccg                                          20
```

-continued

```
SEQ ID NO: 32           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 32
gctggctttt ccgaataaac                                          20

SEQ ID NO: 33           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 33
gtgaccagca cgaccccagc                                          20

SEQ ID NO: 34           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 34
ccaaagtccc cagggtcacc                                          20

SEQ ID NO: 35           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 35
agttggtccc caaagtcccc                                          20

SEQ ID NO: 36           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 36
gccaaagttg gtccccaaag                                          20

SEQ ID NO: 37           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 37
gtccacacag cggccaaagt                                          20

SEQ ID NO: 38           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 38
ggcaaagagg tccacacagc                                          20

SEQ ID NO: 39           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 39
tggaggcacc aatgatgtcc                                             20

SEQ ID NO: 40           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 40
gctgcagtcg ctggaggcac                                             20

SEQ ID NO: 41           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 41
acaaagcagg tgctgcagtc                                             20

SEQ ID NO: 42           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 42
gcatcatggc tgcaatgcca                                             20

SEQ ID NO: 43           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 43
ggcagacagc atcatggctg                                             20

SEQ ID NO: 44           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 44
gaagtggatc agtctctgcc                                             20

SEQ ID NO: 45           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 45
ttggcagaga agtggatcag                                             20

SEQ ID NO: 46           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 46
catctttggc agagaagtgg                                             20

SEQ ID NO: 47           moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 47
tgatgacatc tttggcagag                                                20

SEQ ID NO: 48           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 48
gcctcattga tgacatcttt                                                20

SEQ ID NO: 49           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 49
tcagggaacc aggcctcatt                                                20

SEQ ID NO: 50           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 50
ggtcctcagg gaaccaggcc                                                20

SEQ ID NO: 51           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 51
ggtcagtacc cgctggtcct                                                20

SEQ ID NO: 52           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 52
aaacagctgc caacctgccc                                                20

SEQ ID NO: 53           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 53
ctgcaaaaca gctgccaacc                                                20

SEQ ID NO: 54           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 54
gtaggccccg agtgtgctga                                              20

SEQ ID NO: 55           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 55
agaaactgga gcagctcagc                                              20

SEQ ID NO: 56           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 56
tcctggagaa actggagcag                                              20

SEQ ID NO: 57           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 57
cgttgtgggc ccggcagacc                                              20

SEQ ID NO: 58           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 58
agcaggcagc acctggcaat                                              20

SEQ ID NO: 59           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 59
cacgggtccc catgctggcc                                              20

SEQ ID NO: 60           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 60
ctttgcattc cagacctggg                                              20

SEQ ID NO: 61           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 61
cttgactttg cattccagac                                              20

SEQ ID NO: 62           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 62
ggcagcagat ggcaacggct                                              20

SEQ ID NO: 63           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 63
aaccatttta aagctcagcc                                              20

SEQ ID NO: 64           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 64
tcaagggcca ggccagcagc                                              20

SEQ ID NO: 65           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 65
cccactcaag ggccaggcca                                              20

SEQ ID NO: 66           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 66
atgccccaca gtgagggagg                                              20

SEQ ID NO: 67           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 67
aatggtgaaa tgccccacag                                              20

SEQ ID NO: 68           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 68
catgggaaga atcctgcctc                                              20

SEQ ID NO: 69           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 69
ggagatgagg gccatcagca                                              20
```

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```
SEQ ID NO: 70           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 70
tagatgccat ccagaaagct                                              20

SEQ ID NO: 71           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 71
tctggctaga tgccatccag                                              20

SEQ ID NO: 72           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 72
ggcatagagc agagtaaagg                                              20

SEQ ID NO: 73           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 73
agcctggcat agagcagagt                                              20

SEQ ID NO: 74           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 74
aagtaagaag aggcttggct                                              20

SEQ ID NO: 75           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 75
gctcaaggag ggacagttgt                                              20

SEQ ID NO: 76           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 76
aaagataaat gtctgcttgc                                              20

SEQ ID NO: 77           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 77
acccaaaaga taaatgtctg                                                   20

SEQ ID NO: 78            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 78
tcttcaagtt acaaaagcaa                                                   20

SEQ ID NO: 79            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 79
aaatgcaggg ctaaaatcac                                                   20

SEQ ID NO: 80            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 80
aacccagttc taatgcacct                                                   20

SEQ ID NO: 81            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 81
aagcagggcc tcaggtggaa                                                   20

SEQ ID NO: 82            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 82
aaggaaaggg aggcctagag                                                   20

SEQ ID NO: 83            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 83
aaggaagact tcaggcaggt                                                   20

SEQ ID NO: 84            moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 84
aaggtcacac agttaagagt                                                   20

SEQ ID NO: 85            moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 85
acaaattccc agactcagca                                                 20

SEQ ID NO: 86          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 86
acagcattct tggttaggag                                                 20

SEQ ID NO: 87          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 87
acccgctggt cctcagggaa                                                 20

SEQ ID NO: 88          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 88
actggataca ttggcagaca                                                 20

SEQ ID NO: 89          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 89
agaaccatgg agcacctgag                                                 20

SEQ ID NO: 90          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 90
agactaggag cctgagtttt                                                 20

SEQ ID NO: 91          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 91
agactgatgg aaggcattga                                                 20

SEQ ID NO: 92          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 92
agagacagga agctgcagct                                              20

SEQ ID NO: 93           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 93
agagaggagg gcttaaagaa                                              20

SEQ ID NO: 94           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 94
agcatggcac cagcatctgc                                              20

SEQ ID NO: 95           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 95
agctgccaac ctgcaaaaag                                              20

SEQ ID NO: 96           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 96
aggacccaag tcatcctgct                                              20

SEQ ID NO: 97           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 97
agtcaagctg ctgcccagag                                              20

SEQ ID NO: 98           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 98
agtgtaaaat aaagccccta                                              20

SEQ ID NO: 99           moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 99
ataaatatct tcaagttaca                                              20

SEQ ID NO: 100          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 100
atctcaggac aggtgagcaa                                                20

SEQ ID NO: 101          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 101
atggctgcaa tgccagccac                                                20

SEQ ID NO: 102          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 102
atgtgcagag atcaatcaca                                                20

SEQ ID NO: 103          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 103
atttcataga caaggaaagg                                                20

SEQ ID NO: 104          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 104
cacagtcctg caaaacagct                                                20

SEQ ID NO: 105          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 105
cacattagcc ttgctcaagt                                                20

SEQ ID NO: 106          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 106
ccagtcagag tagaacagag                                                20

SEQ ID NO: 107          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 107
cccactataa tggcaagccc                                                20
```

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```
SEQ ID NO: 108          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 108
cccagcccta tcaggaagtg                                              20

SEQ ID NO: 109          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 109
cccctgcaca gagcctggca                                              20

SEQ ID NO: 110          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 110
cctggaaccc ctgcagccag                                              20

SEQ ID NO: 111          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 111
ctagaggaac cactagatat                                              20

SEQ ID NO: 112          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 112
gaagaggctt ggcttcagag                                              20

SEQ ID NO: 113          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 113
gaataacagt gatgtctggc                                              20

SEQ ID NO: 114          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 114
gagaggttca gatccaggcc                                              20

SEQ ID NO: 115          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 115
gagtaaggca ggttactctc                                            20

SEQ ID NO: 116            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 116
gagtagagat tctcatctca                                            20

SEQ ID NO: 117            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 117
gagtcttctg aagtgccatc                                            20

SEQ ID NO: 118            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 118
gcaccatcca gaccagaatc                                            20

SEQ ID NO: 119            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 119
gcagggcggc caccaggttg                                            20

SEQ ID NO: 120            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 120
gctagttatt aagcacctgc                                            20

SEQ ID NO: 121            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 121
ggagcctaca tgtgcagaga                                            20

SEQ ID NO: 122            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic oligonucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 122
ggagggagct tcctggcacc                                            20

SEQ ID NO: 123            moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 123
ggcactgccc ttccaccaaa                                                20

SEQ ID NO: 124          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 124
ggccatcagc tggcaatgct                                                20

SEQ ID NO: 125          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 125
ggcctgcaga agccagagag                                                20

SEQ ID NO: 126          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 126
ggtgcataag gagaaagaga                                                20

SEQ ID NO: 127          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 127
ggtggtaatt tgtcacagca                                                20

SEQ ID NO: 128          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 128
gtccccaaag tccccagggt                                                20

SEQ ID NO: 129          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 129
gtgccatctg aacagcacct                                                20

SEQ ID NO: 130          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 130
gtgctgacca cacagtcctg                                             20

SEQ ID NO: 131          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 131
gtgttgagca gactgatgga                                             20

SEQ ID NO: 132          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 132
taaaagctgc aagagactca                                             20

SEQ ID NO: 133          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 133
tagacaagga aagggaggcc                                             20

SEQ ID NO: 134          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 134
tagatgtgac taacatttaa                                             20

SEQ ID NO: 135          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 135
tagcacagcc tggcatagag                                             20

SEQ ID NO: 136          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 136
taggagaaag tagggagagc                                             20

SEQ ID NO: 137          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 137
tagggagagc tcacagatgc                                             20

SEQ ID NO: 138          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 138
tcacagctca ccgagtctgc                                              20

SEQ ID NO: 139          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 139
tcagagaaaa cagtcaccga                                              20

SEQ ID NO: 140          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 140
tcagccaggc caaaggaaga                                              20

SEQ ID NO: 141          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 141
tcatggctgc aatgcctggt                                              20

SEQ ID NO: 142          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 142
tcattttaga gacaggaagc                                              20

SEQ ID NO: 143          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 143
tgaaaatcca tccagcactg                                              20

SEQ ID NO: 144          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 144
tgacatccag gagggaggag                                              20

SEQ ID NO: 145          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 145
tgacatcttg tctgggagcc                                              20
```

-continued

```
SEQ ID NO: 146          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 146
tgacatttgt gggagaggag                                              20

SEQ ID NO: 147          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 147
tgagttcatt taagagtgga                                              20

SEQ ID NO: 148          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 148
tggcagcaac tcagacatat                                              20

SEQ ID NO: 149          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 149
tgggcattgg tggccccaac                                              20

SEQ ID NO: 150          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 150
tgtgagctct ggcccagtgg                                              20

SEQ ID NO: 151          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 151
tgtgatgacc tggaaaggtg                                              20

SEQ ID NO: 152          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 152
ttcaggcagg ttgctgctag                                              20

SEQ ID NO: 153          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 153
ttgggagcag ctggcagcac                                                20

SEQ ID NO: 154          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 154
ttttaaagct cagccccagc                                                20

SEQ ID NO: 155          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 155
cttatagtta acacacagaa                                                20

SEQ ID NO: 156          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 156
aagtcaacct tatagttaac                                                20

SEQ ID NO: 157          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 157
aggaacaaag ccaaggtcac                                                20

SEQ ID NO: 158          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 158
gtggtgactt accagccacg                                                20

SEQ ID NO: 159          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 159
gcctggagct gacggtgccc                                                20

SEQ ID NO: 160          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 160
gaccgcctgg agctgacggt                                                20

SEQ ID NO: 161          moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 161
tccaaggtga catttgtggg                                              20

SEQ ID NO: 162          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 162
aaggctagca ccagctcctc                                              20

SEQ ID NO: 163          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 163
caaggctagc accagctcct                                              20

SEQ ID NO: 164          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 164
gcaaggctag caccagctcc                                              20

SEQ ID NO: 165          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 165
cgcaaggcta gcaccagctc                                              20

SEQ ID NO: 166          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 166
aacgcaaggc tagcaccagc                                              20

SEQ ID NO: 167          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 167
gaacgcaagg ctagcaccag                                              20

SEQ ID NO: 168          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 168
ggaacgcaag gctagcacca                                          20

SEQ ID NO: 169          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 169
cggaacgcaa ggctagcacc                                          20

SEQ ID NO: 170          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 170
tcctcctcgg aacgcaaggc                                          20

SEQ ID NO: 171          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 171
gtgctcgggt gcttcggcca                                          20

SEQ ID NO: 172          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 172
tggaaggtgg ctgtggttcc                                          20

SEQ ID NO: 173          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 173
atccttggcg cagcggtgga                                          20

SEQ ID NO: 174          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 174
ggatccttgg cgcagcggtg                                          20

SEQ ID NO: 175          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 175
ccacggatcc ttggcgcagc                                          20

SEQ ID NO: 176          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 176
cgtaggtgcc aggcaacctc                                              20

SEQ ID NO: 177          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 177
tgactgcgag aggtgggtct                                              20

SEQ ID NO: 178          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 178
cagtgcgctc tgactgcgag                                              20

SEQ ID NO: 179          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 179
ggcagtgcgc tctgactgcg                                              20

SEQ ID NO: 180          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 180
cgggcagtgc gctctgactg                                              20

SEQ ID NO: 181          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 181
cggcgggcag tgcgctctga                                              20

SEQ ID NO: 182          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 182
atccccggcg ggcagcctgg                                              20

SEQ ID NO: 183          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 183
aggtatcccc ggcgggcagc                                              20
```

-continued

```
SEQ ID NO: 184          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 184
tgaggtatcc ccggcgggca                                        20

SEQ ID NO: 185          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 185
gtgaggtatc cccggcgggc                                        20

SEQ ID NO: 186          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 186
ggtgaggtat ccccggcggg                                        20

SEQ ID NO: 187          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 187
ttggtgaggt atccccggcg                                        20

SEQ ID NO: 188          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 188
cttggtgagg tatccccggc                                        20

SEQ ID NO: 189          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 189
tcttggtgag gtatccccgg                                        20

SEQ ID NO: 190          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 190
atcttggtga ggtatccccg                                        20

SEQ ID NO: 191          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 191
gatcttggtg aggtatcccc                                                 20

SEQ ID NO: 192          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 192
aggatcttgg tgaggtatcc                                                 20

SEQ ID NO: 193          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 193
gcaggatctt ggtgaggtat                                                 20

SEQ ID NO: 194          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 194
atgcaggatc ttggtgaggt                                                 20

SEQ ID NO: 195          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 195
acatgcagga tcttggtgag                                                 20

SEQ ID NO: 196          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 196
gaagacatgc aggatcttgg                                                 20

SEQ ID NO: 197          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 197
agaaggccat ggaagacatg                                                 20

SEQ ID NO: 198          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 198
gaagccagga agaaggccat                                                 20

SEQ ID NO: 199          moltype = DNA  length = 20
```

-continued

```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 199
tcttcaccag gaagccagga                                              20

SEQ ID NO: 200         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 200
actcatcttc accaggaagc                                              20

SEQ ID NO: 201         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 201
ccactcatct tcaccaggaa                                              20

SEQ ID NO: 202         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 202
cgccactcat cttcaccagg                                              20

SEQ ID NO: 203         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 203
gtcgccactc atcttcacca                                              20

SEQ ID NO: 204         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 204
aggtcgccac tcatcttcac                                              20

SEQ ID NO: 205         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 205
gcaggtcgcc actcatcttc                                              20

SEQ ID NO: 206         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 206
cagcaggtcg ccactcatct                                              20

SEQ ID NO: 207          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 207
tccagcaggt cgccactcat                                              20

SEQ ID NO: 208          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 208
ggcaacttca aggccagctc                                              20

SEQ ID NO: 209          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 209
gcaaagacag aggagtcctc                                              20

SEQ ID NO: 210          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 210
ttcatccgcc cggtaccgtg                                              20

SEQ ID NO: 211          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 211
tattcatccg cccggtaccg                                              20

SEQ ID NO: 212          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 212
tggtattcat ccgcccggta                                              20

SEQ ID NO: 213          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 213
gctggtattc atccgcccgg                                              20

SEQ ID NO: 214          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 214
gggctggtat tcatccgccc                                                20

SEQ ID NO: 215        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 215
atacacctcc accaggctgc                                                20

SEQ ID NO: 216        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 216
gtgtctagga gatacacctc                                                20

SEQ ID NO: 217        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 217
tgctggtgtc taggagatac                                                20

SEQ ID NO: 218        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 218
tcgatttccc ggtggtcact                                                20

SEQ ID NO: 219        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 219
ctcgatttcc cggtggtcac                                                20

SEQ ID NO: 220        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 220
cctcgatttc ccggtggtca                                                20

SEQ ID NO: 221        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 221
ccctcgattt cccggtggtc                                                20
```

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```
SEQ ID NO: 222          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 222
tgccctcgat ttcccggtgg                                              20

SEQ ID NO: 223          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 223
ctgccctcga tttcccggtg                                              20

SEQ ID NO: 224          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 224
cctgccctcg atttcccggt                                              20

SEQ ID NO: 225          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 225
ccctgccctc gatttcccgg                                              20

SEQ ID NO: 226          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 226
atgaccctgc cctcgatttc                                              20

SEQ ID NO: 227          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 227
ccatgaccct gccctcgatt                                              20

SEQ ID NO: 228          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 228
gaccatgacc ctgccctcga                                              20

SEQ ID NO: 229          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 229
cggtgaccat gaccctgccc                                              20

SEQ ID NO: 230           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 230
gtcggtgacc atgaccctgc                                              20

SEQ ID NO: 231           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 231
aagtcggtga ccatgaccct                                              20

SEQ ID NO: 232           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 232
cgaagtcggt gaccatgacc                                              20

SEQ ID NO: 233           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 233
ggcacattct cgaagtcggt                                              20

SEQ ID NO: 234           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 234
tggcctgtct gtggaagcgg                                              20

SEQ ID NO: 235           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 235
ggtgggtgcc atgactgtca                                              20

SEQ ID NO: 236           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 236
ccaggtgggt gccatgactg                                              20

SEQ ID NO: 237           moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 237
cctgccaggt gggtgccatg                                        20

SEQ ID NO: 238         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 238
ccacccctgc caggtgggtg                                        20

SEQ ID NO: 239         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 239
gctgaccacc cctgccaggt                                        20

SEQ ID NO: 240         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 240
tcccggccgc tgaccacccc                                        20

SEQ ID NO: 241         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 241
ggcatcccgg ccgctgacca                                        20

SEQ ID NO: 242         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 242
gccggcatcc cggccgctga                                        20

SEQ ID NO: 243         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 243
gccacgccgg catcccggcc                                        20

SEQ ID NO: 244         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

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```
SEQUENCE: 244
ggccacgccg gcatcccggc                                              20

SEQ ID NO: 245          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 245
tggccacgcc ggcatcccgg                                              20

SEQ ID NO: 246          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 246
ttggccacgc cggcatcccg                                              20

SEQ ID NO: 247          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 247
cttggccacg ccggcatccc                                              20

SEQ ID NO: 248          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 248
ccttggccac gccggcatcc                                              20

SEQ ID NO: 249          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 249
cccttggcca cgccggcatc                                              20

SEQ ID NO: 250          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 250
caccccttggc cacgccggca                                             20

SEQ ID NO: 251          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 251
gcacccttgg ccacgccggc                                              20

SEQ ID NO: 252          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 252
ggcacccttg gccacgccgg                                               20

SEQ ID NO: 253          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 253
tggcaccctt ggccacgccg                                               20

SEQ ID NO: 254          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 254
ctggcaccct tggccacgcc                                               20

SEQ ID NO: 255          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 255
gctggcaccc ttggccacgc                                               20

SEQ ID NO: 256          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 256
cgcatgctgg caccc ttggc                                               20

SEQ ID NO: 257          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 257
cagttgagca cgcgcaggct                                               20

SEQ ID NO: 258          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 258
ggcagttgag cacgcgcagg                                               20

SEQ ID NO: 259          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 259
ccttggcagt tgagcacgcg                                               20
```

-continued

```
SEQ ID NO: 260          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 260
ccttcccttg gcagttgagc                                              20

SEQ ID NO: 261          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 261
gtgcccttcc cttggcagtt                                              20

SEQ ID NO: 262          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 262
ccgtgccctt cccttggcag                                              20

SEQ ID NO: 263          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 263
ggtgccgcta accgtgccct                                              20

SEQ ID NO: 264          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 264
caggcctatg agggtgccgc                                              20

SEQ ID NO: 265          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 265
ccgaataaac tccaggccta                                              20

SEQ ID NO: 266          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 266
tggcttttcc gaataaactc                                              20

SEQ ID NO: 267          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 267
cagctggctt ttccgaataa                                            20

SEQ ID NO: 268          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 268
accagctggc ttttccgaat                                            20

SEQ ID NO: 269          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 269
ggaccagctg gcttttccga                                            20

SEQ ID NO: 270          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 270
ggctggacca gctggctttt                                            20

SEQ ID NO: 271          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 271
gtggccccac aggctggacc                                            20

SEQ ID NO: 272          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 272
agcagcacca ccagtggccc                                            20

SEQ ID NO: 273          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 273
gctgtaccca cccgccaggg                                            20

SEQ ID NO: 274          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 274
cgaccccagc cctcgccagg                                            20

SEQ ID NO: 275          moltype = DNA  length = 20
```

-continued

```
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 275
cggtgaccag cacgacccca                                                  20

SEQ ID NO: 276           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 276
agcggtgacc agcacgaccc                                                  20

SEQ ID NO: 277           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 277
gcagcggtga ccagcacgac                                                  20

SEQ ID NO: 278           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 278
cggcagcggt gaccagcacg                                                  20

SEQ ID NO: 279           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 279
ccggcagcgg tgaccagcac                                                  20

SEQ ID NO: 280           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 280
ttgccggcag cggtgaccag                                                  20

SEQ ID NO: 281           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 281
agttgccggc agcggtgacc                                                  20

SEQ ID NO: 282           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
```

-continued

```
SEQUENCE: 282
gaagttgccg gcagcggtga                                                20

SEQ ID NO: 283          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 283
cggaagttgc cggcagcggt                                                20

SEQ ID NO: 284          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 284
cccggaagtt gccggcagcg                                                20

SEQ ID NO: 285          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 285
gtcccggaag ttgccggcag                                                20

SEQ ID NO: 286          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 286
atgacctcgg gagctgaggc                                                20

SEQ ID NO: 287          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 287
cccaactgtg atgacctcgg                                                20

SEQ ID NO: 288          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 288
ggcattggtg gccccaactg                                                20

SEQ ID NO: 289          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 289
gctggtcttg ggcattggtg                                                20

SEQ ID NO: 290          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 290
cccagggtca ccggctggtc                                              20

SEQ ID NO: 291          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 291
tccccagggt caccggctgg                                              20

SEQ ID NO: 292          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 292
agtccccagg gtcaccggct                                              20

SEQ ID NO: 293          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 293
aaagtcccca gggtcaccgg                                              20

SEQ ID NO: 294          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 294
ccccaaagtc cccagggtca                                              20

SEQ ID NO: 295          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 295
ttggtcccca aagtccccag                                              20

SEQ ID NO: 296          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 296
aaagttggtc cccaaagtcc                                              20

SEQ ID NO: 297          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 297
cggccaaagt tggtccccaa                                              20
```

-continued

```
SEQ ID NO: 298          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 298
agcggccaaa gttggtcccc                                                    20

SEQ ID NO: 299          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 299
acagcggcca aagttggtcc                                                    20

SEQ ID NO: 300          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 300
acacagcggc caaagttggt                                                    20

SEQ ID NO: 301          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 301
ccacacagcg gccaaagttg                                                    20

SEQ ID NO: 302          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 302
aggtccacac agcggccaaa                                                    20

SEQ ID NO: 303          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 303
agaggtccac acagcggcca                                                    20

SEQ ID NO: 304          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 304
aaagaggtcc acacagcggc                                                    20

SEQ ID NO: 305          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 305
caatgatgtc ctcccctggg                                          20

SEQ ID NO: 306          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 306
gaggcaccaa tgatgtcctc                                          20

SEQ ID NO: 307          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 307
gctggaggca ccaatgatgt                                          20

SEQ ID NO: 308          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 308
tcgctggagg caccaatgat                                          20

SEQ ID NO: 309          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 309
agtcgctgga ggcaccaatg                                          20

SEQ ID NO: 310          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 310
gcagtcgctg gaggcaccaa                                          20

SEQ ID NO: 311          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 311
gtgctgcagt cgctggaggc                                          20

SEQ ID NO: 312          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 312
aggtgctgca gtcgctggag                                          20

SEQ ID NO: 313          moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 313
gcaggtgctg cagtcgctgg                                         20

SEQ ID NO: 314         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 314
agcaggtgct gcagtcgctg                                         20

SEQ ID NO: 315         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 315
aaagcaggtg ctgcagtcgc                                         20

SEQ ID NO: 316         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 316
acacaaagca ggtgctgcag                                         20

SEQ ID NO: 317         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 317
tgacacaaag caggtgctgc                                         20

SEQ ID NO: 318         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 318
tcccactctg tgacacaaag                                         20

SEQ ID NO: 319         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 319
tcatggctgc aatgccagcc                                         20

SEQ ID NO: 320         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

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```
SEQUENCE: 320
cagcatcatg gctgcaatgc                                              20

SEQ ID NO: 321          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 321
gacagcatca tggctgcaat                                              20

SEQ ID NO: 322          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 322
cagacagcat catggctgca                                              20

SEQ ID NO: 323          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 323
tcggcagaca gcatcatggc                                              20

SEQ ID NO: 324          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 324
gctcggcaga cagcatcatg                                              20

SEQ ID NO: 325          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 325
cggctcggca gacagcatca                                              20

SEQ ID NO: 326          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 326
tccggctcgg cagacagcat                                              20

SEQ ID NO: 327          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 327
cggccagggt gagctccggc                                              20

SEQ ID NO: 328          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 328
ctctgcctca actcggccag                                                20

SEQ ID NO: 329          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 329
gtctctgcct caactcggcc                                                20

SEQ ID NO: 330          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 330
cagtctctgc ctcaactcgg                                                20

SEQ ID NO: 331          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 331
atcagtctct gcctcaactc                                                20

SEQ ID NO: 332          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 332
ggatcagtct ctgcctcaac                                                20

SEQ ID NO: 333          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 333
gtggatcagt ctctgcctca                                                20

SEQ ID NO: 334          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 334
aagtggatca gtctctgcct                                                20

SEQ ID NO: 335          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 335
gagaagtgga tcagtctctg                                                20
```

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```
SEQ ID NO: 336          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 336
cagagaagtg gatcagtctc                                              20

SEQ ID NO: 337          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 337
ggcagagaag tggatcagtc                                              20

SEQ ID NO: 338          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 338
ctttggcaga gaagtggatc                                              20

SEQ ID NO: 339          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 339
gacatctttg gcagagaagt                                              20

SEQ ID NO: 340          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 340
atgacatctt tggcagagaa                                              20

SEQ ID NO: 341          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 341
attgatgaca tctttggcag                                              20

SEQ ID NO: 342          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 342
tcattgatga catctttggc                                              20

SEQ ID NO: 343          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 343
aggcctcatt gatgacatct                                          20

SEQ ID NO: 344          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 344
ccaggcctca ttgatgacat                                          20

SEQ ID NO: 345          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 345
aaccaggcct cattgatgac                                          20

SEQ ID NO: 346          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 346
ggaaccaggc ctcattgatg                                          20

SEQ ID NO: 347          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 347
gggaaccagg cctcattgat                                          20

SEQ ID NO: 348          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 348
agggaaccag gcctcattga                                          20

SEQ ID NO: 349          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 349
cagggaacca ggcctcattg                                          20

SEQ ID NO: 350          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 350
ctcagggaac caggcctcat                                          20

SEQ ID NO: 351          moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 351
cctcagggaa ccaggcctca                                          20

SEQ ID NO: 352         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 352
tcctcaggga accaggcctc                                          20

SEQ ID NO: 353         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 353
gtcctcaggg aaccaggcct                                          20

SEQ ID NO: 354         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 354
tggtcctcag ggaaccaggc                                          20

SEQ ID NO: 355         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 355
ctggtcctca gggaaccagg                                          20

SEQ ID NO: 356         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 356
gctggtcctc agggaaccag                                          20

SEQ ID NO: 357         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 357
cgctggtcct cagggaacca                                          20

SEQ ID NO: 358         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic oligonucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

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```
SEQUENCE: 358
gtacccgctg gtcctcaggg                                                 20

SEQ ID NO: 359          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 359
cagtacccgc tggtcctcag                                                 20

SEQ ID NO: 360          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 360
acctgcccca tgggtgctgg                                                 20

SEQ ID NO: 361          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 361
caaaacagct gccaacctgc                                                 20

SEQ ID NO: 362          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 362
tcctgcaaaa cagctgccaa                                                 20

SEQ ID NO: 363          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 363
agtcctgcaa aacagctgcc                                                 20

SEQ ID NO: 364          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 364
gctgaccata cagtcctgca                                                 20

SEQ ID NO: 365          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 365
ggccccgagt gtgctgacca                                                 20

SEQ ID NO: 366          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 366
gtgtaggccc cgagtgtgct                                            20

SEQ ID NO: 367          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 367
ccgtgtaggc cccgagtgtg                                            20

SEQ ID NO: 368          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 368
atccgtgtag gccccgagtg                                            20

SEQ ID NO: 369          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 369
ccatccgtgt aggccccgag                                            20

SEQ ID NO: 370          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 370
ggccatccgt gtaggccccg                                            20

SEQ ID NO: 371          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 371
gtggccatcc gtgtaggccc                                            20

SEQ ID NO: 372          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 372
ctggagcagc tcagcagctc                                            20

SEQ ID NO: 373          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 373
aactggagca gctcagcagc                                            20
```

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```
SEQ ID NO: 374          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 374
ggagaaactg gagcagctca                                                20

SEQ ID NO: 375          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 375
ctggagaaac tggagcagct                                                20

SEQ ID NO: 376          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 376
cacctggcaa tggcgtagac                                                20

SEQ ID NO: 377          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 377
gcacctggca atggcgtaga                                                20

SEQ ID NO: 378          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 378
agcacctggc aatggcgtag                                                20

SEQ ID NO: 379          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 379
cagcacctgg caatggcgta                                                20

SEQ ID NO: 380          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 380
gcagcacctg gcaatggcgt                                                20

SEQ ID NO: 381          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 381
ggcagcacct ggcaatggcg                                                20

SEQ ID NO: 382          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 382
aggcagcacc tggcaatggc                                                20

SEQ ID NO: 383          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 383
caggcagcac ctggcaatgg                                                20

SEQ ID NO: 384          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 384
gcaggcagca cctggcaatg                                                20

SEQ ID NO: 385          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 385
tagcaggcag cacctggcaa                                                20

SEQ ID NO: 386          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 386
gtagcaggca gcacctggca                                                20

SEQ ID NO: 387          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 387
tggcctcagc tggtggagct                                                20

SEQ ID NO: 388          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 388
gtccccatgc tggcctcagc                                                20

SEQ ID NO: 389          moltype = DNA  length = 20
```

-continued

```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 389
ggtccccatg ctggcctcag                                              20

SEQ ID NO: 390          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 390
gggtccccat gctggcctca                                              20

SEQ ID NO: 391          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 391
cgggtcccca tgctggcctc                                              20

SEQ ID NO: 392          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 392
acgggtcccc atgctggcct                                              20

SEQ ID NO: 393          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 393
acacgggtcc ccatgctggc                                              20

SEQ ID NO: 394          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 394
gacacgggtc cccatgctgg                                              20

SEQ ID NO: 395          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 395
ggacacgggt ccccatgctg                                              20

SEQ ID NO: 396          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 396
tggacacggg tccccatgct                                              20

SEQ ID NO: 397          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 397
gtggacacgg gtccccatgc                                              20

SEQ ID NO: 398          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 398
agtggacacg ggtccccatg                                              20

SEQ ID NO: 399          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 399
cagtggacac gggtccccat                                              20

SEQ ID NO: 400          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 400
gcagtggaca cgggtcccca                                              20

SEQ ID NO: 401          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 401
ggcagtggac acgggtcccc                                              20

SEQ ID NO: 402          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 402
tggcagtgga cacgggtccc                                              20

SEQ ID NO: 403          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 403
gtggcagtgg acacgggtcc                                              20

SEQ ID NO: 404          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 404
ggtggcagtg gacacgggtc                                             20

SEQ ID NO: 405        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 405
tggtggcagt ggacacgggt                                             20

SEQ ID NO: 406        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 406
tgttggtggc agtggacacg                                             20

SEQ ID NO: 407        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 407
agctgcagcc tgtgaggacg                                             20

SEQ ID NO: 408        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 408
gtgccaaggt cctccacctc                                             20

SEQ ID NO: 409        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 409
tcagcacagg cggcttgtgg                                             20

SEQ ID NO: 410        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 410
ccacgcactg gttgggctga                                             20

SEQ ID NO: 411        moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic oligonucleotide
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 411
actttgcatt ccagacctgg                                             20
```

-continued

```
SEQ ID NO: 412          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 412
gactttgcat tccagacctg                                              20

SEQ ID NO: 413          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 413
tgactttgca ttccagacct                                              20

SEQ ID NO: 414          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 414
ttgactttgc attccagacc                                              20

SEQ ID NO: 415          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 415
tccttgactt tgcattccag                                              20

SEQ ID NO: 416          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 416
cgggattcca tgctccttga                                              20

SEQ ID NO: 417          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 417
gcaggccacg gtcacctgct                                              20

SEQ ID NO: 418          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 418
ggagggcact gcagccagtc                                              20

SEQ ID NO: 419          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 419
cagatggcaa cggctgtcac                                          20

SEQ ID NO: 420          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 420
gcagatggca acggctgtca                                          20

SEQ ID NO: 421          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 421
agcagatggc aacggctgtc                                          20

SEQ ID NO: 422          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 422
cagcagatgg caacggctgt                                          20

SEQ ID NO: 423          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 423
gcagcagatg gcaacggctg                                          20

SEQ ID NO: 424          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 424
cggcagcaga tggcaacggc                                          20

SEQ ID NO: 425          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 425
ccggcagcag atggcaacgg                                          20

SEQ ID NO: 426          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 426
tccggcagca gatggcaacg                                          20

SEQ ID NO: 427          moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 427
ctccggcagc agatggcaac                                             20

SEQ ID NO: 428          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 428
gctccggcag cagatggcaa                                             20

SEQ ID NO: 429          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 429
ccaggtgccg gctccggcag                                             20

SEQ ID NO: 430          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 430
gaggcctgcg ccaggtgccg                                             20

SEQ ID NO: 431          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 431
atgagggcca tcagcacctt                                             20

SEQ ID NO: 432          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 432
gatgagggcc atcagcacct                                             20

SEQ ID NO: 433          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 433
agatgagggc catcagcacc                                             20

SEQ ID NO: 434          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 434
gagatgaggg ccatcagcac                                        20

SEQ ID NO: 435          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 435
tggagatgag ggccatcagc                                        20

SEQ ID NO: 436          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 436
ctggagatga gggccatcag                                        20

SEQ ID NO: 437          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 437
gctggagatg agggccatca                                        20

SEQ ID NO: 438          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 438
agctggagat gagggccatc                                        20

SEQ ID NO: 439          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 439
gctagatgcc atccagaaag                                        20

SEQ ID NO: 440          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 440
ggctagatgc catccagaaa                                        20

SEQ ID NO: 441          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 441
tggctagatg ccatccagaa                                        20

SEQ ID NO: 442          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 442
ctggctagat gccatccaga                                              20

SEQ ID NO: 443          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 443
ctctggctag atgccatcca                                              20

SEQ ID NO: 444          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 444
cctctggcta gatgccatcc                                              20

SEQ ID NO: 445          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 445
gcctctggct agatgccatc                                              20

SEQ ID NO: 446          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 446
agcctctggc tagatgccat                                              20

SEQ ID NO: 447          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 447
acccttggtc acgccggcat                                              20

SEQ ID NO: 448          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 448
ctgcccttcc accaaaatgc                                              20

SEQ ID NO: 449          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 449
actgcccttc caccaaaatg                                              20
```

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```
SEQ ID NO: 450          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 450
cactgccctt ccaccaaaat                                              20

SEQ ID NO: 451          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 451
gcactgccct tccaccaaaa                                              20

SEQ ID NO: 452          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 452
gggcactgcc cttccaccaa                                              20

SEQ ID NO: 453          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 453
tgggcactgc ccttccacca                                              20

SEQ ID NO: 454          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 454
ctgggcactg cccttccacc                                              20

SEQ ID NO: 455          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 455
gctgggcact gcccttccac                                              20

SEQ ID NO: 456          moltype = DNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = Synthetic oligonucleotide
source                  1..19
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 456
gagcaacttc ggaggcagc                                               19

SEQ ID NO: 457          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
```

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                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 457
gcctcagtct gcttcgcacc                                              20

SEQ ID NO: 458           moltype = DNA  length = 14
FEATURE                  Location/Qualifiers
misc_feature             1..14
                         note = Synthetic oligonucleotide
source                   1..14
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 458
gcctatgagg gtgc                                                    14

SEQ ID NO: 459           moltype = DNA  length = 14
FEATURE                  Location/Qualifiers
misc_feature             1..14
                         note = Synthetic oligonucleotide
source                   1..14
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 459
tccaggccta tgag                                                    14

SEQ ID NO: 460           moltype = DNA  length = 14
FEATURE                  Location/Qualifiers
misc_feature             1..14
                         note = Synthetic oligonucleotide
source                   1..14
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 460
cagctcagca gctc                                                    14

SEQ ID NO: 461           moltype = DNA  length = 14
FEATURE                  Location/Qualifiers
misc_feature             1..14
                         note = Synthetic oligonucleotide
source                   1..14
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 461
ttaatcaggg agcc                                                    14

SEQ ID NO: 462           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic oligonucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 462
ggtaaggtgc ggtaagtcct                                              20

SEQ ID NO: 463           moltype = DNA  length = 3512
FEATURE                  Location/Qualifiers
source                   1..3512
                         mol_type = genomic DNA
                         organism = Mus musculus
SEQUENCE: 463
tggggattaa gaggggggaa tgtaacaggt tttgtcctca ctctgagcgt catttgacgc  60
cctttagtac cggggcccgt taatgtttaa tggcgtgatc tcccggcccc caggcgtcca  120
acgtggacgc gcaggctgcc ggtgggctcc ctgagccagt ccttggctcc ccagagacat  180
cgagtcccag gcgtccatgt ccttcccgag cacccactgc tctgcgtggc tgcggtggcc  240
gctgttgctg ctactgtgcc ccaccggcgc agagctgatg ctcgccctcc cgtcccagga  300
ggccaccgcc accttccgcc gttgctccaa tgtggtgctg atggaggaga cccagaggct  360
gacccgggct gcccgccggg gctatgtcat ccctggcttc ttggtgaaga tgagcagtga  420
tgtggagtac attgaggaag actcctttgt gcgaattatc ccagcatggc accagacaga  480
ggtggaggtg tatctcttag ataccagcat ggtcaccatc accgacttca acagcgtgcc  540
ggcgagcaag tgtgacagcc acggcaccca tggtgtggcc aagggcacca gcctgcacag  600
cacagtcagc ggcaccctca taggcctgga cccgtttgca gcccaattag tgtctgggga  660
gggcgaggcc tcagagagga tcttccgatg gtacccacac cccagaaggc cgttctctct  720
ctctttctga cacggcccgc agccccggag gccgcgcgca cctctcctcg gctgttgccg  780
ctgttgccgc tggtgcccag gacgaggatg ggatggcctg gctgatgagg ggaggcctgg  840
aggctgccag acagattgaa caaactgccc caaggttcta catatctttt cctgttgggc  900
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ctggccctga cttcgcccag agcatcccat ggaagaccgc tcccctgatg ccagggtgcc  960
catcgggaga ggaggaggat gggacacgct cctggcaggt gtggtcagcg cctgcgtgtg  1020
ctcaactgtc gtttattcgg aagagtcagc ctcggggcca ctcgtggttc tgctgcccct  1080
ggccggtggg tatagccgca tgcctgccgg cacctggcga ggactggggt ggtgctggtt  1140
gcagcagctg ggacgacgcc tgcctctact ccccagcttc tgctccagag gtcatcacag  1200
gaatgcccag gaccagccag ttaccttggg gactttgggg actaattttg gatttggggt  1260
gaaacctgat gggctcgggg ttccaccttc ggctagagga ccaagtgccc ccccgatggg  1320
cgctgctgct gagattatga ccgcacatgt gaacctacat accgcctgca atgacctctt  1380
agttgcccca ggaacctgga gaagcagcca ttgagggcag tccacagaca gccgggatgc  1440
aagggaaggg taatccagcc tcctcaacgc ggaacttccg tcggggccac gacgctgtgt  1500
ggatctcttt gccccccgga aggacatcat cggagcgtcc agtgactgca gcacatgctt  1560
catgtcacag agtgggacct cacaggctgc tgcccacgtg gccggcattg tggctcggat  1620
gctgagccgg gagcccacac ttaccctggc cgagctgcgg cagaggctga tccacttctc  1680
taccaaagac gtcatcaaca tggcctggtt ccctgaggac cagcaggtgc tgaccccccaa  1740
cctggtggcc acactgcccc ccagcaccca tgagacaggc gggcagctgc tctgtaggac  1800
ggtgtggtcg gcacactcgg ggcccactcg aacagctaca gctacagccc gctgtgcccc  1860
agaagaggag ctgctgagct gctccagctt ctccaggagc gggaggcgtc gtggtgattg  1920
gattgaggcc ataggaggcc agcaggtctg caaggccctc aatgcatttg ggggtgaggg  1980
tgtctatgcc gtcgcgagat gctgcctggt tccccgtgcc aactgcagca tccacaacac  2040
ccctgcagcc agagctggcc tggagaccca tgtccactgc caccagaagg accatgttct  2100
cacaggctgc agcttccatt gggaagtgga agaccttagt gtccggaggc agcctgcgct  2160
gaggtccaga cgtcagcctg gccagtgcgt tggccaccag gcggccagtg tctatgcttc  2220
ctgctgccat gccccagggc tggaatgcaa aatcaaggag catgggatct caggtccttc  2280
agagcaggtc actgtggcct gcgaagcagg atggaccctg actggatgca atgtgctccc  2340
tggggcatcc ctcactctgg gagcctacag cgtggacaac ctgtgtgtgg caagagtcca  2400
tgacactgcc agagcagaca ggaccagtgg agaagccaca gtagctgctg ccatctgctg  2460
ccggagccgg ccttcagcaa aggcctcctg ggttcagtga cagcctcagg cagggatggt  2520
gcttgaggct gggtgcagag atatgcctgc atggctctct tgtagccaaa ggtggggaga  2580
ttctgcgtgg gagaacttgg tgtctcaccc tgggtaccca ttcctggtgt atggaagcac  2640
ctccttcacg gtcagggggc ctgtgcttgg ctttctgcca tcagacatta agctgtagct  2700
ggctctggcc agctgctcca gtgtaccaga acctgaggat gctcgctgca aggcctcagt  2760
tctcaggcct tagggtgtat ttgtctttca ggaagatcat aatggacaga gatccttgga  2820
ggttcaaaga ccaagtacca gactggaaaa ttgagtctga aagccacaag gacagtcaac  2880
tcacagccag ctcacattgc agacaccatt ttgggctccc tgattaaatg cagatcagtt  2940
ctgcacacct ccaggggtgg atccagctgt aaggccatac ctatatcttc cagatgtcct  3000
catctgctgc agggctttgg ccctgctcag gataatgtgc tatgagccct catctgactc  3060
tcagtttgta ctggagaacc atacaggact taccgcacct taccccatcc actaccatgt  3120
gcactgactg gcctcatttt atgaaggaag agacaggacc agagaggcga tgtcacacag  3180
ccagtgatgt caggacataa attcagagtg gctggccctg aataatgcca ggctgggcag  3240
cgagaggaca ggctatggct tgctcctgga cctatactcc cttagcccca gtcccacaga  3300
tcaggtggag actggagtga cagagggcga ctgtaccaag gccacaccag ctgaccagca  3360
cacctctatc cttttgagct cttctgtctt tttatagtaa gcttcctcca cctgtgttgc  3420
ttttgtaact tgatatttat gcagggtttt gtagttttta ttatgtagtg acttttcaga  3480
ataaaagcag ctgatgtgac tgactgcatc cg                                3512

SEQ ID NO: 464          moltype =   length =
SEQUENCE: 464
000

SEQ ID NO: 465          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 465
ccttccctga aggttcctcc                                              20

SEQ ID NO: 466          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic oligonucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 466
gggctcatag cacattatcc                                              20
```

What is claimed is:

1. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-

991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

2. The compound of claim **1**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 1.

3. The compound of claim **2**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 1.

4. The compound of claim **1**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

5. The compound of claim **2**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

6. The compound of claim **3**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-

1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

7. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases fully complementary to an equal length portion of nucleobases 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

8. The compound of claim **7**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 1.

9. The compound of claim **7**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 1.

10. The compound of claim **7**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

11. The compound of claim **8**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

12. The compound of claim **9**, wherein the modified oligonucleotide hybridizes exclusively within 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-

859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

13. The compound of claim **1**, consisting of a single-stranded modified oligonucleotide.

14. The compound of claim **13**, wherein at least one internucleoside linkage is a phosphorothioate internucleoside linkage.

15. The compound of claim **1**, wherein each internucleoside linkage is a phosphorothioate internucleoside linkage.

16. The compound of claim **15**, wherein at least one nucleoside comprises a modified sugar.

17. The compound of claim **16**, wherein at least one modified sugar is a bicyclic sugar.

18. The compound of claim **17**, wherein at least one modified sugar comprises a 2'-O-methoxyethyl.

19. The compound of claim **15**, wherein at least one nucleoside comprises a modified nucleobase.

20. The compound of claim **19**, wherein the modified nucleobase is a 5-methylcytosine.

21. The compound of claim **1**, wherein the modified oligonucleotide comprises:

a gap segment consisting of linked deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; and wherein each nucleoside of each wing segment comprises a modified sugar.

22. The compound of claim **21**, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of five linked nucleosides; and

a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar; and wherein each internucleoside linkage is a phosphorothioate linkage.

23. The antisense compound of claim **21**, wherein the antisense compound is a chimeric oligonucleotide consisting of:

a gap segment consisting of from 10 to 18 linked 2'-deoxynucleosides;

a 5' wing segment consisting of from 1 to 5 linked nucleosides; and

a 3' wing segment consisting of from 1 to 5 linked nucleosides;

wherein each nucleoside of the 5' wing segment is a sugar-modified nucleoside, each nucleoside of the 3' wing segment is a sugar-modified nucleoside, and each internucleoside linkage is a phosphorothioate internucleoside linkage.

24. The compound of claim **2**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

25. A composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, and 3543-3569 of SEQ ID NO: 1, or a salt thereof and a pharmaceutically acceptable carrier or diluent.

26. The composition of claim **25**, wherein the modified oligonucleotide is a single-stranded oligonucleotide.

27. The composition of claim **25**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

28. A method comprising administering to an animal a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-

619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

29. The method of claim **28**, wherein the animal is a human.

30. The method of claim **29**, wherein administering the compound slows progression and/or ameliorates hypercholesterolemia, acute coronary syndrome, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, or improves cardiovascular outcome, or any combination thereof.

31. The method of claim **28** comprising co-administering the compound and at least one additional therapy.

32. The method of claim **31**, wherein the compound and additional therapy are administered concomitantly.

33. The method of claim **31**, wherein the compound and the additional therapy are administered in the same formulation.

34. The method of claim **31**, wherein the additional therapy is a lipid lowering therapy.

35. The method of claim **34**, wherein the lipid lowering therapy is a therapeutic lifestyle change, HMG-CoA reductase inhibitor, cholesterol absorption inhibitor, MTP inhibitor, antisense compound targeted to ApoB, or any combination thereof.

36. The method of claim **35**, wherein the HMG-CoA reductase inhibitor is atorvastatin, rosuvastatin, or simvastatin.

37. The method of claim **35**, wherein the cholesterol absorption inhibitor is ezetimibe.

38. The method of claim **35**, wherein the additional lipid lowering therapy comprises an HMG-CoA reductase inhibitor that is simivastatin, and a cholesterol absorption inhibitor that is ezetimibe.

39. The method of claim **35**, wherein the antisense compound is ISIS 301012.

40. The method of claim **28**, wherein the administering is parenteral administration.

41. The method of claim **40**, wherein the parenteral administration comprises subcutaneous or intravenous administration.

42. The method of claim **28**, wherein the administering comprises oral administration.

43. A method comprising administering to a human with hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, or elevated non-HDL cholesterol, a therapeutically effective amount of a composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases complementary to an equal-length portion of 294-317, 406-440, 406-526, 410-436, 410-499, 446-526, 545-581, 591-619, 591-704, 591-743, 595-622, 600-626, 600-639, 600-670, 601-628, 602-628, 603-630, 611-636, 620-647, 638-665, 648-674, 657-684, 705-743, 782-810, 821-859, 835-859, 835-917, 835-942, 860-887, 860-899, 860-909, 860-917, 869-895, 878-905, 888-909, 923-952, 960-1034, 960-1173, 960-986, 967-991, 970-1023, 970-1064, 970-1117, 970-996, 977-1004, 985-1011, 989-1016, 992-1019, 997-1024, 997-1024, 998-1025, 999-1026, 1000-1027, 1001-1028, 1002-1029, 1003-1029, 1004-1029, 1005-1029, 1006-1029, 1007-1034, 1036-1061, 1045-1072, 1076-1096, 1088-1115, 1098-1123, 1200-1251, 1210-1237, 1219-1245, 1228-1251, 1273-1444, 1295-1316, 1318-1345, 1328-1354, 1337-1361, 1344-1371, 1354-1377, 1380-1406, 1389-1416, 1400-1426, 1409-1434, 1465-1491, 1465-1602, 1474-1499, 1482-1519, 1513-1540, 1523-1549, 1526-1602, 1526-1624, 1532-1558, 1541-1568, 1552-1579, 1560-1587, 1561-1589, 1564-1591, 1565-1592, 1566-1592, 1567-1592, 1570-1597, 1571-1599, 1605-1706, 1628-1706, 1640-1666, 1672-1698, 1681-1706, 1735-1761, 1735-1765, 1740-1765, 1849-1876, 1849-1879, 1850-1877, 1851-1877, 1852-1878, 1852-1879, 1853-1879, 1854-1879, 1905-1955, 1915-1942, 1916-1943, 1917-1944, 1918-1945, 1919-1946, 1920-1939, 1920-1947, 1921-1948, 1922-1949, 1923-1950, 1924-1951, 1925-1952, 1926-1952, 1927-1952, 1928-1955, 1962-2059, 2040-2126, 2100-2126, 2100-2139, 2100-2206, 2101-2126, 2305-2332, 2305-2354, 2306-2333, 2307-2334, 2308-2334, 2309-2334, 2310-2334, 2410-2434, 2504-2528, 2509-2528, 2582-2625, 2606-2668, 2828-2855, 2832-2851, 2900-2927, 2900-2929, 2902-2927, 2983-3007, 2983-3013, 3227-3252, 3227-3456, 3472-3496, or 3543-3569 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 1.

44. The method of claim **28**, wherein the animal has hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated IDL-cholesterol, or elevated non-HDL cholesterol.

45. The method of claim **44**, wherein the elevated LDL-cholesterol level is above a target level of at least about 100 mg/dL, 130 mg/dL, 160 mg/dL, or 190 mg/dL.

46. The method of claim **44**, wherein the administering the compound to the animal reduces LDL-cholesterol levels.

47. The method of claim **46**, wherein administering the compound results in an LDL-cholesterol level below a target level of at least about 190 mg/dL, 160 mg/dL, 130 mg/dL, 100 mg/dL, 70 mg/dL, or 50 mg/dL.

48. The method of claim **44**, wherein the one or more risk factors is selected from age, smoking, hypertension, low HDL-cholesterol, and a family history of early coronary heart disease.

49. The method of claim **44**, wherein the animal failed to meet the LDL-cholesterol target level on lipid-lowering therapy, did not comply with the recommend therapy, experienced side effects of therapy, or any combination thereof.

50. The method of claim **49**, wherein the lipid-lowering therapy is a statin.

51. The method of claim **49**, wherein the lipid-lowering therapy is ezetimibe.

52. The method of claim **28**, wherein administering results in a reduction of ApoB, LDL-cholesterol, VLDL-cholesterol, Lp(a), small LDL-particle, small VLDL-particle, non-HDL-cholesterol, liver triglyceride level, serum triglycerides, serum phospholipids, or any combination thereof.

53. The method of claim **52**, wherein the ApoB reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

54. The method of claim **53**, wherein the ApoB reduction is between 10% and 80%, between 20% and 70%, between 30% and 60%, or between 30% and 70%.

55. The method of claim **52**, wherein the LDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

56. The method of claim **52**, wherein the VLDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

57. The method of claim **52**, wherein the Lp(a) reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

58. The method of claim **52**, wherein the small LDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

59. The method of claim **52**, wherein the small VLDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

60. The method of claim **52**, wherein the non-HDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

61. The method of claim **52**, wherein the liver triglyceride level reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

62. The method of claim **52**, wherein the administering of the pharmaceutical composition results in a serum triglyceride reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

63. The method of claim **52**, wherein the administering of the pharmaceutical composition results in a serum phospholipid reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

64. The method of claim **52** or **53**, wherein the serum phospholipid is serum oxidized phospholipid.

65. The method of claim **28**, wherein administering results in reduced coronary heart disease risk, or slows, stops or prevents the progression of atherosclerosis, or any combination thereof.

66. The method of claim **28**, wherein administering results in improved cardiovascular outcome of the human.

67. The method of claim **66**, wherein the improved cardiovascular outcome is improved carotid intimal media thickness, improved atheroma thickness, increased HDL-cholesterol, or any combination thereof.

68. The method of claim **28**, wherein administering ameliorates hepatic steatosis, results in lipid lowering, improves LDL/HDL ration, or any combination thereof.

69. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

70. The compound of claim **69**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 2.

71. The compound of claim **70**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 2.

72. The compound of claim **69**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

73. The compound of claim **70**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-

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74. The compound of claim **71**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

75. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases fully complementary to an equal length portion of nucleobases 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-

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76. The compound of claim **75**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 2.

77. The compound of claim **75**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 2.

78. The compound of claim **75**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

79. The compound of claim **76**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-

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80. The compound of claim **77**, wherein the modified oligonucleotide hybridizes exclusively within 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

81. The compound of claim **69**, consisting of a single-stranded modified oligonucleotide.

82. The compound of claim **81**, wherein at least one internucleoside linkage is a phosphorothioate internucleoside linkage.

83. The compound of claim **69**, wherein each internucleoside linkage is a phosphorothioate internucleoside linkage.

84. The compound of claim **83**, wherein at least one nucleoside comprises a modified sugar.

85. The compound of claim **84**, wherein at least one modified sugar is a bicyclic sugar.

86. The compound of claim **85**, wherein at least one modified sugar comprises a 2' methoxyethyl.

87. The compound of claim **83**, wherein at least one nucleoside comprises a modified nucleobase.

88. The compound of claim **87**, wherein the modified nucleobase is a 5-methylcytosine.

89. The compound of claim **69**, wherein the modified oligonucleotide comprises:

a gap segment consisting of linked deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; and wherein each nucleoside of each wing segment comprises a modified sugar.

90. The compound of claim **89**, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of five linked nucleosides; and

a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar; and wherein each internucleoside linkage is a phosphorothioate linkage.

91. The antisense compound of claim **89**, wherein the antisense compound is a chimeric oligonucleotide consisting of:

a gap segment consisting of from 10 to 18 linked 2'-deoxynucleosides;

a 5' wing segment consisting of from 1 to 5 linked nucleosides; and

a 3' wing segment consisting of from 1 to 5 linked nucleosides;

wherein each nucleoside of the 5' wing segment is a sugar-modified nucleoside, each nucleoside of the 3' wing segment is a sugar-modified nucleoside, and each internucleoside linkage is a phosphorothioate internucleoside linkage.

92. The compound of claim **70**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

93. A composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, and 27350-27376 of SEQ ID NO: 2, or a salt thereof, and a pharmaceutically acceptable carrier or diluent.

94. The composition of claim **93**, wherein the modified oligonucleotide is a single-stranded oligonucleotide.

95. The composition of claim **93**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

96. A method comprising administering to an animal a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

97. The method of claim **96**, wherein the animal is a human.

98. The method of claim **97**, wherein administering the compound slows progression and/or ameliorates hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, or improves cardiovascular outcome, or any combination thereof.

99. The method of claim **96**, comprising co-administering the compound and at least one additional therapy.

100. The method of claim **99**, wherein the compound and additional therapy are administered concomitantly.

101. The method of claim **99**, wherein the compound and the additional therapy are administered in the same formulation.

102. The method of claim **99**, wherein the additional therapy is a lipid lowering therapy.

103. The method of claim **102**, wherein the lipid lowering therapy is a therapeutic lifestyle change, HMG-CoA reductase inhibitor, cholesterol absorption inhibitor, MTP inhibitor, antisense compound targeted to ApoB or any combination thereof.

104. The method of claim **103**, wherein the HMG-CoA reductase inhibitor is atorvastatin, rosuvastatin, or simvastatin.

105. The method of claim **103**, wherein the cholesterol absorption inhibitor is ezetimibe.

106. The method of claim **103**, wherein the additional lipid lowering therapy comprises an HMG-CoA reductase inhibitor that is simivastatin and a cholesterol absorption inhibitor that is ezetimibe.

107. The method of claim **103**, wherein the antisense compound is ISIS 301012.

108. The method of claim **96**, wherein the administering is parenteral administration.

109. The method of claim **108**, wherein the parenteral administration comprises subcutaneous or intravenous administration.

110. The method of claim **96**, wherein the administering comprises oral administration.

111. A method comprising administering to a human with hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, or elevated non-HDL cholesterol, a therapeutically effective amount of a composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases complementary to an equal-length portion of 2274-2400, 2274-2575, 2433-2570, 2433-2579, 2549-2575, 2552-2579, 2585-2638, 2605-2638, 3056-3075, 4150-5159, 4306-4325, 5590-5618, 5667-5686, 6444-6463, 6482-6518, 6492-6518, 6528-6555, 6528-6623, 6534-6561, 6535-6562, 6536-6563, 6537-6563, 6538-6565, 6539-6565, 6540-6567, 6541-6567, 6542-6569, 6546-6573, 6557-6584, 6575-6602, 6585-6611, 6594-6621, 6596-6623, 6652-6671, 7099-7118, 7556-7584, 8836-8855, 8948-8967, 9099-9118, 9099-9168, 9130-9168, 9207-9233, 9207-9235, 9209-9235, 10252-10271, 10633-10652, 11308-11491, 12715-12734, 12928-12947, 13681-13700, 13746-13779, 13816-13847, 13903-13945, 13977-14141, 14179-14198, 14267-14286, 14397-14423, 14441-14460, 14494-14513, 14494-14543, 14524-14543, 14601-14650, 14670-14700, 14675-14700, 14801-14828, 14877-14912, 14877-14915, 14877-14973, 14916-14943, 14916-14973, 14925-14951, 14934-14963, 14946-14973, 14979-14998, 15254-15280, 15254-15328, 15264-15290, 15279-15305, 15291-15318, 15292-15319, 15293-15320, 15294-15321, 15294-15321, 15295-15322, 15296-15323, 15297-15323, 15298-15323, 15299-15323, 15300-15323, 15301-15328, 15330-15355, 15330-15490, 15339-15366, 15358-15490, 16134-16153, 16668-16687, 17267-17286, 18377-18427, 18561-18580, 18591-18618, 18591-18646, 18591-18668, 18695-18746, 18705-18730, 18709-18736, 18719-18746, 19203-20080, 19931-19952, 19954-19981, 19964-19990, 19973-19999, 19982-20009, 19992-20016, 20016-20042, 20025-20052, 20036-20062, 20045-20070, 20100-20119, 20188-20207, 20624-20650, 20624-20759, 20629-20804, 20633-20660, 20635-20781, 20643-20662, 20657-20676, 20670-20697, 20680-20706, 20683-20781, 20689-20715, 20698-20725, 20709-20736, 20717-20744, 20718-20745, 20719-20746, 20720-20747, 20721-20748, 20722-20749, 20727-20752, 20735-20759, 20762-21014, 20785-21014, 21082-21107, 21082-21152, 21091-21114, 21118-21144, 21127-21152, 21181-21209, 21181-21211, 21183-21211, 21481-21500, 21589-21608, 21692-21719, 22000-22227, 22096-22115, 22096-22223, 22096-22311, 22133-22160, 22133-22163, 22134-22161, 22135-22162, 22136-22163, 22137-22163, 22138-22163, 22189-22239, 22199-22226, 22199-22227, 22200-22227, 22201-22228, 22202-22229, 22203-22230, 22204-22231, 22205-22232, 22206-22233, 22207-22234, 22208-22235, 22209-22236, 22210-22236, 22210-22239, 22211-22236, 22212-22239, 22292-22311, 23985-24054, 24035-24134, 24095-24121, 24858-24877, 24907-24926, 25413-25432, 25994-26013, 26112-26139, 26112-26161, 26112-27303, 26113-26140, 26114-26141, 26115-26141, 26116-26141, 26117-26141, 26117-26475, 26118-26141, 26120-26141, 26132-26151, 26142-26161, 26217-26241, 26311-26335, 26389-26432, 26456-26576, 26635-26662, 26707-26734, 26707-26736, 26790-26820, 27034-27263, 27279-27303, or 27350-27376 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 2.

112. The method of claim **96**, wherein the animal has hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated IDL-cholesterol, or elevated non-HDL cholesterol.

113. The method of claim **112**, wherein the elevated LDL-cholesterol level is above a target level of at least about 100 mg/dL, 130 mg/dL, 160 mg/dL, or 190 mg/dL.

114. The method of claim **112**, wherein the administering the compound to the animal reduces LDL-cholesterol levels.

115. The method of claim **114**, wherein administering the compound results in an LDL-cholesterol level below a target level of at least about 190 mg/dL, 160 mg/dL, 130 mg/dL, 100 mg/dL, 70 mg/dL, or 50 mg/dL.

116. The method of claim **112**, wherein the one or more risk factors is selected from age, smoking, hypertension, low HDL-cholesterol, and a family history of early coronary heart disease.

117. The method of claim **112**, wherein the animal failed to meet the LDL-cholesterol target level on lipid-lowering therapy, did not comply with the recommend therapy, experienced side effects of therapy, or any combination thereof.

118. The method of claim **117**, wherein the lipid-lowering therapy is a statin.

119. The method of claim **117**, wherein the lipid-lowering therapy is ezetimibe.

120. The method of claim **96**, wherein administering results in a reduction of ApoB, LDL-cholesterol, VLDL-cholesterol, Lp(a), small LDL-particle, small VLDL-particle, non-HDL-cholesterol, liver triglyceride level, serum triglycerides, serum phospholipids, or any combination thereof.

121. The method of claim **120**, wherein the ApoB reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

122. The method of claim **121**, wherein the ApoB reduction is between 10% and 80%, between 20% and 70%, between 30% and 60%, or between 30% and 70%.

123. The method of claim **120**, wherein the LDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

124. The method of claim **120**, wherein the VLDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

125. The method of claim **120**, wherein the Lp(a) reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

126. The method of claim **120**, wherein the small LDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

127. The method of claim **120**, wherein the small VLDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

128. The method of claim **120**, wherein the non-HDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

129. The method of claim **120**, wherein the liver triglyceride level reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

130. The method of claim **120**, wherein the administering of the pharmaceutical composition results in a serum triglyceride reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

131. The method of claim **120**, wherein the administering of the pharmaceutical composition results in a serum phospholipid reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

132. The method of claim **120** or **131**, wherein the serum phospholipid is serum oxidized phospholipid.

133. The method of claim **96**, wherein administering results in reduced coronary heart disease risk, or slows, stops or prevents the progression of atherosclerosis, or any combination thereof.

134. The method of claim **96**, wherein administering results in improved cardiovascular outcome of the human.

135. The method of claim **134**, wherein the improved cardiovascular outcome is improved carotid intimal media thickness, improved atheroma thickness, increased HDL-cholesterol, or any combination thereof.

136. The method of claim **96**, wherein administering ameliorates hepatic steatosis, results in lipid lowering, improves LDL/HDL ration, or any combination thereof.

137. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

138. The compound of claim **137**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 3.

139. The compound of claim **138**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 3.

140. The compound of claim **137**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-

1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

141. The compound of claim **138**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

142. The compound of claim **139**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

143. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases fully complementary to an equal length portion of nucleobases 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

144. The compound of claim **143**, wherein the modified oligonucleotide is at least 95% complementary to SEQ ID NO: 3.

145. The compound of claim **143**, wherein the modified oligonucleotide is 100% complementary to SEQ ID NO: 3.

146. The compound of claim **143**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-

2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

147. The compound of claim **144**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

148. The compound of claim **145**, wherein the modified oligonucleotide hybridizes exclusively within 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

149. The compound of claim **137**, consisting of a single-stranded modified oligonucleotide.

150. The compound of claim **149**, wherein at least one internucleoside linkage is a phosphorothioate internucleoside linkage.

151. The compound of claim **137**, wherein each internucleoside linkage is a phosphorothioate internucleoside linkage.

152. The compound of claim **151**, wherein at least one nucleoside comprises a modified sugar.

153. The compound of claim **152**, wherein at least one modified sugar is a bicyclic sugar.

154. The compound of claim **153**, wherein at least one modified sugar comprises a 2' methoxyethyl.

155. The compound of claim **151**, wherein at least one nucleoside comprises a modified nucleobase.

156. The compound of claim **155**, wherein the modified nucleobase is a 5-methylcytosine.

157. The compound of claim **137**, wherein the modified oligonucleotide comprises:

a gap segment consisting of linked deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; and wherein each nucleoside of each wing segment comprises a modified sugar.

158. The compound of claim **157**, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of five linked nucleosides; and

a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar; and wherein each internucleoside linkage is a phosphorothioate linkage.

159. The antisense compound of claim **157**, wherein the antisense compound is a chimeric oligonucleotide consisting of:

a gap segment consisting of from 10 to 18 linked 2'-deoxynucleosides;

a 5' wing segment consisting of from 1 to 5 linked nucleosides; and

a 3' wing segment consisting of from 1 to 5 linked nucleosides;

wherein each nucleoside of the 5' wing segment is a sugar-modified nucleoside, each nucleoside of the 3' wing segment is a sugar-modified nucleoside, and each internucleoside linkage is a phosphorothioate internucleoside linkage.

160. The compound of claim **137**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

161. A composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3 or a salt thereof, and a pharmaceutically acceptable carrier or diluent.

162. The composition of claim **161**, wherein the modified oligonucleotide is a single-stranded oligonucleotide.

163. The composition of claim **161**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

164. A method comprising administering to an animal a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising a portion which consists of at least 8 contiguous nucleobases complementary to an equal-length portion of 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

165. The method of claim **164**, wherein the animal is a human.

166. The method of claim **165**, wherein administering the compound slows progression and/or ameliorates hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, or improves cardiovascular outcome, or any combination thereof.

167. The method of claim **164**, comprising co-administering the compound and at least one additional therapy.

168. The method of claim **167**, wherein the compound and additional therapy are administered concomitantly.

169. The method of claim **167**, wherein the compound and the additional therapy are administered in the same formulation.

170. The method of claim **167**, wherein the additional therapy is a lipid lowering therapy.

171. The method of claim **170**, wherein the lipid lowering therapy is a therapeutic lifestyle change, HMG-CoA reductase inhibitor, cholesterol absorption inhibitor, MTP inhibitor, antisense compound targeted to ApoB, or any combination thereof.

172. The method of claim **171**, wherein the HMG-CoA reductase inhibitor is atorvastatin, rosuvastatin, or simvastatin.

173. The method of claim **171**, wherein the cholesterol absorption inhibitor is ezetimibe.

174. The method of claim **171**, wherein the additional lipid lowering therapy comprises an HMG-CoA reductase inhibitor that is simivastatin, and a cholesterol absorption inhibitor that is ezetimibe.

175. The method of claim **171**, wherein the antisense compound is ISIS 301012.

176. The method of claim **164**, wherein the administering is parenteral administration.

177. The method of claim **176**, wherein the parenteral administration comprises subcutaneous or intravenous administration.

178. The method of claim **164**, wherein the administering comprises oral administration.

179. A method comprising administering to a human with hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, early onset coronary heart disease, acute coronary syndrome, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, or elevated non-HDL cholesterol, a therapeutically effective amount of a composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases complementary to an equal-length portion of 220-253, 290-321, 377-419, 451-615, 653-672, 741-760, 871-897, 915-934, 968-1017, 1075-1124, 1075-1174, 1144-1174, 1275-1302, 1315-1341, 1351-1389, 1351-1447, 1365-1439, 1390-1417, 1390-1429, 1390-1439, 1399-1425, 1408-1435, 1420-1447, 1453-1482, 1490-1516, 1490-1564, 1500-1526, 1515-1541, 1527-1553, 1527-1554, 1528-1554, 1529-1555, 1529-1556, 1530-1556, 1530-1557, 1531-1557, 1532-1558, 1533-1559, 1534-1559, 1535-1559, 1536-1559, 1537-1564, 1566-1602, 1566-1681, 1606-1626, 1618-1645, 1626-1653, 1684-1703, 1730-1781, 1740-1767, 1749-1775, 1758-1781, 1820-1847, 1820-1877, 1822-2198, 1830-1856, 1839-1865, 1840-1867, 1898-1924, 1898-2035, 1903-2127, 1907-1934, 1911-1938, 1946-1971, 1954-1980, 1959-2035, 1959-2057, 1963-1988, 1967-2035, 1972-1999, 1982-2008, 1991-2018, 1993-2019, 1995-2022, 1996-2023, 1997-2024, 1998-2025, 1999-2025, 2000-2025, 2009-2035, 2038-2139, 2061-2139, 2073-2099, 2078-2104, 2105-2131, 2112-2139, 2168-2198, 2170-2177, 2245-2284, 2295-2394, 2355-2381, 2355-2394, 2405-2461, 2560-2587, 2560-2609, 2561-2588, 2562-2589, 2563-2589, 2564-2589, 2565-2589, 2566-2589, 2567-2589, 2568-2589, 2665-2689, 2759-2783, 2837-2880, 2904-2923, 3005-3024, 3005-3174, 3083-3110, 3155-3184, 3238-3268, 3482-3711, 3727-3751, or 3798-3824 of SEQ ID NO: 3, and wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to SEQ ID NO: 3.

180. The method of claim **164**, wherein the animal has hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated IDL-cholesterol, or elevated non-HDL cholesterol.

181. The method of claim **180**, wherein the elevated LDL-cholesterol level is above a target level of at least about 100 mg/dL, 130 mg/dL, 160 mg/dL, or 190 mg/dL.

182. The method of claim **180**, wherein the administering the compound to the animal reduces LDL-cholesterol levels.

183. The method of claim **182**, wherein administering the compound results in an LDL-cholesterol level below a target level of at least about 190 mg/dL, 160 mg/dL, 130 mg/dL, 100 mg/dL, 70 mg/dL, or 50 mg/dL.

184. The method of claim **180**, wherein the one or more risk factors is selected from age, smoking, hypertension, low HDL-cholesterol, and a family history of early coronary heart disease.

185. The method of claim **180**, wherein the animal failed to meet the LDL-cholesterol target level on lipid-lowering therapy, did not comply with the recommend therapy, experienced side effects of therapy, or any combination thereof.

186. The method of claim **185**, wherein the lipid-lowering therapy is a statin.

187. The method of claim **185**, wherein the lipid-lowering therapy is ezetimibe.

188. The method of claim **164**, wherein administering results in a reduction of ApoB, LDL-cholesterol, VLDL-cholesterol, Lp(a), small LDL-particle, small VLDL-particle, non-HDL-cholesterol, liver triglyceride level, serum triglycerides, serum phospholipids, or any combination thereof.

189. The method of claim **188**, wherein the ApoB reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

190. The method of claim **189** wherein the ApoB reduction is between 10% and 80%, between 20% and 70%, between 30% and 60%, or between 30% and 70%.

191. The method of claim **188**, wherein the LDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

192. The method of claim **188**, wherein the VLDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

193. The method of claim **188**, wherein the Lp(a) reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

194. The method of claim **188**, wherein the small LDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

195. The method of claim **188**, wherein the small VLDL-particle reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

196. The method of claim **188**, wherein the non-HDL-cholesterol reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

197. The method of claim **188**, wherein the liver triglyceride level reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

198. The method of claim **188**, wherein the administering of the pharmaceutical composition results in a serum triglyceride reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

199. The method of claim **188**, wherein the administering of the pharmaceutical composition results in a serum phospholipid reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

200. The method of claim **188** or **199**, wherein the serum phospholipid is serum oxidized phospholipid.

201. The method of claim **164**, wherein administering results in reduced coronary heart disease risk, or slows, stops or prevents the progression of atherosclerosis, or any combination thereof.

202. The method of claim **164**, wherein administering results in improved cardiovascular outcome of the human.

203. The method of claim **202**, wherein the improved cardiovascular outcome is improved carotid intimal media thickness, improved atheroma thickness, increased HDL-cholesterol, or any combination thereof.

204. The method of claim **164**, wherein administering ameliorates hepatic steatosis, results in lipid lowering, improves LDL/HDL ration, or any combination thereof.

205. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 12 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in SEQ ID NOs: 4-457.

206. The compound of claim **205** consisting of a single-stranded modified oligonucleotide.

207. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 80% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

208. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 85% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

209. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 90% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

210. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 95% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

211. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 99% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

212. The compound of claim **206**, wherein the nucleobase sequence of the modified oligonucleotide is 100% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

213. The compound of claim **206**, wherein at least one internucleoside linkage is a phosphorothioate internucleoside linkage.

214. The compound of claim **213**, wherein each internucleoside linkage is a phosphorothioate internucleoside linkage.

215. The compound of claim **206**, wherein at least one nucleoside comprises a modified sugar.

216. The compound of claim **215**, wherein at least one modified sugar is a bicyclic sugar.

217. The compound of claim **215**, wherein at least one modified sugar comprises a 2'-O-methoxyethyl.

218. The compound of claim **206**, wherein at least one nucleoside comprises a modified nucleobase.

219. The compound of claim **218**, wherein the modified nucleobase is a 5-methylcytosine.

220. The compound of claim **205**, wherein the modified oligonucleotide comprises:

a gap segment consisting of linked deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; and wherein each nucleoside of each wing segment comprises a modified sugar.

221. The compound of claim **220**, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of five linked nucleosides; and

a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar; and wherein each internucleoside linkage is a phosphorothioate linkage.

222. The antisense compound of claim **220**, wherein the antisense compound is a chimeric oligonucleotide consisting of:

a gap segment consisting of from 10 to 18 linked 2'-deoxynucleosides;

a 5' wing segment consisting of from 1 to 5 linked nucleosides; and

a 3' wing segment consisting of from 1 to 5 linked nucleosides;

wherein each nucleoside of the 5' wing segment is a sugar-modified nucleoside, each nucleoside of the 3' wing segment is a sugar-modified nucleoside, and each internucleoside linkage is a phosphorothioate internucleoside linkage.

223. The compound of claim **206**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

224. A composition comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 12 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in SEQ ID NOs: 4-457 or a salt thereof and a pharmaceutically acceptable carrier or diluent.

225. The composition of claim **224**, wherein the modified oligonucleotide is a single-stranded oligonucleotide.

226. The composition of claim **224**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

227. A method comprising administering to an animal a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 12 contiguous nucleobases of a nucleobase sequence selected from among the nucleobase sequences recited in SEQ ID NOs: 4-457.

228. The method of claim **227**, wherein the animal is a human.

229. The method of claim **228**, wherein administering the compound slows progression and/or ameliorates hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, or improves cardiovascular outcome, or any combination thereof.

230. The method of claim **228**, comprising co-administering the compound and at least one additional therapy.

231. The method of claim **230**, wherein the compound and additional therapy are administered concomitantly.

232. The method of claim **230**, wherein the compound and the additional therapy are administered in the same formulation.

233. The method of claim **230**, wherein the additional therapy is a lipid lowering therapy.

234. The method of claim **233**, wherein the lipid lowering therapy is a therapeutic lifestyle change, HMG-CoA reductase inhibitor, cholesterol absorption inhibitor, MTP inhibitor, antisense compound targeted to ApoB, or any combination thereof.

235. The method of claim **234**, wherein the HMG-CoA reductase inhibitor is atorvastatin, rosuvastatin, or simvastatin.

236. The method of claim **234**, wherein the cholesterol absorption inhibitor is ezetimibe.

237. The method of claim **234**, wherein the antisense compound is ISIS 301012.

238. The method of claim **227**, wherein the administering is parenteral administration.

239. The method of claim **228**, wherein the parenteral administration comprises subcutaneous or intravenous administration.

240. The method of claim **227**, wherein the administering comprises oral administration.

241. The method of claim **227**, wherein the animal has hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, acute coronary syndrome, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, atherosclerosis, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated LDL-cholesterol, elevated non-HDL cholesterol, or any combination thereof.

242. The method of claim **241**, wherein the elevated LDL-cholesterol level is above a target level of at least about 100 mg/dL, at least 130 mg/dL, at least 160 mg/dL, or at least 190 mg/dL.

243. The method of claim **241**, wherein the one or more risk factors is selected from age, smoking, hypertension, low HDL-cholesterol, and a family history of early coronary heart disease.

244. The method of claim **227**, wherein the animal failed to meet a LDL-cholesterol target level on lipid-lowering therapy, did not comply with the recommend therapy, experienced side effects of therapy, or any combination thereof.

245. The method of claim **244**, wherein the lipid-lowering therapy is a statin or ezetimibe.

246. The method of claim **227**, wherein administering results in a reduction of ApoB, LDL-cholesterol, VLDL-cholesterol, Lp(a), small LDL-particle, small VLDL-particle, non-HDL-cholesterol, liver triglyceride level, serum triglycerides, serum phospholipids, or any combination thereof.

247. The method of claim **246**, wherein administering the compound results in a reduction of LDL-cholesterol level below a target level of at least about 190 mg/dL, at least 160 mg/dL, at least 130 mg/dL, at least 100 mg/dL, 70 mg/dL, or 50 mg/dL.

248. The method of claim **247**, wherein the reduction is at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%.

249. The method of claim **248**, wherein the reduction is between 10% and 80%, between 20% and 70%, between 30% and 60%, or between 30% and 70%.

250. The method of claim **246**, wherein the serum phospholipid is serum oxidized phospholipid.

251. The method of claim **229**, wherein the improved cardiovascular outcome is improved carotid intimal media thickness, improved atheroma thickness, increased HDL-cholesterol, or any combination thereof.

252. The method of any of claims **227-251**, comprising administering the pharmaceutical composition every day, every two days, every three days, every four days, every five days, every six days, or every 7 days.

253. The method of any of claims **227-251**, comprising administering the pharmaceutical composition every week, every two weeks, every three weeks, or every four weeks.

254. The method of any of claims **227-251**, comprising administering the pharmaceutical composition every month, every two months, every three months, every four months, or every six months.

255. The method of any of claims **227-251**, wherein the dose is 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 270, 280, 285, 290, 295, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, 400, 405, 410, 415, 420, 425, 430, 435, 440, 445, 450, 455, 460, 465, 470, 475, 480, 485, 490, 495, 500, 505, 510, 515, 520, 525, 530, 535, 540, 545, 550, 555, 560, 565, 570, 575, 580, 585, 590, 595, 600, 605, 610, 615, 620, 625, 630, 635, 640, 645, 650, 655, 660, 665, 670, 675, 680, 685, 690, 695, 700, 705, 710, 715, 720, 725, 730, 735, 740, 745, 750, 755, 760, 765, 770, 775, 780, 785, 790, 795, or 800 mg.

256. The method of any of claims **227-251**, wherein the dose is 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 270, 280, 285, 290, 295, or 300 mg.

257. The method of any of claims **227-251**, wherein the dose is 100, 125, 150, 175, 200, 225, 250, 275, 300, 325, 350, 375, or 400 mg.

258. The method of any of claims **227-251**, wherein the pharmaceutical composition comprises an antisense oligonucleotide concentration of about 50 mg/ml, about 75 mg/ml, about 100 mg/ml, about 125 mg/ml, about 150 mg/ml, about 175 mg/ml, about 200 mg/ml, about 225 mg/ml, or about 250 mg/ml.

259. A method of inhibiting the expression of PCSK9 in human cells or tissues comprising contacting said cells or tissues with a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides targeted to a nucleic acid molecule encoding human PCSK9.

260. A method of treating hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, early onset coronary heart disease, acute coronary syndrome, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated LDL-cholesterol, atherosclerosis, elevated non-HDL cholesterol, or any combination thereof, by administering to a human having hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, coronary heart disease, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated ILDL-cholesterol, elevated non-HDL cholesterol, or any combination thereof, a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides targeted to a nucleic acid molecule encoding human PCSK9.

261. The method of claim **259**, wherein treating results in slowed progression and/or amelioration of hypercholesterolemia, polygenic hypercholesterolemia, mixed dyslipidemia, a history of coronary heart disease, acute coronary syndrome, coronary heart disease, early onset coronary heart disease, one or more risk factors for coronary heart disease, type II diabetes, type II diabetes with dyslipidemia, hepatic steatosis, non-alcoholic steatohepatitis, non-alcoholic fatty liver disease, hypertriglyceridemia, hyperfattyacidemia, hyperlipidemia, metabolic syndrome, elevated ApoB, elevated cholesterol, elevated LDL-cholesterol, elevated VLDL-cholesterol, elevated IDL-cholesterol, atherosclerosis, elevated non-HDL cholesterol, or any combination thereof.

262. The method of claim **259**, wherein the compound is 80% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

263. The method of claim **259**, wherein the compound is 85% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

264. The method of claim **259**, wherein the compound is 90% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

265. The method of claim **259**, wherein the compound is 95% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

266. The method of claim **259**, wherein the compound is 99% complementary to a nucleic acid sequence encoding PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

267. The method of claim **259**, wherein the compound is 100% complementary to a nucleic acid sequence encoding human PCSK9 selected from among the nucleic acid sequences recited in SEQ ID NOs: 1-3.

268. The method of claim **259**, wherein the compound consists of a single-stranded modified oligonucleotide.

269. The method of claim **259**, wherein the compound comprises at least one phosphorothioate internucleoside linkage.

270. The method of claim **269**, wherein each internucleoside linkage in the compound is a phosphorothioate internucleoside linkage.

271. The method of claim **259**, wherein the compound comprises a modified sugar.

272. The method of claim **271**, wherein the modified sugar is a bicyclic sugar.

273. The method of claim **272**, wherein the modified sugar is a 2'-O-methoxyethyl.

274. The method of claim **259**, wherein at least one nucleoside of the compound comprises a modified nucleobase.

275. The method of claim **274**, wherein the modified nucleobase is a 5-methylcytosine.

276. The method of claim **259**, wherein the modified oligonucleotide comprises:

a gap segment consisting of linked deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; and wherein each nucleoside of each wing segment comprises a modified sugar.

277. The method of claim **259**, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of five linked nucleosides; and

a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar; and wherein each internucleoside linkage is a phosphorothioate linkage.

278. The method of claim **259**, wherein the modified oligonucleotide consists of 20 linked nucleosides.

279. The method of claim **259**, wherein the administering is parenteral administration.

280. The method of claim **259**, wherein the paternteral administration comprises subcutaneous or intravenous administration.

281. The method of claim **259**, wherein the administering is an oral administration.

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